US Army Corps
of Engineers

LifeSim

Life Loss Estimation

User's Manual

Version 2.0
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14. ABSTRACT
The U.S. Army Corps of Engineers (USACE) Life Loss Estimation software (LifeSim) is a spatially-distributed dynamic simulation modeling system for estimating potential life loss and direct economic damages from natural and dam and levee failure floods. The main purpose of the LifeSim software is to help USACE study teams better understand the consequences of a flood event. Although flooding can lead to many types of severe consequences, the primary objective of the USACE Dam and Levee Safety program is to manage risk to the public who rely on those structures to keep them reasonably safe from flooding.

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LifeSim, USACE, life loss, spatially; dynamic, simulation, modeling, economic, damages, levee failure, floods, population at risk, PAR, loss, shelter; software; risk assessment; dam; levee safety, analysis, hazard; emergency; hydraulic data; HEC-RAS, grid set; structure inventories; damage category; occupancy type; construction type; stability criteria; submergence criteria, emergency planning zone, EPZ, road network data, shapefile, destination data, alternatives, simulations, results, animation, uncertainty, arrival, attribute, feature, properties, gridded map, vertex

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LifeSim

User's Manual

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LifeSim, User's Manual

Use of the software described by this document is controlled by certain terms and conditions. The user must acknowledge and agree to be bound by the terms and conditions of usage before the software can be installed or used. For reference, a copy of the terms and conditions of usage are included in Appendix L of this document so that they may be examined before obtaining and loading the software. The software described by this document can be downloaded for free from our internet site (https://www.rmc.usace.army.mil/Software/).

RMC cannot provide technical support for this software to non-USACE users; therefore, non-USACE individuals and organizations should use any internet search engine to locate a vendor that can provide support for the software of interest. However, RMC will respond to all documented instances of program errors. Documented errors are bugs in the software due to programming mistakes not model problems due to user-entered data.

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CHAPTER 1

Introduction

The U.S. Army Corps of Engineers (USACE) Life Loss Estimation software (LifeSim) is a spatially-distributed dynamic simulation modeling system for estimating potential life loss and direct economic damages from natural and dam and levee failure floods. LifeSim accounts explicitly for the impact of warning issuance time; warning diffusion; the population at risk's (PAR's) protective action initiation; the PAR's evacuation potential; detailed flood dynamics; and, loss of shelter on loss of life. An agent-based approach has been developed to track individuals throughout the warning and evacuation process. The software can be used to provide inputs for risk assessment and to explore options for improving the effectiveness of a dam owner's emergency plans or a local authority's response plans. While LifeSim was developed for dam and levee safety analyses, the software is not just limited to flood hazards. LifeSim can be applied to any type of hazard that can be converted into a time series of gridded data including fires, toxic gas, and hurricanes.

Monte Carlo sampling techniques are used in LifeSim to capture the natural variability of temporal uncertainty in the warning and evacuation process as well as the potential for life loss when put in threatening flood situations. Many model parameters can be defined with uncertainty in LifeSim, providing a means of capturing knowledge uncertainty. The natural variability and knowledge uncertainty parameters will vary offering a range of potential life loss and economic damage results.

- Knowledge uncertainty (epistemic uncertainty) random variables are not known exactly and exist when it is not possible to make an accurate estimate of the value of an input. A few examples are flood likelihood, hydraulic coefficients, and channel capacities.

- Natural variability (aleatoric uncertainty) random variables vary spatially or event-to-event and exist when the outcome of a natural process is random, or the natural process is so complex it seems random when viewed in isolation. A few examples are flood magnitude and forecasts.

A better understanding of the factors driving consequences are gained by reviewing the distribution of results as opposed to a single deterministic result. LifeSim can be used to provide inputs for risk assessment and to explore options for improving the effectiveness of a dam owner's emergency plans or a local authority's response plans.

1.1 Purpose

The main purpose of the LifeSim software is to help USACE study teams better understand the consequences of a flood event. Although flooding can lead to many types of severe consequences, the primary objective of the USACE Dam and Levee Safety program is to manage risk to the public who rely on those structures to keep them reasonably safe from flooding. Thus, reducing the risk associated with life loss is paramount.

LifeSim was developed to:

- Effectively support reduction of life-safety risks associated with flooding,
- Evaluate existing and residual risks against tolerable risk guidelines,
- Understand life-loss dynamics associated with floods, and
- Create or improve existing emergency action plans.

LifeSim provides a practical life loss estimation approach. The software is designed to facilitate:

- Entry of appropriate data into each of the individual simulation required components,
- The flexibility to apply new research as data inputs to the model,
- Review simulation results from a high level down to individual details, and
- View simulation results as map animations to help facilitate review of results and discussion with emergency managers.

LifeSim evaluates consequences to a study area, and the damageable elements are defined through the addition of user defined agricultural data, structure inventories, road networks and evacuation destinations. These damageable elements can be impacted by hydraulic parameters. Table 1-1 provides a list of the consequences that the LifeSim software can estimate given the following the minimum requirements:

Table 1-1. LifeSim Minimum Requirements for Available Consequence Estimates

Estimate	Minimum Requirements
• Structure damage	To estimate structure damages, LifeSim requires a user-defined structure inventory, and the hydraulic information must include depth for the flood event.
• Life loss	To estimate vehicle damages and life loss LifeSim requires a user-defined structure inventory with population for each structure provided. In addition, they hydraulic information must include maximum depth and arrival time for the flood event. Emergency planning zones (EPZs) are also required.
• Indirect income and job losses (ECAM)	The basic requirements for estimating life loss are necessary for this estimate. In addition, the user must define the total supply of available labor and capital for the entire county (or counties) in the study area.
• Agricultural losses	The basic requirements include importing gridded agricultural data, and the user-provided hydraulic information must contain agricultural arrival time and flood duration.

1.2 Modeling System Assumptions and Limitations

Every modeling system has limitations due to the choices and assumptions made in the design and development of the software. The following list describes the assumptions and limitations inherent to three aspects of the LifeSim modeling system:

Evacuation Modeling:

- When modeling evacuations, LifeSim assumes that people will evacuate to whichever of the designated destinations can be reached in the shortest travel time. This assumption does not account for trip chaining, where an individual might go from work, to pick up a kid at school, go back home, and then evacuate.

  - This is a simplifying assumption made inherent to LifeSim due to data limitations. Specifically, it is prohibitively difficult to predict atypical evacuation behavior to this level of detail.

- Evacuation routes are limited to the user-provided road networks.

  - LifeSim limits evacuation routes to the user-defined road network and assumes that vehicles adhere to the rules of the road.

  - The reason for this assumption and limitation is a lack of available data. Road networks are readily available, but alternative paths, such as those that go through parks or empty lots are not as accessible in a format understandable by LifeSim.

Hydrologic and Hydraulic Uncertainty Modeling:

- LifeSim does not provide means to account for hydrologic and hydraulic uncertainty. Each alternative relies on a constant hydraulic scenario. Therefore, any analysis of hydraulic, or hydrologic uncertainty must be performed outside of the LifeSim software before import.

  - Due to the event based nature of LifeSim, hydraulic and hydrologic uncertainty would need to be run as separate alternatives and performed as a sensitivity analysis as part of the LifeSim study.

Modeling Life Loss:

- In LifeSim population data is linked directly to structures. Therefore, there is no direct accounting for transient or homeless population. Often, the data necessary to model transient or homeless population is not readily available. However, in a situation where the data was available, a modeler could take steps to account for homeless population by manually adding structure data points with assigned population values in locations where structures may not actually exist.

- When a simulation is computed, the road network is assumed empty prior to evacuation.

  - The purpose of this assumption is to prevent double counting of the population both on roads and in structures.

- LifeSim does not account for indirect life loss, or life loss associated with the duration of flooding.

  - This is a simplifying assumption made inherent to LifeSim due to limited resources and available data.

1.3 Organization of Manual

The organization of this manual by chapter is as follows:

Chapter 2	Installation process, getting started with LifeSim, and creating an LifeSim study.
Chapter 3	Overview of the LifeSim interface.
Chapter 4	Overview of hydraulic datasets
Chapter 5	Overview of structure inventories
Chapter 6	Overview of emergency planning datasets
Chapter 7	Overview of road network datasets
Chapter 8	Overview of destination datasets
Chapter 9	Overview of agricultural data
Chapter 10	Overview of Economic Consequence Assessment Model (ECAM) data
Chapter 11	Overview of alternatives
Chapter 12	Overview of simulations
Chapter 13	Provides the user with a thorough understanding of the results that are generated
Appendix A	Details the Map Window and the map layer editing tools
Appendix B	Provides tips and tricks for preparing a road network
Appendix C	References
Appendix D	Terms and Conditions of Use

CHAPTER 2

Installing and Starting LifeSim

This chapter describes the recommended computer system requirements for installing and running LifeSim. The software includes a graphical user interface (GUI), designed to increase the usability and efficiency of the software. The LifeSim GUI provides for file management, data entry and editing, and viewing output results.

2.1 Requirements

- Installation – about 180MB (megabytes) of hard disk space
- Memory – 1GB (gigabytes) RAM (random access memory); recommended 4GB RAM
- Operating System – Microsoft Windows XP, Windows 7, Windows 8/8.1/Windows 10
- Microsoft .NET Framework Version 4.0

2.2 Installation

- Install LifeSim using the self-extracting setup package file downloaded from the Risk Management Center (RMC) website (https://www.rmc.usace.army.mil/Software/).

- The installer (with administrative privileges) will run through a series of steps regarding terms and conditions, install location (e.g., *C:\Program Files*), and shortcut icons.

- Install the LifeSim example dataset from the RMC website (https://www.rmc.usace.army.mil/Software/).

2.3 Starting LifeSim

When starting the LifeSim, double-click the LifeSim icon (shortcut) on the desktop. The main window of the LifeSim will appear (Figure 2-1). If the desktop shortcut icon is not available, the user can start LifeSim from the **Start** menu, click **All Programs**, click **RMC**, and click **LifeSim**.

2.4 Creating a Study

A study in LifeSim is a combination of the data, models, or events required to analyze a specific geographical area. The study represents all the input data and simulation output required to adequately answer the question of estimating consequences from a hazard. There are two ways to create a study.

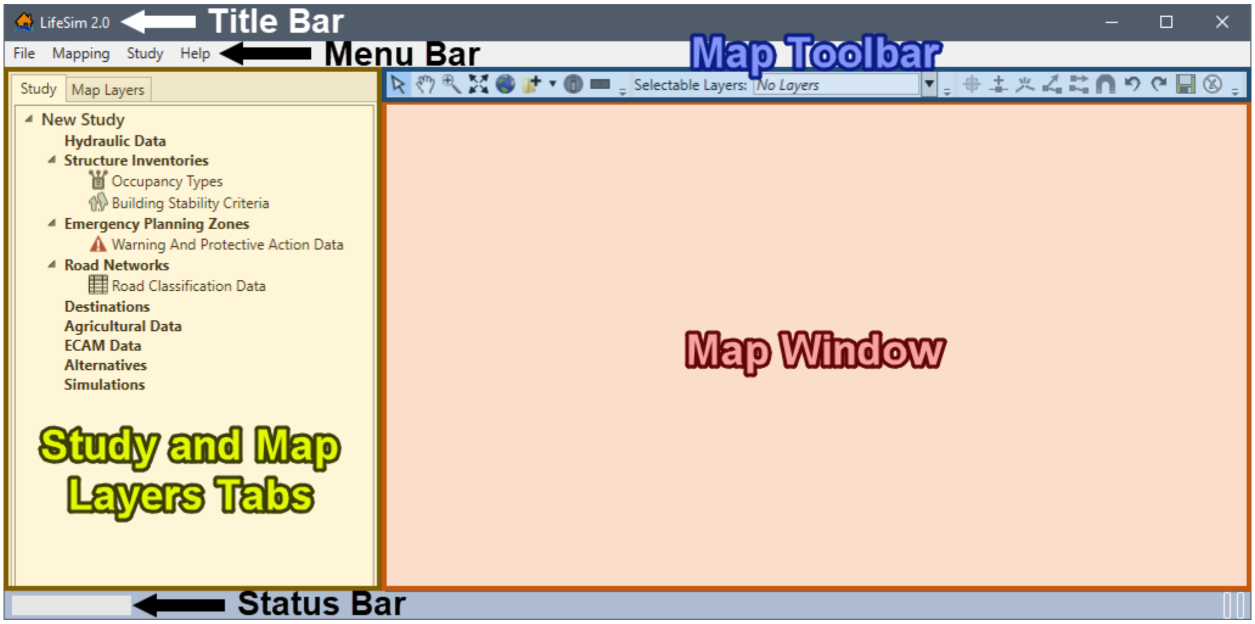


Figure 2-1. LifeSim Main Window

To create a LifeSim study:

1. From the LifeSim main window (Figure 2-1), from the **File** menu (Figure 2-2), click **Create New** (Figure 2-2). Another way is from the **Study Tree**, right-click **New Study**, from the shortcut menu, click **Create New**. Either way, the **Save As** browser will open (Figure 2-3).

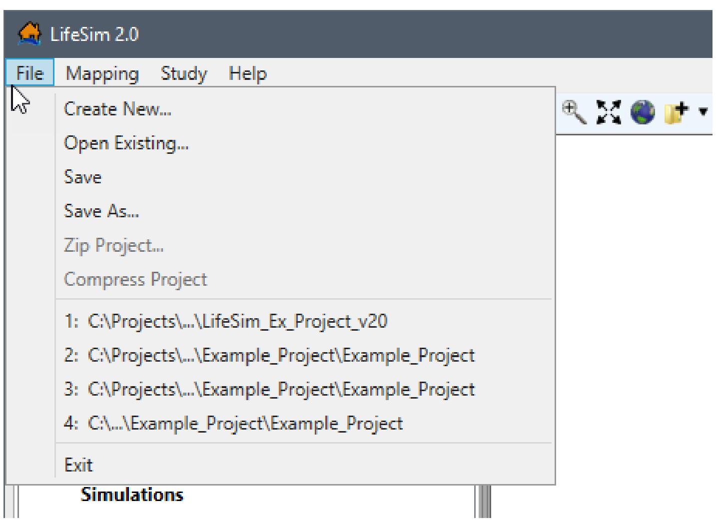


Figure 2-2. LifeSim – File Menu – Create New

2. From the **Save As** browser, navigate to the location (directory) where the LifeSim study will be created. Enter a name (required) in the **File name** box.

3. Click **Save**, the **Save As** browser (Figure 2-3) will close. The LifeSim main window (Figure 2-1) will now have the name of the study in the **Study Tree** and the **Title Bar**.

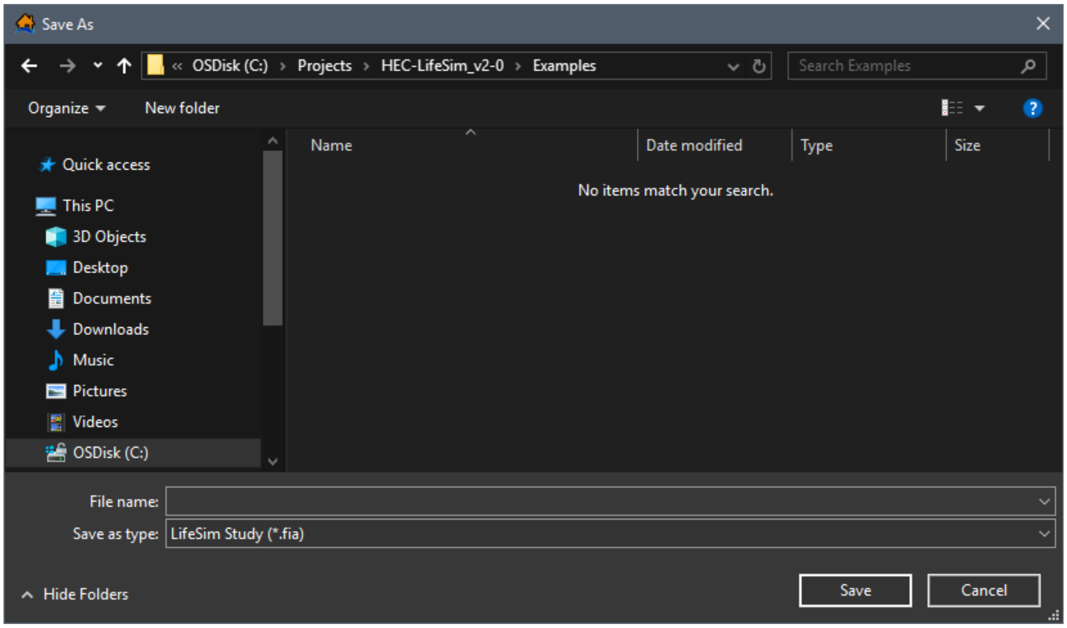


Figure 2-3. Save As Browser

4. A directory with the study name and associated files will be created for the study.

2.5 Opening a Study

It is important to note that at this time LifeSim Version 2.0 cannot open or update LifeSim Version 1.0 studies. To open an existing LifeSim study:

1. From the LifeSim main window (Figure 2-1), from the **File** menu (Figure 2-2), click **Open Existing** (Figure 2-2). Another way is from the **Study Tree**, right-click **New Study**, from the shortcut menu, click **Open Existing**. Either way, an **Open** browser will open (Figure 2-4).

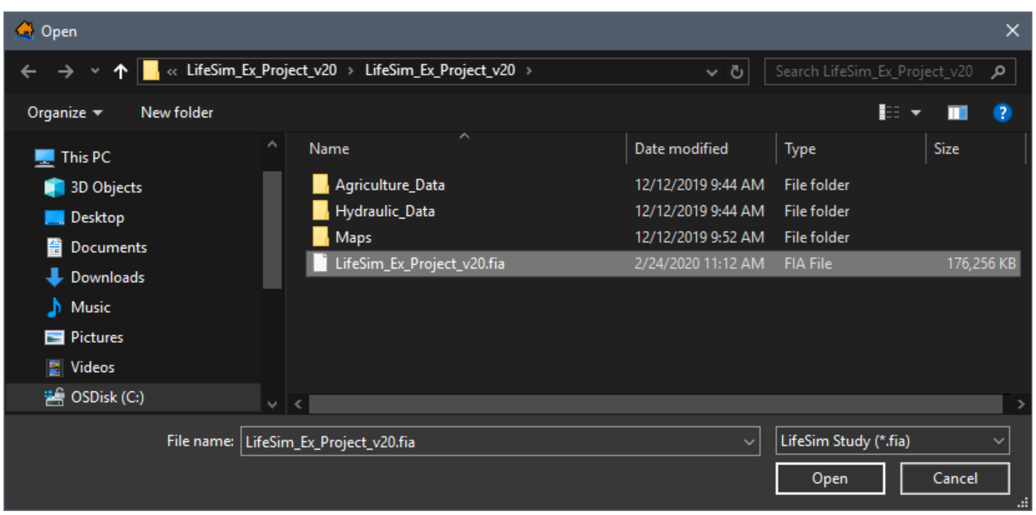


Figure 2-4. Open Browser

2. Browse to the location of the LifeSim study file (**.fia*). Click on the **.fia* file and the **File name** box will now contain the filename (e.g., *LifeSim_Ex_Project_v20.fia*).

3. Click **Open**, the **Open** browser (Figure 2-4) will close and the LifeSim main window (Figure 2-1) will now display the selected study in the **Study Tree** and the study name will display in the **Title Bar**.

2.6 Study Directories and Files

Directories and files are automatically created by LifeSim to store information. As data is entered, subsequent files and subdirectories are created. A description of the created directories is also provided in Table 2-1.

Table 2-1. Example LifeSim Directories and Files

File or Directory Name	Contents
LifeSim Example Model	Study folder that is created when a user creates a new study (Section 4.1.1).
Deming Fake Dam Breach.fia	File created by LifeSim Version 2.0 when a study is created. Contains the name and description of the study, map properties, and references to the LifeSim study elements. The *.fia file also holds the following study data: • Structure Inventory: Contains all of the information about the imported and defined structure inventory, damage categories and structure occupancy types. • Emergency Planning Zones (EPZ) Data: Contains all of the information about the imported and defined EPZ data. • Road Network: Contains all of the information about the imported and defined road network. • Destination Data: Contains the created destination data. • Alternative Data: The created alternative is saved as an XML file. • Simulations: Contains all of the created and computed simulation data.
Hydraulic_Data	Sub-directory created by LifeSim when hydraulic data is imported. Contains files that define data items for imported hydraulic data. Also, sub-directories are created when hydraulic data is defined and are dependent on the hydraulic data import option utilized.
Agricultural_Data	Sub-directory created by LifeSim when agriculture data is imported. Contains the created agriculture data geotiff.

CHAPTER 3

LifeSim Interface

The LifeSim main window (Figure 3-1) displays the framework for the LifeSim software. The interface allows users to enter data, review data, create alternatives, run simulations, and review results. The **Title Bar** (Figure 3-1) displays the LifeSim Version number and the name of a created or opened LifeSim study.

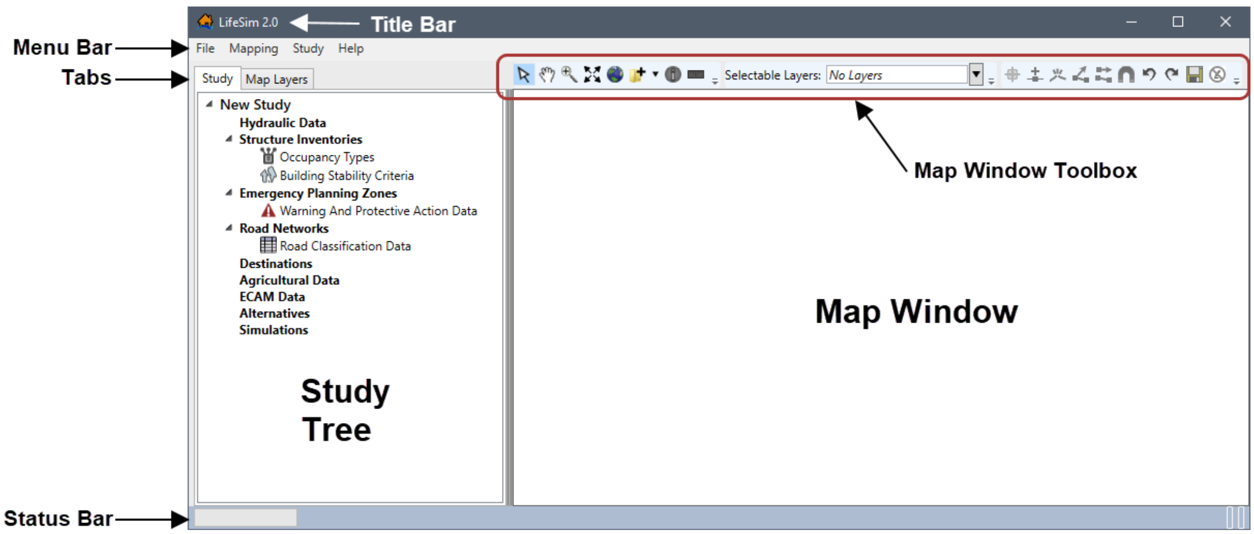


Figure 3-1. LifeSim Main Window – Interface

3.1 Main Window

The main window is organized with a **Title Bar**, a **Menu Bar**, two tabs (**Study** and **Map Layers**), a **Map Window** (with an associated toolbox), and the **Status Bar**. The following describes basic elements of the main window:

Title Bar	Displays the Version number (Version 2.0), for Version 2.0 and above, and the name of the active LifeSim study.
Menu Bar	Contains the LifeSim menus.
Study and Map Layers Tabs	The **Study** tab contains a **Study Tree** that guides users when adding the building blocks of an LifeSim study. The **Map Layers** tab contains a list of all map layers that are currently part of the active study. Clicking a tab opens the tab specific information and commands.
Map Window	Used to graphically display the LifeSim study components and imported map layers, which is geographically referenced (geo-referenced).

Map Toolbar Contains tools used to setup, navigate, and edit within the LifeSim map window.

Status Bar Displays messages related to incoming data, status of the LifeSim software for informational purposes.

3.2 Menu Bar

The menu bar of LifeSim provides the user with many commands to perform various functions. An overview for each menu in LifeSim software is provided.

File The **File** menu allows the user to perform study management functions such as creating, opening, closing, and saving an LifeSim study. Available commands are: **Create New**, **Open Existing**, **Save**, **Save As**, **Zip Project**, **Compress Project** and **Exit**. This menu also displays a list of previously opened LifeSim studies and selecting one automatically opens the selected study.

Mapping From the **Mapping** menu users can add or remove map layers from the **Map Layers** tab and map window. Map layers that can be added to an LifeSim study are - **.shp*, **.flt*, **.tif*, **.vrt*, or web layers. Other available functions from the **Mapping** menu are creating vector layers (**.shp*) and modifying or importing map projections. Available commands are: **Add Map Layer**, **Add Web Layer**, **Create New Vector Layer**, **Remove Map Layer**, and **Map Properties**.

Study From the **Study** menu, users can import, create, modify, and edit the building blocks of an LifeSim study. Commands include **Hydraulic Data**, **Structure Inventories**, **Emergency Planning Data**, **Road Network Data**, **Destinations Data**, **Agricultural Crop Data**, **ECAM Data**, **Alternatives** and **Simulations**.

Help Displays a copy of the LifeSim User's Manual current version; and information about LifeSim.

3.3 Tabs

The LifeSim main window (Figure 3-1) is organized into two tabs that allow the user to view, enter, and edit LifeSim study data or to view and manipulate any map layer geographically in the map window.

Study Tab Allows a user to view, add, and edit data; and add/remove map layers for an LifeSim study. The **Study** tab is the default tab and provides a view of the study data in a tree (**Study Tree**). Also, from the **Study Tree** a user can create, edit, and delete datasets, alternatives and simulations. The **Study Tree** lists datasets, alternatives and simulations that have been created.

Map Layers Tab The **Map Layers** tab provides a list of the map layers in a tree (**Map Layers Tree**). From the **Map Layers** tab, a user can import, create, edit and delete map layers. The **Map Layers Tree** allows users to also display map layers in the map window. However, only selected layers in the **Map Layers** tab will display in the map window. The user can turn map layers on or off, adjust the properties of the layers, modify the order of the layers for viewing in the map window. Appendix A provides greater detail on the **Map Layers** tab and associated map window.

3.4 Map Window

The **Map Window** (Figure 3-1) in LifeSim provides a way to graphically display LifeSim components. This section is designed to give a brief overview of what is available in the map window and map layers. For more detail on the mapping functionality in LifeSim, refer to Appendix A.

The **Map Window Toolbox** (Figure 3-1) contains three toolbars. The first toolbar is the **General Toolbar**, second is the **Selectable Layers Toolbar**, and third is the **Editor Toolbar**. If a tool has a parenthesis after the tooltip, the letter in the parentheses represents a hot key that allows a user to access the tool through the keyboard. For example, to access the **Query** tool **(Q)** through keyboard selections, the user will click the **Shift-Q** keys over the map window. This will activate the **Query** tool. A list of all of the hot keys is located in Appendix A.

3.4.1 General Toolbar

The **General Toolbar** includes the **Select Tool (B)**, **Pan Tool (P)**, **Zoom In Tool (Z)**, **Zoom to full extents**, **Add Data**, **Query Features Tool (Q)**, and **Measure Distance Tool (M)**. These tools change the appearance of the cursor, as well as the functionality of the mouse cursor in the map window.

 **Select Tool (B)**

The **Select** tool allows the user to select elements displayed in the map window that are currently selected in the **Selectable Layers**. More details on **Selectable Layers** can be found in the next section. To select features, click on a feature or click, hold, and drag a rectangle around features to select them in the map window. Additionally, the **Select** tool allows users to access shortcut menus which can be used to customize features when in editing mode. Refer Appendix A for more information regarding editing mode.

 **Pan Tool (P)**

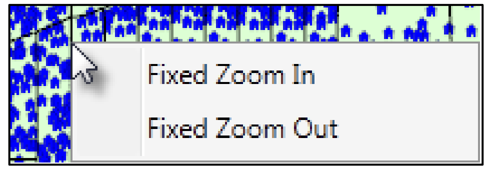
The **Pan** tool allows the user to pan the map window; click the **Pan** tool and click, hold and move the mouse around the map window. Also, if the mouse has a wheel then the user can pan the map window by using the wheel (click and hold the wheel to pan).

 **Zoom In Tool (Z)**

The **Zoom In** tool allows the user to zoom in to areas of the map window. To zoom in, hold the mouse button down and outline the area of interest to enlarge. Also, if the mouse has a wheel then the user can zoom in or out using the wheel (wheel up zooms in and wheel down zooms out).

Fixed Zoom Out Tool (O)

The **Fixed Zoom Out** tool zooms out to a fixed distance. Click the **Fixed Zoom Out** icon once and the LifeSim map window view will zoom out a fixed distance. Double-click on the tool icon and the map window view will zoom out twice the fixed distance.

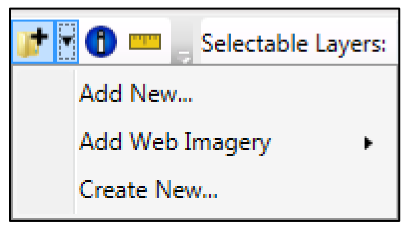


Additionally, the user can utilize the shortcut menu option, right-click anywhere in the map window, to zoom in or out a fixed distance.

Zoom to full extents Tool

The **Zoom to full extents** tool is a quick way to re-center the map by zooming out to the full geo-graphical extent of the map layers in the map window. Web layers do not affect the map extents.

Add Data Tool



Click the **Add Data** tool, the **Browse for files** browser will open. This browser allows users to navigate to layers (including **.shp*, **.flt*, **.tif*, and **.vrt*) of interest and add the layer(s) to the map window.

If the user clicks the down arrow next to the **Add Data** tool a shortcut menu will appear which allows the user to add data, add web imagery, or create new data that is added to the map window.

Query Features Tool (Q)

The **Query Features** tool identifies the geographic feature or place currently selected in the map window. Alternatively, multiple features can be identified by dragging a box over the desired area. Once a selection is made the **Feature Query** dialog box will open with the information presented in table format separated by **Feature** type. Only visible layers in the map window will be queried and the **Selectable Layers** option does not modify the **Query Features** tool.

Measure Distance Tool (M)

The **Measure Distance** tool allows the user to measure the distance along a specified line. With the tool selected, click to initiate the measure line. Each subsequent click will add a new point to the measure line. When finished, double-click on the location of the final point. The **Measure Results** dialog box will open and display the distance in the units the user has chosen (feet, miles, meters, or kilometers).

3.4.2 Selectable Layers Toolbar

Selectable Layers Toolbar allows the user to turn on and off map layers of a study using the **Select** tool from the **Selectable Layers Toolbar**. The list of options depends on the maps layers located in the **Map Layers** tab, except Internet (web) layers. For further explanation on the use of **Selectable Layers**, refer to Appendix A, Section A.13.1.

3.4.3 Editor Toolbar

The set of tools located in the **Editor Toolbar** initiate an editing session for the selected layer. When a user right-clicks on a map layer in the **Map Layers** tab, from the shortcut menu (Figure 3-2), click **Edit**, and the tools in the **Editor Toolbar** become active. Only one map layer can be edited at one time. For lines and polygons, there are two editing modes: feature edit, and vertex edit. Clicking on a selected feature puts the user into the feature edit mode (move, delete, reverse), double-clicking on a selected feature puts the user in the vertex edit mode (add, delete, edit vertex points).

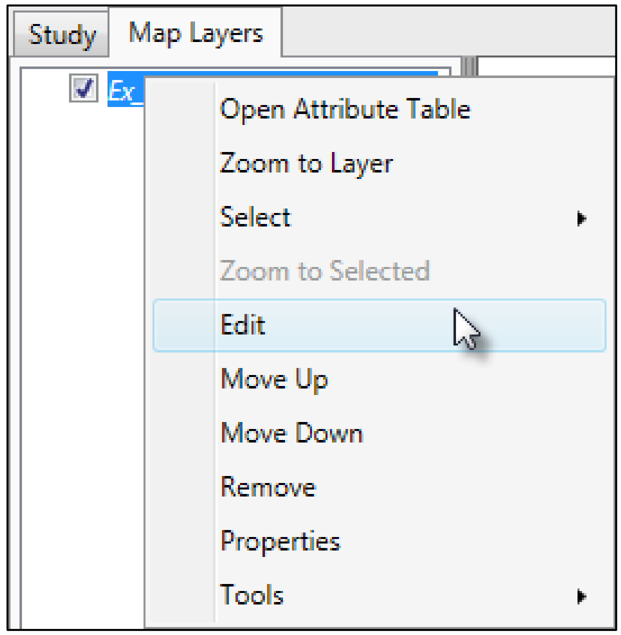
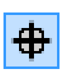
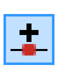


Figure 3-2. Map Layers Tree Shortcut Menu

There are some tools available during an edit session which are not available from the shortcut menu (Figure 3-2) or the **Editor Toolbar**. Clicking **Delete** will delete all currently selected features or currently selected vertices depending on which mode the user is in. Clicking the **R** key will reverse the direction of a currently selected line feature. Also, right-clicking during an edit session allows the user to do multiple tasks depending on what is clicked. Detailed information on all of the functionality of the editing tools is available in Appendix A.

 **Add new features Tool (A)**

The **Add new features** tool allows the user to add a new feature to the editable map layer by left-clicking to add feature data. Selected features can be deleted by either clicking the **Delete** key, or right-click on the selected feature(s) and from the shortcut menu click **Delete Selected Features**.

 **Insert vertex Tool (I)**

The **Insert vertex** tool allows the user to add a vertex to an existing line or polygon map layer. The **Insert vertex** tool becomes available when the selected feature is in the vertex edit mode. As the cursor gets close to the line or polygon edge the map window will display what the updated feature will look like if a vertex is added at the cursor. Left-click to insert the new vertex.

Break line feature into two line features at a defined location Tool (K)

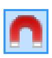
The **Break line feature** tool will split a line feature into two features at the point the user clicked on the line. The user needs to be in the feature edit mode for this tool to be enabled.

Continue from end Tool (lines only)

The **Continue from end (lines only)** tool will continue a line feature from the end point of the line feature to the point to a second point clicked in the map window. Double-click to finish the line continuation edits or right-click and choose **Finish Editing Vertices** or choose a different tool to finish the line continuation. The user needs to be in the vertex edit mode for this tool to be enabled.

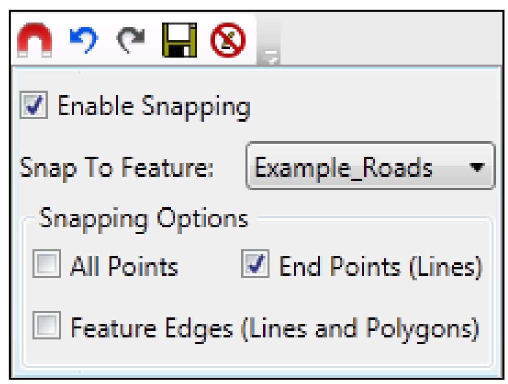
Add new line or polygon to existing feature Tool

The **Add new line or polygon to existing feature** tool will allow the user to add a polygon or line to an existing selected polygon or line feature. For example, to add a hole to a polygon, click to add points defining the hole and double-click to finish. The user needs to be in the vertex edit mode for this tool to be enabled.



Feature Snapping Settings Tool

The **Feature Snapping Settings** tool allows the user to define snapping rules when editing. Snapping will automatically move the desired action to the nearest snap point when the cursor is close. This can be very useful when trying to line up endpoints of line features or place points directly on top of other features. Enable snapping by clicking **Enable Snapping**. From the **Snap to Feature** list, select the desired feature to snap to, then select how snapping will be applied (all points, end points, or edges).

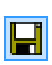


Undo Tool (Ctrl + Z)

The **Undo** tool allows the user to undo changes made during an edit session. Only enabled if there are edits that can be undone.

Redo Tool (Ctrl + Y)

The **Redo** tool allows the user to redo changes made. Only enabled if there are edits to redo.



Save Edits Tool (Ctrl + S)

The **Save Edits** tool will save all edits made during the editing session. Only enabled if edits have been made.

Stop Editing Tool

The **Stop Editing** tool will stop the edit session. At this point a new map layer can be selected for editing. If edits have been made, then the user will be prompted to save them before stopping the edit session.

3.5 Table Tools

Several tables in LifeSim contain the **Export and Copy Tools** (Figure 3-3) and the **Selection Tools** (Figure 3-4) toolbars. For example, the **Results Summary Tables** (Figure 3-5) includes the **Export and Copy Tools** and the **Selection Tools** toolbars (review Section 13.2.2).

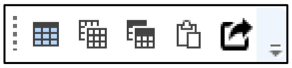


Figure 3-3. Export and Copy Tools

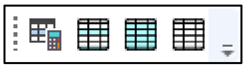
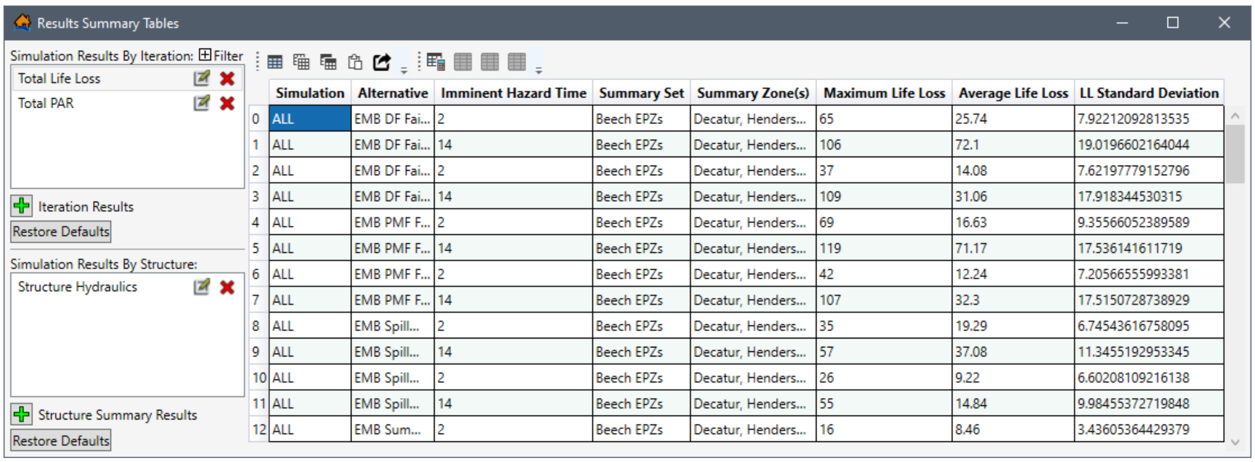


Figure 3-4. Selection Tools



	Simulation	Alternative	Imminent Hazard Time	Summary Set	Summary Zone(s)	Maximum Life Loss	Average Life Loss	LL Standard Deviation
0	ALL	EMB DF Fai...	2	Beech EPZs	Decatur, Henders...	65	25.74	7.92212092813535
1	ALL	EMB DF Fai...	14	Beech EPZs	Decatur, Henders...	106	72.1	19.0196602164044
2	ALL	EMB DF Fai...	2	Beech EPZs	Decatur, Henders...	37	14.08	7.62197779152796
3	ALL	EMB DF Fai...	14	Beech EPZs	Decatur, Henders...	109	31.06	17.918344530315
4	ALL	EMB PMF F...	2	Beech EPZs	Decatur, Henders...	69	16.63	9.35566052389589
5	ALL	EMB PMF F...	14	Beech EPZs	Decatur, Henders...	119	71.17	17.536141611719
6	ALL	EMB PMF F...	2	Beech EPZs	Decatur, Henders...	42	12.24	7.20566555993381
7	ALL	EMB PMF F...	14	Beech EPZs	Decatur, Henders...	107	32.3	17.5150728738929
8	ALL	EMB Spill...	2	Beech EPZs	Decatur, Henders...	35	19.29	6.74543616758095
9	ALL	EMB Spill...	14	Beech EPZs	Decatur, Henders...	57	37.08	11.3455192953345
10	ALL	EMB Spill...	2	Beech EPZs	Decatur, Henders...	26	9.22	6.60208109216138
11	ALL	EMB Spill...	14	Beech EPZs	Decatur, Henders...	55	14.84	9.98455372719848
12	ALL	EMB Sum...	2	Beech EPZs	Decatur, Henders...	16	8.46	3.43605364429379

Figure 3-5. Example – Results Summary Table – Table Tools

3.5.1 Table Export and Copy Tools

Select all cells Tool

This tool selects all cells in the table.

Copy Selected Cells Tool

This tool copies cells selected in the table. The information is copied to the clipboard and needs to be pasted to another application (e.g., Microsoft Excel®) for saving.

Copy Selected Cells with Table Headers Tool

This tool copies cells selected in the table and the table headers. The information is copied to the clipboard and needs to be pasted to another application (e.g., Microsoft Excel®) for saving.

Paste from Clipboard into Table
This tool pastes data copied from another table into the current table.

Export Table
The **Export Table** tool opens a **Save As** browser window (Figure 3-6), which allows the user to export the data in the table to a Microsoft Excel® (*.*xlsx*) file.

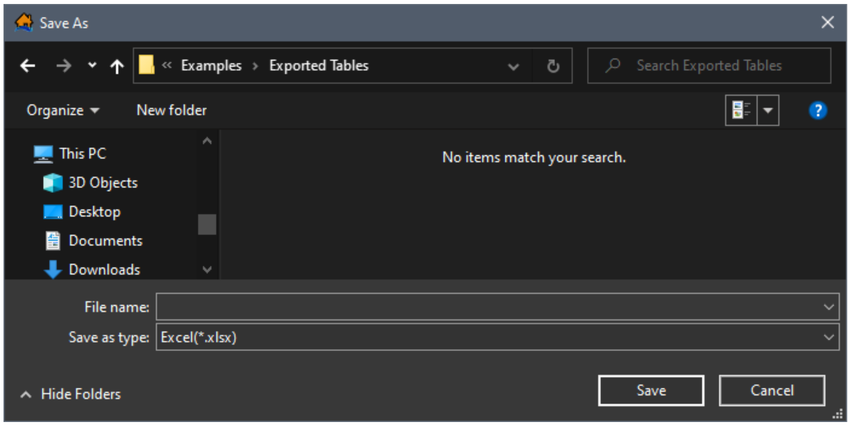


Figure 3-6. Example – Export Table Tool – Save As Browser

3.5.2 Table Selection Tools

Select Rows By Attribute Tool
This tool opens the **Select By Attribute** dialog box (Figure 3-7), which allows the user to perform queries, make selections, and locate features in the table.

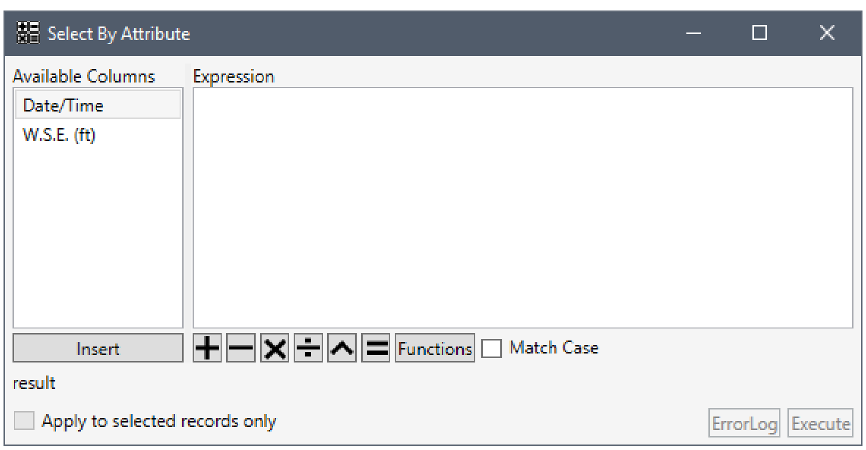


Figure 3-7. Select By Attribute Dialog Box

Show All Rows Tool
This tool allows the user to display all rows (selected or unselected) in the table.

Show Selected Rows Only Tool
This tool allows the user to only display features selected in the table.

 **Deselect All Rows**
This tool unselects all rows which were selected in the table.

3.6 Customizing Plots and Plotting Tools

Several plots include the general plotting tools (described in Section 3.6.1). For the example, in Figure 3-8, the Time Series Data plotting window (included in the Hydraulic Data importers, as described in Section 4.1) includes the general plotting tools.

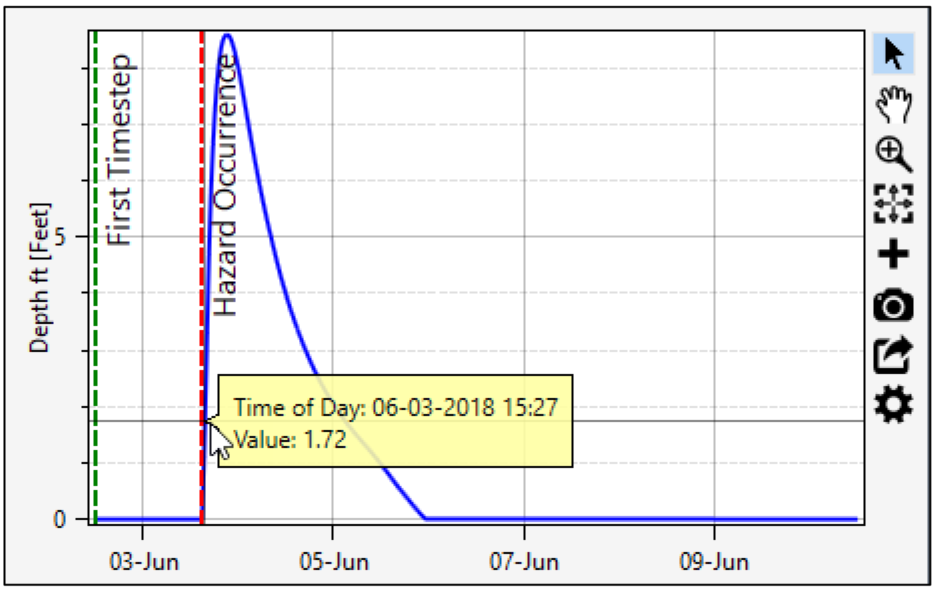


Figure 3-8. Time Series Data – Example Plot Window – Plotting Tools

3.6.1 General Plotting Tools

The **General Plotting Tools** includes the **Track Data**, **Pan**, **Zoom to Extents**, **Add Annotation**, and **Plot Properties** tools. Through the use of these tools, plots are highly customizable and offer an array of information to assist in reviewing the data, such as the time series data plot in Figure 3-8.

 **Track Data Tool**
Using the **Track Data** tool, click and hold inside the plot window (Figure 3-8), a yellow callout box will display that provides the data at the specific x and y location. For example, in Figure 3-8 the callout displays the **Time of Day** and corresponding **Value** for the specified location along the hydrograph. Click and drag to get information in a yellow callout box anywhere along the graph.

 **Pan Tool (P)**
The **Pan** tool allows the user to pan the plot window (Figure 3-8). To pan, select the **Pan** tool, click and hold while moving the mouse to pan. Another way to pan, is press down on the scroll wheel (middle mouse button) of the mouse (if the mouse has a scroll wheel).

Zoom to Extents Tool

The **Zoom to Extents** tool (Figure 3-8) is a quick way to re-center the data displayed in the plot window by zooming out to the full extent of the plot.

Add Annotation Tool

The **Add Annotation** tool (Figure 3-8) opens a list of annotations (Figure 3-9) that can be added to the plot. Three annotation types are available (Figure 3-9). To add the first type (Text, Vertical Line, Horizontal Line and Point), simply click and drag to the desired location. To add the second type (Arrow, Rectangle, and Ellipse), click, drag and release. To add the third type, click a starting point, additional clicks continue the polyline or polygon, and double-click to complete the polyline or polygon annotation. Edit the default text for the added annotation by double-clicking to select the text and enter.

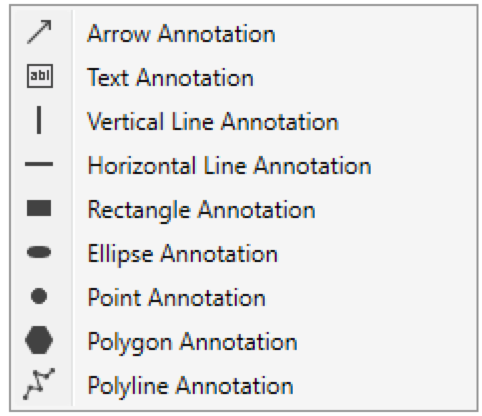


Figure 3-9. Add Annotation List

Default annotation properties can be modified using the **Plot Properties** tool. Note, the **Save Plot** tool will include all annotations added to the time series plot.

Save Plot Tool

The **Save Plot** tool allows the user to save the data displayed in the plot window (Figure 3-8). The **Save Plot** tool opens the **Save Plot** dialog (Figure 3-10). From the **Save Plot** dialog set the pixel (px) width and height of the plot, click **Save** (Figure 3-10). A **Save As** browser window opens (Figure 2-3). Navigate to the location (directory) to save to plot, enter a name (required) in the **File name** box for the plot and select the file type from the available options (*.*png*, *.*pdf*, *.*svg*). Click **Save**, the **Save As** browser (Figure 2-3) closes. Note, the **Save Plot** tool will include all annotations added to the time series plot and all alterations made to the plot using the **Plot Properties** tool.

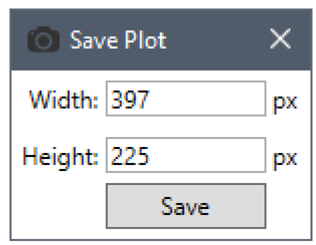


Figure 3-10. Save Plot Dialog

Plot Properties Tool

The **Plot Properties** tool allows the user to configure default properties for customizing LifeSim Version 2.0 plot window, for example the **Time Series Data** plot (Figure 3-8) can be customized by clicking the **Plot Properties** tool to open the **Plot Properties** dialog (Figure 3-12). Section 3.6.2 provides a detailed explanation regarding customizing plots in LifeSim Version 2.0.

Export Series Data Tool

The **Export Series Data** tool opens a **Save As** browser window (Figure 3-11), which allows the user to export the data as a comma delimited (*.*csv*), a Microsoft Excel® (*.*xlsx*) or Sqlite (*.*sqlite*) file.

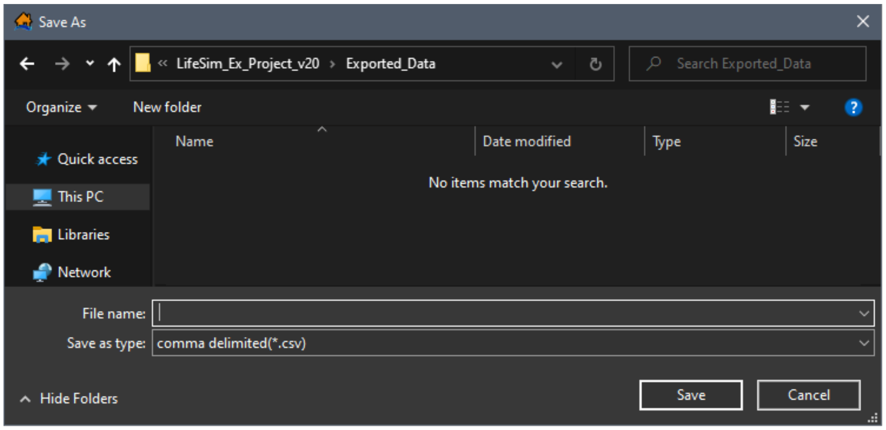


Figure 3-11. Save As Browser Window – Export Series Data

3.6.2 Customizing Plots Overview

From the **Plot Properties** dialog, opened by clicking the **Plot Properties** tool or right clicking on a plot and selecting an edit option, from specific plot windows. The **Plot Properties** dialog allows the user to configure multiple display properties for the individual plot, including **GENERAL PLOT SETTINGS** (Figure 3-12), the **LEGEND** (Figure 3-12), **AXES** (Figure 3-14), data **SERIES** curves (Figure 3-15), and plot **ANNOTATIONS** (Figure 3-15 and Figure 3-16). Access different customization editors (e.g., **LEGEND** in Figure 3-12), by toggling through the plot properties dropdown menu to access the different editing options.

Depending on the plot properties dropdown menu selection, the **Plot Properties** dialog updates to include a selector dropdown menu and expandable sections. For example, the **AXES** (Figure 3-14) editor includes a dropdown menu to allow users to select the x- or y-axis and also contains three tabs for editing the general, labels, or grid lines. Click the section ▷ arrow to expand the section, or collapse the section by clicking the expand ◢ button (Figure 3-12).

Depending on the expanded editing section users can enter text labels, choose the color, and/or use dropdown menus to modify the size, type, weight, padding, and/or alignment of the specified component of the plot. In general, checkboxes ☐ allow users to turn on or off the display of an item (e.g., turn on or off the visibility of the legend from the **LEGEND AREA** section displayed in Figure 3-12).

Once customization of the plot properties has been completed click the ← **Close Plot Properties** button, or the **Close** dialog button (Figure 3-12).

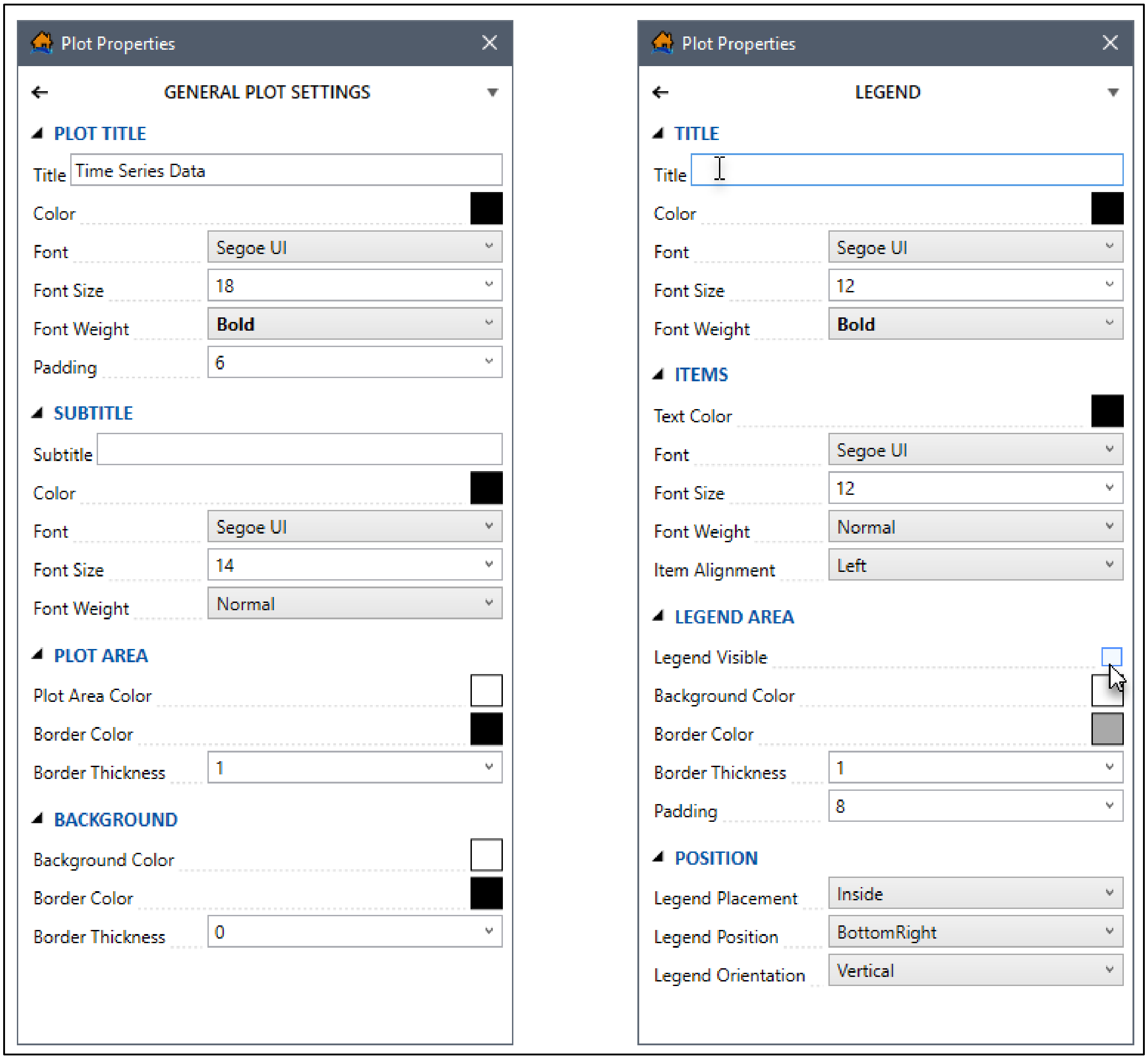


Figure 3-12. Plot Properties – General Settings (*left*) – Legend (*right*)

To edit the default color selection, click the color box to open a color chooser window (Figure 3-13). The color chooser window provides a large color picker box (tint varies along the left axis, tone varies along the right axis, and saturation varies along the top axis) a vertical color hue slider bar, a **Transparency** horizontal slider bar, and a **Selected Colors** section that provides several default color swatches (Figure 3-13).

Use the vertical hue slider bar to select a color (e.g., blue in Figure 3-13) and use the color picker box to choose the desired shade, tint and tone (adding black, white and gray respectively) for the chosen color. Selecting a color in the color chooser window (Figure 3-13) automatically updates the color box in **Plot Properties** dialog (Figure 3-12).

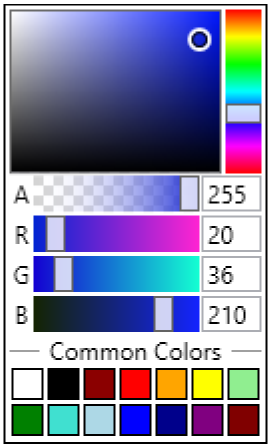


Figure 3-13. Color Chooser Window

General Plot Settings

By default, the **Plot Properties** dialog opens with the **GENERAL PLOT SETTINGS** (Figure 3-12) selected from the dropdown menu. Options for modifying the plot general settings includes the plot title, subtitle, plot area and background. To access the general settings options, click the section ▷ arrow to expand ◢ the section (Figure 3-12).

Legend

To modify the legend properties, select **LEGNED** from the **Plot Properties** dropdown menu (Figure 3-12). Users can modify the title, items, area, and position of the plot legend. Note, if making modifications to the legend make sure the **Legend Visible** (Figure 3-12) checkbox is ☑ checked or the legend will not display in the plot window.

Axes

To edit the plot axes, select **AXES** from the **Plot Properties** dropdown menu (Figure 3-14). When modifying the plot axes, choose the axis (e.g., ***Y Values*** in Figure 3-14) and the component to modify, by selecting the **GENERAL**, **LABELS**, or **GRID LINES** tab (Figure 3-14). From the **GENERAL** tab (Figure 3-14), modify the axis minimum, maximum, axis type, or axis display properties. From the **LABELS** tab (Figure 3-14), edit the axis label, text color, font, distance from the axis and/or enter the units for the axis. From the **GRID LINES** tab (Figure 3-14), modify the major and minor grid lines and the tick mark options.

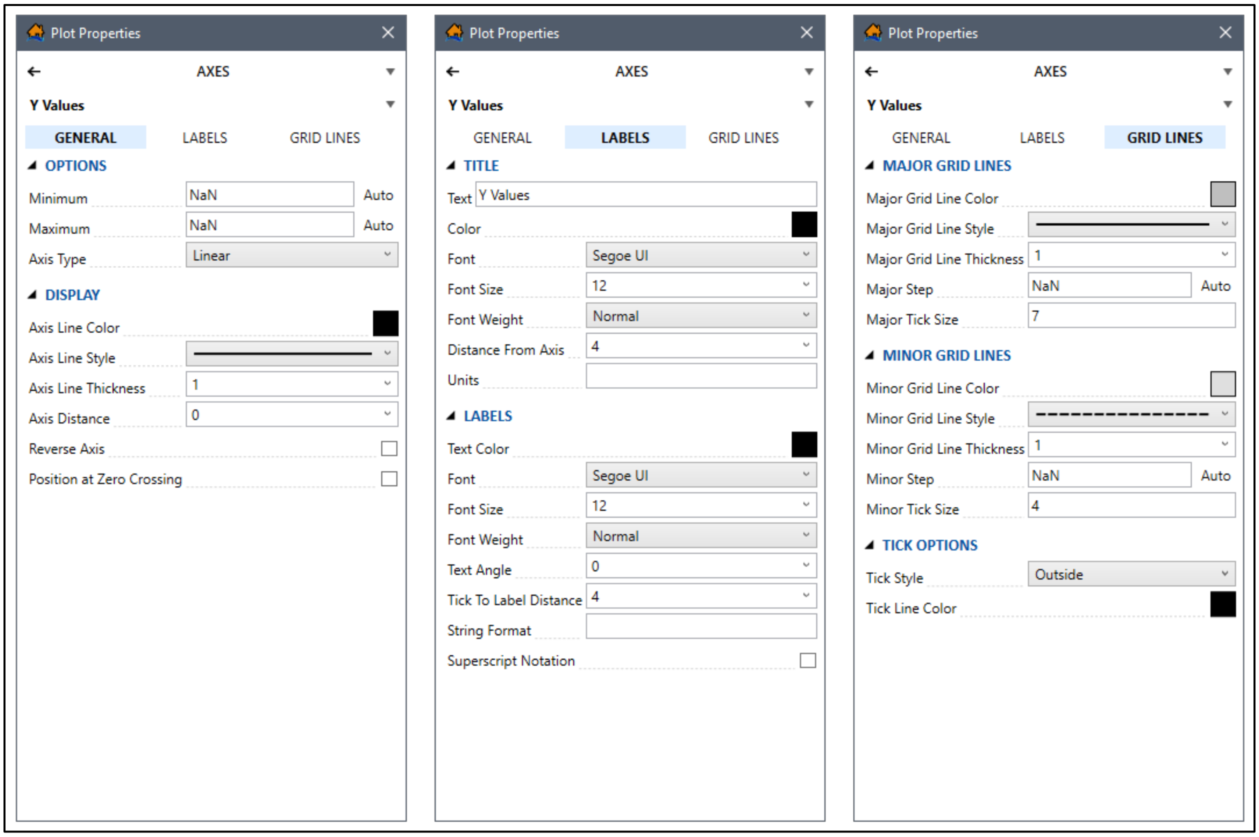


Figure 3-14. Plot Properties – AXES Options – GENERAL (*left*) – LABELS (*middle*) – GRID LINES (*right*)

Data Series Curves

The data series curves can be modified, by selecting **SERIES** from the **Plot Properties** dropdown menu (Figure 3-15). Users can select the specific data series in the plot (e.g., Time Series in Figure 3-15) when more than one data series is displayed in a plot window. Users can modify the series display and markers (if applicable). For example, if the series is a time series hydrograph, then the display and marker sections are available to modify the line and marker style, color, thickness and size can be modified. However, if the series is box and whisker then only the display section is available to modify the box and line style, color, thickness and size can be modified.

Annotations

To add annotations, from the **Plot Properties** dropdown menu, select **ANNOTATIONS** (Figure 3-15). Depending on the plot, default annotations will already be added to the plot. For the example provided in Figure 3-8, the default annotations included in the time series hydrograph are the First Timestep (green vertical dashed line with text) and Hazard Occurrence (red vertical dashed line with label). Users can modify the text and display properties of existing or added annotations. For the example provided in Figure 3-15, the ***First Timestep*** annotation is selected from the annotation dropdown menu and the **TEXT** and **DISPLAY** sections are expanded to display the default First Timestep annotation properties.

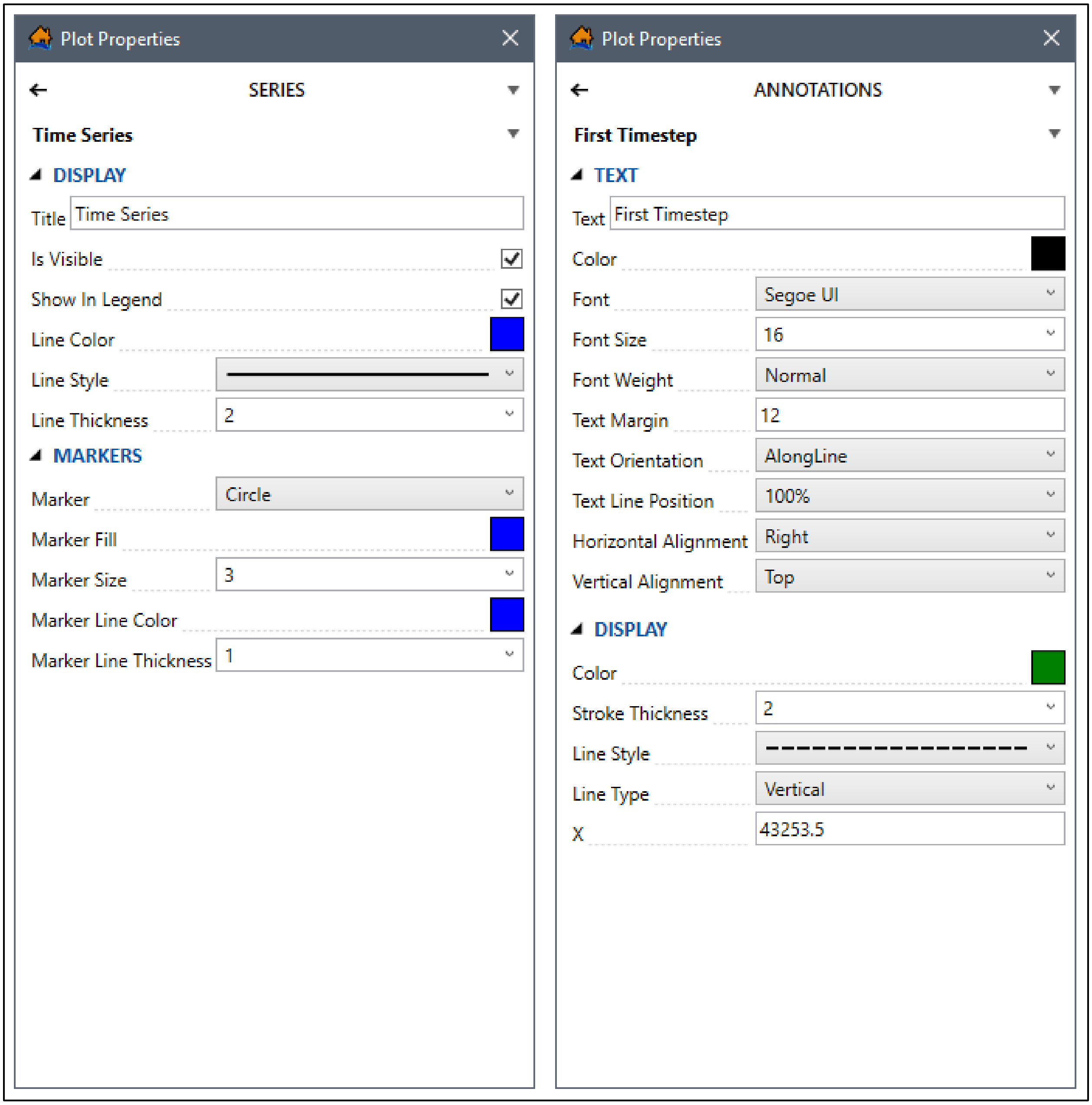


Figure 3-15. Plot Properties – SERIES (*left*) – ANNOTATIONS (*right*)

Add an additional annotation to the plot by selecting the desired annotation type from the dropdown menu (Figure 3-16). Annotation types include arrow, text, line, rectangle, ellipse, point, polygon, or polyline. Users also have the option to removed added annotations (including default annotations) by clicking the **Delete Annotation** button.

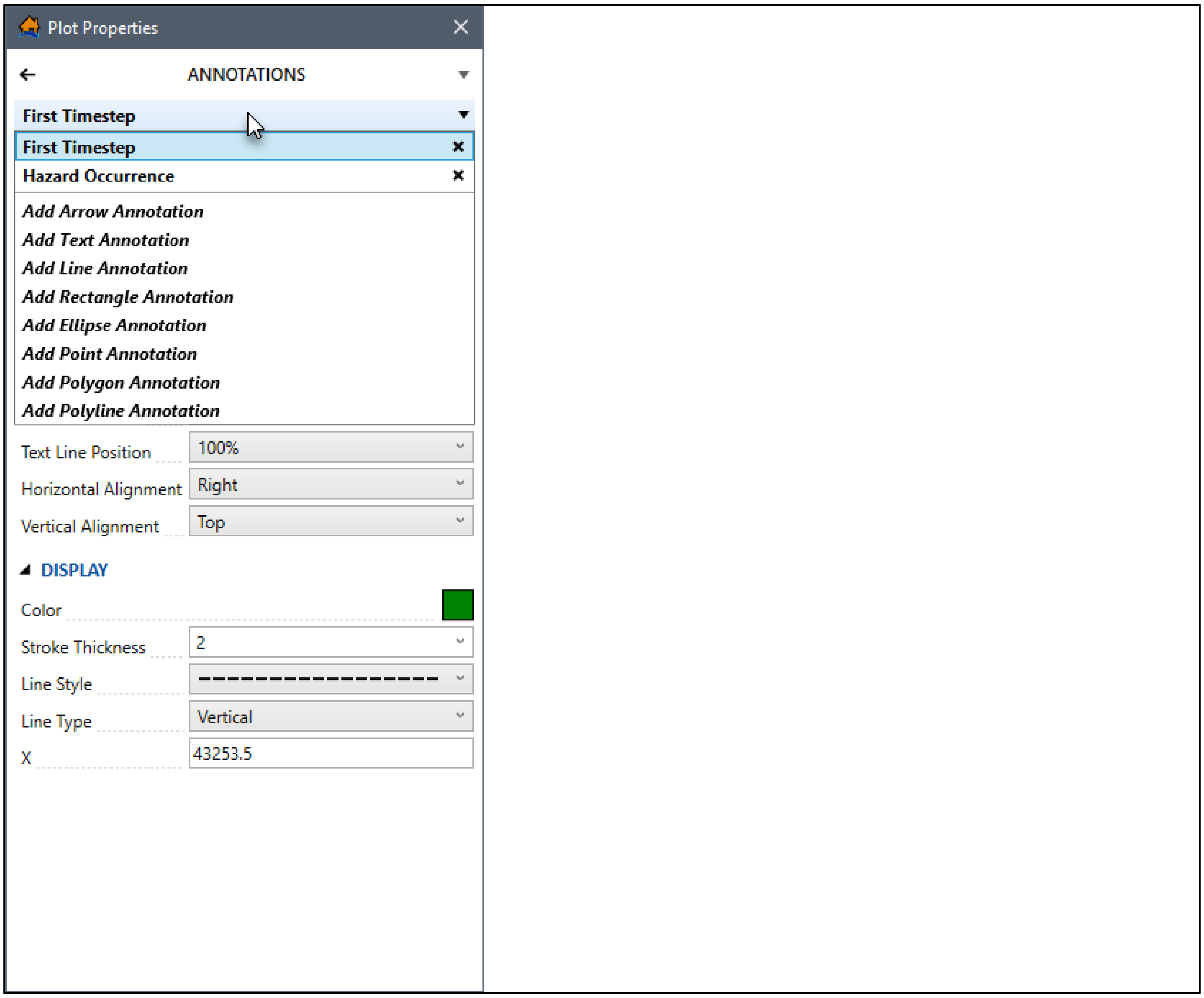


Figure 3-16. Plot Properties – Annotation Types Dropdown Menu

3.7 Animation Toolbar

When animations are viewed in a map window an **Animation Toolbar** (Figure 3-17) appears. For example, from the **RAS Map Data Selector** dialog box (Figure 3-18) users can import hydraulic data from a RAS map. Another example is when users view hydraulic or evacuation animations results in the map window (refer to Chapter 13, Section 13.5.2).

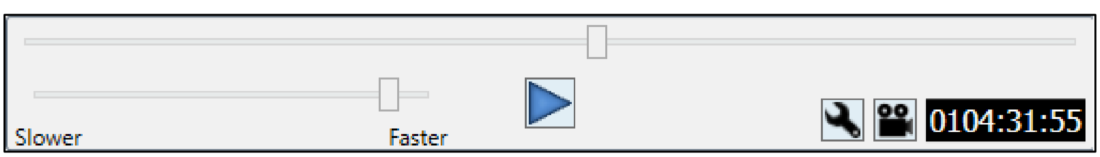


Figure 3-17. Animation Toolbar

From the **Animation Toolbar** at the bottom of the **RAS Map Data Selector** (Figure 3-18), the user can view (animated) the HEC-RAS depth simulation for the specified plan. The controls for the animation bar function the same as the controls to a video player. Use the slider bars to modify the speed (faster or slower) and position of the animation playback. To modify the animation settings, click , the **Animation Settings** dialog box (Figure 3-19) opens.

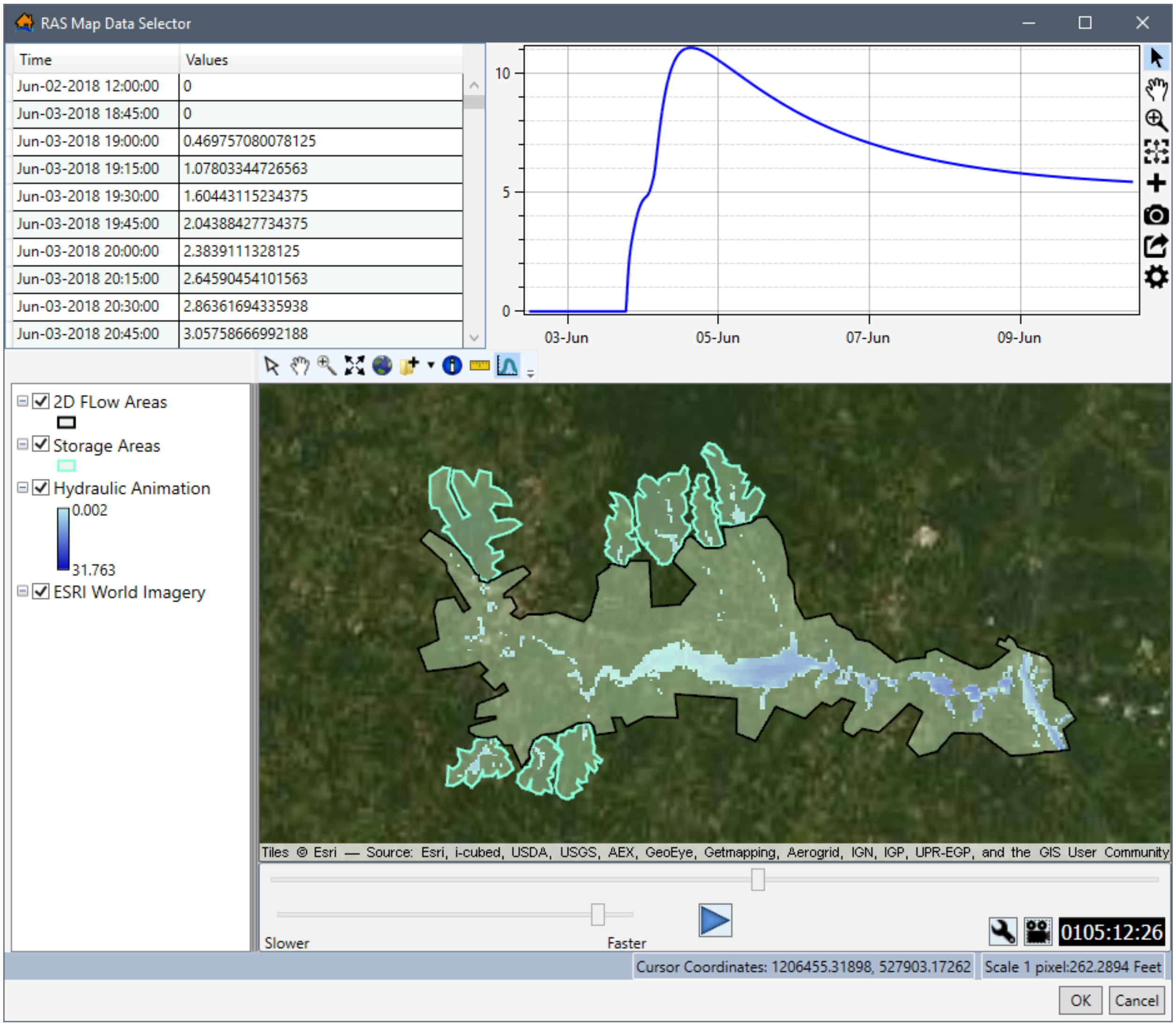


Figure 3-18. RAS Map Data Selector Dialog Box – Animation Toolbar

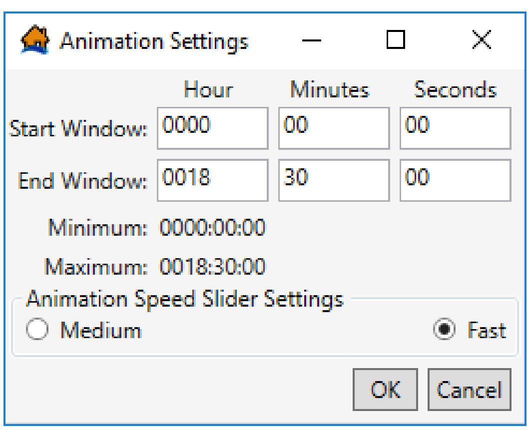


Figure 3-19. Animation Settings Dialog Box

The user can export a copy of the animation, click , the **Export Animation Wizard** (Figure 3-20) opens. To save an animation output file, click , and a **Save As** browser window opens (Figure 2-3). Navigate to the location (directory) to save to plot, enter a name (required) in the **File name** box for the plot and select the file type from the available options (**.mp4*, **.wmv*). Click **Save**, the **Save As** browser (Figure 2-3) will close. From the **Export Animation Wizard** (Figure 3-21), set the desired **Start** and **End Window**, **Timer Position**, and **Frame Rate**. To complete the export, click **OK**. The progress bar at the bottom of the **Export Animation Wizard** (Figure 3-21) provides the progress of creating the video file, and once complete the **Export Animation Wizard** (Figure 3-21) closes.

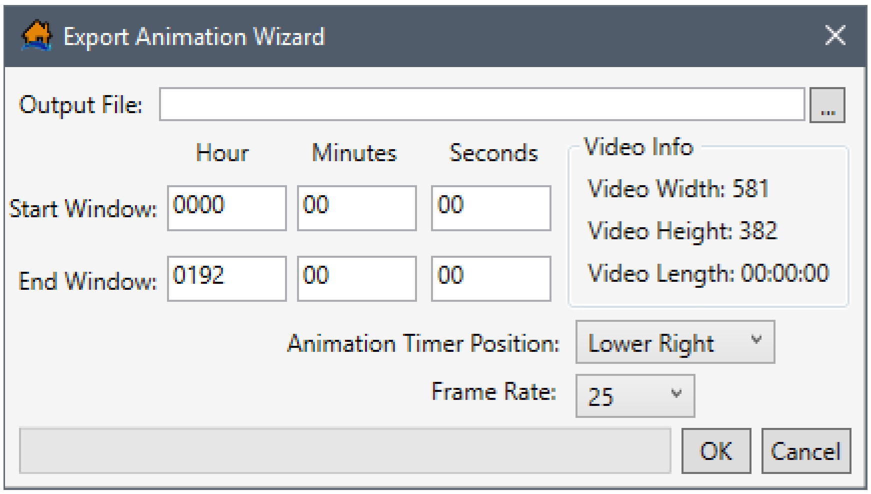


Figure 3-20. Export Animation Wizard

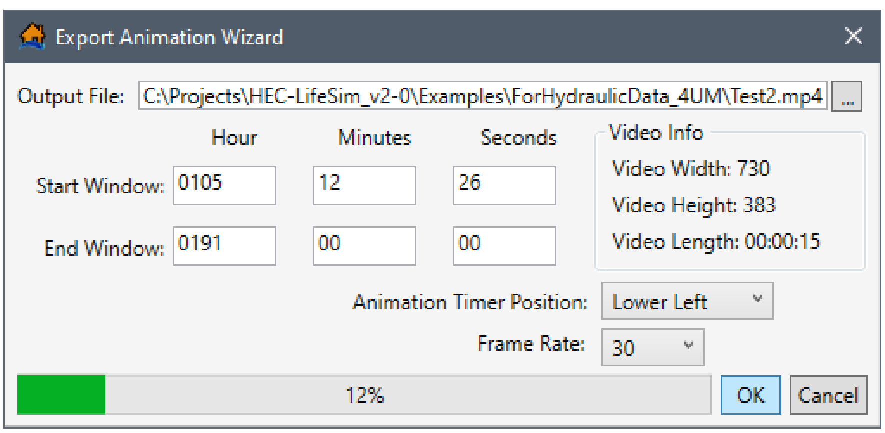


Figure 3-21. Example – Export Animation Wizard

CHAPTER 4

Hydraulic Data

Hydraulic data is a critical component to computing life loss in LifeSim. Hydraulic data sources in LifeSim generally come from a two-dimensional hydraulic model, such as HEC-RAS (Hydrologic Engineering Center's (HEC) River Analysis System software), FLO-2D (FLO-2D Software, Inc., Nutrioso, AZ), or MIKE 21 (Danish Hydraulic Institute, Denmark (DHI), see Appendix C). New to LifeSim Version 2.0, hydraulic data can also be imported from USACE Engineer Research and Development Center's (ERDC) Adaptive Hydraulics Model (ADH) system (see Appendix C). Also new to LifeSim 2.0 is the ability to import summary grids. LifeSim will interpolate the depths and velocities between hydraulic time steps. A shorter hydraulic time step will generally provide more hydraulically accurate computations.

In addition to providing the time series of depth and velocity, if life loss is being assessed then a hydrograph representing the hydraulic event must be provided as well. The hydrograph is critical in developing the warning issuance timing relative to a specific condition (e.g., dam or levee breach). This chapter covers the various ways to import, edit, and view hydraulic data.

4.1 Hydraulic Data Import Options

The four available options to import hydraulic data in LifeSim Version 2.0 are from HEC-RAS (*.pXX.hdf or *.hdf) files, FLO-2D files, gridded data (*.) files for depth and velocity, and summary grids (new to LifeSim Version 2.0). Open any of the hydraulic data import options, from the LifeSim main window (Figure 2-1**Error! Reference source not found.**), from the **Study** tab, from the **Study Tree**, right-click on **Hydraulic Data**, to open the shortcut menu (Figure 4-1). Another way is from the LifeSim main window, from the **Study** menu, point to **Hydraulic Data** (Figure 4-2), and select the desired import option.

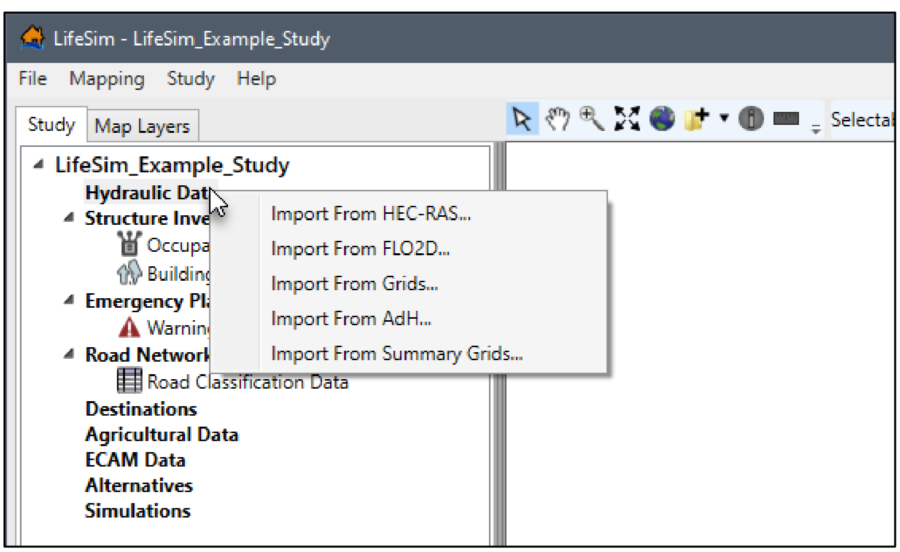


Figure 4-1. LifeSim Main Window – Study Tree – Hydraulic Data Shortcut Menu

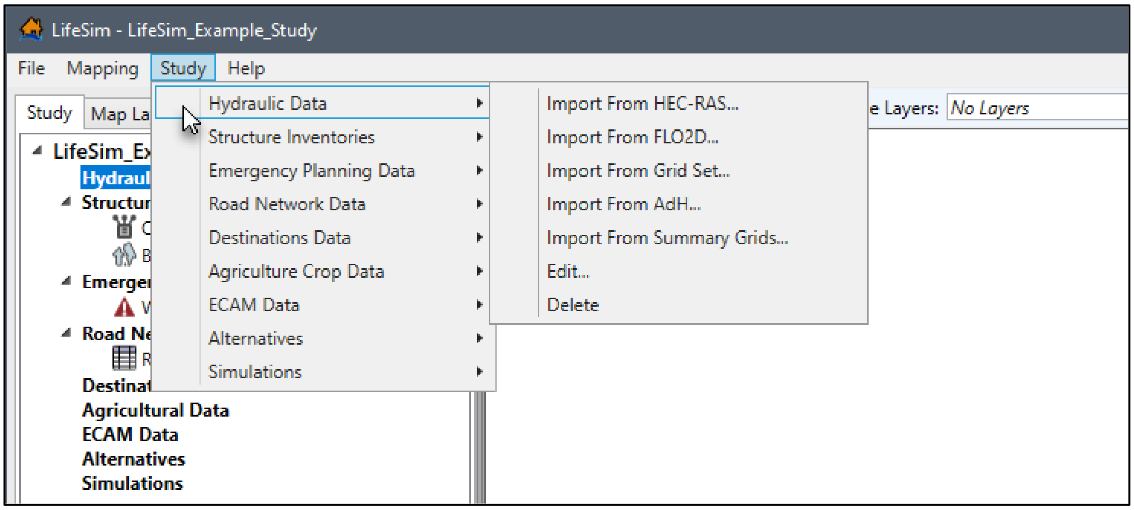


Figure 4-2. LifeSim Main Window – Study Menu – Hydraulic Data Sub-menu

The import dialog for each hydraulic source contains the same general structure. The top portion of the hydraulic data import dialogs provide browser buttons for selecting the source file(s) and the bottom portion of the import dialog contains the **Hydrograph Definition** panel (Figure 4-3). Depending on the hydraulic data importer the hydrograph definition panel provides two of three buttons as identified in Table 4-1.

Table 4-1. Hydraulic Data Import Option and Hydrograph Definition Options

Hydraulic Summary Grid	Import From RAS	Import From Map	Manual Entry
Import from HEC-RAS	Available	Available	Available
Import from FLO-2D	Not Available	Not Available	Available
Import from Grid Set	Not Available	Not Available	Available
Import from AdH	Not Available	Not Available	Available
Import from Summary Grids	Available	Not Available	Available

4.1.1 Import from HEC-RAS

LifeSim has the ability to import hydraulic results directly from an HEC-RAS study as long as the HEC-RAS results were produced with HEC-RAS Version 5.0 or later. The ability to link directly with the hydraulic results, and convert the hydraulic information, is an efficient step in creating hydraulic data for an LifeSim study. LifeSim uses the HEC-RAS RASMapper.dll library to sample hydraulic information and create layers that can be viewed in the map window (Figure 2-1).

Creating hydraulic data from an HEC-RAS study:

1. From the LifeSim main window (Figure 2-1) from the **Study** tab, from the **Study Tree**, right-click on **Hydraulic Data**, from the shortcut menu click **Import From HEC-RAS** (Figure 4-1). Another way is from the LifeSim main window, from the **Study** menu, point to **Hydraulic Data** (Figure 4-2), click **Import From HEC-RAS**.

2. Either way, the **Import From HEC-RAS (5.0 and later)** dialog box (Figure 4-3) opens.

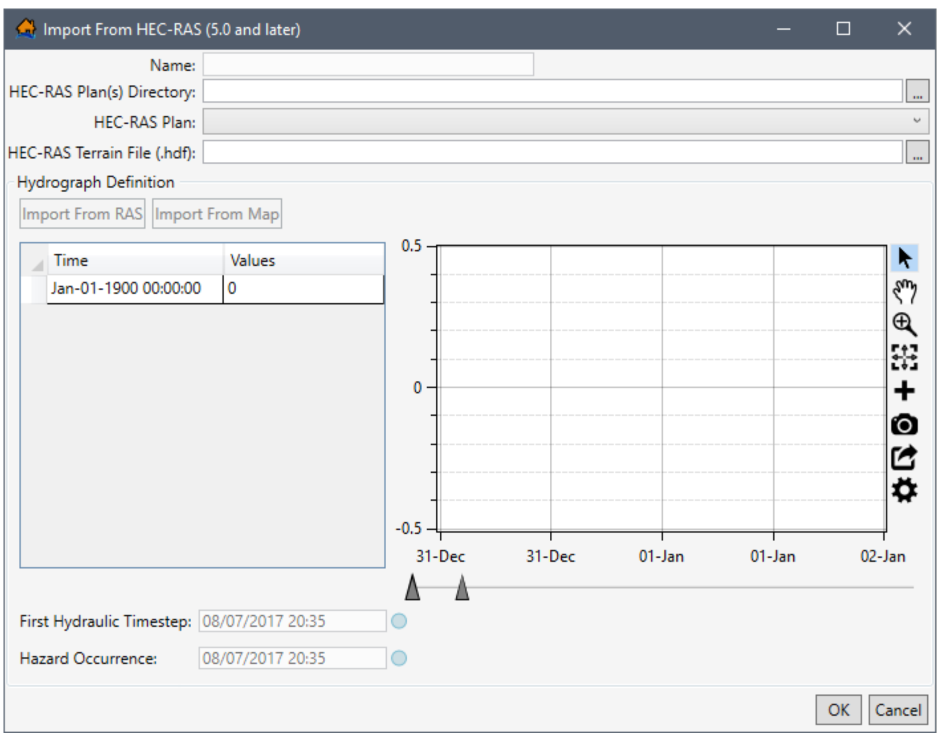


Figure 4-3. Import From HEC-RAS (5.0 and later) Dialog Box

3. To select the HEC-RAS output file, from the **RAS Output Set Files Directory** box click , an **Open Directory Containing RAS Output** browser (Figure 4-4) will open. Navigate to the location of an HEC-RAS plan output files; select the correct HEC-RAS plan output folder. The selected folder should contain the output files with the file extension *.*hdf*. Click **Select Folder**, the **Open Directory Containing RAS Output** browser closes (Figure 4-4) and the user is returned to the **Import From HEC-RAS (5.0 and later)** dialog box (Figure 4-6). Note, if the wrong folder is selected then the **HEC-RAS Output File** dropdown menu will be empty.

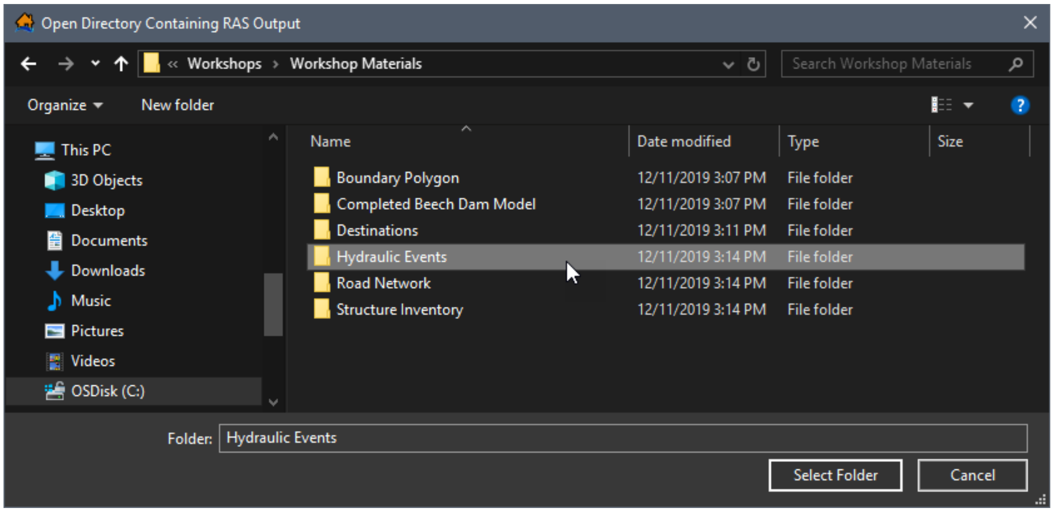


Figure 4-4. Open Directory Containing RAS Output Browser

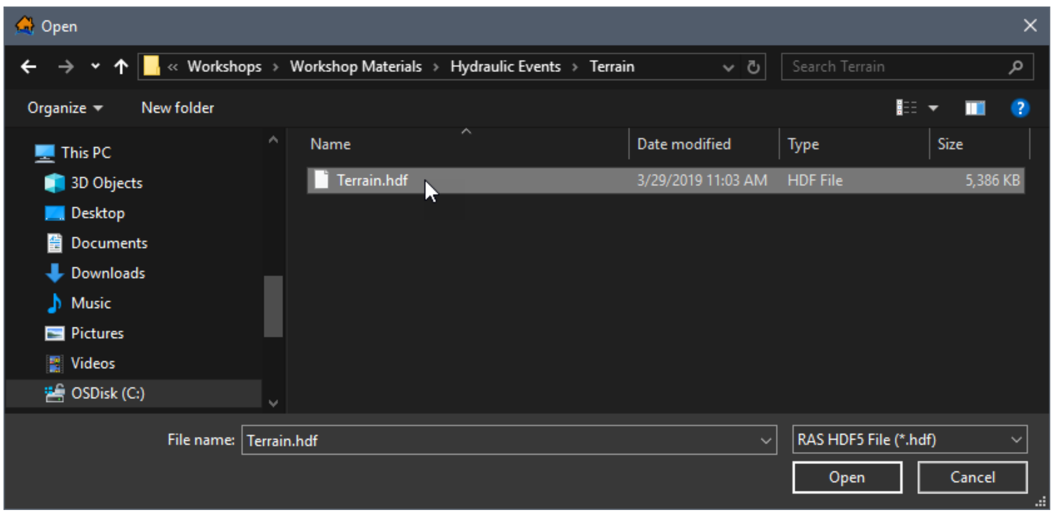


Figure 4-5. Open Browser – Select RAS HDF5 File

The **RAS Output Set Files Directory** box (Figure 4-5) updates with the location and the name of the HEC-RAS plan output file folder. Also, if there is an associated terrain file in the same location, then the **HEC-RAS Terrain File (*.*hdf***)** box will be automatically populated.

4. Select the desired HEC-RAS file from the **HEC-RAS Output File** dropdown (Figure 4-5) which contains the names of the HEC-RAS plan output files contained in the selected HEC-RAS output directory. By default, the name of the HEC-RAS plan output file is used as the name of the hydraulic dataset. The name appears in the **Name** box (Figure 4-5), the user can change the default name.

5. If the HEC-RAS terrain file is not located in the same place as the HEC-RAS plan output files, to the right of the **HEC-RAS Terrain File (.*hdf*)** box click ... . An **Open** browser (Figure 4-4) opens, navigate to the directory where the terrain file is located. Select the correct terrain file (e.g., *.*hdf*), click **Open**, the **Open** browser closes (Figure 4-4) and the

Import From HEC-RAS (5.0 and later) dialog box (Figure 4-5) updates with the selected file.

6. Next, the user can import the representative hydrograph (time series). Depending on the HEC-RAS output results file two options are available for selecting a representative hydrograph. The first option, **Import From RAS**, only works for HEC-RAS Version 5.0 and later output files (review Section 4.2.1). The second option, **Import From Map** only works for HEC-RAS Version 5.0 output files (review Section 4.2.2). The representative hydrograph can be created from a cross section location or a storage area location. Users can also add the representative hydrograph manually (review Section 4.2.3).

7. The **Hazard Occurrence** information need to be defined, Section 4.3 provides further details. Note, the first hydraulic timestep is defined by the model output for only hydraulic data imported from the **Import From HEC-RAS (5.0 and later)** dialog box (Figure 4-5) option.

8. Once all the data required to build a hydraulic dataset from HEC-RAS has been completed click **OK** (Figure 4-5).

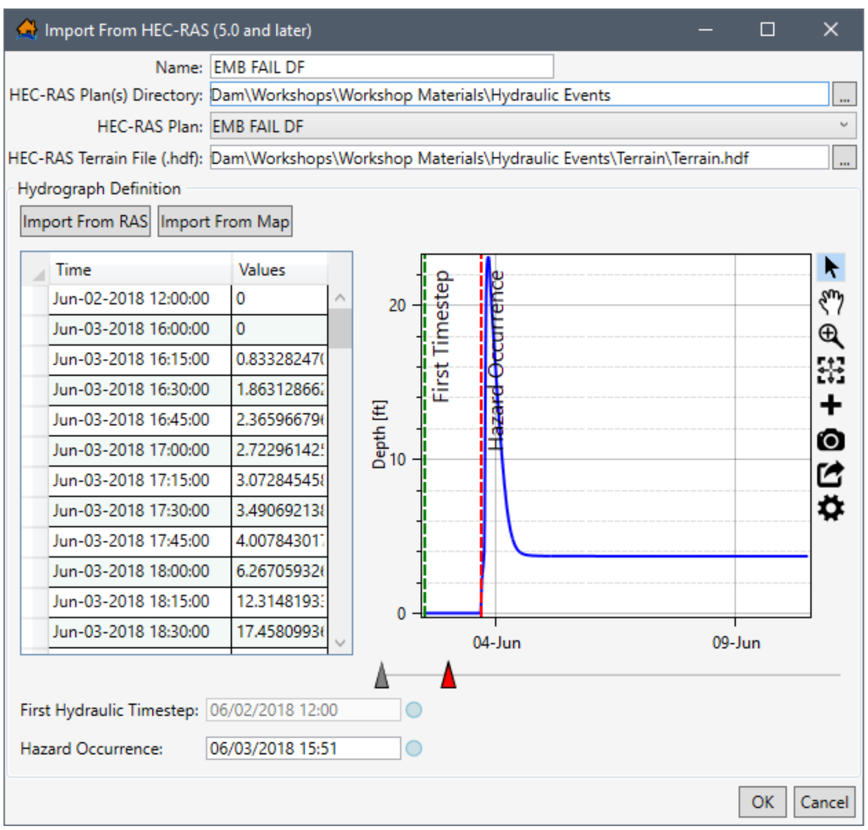


Figure 4-6. Example – Import From HEC-RAS (5.0 and later) Dialog Box – Completed

9. The **Import From HEC-RAS (5.0 and later)** dialog box (Figure 4-5) closes, a **Progress Window** opens displaying the progress of the import and once complete the hydraulic data is added to the study and displays in the **Study Tree** (Figure 4-1), under **Hydraulic Data**.

4.1.2 Import from FLO-2D

LifeSim has the ability to import hydraulic results directly from FLO-2D output files. The required FLO-2D output files are DEPTH.OUT and TIMDEP.OUT which can be generated automatically from a FLO-2D simulation. LifeSim will take the FLO-2D results files and convert them into a time series of depth and velocity compressed geotiff files. The geotiff files are stored in the hydraulics directory of the LifeSim study. The ability to link directly with the hydraulic results saves a huge step in converting hydraulic information to a software specific format.

To import hydraulic data, from FLO-2D, to an existing LifeSim study:

1. From the LifeSim main window (Figure 2-1), from the **Study** tab, from the **Study Tree**, right-click on **Hydraulic Data**, from the shortcut menu click **Import From FLO2D**. Another way is from the LifeSim main window, from the **Study** menu, point to **Hydraulic Data** (Figure 4-2), and click **Import From FLO2D**.

2. From either way, the **Import From FLO2D** dialog box (Figure 4-7) opens. In the **Name** box (Figure 4-7), enter a name for the hydraulic data.

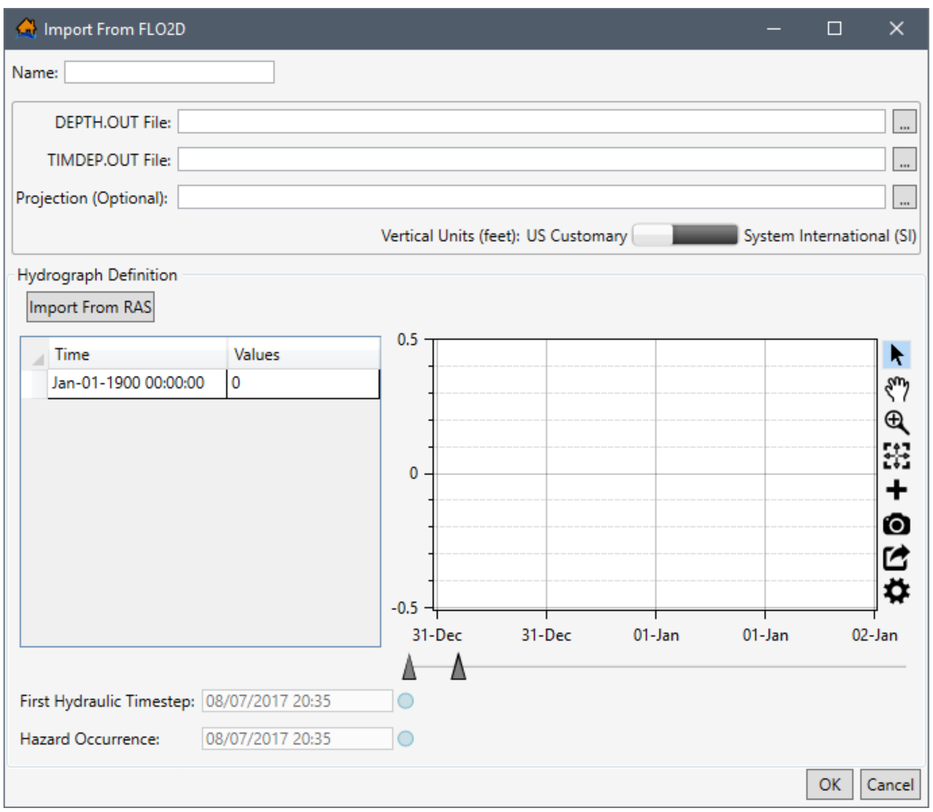


Figure 4-7. Import From FLO2D Dialog Box

3. In the **DEPTH.OUT File** box click , an **Open** browser (Figure 4-4) will open. Navigate to the directory where the hydraulic data (**DEPTH.OUT**) for the FLO-2D model is located. Select the correct FLO-2D output file (e.g., **.OUT*). Click **Open**, the **Open** browser closes (Figure 4-4), and the **Import From FLO2D** dialog box (Figure 4-8) updates to display the selected file location in the **DEPTH.OUT File** box.

4. If the FLO-2D depth file (DEPTH.OUT) is not located in the same directory as the time of depth file (TIMDEP.OUT), to right of the **TIMDEP.OUT File** box click ..., an **Open** browser will open (Figure 4-4). Navigate to the directory where the hydraulic file (**TIMDEP.OUT**) is located, select the file (**.OUT)*. Click **Open**, the **Open** browser closes (Figure 4-4), and the **Import From FLO2D** dialog box (Figure 4-8) updates to display the selected file location in the **TIMEDEP.OUT File** box.

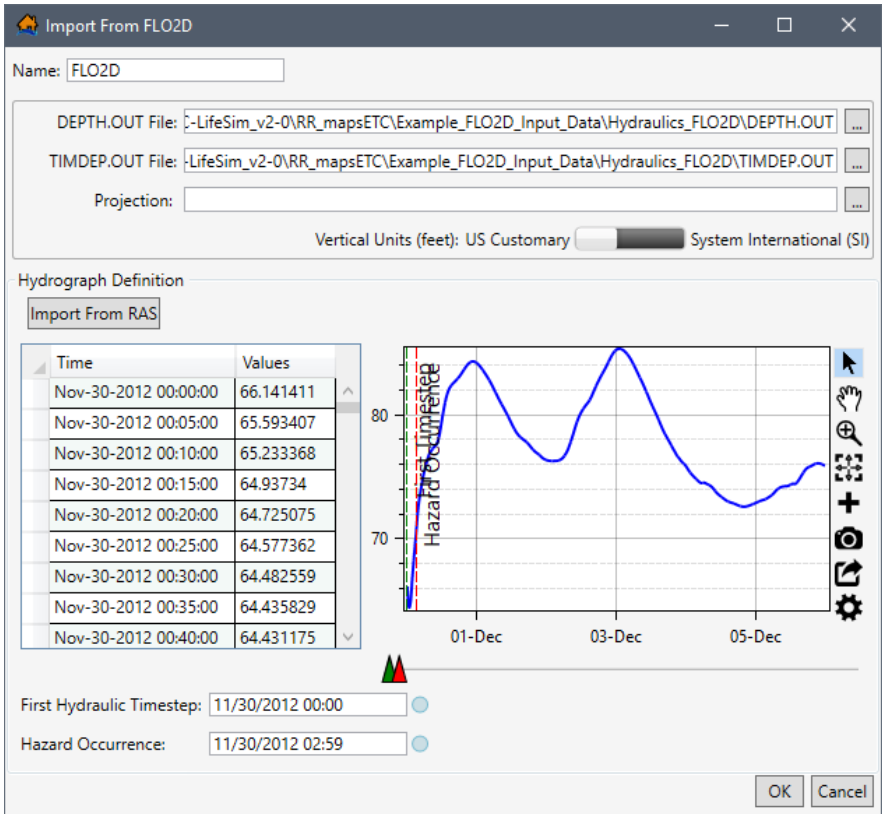


Figure 4-8. Import From FLO2D Dialog – Example Selected Time Series Data

5. The **Projection** box is available for selecting the geographic projection for the FLO-2D hydraulic data to be imported using an ESRI®'s projection file format. The user may add the geographic projection by clicking ..., an **Open** browser will open (Figure 4-4). Navigate to the location of the projection file *(*.prj)*, click **Open**, the **Open** browser closes (Figure 4-4) and updates the **Import From FLO2D** dialog box (Figure 4-8) to display the selected projection file in the **Projection** box.

6. Set the vertical units for the hydraulic data (either **US Customary** or **System International**) from the **Vertical Units** list (Figure 4-8).

7. Next, enter the representative hydrograph (time series). Users can enter the representative hydrograph manually (review Section 4.2.3).

8. The **First Hydraulic Timestep** and the **Hazard Occurrence** information needs to be defined next, refer to Section 4.3 for further details.

9. Click **OK**, the **Import From FLO2D** dialog box (Figure 4-8) closes. The hydraulic data is added to the study and displays in the **Study Tree** (Figure 4-1), under **Hydraulic Data**.

4.1.3 Import from Grid Set

LifeSim has the ability to import hydraulic results (e.g., MIKE.21, DHI) directly from a time series of gridded files (grid set). Most hydraulic simulation engines can store depth and velocity grids as time series, which is an easy and convenient way to transfer hydraulic simulation data among various groups that may not have the hydraulic modeling software. LifeSim can import depth and velocity grids in several formats: **.flt*, **.tif*, and binary ESRI® grids. When importing a grid set, LifeSim converts the gridded data into geotiff files which are stored in the hydraulics study directory.

To import hydraulic data, in a gridded format, to an existing LifeSim study:

1. From the LifeSim main window (Figure 2-1), from the **Study** tab, from the **Study Tree**, right-click on **Hydraulic Data**, from the shortcut menu click **Import From Grids**. Another way is from the LifeSim main window, from the **Study** menu, point to **Hydraulic Data** (Figure 4-2), and click **Import From Grid Set**. Either way, the **Import From Grid Set** dialog box (Figure 4-9) opens.

2. In the **Name** box (Figure 4-9), enter a name for the hydraulic gridded data.

3. To the right of the **Grid Set Files Directory** box click , an **Open Directory Containing grids** browser opens (Figure 4-10). Navigate to the folder where the hydraulic grid data is located, select the correct folder, click **Select Folder**. The **Open Directory Containing grids** browser closes (Figure 4-10), and the **Import From Grid Set** dialog box (Figure 4-11) updates to display the selected folder in the **Grid Set Files Directory** box.

4. Next, if the depth grids are spaced at regular time intervals and are in a specific naming convention and numerical order, click **Grids are in numeric order with the first depth grid = dp1 or dp001 and velocity = vp1 or vp001**. Only select this option, if the **Grid Set Timestep (minutes)** is the same between each grid and the grids follow the specified naming convention of "depth grids being dp# and velocity grids being vp#" where # is 1 or 001 for the first grid and incrementing by one for each subsequent timestep. Enter the time interval separating the grids in the **Grid Set Timestep (minunte)** box.

5. If the gridded files do not satisfy the regular interval naming format, do not select **Grids are in numeric order with the first depth grid = dp1 or dp001 and velocity = vp1 or vp001** and the **Grid Set Timestep** (**minutes**) box will remain disabled (grayed out). The user then must enter the **Depth**, **Velocity** and **Time (hours)** in the table. Click the **Add Row** and **Remove Row** buttons to add a row or remove a row in the **Depth**, **Velocity**, and **Time (hours)** table. Rows will always be either added or removed from the bottom of the table.

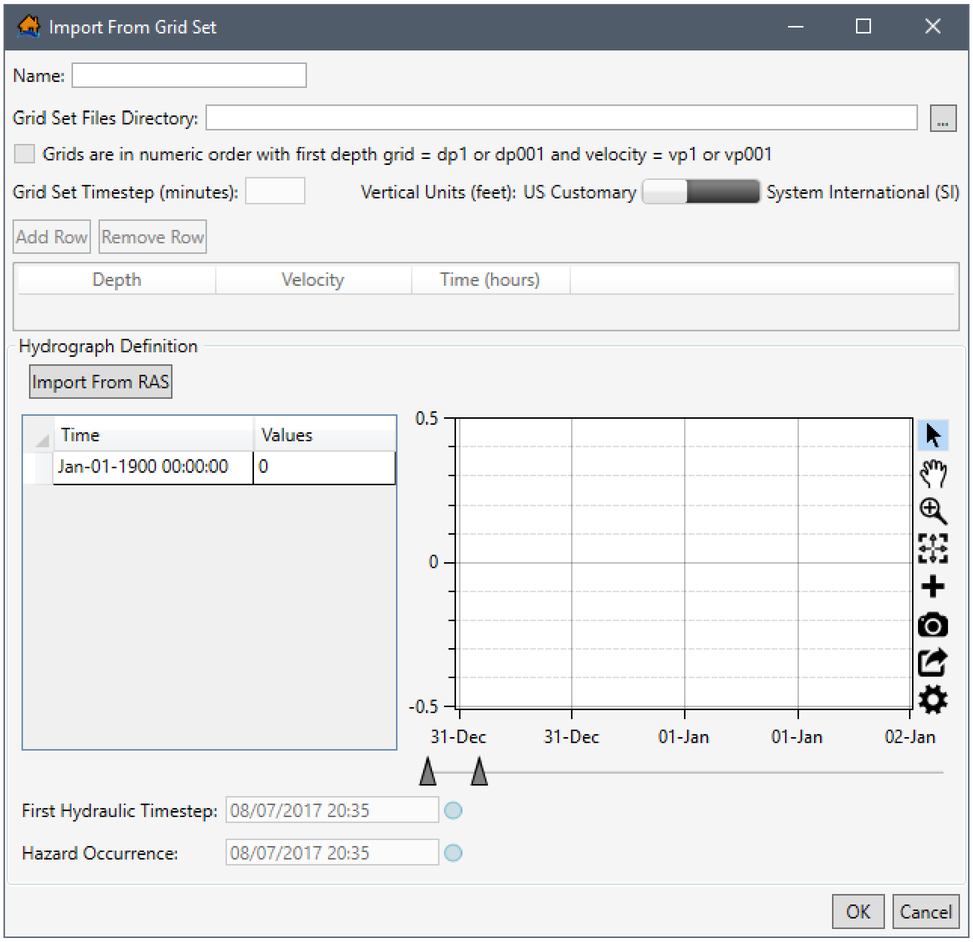


Figure 4-9. Import From Grid Set Dialog Box

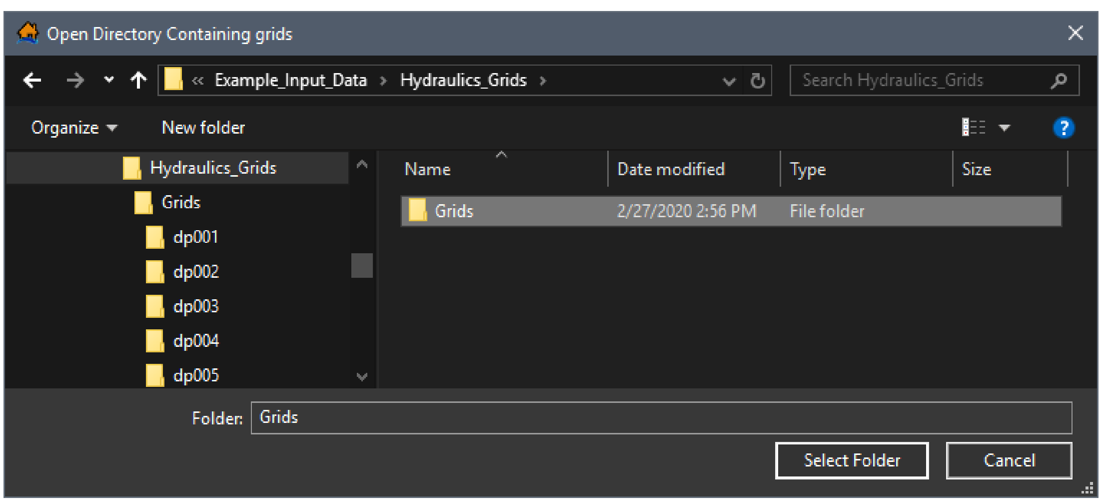


Figure 4-10. Open Directory Containing grids Browser Window

6. Set the **Vertical Units** (either **US Customary**, feet, or **System International (SI)**, meters) for the hydraulic data (Figure 4-11).

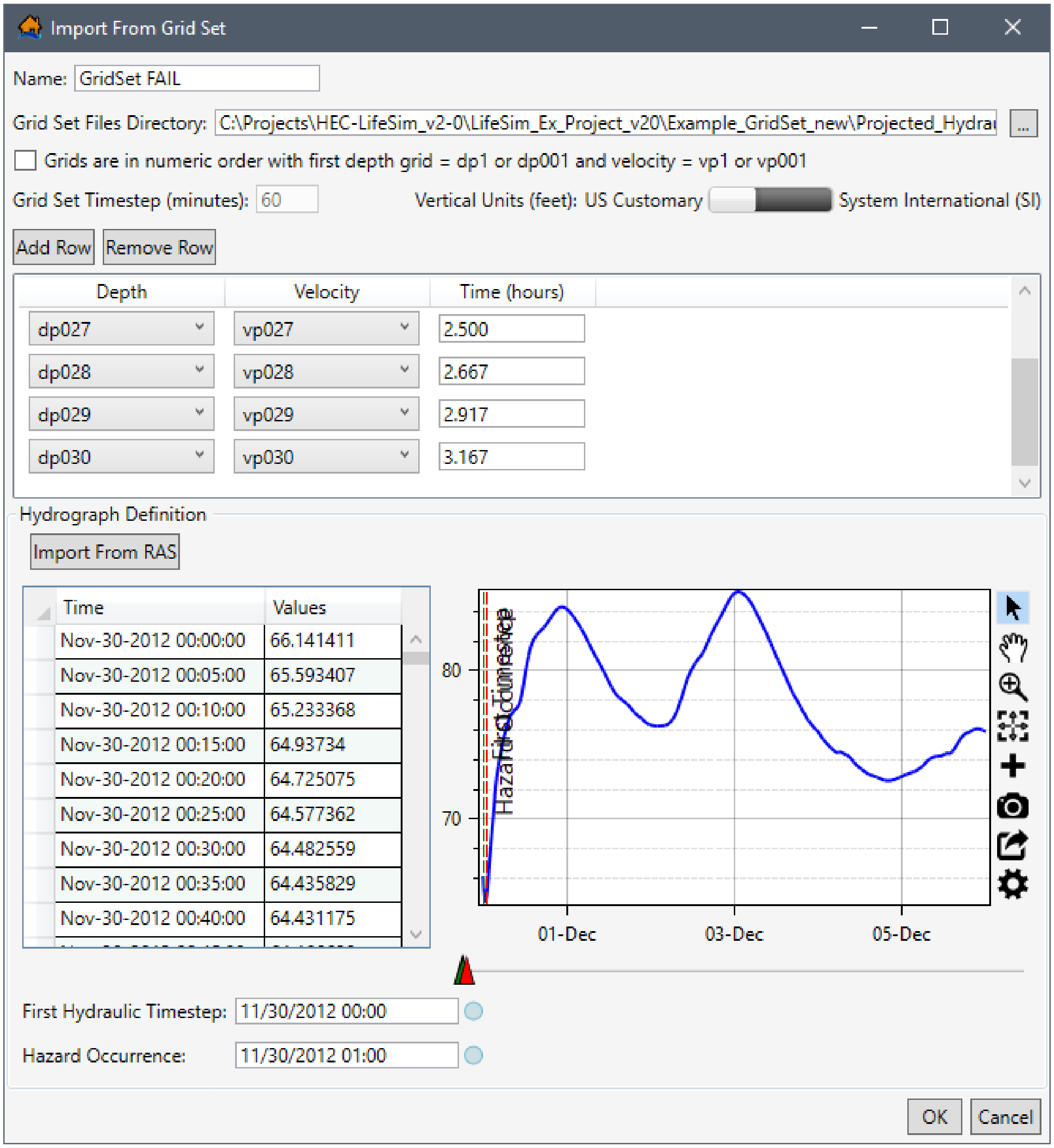


Figure 4-11. Import From Grid Set Dialog Box – Completed Example

7. Next, the user can import the representative hydrograph (time series). The first option, **Import From RAS**, only works for HEC-RAS Version 5.0 and later output files (review Section 4.2.1). The second option is to add the representative hydrograph manually (review Section 4.2.3).

8. The **First Hydraulic Timestep** and the **Hazard Occurrence** information needs to be defined next, refer to Section 4.3 for further details.

9. Once all of the data has been entered click **OK**, the **Import From Grid Set** dialog box (Figure 4-11) closes. The hydraulic data is added to the study and displays in the **Study Tree** (Figure 4-1), under **Hydraulic Data**.

4.1.4 Import from AdH

New to Version 2.0, LifeSim has the ability to import hydraulic results from ERDC's Adaptive Hydraulics (AdH) computational tool simulation output files. LifeSim will take the AdH results files and convert them into a TIN mesh time series of depth and velocity. The TIN and time series data are stored in the hydraulics directory of the LifeSim study. The ability to link directly with the hydraulic results saves a huge step in converting hydraulic information to a software specific format.

To import hydraulic data, from AdH, to an existing LifeSim study:

1. From the LifeSim main window (Figure 2-1), from the **Study** tab, from the **Study Tree**, right-click on **Hydraulic Data**, from the shortcut menu click **Import From AdH**. Another way is from the LifeSim main window, from the **Study** menu, point to **Hydraulic Data** (Figure 4-2), and click **Import From AdH**.

2. From either way, the **Import From AdH** dialog box (Figure 4-12) opens. In the **Name** box (Figure 4-12), enter a name for the hydraulic data.

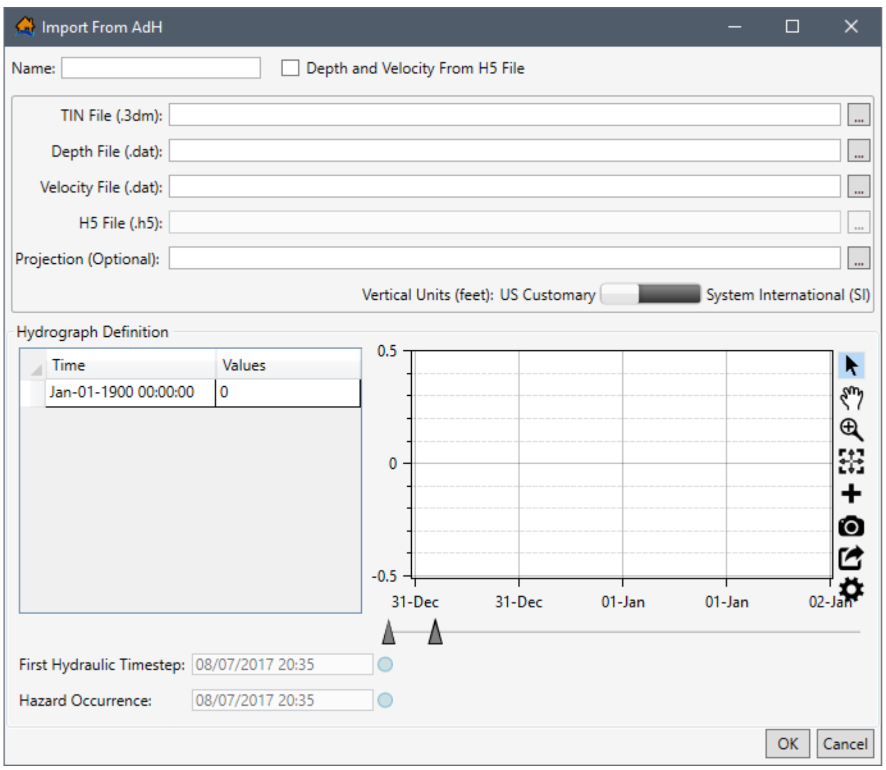


Figure 4-12. Import From AdH Dialog Box

3. Next to the **TIN File (.3dm)** box, click [...], an **Open** browser (Figure 4-13) opens. Navigate to the folder where the AdH mesh data is located, select the correct TIN File (*.3dm) file, click **Open**. The **Open** browser closes (Figure 4-13), and the **Import From AdH** dialog box (Figure 4-14) updates the **TIN File (.3dm)** box (Figure 4-14).

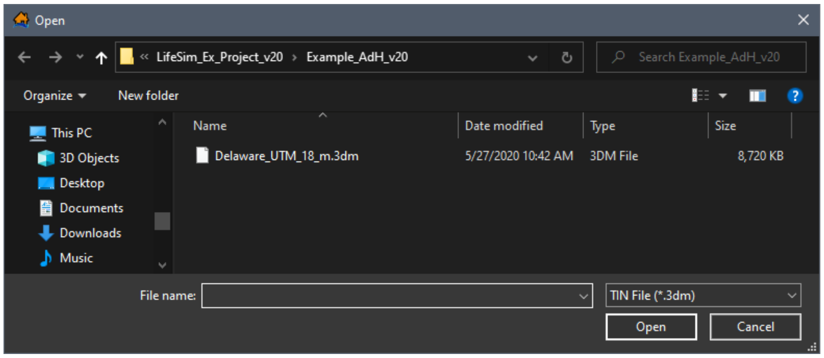


Figure 4-13. Open Browser Window

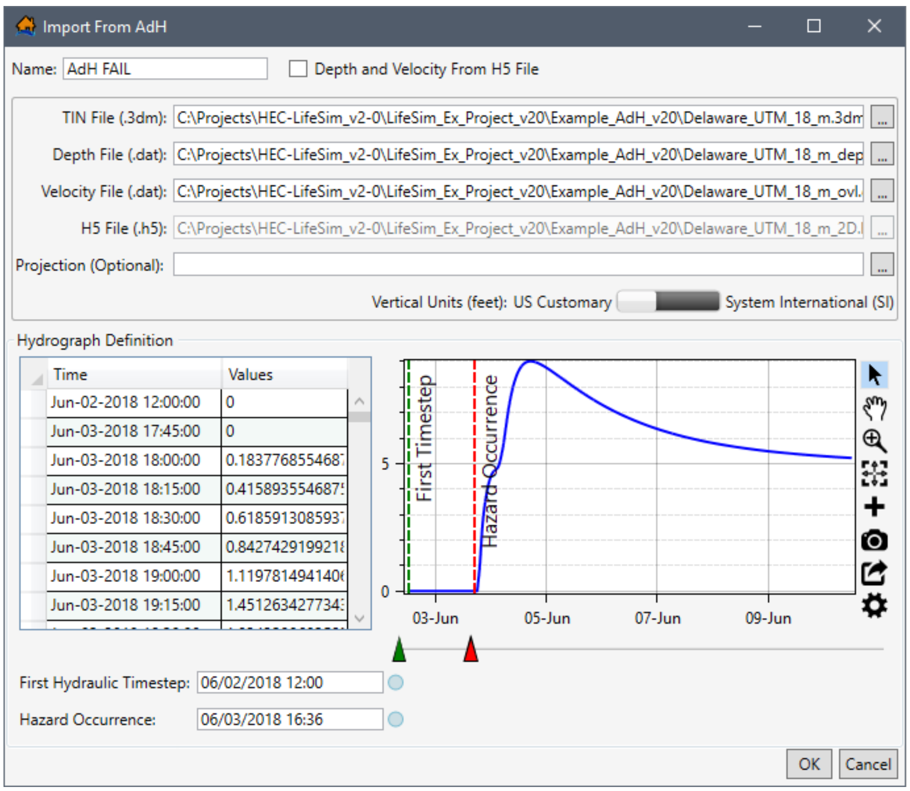


Figure 4-14. Import From AdH – Example Completed

4. If the **Depth File (.dat)** and **Velocity File (.dat)** or **H5 File (.h5)** files (Figure 4-14) are saved in the same location as the TIN file, then the input files populate automatically after selecting the TIN file. If not, click ... for each file and from the **Open** browser window (Figure 4-13), locate and select the appropriate files, and click **Open**.

5. Alternatively, if the depth and velocity data is contained in the H5 (*.h5) AdH output file, check ☑ the **Depth and Velocity From H5 File** (Figure 4-14) checkbox. Then next to the **H5 File (.h5)** box, click .... From the **Open** browser window (Figure 4-13), locate and select the appropriate H5 (*.h5) file, and click **Open**. The **Open** browser closes (Figure 4-13), and the **Import From AdH** dialog box (Figure 4-14) updates to display the selected folder in the **H5 File (.h5)** box (Figure 4-14).

6. Set the **Vertical Units** (either **US Customary**, feet, or **System International (SI)**, meters) for the hydraulic data (Figure 4-14).

7. Next, add the representative hydrograph manually (review Section 4.2.3).

8. Define the **First Hydraulic Timestep** and the **Hazard Occurrence** information, refer to Section 4.3 for further details.

9. Once all of the data has been entered click **OK**, the **Import From AdH** dialog box (Figure 4-14) closes. The hydraulic data is added to the study and displays in the **Study Tree** (Figure 4-1), under **Hydraulic Data**.

4.1.5 Import from ADCIRC

New to Version 2.0, LifeSim has the ability to import hydraulic results from the University of North Carolina's Advanced Circulation Model (ADCIRC) simulation output files. LifeSim will take the ADCIRC results files and convert them into a TIN mesh time series of depth and velocity. The TIN and time series data are stored in the hydraulics directory of the LifeSim study. The ability to link directly with the hydraulic results saves a huge step in converting hydraulic information to a software specific format.

To import hydraulic data, from ADCIRC, to an existing LifeSim study:

1. From the LifeSim main window (Figure 2-1), from the **Study** tab, from the **Study Tree**, right-click on **Hydraulic Data**, from the shortcut menu click **Import From ADCIRC**. Another way is from the LifeSim main window, from the **Study** menu, point to **Hydraulic Data** (Figure 4-2), and click **Import From ADCIRC**.

2. From either way, the **Import From ADCIRC** dialog box (Figure 4-15) opens. In the **Name** box (Figure 4-15), enter a name for the hydraulic data.

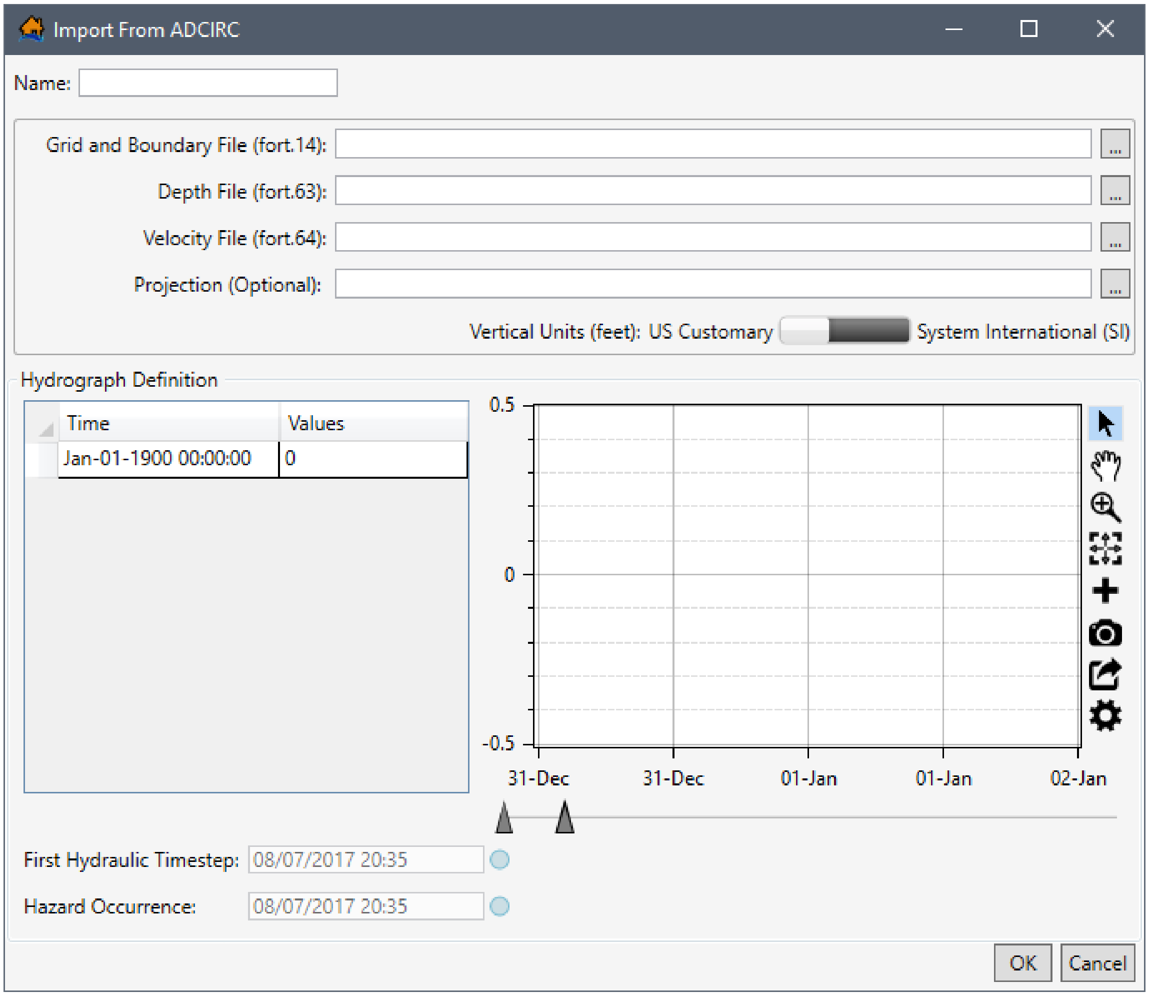


Figure 4-15. Import From ADCIRC Dialog Box

3. Next to the **Grid and Boundary File (fort.14)** box, click ..., an **Open** browser (Figure 4-16) opens. Navigate to the folder where the ADCIRC mesh data is located, select the correct TIN File (*.14) file, click **Open**. The **Open** browser closes (Figure 4-16), and the **Import From ADCIRC** dialog box (Figure 4-14) updates the **Grid and Boundary File (fort.14)** box (Figure 4-17).

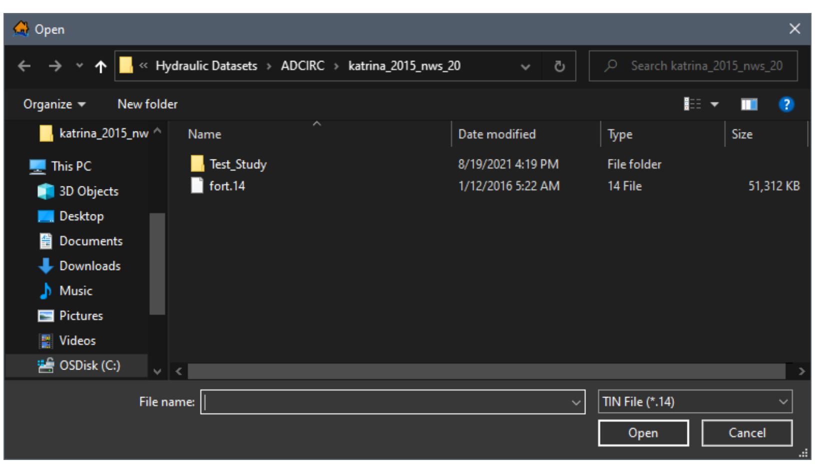


Figure 4-16. Open Browser Window

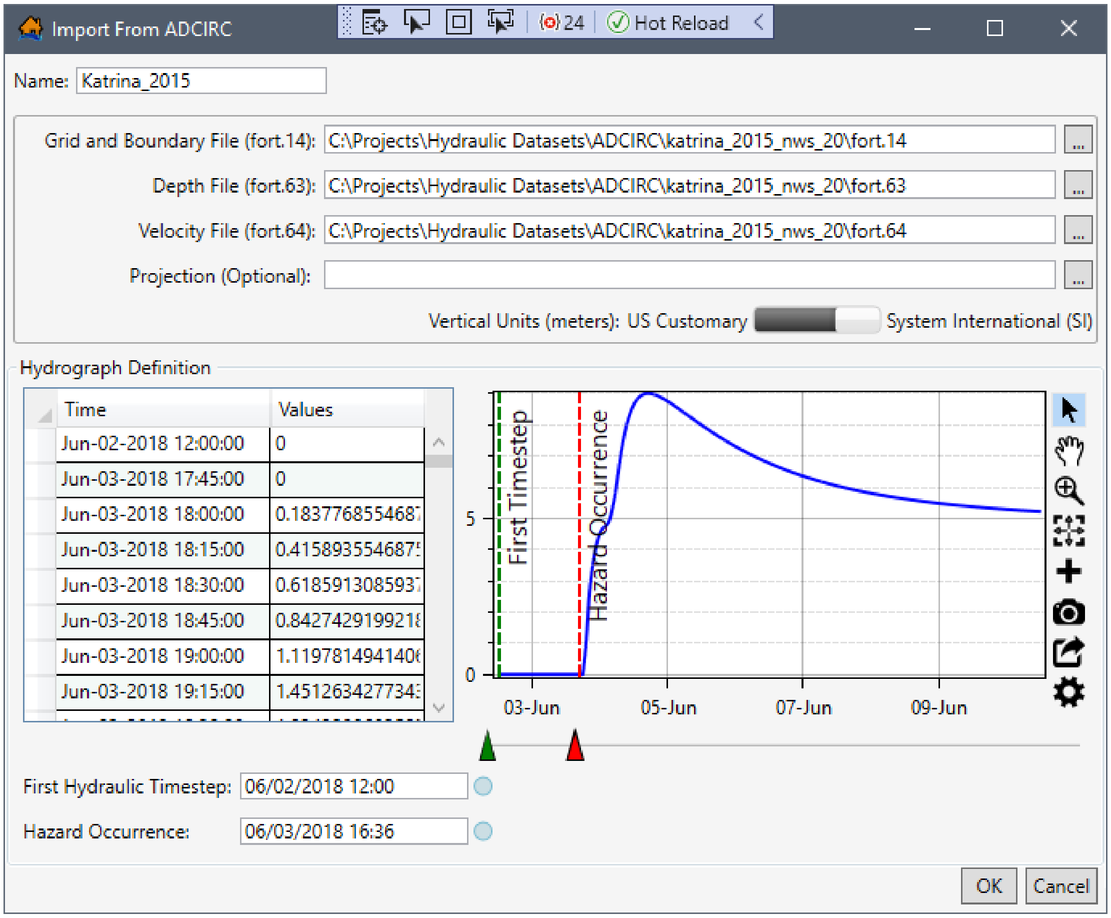


Figure 4-17. Import From ADCIRC – Example Completed.

4. If the **Depth File (fort.63)** and **Velocity File (fort.64)** files (Figure 4-17) are saved in the same location as the TIN file, then the input files populate automatically after selecting the TIN file. If not, click ... for each file and from the **Open** browser window (Figure 4-16), locate and select the appropriate files, and click **Open**.

5. Set the **Vertical Units** (either **US Customary**, feet, or **System International (SI)**, meters) for the hydraulic data (Figure 4-17).

6. Next, add the representative hydrograph manually (review Section 4.2.3).

7. Define the **First Hydraulic Timestep** and the **Hazard Occurrence** information, refer to Section 4.3 for further details.

8. Once all of the data has been entered click **OK**, the **Import From ADCIRC** dialog box (Figure 4-17) closes. The hydraulic data is added to the study and displays in the **Study Tree** (Figure 4-1), under **Hydraulic Data**.

4.1.6 Import from Summary Grids

New to Version 2.0, LifeSim has the ability to import hydraulic data from summary grids (raster files). Importing summary grids bypasses the need for LifeSim to calculate hydraulic characteristics. LifeSim can import summary grids in several formats: **.flt*, **.tif*, and **.vrt*. The five hydraulic characteristics that can be represented by summary grids are maximum depth, maximum velocity, non-evacuation arrival time, first-inundated arrival time, and duration inundated. Depending on the target consequence estimate (e.g., structure damages, life loss, agricultural damages) specific hydraulic characteristics are required (Table 4-2).

LifeSim stores imported summary grid files in the hydraulics study directory.

Table 4-2. Required Summary Grid Hydraulic Characteristics by Target Consequence Estimate

Hydraulic Summary Grid	Structural Damages	Life Loss	Agricultural Damages
Maximum Depth	Required	Required	Not Applicable
Maximum Velocity	Optional	Optional	Not Applicable
Non-Evacuation Depth Arrival Time	Not Applicable	Required	Not Applicable
First-Inundated Arrival Time	Not Applicable	Optional	Required
Duration Inundated	Not Applicable	Not Applicable	Required

To import hydraulic data, in a gridded format, to an existing LifeSim study:

1. From the LifeSim main window (Figure 2-1), from the **Study** tab, from the **Study Tree**, right-click on **Hydraulic Data**, from the shortcut menu click **Import From Summary Grids**. Another way is from the LifeSim main window, from the **Study** menu, point to **Hydraulic Data**, click **Import From Summary Grids**.

2. Either way, the **Import From Summary Grids** dialog box (Figure 4-18) opens.

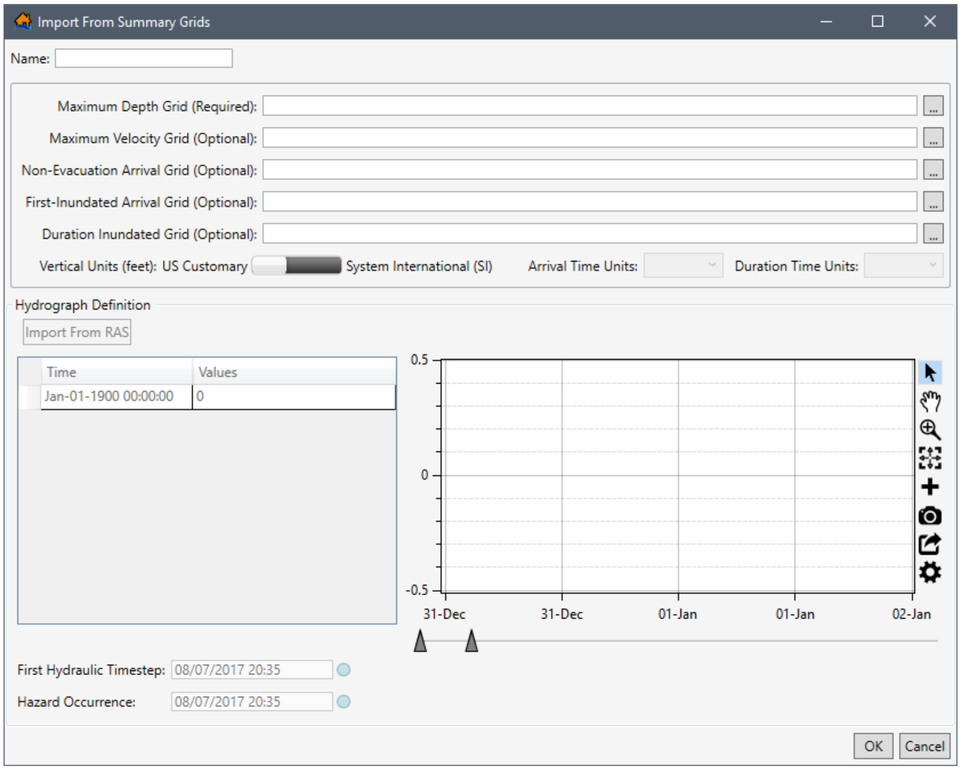


Figure 4-18. Import From Summary Grids Dialog Box

3. In the **Name** box (Figure 4-18), enter a name for the hydraulic data.

4. To the right of the **Maximum Depth Grid (Required)** box click ..., an **Open** browser opens (Figure 4-5). Navigate to the folder containing the hydraulic summary grid (*.tif, *.flt, or *.vrt), select the correct file and file type (e.g., FLOAT File (*.flt)), and click **Open**. The **Open** browser closes (Figure 4-5), the **Import From Summary Grids** dialog box (Figure 4-19) displays the selected file in the **Maximum Depth Grid** box.

5. Continue to select the desired summary grids to import by clicking the browse to ... button, navigating to the desired summary grid, selecting the grid and clicking **Open**. The **Import From Summary Grids** dialog box (Figure 4-19) displays the selected file.

6. Set the **Vertical Units** (either **US Customary**, feet, or **System International (SI)**, meters) for the hydraulic data (Figure 4-19).

7. If the summary grids selected include the Non-Evacuation Arrival and/or First-Inundated Arrival grids, then the **Arrival Time Units** is enabled. From the **Arrival Time Units** list (Figure 4-19), select the correct units (**Seconds**, **Minutes**, **Hours**, **Days**) for the selected arrival grids. If both Non-Evacuation and First-Inundated grids are supplied, it will be assumed that they both have the same time units.

8. If a summary grid is selected for the Duration Inundated Grid, then **Duration Time Units** is enabled. From the **Duration Time Units** list (Figure 4-19), select the correct units (**Seconds**, **Minutes**, **Hours**, **Days**) for the selected arrival grids.

9. Next, create the representative hydrograph by clicking the **Import From RAS** (refer to Section 4.2.1) button, or by adding the hydrograph time series manually (review Section 4.2.3).

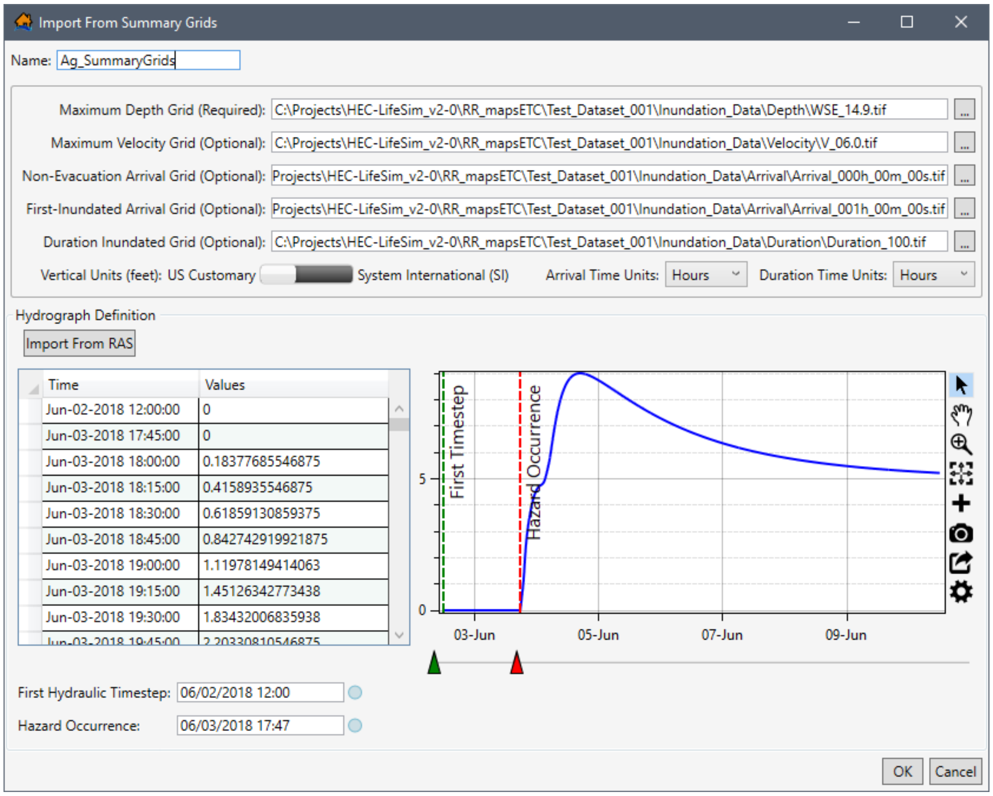


Figure 4-19. Import From Summary Grids – Completed Example

10. Then define the **Hazard Occurrence** and the **First Hydraulic Timestep** information next, refer to Section 4.3 for further details.

11. Once all of the data has been entered in the **Import From Summary Grids** dialog box (Figure 4-19), click **OK**. The **Import From Summary Grids** dialog box (Figure 4-19) closes. The hydraulic data is added to the study and displays in the **Study Tree** (Figure 4-1), under **Hydraulic Data**.

4.2 Representative Hydrograph

In addition to providing the time series of depth and velocity (review Section 4.1), a hydrograph representing the hydraulic event must be provided as well for performing life loss calculations. The representative hydrograph is critical in developing the warning issuance timing relative to a specific condition (e.g., dam or levee breach). However, the hydrograph is for visualization purposes only and is not used in the simulation engine. The representative hydrograph can be created by importing the hydrograph from HEC-RAS data, import from a map (created from the HEC-RAS hydraulic data that has been imported), or by entering the hydrograph time series data manually.

4.2.1 Import From RAS

The representative hydrograph can be created by importing the hydrograph from HEC-RAS data. Many of the hydraulic data import options described in Section 4.1, include the option to select the representative hydrograph from HEC-RAS Version 5.0 (and later) output data by clicking the **Import From RAS** button (Figure 4-20) to open the **RAS Results Selector** (Figure 4-21) dialog box.

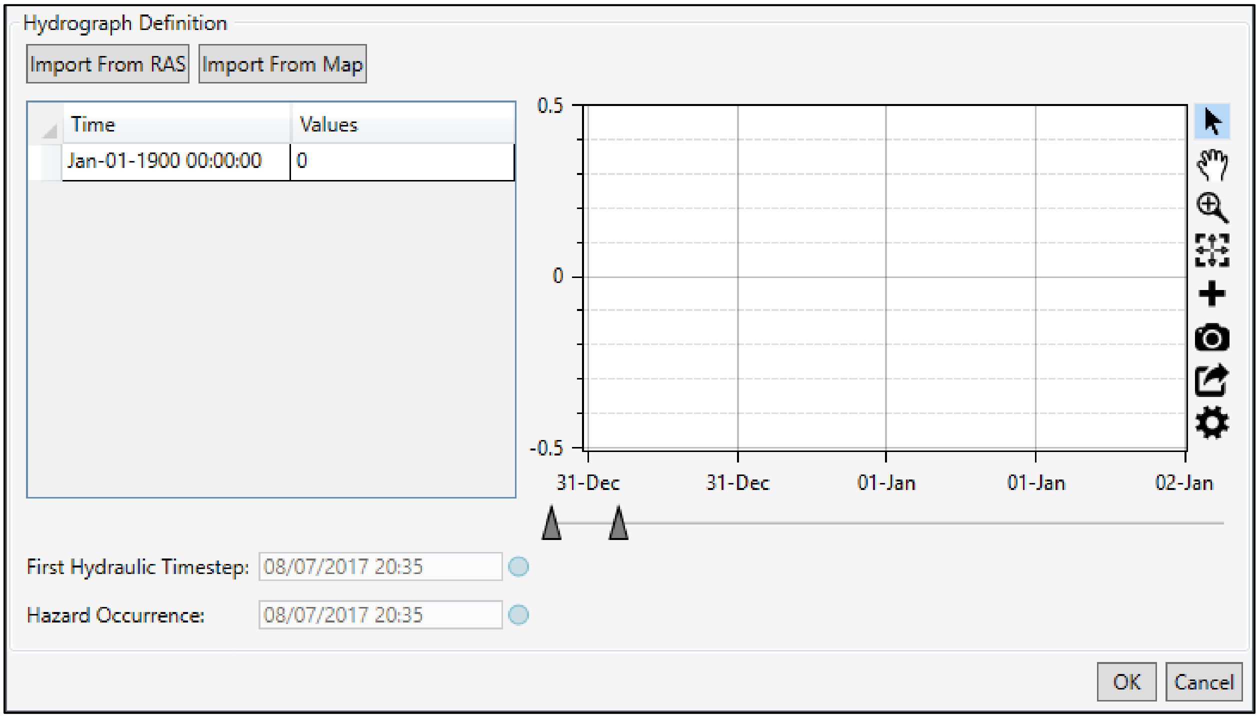


Figure 4-20. Example – Hydraulic Data Importer – Hydrograph Definition Panel

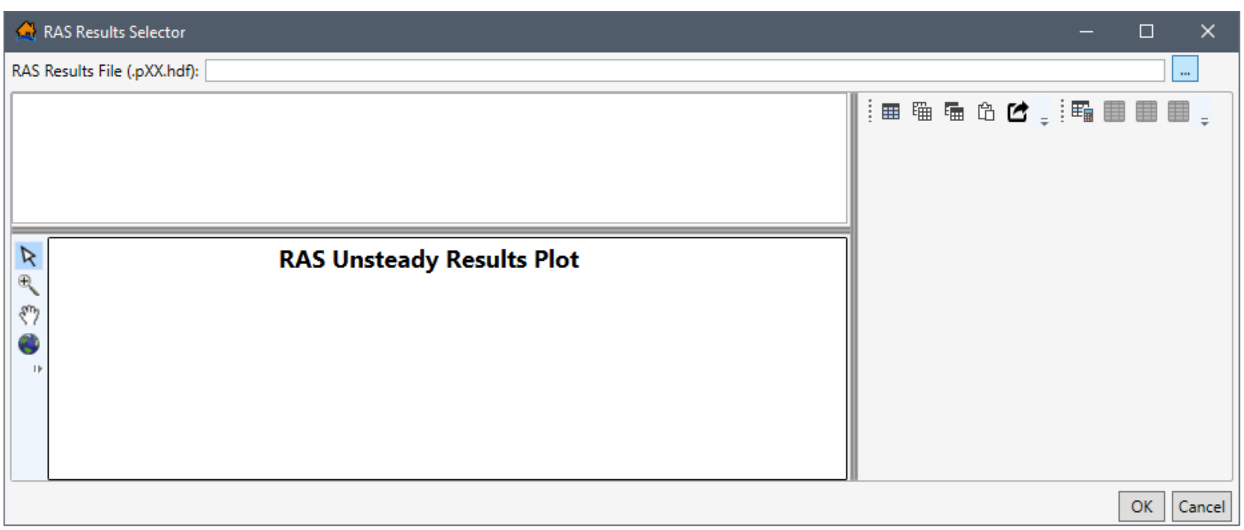


Figure 4-21. RAS Results Selector Dialog Box

To select the representative hydrograph from HEC-RAS Version 5.0 results:

1. From the **RAS Results File (*.pXX.hdf)** box click [...]. An **Open** browser (Figure 4-4) opens, navigate to the directory where the HEC-RAS output file is located. Select the correct RAS results file (*.pXX.hdf), click **Open**, the **Open** browser will close (Figure 4-4) and the selected file opens in the **RAS Results Selector** (Figure 4-21). The representative hydrograph can be created from a cross section location or a storage area location.

2. To create the representative hydrograph from an HEC-RAS cross section location, from the **RAS Results Selector** (Figure 4-21) dialog box, click ⌄ to expand the **Cross Sections** (Figure 4-22) list. From the ⌃ expanded list, select the HEC-RAS cross section that contains the representative hydrograph (e.g., ***Russian - DCtoOcean - 8693.624*** in Figure 4-22).

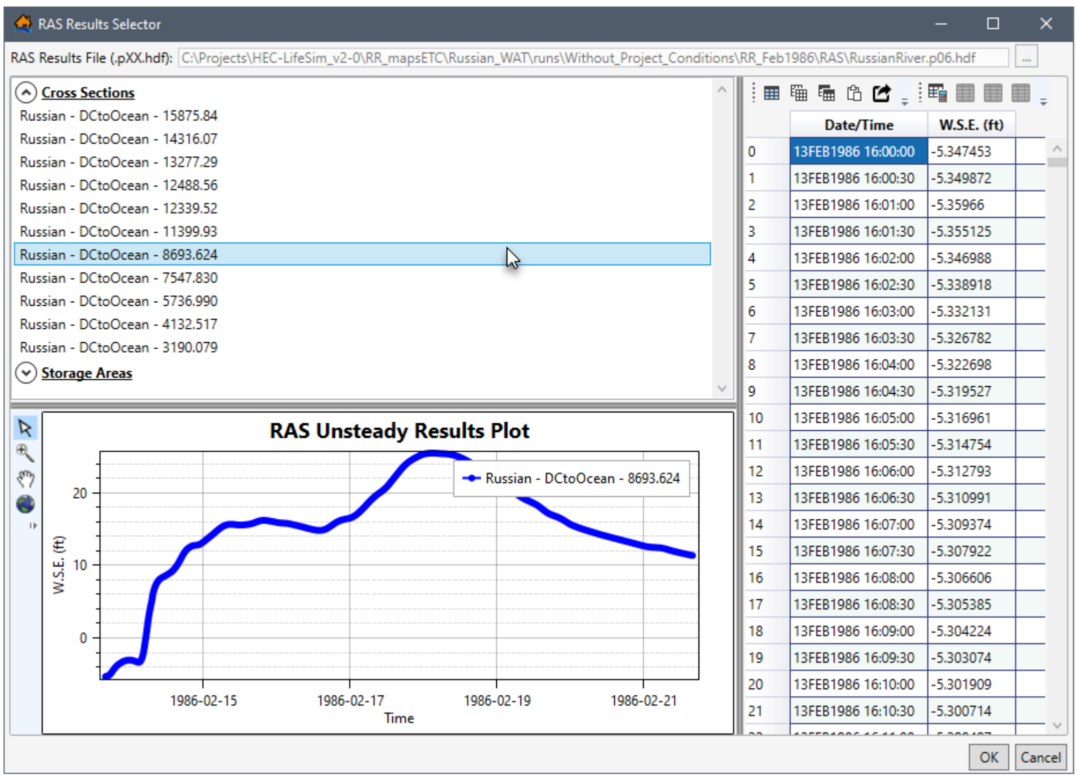


Figure 4-22. RAS Results Selector Dialog Box – Hydrograph from a Cross Section

3. Alternatively, to create the representative hydrograph from an HEC-RAS storage area location, from the **RAS Results Selector** (Figure 4-21), expand **Storage Areas** (Figure 4-23), and select the HEC-RAS storage area that contains the representative hydrograph (e.g., ***367*** in Figure 4-23).

4. The **RAS Unsteady Results Plot** (Figure 4-22) located at the bottom of the **RAS Results Selector** dialog box, displays the hydrograph for the selected cross section or storage area (Figure 4-23). The table section, located on the right side of the **RAS Results Selector**, displays the time series data for the representative hydrograph (Figure 4-23). Refer to Chapter 3 for details regarding map window tools, table tools, and customizing plots and plotting tools.

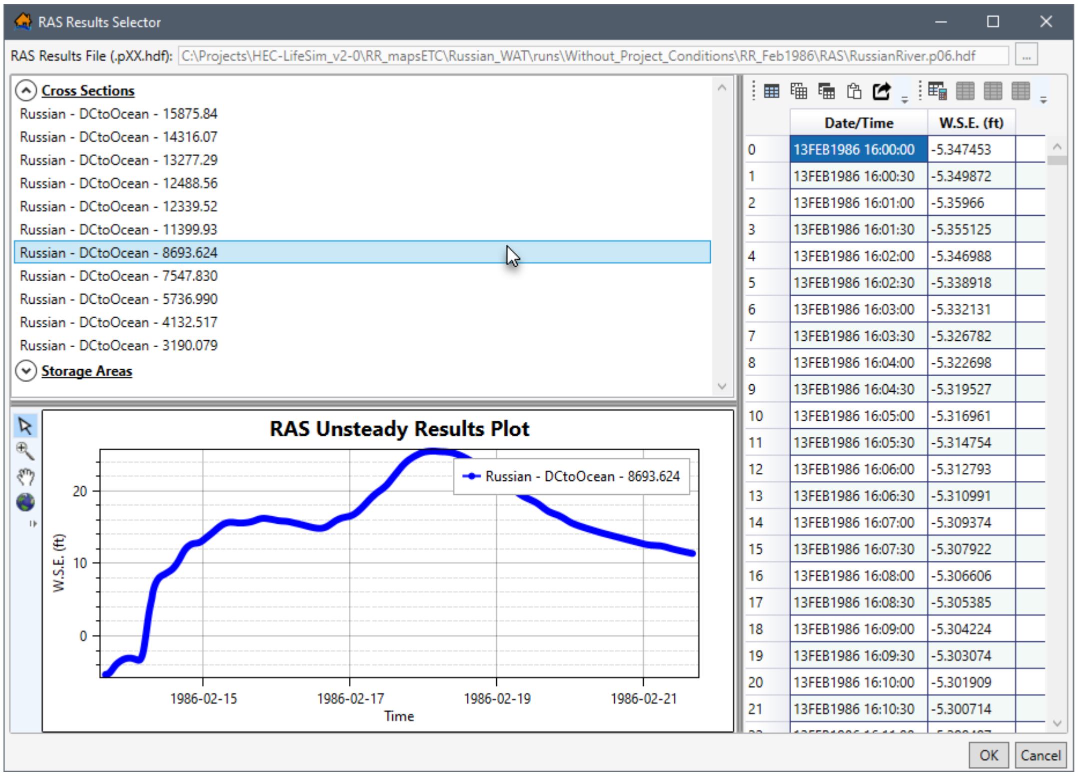


Figure 4-23. RAS Results Selector Dialog – Hydrograph from a Storage Area

5. Once the desired representative hydrograph is located and displayed, click **OK**, the **RAS Results Selector** dialog box closes (Figure 4-23) and the selected hydrograph displays in the hydraulic data import dialog, in the **Hydrograph Definition** panel.

6. From the hydraulic data import dialog, from the **Hydrograph Definition** panel (Figure 4-20), users can adjust the first hydraulic timestep and hazard occurrence for the representative hydrograph. To complete the **First Hydraulic Timestep** and the **Hazard Occurrence** start time adjustment refer to the instructions provided in Section 4.3.

4.2.2 Import From Map

Another way to import the representative hydrograph from HEC-RAS Version 5.0 (and later) data, is from a "map" view of the data. Import from HEC-RAS map data is only available from the **Import From HEC-RAS (5.0 and later)** dialog (Figure 4-3) described in Section 4.1.1. To select the representative hydrograph from HEC-RAS map data, from the **Import From HEC-RAS (5.0 and later)** dialog (Figure 4-3), click the **Import From Map** button (Figure 4-20) to open the **RAS Map Data Selector** dialog (Figure 4-24).

The **RAS Map Data Selector** (Figure 4-24) allows the user to specify a time series of depth at any point in the HEC-RAS study area. This method is generally used when no detailed information is given regarding hazard occurrence timing relative to an HEC-RAS geometric feature. Using the **Select Hydrograph** tool, the user can select a cross section or storage area on the map and the data for the hydrograph from that selection area will appear (Figure 4-25).

Refer to Chapter 3 for details regarding map window tools, table tools, customizing plots and plotting tools, and the animation toolbar. However, the **Select Hydrograph** tool is unique to the RAS Map Data Selector (Figure 4-24) dialog box. This tool is the default selection upon opening the **RAS Map Data Selector** dialog box (Figure 4-24). This tool is what allows the user to specify a time series of depth at any point in the HEC-RAS study area.

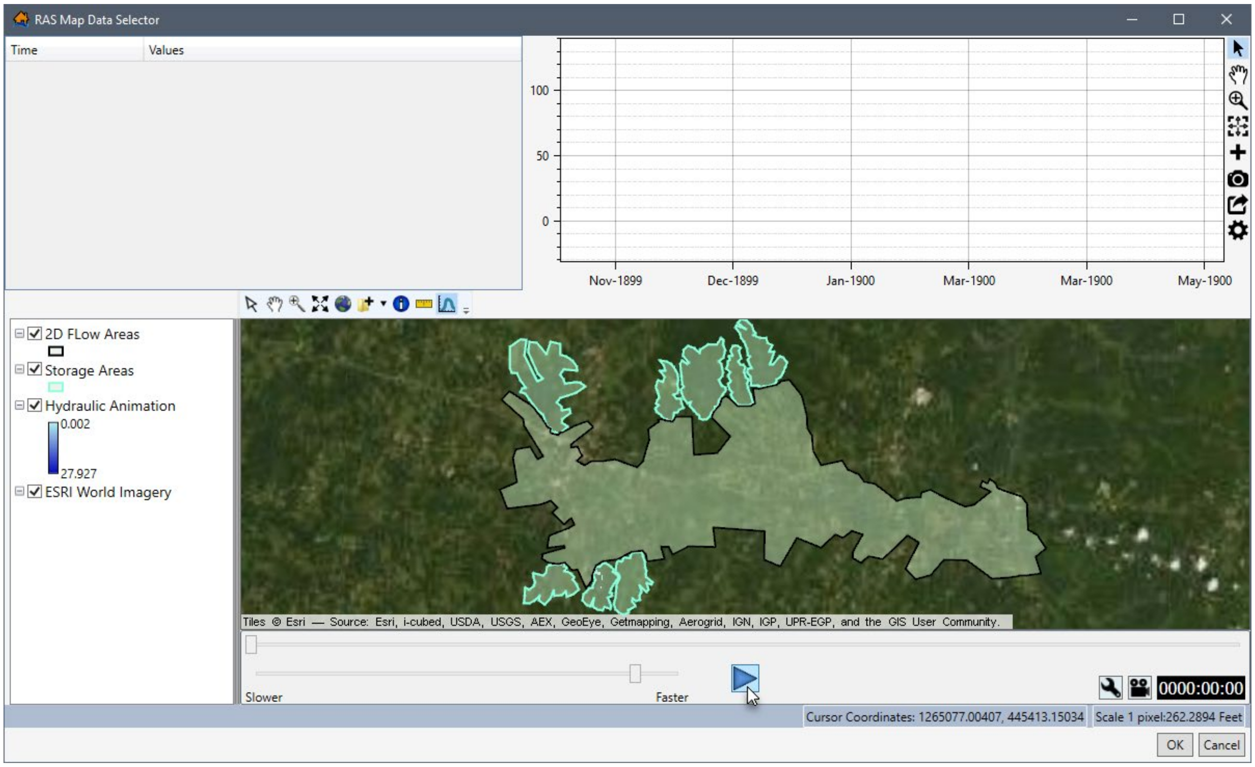


Figure 4-24. RAS Map Data Selector Dialog Box

To select the representative hydrograph from HEC-RAS Version 5.0 results, from the **RAS Map Data Selector** (Figure 4-25) dialog box, click the button to start the animation in the map window. At the desired water depth, pause the animation and click the **Select Hydrograph** tool, from the map window, click on a location in the study area, a **Progress Window** (Figure 4-26) opens. The **Progress Window** shows the collection of depth time series data for that selected point.

Once the desired representative hydrograph is displayed in the table and plot window, click **OK**, the **RAS Map Data Selector** (Figure 4-25) closes and the selected representative hydrograph is used for the hydraulic import. From the **Import From HEC-RAS (5.0 and later)** dialog box (Figure 4-29) the user can adjust the **First Hydraulic Timestep** and/or the **Hazard Occurrence** start time (Section 4.3).

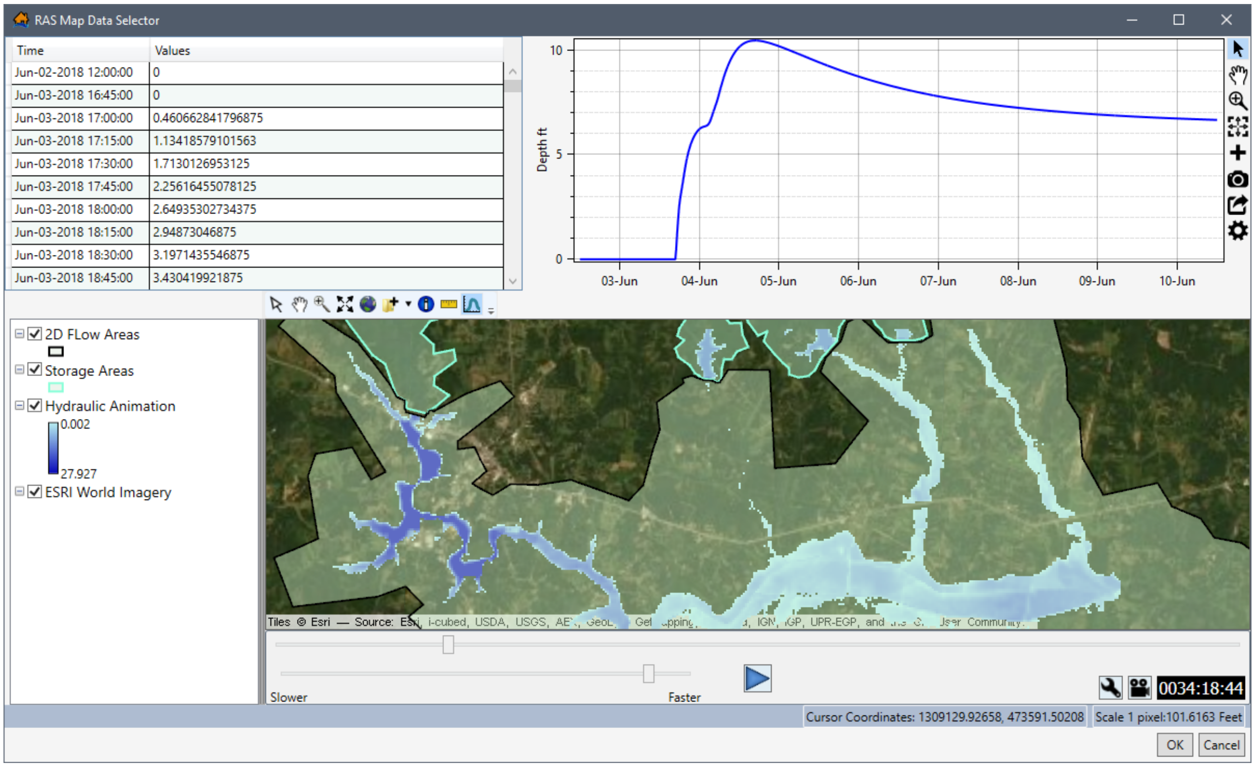


Figure 4-25. Example – RAS Map Data Selector Dialog Box

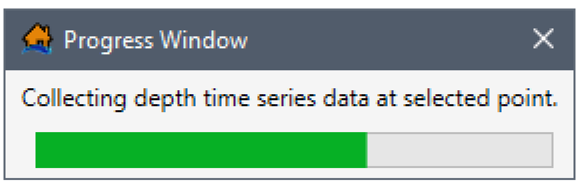


Figure 4-26. Progress Window

4.2.3 Create Hydrograph Manually

Regardless of the hydraulic data import option used, the representative hydrograph can be created manually by entering the data by hand or copying and pasting the data from external source (e.g., Microsoft Excel, HEC-DSSVue). From the **Hydrograph Definition** panel (Figure 4-27), available from all hydraulic data import options described in Section 4.1, users can enter the representative hydrograph manually in the **Time** and **Values** table (Figure 4-27). The date and time variable must be entered as: MMM-DD-YYYY HH:MM:SS (e.g., ***Jan-01-1900 00:00:00*** in Figure 4-27).

The time series data can be entered by hand, copied from another hydraulic dataset or from another application (e.g., Microsoft Excel®) and pasted into the table (i.e., entire table, individual cells, columns, rows). From the **Time** and **Values** table (Figure 4-27), right-click to open the shortcut menu (Figure 4-27), from the shortcut menu the user can add, insert, and delete row(s); select all; copy or copy with headers; and paste.

For example, from another application (e.g., Microsoft Excel®) select the date and value range of cells in the Excel spreadsheet and use **Ctrl + C** (or the **Copy** shortcut menu command) to copy the data to the clipboard. Return to the **Time** and **Values** table (Figure 4-27) and select the first cell in the first row and use **Ctrl + V** to paste all the copied data (Figure 4-28).

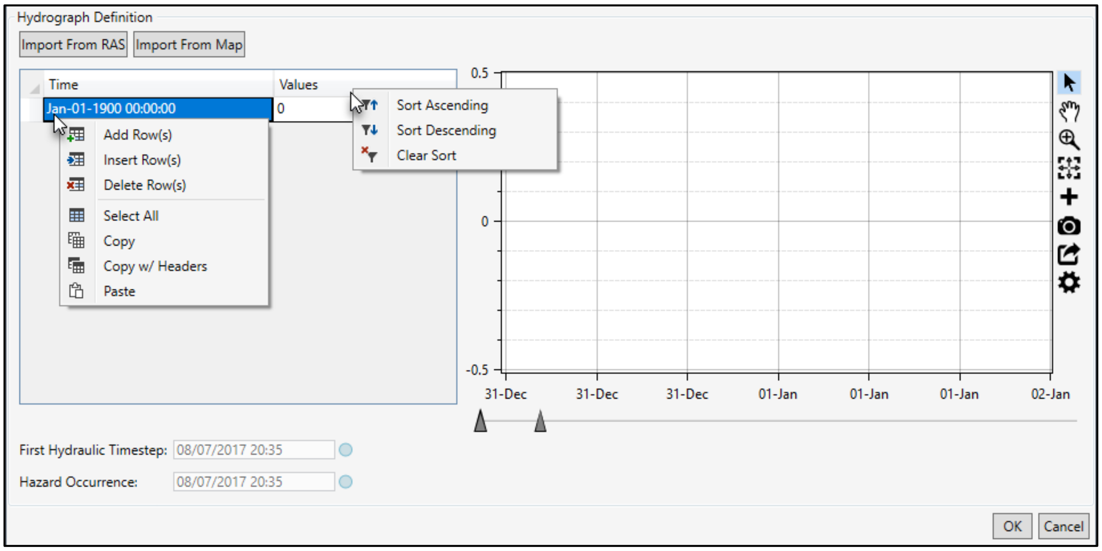


Figure 4-27. Example Hydraulic Data Importer – Hydrograph Definition Panel – Table Shortcuts

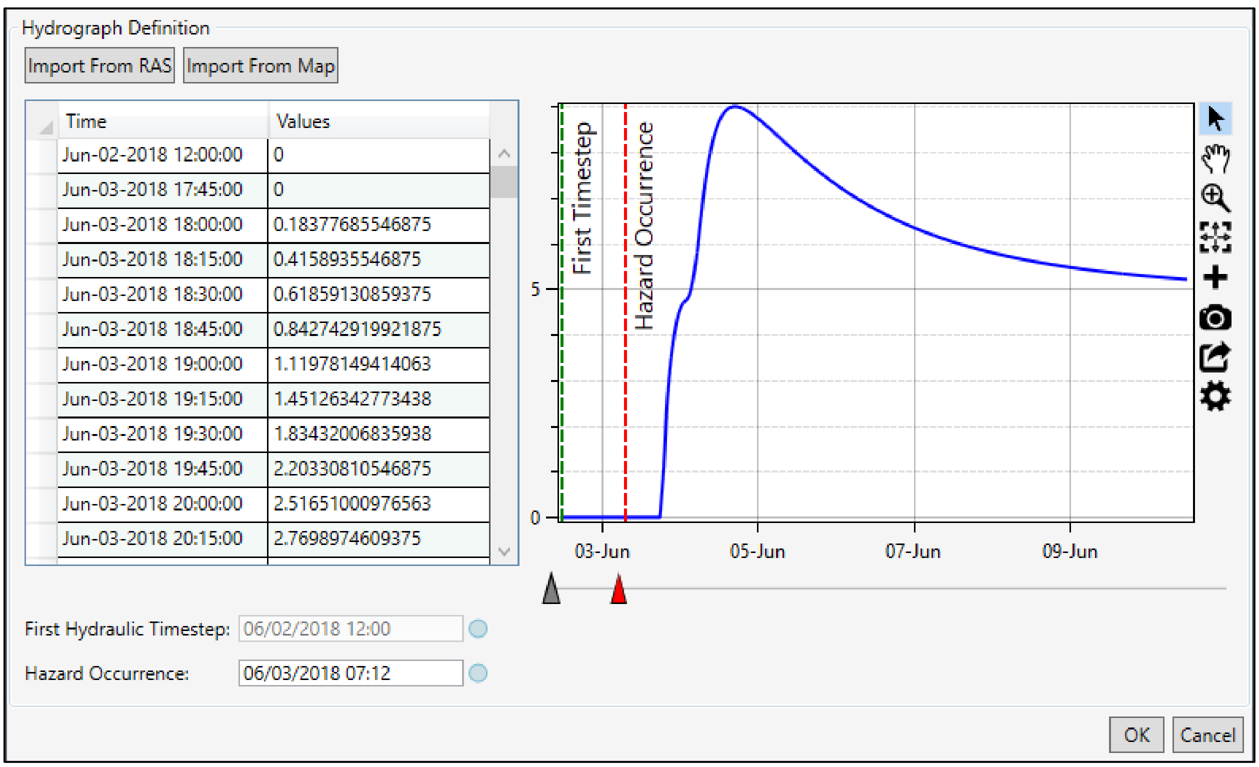


Figure 4-28. Example Completed – Hydraulic Data Importer – Hydrograph Definition Panel

The hydrograph time series data can also be copied (with or without headers) into another application (e.g., Microsoft Excel®) for saving. Use **Ctrl + A** (or the **Select All** command from the shortcut menu), right-click and select **Copy w/ Headers**. Then paste the data in the desired application to save.

The **Hydrograph Definition** panel (Figure 4-27) contains a plot window (Figure 4-28), that provides a live update for the data entered in the table. Review Section 3.6 for instructions on using the general plotting tools (located on the right side of the plot window) and for customizing plots.

When the desired hydrograph data has been entered manually in the hydraulic data import dialog, from the **Hydrograph Definition** panel (Figure 4-20), users can adjust the first hydraulic timestep and hazard occurrence for the representative hydrograph. To continue and complete the **First Hydraulic Timestep** and the **Hazard Occurrence** start time adjusted refer to the instructions provided in Section 4.3.

4.3 First Hydraulic Timestep and Hazard Occurrence

For all hydraulic data import options described in Section 4.1, following the hydraulic data import selection and selection of the representative hydrograph, from the **Hydrograph Definition** panel (Figure 4-29), the first hydraulic timestep and time of hazard occurrence (e.g., dam breach) must be set. Defining the **First Hydraulic Timestep** and **Hazard Occurrence** (Figure 4-29) are critical for LifeSim to correctly calculate hydraulic timing in relation to warning and evacuation information during the simulation. The first hydraulic timestep represents the start of the hydraulic data (time 0) that has been imported. All hydraulic timing information will be calculated relative to the first hydraulic timestep. The hazard occurrence time represents the time that the hazard occurs in the timeline. All warning and evacuation assumptions will be relative to the hazard occurrence time.

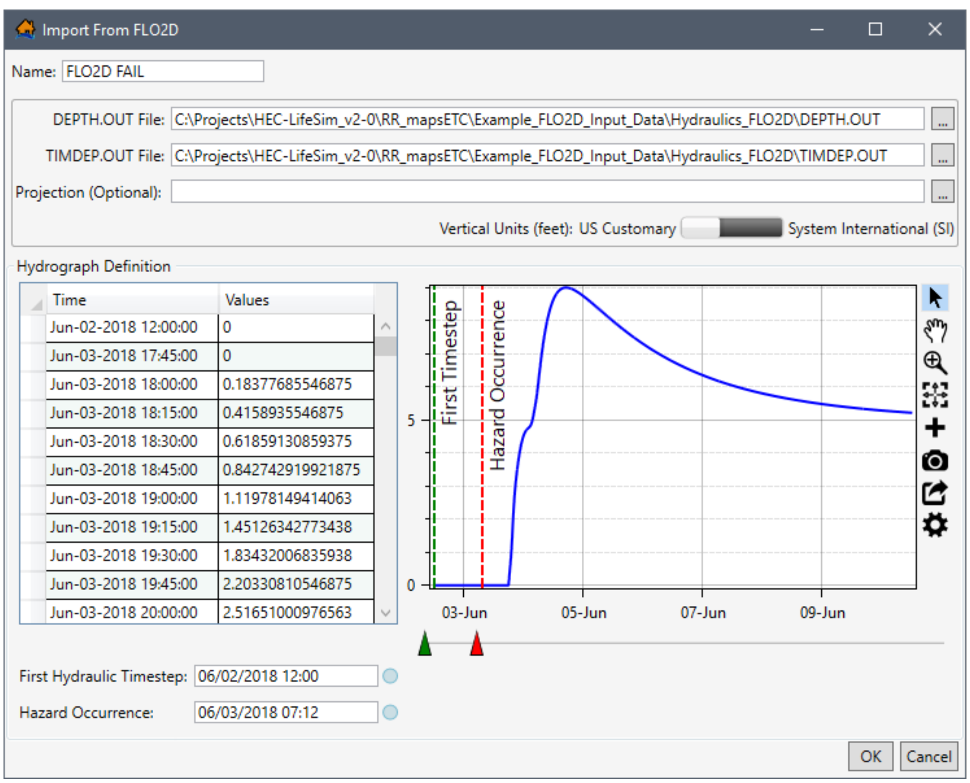


Figure 4-29. Example – Import From FLO2D Dialog Box – First Hydraulic Timestep and Hazard Occurrence

Once the representative hydrograph has been entered, the first hydraulic timestep can be modified. The first hydraulic timestep can be modified for all hydraulic data import dialogs (Figure 4-29), except the **Import From HEC-RAS (5.0 and later)** (Figure 4-30). For the example in Figure 4-29, note that the **First Timestep** plot slider is green and can be moved. However, for the example provided in Figure 4-30, note that the **First Timestep** plot slider is gray and cannot be moved. Likewise, the **First Hydraulic Timestep** date and time box is disabled in Figure 4-30, but is not disabled (can be edited) in Figure 4-29. Instead, when the representative hydrograph is set using the **Import From RAS** option the first hydraulic timestep is defined by the model output. All other hydraulic data import options allow the **First Hydraulic Timestep** and the **Hazard Occurrence** start time to be adjusted by the user.

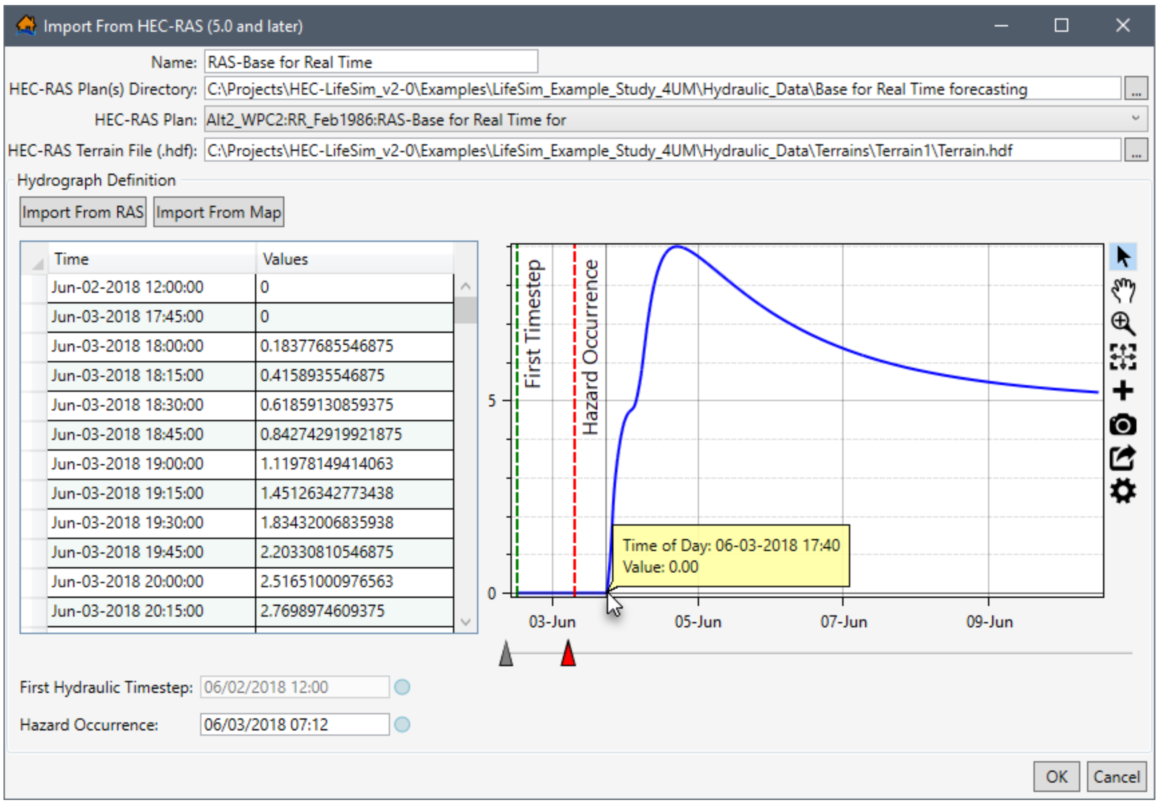


Figure 4-30. Example – Import From HEC-RAS (5.0 and later) Dialog Box – First Hydraulic Timestep and Hazard Occurrence

Users can set the first hydraulic timestep and hazard occurrence using the following options:

1. Enter the date and time manually. The date and time variable must be entered as: MM/DD/YYYY HH:MM (e.g., ***06/03/2018 07:12*** in Figure 4-29).

2. Alternatively, use the green **First Timestep** plot slider and red **Hazard Occurrence** plot slider (Figure 4-29) to adjust the green and red dotted lines on the time series plot.

3. Users can also adjust the first hydraulic timestep and hazard occurrence by clicking the button, that opens a calendar and time adjustment window (Figure 4-31). From the calendar panel (Figure 4-31), select the date. Use the left and right arrows to select a preceding or subsequent month. From the time panel (Figure 4-31), first select the hour

(displayed in military time), then select the minutes. To modify the hour or minutes selection click the displayed hour header (e.g., ***17*** in Figure 4-31) or the displayed minute header (e.g., ***14*** in Figure 4-31).

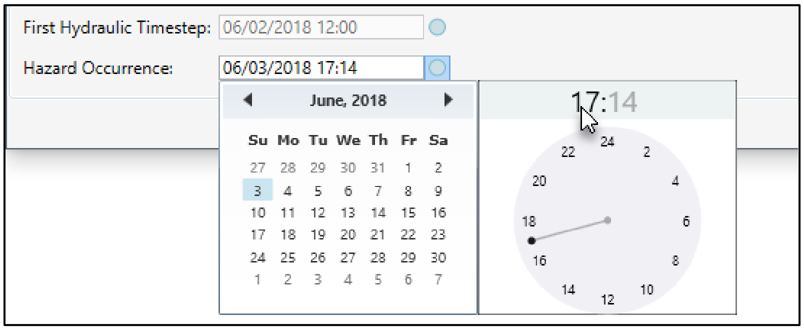


Figure 4-31. Calendar and Time Adjustment Window

Once all of the appropriate information has been entered click **OK**, the hydraulic data importer dialog box closes (Figure 4-30). The hydraulic data is added to the study and displays in the **Study Tree** (Figure 4-1), under **Hydraulic Data**.

4.4 Editing, Viewing, and Removing Hydraulic Datasets

Once hydraulic data has been added to the LifeSim study, a hydraulic dataset can be edited, deleted, viewed, moved, or a hydraulic summary of the data can be generated. Access the editing options, from the LifeSim main window (Figure 2-1), from the **Study** menu, point to **Hydraulic Data**, point to **Edit** (Figure 4-32), and select the appropriate hydraulic dataset. Alternatively, users can access the edit options for an individual hydraulic dataset from the LifeSim main window (Figure 2-1), from the **Study** tab, from the **Study Tree**, under **Hydraulic Data**, right-click on a hydraulic dataset (e.g., ***EMB FAIL DF*** in Figure 4-33). From the shortcut menu (Figure 4-33), click **Edit Hydrograph Data and Timing**.

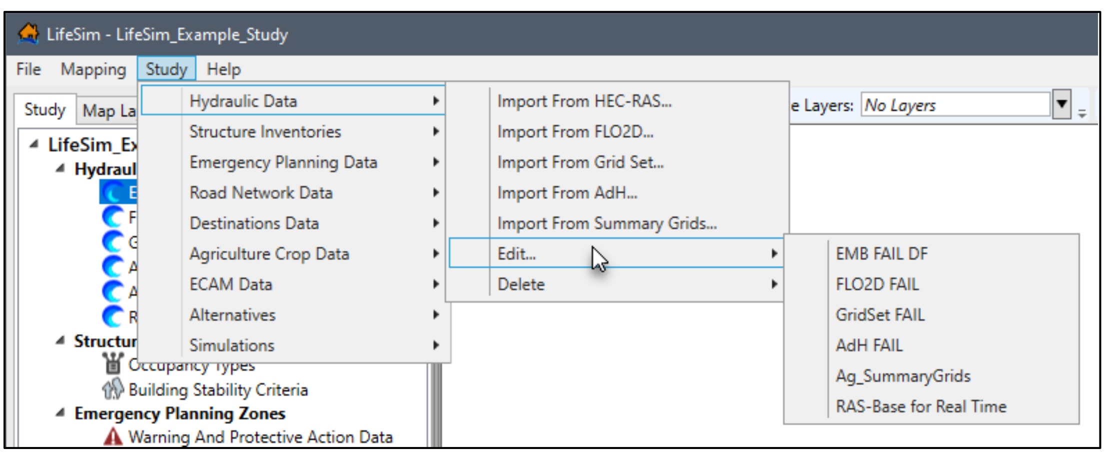


Figure 4-32. LifeSim Main Window – Study Menu – Hydraulic Data Sub-menu

To delete, view, move, or generate a hydraulic summary for an individual hydraulic dataset, from the **Study** tab, from the **Study Tree**, under **Hydraulic Data**, right-click on a hydraulic dataset (e.g., ***EMB FAIL DF*** in Figure 4-33) to access the shortcut menu (Figure 4-33) commands.

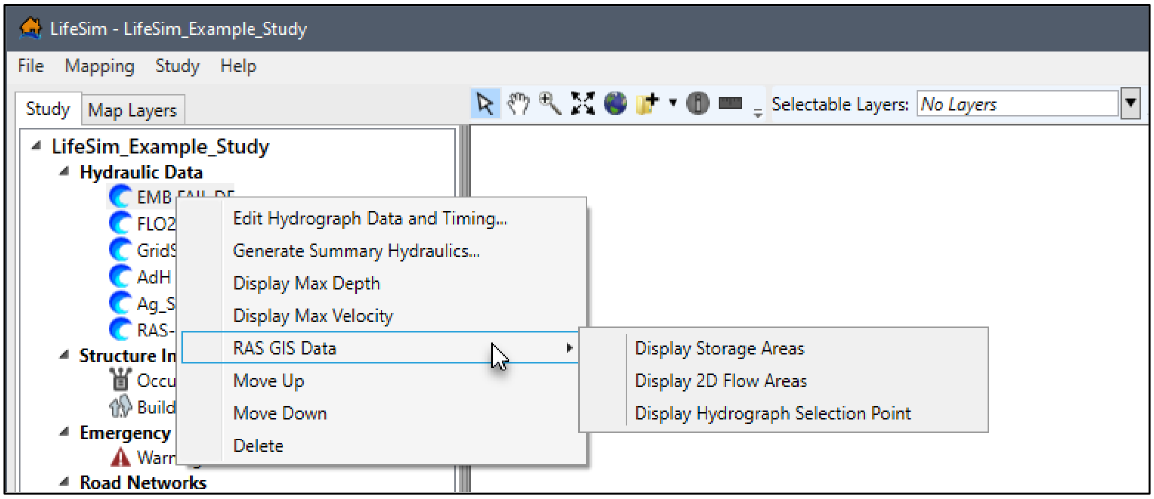


Figure 4-33. LifeSim Main Window – Study Tree – Individual Hydraulic Dataset – Shortcut Menu

The following sections describe how to edit, delete, view, move, or generate a hydraulic summary for an individual hydraulic dataset.

4.4.1 Edit a Hydraulic Dataset

To edit a hydraulic dataset:

1. From the LifeSim main window (Figure 2-1), from the **Study** tab, from the **Study Tree**, under **Hydraulic Data**, right-click on a hydraulic dataset (e.g., ***EMB FAIL DF*** in Figure 4-33). From the shortcut menu (Figure 4-33), click **Edit Hydrograph Data and Timing**. Another way is from the LifeSim main window (Figure 2-1), from the **Study** menu, point to **Hydraulic Data**, point to **Edit** (Figure 4-32), and select the appropriate hydraulic dataset. Alternatively, from the **Study Tree**, under **Hydraulic Data**, double-click on a hydraulic dataset.

2. The **Select Hydrograph and Timing Information For** *"hydraulic dataset"* dialog box opens (Figure 4-34).

3. From the **Select Hydrograph and Timing Information For** *"hydraulic dataset"* dialog box (Figure 4-34) users can replace existing representative hydrographs. Depending on the hydraulic data source, the user can click **Import From RAS** (Section 4.2.1), **Import From Map** (Section 4.2.2), or can be entered manually (Section 4.2.3).

4. The time series data table, **First Hydraulic Timestep** (when enabled), and **Hazard Occurrence** can be modified (Section 4.3).

5. Once editing of the hydraulic data is complete click, **OK**, and the **Select Hydrograph and Timing Information For** *"hydraulic dataset"* dialog box closes (Figure 4-34).

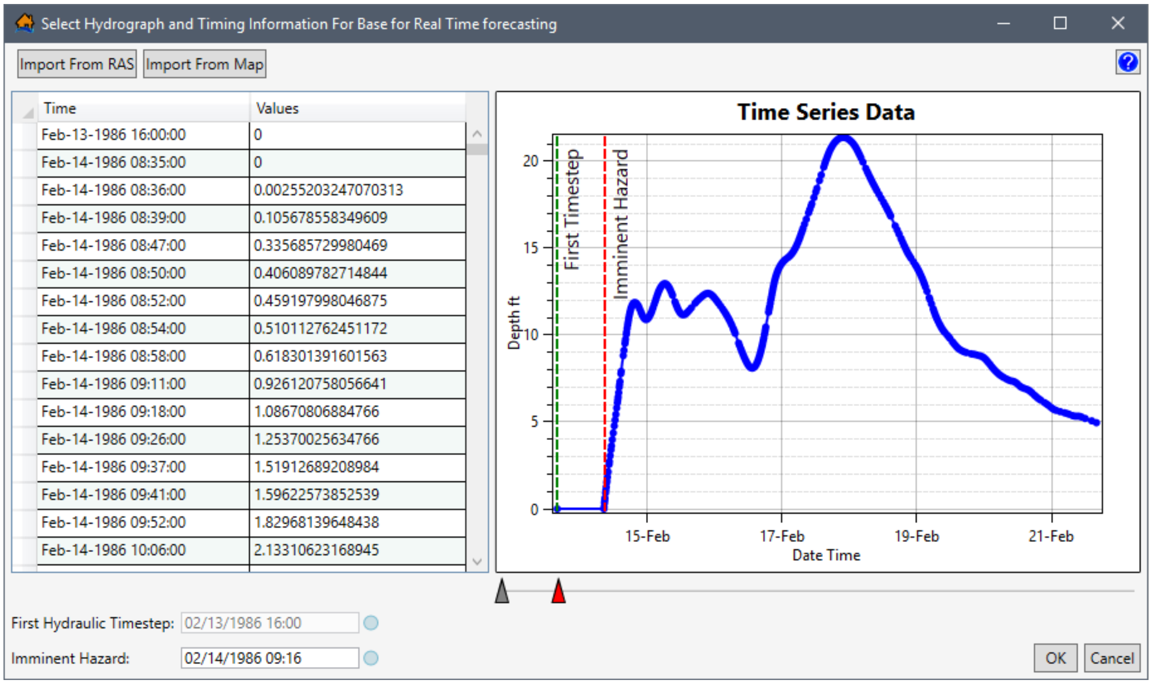


Figure 4-34. Select Hydrograph and Timing Information for *"hydraulic dataset"* Dialog Box

4.4.2 Generate Hydraulic Summary

The user can create an output shapefile based on any selected hydraulic scenario for point or polyline shapefiles. The created output shapefile contains hydraulic information based on the selected scenario such as maximum depth, velocity, and key hydraulic threshold arrival times at each feature in the selected point (e.g., structure inventory) or polyline (e.g., roads) shapefile. The hydraulic summary data can be useful in identifying potential issues in a structure inventory or road network; for example, structures inundated too quickly due to bad placement, or bridges that have water underneath prior to the start of the flood event. To generate the hydraulic summary:

1. From the LifeSim main window (Figure 2-1), from the **Study** tab, from the **Study Tree**, under **Hydraulic Data**, right-click on a hydraulic dataset (e.g., ***EMB FAIL DF*** in Figure 4-33). From the shortcut menu (Figure 4-33), click **Generate Summary Hydraulics**. The **Generate Hydraulic Summary** dialog box opens (Figure 4-35).

2. The user needs to provide a name and a location to save the output shapefile. To the right of the **Output Shapefile** box click ..., a **Save As** browser opens (Figure 4-36). Navigate to the directory to save the output shapefile. Enter a name for the output shapefile in the **File name** box (**.shp*) and click **Save**. The **Save As** browser closes (Figure 4-36) and the from the **Generate Hydraulic Summary** dialog box (Figure 4-35), the **Output Shapefile** box updates to display the pathname and shapefile name of the output shapefile.

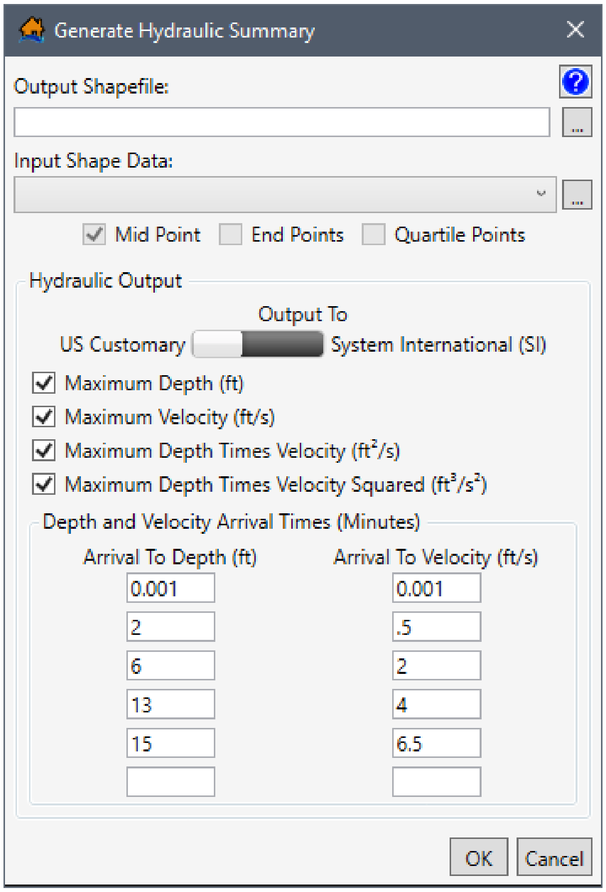


Figure 4-35. Generate Hydraulic Summary Dialog Box

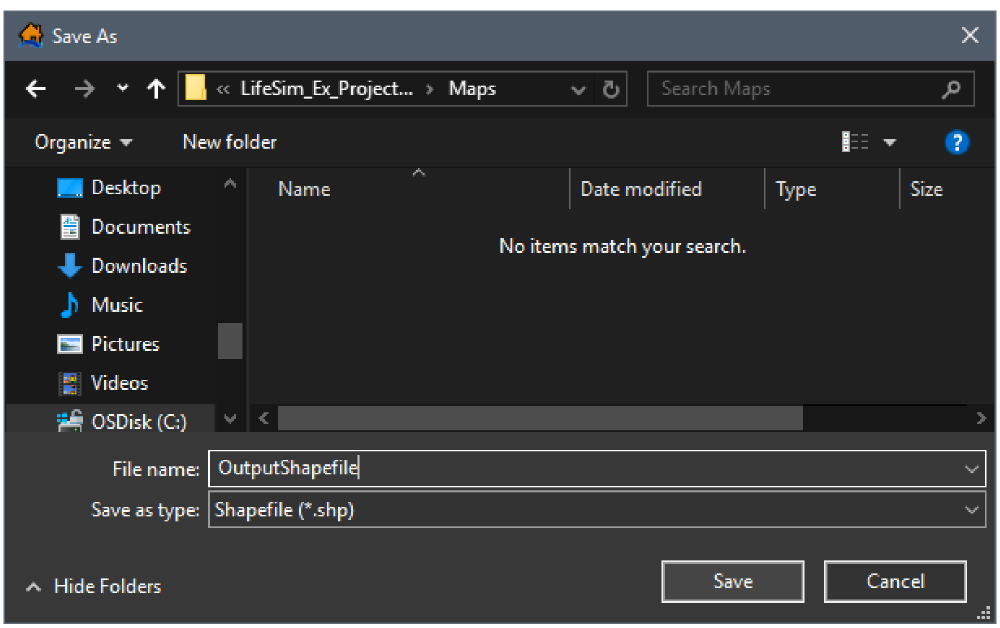


Figure 4-36. Save As Browser

3. From the **Input Shape Data** list (Figure 4-37) select the desired point or polyline shapefile. If no point or polyline dataset exists in the map window, then to select the **Input Shape Data**, to the right of the list (Figure 4-35), click . An **Open** browser

(Figure 2-4) opens, navigate to the directory where the point or polyline shapefile is located. Select the correct shapefile (e.g., **.shp*), click **Open**, the **Open** browser closes (Figure 2-4) and the user returns to the **Generate Hydraulic Summary** dialog box (Figure 4-37).

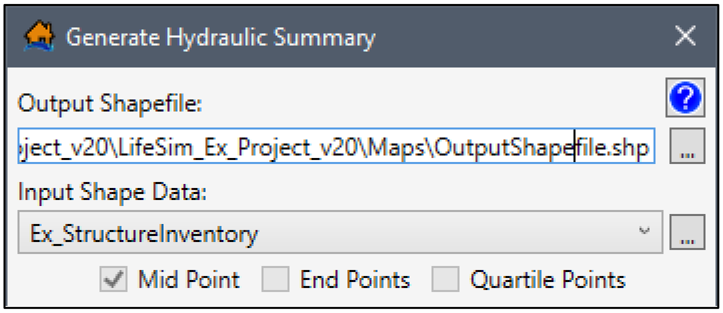


Figure 4-37. Generate Hydraulic Summary – Example

4. For the selected polyline **Input Shape Data** identify how the hydraulic summary should be generated either at the **Mid Point** (default, and automatically selected for point shapefiles), **End Point**, and/or **Quartile Points** (Figure 4-37).

5. In the **Hydraulic Output** panel (Figure 4-35), set the **Output to** units to either **US Customary** or **System International (SI)**. Units are either in feet or meters and depend on the **Output to** (Figure 4-35) selection. In other words, if the hydraulic data is in feet and System International (SI) is selected then the resulting output will be converted to meters.

6. By default, in the **Hydraulic Output** panel (Figure 4-35), the **Maximum Depth**, **Maximum Velocity**, **Maximum Depth Times Velocity**, and **Maximum Depth Times Velocity Squared** hydraulic output options have been selected. To unselect a default output selection, uncheck ☐ the checkbox for the option.

7. From the **Depth and Velocity Arrival Time (Minutes)** panel (Figure 4-35), default values for **Arrival To Depth** and **Arrival To Velocity** have been set. The default values can be modified, but no more than six boxes per arrival time type can be identified per output shapefile. Not all six boxes require data.

8. Once all of the necessary and desired parameters have been set, click **OK**, a **Progress Window** opens (Figure 4-38) showing the progress of the hydraulic summary output shapefile creation. When processing of the output shapefile is complete the **Progress Window** closes (Figure 4-38). The created hydraulic summary output shapefile is saved in the appropriate location and displays in the map window.

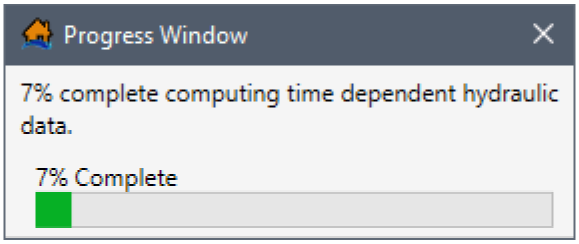


Figure 4-38. Progress Window

4.4.3 View a Hydraulic Dataset

Depending on which method was chosen to import hydraulic datasets for a LifeSim study, the user can view hydraulic datasets in the map window. To display a hydraulic dataset feature in the map window, the LifeSim main window (Figure 2-1), from the **Study** tab, from the **Study Tree**, under **Hydraulic Data**, right-click on a hydraulic dataset (e.g., ***EMB FAIL DF*** in Figure 4-33). From the shortcut menu (Figure 4-33) click **Display** *"hydraulic dataset feature"*.

In general, the available display options are:

- Display Max Depth (LifeSim created *Max_Depth.vrt* and **.tif* files)
- Display Max Velocity (LifeSim created *Max_Velocity.vrt* and **.tif* files)

If the hydraulic data is imported from HEC-RAS data then from the LifeSim main window (Figure 2-1), from the **Study** tab, from the **Study Tree**, under **Hydraulic Data**, right-click on a hydraulic dataset (e.g., ***EMB FAIL DF*** in Figure 4-33). From the shortcut menu (Figure 4-33), point to **RAS GIS Data** (Figure 4-33) and the shortcut sub-menu will display one or more of the following options:

- Display Cross Sections (LifeSim created shapefile (**.shp*) file)
- Display Storage Areas (LifeSim created shapefile (**.shp*) file)
- Display 2D Flow Areas (LifeSim created shapefile (**.shp*) file)
- Display Hydrograph Selection Point (LifeSim created shapefile (**.shp*) file)

Note, the above files are created in the Hydraulic_Data sub-directory when the files are added to the map window.

To remove a hydraulic dataset feature displayed in the map window, from the LifeSim main window (Figure 2-1), from the **Study** tab, from the **Hydraulic Data** folder, right-click on a hydraulic dataset, from the shortcut menu (Figure 4-33), and click **Remove** *"hydraulic dataset feature"* **From the Map Window**. Or remove the map layer from the map window (Appendix A).

4.4.4 Move a Hydraulic Dataset

To move specific hydraulic datasets up or down in the **Study Tree**:

1. From the LifeSim main window (Figure 2-1), from the **Study** tab, from the **Study Tree**, under **Hydraulic Data**, right-click on a hydraulic dataset (e.g., ***EMB FAIL DF*** in Figure 4-33). From the shortcut menu (Figure 4-33), click **Move Up**, the selected dataset will move up one layer in the **Study Tree**.

2. To move the selected dataset down, from the shortcut menu (Figure 4-33), click **Move Down**.

3. Datasets can also be moved in the **Study** tab by utilizing the drag and drop method. From the **Study Tree** on the LifeSim main window (Figure 2-1), click and hold on the dataset of interest. The user can move the selected dataset up or down the list by moving the mouse. A bar will show the new location where selected dataset will be moved.

4. Release the mouse to set the selected dataset at the desired location.

4.4.5 Delete a Hydraulic Dataset

To delete a hydraulic dataset, from the LifeSim main window (Figure 2-1), from the **Study** tab, from the **Study Tree**, under **Hydraulic Data**, right-click on a hydraulic dataset (e.g., ***EMB FAIL DF*** in Figure 4-33). From the shortcut menu (Figure 4-33), click **Delete**. Another way is from the LifeSim main window (Figure 2-1), from the **Study** menu, point to **Hydraulic Data** (Figure 4-32), point to **Delete**, and select the appropriate hydraulic dataset.

The **Verify Deletion of Hydraulic Data** message dialog (Figure 4-39) opens. To verify the deletion of the selected hydraulic data, from the **Verify Deletion of Hydraulic Data** message dialog (Figure 4-39), click **Yes**. The selected dataset will be removed from the study. Alternatively, click **No**, to close the **Verify Deletion of Hydraulic Data** message dialog (Figure 4-39) without deleting the hydraulic data.

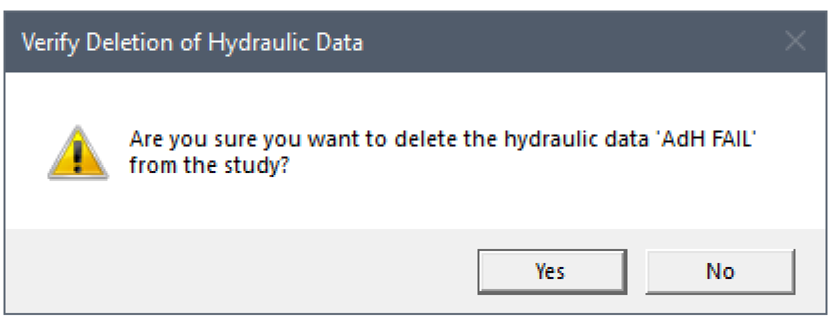


Figure 4-39. Verify Deletion of Hydraulic Data Message Dialog

CHAPTER 5

Structure Inventories

The next step after adding hydraulic data to the study is to add **structure inventories**. A structure inventory represents geospatial points that contain damageable elements (typically a structure such as a home). Structure inventories are required and allow LifeSim to estimate direct economic damage for a flood event. To estimate life loss for an event, a structure inventory must be supplied with population data.

Structure occupancy types provide a level of aggregation for structures in LifeSim, and are imperative for any consequence calculation. The occupancy types describe depth-damage relationships, life loss parameters, general information about the structures, and contain information about uncertainty computations associated with the structure inventory. Each structure is required to have an occupancy type.

LifeSim Version 2.0 can accept structure inventories in the form of point shapefiles and the National Structure Inventory database (NSI). This chapter describe the ways structure inventories in different formats can be added to, edited and displayed in an LifeSim study.

5.1 Import from Shapefile

The point shapefile that represents a structure inventory for an area will contain information on structure occupancy types (Appendix C), population, structure values, foundation height, construction type, and other information. The LifeSim Version 2.0 Technical Reference Manual, Chapter 4, provides the definitions for the attributes required for importing a structure inventory from a shapefile.

To create a structure inventory from a point shapefile:

1. From the LifeSim main window (Figure 5-1), from the **Study** tab, from the **Study Tree**, right-click on **Structure Inventories**, from the shortcut menu click **Import Structures From Shapefile** (Figure 5-1).

   Another way is from the LifeSim main window, from the **Study** menu, point to **Structure Inventories**, click **Import Structures From Shapefile** (Figure 5-2).

2. Either way, the **Import From Point Shapefile** dialog box (Figure 5-3) will open.

3. In the **Name** box (Figure 5-3), enter a name for the structure inventory.

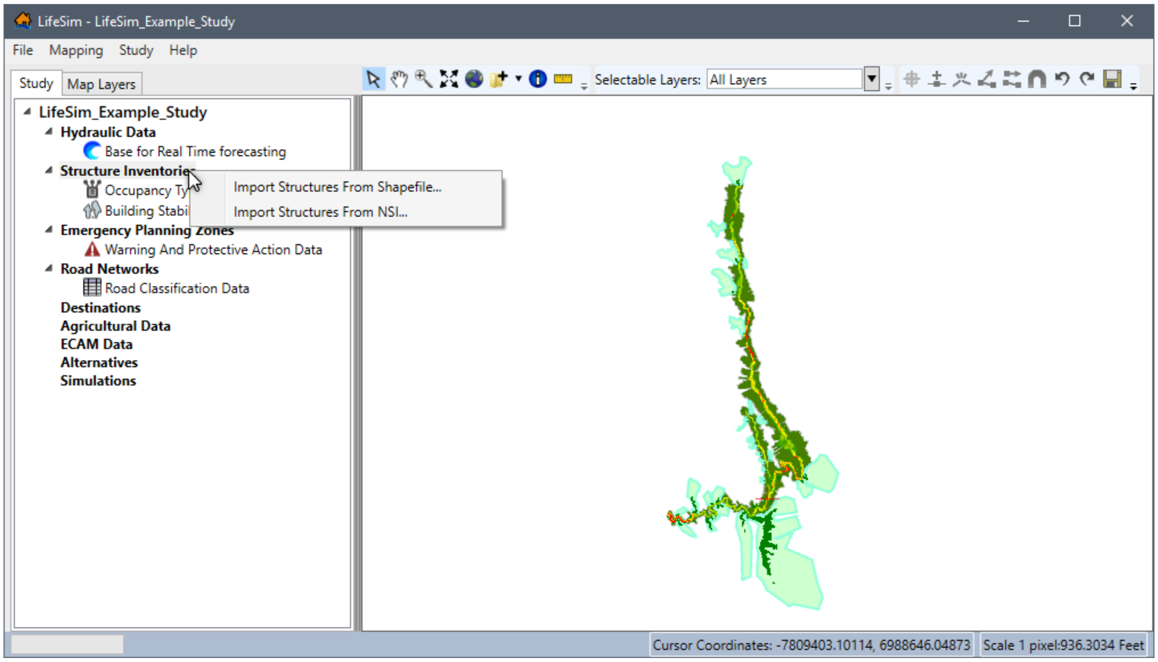


Figure 5-1. LifeSim Main Window – Study Tree – Structure Inventories Shortcut Menu

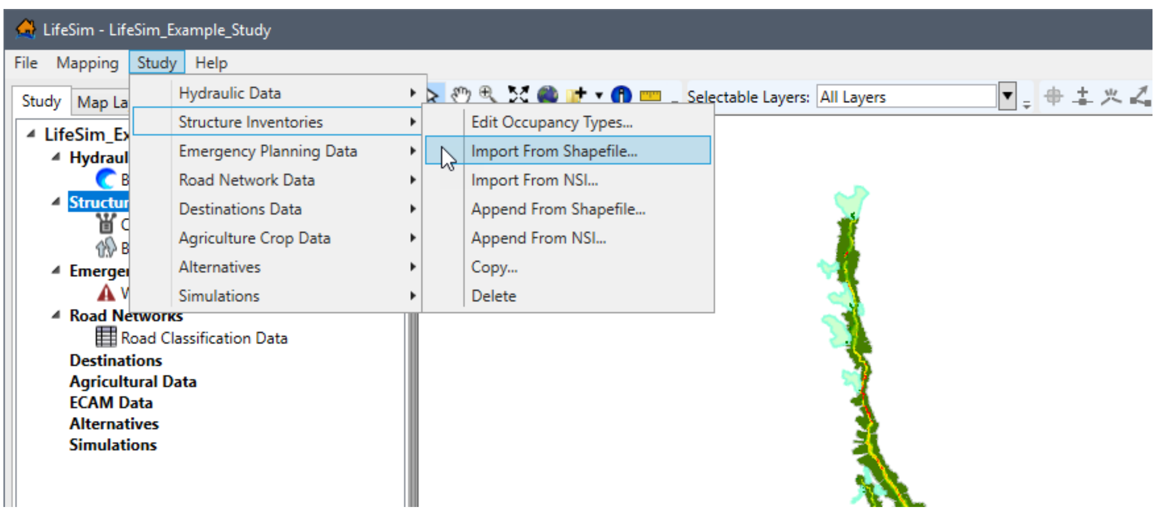


Figure 5-2. LifeSim Main Window – Study Menu – Structure Inventories Sub-menu

4. To select the shapefile that contains information about the structure inventory, to the right of the **Structure Inventory Shapefile** dropdown menu (Figure 5-3), click ... . An **Open** browser (Figure 5-4) opens, navigate to the directory where the point shapefile is located. Select the correct point shapefile (e.g., **.shp*), click **Open**, the **Open** browser will close (Figure 5-4) and the user will be returned to the **Import From Point Shapefile** dialog box (Figure 5-3). Alternatively, if the desired shapefile has already been imported as a map layer into the LifeSim study, then from the **Structure Inventory Shapefile** dropdown menu (Figure 5-5), select the desired map layer.

5. The **Structure Inventory Shapefile** box (Figure 5-3) will now contain the name and location of the selected point shapefile.

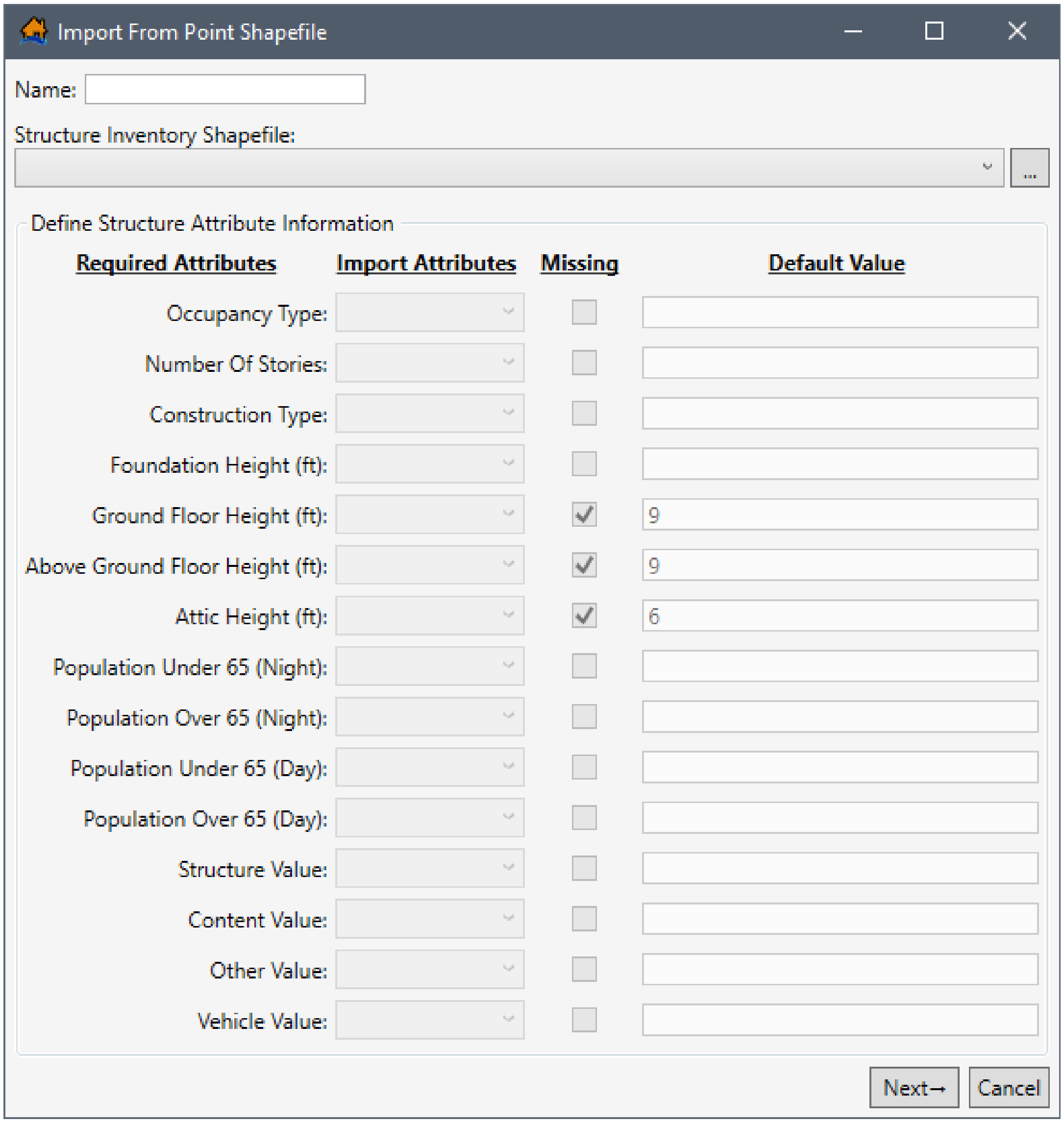


Figure 5-3. Import From Point Shapefile Dialog Box

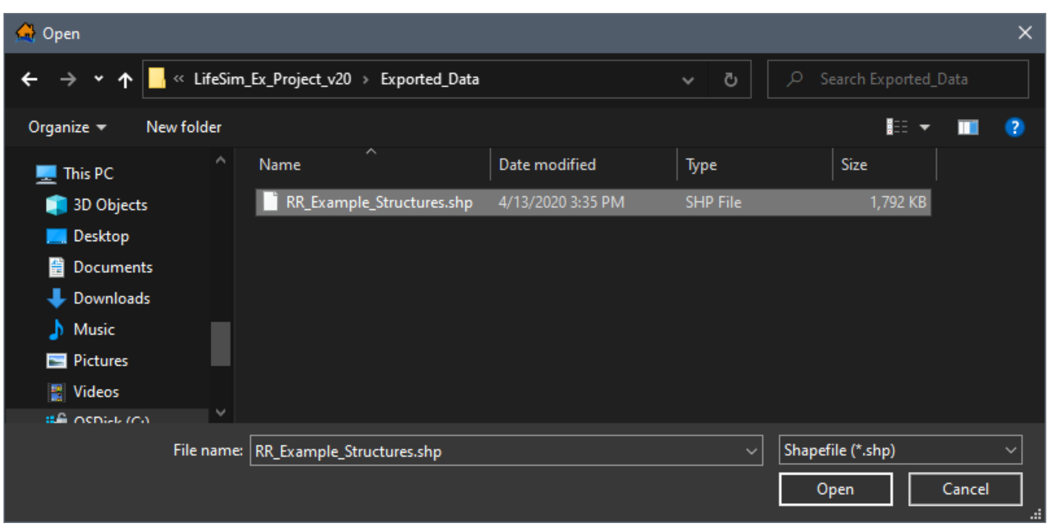


Figure 5-4. Open Browser

6. A structure inventory has required attributes that will be used in an LifeSim study, so the user must provide information about those attributes in the selected point shapefile. From the **Import Attributes** column (Figure 5-3), for each listed **Required Attribute**, the user must select from a list the field name of the attribute in the selected point shapefile.

7. For example, in Figure 5-5, for **Occupancy Type** (required attribute), from the **Import Attributes** list for **Occupancy Type**, possible field names are presented for the user. To set the **Import Attributes**, use the dropdown menu to select the appropriate field name – e.g., *ST_DMGFUN* in Figure 5-5.

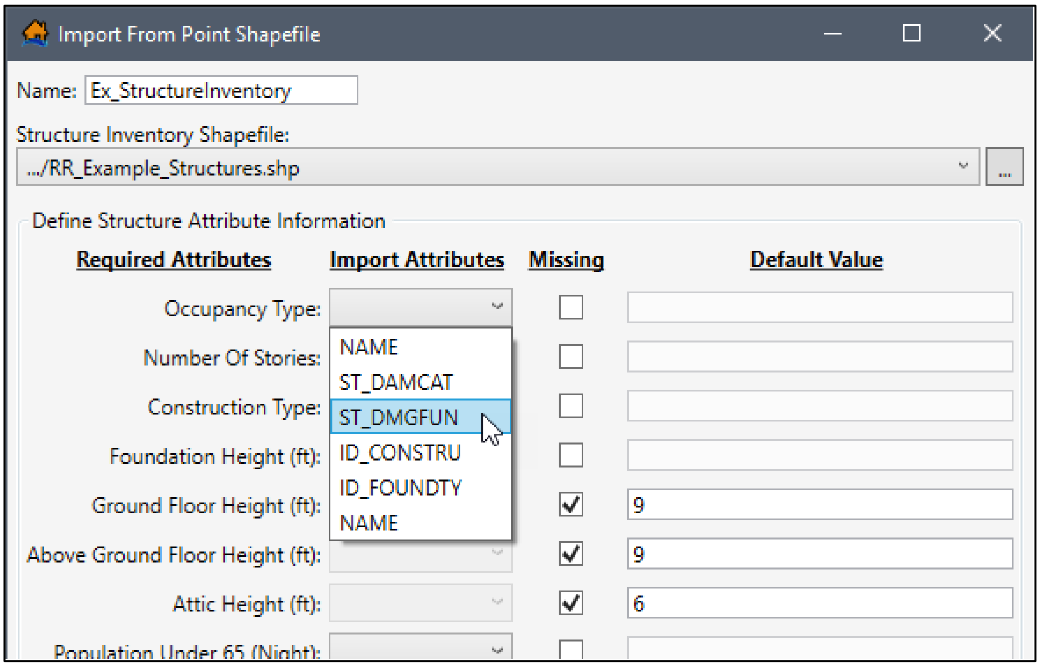


Figure 5-5. Define Structure Attribute Information – Occupancy Type

8. Another example, in Figure 5-6, for **Population Over 65 (Night)** (required attribute), from the **Import Attributes** list for **Population Over 65 (Night)**, possible field names are presented for the user. Use the dropdown menu to select the appropriate field name – e.g., *POP_O65DAY* in Figure 5-6.

9. If the point shapefile does not contain a required attribute, the user must click the **Missing** checkbox (Figure 5-6) for that required attribute (e.g., *Ground Floor Height*). In the **Default Value** column (Figure 5-6) for that required attribute, the value box activates, and the user must enter a value for the required attribute (e.g., *9*). The user defined default value will then be applied to all structures in the inventory.

10. Once all of the required structure attribute information has been completed, from the **Import From Point Shapefile** dialog box, click **Next** (Figure 5-6).

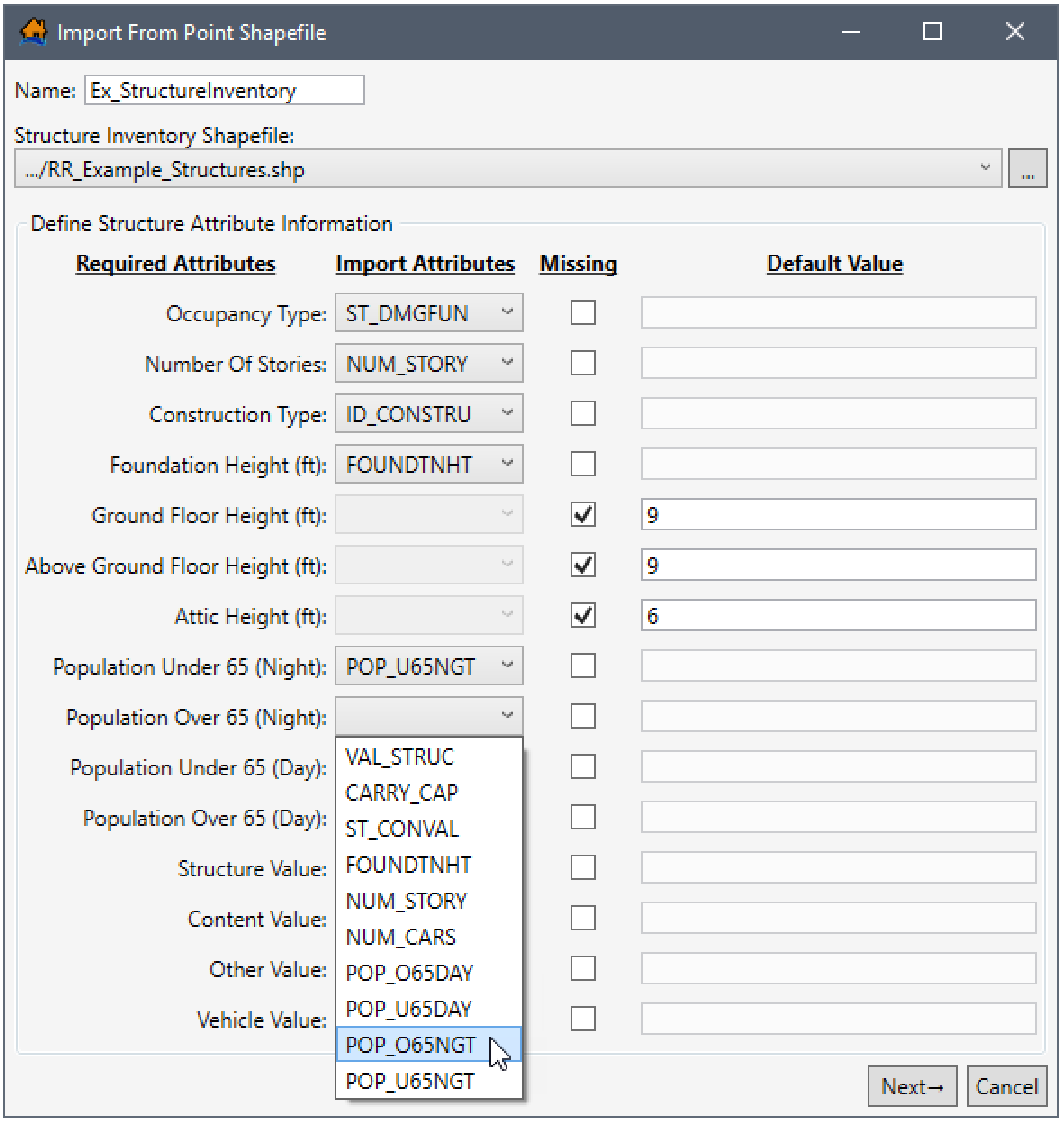


Figure 5-6. Example – Define Structure Attribute Information – Population Over 65 (Night)

5.1.1 Default Occupancy Type Assignments

Next the user must confirm the occupancy type names from the structure shapefile associated with the occupancy type IDs (Appendix C) that have been defined in the study. Occupancy types are another level of aggregation for structures. Occupancy types are imperative for the consequence calculations. Refer to Section 5.4 for more information on occupancy types.

1. From the point shapefile, LifeSim tries to match up the occupancy type names to the occupancy type IDs (Figure 5-7) that exist in the model. For example, in Figure 5-7, for occupancy type ID ***RES3BI***, LifeSim selected ***RES3B*** from the existing occupancy types.

2. If the default selection is not correct, from the list the user can change the assignment from the **Default Assignment** dropdown menu (Figure 5-7).

3. If the selected point shapefile does not contain a default occupancy type in the assigned field, then the dropdown menu will be empty. Users can check for different occupancy type names from the **Default Assignment** dropdown menu (Figure 5-7); however, as the example provided in Figure 5-7 illustrates the selected shapefile does not contain the **RES1-3SNB** occupancy type.

4. Once the user has reviewed the default occupancy type assignments, from the **Import From Point Shapefile** dialog box, click **Next** (Figure 5-7).

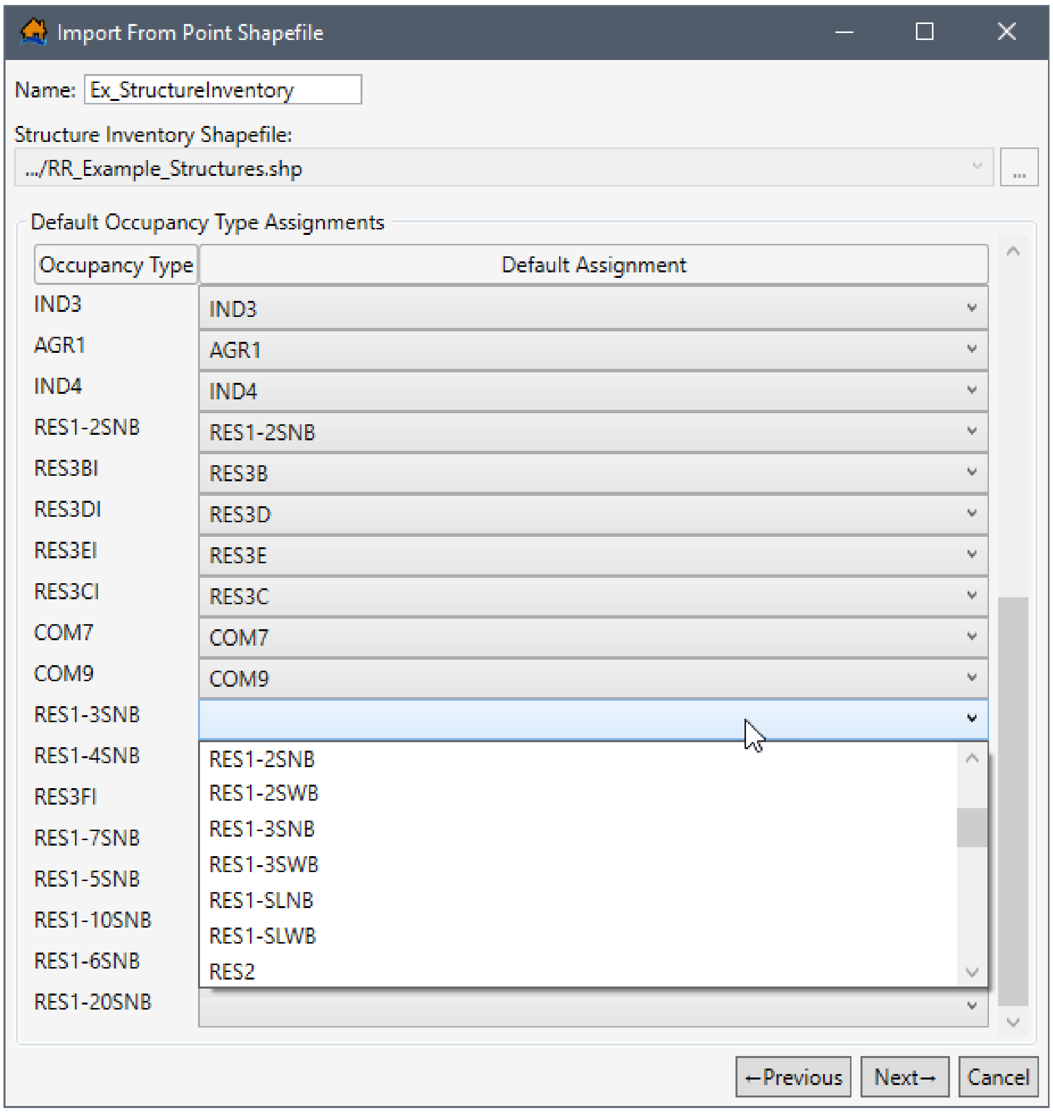


Figure 5-7. Default Occupancy Type Assignments

5.1.2 Stability Criteria Assignments

Next the user must assign structure stability functions. Stability criteria are assigned to a structure based on rules that relate stability functions to identified attribute values. For example, if a rule states that all structures that are construction type 'Wood' should be assigned the 'Wood-Framed Attached' stability criteria. Then once Finish is clicked every structure that meets the criteria will be assigned the Wood-Framed Attached stability function. These rules are not retained for future use. They are a one-time application to define the stability criteria that will be used by structures. The assigned stability criteria can be viewed through the structure inventory attribute table (Section 5.3.1). Refer to Section 5.5 for further discussion on structure stability criteria.

LifeSim Version 2.0 has default rules that assign stability criteria for structures with specific construction and/or occupancy types. Alternatively, the user may edit an existing default rule or create and define a new rule to assign stability criteria to structures based on any of the structure attributes within the selected point shapefile. Note, the created rules are a 'rule stack' meaning the order in which the criteria are defined is important. If two rules apply to the same structure, then the higher (top) rule will dominate. Users should make certain all structures are covered by the selected/created rules, for missed structures will not have a stability criteria assignment.

To assign default stability criteria:

1. LifeSim provides a list default rules for assigning stability criteria (Figure 5-8). For example, in Figure 5-8, the **Rule List** includes ***Steel Buildings Collapse Criteria***. If a default rule in the **Rule List** is not desired, then the user can remove the rule. Default rules can also be modified.

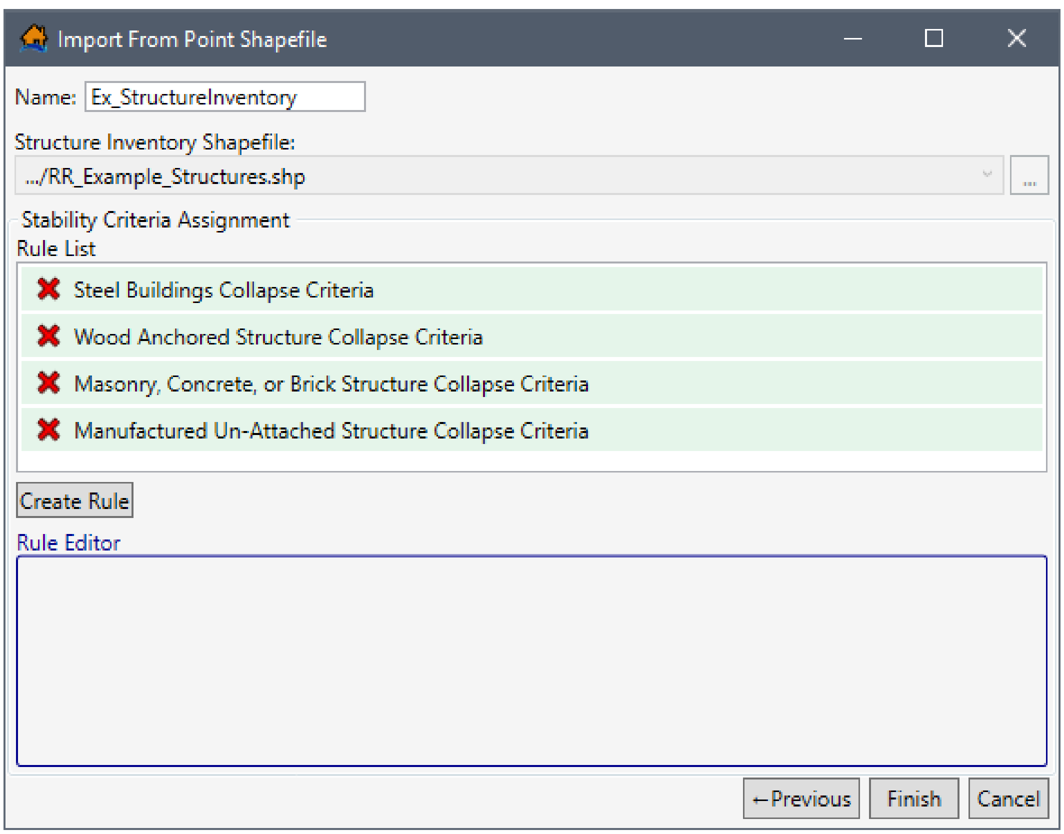


Figure 5-8. Default Construction Type Assignments

2. Default or created rules can be deleted from the **Rule List**, by clicking the **Delete** ❌ button next to the name of undesired rule. Note: To add all default rules removed by

accident, (this also removes all rules created by the user), click the **Previous** button twice, to return to the first window in the import wizard (**Define Structure Attribute Information** window), and then click the **Next** button twice, to return to the **Stability Criteria Assignment** window.

3. To view the conditions for any rules in the **Rule List**, select the desired rule (e.g., ***Manufactured Un-Attached Structure Collapse Criteria***) and view the rule conditions in the **Rule Editor** (Figure 5-9).

4. Rules displayed in the **Rule Editor** (Figure 5-9) can be modified. Click the **Add** button to add a new row to the bottom of the rule stack. Click the **Delete** button to remove a row. To move a row in the rule stack, click and hold the **Move** button and drag the row to the desired location and drop the row.

5. The first list in the row is populated by the Import Attributes in the selected shapefile (e.g., *ST_DMGFUN*). The second list contains the appropriate comparison operators (e.g., *Equals*, *Not Equal*, *Greater Than*, *Less Than*, *Contains*) for the selected Import Attribute. The final list in the row is populated with the selected import attribute fields (e.g., *RES2*).

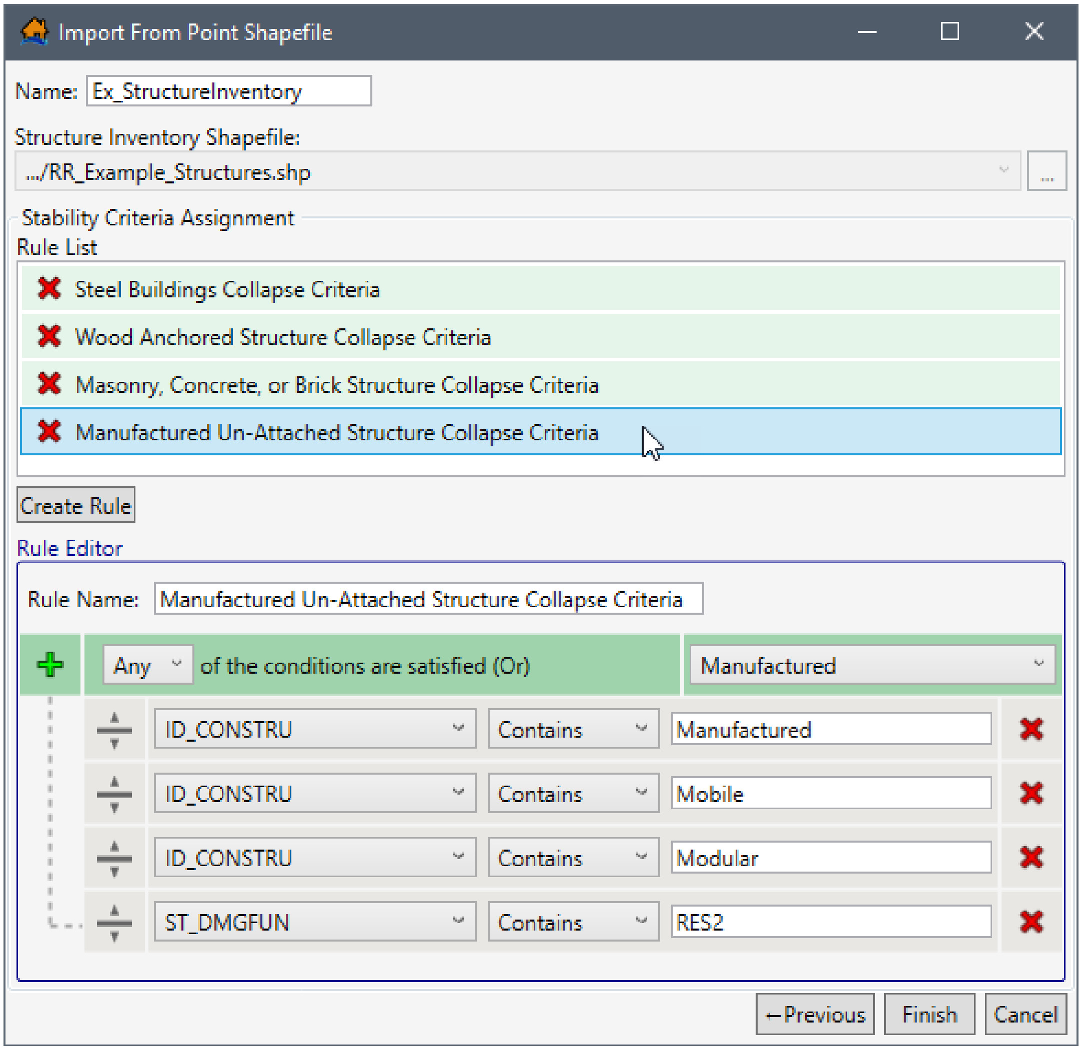


Figure 5-9. Import From Point Shapefile – Example Default Rule – Rule Editor

To create and define a new rule:

6. For new rules, click the **Create Rule** button. The **Rule Editor** will populate with the base rule stack (Figure 5-10). Enter a name for the new rule in the **Rule Name** box.

7. In the top row, from the list, select whether the rule is applied if **Any** or **All** (Figure 5-10) of the criteria are satisfied. From the list, from the top row, select a construction type from the default list (e.g., *RESCDAM – Wood Unanchored* in Figure 5-10).

8. Complete the rule definition by adding and defining new conditions for the rule stack. For each row added selected the Import Attribute (e.g., *ID_CONSTRU* in Figure 5-10), the comparison operator (e.g., *Contains* in Figure 5-10), and the condition to search for based on the selected Import Attribute (e.g., *Wood* in Figure 5-10).

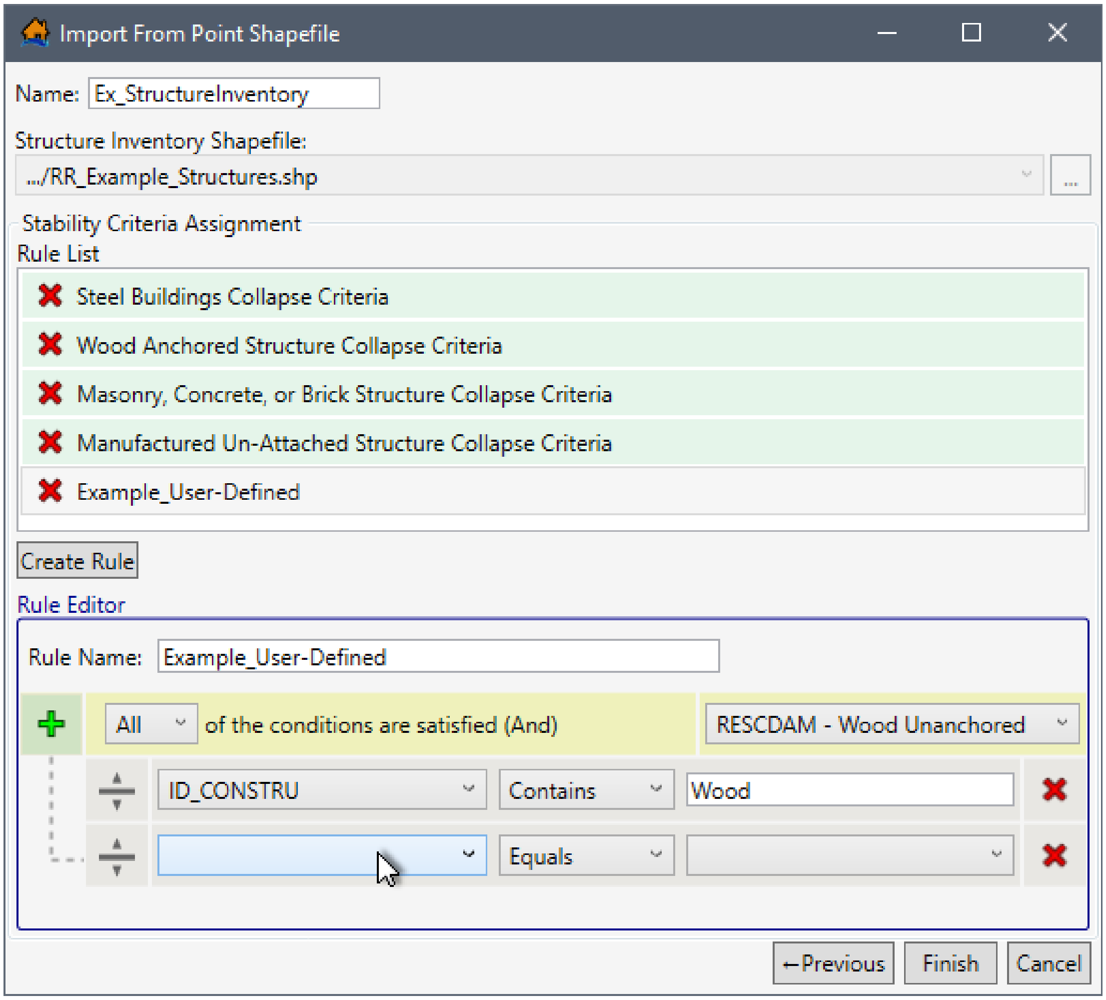


Figure 5-10. Import From Point Shapefile – Create New Rule – Rule Editor Example

9. Once the user has the desired stability criteria assignments rules in the **Rule List**, from the **Import From Point Shapefile** dialog box, click **Finish** (Figure 5-10).

10. The **Import From Point Shapefile** dialog box will close (Figure 5-10). From the LifeSim main window (Figure 5-1), from the **Study** tab, from the **Study Tree**, under the **Structure Inventories** folder, the name of the imported **Structure Inventory** data set is listed.

5.2 Import from National Structure Inventory (NSI)

The National Structure Inventory (NSI) for an area, will contain information on structure occupancy types, population, structure values, foundation height, build type, and other information. The NSI is currently only available to U.S. Army Corps of Engineer (USACE) users. The NSI is a database of structure locations and attributes and was developed using information from the HAZUS database and other sources including the Longitudinal Employer-Household Dynamics database (https://lehd.ces.census.gov/). To create a structure inventory from the NSI:

1. From the LifeSim main window (Figure 5-1), from the **Study** tab, from the **Study Tree**, right-click on **Structure Inventories**, from the shortcut menu click **Import Structures From NSI** (Figure 5-1). Another way is from the LifeSim main window, from the **Study** menu, point to **Structure Inventories**, click **Import Structures From NSI**. Either way, the **Import Structures From NSI** dialog box (Figure 5-11) opens.

2. In the **Structure Inventory Name** box (Figure 5-11), enter a name for the structure inventory.

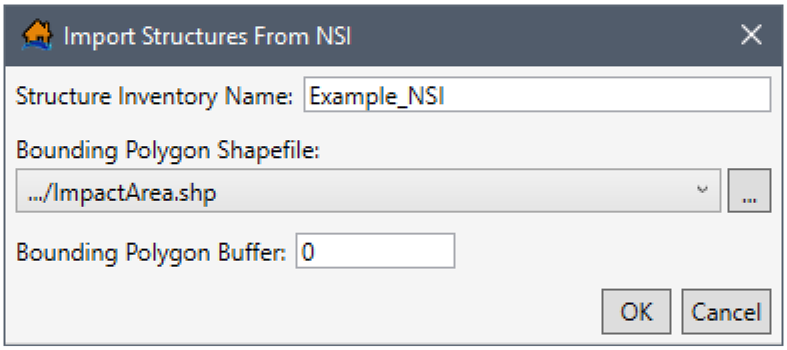


Figure 5-11. Import Structures From NSI Dialog Box

3. To select the bounding polygon shapefile that defines the boundaries of the structure inventory, to the right of the **Bounding Polygon Shapefile** dropdown menu (Figure 5-11), click . An **Open** browser will open (Figure 5-4), navigate to the directory where the bounding polygon shapefile is located. The bounding polygon shapefile must have a projection in order to properly access the NSI database. Select the correct bounding polygon shapefile (e.g., **.shp*), click **Open**, the **Open** browser will close (Figure 5-4), and the user will be returned to the **Import Structures From NSI** dialog box (Figure 5-11). Alternatively, if the desired shapefile has already been imported as a map layer into the LifeSim study, then from the **Bounding Polygon Shapefile** dropdown menu (Figure 8-3), select the desired map layer.

4. In the **Bounding Polygon Buffer** (units set by the **Bounding Polygon Shapefile** selection) box, enter a buffer distance for the bounding polygon shapefile (optional). The default value is zero.

5. Click **OK**, the **Import Structures From NSI** dialog box will close (Figure 5-11). The LifeSim **Status Bar** will display the progress for the NSI structures download.

6. Once the download is complete the **Status Bar** will display the message, "Structure Inventory successfully imported from NSI". From the LifeSim main window (Figure 5-1), from **Study** tab, from the **Study Tree**, under the **Structure Inventories** folder, the name of the imported **Structure Inventory** dataset is listed.

5.3 View and Edit a Structure Inventory

An imported structure inventory can be copied or deleted, viewed in the map window, appended with new structures, population information can be imported, and new stability criteria rules can be created. Sections 5.3.1 through 5.3.6 provide detailed instructions for viewing and editing the structure inventory.

To access the **Study Tree** shortcut commands and edit an imported structure inventory, from the LifeSim main window (Figure 5-1), from the **Study** tab, from the **Study Tree**, from the **Structure Inventories** folder, right-click on a structure inventory. The structure inventory shortcut menu opens (Figure 5-12). Commands include show in (or remove from) map window, tools, move up, move down, copy, and delete. The **Tools** shortcut submenu (Figure 5-12) contain commands for appending the structure inventory, importing population, and adding stability assignment criteria rules.

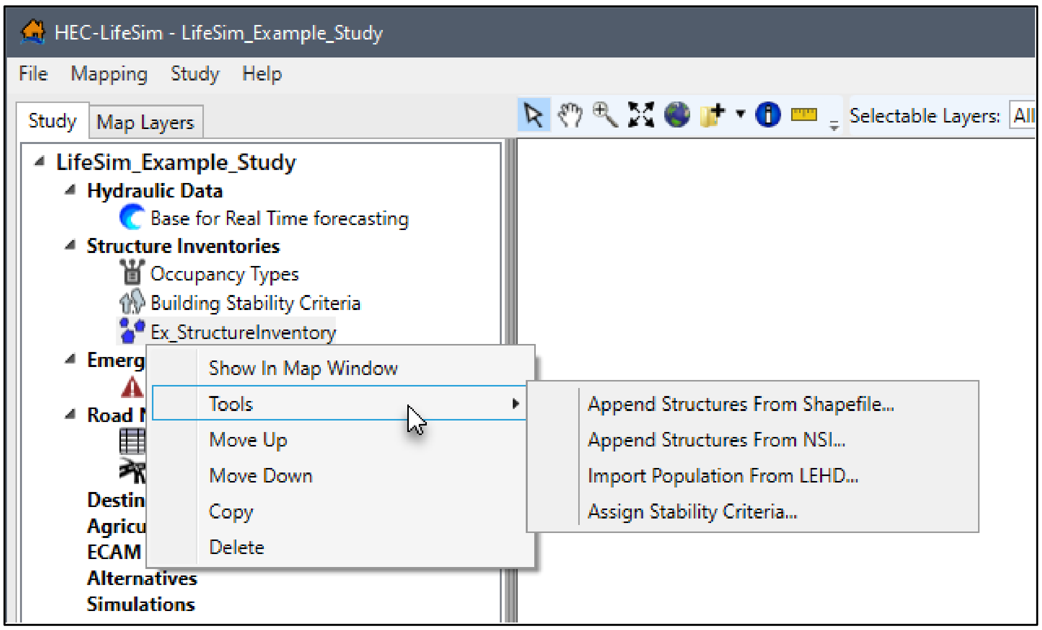


Figure 5-12. LifeSim Study Tree – Imported Structure Inventory Shortcut Commands

When a structure inventory is imported, default criteria for damage, stability, and submergence are assigned to structures based on the occupancy type attribute information and the stability criteria rules. The criteria for structure damage, stability, and submergence can be edited, created, and assigned. Sections 5.4 and 5.5 describe how to edit the occupancy types and structure inventory criteria, respectfully.

Editing individual structures or attributes of the imported structure inventory map layer can be accomplished from the **Map Layers Tab**; review Appendix A for information regarding editing the structure inventory map layer.

5.3.1 View a Structure Inventory

Users can view the structure inventory in the map window and can view the structure inventory attributes.

Show in Map Window

From the LifeSim main window (Figure 5-1), from the **Study** tab, from the **Study Tree**, from the **Structure Inventories** folder, right-click on a structure inventory. From the shortcut menu (Figure 5-12), click **Show In Map Window**. The selected structure inventory will display in the map window of the LifeSim main window (Figure 5-13). Note, right-click a displayed structure inventory and click **Remove From Map Window**, from the shortcut menu to stop displaying the structure inventory.

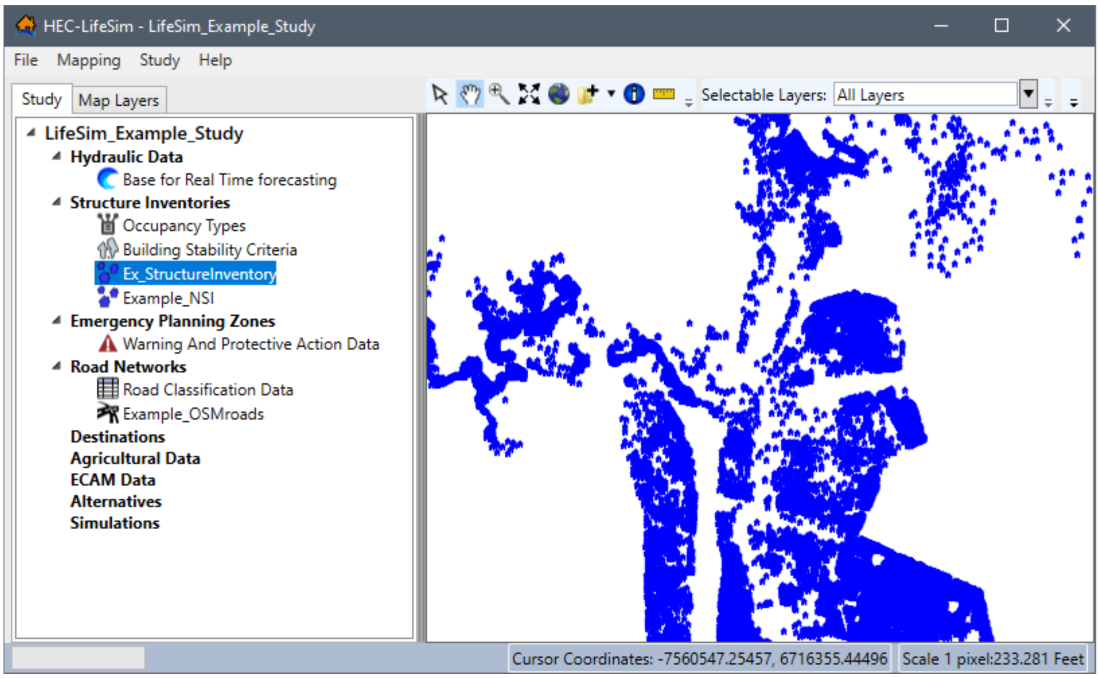


Figure 5-13. Example -- LifeSim Main Window – Structure Inventory Displayed in Map Window

View Attributes Table

Map layer attributes can be accessed for each map layer and is provided in a table for the dataset referenced by the map layer. The structure inventory dataset must be displayed in the map window before the attribute table can be accessed. To access a specific map layer **Attribute** table, from the **Map Layers** tab, right-click on a map layer of interest, from the shortcut menu, click **Open Attribute Table**. The *Map Layer* **Attributes** dialog box opens; review Appendix A, Section A.4 for more information. Editing of the map layer attributes is detailed in Section A.10 of Appendix A.

5.3.2 Append Structures

Structures can be appended to from a point shapefile or from NSI.

Append from a Shapefile

To append the structure inventory from a shapefile, from the LifeSim main window (Figure 5-1), from the **Study** tab, from the **Study Tree**, from the **Structure Inventories** folder, right-click on a structure inventory (e.g., *Ex_StructureInventory* in Figure 5-12). From the structure inventory shortcut menu, point to **Tools** and select **Append Structures From Shapefile** (Figure 5-12) submenu command. The **Append From Point Shapefile** dialog box opens (Figure 5-14).

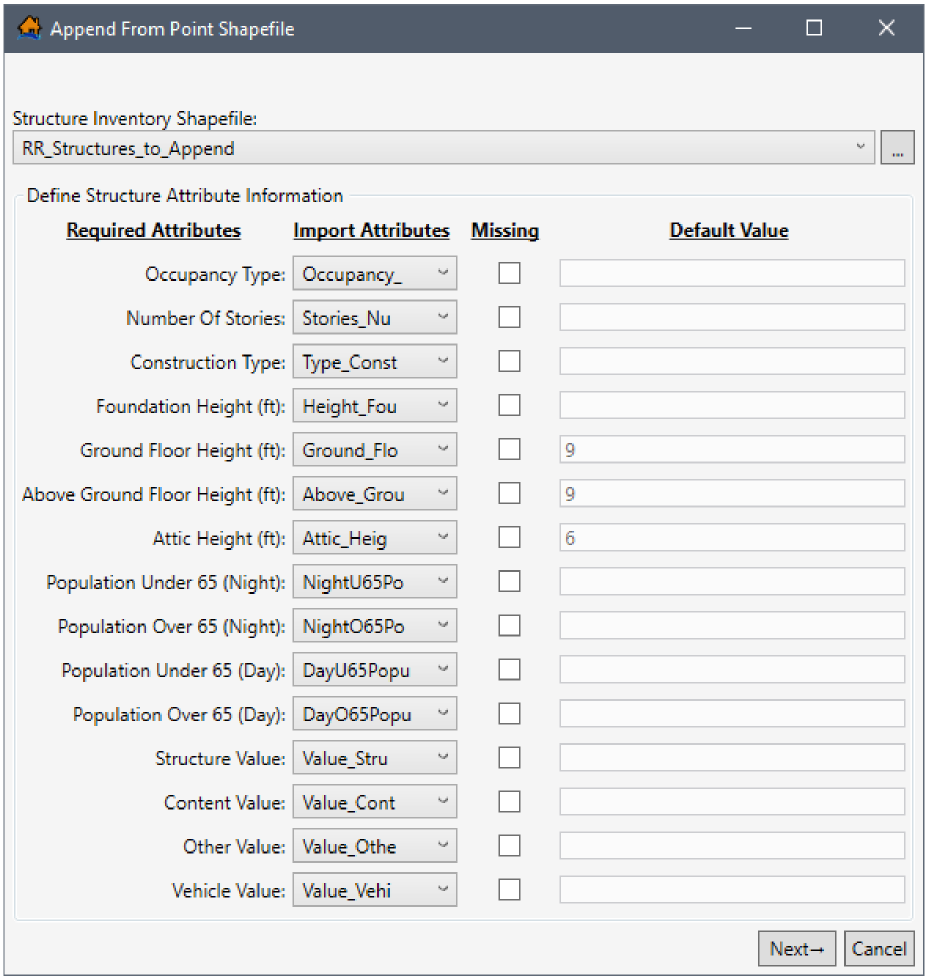


Figure 5-14. Example – Append From Point Shapefile Dialog Box

Alternatively, to open the **Append From Point Shapefile** dialog box (Figure 5-14) from the **Study** menu, point to **Structure Inventories**, point to **Append From Shapefile** (Figure 5-2) and select the structure inventory.

Select the appropriate shapefile, define the structure attribute information, the default occupancy type assignments, and the stability criteria assignment for the point shapefile. Except for adding a name, the **Append From Point Shapefile** dialog box (Figure 5-14) requires the same steps as the **Import From Point Shapefile** dialog box (Figure 5-3); refer to Section 5.1 for detailed instructions on completing the **Import From Point Shapefile** dialog box.

Append from NSI

Alternatively, append structures from NSI. To append the structure inventory from NSI, from the LifeSim main window (Figure 5-1), from the **Study** tab, from the **Study Tree**, from the **Structure Inventories** folder, right-click on a structure inventory. From the structure inventory shortcut menu, point to **Tools** and select **Append Structures From NSI** (Figure 5-12) submenu command. Alternatively, to open the **Append From NSI** dialog box (Figure 5-14) from the **Study** menu, point to **Structure Inventories**, point to **Append From NSI** (Figure 5-2) and select the structure inventory.

Either way, the **Import Structures From NSI** dialog box (Figure 5-15) opens. Except for adding a name, the **Import Structures From NSI** dialog box (Figure 5-14) for appending structures requires the same steps as the **Import Structures From NSI** dialog box (Figure 5-11); refer to Section 5.2 for detailed instructions on completing the **Import Structures From NSI** dialog box.

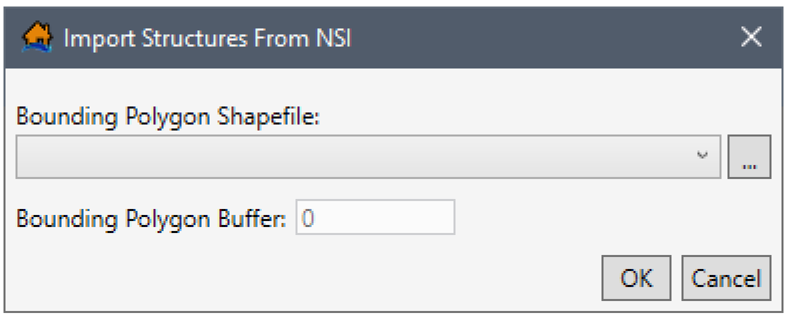


Figure 5-15. Import Structures From NSI – Appending a Structure Inventory

5.3.3 Import Population

Users can import population information for a structure inventory from the Longitudinal Employer-Household Dynamics (LEHD) database (https://lehd.ces.census.gov/). To append the structure inventory from NSI, from the LifeSim main window (Figure 5-1), from the **Study** tab, from the **Study Tree**, from the **Structure Inventories** folder, right-click on a structure inventory (e.g., *Ex_StructureInventory* in Figure 5-12). From the structure inventory shortcut menu, point to **Tools** and select **Import Population From LEHD** (Figure 5-12) submenu command. The **Import Population From LEHD** dialog box (Figure 5-16) opens.

The **Import Population From LEHD** dialog box (Figure 5-16) contains a table providing two editable columns, **Day** and **Night Households**, for users to define the household size for each occupancy type. The entered household size is used to distribute the imported population. Note: all household values must be greater than zero. Cells containing a value of zero are highlighted in red (Figure 5-16). Shortcut menu commands provide users with the ability to add, insert and delete row(s); select all; copy, copy with headers and paste (Figure 5-16). Open the shortcut menu by highlighting and right-clicking the row(s) of interest.

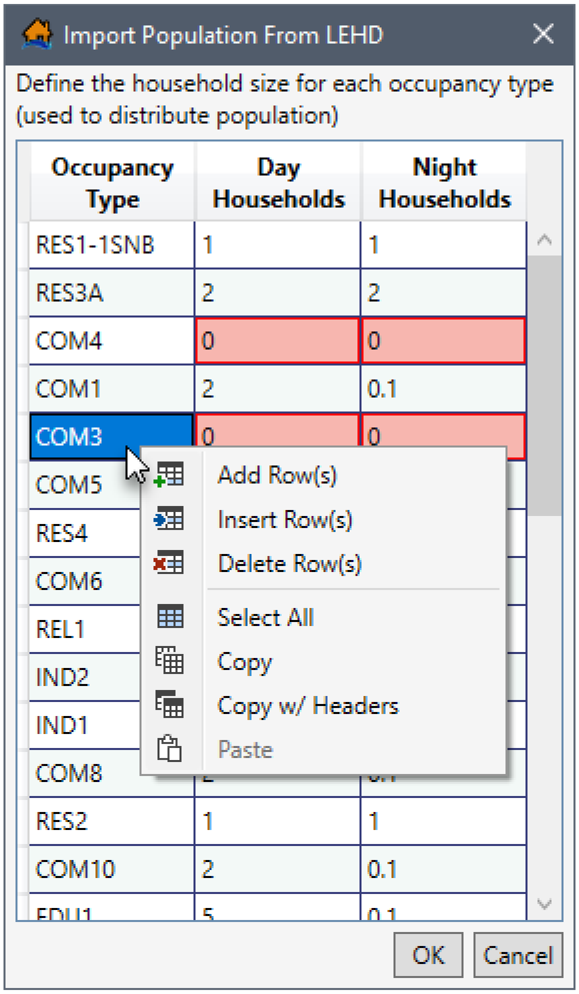


Figure 5-16. Import Population From LEHD Dialog Box – Shortcut Menu – Example with Errors

Once the day and night household cells contain the correct household size, click **OK** to import the population. **Note:** An **LifeSim** error message opens (Figure 5-17) when cells do not contain values greater than zero and the **OK** button is clicked.

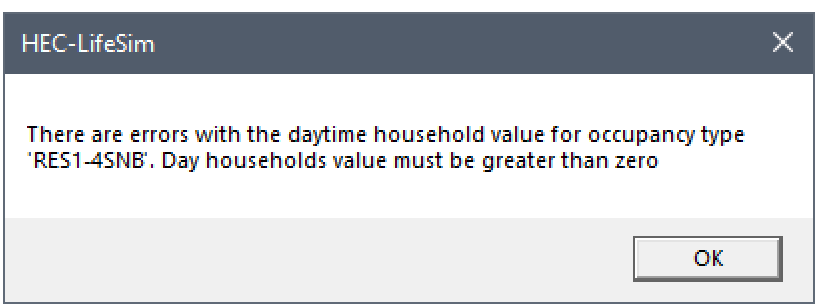


Figure 5-17. LifeSim Error Message

Click **OK** to close the **LifeSim** error message (Figure 5-17) and return to the **Import Population From LEHD** dialog box (Figure 5-16). Fix the cell(s) with zero values and click **OK** to import the population.

5.3.4 Assign Stability Criteria

Stability criteria can be assigned to a structure based on rules that relate stability functions to identified attribute values of the structure. The stability criteria assignment tool will take the rules that are defined and use them to assign the selected stability criteria to each structure that meets the rules. For example, if a rule states that all structures that are construction type 'Wood' should be assigned the 'Wood-Framed Attached' stability criteria. Then once OK is clicked every structure that meets the criteria will be assigned the Wood-Framed Attached stability function. These rules are not retained for future use. They are a one-time application to define the stability criteria that will be used by structures. The assigned stability criteria can be viewed through the structure inventory attribute table (Section 5.3.1). Refer to Section 5.5 for further discussion on structure stability criteria.

Users can create structure inventory specific stability criteria assignment rules from the **Stability Criteria Assignment Tool** dialog box (Figure 5-18). To open the dialog box, from the **Study Tree** right-click the structure inventory name (e.g., *Ex_StructureInventory* in Figure 5-12), from the shortcut menu point to **Tools** and select **Assign Stability Criteria**. The **Stability Criteria Assignment Tool** dialog box (Figure 5-18) opens.

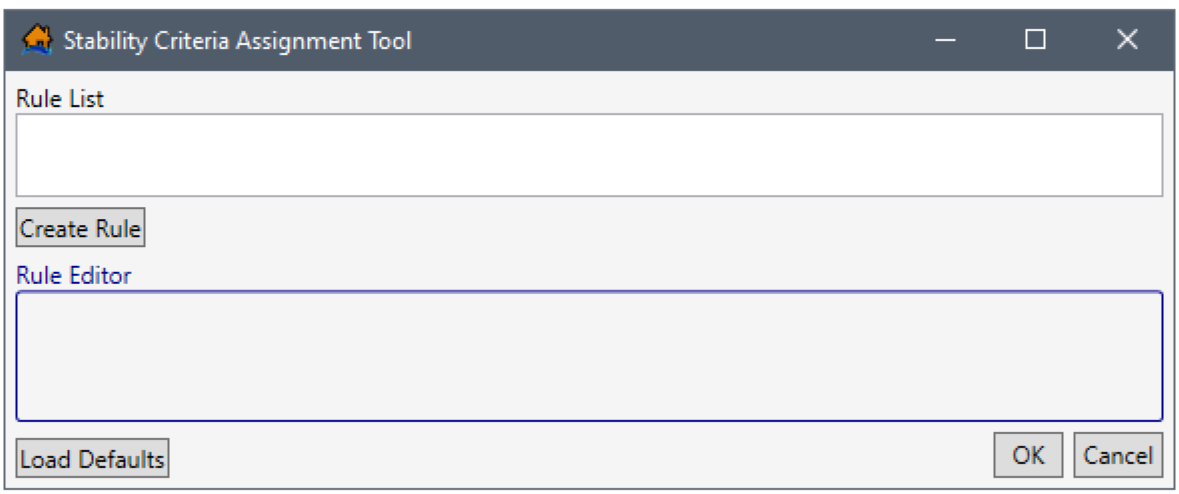


Figure 5-18. Stability Criteria Assignment Tool Dialog Box

LifeSim Version 2.0 has default rules that assign stability criteria for structures with specific construction and/or occupancy types. To load the default rules, click the **Load Defaults** button. Alternatively, the user may edit an existing default rule or create and define a new rule based on any of the structure attributes within the selected structure inventory. Note, the created rules are a 'rule stack' meaning the order in which the criteria are defined is important. If two rules apply to the same structure, then the higher (top) rule will dominate. Users should make certain all structures are covered by the selected/created rules, for missed structures will not have a stability criteria assignment.

The **Import From Point Shapefile** dialog box (Figure 5-8) requires the same steps as the **Stability Criteria Assignment Tool** dialog box (Figure 5-18); refer to Section 5.1.2 for detailed instructions on completing the **Stability Criteria Assignment Tool** dialog box (Figure 5-18).

5.3.5 Copy a Structure Inventory

To copy an existing structure inventory, from the LifeSim main window (Figure 5-1), from the **Study** tab, from the **Study Tree**, from the **Structure Inventories** folder, right-click on a structure inventory (e.g., *Ex_StructureInventory* in Figure 5-12). From the shortcut menu (Figure 5-12) click **Copy**. Another way is from the LifeSim main window, from the **Study** menu, point to **Structure Inventories** (Figure 5-2), point to **Copy**, and click on a structure inventory.

Either way, the **Name of New Study Item** dialog box (Figure 5-19) opens. Enter a new name in the **Name** box. Click **OK**, the **Name of New Study Item** dialog box (Figure 5-19) closes and the copied structure inventory is added to the **Study Tree** under the **Structure Inventories** folder.

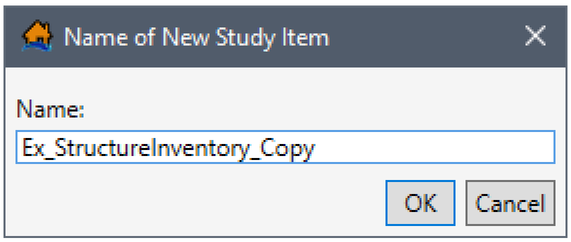


Figure 5-19. Name of New Study Item Dialog Box

5.3.6 Delete a Structure Inventory

To delete an existing structure inventory, from the LifeSim main window (Figure 5-1), from the **Study** tab, from the **Study Tree**, from the **Structure Inventories** folder, right-click on a structure inventory. From the shortcut menu (Figure 5-12) click **Delete**. Another way is from the LifeSim main window, from the **Study** menu, point to **Structure Inventories** (Figure 5-2), point to **Delete**, and click on a structure inventory. Either way, a **Confirm Delete** window (Figure 5-20) will open asking for confirmation to delete the selected structure inventory.

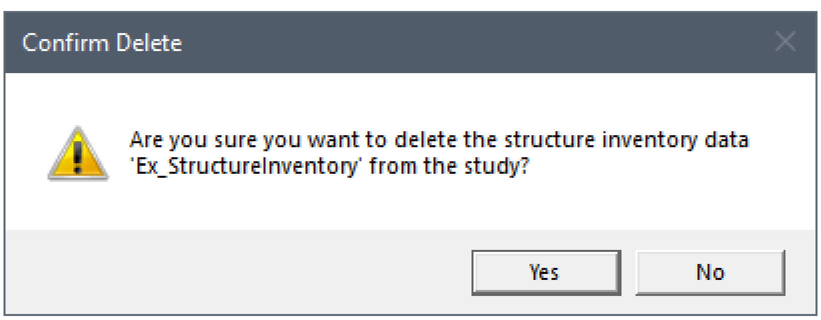


Figure 5-20. Confirm Delete Message Box – Structure Inventory Data

Click **Yes**, the **Confirm Delete** message window will close (Figure 5-20), and the selected structure inventory is deleted, and no longer displays in the **Study Tree** under the **Structure Inventories** folder.

5.4 Occupancy Type Editor

Each structure is required to have an **occupancy type**. Occupancy types are a subcategory of the default **damage categories** assigned to a study and describe a class of structures (e.g., RES1-1SNB which is a single family residential structure that is one story tall with no basement). **Damage categories** represent a high level grouping for structures. They are used in the Economic Consequence Assessment Model (ECAM) process to determine population and capital impacted for each event. In other words, damage categories represent a grouping of occupancy types. By default, a LifeSim study is pre-loaded with default damage categories and occupancy types. Default damage categories are residential, commercial, industrial, and public. Default occupancy types follow the naming convention developed for HAZUS (FEMA 2019) and are listed in the LifeSim Version 2.0 Technical Reference Manual, Appendix D.

Users can modify default occupancy types or create and define new occupancy types with unique depth-damage functions and uncertainty parameters. However, entered data is applied to all the structures assigned to a created or modified occupancy type. Several occupancy types can be assigned to a single damage category. For example, single-story structures with no basements, single-story structures with basements, mobile homes, and duplex apartments are different structure occupancy types assigned to a residential damage category. Structure inventories imported from NSI will automatically have the LifeSim default occupancy types defined as attributes. Structures imported from point shapefiles require users to match the shapefiles occupancy types with the LifeSim preloaded default occupancy type assignment (review Section 5.1.1).

Data describing an occupancy type include: the depth-percent damage functions (structure, content, vehicle and other); the evacuation parameters; submergence criteria; and the uncertainties in the first foundation height offset, submergence criteria, variation in values (structure, content vehicle and other) as a percent; and the damage in the depth-damage functions.

The **Occupancy Type Editor** (Figure 5-21) allows the user to revise and create the evacuation parameters, submergence criteria, economic damage depth-damage functions, value uncertainty, and foundation height uncertainty associated with structures that share common attributes (and therefore grouped by damage category). Also, from the editor, a user can assign damage criteria based on any combination of structure damage category, occupancy class, number of stories, basement information and construction type.

5.4.1 Occupancy Type Editor Structure

To open the **Occupancy Type Editor** (Figure 5-21), from the LifeSim main window (Figure 5-1), from the **Study** menu, point to **Structure Inventories** (Figure 5-2), and click **Edit Occupancy Types**. Alternatively, from the **Study Tree**, under **Structure Inventories** right-click on **Occupancy Types** and from the shortcut menu click **Edit Occupancy Types** (Figure 5-22). Either way, the **Occupancy Type Editor** (Figure 5-21) opens.

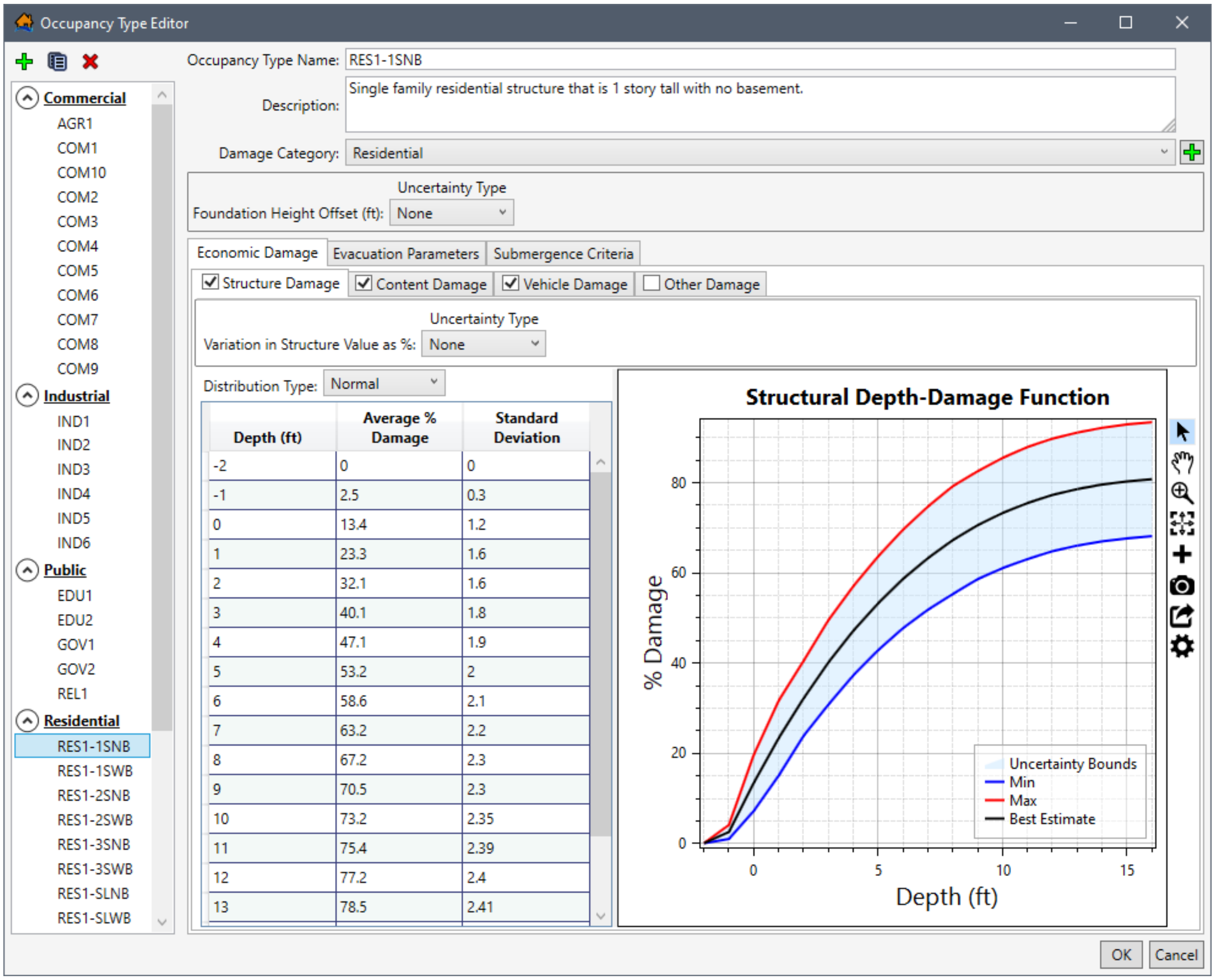


Figure 5-21. Occupancy Type Editor – Example Residential (RES1-1SNB) Selected

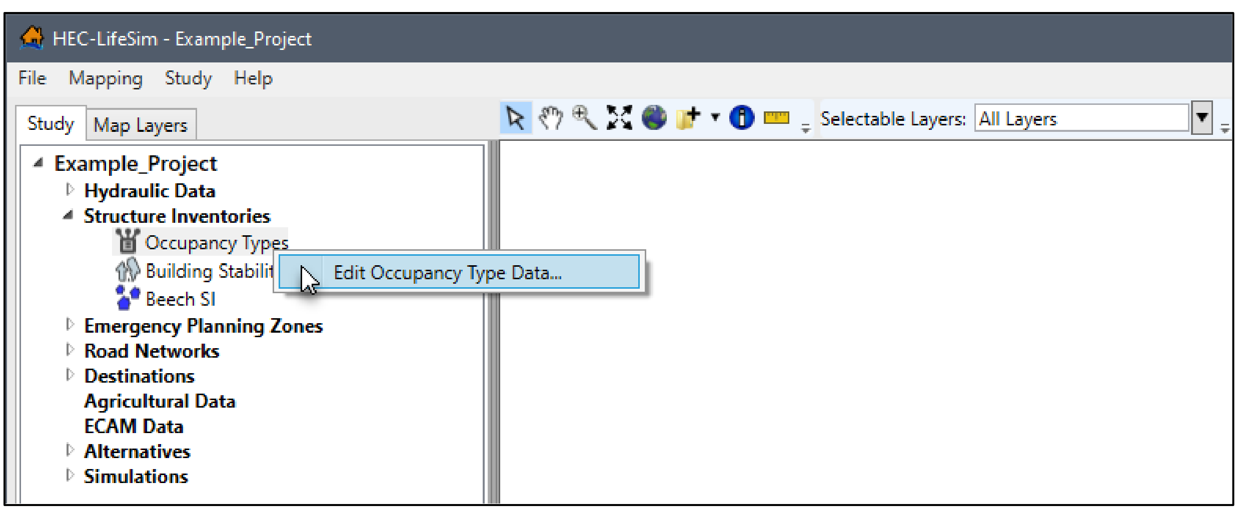


Figure 5-22. LifeSim Study Tree – Structure Inventories – Edit Occupancy Type Data Shortcut Command

Tree

The left side of the **Occupancy Type Editor** (Figure 5-21) provides a view of the occupancy types grouped by damage category in a tree (**Occupancy Types Tree**). From the **Occupancy Types Tree**, select an occupancy type name to display the data describing the selected occupancy type.

Buttons

The **Occupancy Type Editor** (Figure 5-21) has buttons that allow for creating, deleting, and copying of items that are available for a structure occupancy type. The function of each of the buttons is listed below:

Add New Damage Category allows a user to add a new damage category for grouping occupancy types. Click the **Add New Damage Category** button, the **Name of New Damage Category** dialog box opens (Figure 5-23). Enter the name for the new damage category in the **Name** field and click **OK** to create the new damage category. Note, existing and created damage categories cannot be renamed, copied, or deleted.

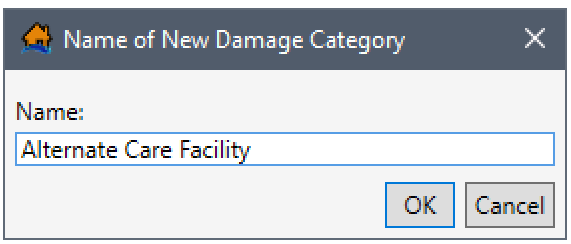


Figure 5-23. Name of New Damage Category

Create New Occupancy Type allows a user to create a new occupancy types. From the **Occupancy Type Editor** (Figure 5-21) click **Add New Occupancy Type** button, the **Name of New Occupancy Type** dialog box (Figure 5-24) opens. In the **Name** box, enter a name for the new occupancy type.

Click **OK**, the **Name of New Occupancy Type** dialog box will close (Figure 5-24). Users will return to the **Occupancy Type Editor** (Figure 5-21), with the new occupancy type displayed.

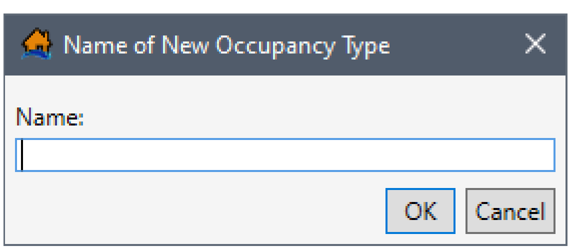


Figure 5-24. Name of New Occupancy Type Dialog Box

Delete Selected Occupancy Type button allows the user to delete the selected occupancy types. From the **Occupancy Type Editor** (Figure 5-21) select an occupancy type from the

Occupancy Type Name list. Click the **Delete** button and the selected occupancy type is deleted. **NOTE:** this action is immediate and cannot be undone.

Copy Selected Occupancy Type allows the user to copy an existing occupancy type with a new name. From the **Occupancy Type Editor** (Figure 5-21) select an occupancy type from the **Name** list. Click **Copy**, the **Save Occupancy Type As** dialog will open (Figure 5-25). Enter a new name in the **Name** box. Click **OK**, the **Save Occupancy Type As** dialog box will close (Figure 5-25). Users will return to the **Occupancy Type Editor** (Figure 5-21) with the new occupancy type displayed.

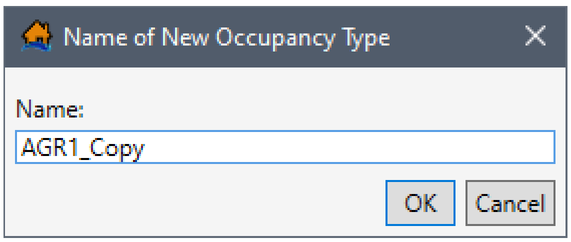


Figure 5-25. Name of New Occupancy Type Dialog Box

Collapse the damage category group in the **Occupancy Types Tree**. Click the collapse icon to expand the damage category group and view the list of occupancy types. Note, clicking on the expanded header collapses the group or vice versa collapses an expanded group.

Expand the damage category group in the **Occupancy Types Tree**. Click the expand icon to collapse the damage category group. Note, clicking on the expanded header collapses the group or vice versa collapses an expanded group.

Occupancy Type Data

From the **Occupancy Type Editor** (Figure 5-21), the right side of the editor contains the data entered for the selected occupancy type (e.g., *RES1-1SNB* in Figure 5-21). Directly below the **Occupancy Type Name** field is the **Description** field; both fields can be edited. Modify or set the damage category for selected occupancy type using the **Damage Category** dropdown menu; the menu contains the list of default damage categories (and any user added damage categories). Edit or set the uncertainty in the foundation height offset, from the **Foundation Height Offset** panel. Add or edit the structure, content, vehicle, and other depth-damage functions from the set of tabs located within the **Economic Damage** tab; set the warning and protective action behavior from the **Evacuation Parameters** tab; and set the uncertainty in the high hazard heights from the **Submergence Criteria** tab. Review the next section to learn how to create and define a new occupancy type in the **Occupancy Type Editor** (Figure 5-21).

5.4.2 Create or Edit an Occupancy Type

The following sections describe how to create a new occupancy type and define the individual components of the occupancy type. Note, skip the instructions for creating a new occupancy type and start from the damage category assignment section to learn how to editing an existing or copied occupancy type.

Create New Occupancy Type

To create a new occupancy type:

1. From the LifeSim main window's **Study** tab (Figure 5-1), from the **Study Tree**, right-click on **Structure Inventories**, from the shortcut menu click **Edit Occupancy Types** (Figure 5-22). Another way is from the LifeSim main window, from the **Study** menu, point to **Structure Inventories** (Figure 5-2), click **Edit Occupancy Types**. Either way the **Occupancy Type Editor** (Figure 5-21) opens.

2. From the **Occupancy Type Editor** (Figure 5-21), click the **Create New Occupancy Type** button. The **Name of New Occupancy Type** dialog box opens (Figure 5-24).

3. Enter a unique name (required) in the **Name** box and click **OK**. The **Name of New Occupancy Type** dialog box closes (Figure 5-24), and the **Occupancy Type Editor** (Figure 5-21) will create and display the new undefined occupancy type.

Assign the Damage Category

4. By default, newly created occupancy types are added alphabetically to the occupancy type list for the damage category selected at the time that the new occupancy type was created. For example, if the user has **AGR1** occupancy type name selected (in the **Occupancy Type Tree**) and creates the new occupancy type, then the new occupancy type will automatically be added to the **Commercial** damage category (Figure 5-21). Select the appropriate damage category, from the **Damage Category** dropdown menu (Figure 5-21).

   Alternatively, create an new damage category by clicking the **Add New Damage Category** button to open the **Name of New Damage Category** dialog box (Figure 5-23). Enter the name for the new damage category in the **Name** field and click **OK** to create the new damage category. Note, existing and created damage categories cannot be renamed, copied, or deleted.

Enter the Foundation Height Offset

5. The user can set or modify the uncertainty around the foundation height using the Foundation Height Offset. The user needs to define the uncertainty about foundation heights at the occupancy type level using the structure's foundation height attribute as the most likely or mean estimate. For example, if all RES1-1SNB occupancy type structures have uncertainty about foundation height of 1' then the Foundation Height Offset could be set to a uniform distribution of minimum -1' and maximum of 1'. The default uncertainty is **None**, from the **Uncertainty Type** dropdown menu (Figure 5-26), the user can chose **Triangular**, **Normal** (standard deviation), and **Uniform** (minimum, maximum). For the example provided in Figure 5-26, **Triangular** has been selected; therefore, the **Minimum** and **Maximum** values need to be entered.

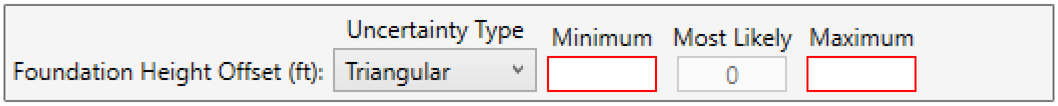


Figure 5-26. Foundation Height Offset Uncertainty – Triangular Example

Enter the Economic Depth-Damage Functions

The **Economic Damage** tab holds the **Structure Damage**, **Content Damage**, **Vehicle Damage** and **Other Damage** sub-tabs (Figure 5-27). Each sub-tab provides the same configuration for entering and defining the depth-damage functions. Therefore, the following instructions, although written from the perspective of the **Structure Damage** sub-tab, can be used to complete any of the depth-percent damage functions (structure, content, vehicle and other).

To enter the structure damage depth-damage function:

6. Enable the **Structure Damage** (Figure 5-27) sub-tab by checking the ☑ checkbox for the tab.

7. The user can set the **Variation in Structure Value as %** (Figure 5-27) by defining the uncertainty about the structure value associated with a structure based on the selected occupancy type. From the **Uncertainty Type** dropdown menu, the default selection is **None** (Figure 5-27). The uncertainty types available are the **Triangular** (minimum, maximum), **Normal** (standard deviation), and **Uniform** (minimum, maximum).

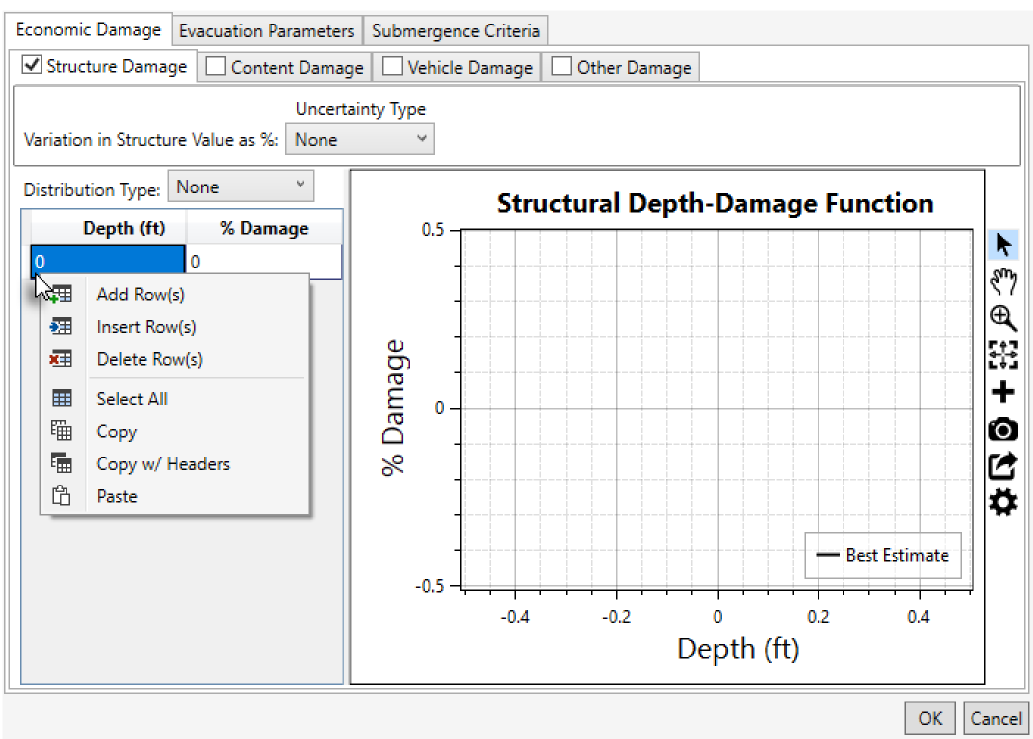


Figure 5-27. Occupancy Type Editor – Economic Damage Tab – New Occupancy Type

8. The user can also set the uncertainty associated with how much damage can occur from depths of water at the structure for the **Structure Depth-Damage Function** (Figure

5-27). By default, the uncertainty in the structure depth-damage function is **None** (Figure 5-27). Uncertainty types available are the **Triangular** (minimum, maximum), **Normal** (standard deviation), and **Uniform** (minimum, maximum).

9. Depth-damage function values can be entered into the **Depth-Damage Function Table** (Figure 5-27) by hand, copied from another occupancy type or from another application (e.g., Microsoft Excel®) and pasted into the table (i.e., entire table, individual cells, columns, rows). From the **Depth-Damage Function Table**, right-click to open the shortcut menu (Figure 5-27), from the shortcut menu the user can add, insert, and delete row(s); select all; copy or copy with headers; and paste.

   For example, from another occupancy type select a cell in the table and use **Ctrl + A** (or right-click and select the **Select All** command from the shortcut menu), to select the entire table. Use **Ctrl + C** (or the **Copy** shortcut menu command) to copy the data to the clipboard. Return to the newly created occupancy type and select the first cell in the first row and use **Ctrl + V** to paste all the copied data.

   The depth-damage function table can also be copied (with or without headers) into another application (e.g., Microsoft Excel®) for saving. Use **Ctrl + A** (or the **Select All** command from the shortcut menu), right-click and select **Copy w/ Headers**. Then paste the data in the desired application to save.

10. Errors in the depth-damage function are highlighted in red and a tooltip displays a message describing the error (Figure 5-28). Right-click on a header row to open the shortcut menu (Figure 5-28) to sort the table in ascending or descending order or clear the table sorting.

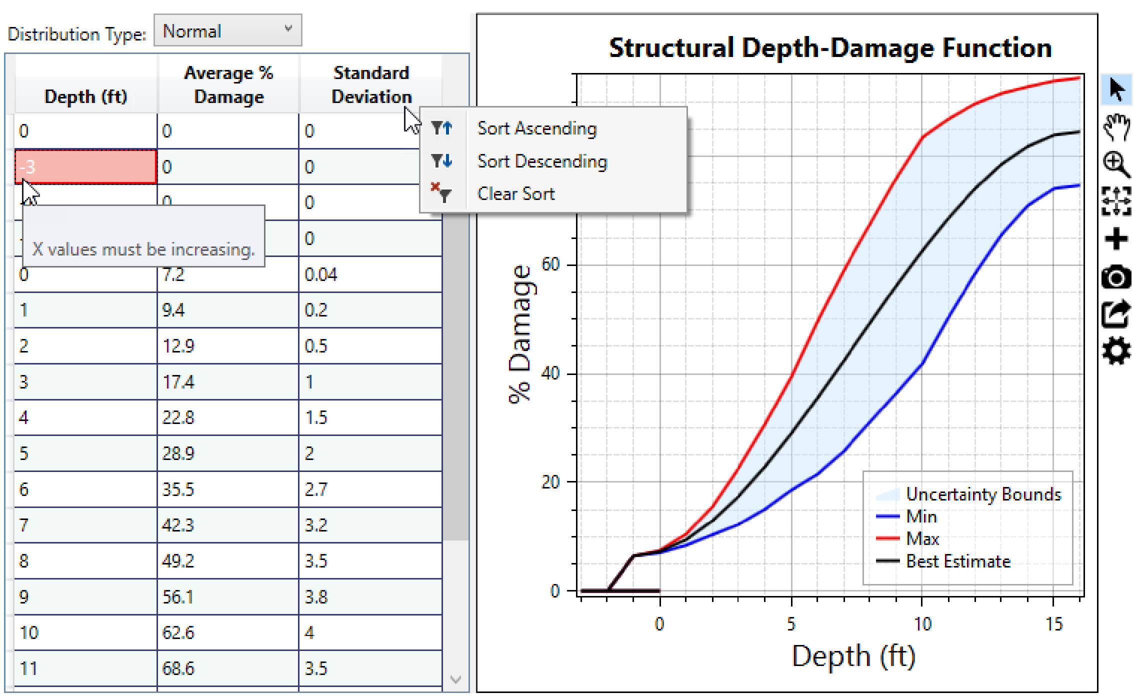


Figure 5-28. Example – Occupancy Type Editor – Depth-Damage Function

The **Structural Depth-Damage Function** plot (Figure 5-28) provides a live update for the data entered in the table. Review Section 3.6 for instructions on using the general plotting tools (located on the right side of the plot window) and for customizing plots.

11. Complete the economic depth-damage functions by repeating the above steps to enter the content, vehicle, and other depth-damage functions from the **Content Damage**, **Vehicle Damage** and **Other Damage** sub-tabs (Figure 5-27), respectfully.

Set the Evacuation Parameters

After completing the economic depth-damage functions for the new occupancy type, select the **Evacuation Parameters** tab (Figure 5-29). This tab contains evacuation parameters set for all structures assigned to the selected occupancy type. Review the LifeSim Technical Reference Manual for more information regarding the evacuation parameters defined by the structure occupancy type.

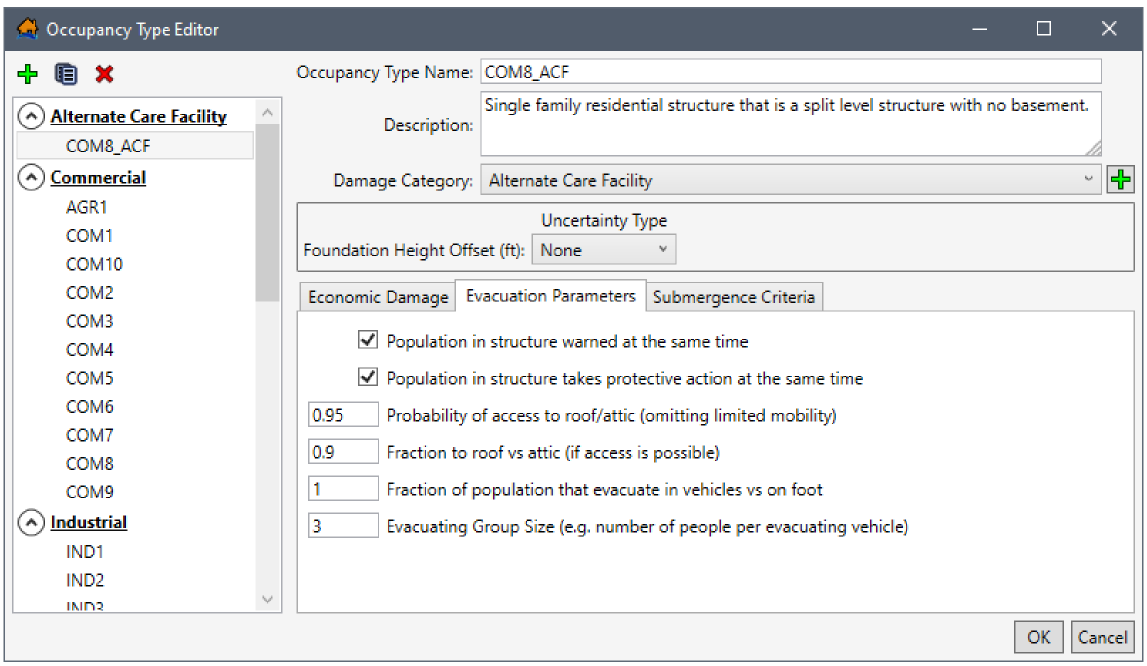


Figure 5-29. Occupancy Type Editor – Evacuation Parameters Tab

12. From the **Evacuation Parameters** tab (Figure 5-29), define the following evacuation parameters for the new occupancy type:

   - The user can set warning and protective action behavior for the selected occupancy type. If the **Population in structure warned at the same time** checkbox is selected (Figure 5-29) then the warning will reach all people in the structure at once instead of at different times. For example, in an apartment building, residents could receive a warning at different times but for a single family residential structure the warning will reach all people in the structure at the same time.

   - The user can also select **Population in structure takes protective action at the same time** (Figure 5-29), this is where all of the people within the structure will try to evacuate at the same time. For example, in a single-family residential structure everyone will try to evacuate at the same time but in a commercial structure everyone might try to evacuate at different times.

- Enter the probability that the population of the structure can access the roof or attic of the structure in the **Probability of access to roof/attic (omitting limited mobility)** (Figure 5-29) field. By default, the probability is set to ***0.9***, and the range of possible probabilities is 0.0 to 1.0.

- In the **Fraction to roof vs attic (if access if possible)** (Figure 5-29) field, set the fraction of the population of the structure that can access the roof. By default, the fraction for roof vs. attic is ***1***; enter a value between 0.0 and 1.0.

- For the **Fraction of population that evacuate in vehicles vs on foot** (Figure 5-29) field, enter the fraction of the structure's population that evacuate in vehicles vs. on foot. By default, the fraction for vehicle vs. on foot is ***1***; enter a value between 0.0 and 1.0.

- By default, the **Evacuating Group Size (e.g., number of people per evacuating vehicle)** (Figure 5-29) field is ***3***. Enter the appropriate group size for the created occupancy type. For example, RES5 (institutional dormitories) has a set group size of 48 to simulate evacuation by bus.

Enter the Submergence Criteria

Submergence is defined as a water level at a structure that can affect probability of survival. The **submergence criteria** in LifeSim Version 2.0 is used to define the threshold between high hazard and low hazard conditions when people are trapped in a flooded structure.

Note, in LifeSim Version 1.0, submergence criteria were divided into three zones: chance, compromised and safe. However, in LifeSim Version 2.0, the original three zones were replaced with two new zones: low and high hazard. The LifeSim Version 2.0 Technical Reference Manual, Chapter 9, Section 9.2, provides a detailed explanation regarding the development of the two new zones. LifeSim Version 2.0 characterizes the high hazard or low hazard conditions as described below:

High Hazard Zone:	Refers to those conditions where the stability criteria or submergence criteria of the person, the vehicle (if caught while evacuating), or the structure (if not mobilized) has been exceeded. When the stability (or submergence) criteria has been exceeded survival depends largely on chance.
Low Hazard Zone:	Refers to those conditions where the person or group of people are exposed to relatively calm floodwaters, where the stability of the individual, group, or shelter is not at risk. A hazard exists due to the potential for bad things to happen when people come in contact with water in locations not meant for such an interaction.

The threshold of submergence is an uncertain parameter sampled uniquely for each structure. Review the LifeSim Technical Reference Manual for more information regarding the submergence criteria. To enter the submergence criteria for the High Hazard zone for the new occupancy type:

13. Select the **Submergence Criteria** tab (Figure 5-30) and set the uncertainty in the threshold of submergence from the **Uncertainty Type** dropdown menu (Figure 5-30), the user can chose **Triangular**, **Normal** (standard deviation), and **Uniform** (minimum, maximum); or select **None** for no uncertainty. For the example provided in Figure 5-30, **Triangular** has been selected for all three criteria; therefore, the **Minimum**, **Most Likely** and **Maximum** values need to be entered.

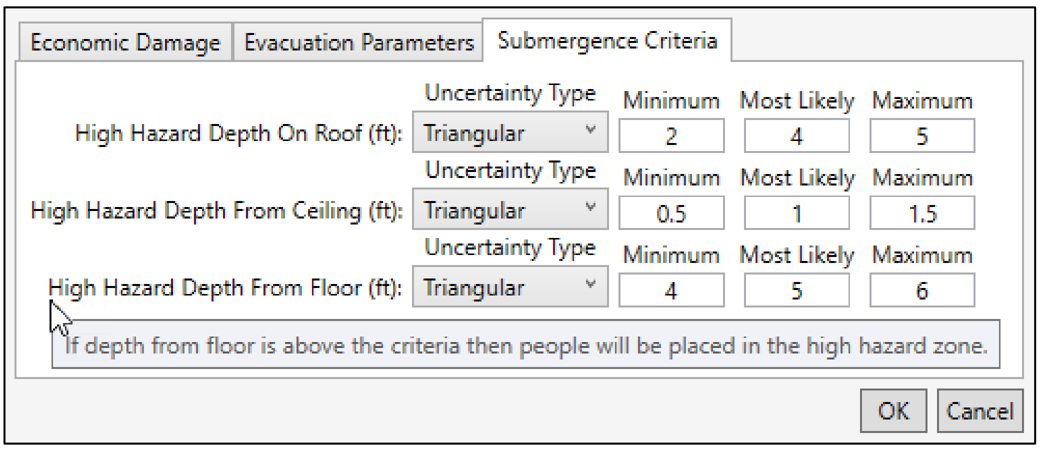


Figure 5-30. Occupancy Type Editor – Submergence Criteria Tab – Default Values

Users set the threshold values for the three potential high hazard zones. If the depth of flooding is above the defined high hazard zone, then people caught in the structure will be placed in the high hazard zone. The three submergence criteria are:

- **High Hazard Depth On Roof (ft)** – users set the threshold for this high hazard zone with or without uncertainty by entering depth (ft) as measured from the roof up. For the example provided in Figure 5-30, the most likely value (or best estimate) is 4 ft above the roof, with a minimum of 2 ft and maximum of 5 ft above the roof. In this case if the depth of flooding is above the **High Hazard Depth On Roof** submergence criteria, then able-bodied people caught on the roof will be placed in the high hazard zone.

- **High Hazard Depth From Ceiling (ft)** – users set the threshold for this high hazard zone with or without uncertainty by entering depth (ft) as measured from the top of the ceiling (of the top floor of the building). For the example provided in Figure 5-30, the most likely value (or best estimate) is 1 ft below the ceiling, with a minimum of 0.5 ft and maximum of 1.5 ft below the ceiling. In this case if the depth of flooding is above the **High Hazard Depth From Ceiling** submergence criteria, then able-bodied people caught in the structure will be placed in the high hazard zone.

- **High Hazard Depth From Floor (ft)** – users set the threshold for this high hazard zone with or without uncertainty by entering depth (ft) as measured from floor towards the ceiling (of the top floor of the building). For the example provided in Figure 5-30, the most likely value (or best estimate) is 5 ft above the floor, with a minimum of 4 ft and maximum of 6 ft above the floor. In this case if the depth of flooding is above the **High Hazard Depth From Floor** submergence criteria, then limited mobility occupants caught in the structure will be placed in the high hazard zone.

14. Once information about the selected occupancy type is entered, from the **Occupancy Type Editor** (Figure 5-21), click **OK**. The **Occupancy Type Editor** will close and the information for the new (or edited) occupancy type is saved.

5.4.3 Delete an Occupancy Type

From the **Occupancy Type Editor** (Figure 5-21) select an occupancy type from the **Occupancy Type Name** list. Click the **Delete** button and a **Confirm Delete** window opens asking for confirmation to delete the selected item. Click **Yes**, the **Confirm Delete** message window closes and the selected stability criteria is deleted.

5.4.4 Copy an Occupancy Type

From the **Occupancy Type Editor** (Figure 5-21) select an occupancy type from the **Name** list. Click **Copy**, the **Save Occupancy Type As** dialog will open (Figure 5-25). Enter a new name in the **Name** box. Click **OK**, the **Save Occupancy Type As** dialog box will close (Figure 5-25). Users will return to the **Occupancy Type Editor** (Figure 5-21) with the new occupancy type displayed.

5.5 Building Stability Curve Editor

Stability criteria defines the depth and velocity threshold for total structure collapse due to a hazard. The threshold is defined by a depth – velocity curve. If at any time during a simulation the depth and velocity are at a point above the velocity curve then the structure is assumed to be collapsed which can have significant impacts on estimated life loss. The building stability criteria is generally based on the construction type. The user can add, edit, and assign stability criteria based on the structure attributes: construction type, number of stories, occupancy type. The **Building Stability Curve Editor** (Figure 5-31) allows the user to define stability criteria for the structure inventory.

5.5.1 Building Stability Curve Editor Overview

Similar to the **Occupancy Type Editor** (Figure 5-21), the **Building Stability Curve Editor** (Figure 5-31) contains a tree on the left side to allow users to select a building stability criteria from the list of defaults. Selected stability criteria data includes a **Structural Stability Threshold** plot and is displayed on the right side of the editor. The **Structural Stability Threshold** plot has a toolbar; review Section 3.6 for instructions on using the general plotting tools (located on the right side of the plot window) and for customizing plots. The **Building Stability Curve Editor** (Figure 5-31) has buttons for creating, deleting and copying a selected stability criteria. The following sections describes the **Building Stability Curve Editor** (Figure 5-31) in more detail.

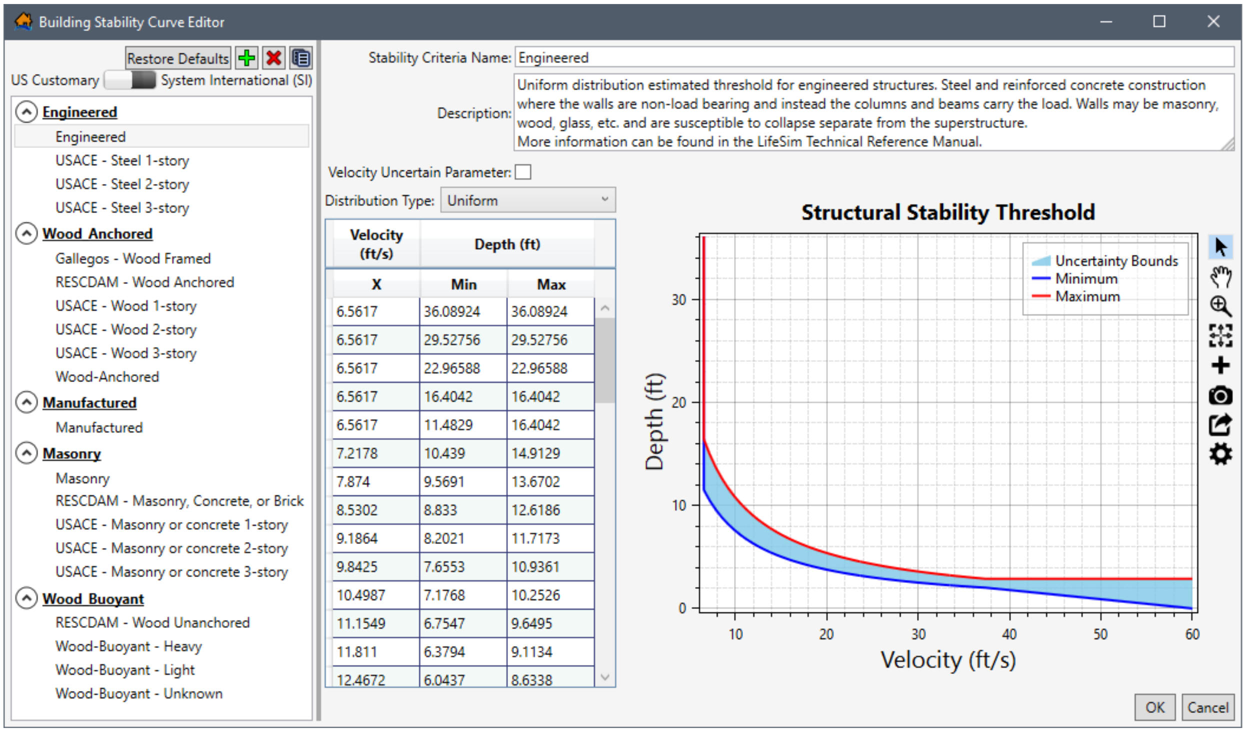


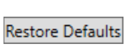
Figure 5-31. Building Stability Curve Editor

Tree

The left side of the **Building Stability Curve Editor** (Figure 5-31) provides a view of the stability criteria grouped by construction type in a tree (**Stability Criteria Tree**). From the **Stability Criteria Tree**, select a stability criteria name to display the data describing the selected stability criteria in the right side of the editor.

Buttons

The **Building Stability Curve Editor** (Figure 5-31) has buttons that allow for creating, deleting, and copying of stability criteria. In addition to the buttons, the **Building Stability Curve Editor** allows the user to set the display units; slide the bar to either **US Customary** or **System International (SI)** (Figure 5-31) to the desired display units. The button functions are listed:



Restore Defaults

The **Restore Defaults** button removes all edits made to the default building stability curves. **NOTE:** Selecting the **Restore Defaults** button also removes all custom created stability criteria thresholds. To restore the default building stability criteria, click the **Restore Defaults** button. The **Clear Existing** message window (Figure 5-32) opens. Click **Yes** to close the **Clear Existing** message window (Figure 5-32) and restore defaults. Click **No** to close the **Clear Existing** message window and load defaults without removing the existing occupancy types. Or Click **Cancel** to close the **Clear Existing** message window without restoring defaults.

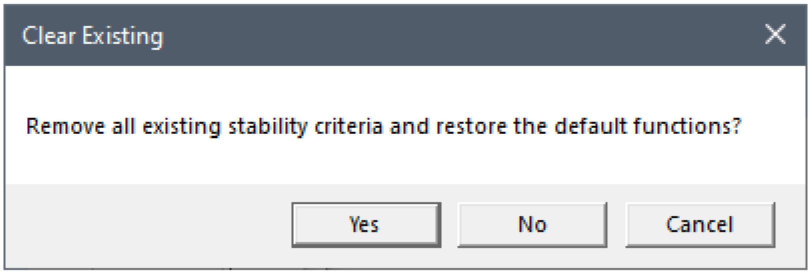


Figure 5-32. Clear Existing Message Window

Add New

Add New allows a user to create a stability criteria. Click the **Add New** button, the **Name of New Stability Criteria Threshold** dialog box (Figure 5-33) opens. In **Name** box, enter a name for the new stability criteria. Click **OK**, the **Name of New Stability Criteria Threshold** dialog box (Figure 5-33) closes. The **Building Stability Curve Editor** (Figure 5-31) displays the new blank stability criteria.

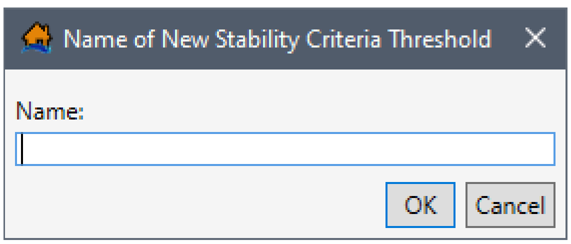


Figure 5-33. Name of New Stability Criteria Threshold Dialog Box

Delete

Delete allows the user to delete selected stability criteria. From the **Building Stability Curve Editor** (Figure 5-31) select a stability criteria from the **Name** list. Click **Delete**, a **Confirm Delete** window (Figure 5-34) opens asking the user for confirmation to delete the selected item. Click **Yes**, to close the **Confirm Delete** message window (Figure 5-34) and delete the selected stability criteria.

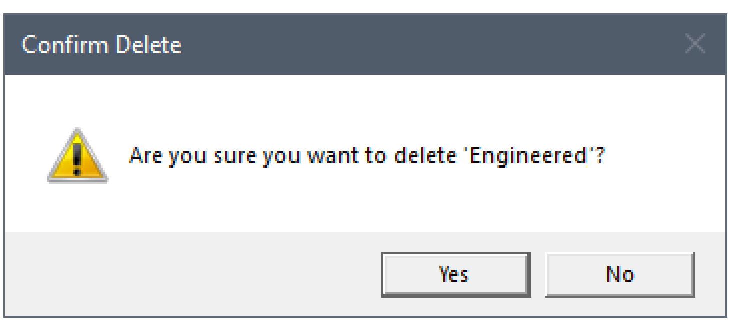


Figure 5-34. Confirm Delete Message Window

Copy

Copy allows the user to copy an existing stability criteria with a new name. From the **Building Stability Curve Editor** (Figure 5-31) select a stability criteria from the **Name** list.

Click **Copy**, the **Name of New Stability Criteria Threshold** dialog box (Figure 5-35) opens. By default, the selected stability criteria name is entered in the **Name** box and appended with "***_Copy***". Keep the default name or enter a new name in the **Name** box. Click **OK**, the **Name of New Stability Criteria Threshold** dialog box (Figure 5-35) closes. The **Building Stability Curve Editor** (Figure 5-31) the displays the copied stability criteria.

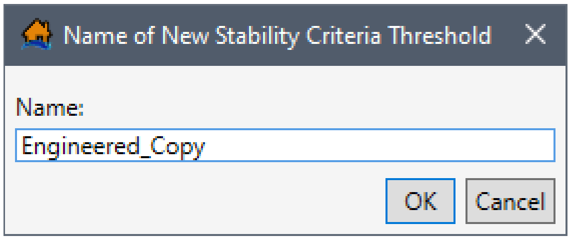


Figure 5-35. Copy – Name of New Stability Criteria Threshold Dialog Box

Collapse the construction type group in the **Stability Criteria Tree**. Click the collapse icon to expand the construction type group and view the list of stability criteria names. Note, clicking on the expanded header collapses the group or vice versa collapses an expanded group.

Expand the construction type group in the **Stability Criteria Tree**. Click the expand icon to collapse the construction type group. Note, clicking on the expanded header collapses the group or vice versa collapses an expanded group.

5.5.2 Create or Edit a Building Stability Criteria

To create a new stability criteria:

1. To open the **Building Stability Curve Editor** (Figure 5-31), from the LifeSim main window (Figure 5-1), from the **Study Tree**, under **Structure Inventories** right-click on **Building Stability Criteria** and from the shortcut menu click **Building Stability Criteria Data** (Figure 5-36). The **Building Stability Curve Editor** (Figure 5-31) opens.

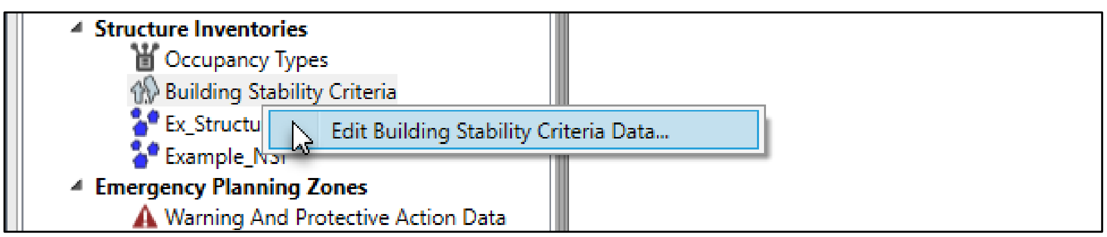


Figure 5-36. LifeSim Study Tree – Structure Inventories – Edit Building Stability Criteria Data Shortcut Command

2. From the **Building Stability Curve Editor** (Figure 5-31), click the **Add New** button. The **Name of New Stability Criteria Threshold** dialog box (Figure 5-33) opens.

3. Enter a name (required) in the **Name** box for the new stability criteria. Click **OK**, the **Name of New Stability Criteria Threshold** dialog box (Figure 5-33) closes. The **Building Stability Curve Editor** (Figure 5-37) displays the new blank stability criteria.

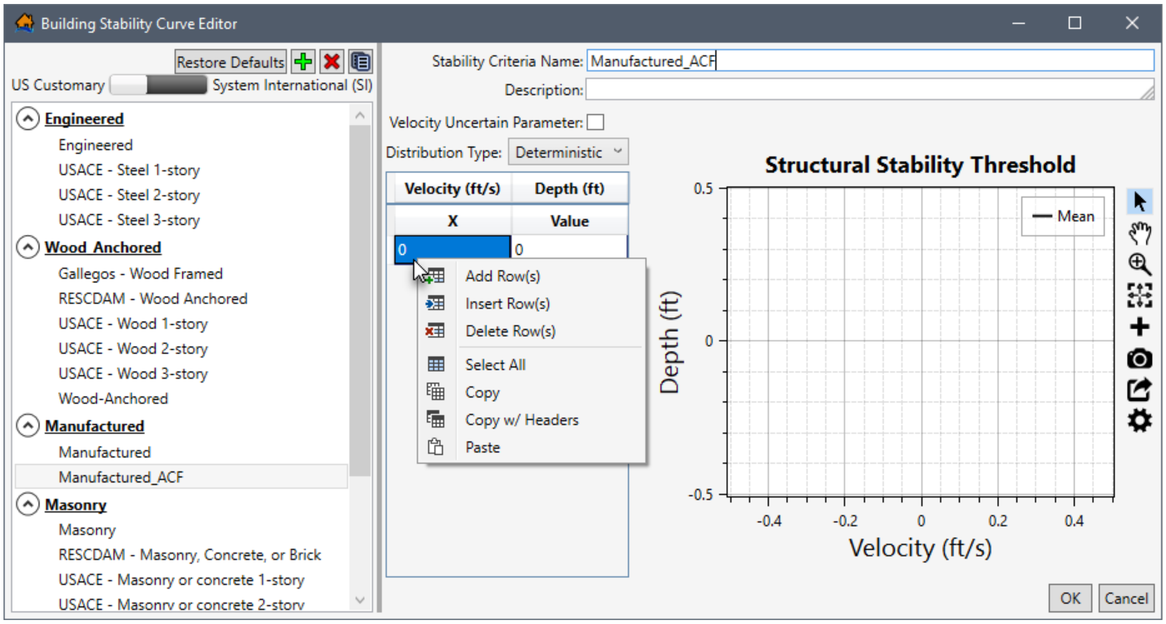


Figure 5-37. Building Stability Curve Editor – New Structure Criteria

An existing stability criteria can be edited or a newly created criteria can be defined following the remaining instructions in this section.

4. Enter a description (optional but recommended) in the **Description** field (Figure 5-37).

5. Next, enter the structural stability threshold curve in the table on the **Building Stability Curve Editor** (Figure 5-35). If adding uncertainty for the depth (ft) parameter, uncheck the ☐ checkbox to disable the **Velocity Uncertain Parameter** (Figure 5-37), by default the checkbox is unchecked.

   Alternatively, add uncertainty for the velocity parameter, check the ☑ checkbox to enable the **Velocity Uncertain Parameter** (Figure 5-38). Note, Depth (ft) is now the x-variable and Velocity (ft/s) is now the y-variable.

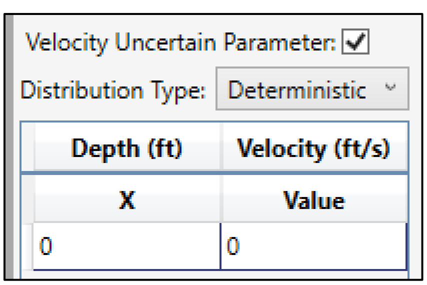


Figure 5-38. Velocity Uncertain Parameter Enabled

6. From the **Distribution Type** dropdown menu (Figure 5-37), select the desired distribution type for the y-variable. By default, **Deterministic** (Figure 5-38) is selected. The distribution types available are:

   - **Normal** – mean (μ), standard deviation (σ).

- **Truncated Normal** – mean (μ), standard deviation (σ), minimum, maximum.
- **Log-Normal** – mean (of log, μ), standard deviation (of log, σ).
- **Triangular** – minimum (a), most likely (c), maximum (b).
- **Uniform** – minimum, maximum.

7. The threshold depth – velocity curve can be entered into the **Structural Stability Threshold Table** (Figure 5-37) by hand, copied from another stability criteria or from another application (e.g., Microsoft Excel®) and pasted into the table (i.e., entire table, individual cells, columns, rows). From the **Structural Stability Threshold Table**, right-click to open the shortcut menu (Figure 5-37), from the shortcut menu the user can add, insert, and delete row(s); select all; copy or copy with headers; and paste.

   For example, from another stability criteria select a cell in the table and use **Ctrl + A** (or right-click and select the **Select All** command from the shortcut menu), to select the entire table. Use **Ctrl + C** (or the **Copy** shortcut menu command) to copy the data to the clipboard. Return to the newly created stability criteria and select the first cell in the first row and use **Ctrl + V** to paste all the copied data.

   The stability threshold table can also be copied (with or without headers) into another application (e.g., Microsoft Excel®) for saving. Use **Ctrl + A** (or the **Select All** command from the shortcut menu), right-click and select **Copy w/ Headers**. Then paste the data in the desired application to save.

   Errors in the stability threshold table are highlighted in red and a tooltip displays a message describing the error (Figure 5-28). Right-click on a header row to open the shortcut menu (Figure 5-28) to sort the table in ascending or descending order, or clear the table sorting.

   The **Structural Stability Threshold** plot (Figure 5-31) provides a live update for the data entered in the table. Review Section 3.6 for instructions on using the general plotting tools (located on the right side of the plot window) and for customizing plots.

8. Click **OK**, the **Building Stability Curve Editor** (Figure 5-31) closes and saves all edits to building stability curves.

5.5.3 Delete a Stability Criteria

To delete an existing stability criteria, from the **Building Stability Curve Editor** (Figure 5-31), select a stability criteria from the **Stability Criteria Name** list. Click the **Delete** button. A **Confirm Delete** window (Figure 5-34) opens asking for confirmation to delete the selected item. Click **Yes**, the **Confirm Delete** message window closes (Figure 5-34), and the selected stability criteria is deleted.

5.5.4 Copy a Stability Criteria

To copy an existing stability criteria, from the **Building Stability Curve Editor** (Figure 5-31), select a stability criteria from the **Name** list. Click **Copy**, the **Name of New Stability Criteria**

Threshold dialog box (Figure 5-35) opens. By default, the selected stability criteria name is entered in the **Name** box and appended with "***_Copy***". Keep the default name or enter a new name in the **Name** box. Click **OK**, the **Name of New Stability Criteria Threshold** dialog box (Figure 5-35) closes. The **Building Stability Curve Editor** (Figure 5-31) displays the copied stability criteria.

CHAPTER 6

Emergency Planning Data

The next step in building an LifeSim study is to add **emergency planning data** which represents an area (or areas) with potentially unique overall evacuation effectiveness such as evacuation preparedness, community awareness, and the types of flood warning systems available. Refer to Sorensen et. al. (2014a; 2014b; and 2014c) for examples of how to define an emergency planning zone (EPZ) or area (Appendix C). The effectiveness of the warning systems available and the anticipated community responses is captured in community diffusion curves for first warning alert and protective action initiation (sometimes called mobilization).

Central to the evacuation process in LifeSim are the first alert and Protective Action Initiation (PAI) community diffusion curves (Figure 6-1). The first alert curves represent the percentage of the population that receives a notification of the warning over time (e.g., fifty percent of the population will receive a warning alert within forty minutes of warning issuance). First alert curves are entered for daytime and night-time scenarios because the speed at which a first alert can be received can vary significantly between daytime and night-time. The PAI curves represent the percent of population that takes protective action over time. For example, fifty percent of the population will have taken protective action within eight minutes of a first alert (Sorensen et. al., 2014c; Appendix C). The diffusion curves can be entered with uncertainty and LifeSim performs curve sampling on an iteration basis in the Monte Carlo simulation.

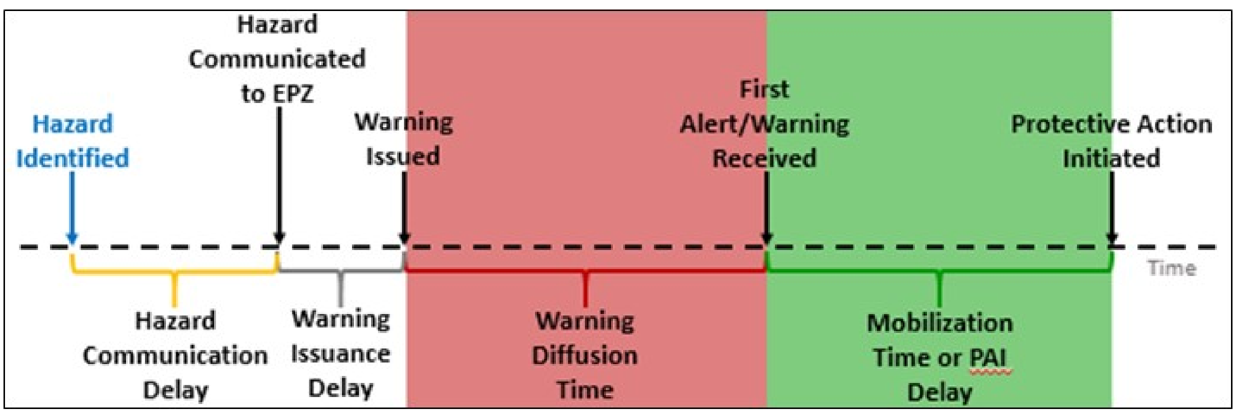


Figure 6-1. Conceptual diagram illustrating the flood warning and evacuation timeline. Highlighted sections relevant to emergency planning zone input data are, warning diffusion time from warning issuance (light red) and protective action initiation delay from first alert (light green)

6.1 Warning And Protective Action Data Editor

The **Warning And Protective Action Data Editor** (Figure 6-2) allows users to edit pre-set warning and evacuation parameters that were developed using the findings of Mileti and Sorensen (Appendix C). Users are able to adjust these parameters or create new warning and evacuation parameters to better reflect the specific situation in the study area. Modifications made in the **Warning And Protective Action Data Editor** (Figure 6-2) is available for all emergency planning zone datasets in the **Emergency Planning Zone Editor** (discussed in Section 6.2).

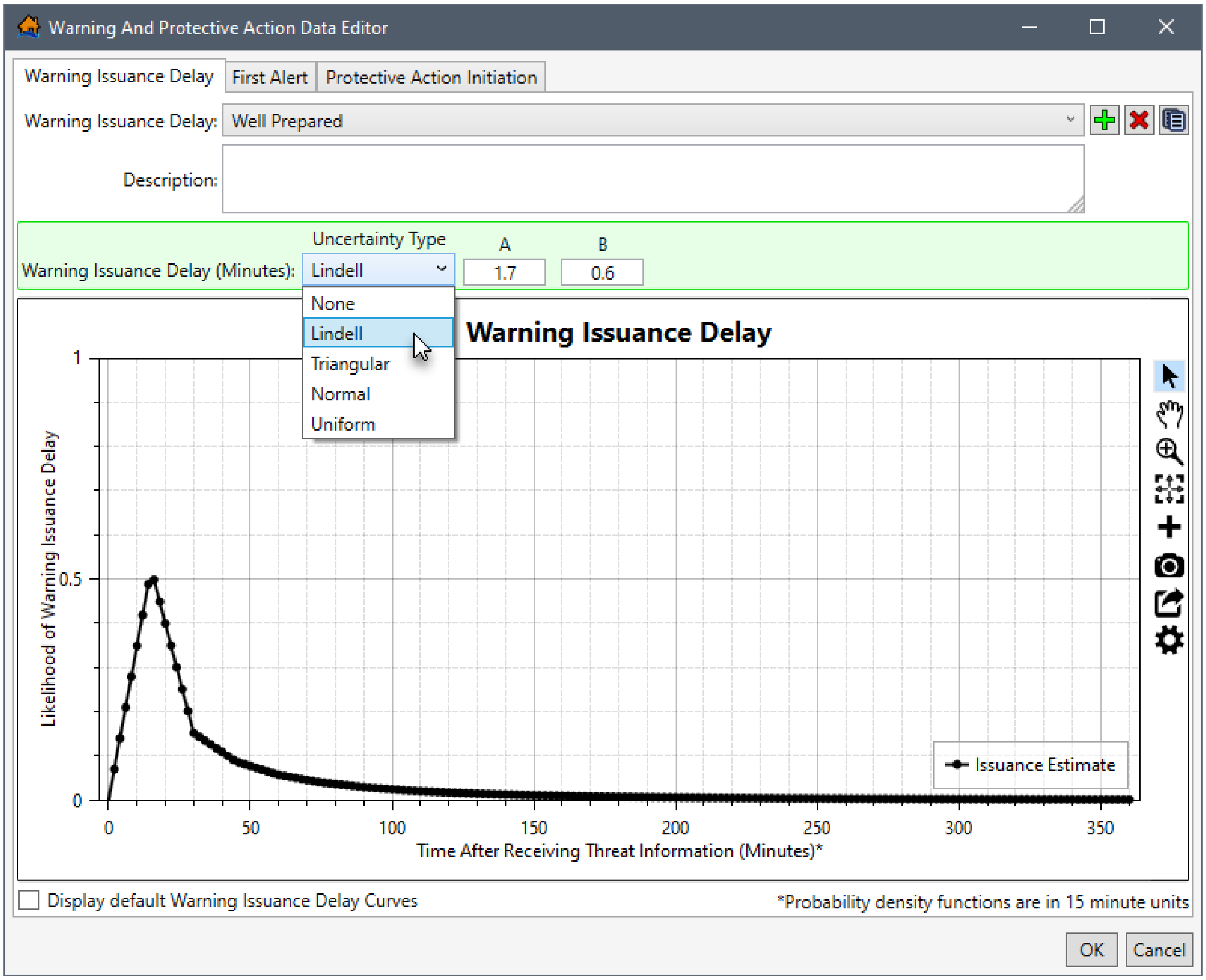


Figure 6-2. Warning And Protective Action Data Editor – Warning Issuance Delay Tab

6.1.1 Overview

Open the **Warning And Protective Action Data Editor** (Figure 6-2), from the **Study Tree**, under **Emergency Planning Zones**, right-click **Warning And Protective Action Data** and from the shortcut menu, click **Edit Warning and PAI Data** (Figure 6-3).

The **Warning And Protective Action Data Editor** (Figure 6-2) contains three tabs: **Warning Issuance Delay**, **First Alert**, and **Protective Action Initiation**; each tab is discussed in Sections 6.1.2, 6.1.3, and 6.1.4; respectfully. The tabs contain the same buttons for adding, deleting, and copying existing warning and evacuation parameters and plot windows for viewing the warning and protective action curves. Review Section 3.6 for instructions on using the general plotting tools (located on the right side of the plot window) and for customizing the plots.

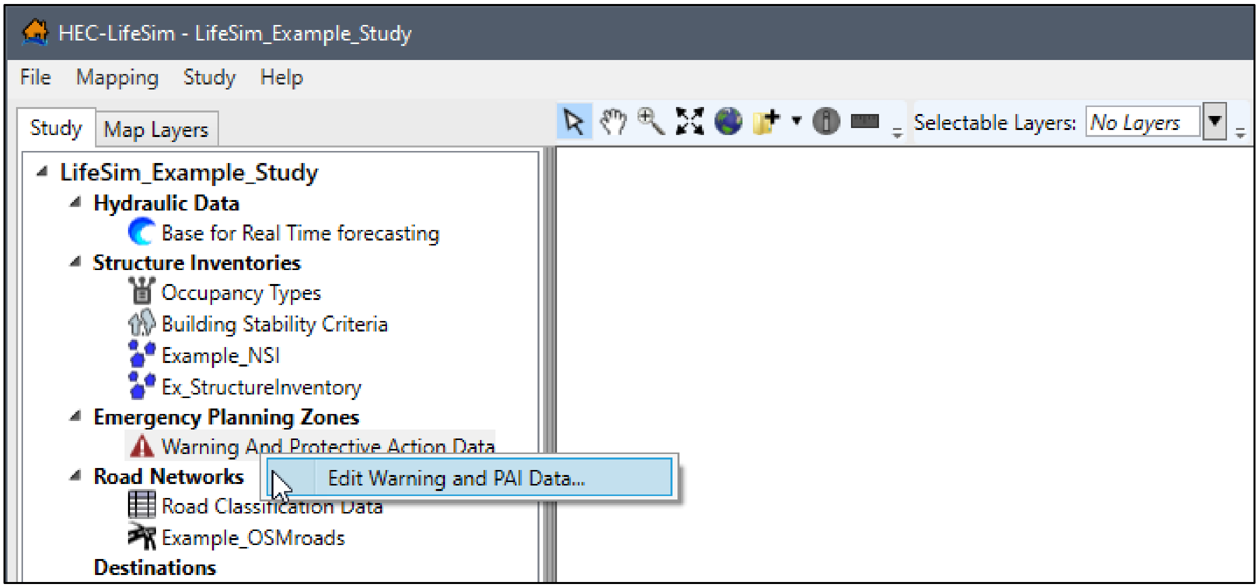


Figure 6-3. LifeSim Study Tree – Emergency Planning Zones – Edit Warning and PAI Data Shortcut Command

Buttons

Add New

Add New allows a user to create a warning issuance delay, first alert relationship, or new protective action initiation relationship. For example, click the **Add New** button from the **First Alert** tab (Figure 6-2) opens the **Name of New First Alert Relationship** dialog box (Figure 6-4). In **Name** box, enter a name for the new warning issuance delay, first alert relationship, or new protective action initiation relationship. Click **OK**, the **Name of New First Alert Relationship** dialog box (Figure 6-4) closes and the new warning issuance delay, first alert relationship, or new protective action initiation relationship is created.

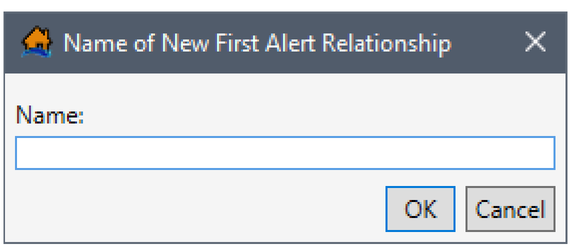


Figure 6-4. Name of New First Alert Relationship Dialog Box

Delete

Delete allows the user to delete selected warning issuance delay, first alert relationship, or new protective action initiation relationship. From the **Warning And Protective Action Data Editor** (Figure 6-2) select a warning issuance delay, first alert relationship, or new protective action initiation relationship from the dropdown menu (e.g., *Well Prepared* in Figure 6-2). Click **Delete**, a **Confirm Delete** window (like Figure 5-34) opens asking the user for confirmation to delete the selected item.

Click **Yes**, to close the **Confirm Delete** message window (Figure 5-34) and delete the selected warning issuance delay, first alert relationship, or new protective action initiation relationship.

Copy
Copy allows the user to copy a warning issuance delay, first alert relationship, or new protective action initiation relationship with a new name. From the **Warning And Protective Action Data Editor** (Figure 6-2) select a warning issuance delay, first alert relationship, or new protective action initiation relationship from the dropdown menu (e.g., *Well Prepared* in Figure 6-2).

For example, from the **Warning Issuance Delay** tab click **Copy**, the **Name of New Warning Issuance Delay** dialog box (Figure 6-5) opens. By default, the selected warning and evacuation parameters name is entered in the **Name** box and appended with "***_Copy***". Keep the default name or enter a new name in the **Name** box. Click **OK**, the **Name of New Warning Issuance Delay** dialog box (Figure 6-5) closes and the new copied warning issuance delay, first alert relationship, or new protective action initiation relationship is created.

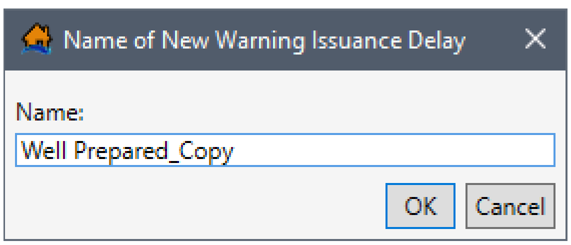


Figure 6-5. Copy – Name of New Warning Issuance Delay Dialog Box

6.1.2 Warning Issuance Delay

The **Warning Issuance Delay** tab (Figure 6-2) allows the user to create a new or edit an existing pre-set warning issuance delay. The warning issuance delay is the time it takes from when the emergency managers receive the notification of the imminent hazard, or imminent hazard ID time (Section 11.1.2), to when an evacuation order is issued to the public.

Create a New Warning Issuance Delay

To create a new warning issuance delay:

1. Click the **Add New** button (Figure 6-2). The **Name of New Warning Issuance Delay** dialog box (similar to Figure 6-4) opens.

2. In **Name** box, enter a name for the new warning issuance delay. Click **OK**, the **Name of New Warning Issuance Delay** dialog box (similar to Figure 6-4) closes and the new warning issuance delay is created.

3. From the **Warning Issuance Delay (Minutes)** panel, select the desired uncertainty type from the **Uncertainty Type** (Figure 6-2) dropdown menu and define the amount of time

(in minutes) it could take for the emergency managers to issue the warning to the public. The uncertainty options include:

- **None** – (most likely),
- **Lindell** – (A, B),
- **Triangular** – (minimum, most likely, maximum),
- **Normal** – (most likely, standard deviation), and
- **Uniform** – (minimum, maximum).

For example, if the emergency manager for an EPZ took anywhere between 15 to 30 minutes to verify the hazard and initiate a public evacuation order, then the user would enter a warning issuance delay function with a uniform distribution (Figure 6-6).

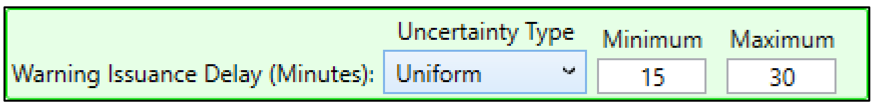


Figure 6-6. Warning Issuance Delay – Uniform Uncertainty Type Example

4. Click **OK** to save the new warning issuance delay and close the **Warning And Protective Action Data Editor** (Figure 6-2).

Edit a Warning Issuance Delay

To edit an existing warning issuance delay, select the desired warning issuance delay name, from the **Warning Issuance Delay** dropdown menu (e.g., *Well Prepared* in Figure 6-2). From the **Warning Issuance Delay (Minutes)** panel, select the desired uncertainty type from the **Uncertainty Type** (Figure 6-2) dropdown menu and define the amount of time (in minutes) it could take for the emergency managers to issue the warning to the public.

To view all of the default warning issuance delay curves, check the ☑ checkbox to **Display default Warning Issuance Delay Curves**; all the default curves will display in the **Warning Issuance Delay** plot window. Click **OK** to save the new warning issuance delay and close the **Warning And Protective Action Data Editor** (Figure 6-2).

6.1.3 First Alert Curves

From the **Warning And Protective Action Data Editor** (Figure 6-2), select the **First Alert** tab (Figure 6-7). The first alert curves, alternatively known as warning diffusion (Figure 6-1), represent the percentage of the population that receives a notification of the warning over time (e.g., fifty percent of the population will receive a warning alert within forty minutes of warning issuance). First alert curves are entered for daytime (from the **Daytime First Alert** tab) and night-time (from the **Night-time First Alert** tab) scenarios because the speed at which a first alert can be received can vary significantly between daytime and night-time.

Daytime First Alert

The **Daytime First Alert** tab (Figure 6-7) allows the user to set a unique (for multiple zones, when present) warning system with uncertainty to define the diffusion curve. The diffusion curve represents the percentage of the population which will receive a first alert warning over time during daytime hours from when the warning was issued and can be described with uncertainty (e.g., ***Triangular*** in Figure 6-7).

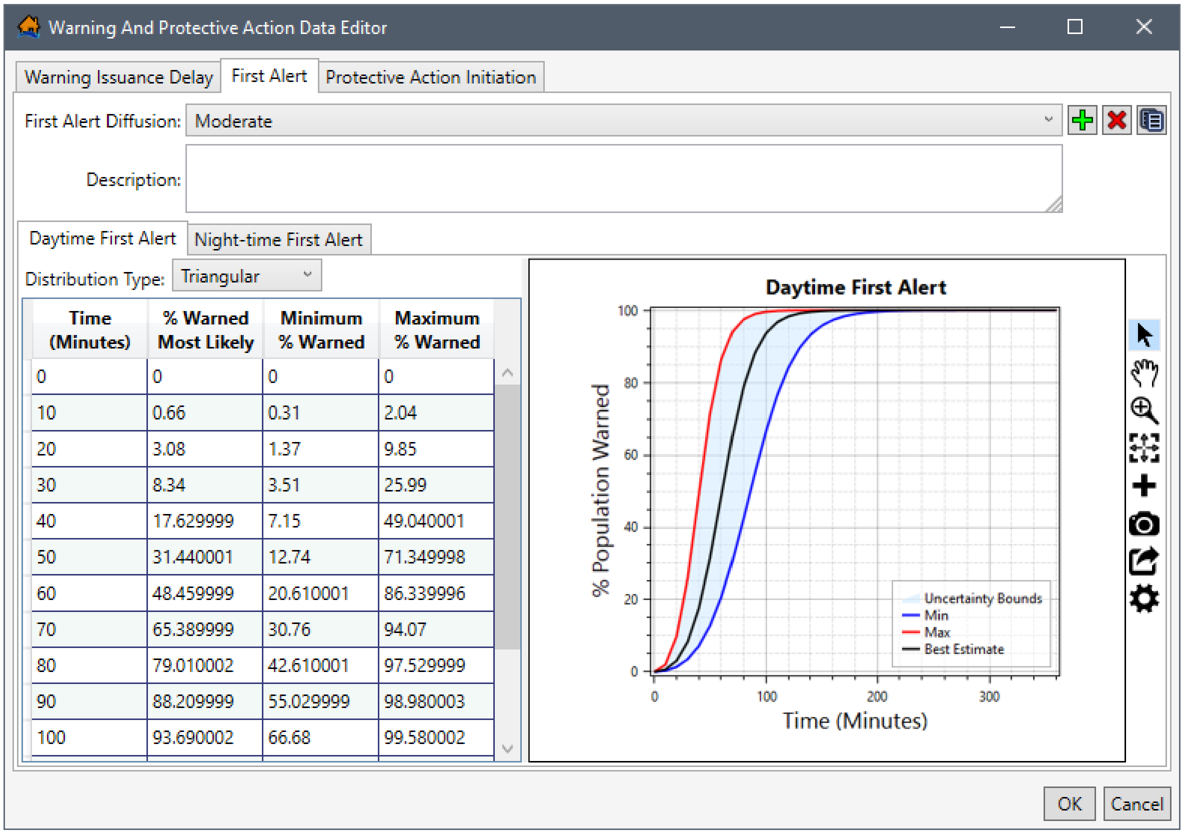


Figure 6-7. Warning And Protective Action Data Editor – First Alert Tab – Daytime First Alert

Night-time First Alert

Night-time First Alert tab (Figure 6-8) allows the user to set a unique (for multiple zones, when present) warning system with uncertainty to define the diffusion curve. The diffusion curve represents the percentage of the population which will receive a first alert warning over time during night-time hours from when the warning was issued and can be described with uncertainty. The default curves take into account the challenges with sending out a warning at night (Figure 6-8).

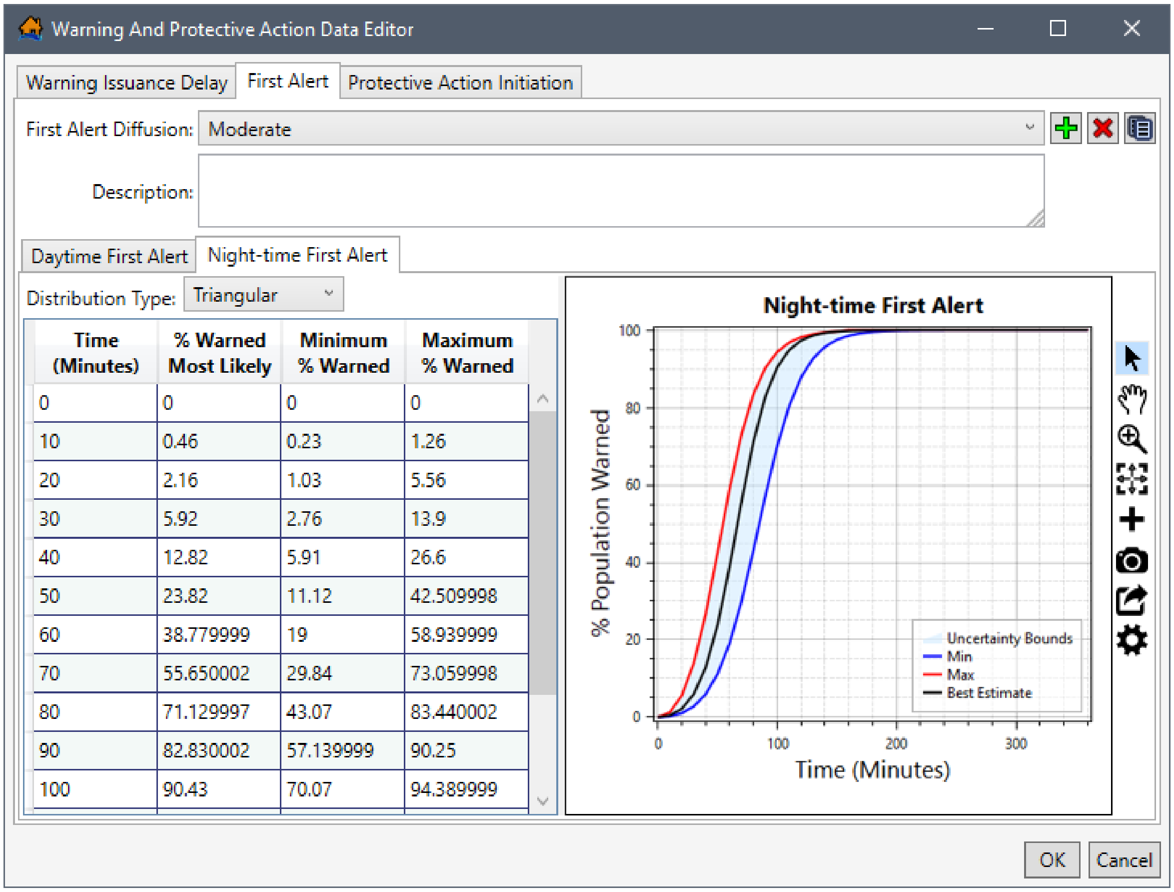


Figure 6-8. Warning And Protective Action Data Editor – First Alert Tab – Night-time First Alert

Create New First Alert Diffusion Curves

To create a new set of first alert diffusion curves:

1. From the **First Alert** tab (Figure 6-7), click the **Add New** ✚ button (Figure 6-2). The **Name of New First Alert Relationship** dialog box (Figure 6-4) opens.

2. In **Name** box, enter a name for the new first alert relationship. Click **OK**, the **Name of New First Alert Relationship** dialog box (Figure 6-4) closes and the new first alert relationship is created.

3. From the **Daytime First Alert** tab (Figure 6-7), select the desired distribution from the **Distribution Type** (Figure 6-7) dropdown menu. The distribution options include:

   - **None** – (percent warned),
   - **Triangular** – (most likely, minimum, maximum),
   - **Uniform** – (minimum, maximum),
   - **Normal** – (average, standard deviation), and
   - **Lognormal** – (average, standard deviation).

4. Define the daytime diffusion curve by entering the time (minutes) and percent warned in the **Daytime First Alert Table** (Figure 6-7). Diffusion curve values can be entered by hand, copied from another application (e.g., Microsoft Excel®) and pasted into the table (i.e., entire table, individual cells, columns, rows). From the **Daytime First Alert Table** (Figure 6-7), right-click to open the shortcut menu (Figure 5-27), from the shortcut menu the user can add, insert, and delete row(s); select all; copy or copy with headers; and paste.

   The daytime first alert table can also be copied (with or without headers) into another application (e.g., Microsoft Excel®) for saving. Use **Ctrl + A** (or the **Select All** command from the shortcut menu), right-click and select **Copy w/ Headers**. Then paste the data in the desired application to save.

   Errors in the daytime diffusion curve are highlighted in red in the table and a tooltip displays a message describing the error (Figure 5-28). Right-click on a header row to open the shortcut menu (Figure 5-28) to sort the table in ascending or descending order, or clear the table sorting.

5. Next, from the **Night-time First Alert** tab (Figure 6-8), select the desired distribution from the **Distribution Type** (Figure 6-8) dropdown menu. The distribution options are the same as in the **Daytime First Alert** tab.

6. Define the night-time diffusion curve by entering the time (minutes) and percent warned in the **Night-time First Alert Table** (Figure 6-8).

7. Click **OK** to save the new first alert diffusion curves and close the **Warning And Protective Action Data Editor** (Figure 6-8).

Edit First Alert Diffusion Curves

Diffusion curves can be edited for both the daytime and night-time first alert curves, or individually modified. To edit existing first alert diffusion curves, select the desired diffusion curve set, from the **First Alert Diffusion** dropdown menu (e.g., *Moderate* in Figure 6-8). Select the appropriate tab (**Daytime First Alert** tab or **Night-time First Alert** tab), either begin editing the values in the table or select the desired distribution from the **Distribution Type** dropdown menu and define the first alert diffusion curve. Click **OK** to save edits and close the **Warning And Protective Action Data Editor** (Figure 6-8).

6.1.4 Protective Action Initiation (PAI)

The **Protective Action Initiation** tab (Figure 6-9) allows the user to define new or edit existing mobilization protective action initiation (PAI) delay diffusion curve with or without uncertainty. The PAI delay represents the percentage of the population which will take protective action over time from when the first alert is received (Figure 6-1). During the PAI delay, people are confirming the severity of the situation, packing, notifying others, protecting property, gathering family and pets, etc.

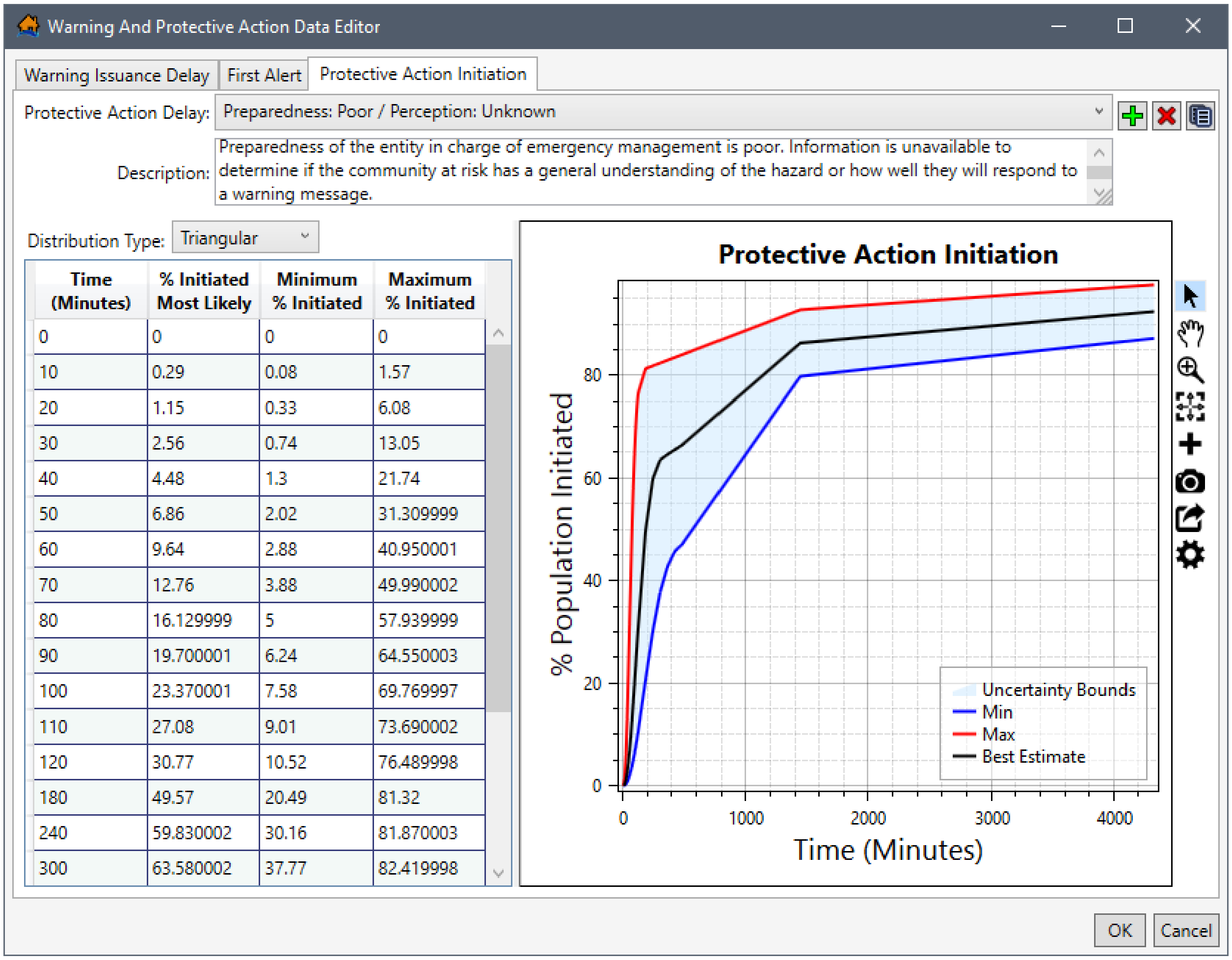


Figure 6-9. Warning And Protective Action Data Editor – Protective Action Initiation Tab

LifeSim provides nine default PAI delay curves; the curves are categorized based on a combination of three emergency management agency preparedness categories and three categories defining the public perception of the level of risk. The emergency management agency preparedness categories are:

- **Good** – The emergency management agency is well-prepared to issue a well-crafted and convincing message to the public that motivates the population receiving the message to act.

- **Poor** – The emergency management agency is not well-prepared to issue an effective message to the public.

- **Unknown** – This is an average condition between good and poor is assumed with the full range of uncertainty of both categories.

The public perception of the level of risk is generally a function of the individual or group's distance from the hazard (e.g., dam or river). Other factors that can influence public perception is the frequency and severity of recent events and extremely effective and persistent public awareness campaigns.

The public perception of the level of risk categories are:

- **Unlikely to impact** – The community at risk has a general perception that the warning will not affect them. When considering distance to the hazard location, this category could be used when the population is greater than 0.25 miles from the river or greater than 0.5 miles from the dam.

- **Likely to impact** – The community at risk has a general perception that the event will likely affect them. When considering distance to the hazard location, this category could be used when the population is less than 0.25 miles from the river or less than 0.5 miles from the dam.

- **Unknown** – This is an average condition between unlikely to impact and likely to impact is assumed with the full range of uncertainty of both categories.

Create New Protective Action Initiation

To create a new protective action initiation relationship:

1. From **Protective Action Initiation** tab (Figure 6-9), click the **Add New** button. The **Name of New Protective Action Initiation** dialog box (Figure 6-10) opens.

2. In **Name** box, enter a name for the new protective action initiation relationship. Click **OK**, the **Name of New Protective Action Initiation** dialog box (Figure 6-10) closes and the new PAI relationship is created.

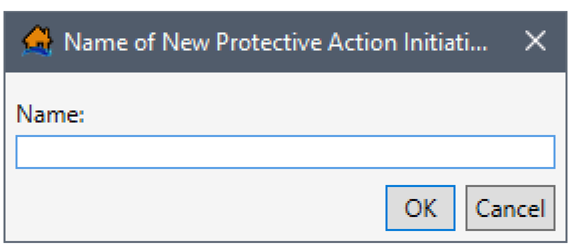


Figure 6-10. Name of New Protective Action Initiation Dialog Box

3. From the **Protective Action Initiation** tab (Figure 6-9), select the desired distribution from the **Distribution Type** (Figure 6-9) dropdown menu. The distribution options include:

   - **None** – (percent initiated),
   - **Triangular** – (most likely, minimum, maximum),
   - **Uniform** – (minimum, maximum),
   - **Normal** – (average, standard deviation), and
   - **Lognormal** – (average, standard deviation).

4. Define the daytime diffusion curve by entering the time (minutes) and percent initiated to take action in the **PAI Table** (Figure 6-9). Diffusion curve values can be entered by hand, copied from another application (e.g., Microsoft Excel®) and pasted into the table (i.e.,

entire table, individual cells, columns, rows). From the **PAI Table** (Figure 6-9), right-click to open the shortcut menu (Figure 5-27), from the shortcut menu the user can add, insert, and delete row(s); select all; copy or copy with headers; and paste.

The PAI table can also be copied (with or without headers) into another application (e.g., Microsoft Excel®) for saving. Use **Ctrl + A** (or the **Select All** command from the shortcut menu), right-click and select **Copy w/ Headers**. Then paste the data in the desired application to save.

Errors in the PAI diffusion curve are highlighted in red in the table and a tooltip displays a message describing the error (Figure 5-28). Right-click on a header row to open the shortcut menu (Figure 5-28) to sort the table in ascending or descending order, or clear the table sorting.

5. Click **OK** to save the new PAI diffusion curve and close the **Warning And Protective Action Data Editor** (Figure 6-9).

Edit Protective Action Initiation

To edit an existing PAI diffusion curve, select the desired protective action delay relationship, from the **Protective Action Delay** dropdown menu (e.g., *Preparedness: Poor / Perception: Unknown* in Figure 6-9). Either begin editing the values in the PAI table or select the desired distribution from the **Distribution Type** dropdown menu (Figure 6-9) and define the PAI diffusion curve. Click **OK** to save edits and close the **Warning And Protective Action Data Editor** (Figure 6-9).

6.2 Emergency Planning Zone (EPZ) Editor

The **Emergency Planning Zone Editor** (Figure 6-11) allows the user to import an emergency planning zones (EPZ) dataset and enter information defining separate warning and evacuation parameters per EPZ. In most cases, separate emergency management agency areas within the study area are modeled as unique emergency planning zones since the warning issuance efficiency, message content, and public response can vary (Figure 6-1).

Users import the emergency planning zone dataset from a polygon shapefile where each polygon feature in the shapefile represents an emergency planning zone. Multiple EPZ datasets can be imported and defined. The warning and evacuation parameters cannot be modified in the **Emergency Planning Zone Editor** (Figure 6-11); however, selecting the **Edit Data** button opens the **Warning And Protective Action Data Editor** (Figure 6-2) which allows users to edit and define warning and protective action parameters.

Note, the **Emergency Planning Zone Editor** (Figure 6-11) opens when editing imported EPZ datasets (refer to Section 6.2.3), but the **Import Emergency Planning Zone** editor (Figure 6-14) opens when importing a new EPZ dataset (refer to Section 6.2.2).

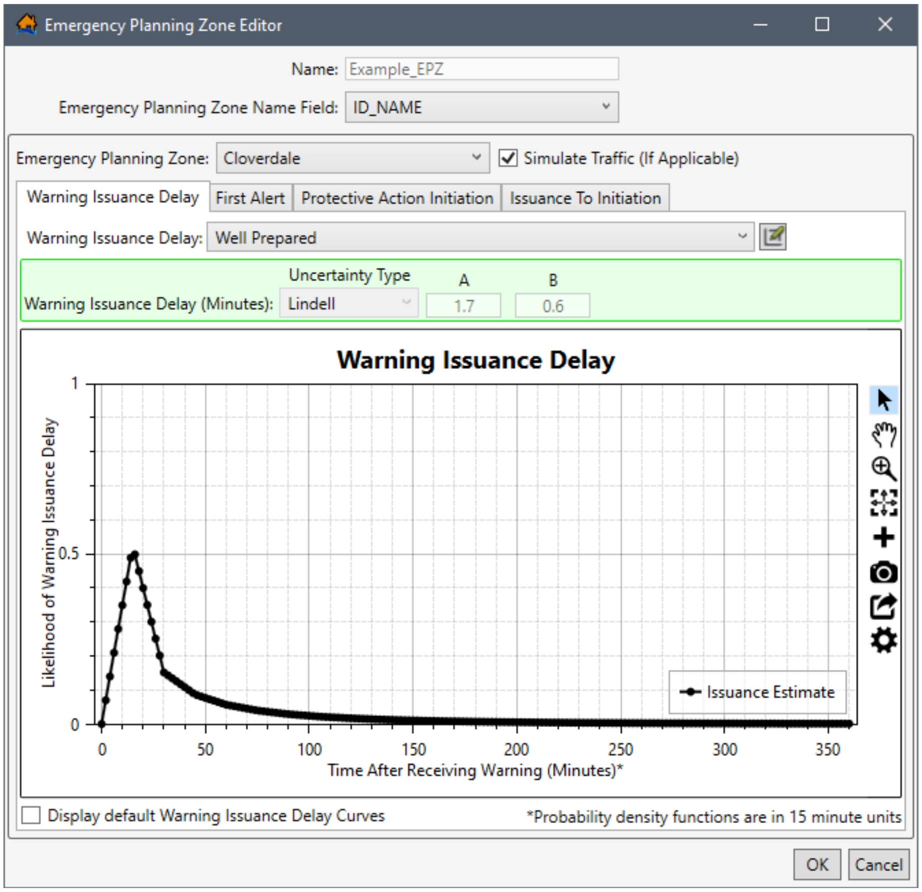


Figure 6-11. Example – Emergency Planning Zone Editor

6.2.1 EPZ Editor Overview

The tabs on the **Emergency Planning Zone Editor** (Figure 6-11) represent the warning delay and the first alert and PAI diffusion curves that need to be created for the emergency planning zones. The **Emergency Planning Zone Editor** (Figure 6-11) contains four tabs: **Warning Issuance Delay**, **First Alert**, **Protective Action Initiation**, and **Issuance To Initiation**. The first three tabs contain the same buttons for editing the warning and evacuation parameters. The **Edit Data** button opens the **Warning And Protective Action Data Editor** (Figure 6-2). All four tabs contain the same general plotting tools (located on the right side of the plot window) for customizing the plots (review Section 3.6 for more information).

6.2.2 Create and Define an EPZ Dataset

In most cases, separate emergency management agency areas within the study area are modeled as unique emergency planning zones since the warning issuance efficiency, message content, and public response can vary.

Import EPZ from Polygon Shapefile

To create emergency planning zones from a point shapefile:

1. From the LifeSim main window, from the **Study** tab, from the **Study Tree**, right-click on **Emergency Planning Zones**, from the shortcut menu click **Import Emergency Planning Zones From Shapefile** (Figure 6-12). Another way from the LifeSim main window is from the **Study** menu, point to **Emergency Planning Data**, click **Import From Shapefile** (Figure 6-13).

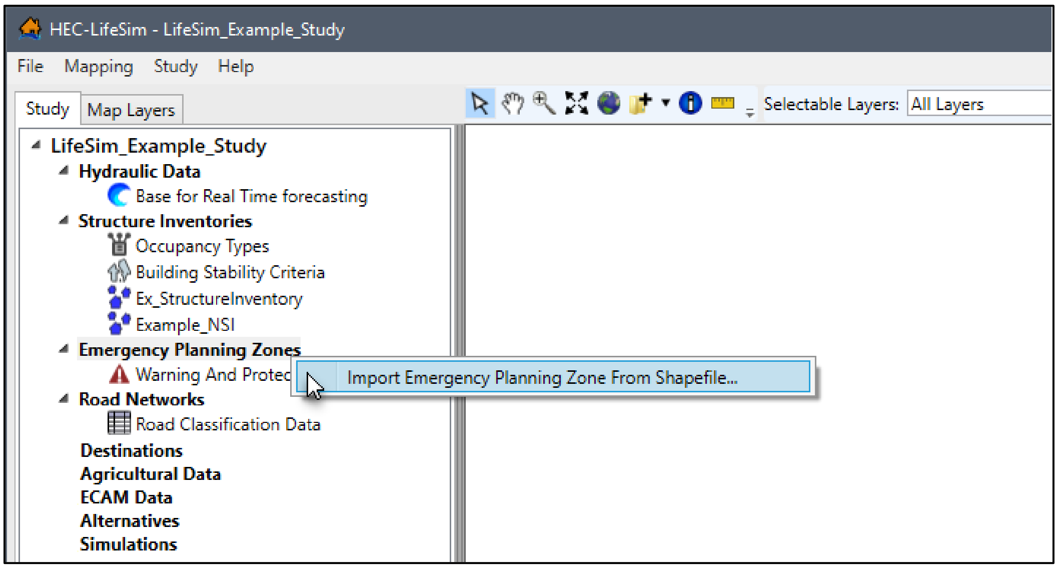


Figure 6-12. LifeSim Study Tree – Emergency Planning Zones – Import Emergency Planning Zone From Shapefile Shortcut Command

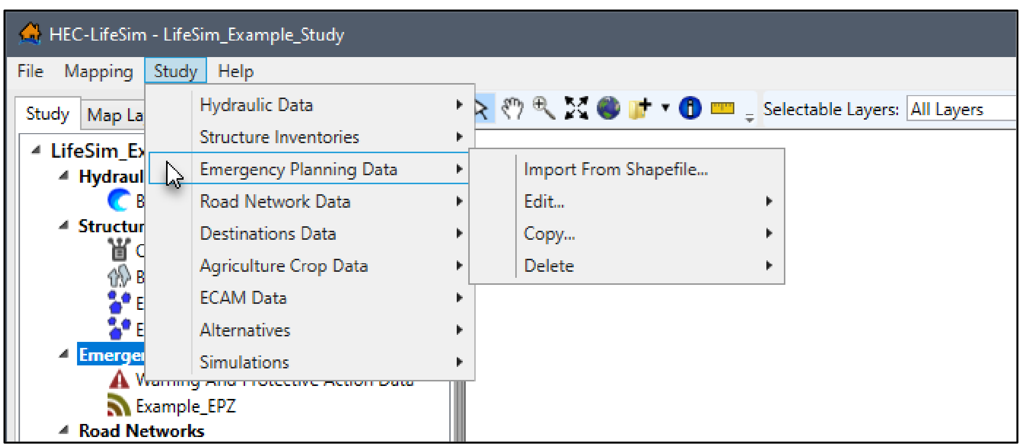


Figure 6-13. LifeSim Main Window – Study Menu – Emergency Planning Zones Sub-menu

2. Either way, the **Import Emergency Planning Zone** editor (Figure 6-14) opens. Enter a name for the emergency planning zone in the **Name** box (Figure 6-14).

3. To select the shapefile that contains information about the emergency planning zone, to the right of the **Emergency Planning Zone Polygon Shapefile** dropdown menu (Figure 6-14), click ... . An **Open** browser (Figure 6-15) opens, navigate to the directory where the polygon shapefile is located. Select the correct polygon shapefile (e.g., **.shp*) and click **Open**; the **Open** browser (Figure 6-15) closes and the user will be returned to the **Import Emergency Planning Zone** dialog box (Figure 6-14). Alternatively, if the

desired shapefile has already been imported as a map layer into the LifeSim study, then from the **Emergency Planning Zone Polygon Shapefile** dropdown menu (Figure 6-14), select the desired map layer.

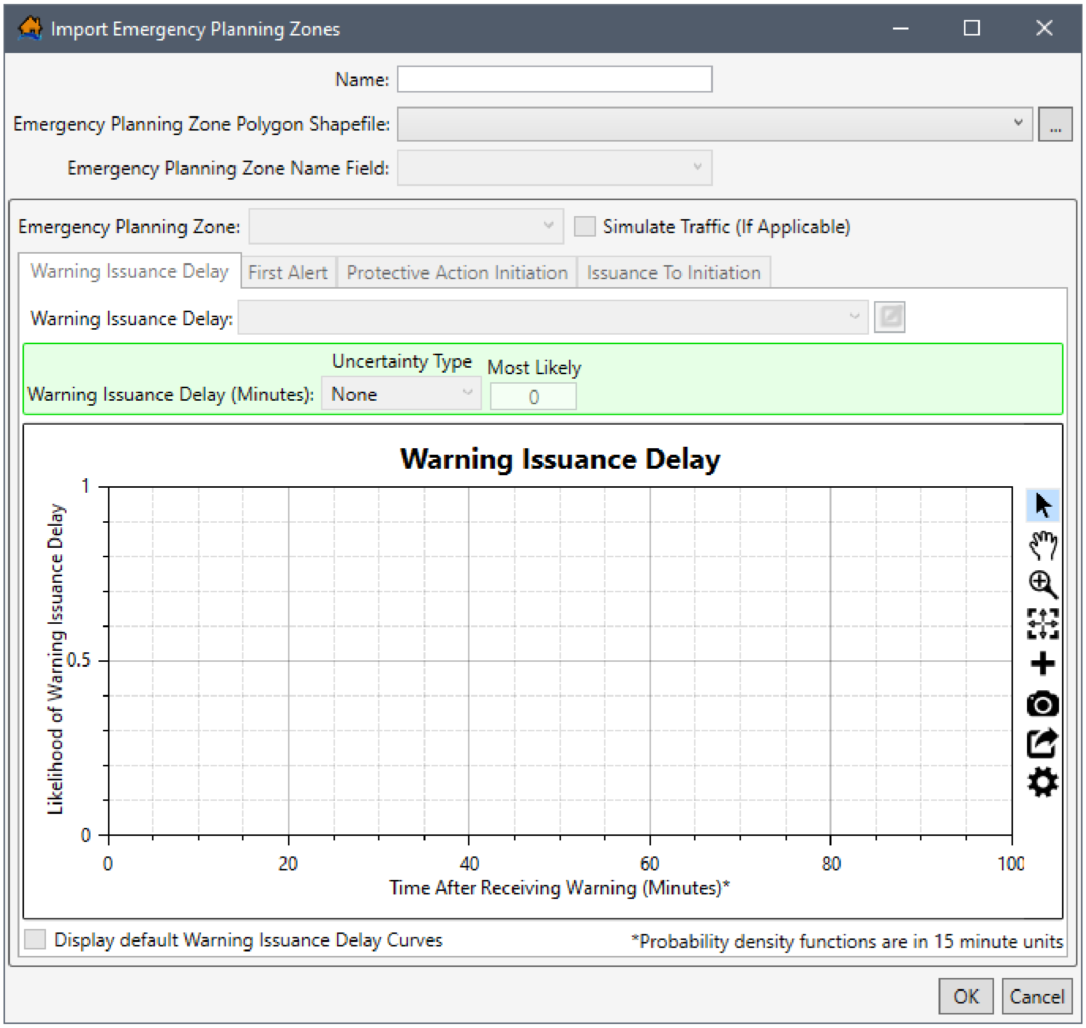


Figure 6-14. Import Emergency Planning Zone Dialog Box

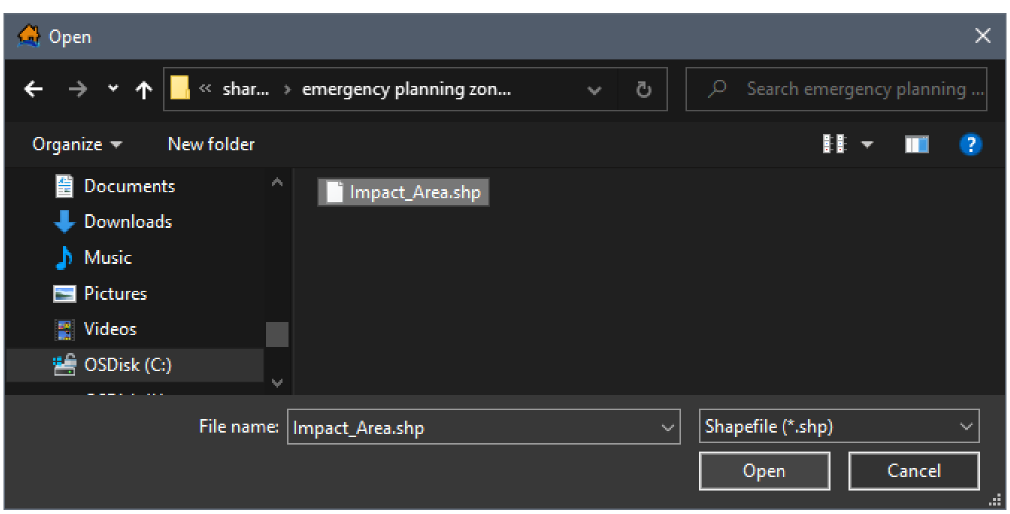


Figure 6-15. Open Browser Window – Select Polygon Shapefile

Select EPZ Name Field

4. From the **Emergency Planning Zone Name Field** (Figure 6-16) dropdown menu select the shapefile attribute that is representative of the unique emergency planning zone (e.g., *ID_NAME* in Figure 6-16). The list of items in the **Emergency Planning Zone Name Field** (Figure 6-16) dropdown menu is generated from the selected emergency planning zone shapefile.

5. The **Emergency Planning Zone Name Field** dropdown menu (Figure 6-14) selection updates the **Emergency Planning Zone** dropdown menu (Figure 6-14) to contain the names of all of the zones (polygons) in the selected shapefile. The user must define the warning and evacuation parameters to reflect the specific situation in each zone in the study area.

The following steps must be completed for every EPZ area in the **Emergency Planning Zone** dropdown menu (Figure 6-14).

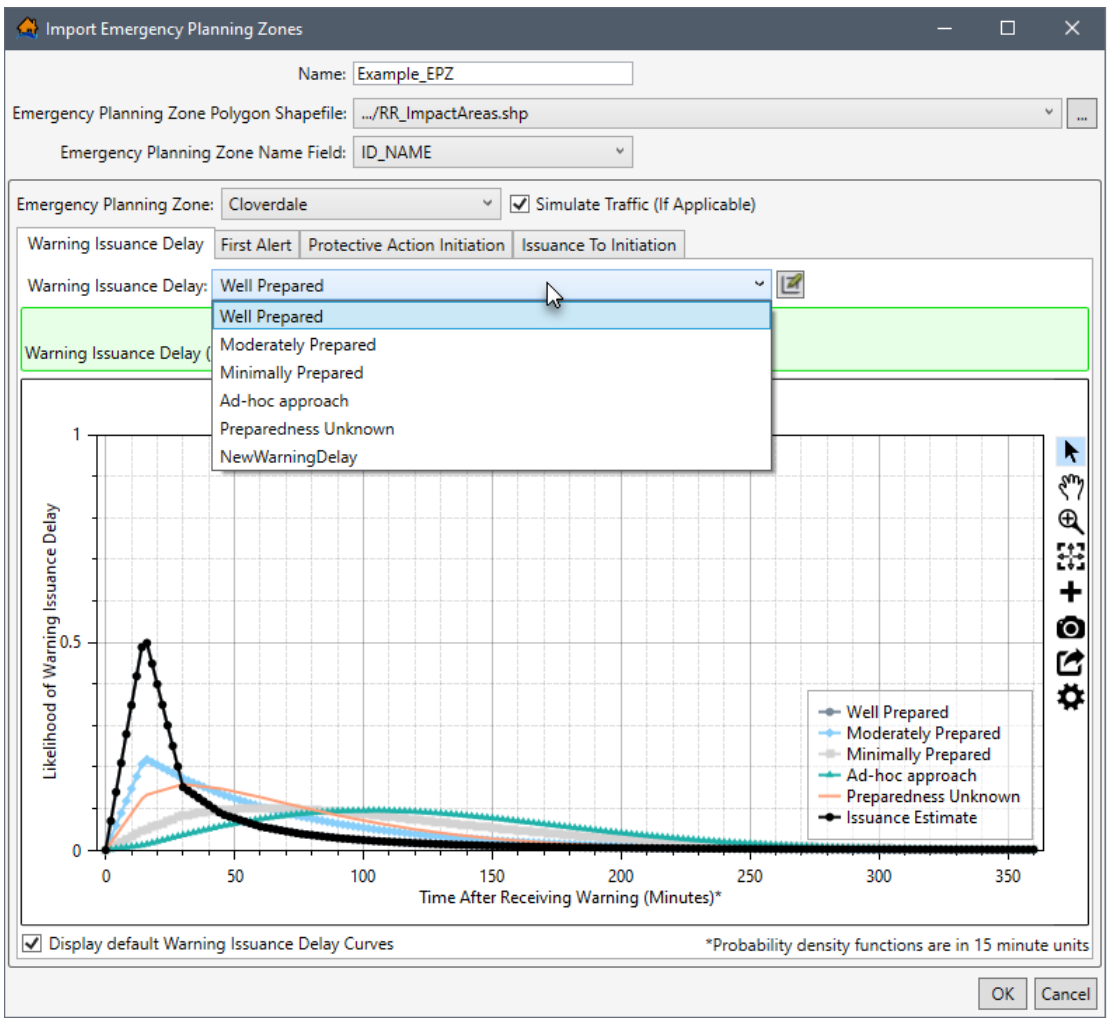


Figure 6-16. Import Emergency Planning Zones – Warning Issuance Delay Tab

6. By default, the **Simulation Traffic (If Applicable)** option is turned on (the checkbox is ☑ checked) to simulate traffic in the selected EPZ. However, the traffic simulation process can be intensive and take a long time to compute. Therefore, if an area is unlikely to require simulating traffic along evacuation routes, for groups taking protective action, then turn off the **Simulation Traffic (If Applicable)** option by unchecking ☐ the checkbox.

Define Warning Issuance Delay

7. From the **Warning Issuance Delay** tab (Figure 6-16) the user will provide parameters that define the amount of time it could take for emergency managers to issue a warning to the public. From the **Warning Issuance Delay** list (Figure 6-16) a select the appropriate warning issuance delay curve for the selected EPZ area (e.g., *Cloverdale* in Figure 6-16).

8. The user can also edit and delete existing curves; and/or create, delete, and add user defined curves by clicking the **Edit Data** button to open the **Warning And Protective Action Data Editor** (Figure 6-2); refer to Section 6.1 for more information.

9. To view all of the default warning issuance delay curves, from the **Import Emergency Planning Zone** dialog box (Figure 6-16), check the ☑ checkbox to **Display default Warning Issuance Delay Curves**; all the default curves will display in the **Warning Issuance Delay** plot window (Figure 6-16).

Select First Alert Diffusion Curves

From the **First Alert** tab (Figure 6-17) the user will provide parameters that define the warning diffusion curves for daytime (from the **Daytime First Alert** tab) and night-time (from the **Night-time First Alert** tab) scenarios. First alert curves are separated into daytime and night-time scenarios because the speed at which a first alert can be received can vary significantly between daytime and night-time.

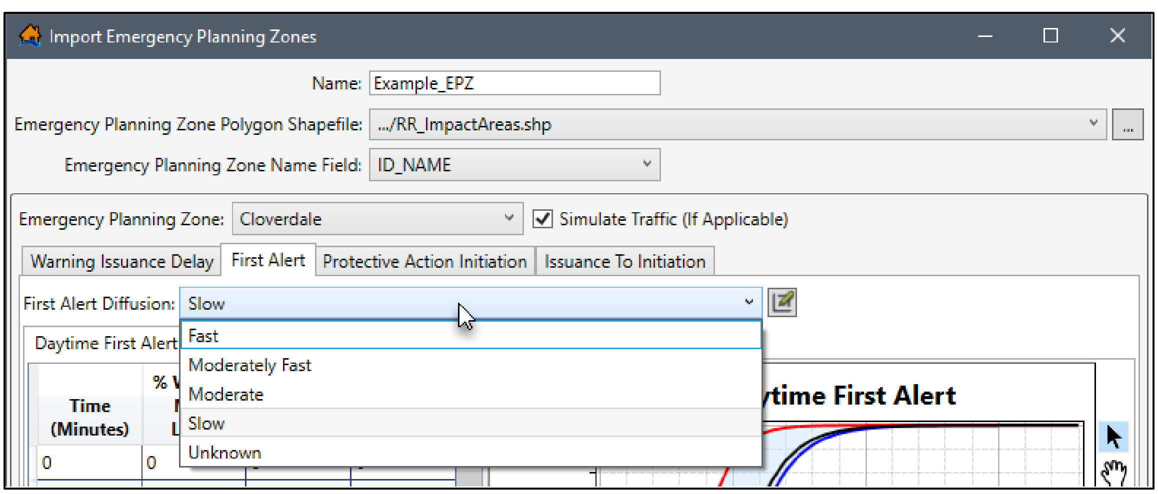


Figure 6-17. Import Emergency Planning Zones – Daytime First Alert Tab – Warning Diffusion Curves

10. From the **First Alert Diffusion** dropdown menu (Figure 6-17), select the appropriate first alert diffusion curves; the selection sets the daytime and night-time curves for the selected EPZ area (e.g., *Cloverdale* in Figure 6-17).

11. The user can also edit and delete existing curves; and/or create, delete, and add user defined curves by clicking the **Edit Data** button to open the **Warning And Protective Action Data Editor** (Figure 6-2); refer to Section 6.1 for more information.

Set Protective Action Initiation Function

12. From the **Protective Action Initiation** tab (Figure 6-18) select the appropriate curve from the **Protective Action Delay** dropdown menu (Figure 6-18) to define the PAI response curve for the selected EPZ area (e.g., *Cloverdale* in Figure 6-18).

13. The user can also edit and delete existing curves; and/or create, delete, and add user defined curves by clicking the **Edit Data** button to open the **Warning And Protective Action Data Editor** (Figure 6-2); refer to Section 6.1 for more information.

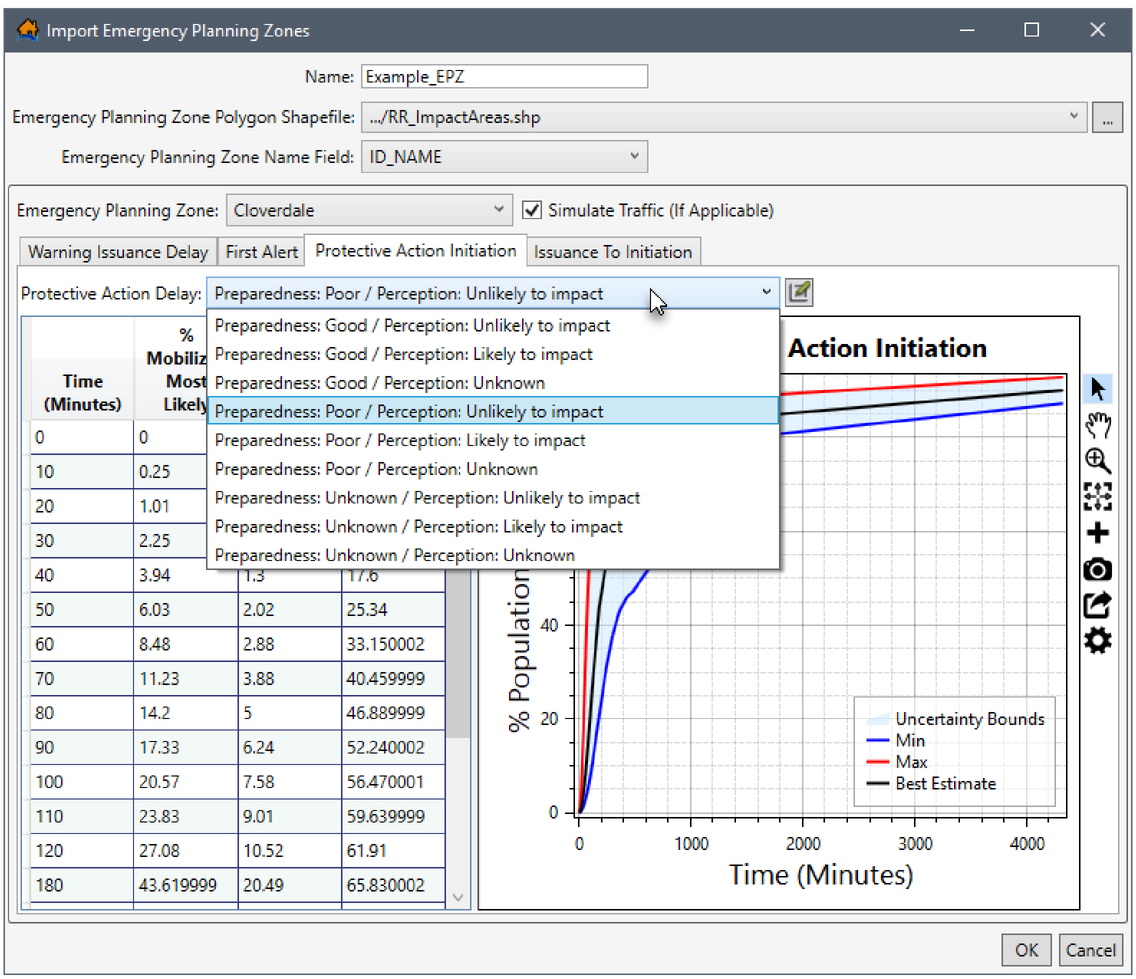


Figure 6-18. Import Emergency Planning Zones – Protective Action Initiation Tab – PAI Response Curves

View the Combined Warning and Evacuation Curve

The issuance to initiation tab provides a visualization of the combined warning and mobilization function. It is the % of population that will take protective action over time once the warning is issued. It also provides a way for the user to display the combined uncertainty of the warning and protective action functions. The combined function is a concise way to present the evacuation response to warning issuance.

14. View the combined warning and evacuation curve for the selected EPZ area (e.g., *Cloverdale* in Figure 6-18) by selecting the **Issuance To Initation** tab (Figure 6-19).

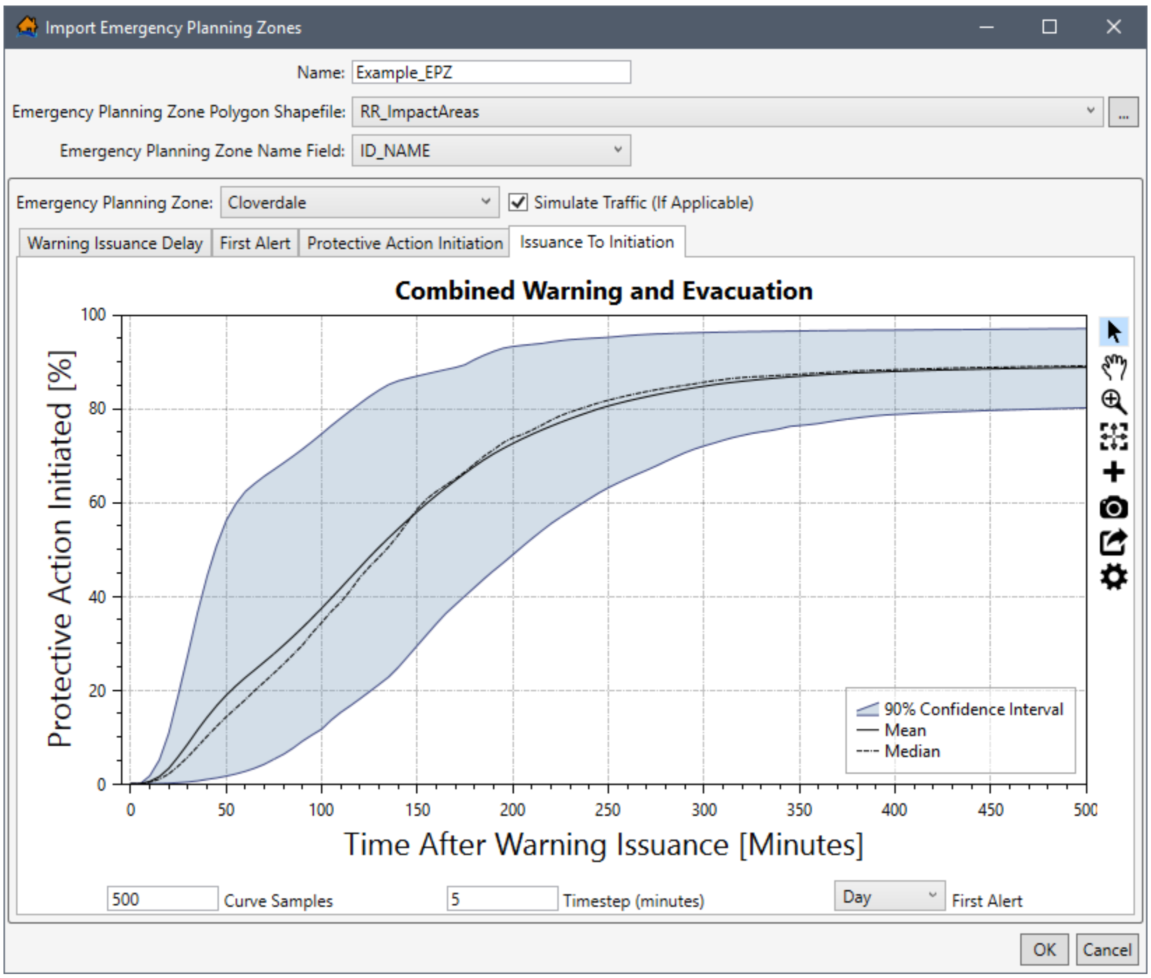


Figure 6-19. Import Emergency Planning Zones – Issuance To Initiation Tab

15. From the bottom of **Issuance To Initation** tab (Figure 6-19), modify the combined warning and evacuation plot by entering values for the **Curve Samples**, **Timestep (minutes),** and **First Alert** time. View the summary plot for the daytime or night-time scenarios by selecting ***Day*** or ***Night***, from the **First Alert** dropdown menu (Figure 6-19). Modifying the values does not impact the simulation, only the combined warning and evacuation visualization is impacted.

Save the New EPZ Dataset

16. Continue to set the warning and protective action parameters for every emergency planning zone in the selected shapefile; and therefore, listed in the **Emergency Planning Zone** dropdown menu (e.g., *Cloverdale* in Figure 6-19).

17. Once all of the correct information for the EPZ dataset has been entered (e.g., *Example_EPZ* in Figure 6-19), click **OK**. The **Import Emergency Planning Zone** dialog box (Figure 6-19) closes. From the LifeSim main window (Figure 6-13), from the **Study Tree**, under the **Emergency Planning Zones** folder, the name of the EPZ dataset will display (e.g., *Example_EPZ* in Figure 6-13).

6.2.3 Edit an EPZ Dataset

Once an EPZ dataset has been created, the user can edit that EPZ dataset from the **Emergency Planning Zone Editor** (Figure 6-11). From the LifeSim main window (Figure 6-13), from the **Study** tab, from the **Study Tree**, from the **Emergency Planning Zones** folder, right-click on an EPZ dataset, from the shortcut menu (Figure 6-20), click **Edit**. Another way is from the LifeSim main window, from the **Study** menu, point to **Emergency Planning Data**, point to **Edit** (Figure 6-13), click an EPZ dataset. Either way, the **Emergency Planning Zone Editor** (Figure 6-11) opens. Refer to Section 6.2 which provides information on defining warning and evacuation parameters for a selected EPZ in the **Emergency Planning Zone Editor** (Figure 6-11).

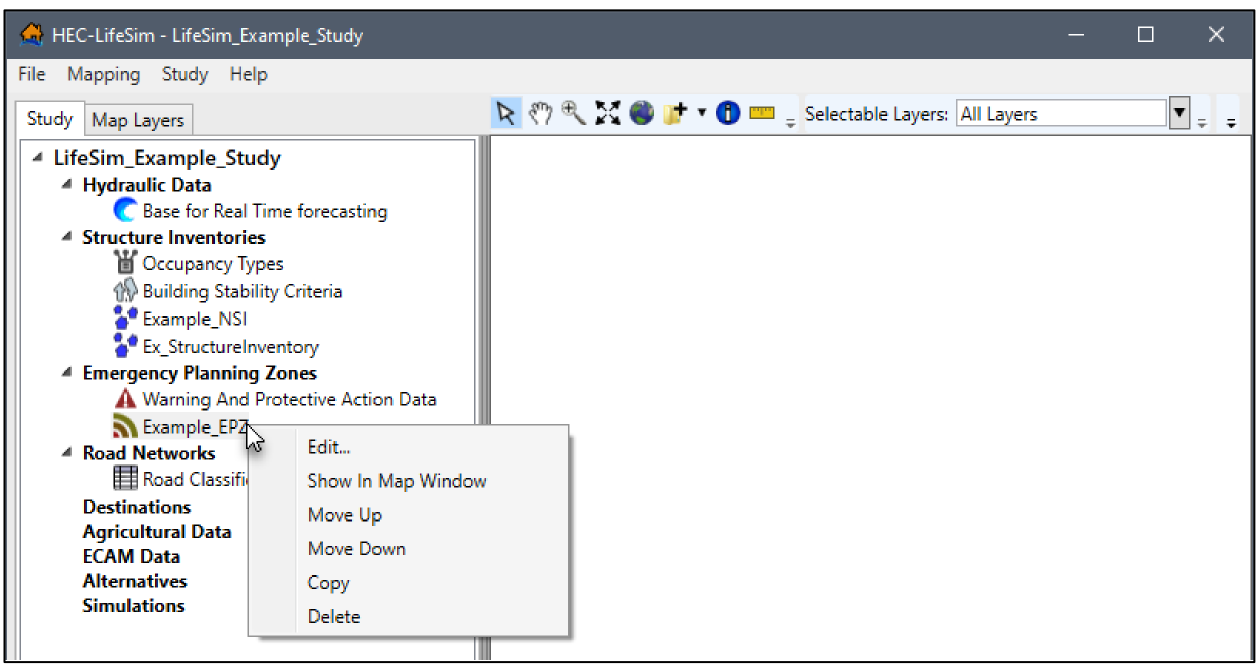


Figure 6-20. LifeSim Study Tree – Emergency Planning Zones Dataset – Shortcut Menu

6.2.4 Display an EPZ Dataset

To display an EPZ dataset, from the LifeSim main window (Figure 6-13), from the **Study** tab, from the **Study Tree**, from the **Emergency Planning Zones** folder, right-click on an EPZ dataset, from the shortcut menu (Figure 6-20), click **Show In Map Window**. The selected EPZ dataset will display in the map window of the LifeSim main window.

Map layer attributes can be accessed for each map layer and is provided in a table for the dataset referenced by the map layer. The EPZ dataset must be displayed in the map window before the attribute table can be accessed. To access a specific map layer **Attribute** table, from the **Map Layers** tab, right-click on a map layer of interest, from the shortcut menu, click **Open Attribute Table**. The *Map Layer* **Attributes** dialog box opens; review Appendix A, Section A.4 for more information. Editing of the map layer attributes is detailed in Section A.10 of Appendix A.

6.2.5 Copy an EPZ Dataset

To copy an EPZ dataset from an existing dataset, from the LifeSim main window (Figure 6-13), from the **Study** tab, from the **Study Tree**, from the **Emergency Planning Zones** folder, right-click on an EPZ dataset, from the shortcut menu (Figure 6-20), click **Copy**. Another way is from the LifeSim main window, from the **Study** menu, point to **Emergency Planning Data**, point to **Copy** (Figure 6-13), click an EPZ dataset.

Either way, the **Name of New Study Item** dialog box (Figure 5-19) opens. Enter a new name in the **Name** box. Click **OK**, the **Name of New Study Item** dialog box (Figure 5-19) closes and the copied EPZ dataset will display in the **Study Tree** under the **Emergency Planning Zones** folder.

6.2.6 Delete an EPZ Dataset

To delete an EPZ dataset from the LifeSim study and study directory files, from the LifeSim main window (Figure 6-13), from the **Study** tab, from the **Study Tree**, from the **Emergency Planning Zones** folder, right-click on an EPZ dataset, from the shortcut menu (Figure 6-20), click **Delete**. Another way is from the LifeSim main window, from the **Study** menu, point to **Emergency Planning Data**, point to **Delete** (Figure 6-20), click an EPZ dataset.

Either way, a **Confirm Delete** window (similar to Figure 5-20) opens asking for confirmation to delete the selected EPZ dataset. Click **Yes**, the **Confirm Delete** message window closes (similar to Figure 5-20), and the selected EPZ dataset is deleted from the study and no longer displays in the **Study Tree** under the **Emergency Planning Zones** folder.

CHAPTER 7

Road Network Data

If traffic is being simulated (set from the **Emergency Planning Zone Editor**, see Section 6.2), **road network** data and destination data (see Chapter 8) are required in a LifeSim study to represent possible evacuation routes that people can take during an evacuation. The road network features can be categorized into multiple road types with attributes defining their census feature classification codes (CFCC), flow direction, and vertical offset (Appendix C).

- The CFCC classification is a three-character system that is used to define feature data. The first character *A* denotes roads, the second character is an integer that denotes road type (e.g., Primary Highway), and the third character further classifies within the road type (e.g., underpass). The CFCC codes are used in LifeSim to define attributes about the road that affect transportation (e.g., free flow speed). Further information on CFCC classification is available in Appendix C.

- Roads can be defined as one-way directional in LifeSim. If roads are defined in this way, care must be taken to make sure that the direction and links are set up correctly.

- The vertical offset attribute is designed for roads that are elevated above the ground such as a bridge that would have water running under the road instead of on the road.

Appendix B is designed to help users prepare and edit road networks for use in LifeSim. The LifeSim Version 2.0 Technical Reference Manual, Chapter 6, describes the computational input parameters required for the traffic simulation model used to determine the speed of vehicles at each time step. There are two ways to add road network data to an LifeSim study and only one format is supported (polygon shapefiles) to be utilized in LifeSim simulations.

7.1 Road Classification Data

In LifeSim, roads are classified using census feature classification codes (CFCC). Users can view and edit the default CFCC codes; however, care should be taken as the default road parameters have a significant impact on the traffic simulation model. To view and/or edit the default CFCC road attributes:

1. From the LifeSim main window, from the **Study** tab, from the **Study Tree**, from the **Road Networks** folder, right-click on **Road Classification Data** (Figure 7-1), and click **Edit Road Classification Data** from the shortcut menu. The **Road Classification Editor** (Figure 7-2) opens.

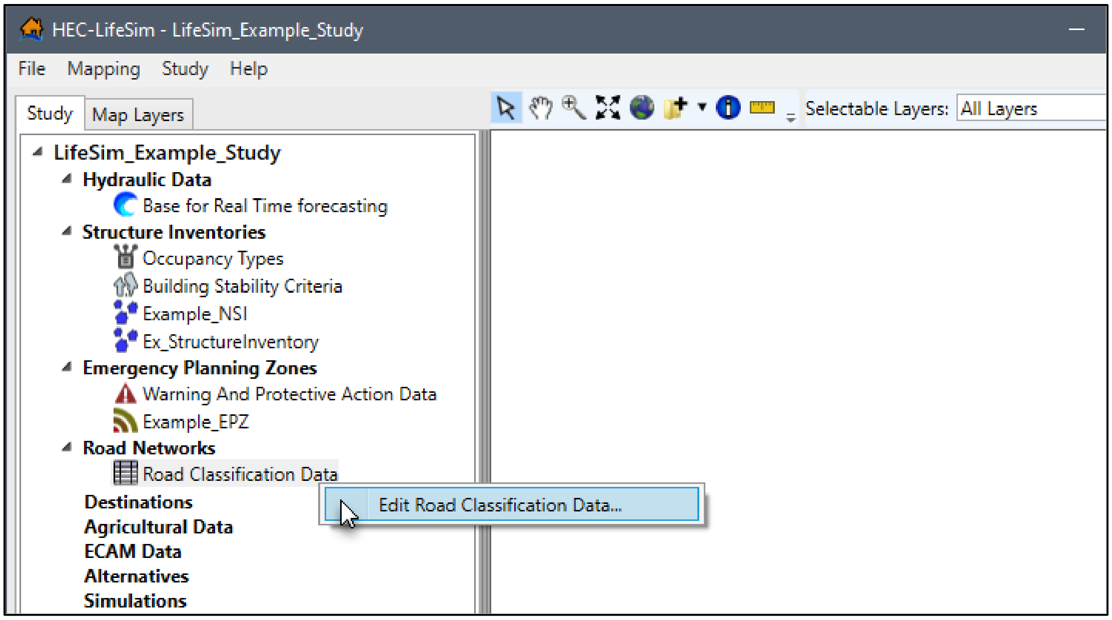


Figure 7-1. LifeSim Study Tree – Road Classification Data Shortcut Menu

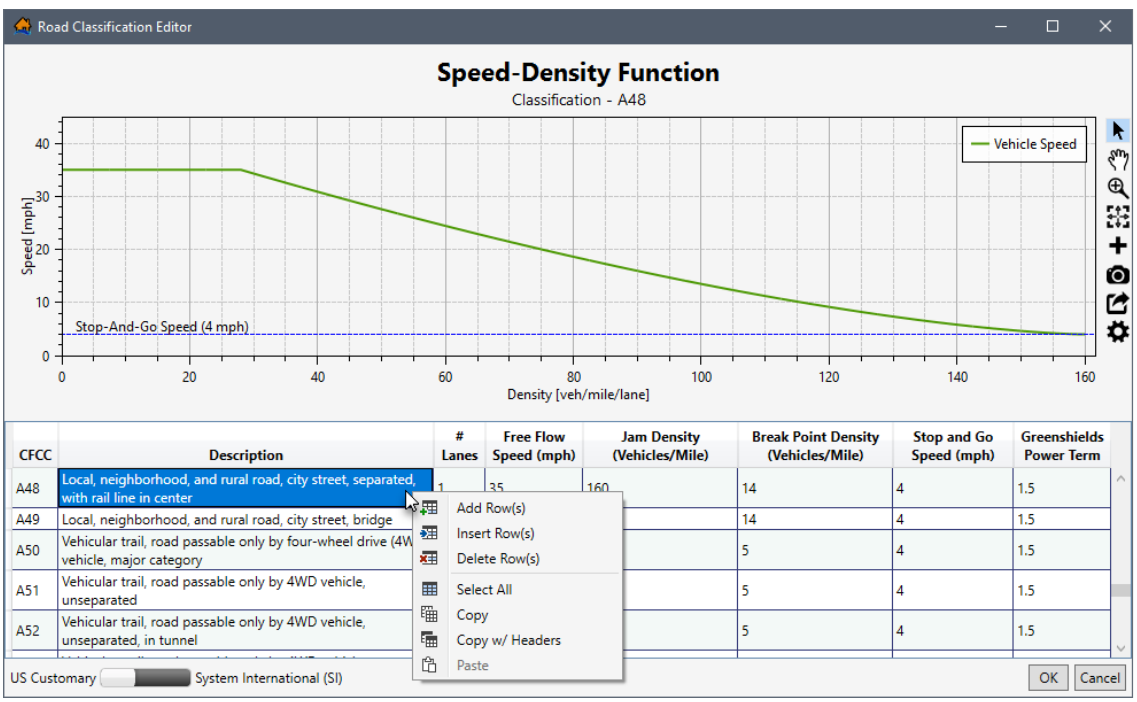


Figure 7-2. Road Classification Editor

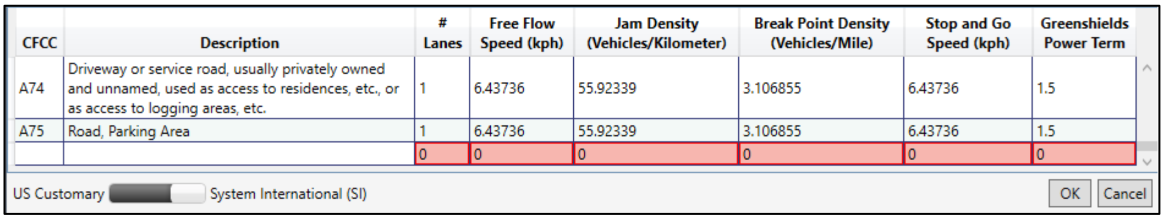
2. A plot of the speed-density function is displayed at the top of the **Road Classification Editor** (Figure 7-2). Review Section 3.6 for instructions on using the general plotting tools (located on the right side of the plot window) and for customizing plots.

   The editable road classification table below the plot window lists the default CFCC data. The road classification table lists the number of lanes, free flow speed, jam density, break point density, stop and go speed, and Greenshields power term.

Users set the display units for the plot window and road classification table to either **US Customary** (Figure 7-2) or **System International (SI)** (Figure 7-3) using the slider bar at the bottom of the editor.

An existing CFCC road classification code can be edited, or new road classification codes can be defined following the remaining instructions in this section.

3. To add a new road classification code, right-click a cell to open the shortcut menu (Figure 7-2), from the shortcut menu click, **Add Row(s)**. By default, one row is added to the bottom of the road classification table (Figure 7-3).



CFCC	Description	# Lanes	Free Flow Speed (kph)	Jam Density (Vehicles/Kilometer)	Break Point Density (Vehicles/Mile)	Stop and Go Speed (kph)	Greenshields Power Term
A74	Driveway or service road, usually privately owned and unnamed, used as access to residences, etc., or as access to logging areas, etc.	1	6.43736	55.92339	3.106855	6.43736	1.5
A75	Road, Parking Area	1	6.43736	55.92339	3.106855	6.43736	1.5
		0	0	0	0	0	0

Figure 7-3. Example Road Classification Editor – Row Added for New Road Classification

4. For the new row, double-click the **CFCC** column and enter the road classification code. Double-click to enter the description of the new road in the **Description** column. Continue to enter the following road classification parameters in the new row:

   - **# Lanes** – total number of lanes going in the same direction,
   - **Free Flow Speed** – this is the speed of vehicles on the road with no impediments,
   - **Jam Density** – this is the density of vehicles per mile required to create traffic jam conditions,
   - **Break Point Density** – this is the density of vehicles on the road required to start impacting the overall speed,
   - **Stop and Go Speed** – is the speed that vehicles travel when under stop and go conditions, and
   - **Greenshields Power Term** – this parameter is used to help define the relationship between speed and density of vehicles on the road network.

   Errors in the road classification table are highlighted in red (Figure 7-3) and a tooltip displays a message describing the error. Right-click on a header row to open the shortcut menu (Figure 5-28) to sort the table in ascending or descending order, or clear the table sorting.

5. To edit an existing CFCC code, select the row for the desired code to view the speed-density function in the plot window (e.g., ***A48*** in Figure 7-2). Double-click the desired cell to edit the information for the selected CFCC code.

6. Click **OK**, the **Road Classification Editor** (Figure 7-2) closes and saves all edits to road classification data.

7.2 Create Road Network Datasets

Road network datasets can be created by importing information from a polyline shapefile that represents a road network. Another way to import a road network dataset is using a bounding polygon shapefile to generate a road network using all available data within the bounding polygon from the OpenStreetMap database (Appendix C).

7.2.1 Import from a Shapefile

To create a road network dataset from a polygon shapefile:

1. From the LifeSim main window, from the **Study** tab, from the LifeSim main window, from the **Study** menu, point to **Road Network Data** (Figure 7-4), click **Import From Shapefile**. Another way is from the **Study Tree**, right-click on **Road Networks**, from the shortcut menu (Figure 7-5), click **Import Road Network From Shapefile**.

2. Either way, the **Import Road Network** dialog box (Figure 7-6) will open.

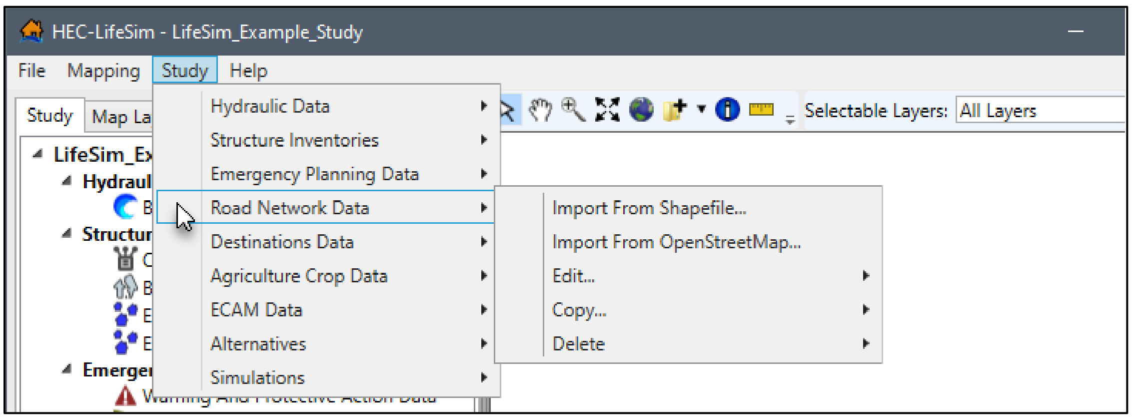


Figure 7-4. LifeSim Main Window – Study Menu – Road Network Data Sub-menu

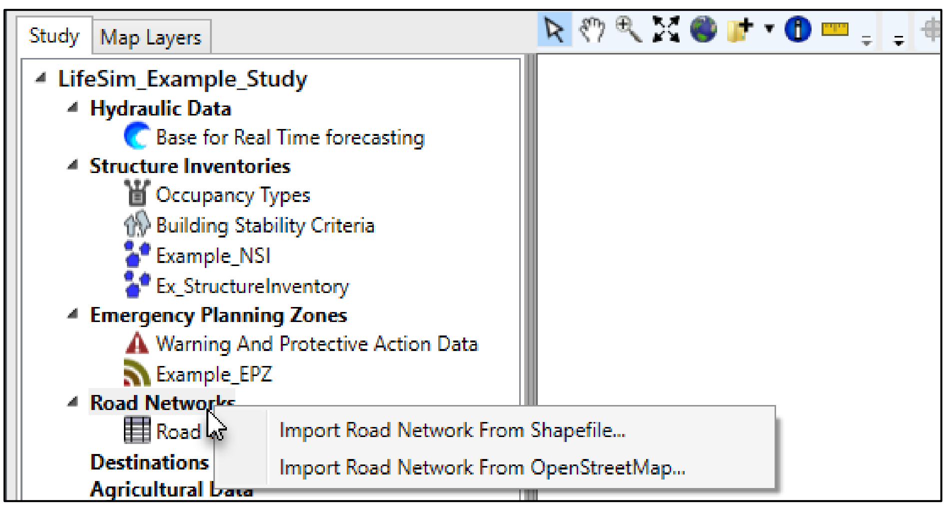


Figure 7-5. LifeSim Study Tree – Road Networks – Shortcut Menu

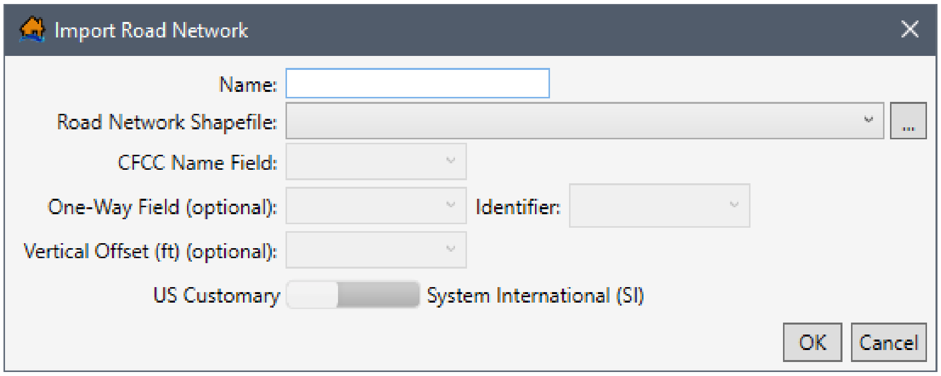


Figure 7-6. Import Road Network Dialog Box

3. Enter a name for the road network dataset in the **Name** box (Figure 7-6).

4. To select the shapefile that contains information about the road network, to the right of the **Road Network Shapefile** dropdown menu (Figure 7-6), click . An **Open** browser window (Figure 7-7) opens; navigate to the directory where the road network polyline shapefile is located. Select the correct polyline shapefile (e.g., **.shp*), click **Open**; the **Open** browser (Figure 7-7) closes and the user is returned to the **Import Road Network** dialog box (Figure 7-8). The location and name of the selected polyline shapefile displays in the **Road Network Shapefile** box (Figure 7-8). Alternatively, if the desired shapefile has already been imported as a map layer into the LifeSim study, then from the **Road Network Shapefile** dropdown menu (Figure 7-6), select the desired map layer.

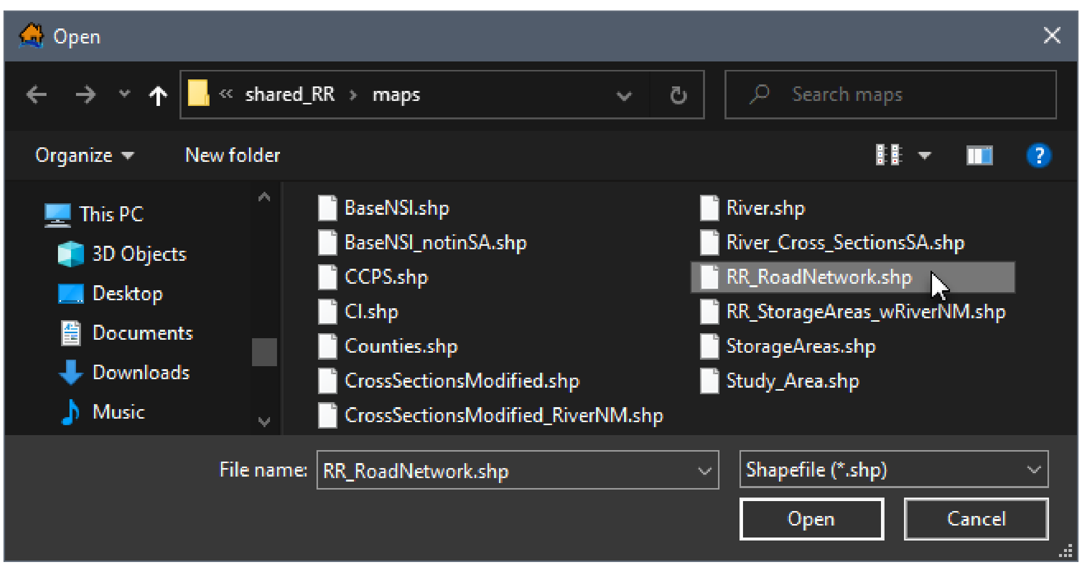


Figure 7-7. Open Browser Window

5. From the **CFCC Name Field** dropdown menu (Figure 7-8), select the appropriate attribute field in the selected shapefile that lists the CFCC codes (e.g., ***CFCC*** in Figure 7-8). The CFCC field is used to identify the most noticeable characteristic of a road.

   If this information is not known (CFCC is not available from the **CFCC Name Field** list), then the user could access the shapefile attribute table (see Appendix A, Section A.4 for details on accessing the attribute table) to determine which column header contains the CFCC road identifications (e.g., A73, A15, A63, etc.). Review Section 7.1 for more information regarding the default CFCC codes in LifeSim.

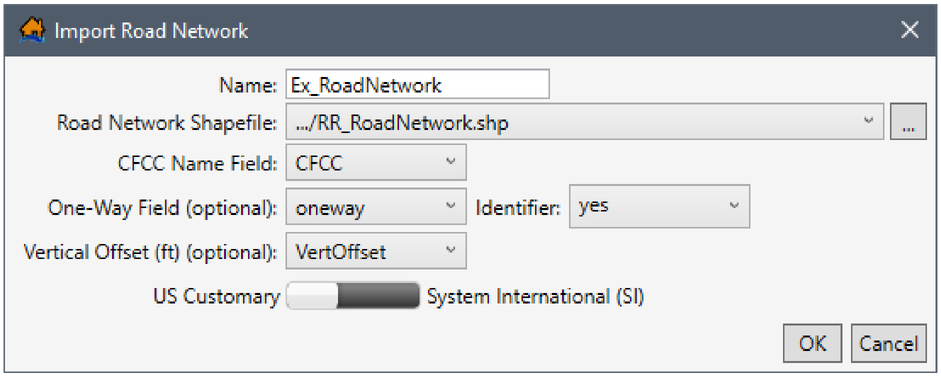


Figure 7-8. Example – Completed Import Road Network Dialog Box

6. If the road network shapefile contains data classifying one-way streets and the attribute identification (e.g., true or false; yes or no; 0 or 1), then the user can set the information for one-way streets (optional). For the example provided in Figure 7-8, the **One-Way Field** attribute is ***oneway***, and the one-way field **Identifier** is ***yes***.

7. Another optional field which depends on the road networks shapefile attributes is the **Vertical Offset** dropdown menu (Figure 7-8), which sets the vertical offset.

8. Set the units for the road network to either **US Customary** or **System International (SI)** by using the slider bar (e.g., ***US Customary*** is selected in Figure 7-8).

9. Once all of the information has been entered click **OK**, the **Import Road Network** dialog will close (Figure 7-8). From the LifeSim main window (Figure 7-4), from the **Study Tree**, under the **Road Networks** folder, the name of the road network dataset is displayed.

7.2.2 Import from OpenStreetMap

OpenStreetMap is a collaborative project to create a free editable map of the world (Appendix C). It is supported by a large community of dedicated members to provide global geographic information system (GIS) data. LifeSim can take the road network information stored within the OpenStreetMap database to create a road network for use in the software. To create a road network dataset from a bounding polygon shapefile:

1. From the LifeSim main window (Figure 7-4), from the **Study** tab, from the **Study Tree**, right-click on **Road Networks**, from the shortcut menu (Figure 7-5), click **Import Road Network From OpenStreeetMap**. Another way is from the LifeSim main window, from the **Study** menu, point to **Road Network Data** (Figure 7-4), click **Import From OpenStreetMap**.

2. Either way, the **Import Road Network From OpenStreetMap** dialog box (Figure 7-9) opens. Enter the name of the road network dataset in the **Road Network Name** box (Figure 7-9).

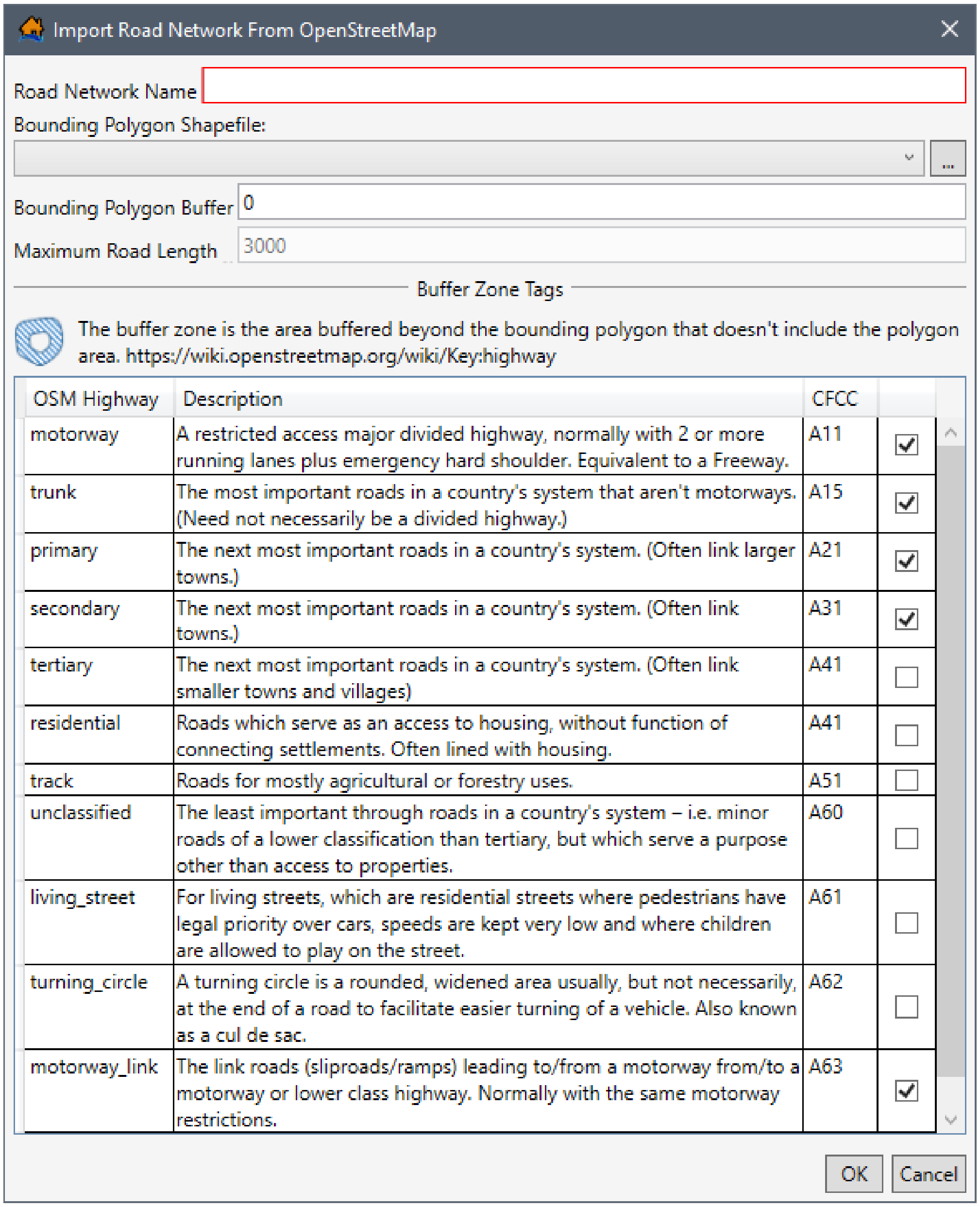


Figure 7-9. Import Road Network From OpenStreetMap Dialog Box

3. If the bounding polygon shapefile has been imported as a map layer, then select the map layer from the **Bounding Polygon Shapefile** dropdown menu (Figure 7-9). Otherwise, click the ... button (Figure 7-9). An **Open** browser window (Figure 7-7) opens, navigate to the directory where the bounding polygon shapefile is located. Select the correct polygon shapefile (e.g., **.shp*), click **Open**, the **Open** browser will close (Figure 7-7).

4. The selected shapefile displays in the **Bounding Polygon Shapefile** dropdown menu (Figure 7-9). Enter the **Bounding Polygon Buffer** distance (positive value) if the default

value, ***0***, is not desired. Note, the units displayed in the **Import Road Network From OpenStreetMAP** dialog box (Figure 7-9) is relative to the horizontal unit of the underlying projection (typically feet). The buffer zone adds a consistent distance around the perimeter of the polygon area.

5. Enter the **Maximum Road Length** if the default value, ***3000***, is not desired. Note, the units displayed in the **Import Road Network From OpenStreetMAP** dialog box (Figure 7-9) is relative to the horizontal unit of the underlying projection (typically feet).

6. The OSM Highway and CFCC table (Figure 7-9), allows users to select roads, classified by CFCC codes, to import. Import the desired road(s) by checking the ☑ checkboxes for the road(s); alternatively, uncheck the ☐ checkboxes to prevent specific road(s) from importing.

7. Click **OK**, and the **Import Road Network From OpenStreetMap** dialog box (Figure 7-9) closes and a **Progress Window** opens (Figure 7-10). The **Progress Window** displays the import process where LifeSim contacts the OpenStreetMap Data Server and retrieves data based on the selected bounding polygon shapefile and import conditions (buffer distance, max road length, CFCC selected roads).

8. Once the import is completed the **Progress Window** closes (Figure 7-10). From the LifeSim main window (Figure 7-4), from the **Study Tree**, under the **Road Networks** folder, the name of the road network dataset is displayed.

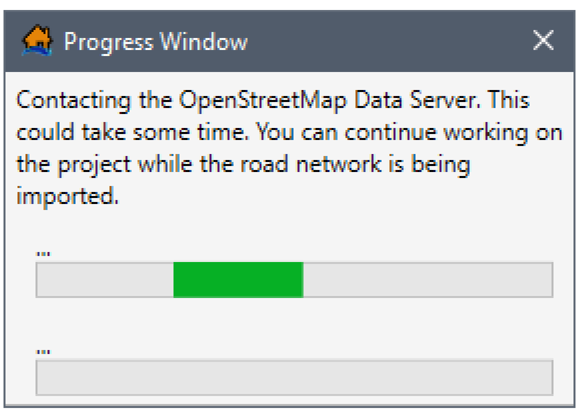


Figure 7-10. Progress Window

7.3 Edit a Road Network Dataset

Once a road network dataset has been created, the user can edit that road network dataset. To edit a road network dataset, from the LifeSim main window (Figure 7-4), from the **Study** tab, from the **Study Tree**, from the **Road Networks** folder, right-click on a road network dataset, and click **Edit** from the shortcut menu (Figure 7-11). Another way is from the LifeSim main window, from the **Study** menu, point to **Road Network Data** (Figure 7-4), point to **Edit**, click a road network dataset.

Either way, the **Edit Road Network** dialog box (Figure 7-12) opens. Section 7.2.1 provides information on the editor.

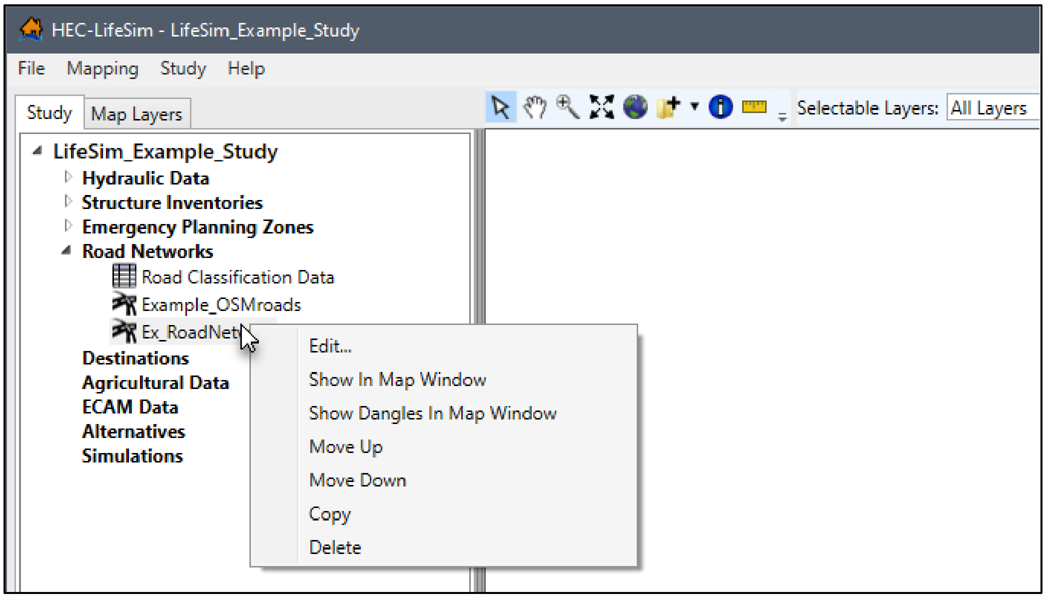


Figure 7-11. LifeSim Study Tree – Road Network Dataset – Shortcut Menu

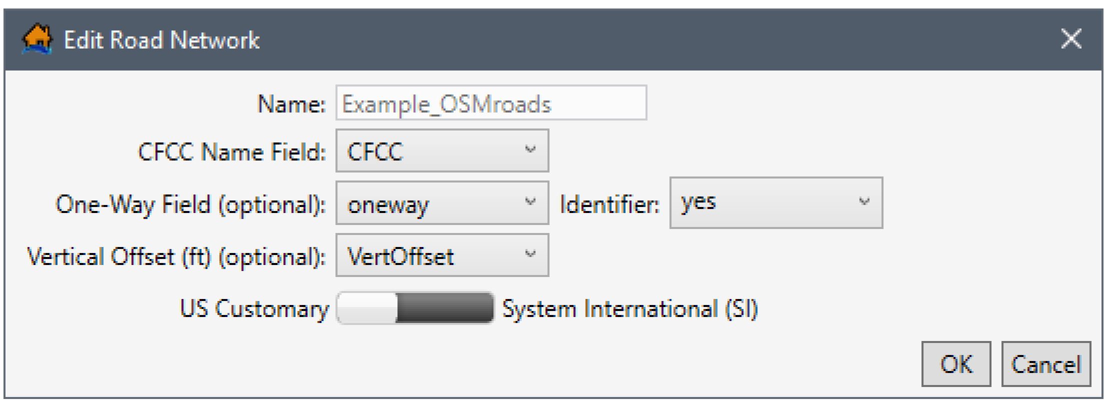


Figure 7-12. Edit Road Network Dialog Box

7.4 Display a Road Network Dataset

To display a road network dataset, from the LifeSim main window (Figure 7-4), from the **Study** tab, from the **Study Tree**, from the **Road Networks** folder, right-click on a road network dataset, from the shortcut menu (Figure 7-11), click **Show In Map Window**. The selected road network dataset displays in the map window of the LifeSim main window.

Also, road network dangles (or line ends that do not share an endpoint with any other line) can be displayed. Dangles are useful for determining incomplete road networks by being able to quickly identify locations where roads are disconnected when they should be connected (refer to Appendix B for details on modifying road networks using dangles). From the LifeSim main window (Figure 7-4), from the **Study** tab, from the **Study Tree**, from the **Road Networks** folder, right-click on a road network dataset, from the shortcut menu (Figure 7-11), click **Show Dangles in Map Window**. The selected road network dataset's dangles will display in the map window of the LifeSim main window (Figure 7-4).

Map layer attributes can be accessed for each map layer and is provided in a table for the dataset referenced by the map layer. The road network dataset must be displayed in the map window before the attribute table can be accessed. To access a specific map layer **Attribute** table, from the **Map Layers** tab, right-click on a map layer of interest, from the shortcut menu, click **Open Attribute Table**. The *Map Layer* **Attributes** dialog box opens; review Appendix A, Section A.4 for more information. Editing of the map layer attributes is detailed in Section A.10 of Appendix A.

7.5 Copy a Road Network Dataset

To copy a road network dataset from an existing dataset, from the LifeSim main window (Figure 7-4), from the **Study** tab, from the **Study Tree**, from the **Road Networks** folder, right-click on a road network dataset, from the shortcut menu (Figure 7-11), click **Copy**. Another way is from the LifeSim main window, from the **Study** menu, point to **Road Network Data** (Figure 7-4), point to **Copy**, and click a road network dataset.

Either way, the **Name of New Study Item** dialog box (Figure 5-19) opens. Enter a new name in the **Name** box. Click **OK**, the **Name of New Study Item** dialog box (Figure 5-19) closes and the copied road network dataset will display in the **Study Tree** under the **Road Networks** folder.

7.6 Delete a Road Network Dataset

To delete a road network dataset from the LifeSim study and study directory files, from the LifeSim main window (Figure 7-4), from the **Study** tab, from the **Study Tree**, from the **Road Networks** folder, right-click on a road network dataset, from the shortcut menu (Figure 7-11), click **Delete**. Another way is from the LifeSim main window, from the **Study** menu, point to **Road Network Data** (Figure 7-4), point to **Delete**, click a road network dataset.

Either way, a **Confirm Delete** window (similar to Figure 5-34) opens asking for confirmation to delete the selected road network dataset. Click **Yes**, the **Confirm Delete** message window closes (similar to Figure 5-34), and the selected road network dataset is deleted, and no longer displays in the **Study Tree** under the **Road Networks** folder.

CHAPTER 8

Destination Data

Destination data is required in a LifeSim study to represent possible evacuation locations during flooding events. The destination can either be a location within the flooded area that is considered a safe haven for evacuees or locations outside the flooding extents. The local Emergency Management Agency (EMA) can be a helpful resource for locating safe havens and for information on defining these types of locations. The Federal Emergency Management Agency (FEMA) provides contact information for EMAs (https://www.fema.gov/emergency-management-agencies).

The destination data is represented as points in space, LifeSim will find the nearest road from a destination point and consider that the evacuation location. The destinations do not have to be directly on a road; however, only one destination per road segment is allowed. A method to create destinations and define their location is detailed in Appendix A, Section A.9.5.

8.1 Create Destination Datasets

To create a destination dataset from a shapefile:

1. From the LifeSim main window (Figure 8-1), from the **Study** tab, from the **Study Tree**, right-click on **Destinations**, from the shortcut menu, click **Import Destinations From Shapefile**. Another way is from the LifeSim main window, from the **Study** menu, point to **Destinations Data** (Figure 8-2), click **Import From Shapefile**.

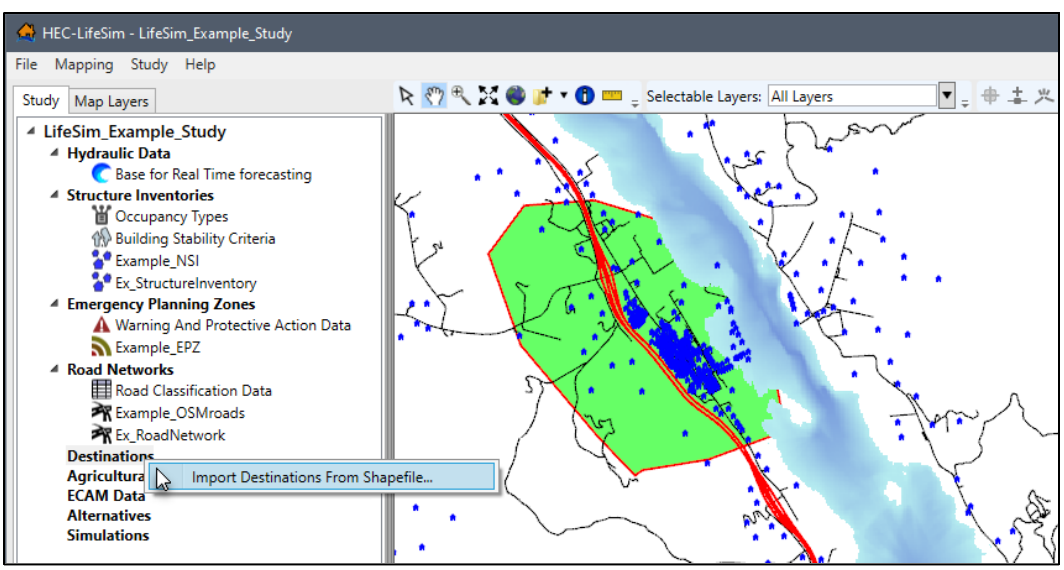


Figure 8-1. LifeSim Study Tree – Destinations Shortcut Menu

2. Either way, the **Import Destinations From Point Shapefile** dialog box (Figure 8-3) will open.

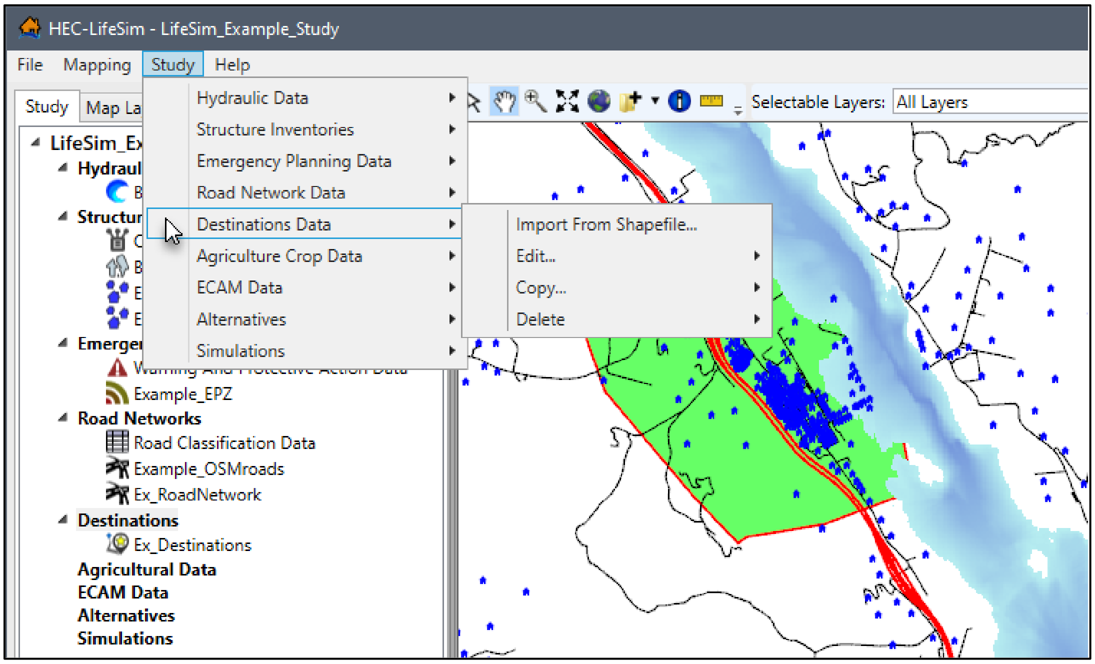


Figure 8-2. LifeSim Main Window – Study Menu – Destinations Data Sub-menu

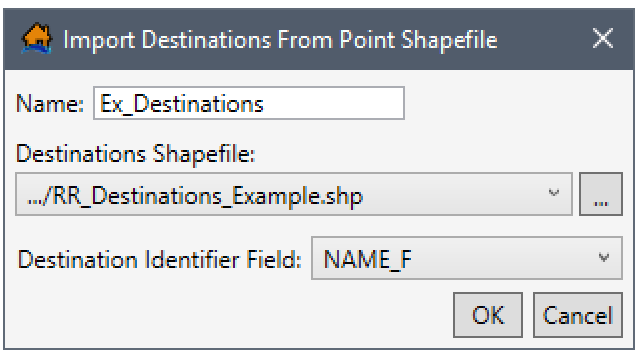


Figure 8-3. Import Destinations From Point Shapefile Dialog Box

3. Enter a name for the destination dataset in the **Name** box (Figure 8-3).

4. To select the shapefile that contains information about destinations, to the right of the **Destinations Shapefile** dropdown menu (Figure 8-3), click . An **Open** browser window (Figure 8-4) opens; navigate to the directory where the desired point shapefile is located. Select the correct point shapefile (e.g., **.shp*), click **Open**, the **Open** browser closes (Figure 8-4) and the user will be returned to the **Import Destinations From Point Shapefile** (Figure 8-3). The location and name of the selected point shapefile will display in the **Destinations Shapefile** dropdown menu (Figure 8-3).

   Alternatively, if the desired shapefile has already been imported as a map layer into the LifeSim study, then from the **Destinations Shapefile** dropdown menu (Figure 8-3), select the desired map layer.

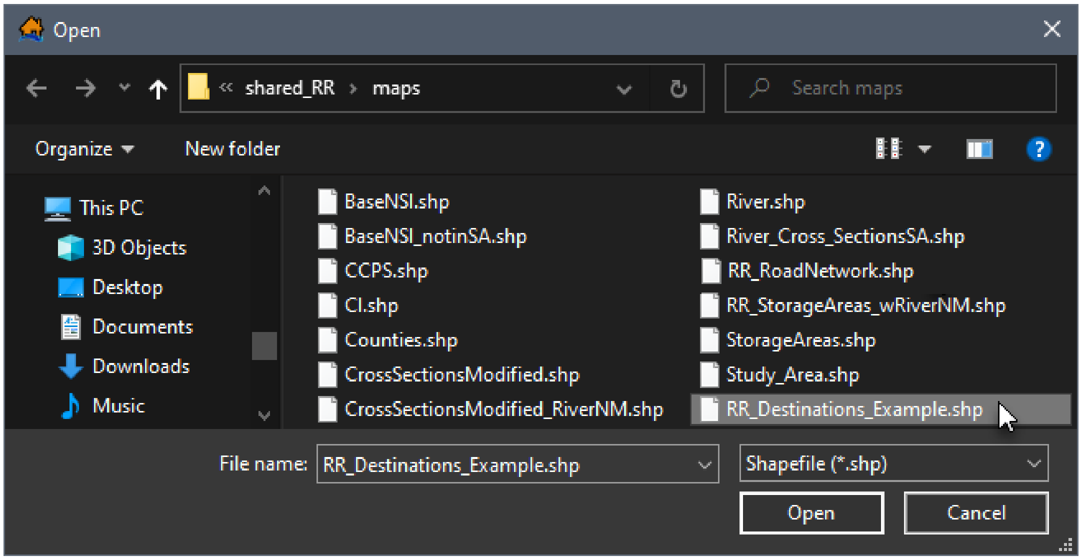


Figure 8-4. Open Browser Window

5. From the **Destination Identifier Field** dropdown menu (Figure 8-3), select the attribute field that identifies the destinations (e.g., destination names) in the selected shapefile.

6. Click **OK**, the **Import Destinations From Point Shapefile** dialog closes (Figure 8-3). From the LifeSim main window (Figure 8-1), from the **Study Tree**, under the **Destinations** folder (Figure 8-2), the name of the destination dataset is displayed.

8.2 Edit a Destination Dataset

To copy a destination dataset from an existing dataset, from the LifeSim main window (Figure 8-1), from the **Study** tab, from the **Study Tree**, from the **Destinations** folder, right-click on a destination dataset, from the shortcut menu (Figure 8-5), click **Edit**. Another way is from the LifeSim main window, from the **Study** menu, point to **Destinations Data** (Figure 8-2), point to **Edit**, and click a destination dataset.

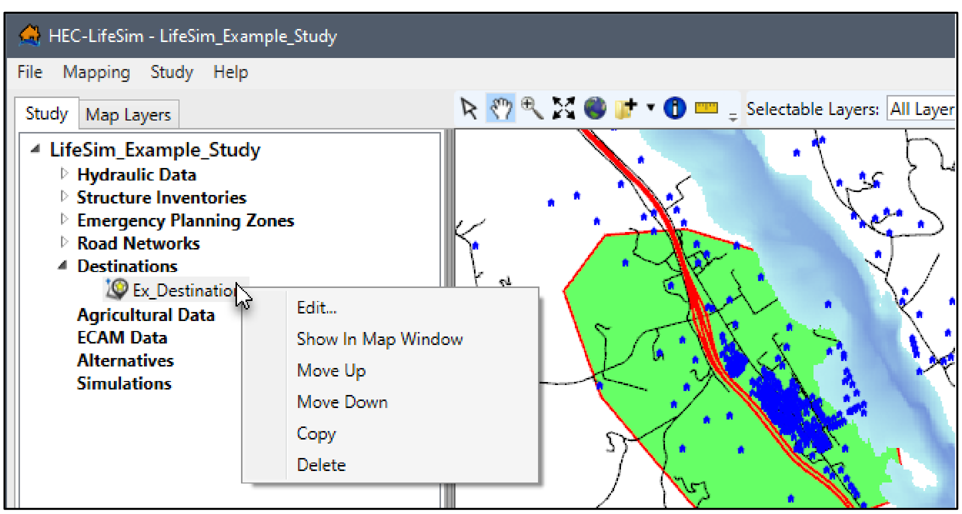


Figure 8-5. LifeSim Study Tree – Destination Dataset – Shortcut Menu

Either way, the **Import Destinations From Point Shapefile** dialog box (Figure 8-3) opens. From the **Destination Identifier Field** dropdown menu (Figure 8-3), select the attribute field that identifies the destinations (e.g., destination names) in the selected shapefile. Click **OK**, the **Import Destinations From Point Shapefile** dialog closes (Figure 8-3).

8.3 Display a Destination Dataset

To display a destination dataset, from the LifeSim main window (Figure 8-1), from the **Study** tab, from the **Study Tree**, from the **Destinations** folder, right-click on a destination dataset, from the shortcut menu (Figure 8-5), click **Show In Map Window**. The selected destination dataset will display in the map window of the LifeSim main window.

Map layer attributes can be accessed for each map layer and is provided in a table for the dataset referenced by the map layer. The destinations dataset must be displayed in the map window before the attribute table can be accessed. To access a specific map layer **Attribute** table, from the **Map Layers** tab, right-click on a map layer of interest, from the shortcut menu, click **Open Attribute Table**. The *Map Layer* **Attributes** dialog box opens; review Appendix A, Section A.4 for more information. Editing of the map layer attributes is detailed in Section A.10 of Appendix A.

8.4 Copy a Destination Dataset

To copy a destination dataset from an existing dataset, from the LifeSim main window (Figure 8-1), from the **Study** tab, from the **Study Tree**, from the **Destinations** folder, right-click on a destination dataset, from the shortcut menu (Figure 8-5), click **Copy**. Another way is from the LifeSim main window, from the **Study** menu, point to **Destinations Data** (Figure 8-2), point to **Copy**, and click a destination dataset.

Either way, the **Name of New Study Item** dialog box (Figure 5-19) opens. Enter a new name in the **Name** box. Click **OK**, the **Name of New Study Item** dialog box (Figure 5-19) closes and the copied destination dataset displays in the **Study Tree** under the **Destinations** folder.

8.5 Delete a Destination Dataset

To delete a destination dataset from the LifeSim study and study directory files, from the LifeSim main window (Figure 8-1), from the **Study** tab, from the **Study Tree**, from the **Destinations** folder, right-click on a destination dataset, from the shortcut menu (Figure 8-5), click **Delete**. Another way is from the LifeSim main window, from the **Study** menu, point to **Destinations Data** (Figure 8-2), point to **Delete**, click a destination dataset.

Either way, a **Verify Deletion of Destination Data** message window (Figure 8-6) opens asking for confirmation to delete the selected destination dataset. Click **Yes**, the **Verify Deletion of Destination Data** message window closes (Figure 8-6), and the selected destination dataset is deleted, and no longer displays in the **Study Tree** under the **Destinations** folder.

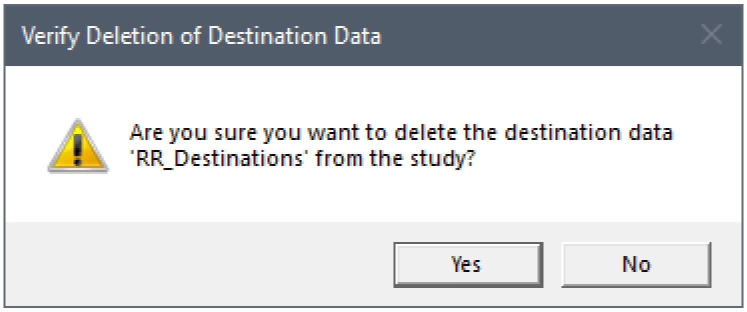


Figure 8-6. Verify Deletion of Destination Data Message Window

CHAPTER 9

Agricultural Data

New to LifeSim Version 2.0 is **agricultural data**. Agriculture data was added for the purpose of estimating economic impacts from damage to existing crops and interruptions to the planting, growing, and harvesting of crops due to flooding in agricultural areas. Agricultural data is not typically used in a consequence assessment; however, it may assist the user in understanding the amount and type of agriculture in the floodplain and potential flood impacts to it. To compute agricultural damages hydraulic data (including duration and event arrival), agricultural data, crop budgets, crop values, and crop duration-damage relationships are required. Review the LifeSim Version 2.0 Technical Reference Manual, Chapter 7, for more information regarding the agricultural inventory and the computational procedures used to calculate agricultural flood damages.

9.1 Import Agricultural Data

Users can import agricultural data directly from the U. S. Department of Agriculture (USDA) National Agricultural Statistics Service (NASS) Cropland Data Layer (CDL) database through LifeSim (Version 2.0). Alternatively, users can download a NASS GeoTiff file from the NASS website: https://www.nass.usda.gov/Research_and_Science/Cropland/SARS1a.php. The NASS website provides the CDL in GeoTiff (**.tif*) format, where each cell represents a crop type (review the HEC-FIA Version 3.0 User's Manual, Appendix B for tips preparing NASS data for import into LifeSim). Then users can import the prepared NASS GeoTiff file into LifeSim (Version 2.0). This chapter describe the ways agricultural data in different formats can be added to, edited, and displayed in an LifeSim study.

In LifeSim Version 2.0 the agricultural data importers, **Import From NASS** (Figure 9-5) and **Import From File** (Figure 9-6) exclude several non-crop codes (Table 9-1).

Table 9-1. NASS Codes and Categories LifeSim Excludes during Import

NASS Code	NASS Category
37	Other Hay/Non Alfalfa
59	Sod/Grass Seed
60	Switchgrass
63	Forest
64	Shrubland
65	Barren
81	Clouds/No Data
82	Developed
83	Water
87	Wetlands
88	Nonag/Undefined

NASS Code	NASS Category
92	Aquaculture
111	Open Water
112	Perennial Ice/Snow
121	Developed/Open Space
122	Developed/Low Intensity
123	Developed/Med Intensity
124	Developed/High Intensity
131	Barren
141	Deciduous Forest
142	Evergreen Forest
143	Mixed Forest
152	Shrubland
176	Grassland/Pasture
190	Woody Wetlands
195	Herbaceous Wetlands

9.1.1 Import from NASS

To create an agricultural grid from the NASS database from LifeSim:

1. From the LifeSim main window (Figure 9-1), from the **Study** tab, from the **Study Tree**, right-click on **Agricultural Data**, from the shortcut menu (Figure 9-1), click **Import Agricultural Data From NASS**. Another way is from the LifeSim main window, from the **Study** menu, point to **Agricultural Crop Data** (Figure 9-2), click **Import From NASS**. Either way, the **Import From NASS** dialog box (Figure 9-3) opens.

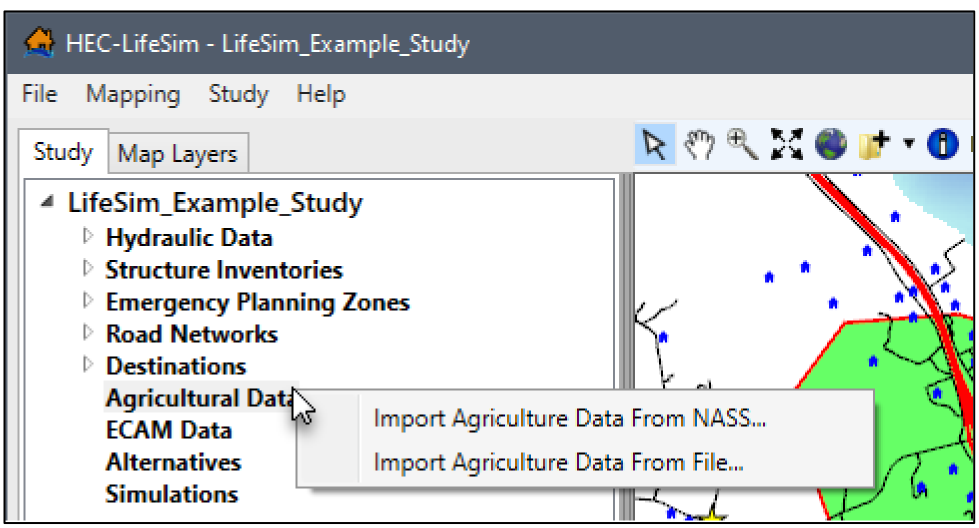


Figure 9-1. LifeSim Study Tree – Agricultural Data Shortcut Menu

2. From the **Import From NASS** dialog box (Figure 9-3), provide a name for the agricultural grid in the **Agriculture Inventory Name** box (Figure 9-3).

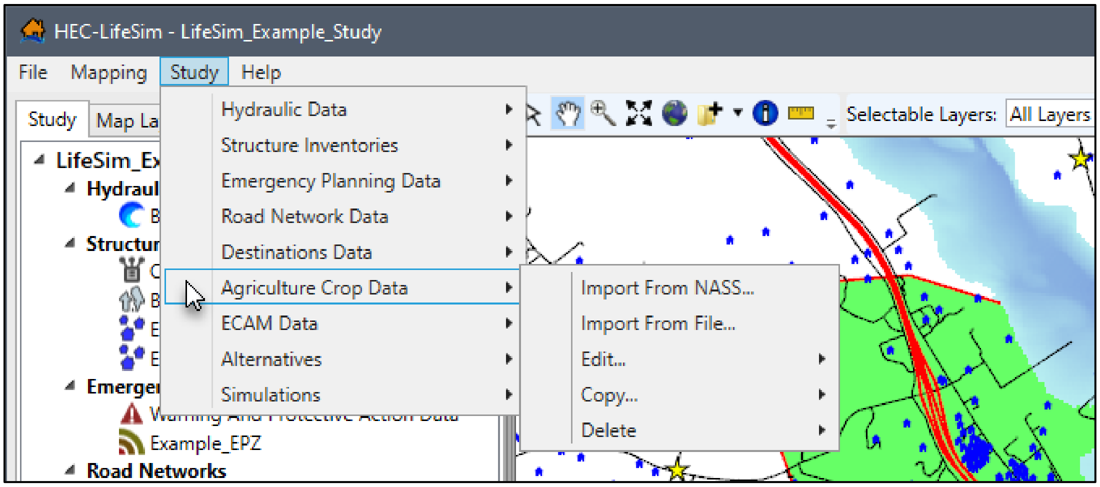


Figure 9-2. LifeSim Main Window – Study Menu – Agricultural Data Sub-menu

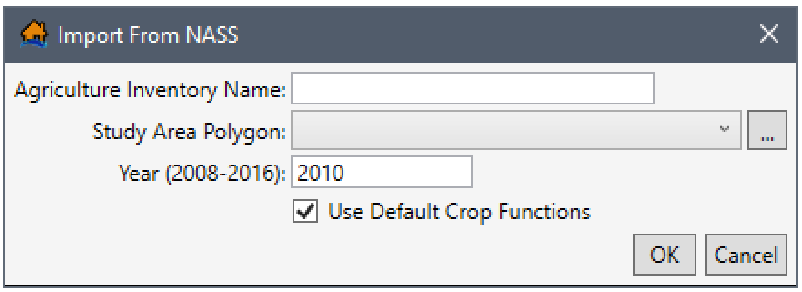


Figure 9-3. Import From NASS Dialog Box

3. Click the **Browse To** ... button (Figure 9-3) to select the study area polygon shapefile that provides the boundary polygon for limiting the agricultural data imported from the NASS database. An **Open** browser window (Figure 9-4) opens; navigate to the directory where the desired polygon shapefile is located. Select the correct shapefile (e.g., **.shp*), click **Open**, the **Open** browser closes (Figure 9-4) and the user will be returned to the **Import From NASS** dialog box (Figure 9-3).

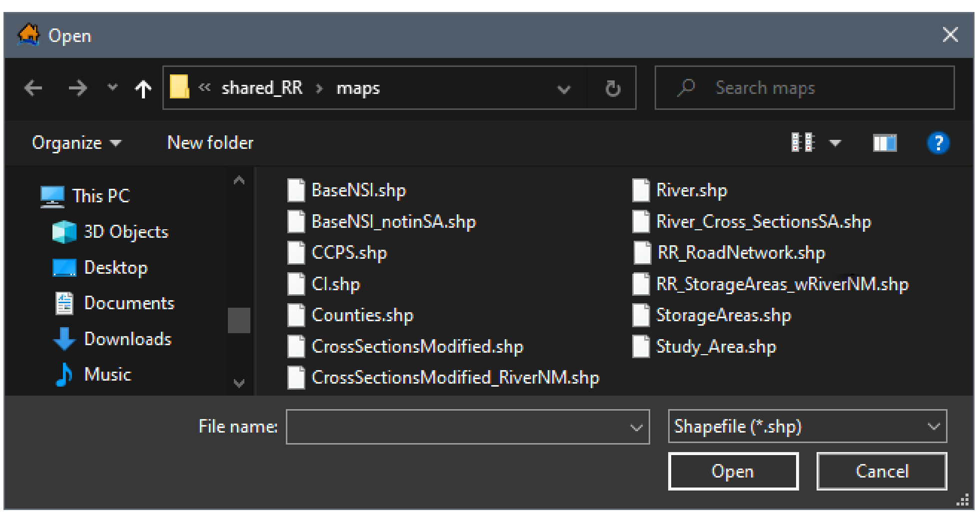


Figure 9-4. Open Browser Window

The location and name of the selected polygon shapefile will display in the **Study Area Polygon** dropdown menu (Figure 9-5). Alternatively, if the shapefile has already been imported as a map layer, select the appropriate map layer from the **Study Area Polygon** dropdown menu (Figure 9-3).

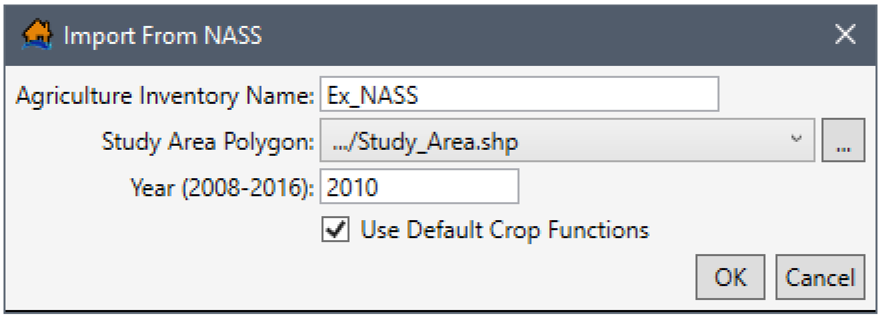


Figure 9-5. Example – Import From NASS Dialog Box

4. From the **Year (2008-2016)** box, enter the desired year (e.g., ***2010*** in Figure 9-5) that corresponds to the year of the flood event. Users can enter any year between 2008 and 2016.

5. Check the **Use Default Crop Functions** ☑ checkbox (Figure 9-5) to import the default crop functions.

6. Click **OK**, the **Import From NASS** dialog box (Figure 9-5) closes. LifeSim then filters the NASS CDL (**.tif*) to only include cells within a boundary polygon created based on the selected study area polygon's top, bottom, left and right geometric extent. **NOTE:** If the study area polygon only contains non-crop NASS categories (listed in Table 9-1), then a message window opens stating that "There are no valid crop ID's in this gridded data." Users must select a more suitable **Study Area Polygon** (Figure 9-5) before continuing.

   Once the import is completed, from the LifeSim main window (Figure 9-8), from the **Study Tree**, under **Agricultural Data**, the name of the agricultural grid appears.

9.1.2 Import from File

Access and download the NASS Cropland Data Layer (CDL) from the following website: https://www.nass.usda.gov/Research_and_Science/Cropland/SARS1a.php. The NASS CDL agricultural grid is in provided as a zipped raster based GeoTiff (**.tif*) file. The GeoTiff file contains a single CDL data layer, where each cell represents a crop type (review the HEC-FIA Version 3.0 User's Manual, Appendix B for tips preparing NASS data for import into LifeSim). To create an agricultural grid from a NASS CDL GeoTIFF file:

1. From the LifeSim main window (Figure 9-1), from the **Study** tab, from the **Study Tree**, right-click on **Agricultural Data**, from the shortcut menu (Figure 9-1), click **Import Agricultural Data From File**. Another way is from the LifeSim main window, from the **Study** menu, point to **Agricultural Crop Data** (Figure 9-2), click **Import From File**. Either way, the **Import From File** dialog box (Figure 9-6) opens.

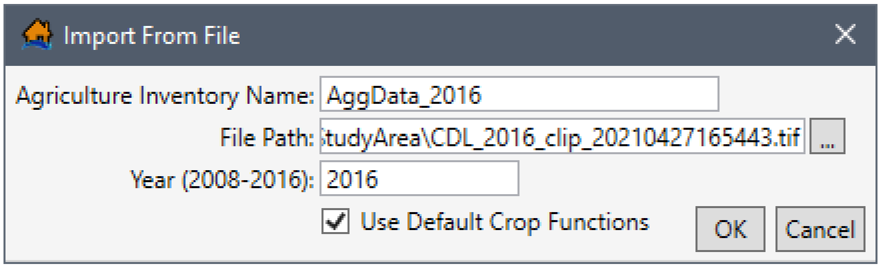


Figure 9-6. Example – Import From File Dialog Box

2. From the **Import From File** dialog box (Figure 9-6), provide a name for the agricultural grid in the **Agriculture Inventory Name** box (Figure 9-6).

3. Click the **Browse To** ... button (Figure 9-6). An **Open** browser window (Figure 9-7) opens; navigate to the directory where the desired NASS CDL GeoTiff (**.tif*) file is located. Select the correct GeoTiff file, click **Open**, the **Open** browser closes (Figure 9-7) and the user will be returned to the **Import From File** dialog box (Figure 9-6). The path and file name will display in the **File Path** box (Figure 9-6).

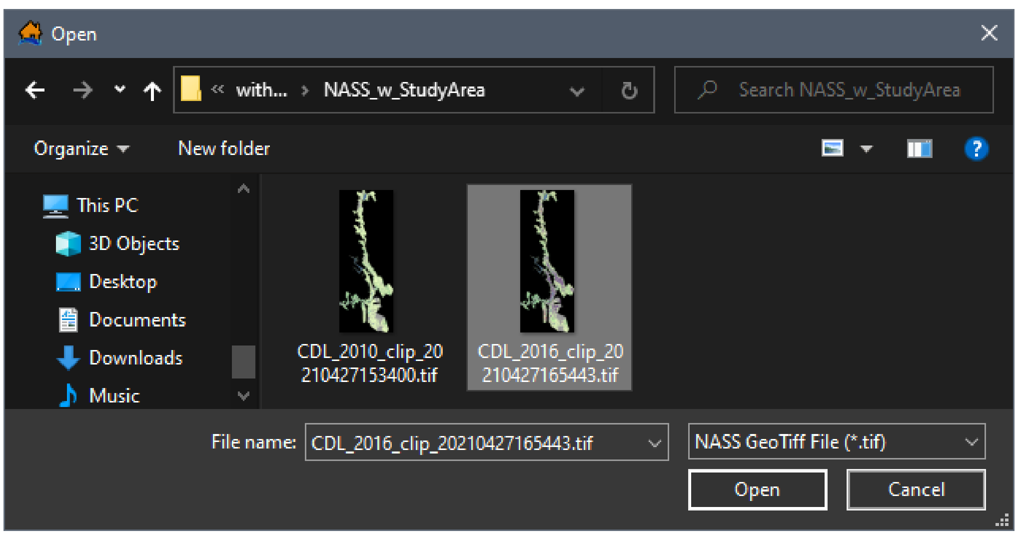


Figure 9-7. Open Browser Window

4. From the **Year (2008-2016)** box, enter the desired year (e.g., ***2016*** in Figure 9-6) that corresponds to the year of the flood event. Users can enter any year between 2008 and 2016.

5. Check the **Use Default Crop Functions** ☑ checkbox (Figure 9-6) to import the default crop functions.

6. Click **OK**, the **Import From File** dialog box (Figure 9-6) closes and LifeSim imports the NASS CDL GeoTiff file. **NOTE:** If the study area polygon only contains non-crop NASS categories (listed in Table 9-1), then a message window opens stating that "There are no valid crop ID's in this gridded data." Users must select a GeoTiff file that contains valid NASS crop categories before continuing.

Once the import is completed, from the LifeSim main window (Figure 9-8), from the **Study Tree**, under **Agricultural Data**, the name of the agricultural grid appears.

9.2 Display an Agricultural Grid

Users can view the agricultural data in the map window. From the **Study Tree**, under **Agricultural Data**, right-click on the name of the created agricultural data (e.g., ***AggData_2016*** in Figure 9-8), and from the shortcut menu, click **Show In Map Window** (Figure 9-8). The selected agricultural grid displays in the map window.

From the agricultural data **Study Tree** shortcut menu (Figure 9-8), the user can display or remove the agricultural grid from the active map window, edit the agricultural grid, move the agricultural data up or down in the **Study Tree**, copy, and/or delete the agricultural grid from the study.

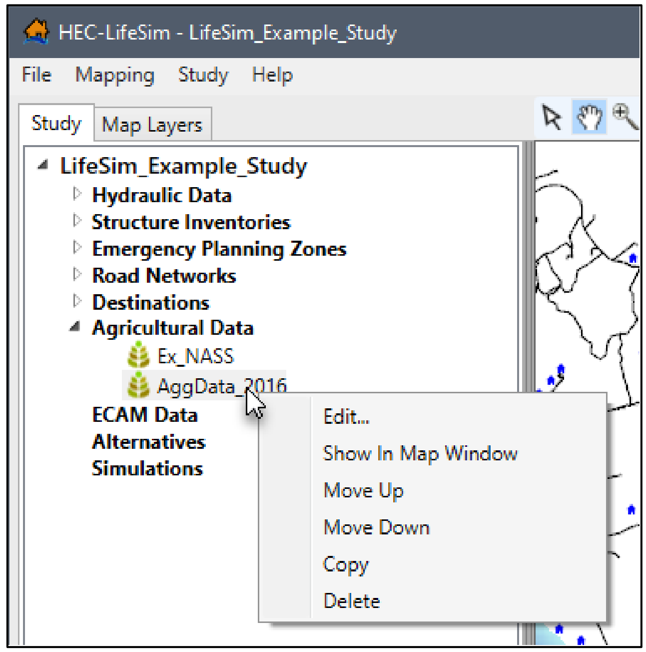


Figure 9-8. LifeSim Study Tree – Agricultural Data – Shortcut Menu

Note, right-click a displayed agricultural grid and from the shortcut menu (Figure 9-8), click **Remove From Map Window**, to stop displaying the agricultural data in the map window.

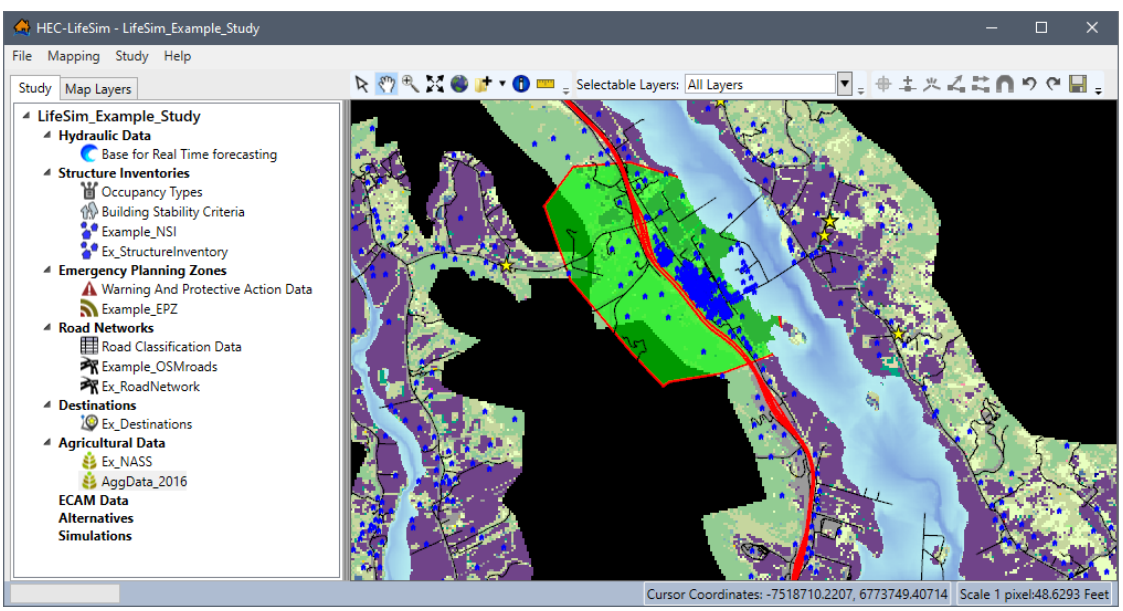


Figure 9-9. Example -- LifeSim Main Window – Agricultural Data Displayed in Map Window

9.3 Edit Agricultural Grid

When completing an agricultural economic loss assessment in LifeSim, Economic Guidance Memorandum (EGM) 14-02 states that current normalized agricultural prices can be found at the U.S. Department of Agriculture (USDA) website (Appendix C). The EGM also states that the USACE website will no longer disseminate current normalized agricultural prices, but its location will be provided as a memorandum. Each crop category selected (from the **Crop Editor**, Figure 9-10) to be used in the damage compute requires planting date information and monthly crop production costs. Typically, crop planting and production information can be obtained from a local land-grant university through an agricultural extension agency. For example, the USDA National Institute of Food and Agriculture's land-grant University directory. Additionally, price, yield, and harvest cost and date information need to be populated. This information typically comes from the agriculture extension service or from the NASS website.

Note, when agriculture data is imported with the option to **Use Default Crop Functions** ☑ (Figure 9-5 and Figure 9-6) then several crops in the **Crop Editor** (e.g., ***Alfalfa*** in Figure 9-10) contain pre-populated crop budgets, crop values, and crop duration-damage relationships.

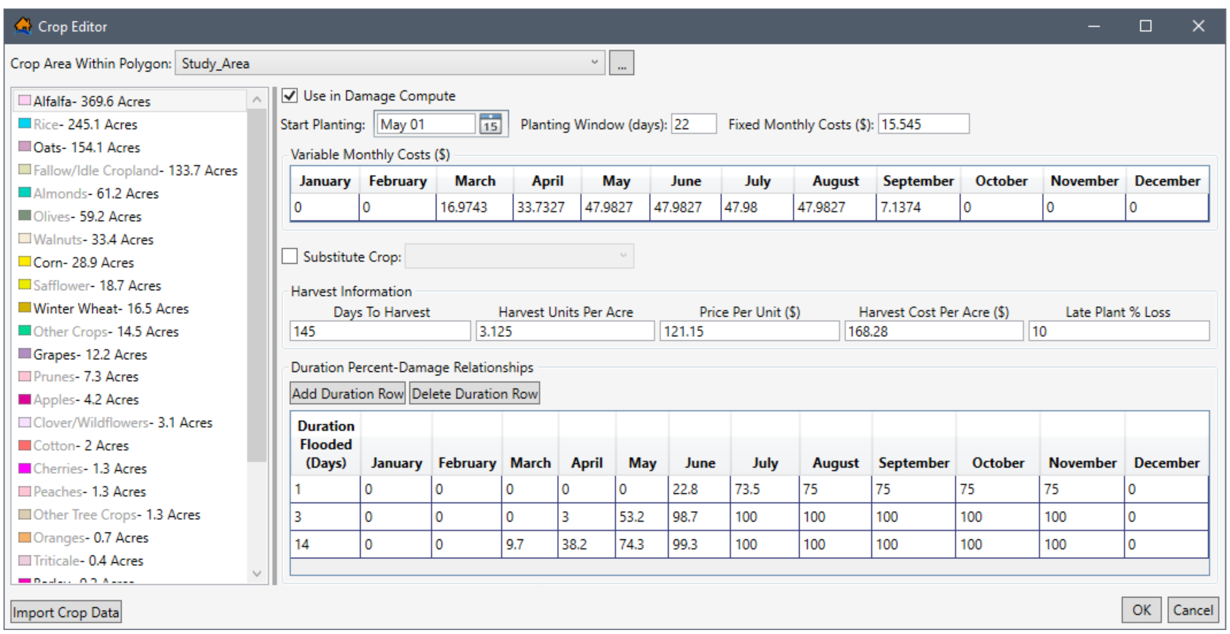


Figure 9-10. Example – Crop Editor with Default Crop Functions Loaded

9.3.1 Open the Crop Editor

To edit agricultural data:

1. From the **Study Tree**, under **Agricultural Data**, right-click on the name of the created agricultural data (e.g., ***AggData_2016*** in Figure 9-8), and from the shortcut menu, click **Edit** (Figure 9-8). Another way is from the LifeSim main window, from the **Study** menu, point to **Agricultural Crop Data** (Figure 9-2), point to **Edit**, and select the appropriate agricultural dataset name. Either way, the **Crop Editor** opens (Figure 9-10).

2. From the **Crop Area Within Polygon** dropdown menu, select the appropriate study area boundary polygon (e.g., ***Study_Area*** in Figure 9-10).

   The left side of the **Crop Editor** (Figure 9-10) provides a view of the crops in a tree (**Crop Data Tree**) for the selected agricultural data. The boundary polygon updates the **Crop Data Tree** to display the total acreage of each crop within the selected polygon.

   Select a crop name (e.g., ***Alfalfa*** in Figure 9-10) from the **Crop Data Tree** to display the data describing the selected crop in the editing panel located on the right side of the editor. For the example provided in Figure 9-10, ***Alfalfa*** is selected and the editing panel displays the pre-populated default crop functions.

9.3.2 Include Crop in Damage Compute

1. Navigate through the list of crops in the **Crop Data Tree** (Figure 9-10), and check ☑ the **Use in Damage Compute** checkbox to select the crops to include in the agricultural flood damages calculated during the economic damage assessment. By default, all crops listed in the **Crop Editor** (Figure 9-10) have the **Use in Damage Compute** checkbox ☐ unchecked; therefore, the user must select the specific crops to include in the damage assessment.

   After selecting the crop type from the **Crop Data Tree** and enabling the **Use in Damage Compute** option, enter (or edit default) crop functions (planting, harvesting, and duration percent-damage relationships) in the editing panel.

9.3.3 Define the Crop Budget

2. Enter (or edit) the crop planting data. Click 15 to select the **Start Planting** date and enter the **Planting Window (days)** in the textbox (Figure 9-10). The start planting date and planting window define the first and last possible plantings for the crop. Define the monthly crop planting costs by entering either a **Fixed Monthly Costs ($)** value, or by completing the **Variable Monthly Costs ($)** table (Figure 9-10).

3. If desired, check ☑ the **Substitute Crop** checkbox to enable the dropdown menu, and select a secondary crop (Figure 9-10). The **Substitute Crop** option allows users to identify a secondary crop which LifeSim will use during the agricultural flood damages calculation if flooding is determined to limit the ability to plant the primary crop. Review the LifeSim Version 2.0 Technical Reference Manual, Chapter 7, for more information regarding the computational procedures used to calculate agricultural flood damages. **NOTE:** When using a substitute crop, users must include the secondary crop in the damage compute and define the crop functions.

9.3.4 Define Crop Harvest Data

4. From the **Harvest Information** panel (Figure 9-10) enter the days required to harvest the crop, the yield (units per acre), the unit price, the harvest cost (per acre), and the

percentage of the total crop value lost due to late planting. To enter the harvest data, select the textbox and enter the appropriate value.

9.3.5 Define Crop Percent-Damage Relationships

5. The **Duration Percent-Damage Relationships** panel (Figure 9-10) provides a table for defining the crop damage curve. The damage curve defines the percentage of crop loss per month based on the duration (i.e., the number of days) crops are inundated each month.

6. By default, un-defined crops contain one row for adding a duration percent-damage relationship. However, users can add damage curves by clicking the **Add Duration Row** (Figure 9-10) button. Double-click a cell in the table to enter the duration flooded (days) and/or percent crop loss per month. Alternatively, right-click a row and from the shortcut menu (Figure 7-2), click **Insert Row(s)** to insert a row above the selected row or click **Add Row(s)** to add a row at the bottom of the table.

7. To remove damage curves, select the appropriate row and click the **Delete Bottom Duration Row** (Figure 9-10) button. Alternatively, right-click the appropriate row and from the shortcut menu (Figure 7-2), click **Delete Row(s)** to remove the selected row.

8. Click **OK** to close the **Crop Editor** (Figure 9-10) and save all entered crop data.

9.3.6 Import Crop Data Option

If crop functions (planting, harvesting, and duration percent-damage relationships) have been defined for a different agricultural dataset, then users have the ability to select specific crop data to import. However, use caution as the import overwrites any existing crop data in the selected agricultural dataset, and this action cannot be undone.

To import crop data from another LifeSim Version 2.0 study:

1. From the **Crop Editor** (Figure 9-10), click the **Import Crop Data** button (Figure 9-10). An **Open** browser window (Figure 9-11) opens. Navigate to the directory where the desired LifeSim Study (*.*fia*) file is located, select the appropriate file, and click **Open** (Figure 9-11).

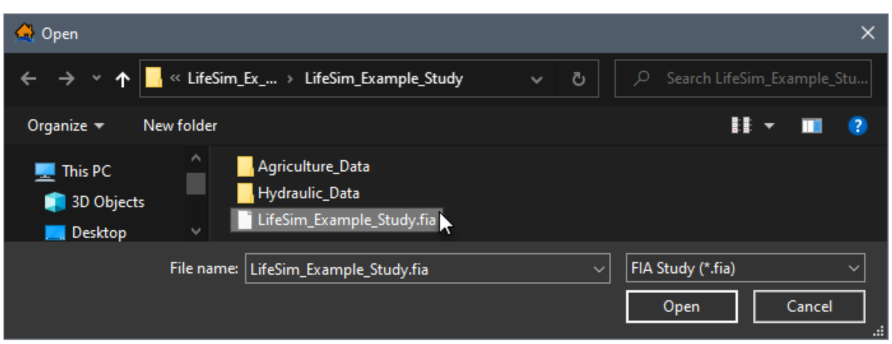


Figure 9-11. Open Browser Window

2. The **Open** browser closes (Figure 9-11) and the **Crop Import Dialog** (Figure 9-12) opens. From the **Crop Data Sets** panel (Figure 9-12), select the desired crop data set to import the crop data from (e.g., ***AggData_2016***).

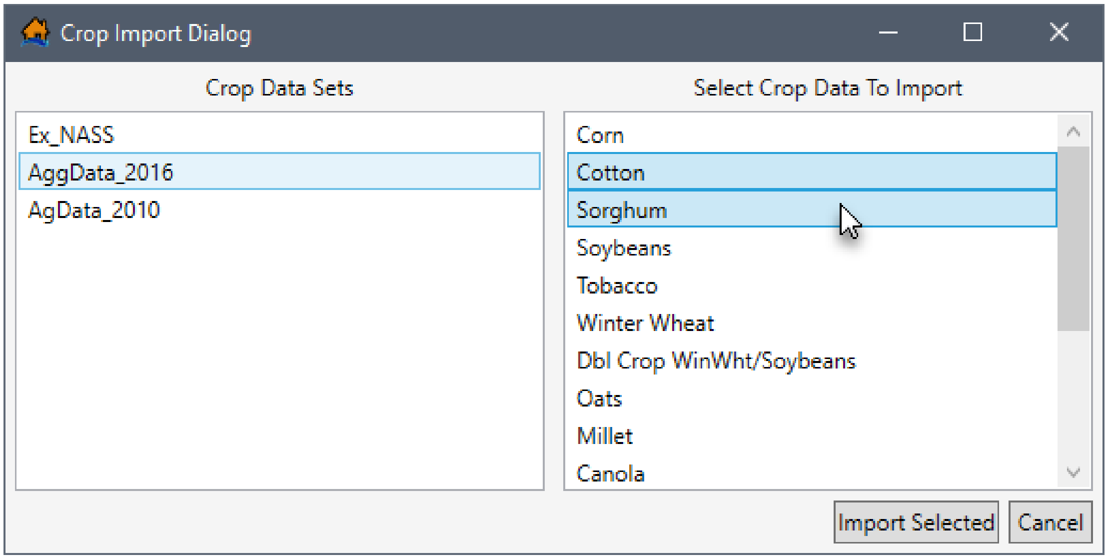


Figure 9-12. Crop Import Dialog – Example Crop Data Set Selected

3. From the **Select Crop Data To Import** panel (Figure 9-12), click to select (click a second time to unselect) the desired crop functions to import.

4. Click **the Import Selected** (Figure 9-12) button to import the selected crop functions. Note, the import overwrites any existing crop data in the selected agricultural dataset, and this action cannot be undone. The **Crop Import Dialog** (Figure 9-12) closes and the crop data for the selected crops is updated in the **Crop Editor** (Figure 9-10).

5. Click **OK** to close the **Crop Editor** (Figure 9-10) and save all entered crop data.

9.4 Copy an Agricultural Grid

To create a copy of the agricultural grid, from the **Study Tree**, under **Agricultural Data**, right-click on the name of the created agricultural data (e.g., ***AggData_2016*** in Figure 9-8), and from the shortcut menu, click **Copy** (Figure 9-8). Another way is from the LifeSim main window, from the **Study** menu, point to **Agricultural Crop Data** (Figure 9-2), point to **Copy**, and select the appropriate agricultural dataset name.

Either way, the **Name of New Study Item** dialog box (Figure 5-19) opens. Enter a new name in the **Name** box. Click **OK**, the **Name of New Study Item** dialog box (Figure 5-19) closes and the copied agricultural dataset displays in the **Study Tree** under the **Agricultural Data** folder.

9.5 Delete an Agricultural Grid

To delete an agricultural grid dataset from the LifeSim study and study directory files, from the **Study Tree**, under **Agricultural Data**, right-click on the name of the created agricultural data, and from the shortcut menu, click **Delete** (Figure 9-8). Another way is from the LifeSim main window, from the **Study** menu, point to **Agricultural Crop Data** (Figure 9-2), point to **Delete**, and select the appropriate agricultural dataset name.

Either way, a **Confirm Delete** message window (Figure 9-13) opens asking for confirmation to delete the selected agricultural dataset. Click **Yes**, the **Confirm Delete** message window (Figure 9-13) closes, and the selected agricultural dataset is deleted, and no longer displays in the **Study Tree** under the **Agricultural Data** folder.

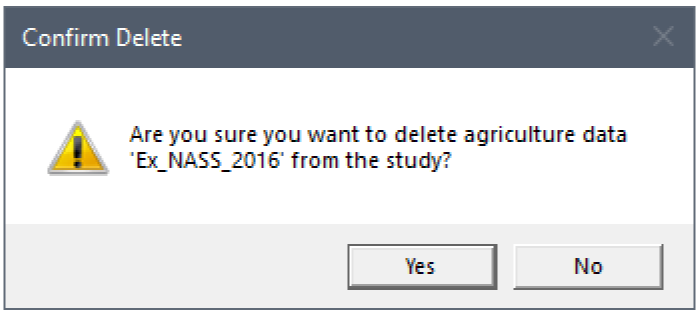


Figure 9-13. Confirm Delete Message Window

CHAPTER 10

ECAM Data

Economic Consequence Assessment Model (**ECAM**) data is a computable general equilibrium model used to evaluate the impact of shocks to the economy in terms of reductions in labor and/or capital. LifeSim allows users to access the ECAM model through the internet during simulations. To use ECAM during an LifeSim simulation, an internet connection is required. In the context of LifeSim, the model is used to evaluate the indirect economic damages associated with a flood event.

As stated above, the ECAM model requires estimates of the reductions in labor and capital due to flood events. Since labor reduction calculations take life loss risks into account, ECAM cannot be run unless the life loss compute option is turned on (review Section 11.1). The labor reductions are based on the number of work hours reduced due to people being displaced from their homes due to flooding and life loss. Capital reductions are based on the damages to commercial, industrial, and public structures and their contents. To convert these to ratios, a total supply of available labor and capital must be provided.

Because ECAM runs each county independently, values for labor and capital must be provided for each county in the study area. The total labor and capital must represent the total for the whole county, not the portion of the county in the study area. Available capital is entered as the total value of commercial, industrial and public structures. Labor should be entered as the total under 65 years of age population at all structures at 2AM. For more detail on computing indirect damages with ECAM, please refer to the LifeSim Version 2.0 Technical Reference Manual, Chapter 8.

10.1 Import ECAM Data

Next, from the **County Data Source** panel, select the polygon shapefile from the **County Shapefile** list that represents the counties in the study area. Select the polygon shapefile from the dropdown list, which contains all valid shapefiles added as map layers to the project; alternatively, click the button to browse to and open the appropriate shapefile. Following the county shapefile selection from the **5 Digit FIPS** list, select the appropriate field name (e.g., *GEOID*) that contains the county's 5-digit Federal Information Processing Standard (FIPS) code ("SSCCC", where SS is a two-digit state code and CCC is a three-digit county code). The county data table (located at the bottom of the ECAM tab) updates with the County FIPS and County Name attributes contained in the selected shapefile. Enter the Total Capital and Total Labor values for each county in the study area.

ECAM also requires users to enter the flood duration (hours), the reconstruction period duration (hours), and the cleanup period duration (hours). The flood duration depends on how quickly the flood waters recede. This data can usually be obtained from a hydraulic model. Currently, the user must provide a single value for the entire flooded area, so thought and judgment must be given to the definition of this duration time.

The flooding **Duration** depends on characteristics of the river, and the timing of floodwaters receding. Users can get a general idea of the flood duration for the event from the hydraulic model. The **Reconstruction Period** is the average number of days required to rebuild a damaged area to pre-flood economic conditions. The **Cleanup Period** is the average number of days required to remove debris from flooded area. The reconstruction and cleanup period durations depend on the types of structures located in the flooded area.

To import and create an ECAM dataset:

1. From the LifeSim main window (Figure 10-1), from the **Study** tab, from the **Study Tree**, right-click on **ECAM Data**, from the shortcut menu click **Import ECAM Data** (Figure 10-2). Another way is from the LifeSim main window, from the **Study** menu, point to **ECAM Data**, click **Import ECAM Data**. Either way, the **ECAM Data Importer** dialog box (Figure 10-3) opens.

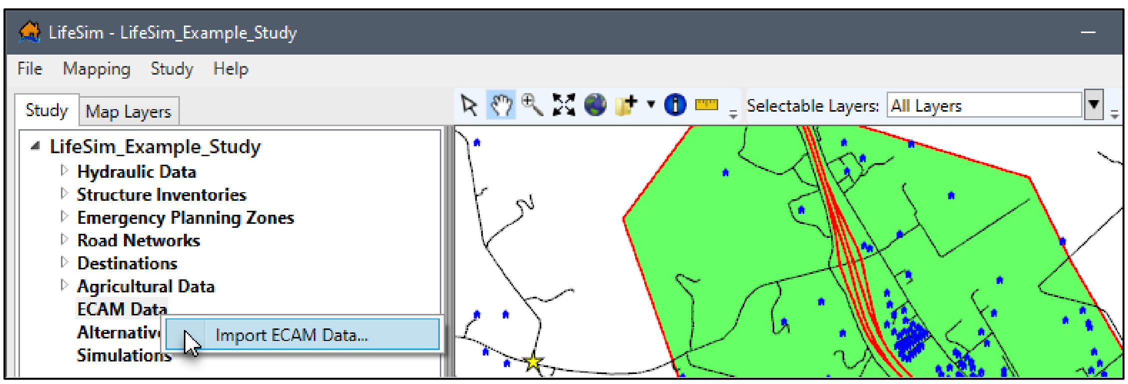


Figure 10-1. LifeSim Study Tree – ECAM Data Shortcut Menu

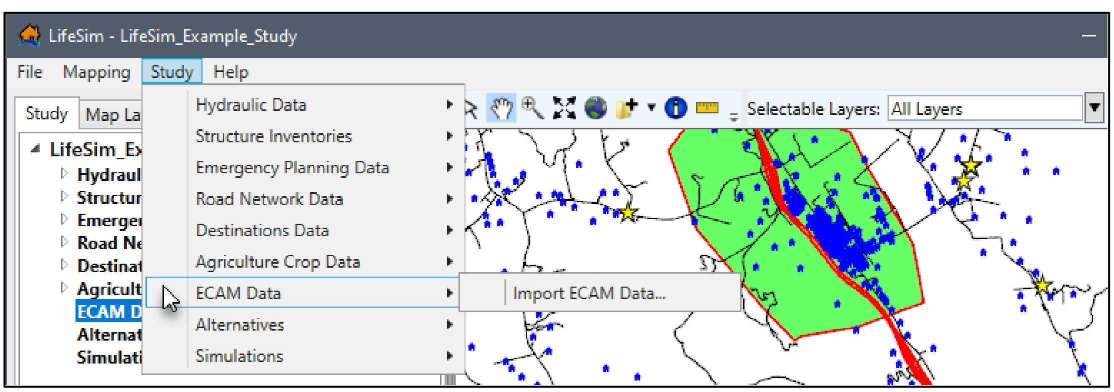


Figure 10-2. LifeSim Main Window – Study Menu – ECAM Data Sub-menu

2. In the **ECAM Data Name** box (Figure 10-3), enter a name for the ECAM dataset.

3. From the **Study Shape Data** dropdown list (Figure 10-3) select the desired imported map layer (e.g., *Ex_Destinations* in Figure 10-4). Alternatively, click the **Browse To** ... button to open an **Open** browser window (Figure 8-4). Navigate to the directory where

the desired shapefile (e.g., **.shp*) is located, select the shapefile and click **Open**. The **Open** browser window closes (Figure 8-4), and the **ECAM Data Importer** displays the selected shapefile in the **Study Shape Data** (Figure 10-3) list.

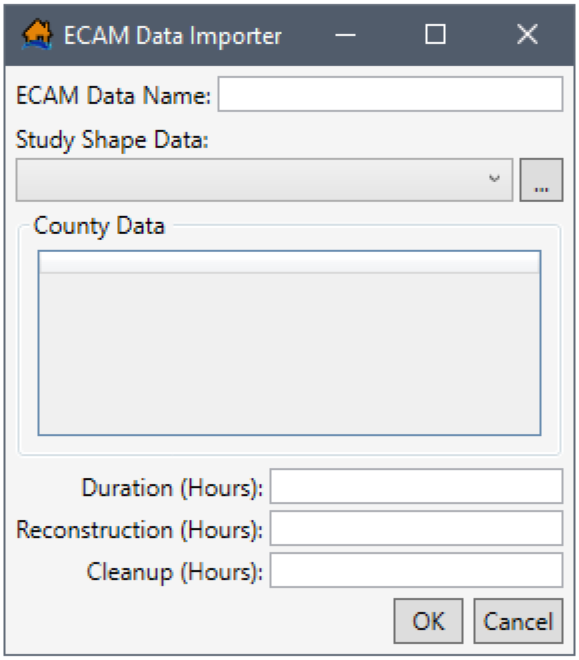


Figure 10-3. ECAM Data Importer

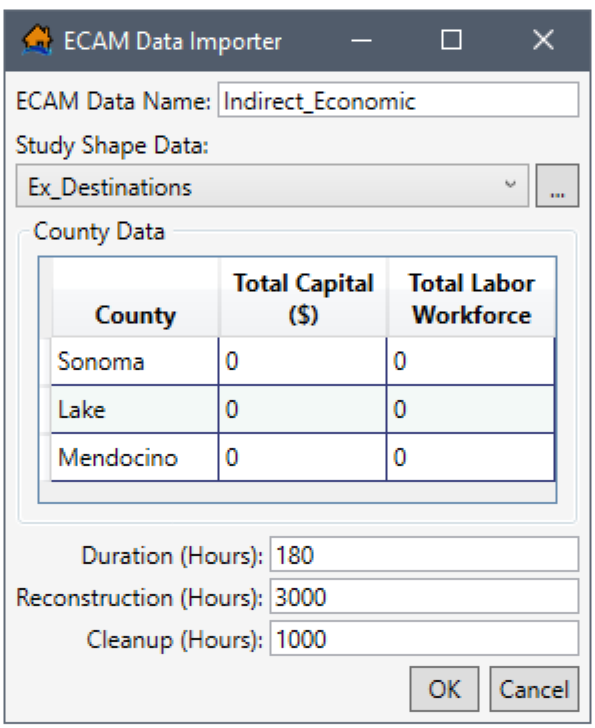


Figure 10-4. Example – ECAM Data Importer

4. The **Study Shape Data** selection updates the table in the **County Data** panel (Figure 10-4) with a list of the counties retrieved from the selected **Study Shape Data** (e.g.,

Ex_Destinations in Figure 10-4). The county data is retrieved based on the selected bounding polygon shapefile.

In other words, LifeSim filters the list of counties to only include areas within a boundary polygon created based on the selected study area's top, bottom, left and right geometric extent.

5. The ECAM model requires estimates of the reductions in labor and capital due to flood events and runs each county independently. Therefore, for the desired counties in the **County Data** table (Figure 10-4), enter the **Total Capital ($)** and **Total Labor Workforce** values. Note, labor should be entered as the total population under 65 years of age at all structures at night (2AM).

6. At the bottom of the **ECAM Data Importer** (Figure 10-4), enter the **Duration (Hours)** of the flood, and the **Reconstruction (Hours)** and **Cleanup (Hours)** after the flood event. These estimates are used to calculate the labor reductions based on the number of work hours reduced due to and following the event.

7. Click **OK**, the **ECAM Data Importer** dialog box (Figure 10-4) closes and LifeSim imports the ECAM data. **NOTE:** In LifeSim Version 2.0 users can view but cannot modify the entered duration, reconstruction, and cleanup hours after creating the ECAM dataset.

10.2 View ECAM Data

Users can view the ECAM dataset from the Map Layers tab in the map window. Created ECAM data can be copied or deleted, viewed, and/or displayed in the map window.

10.2.1 View ECAM Data

To view the duration, reconstruction, and cleanup hours for an ECAM dataset, from the LifeSim main window (Figure 10-5), from the **Study** tab, from the **Study Tree**, under **ECAM Data**, right-click an ECAM dataset (e.g., ***Indirect_Economic*** in Figure 10-5).

From the shortcut menu (Figure 10-5), click **View**. The **ECAM Data Viewer** (Figure 10-6) opens and displays the name, duration, reconstruction and cleanup information for the selected dataset. Click **Close** (Figure 10-6), to close the **ECAM Data Viewer** (Figure 10-6).

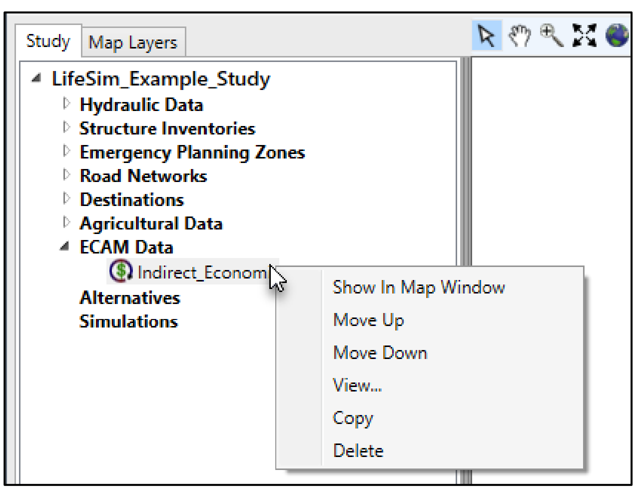


Figure 10-5. LifeSim Study Tree – ECAM Data – Shortcut Menu

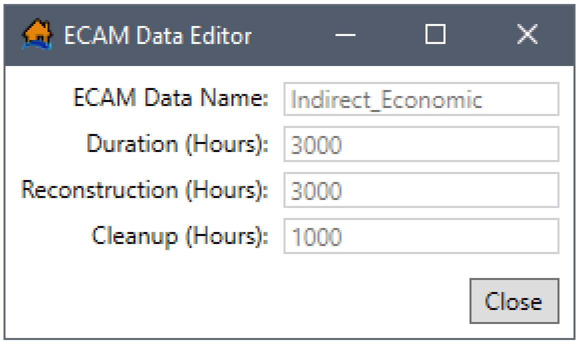


Figure 10-6. Example – ECAM Data Editor

10.2.2 Show in Map Window

To view the ECAM dataset in the map window, from the LifeSim main window (Figure 10-5), from the **Study** tab, from the **Study Tree**, from the **ECAM Data** folder, right-click on an ECAM dataset name. From the shortcut menu (Figure 10-5), click **Show In Map Window**. The selected ECAM dataset is added to the **Map Layer** tab and displays in the map window (Figure 10-7). Note, from the **Study** tab, right-click the ECAM dataset and click **Remove From Map Window** from the shortcut menu to stop displaying the dataset. Alternatively, from the **Map Layers** tab, right-click the ECAM dataset and click **Remove** (Figure 10-7) from the shortcut menu.

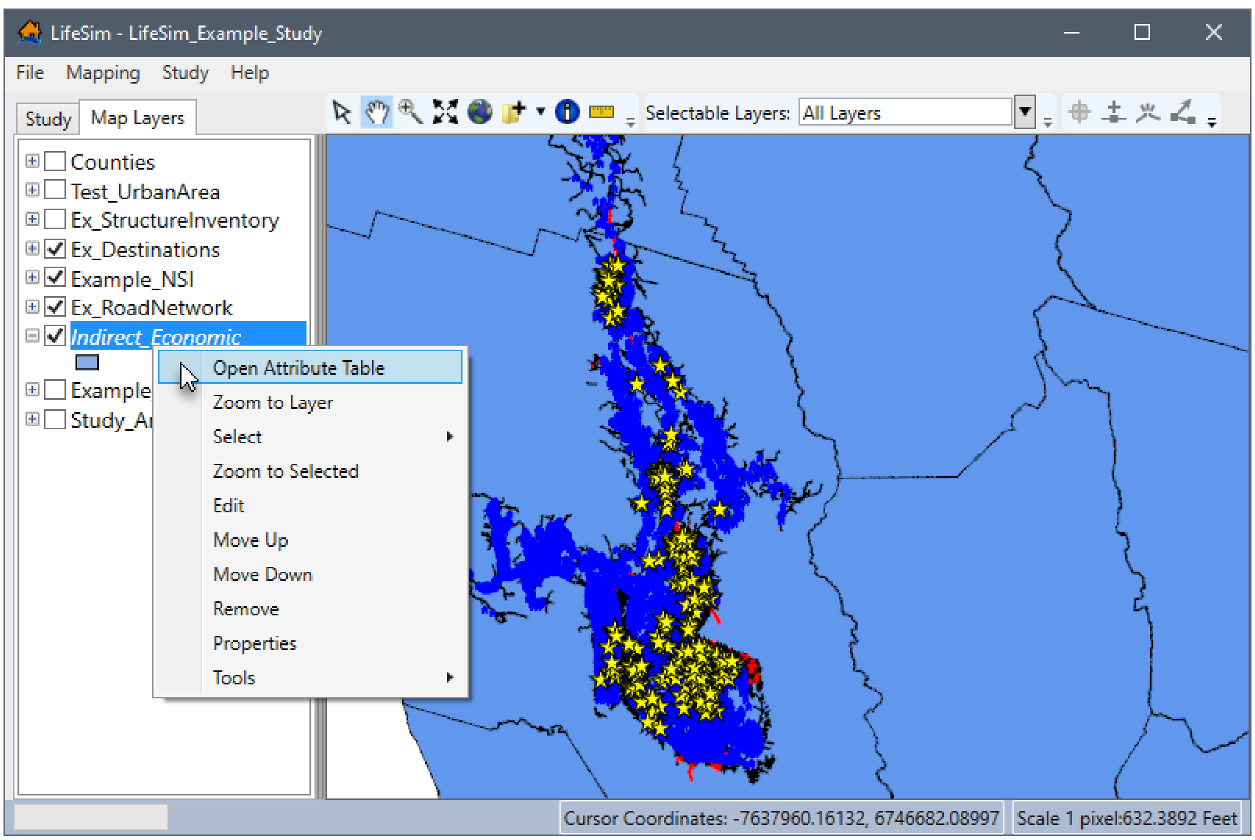


Figure 10-7. Example -- LifeSim Main Window – ECAM Data Displayed in Map Window

10.2.3 View Attributes Table

Map layer attributes can be accessed for each map layer and is provided in a table for the dataset referenced by the map layer. The ECAM dataset must be displayed in the map window before the attribute table can be accessed. To access a specific map layer **Attribute** table, from the **Map Layers** tab, right-click on a map layer of interest, from the shortcut menu, click **Open Attribute Table** (Figure 10-7). The *Map Layer* **Attributes** dialog box opens; review Appendix A, Section A.4 for more information.

10.2.4 Edit Attributes Table

From the opened **Attribute** table for the ECAM dataset, click the Start Feature Edit Session to edit the total capital ($) and total labor values by county. Further information regarding editing the map layer attributes is provided in this manual in Appendix A, Section A.10.

10.3 Copy an ECAM Dataset

To copy an ECAM dataset from an existing dataset, from the LifeSim main window (Figure 10-5), from the **Study** tab, from the **Study Tree**, under **ECAM Data**, right-click an ECAM dataset (e.g., ***Indirect_Economic*** in Figure 10-5), from the shortcut menu (Figure 10-5), click **Copy**. The **Name of New Study Item** dialog box (Figure 5-19) opens. Enter a new name in the **Name** box. Click **OK**, the **Name of New Study Item** dialog box (Figure 5-19) closes and the copied ECAM dataset displays in the **Study Tree** under the **ECAM Data** folder.

10.4 Delete an ECAM Dataset

To delete a ECAM dataset from the LifeSim study and study directory files, from the LifeSim main window (Figure 10-5), from the **Study** tab, from the **Study Tree**, under **ECAM Data**, right-click an ECAM dataset (e.g., ***Indirect_Economic*** in Figure 10-5), from the shortcut menu (Figure 10-5), click **Delete**. A **Confirm Delete** message window (similar to Figure 9-13) opens asking for confirmation to delete the selected ECAM dataset. Click **Yes**, the **Confirm Delete** message window (similar to Figure 9-13) closes, and the selected ECAM dataset is deleted, and no longer displays in the **Study Tree** under the **ECAM Data** folder.

CHAPTER 11

Alternatives

An **alternative** in LifeSim allows the user to define what input data sources and parameters will make up the simulation computations. Any combination of input data sources can be used to define an alternative. For example, one alternative may use a road network that allows for contra-flow and another may use a different hydraulic event or different warning issuance times. Setting up multiple alternatives allows the user to quantify and compare the effects of operational changes on the hazard analysis. Alternatives in LifeSim represent the parameters that define a simulation and include defining:

- Input data sources (e.g., hydraulics, structure inventory, road network, ECAM data),
- Warning issuance times,
- Destination locations by Emergency Planning Zone (EPZ), and
- Various other parameters (e.g., evacuation parameters, life loss probability, stability criteria).

11.1 Create an Alternative

To create an alternative in LifeSim the user will provide a name, define the input data sources, and provide additional information (e.g., evacuation parameters).

1. From the LifeSim main window (Figure 11-1), from the **Study** tab, from the **Study Tree**, right-click on **Alternatives**, and click **Create New Alternative** from the shortcut menu. Alternatively, from the LifeSim main window, from the **Study** menu, point to **Alternatives**, and click **Create New**.

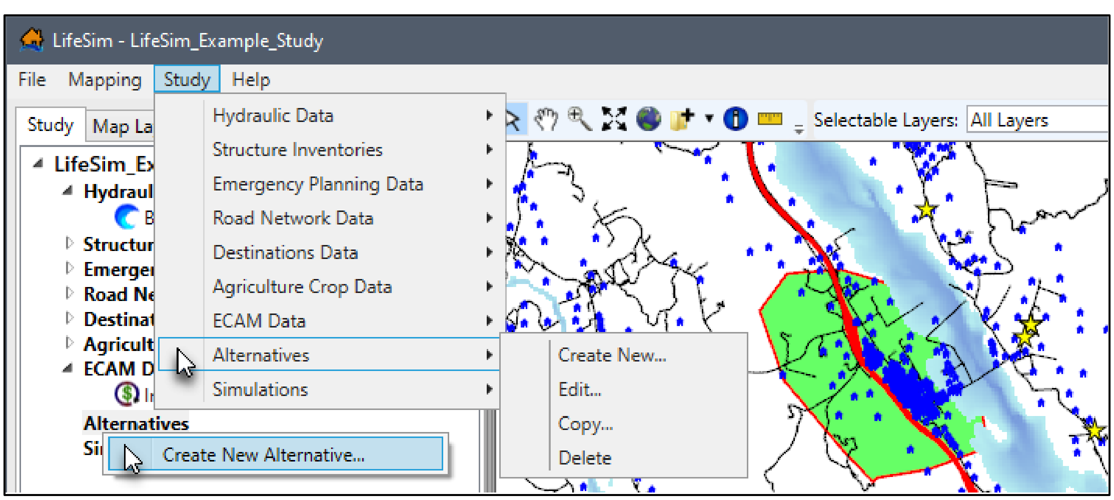


Figure 11-1. LifeSim Main Window – Study Menu Alternatives Sub-menu – Alternatives Study Tree Shortcut Menu

2. Either way, the **Create New Alternative** dialog box (Figure 11-2) opens. Enter a name for the alternative in the **Name** box (Figure 11-2).

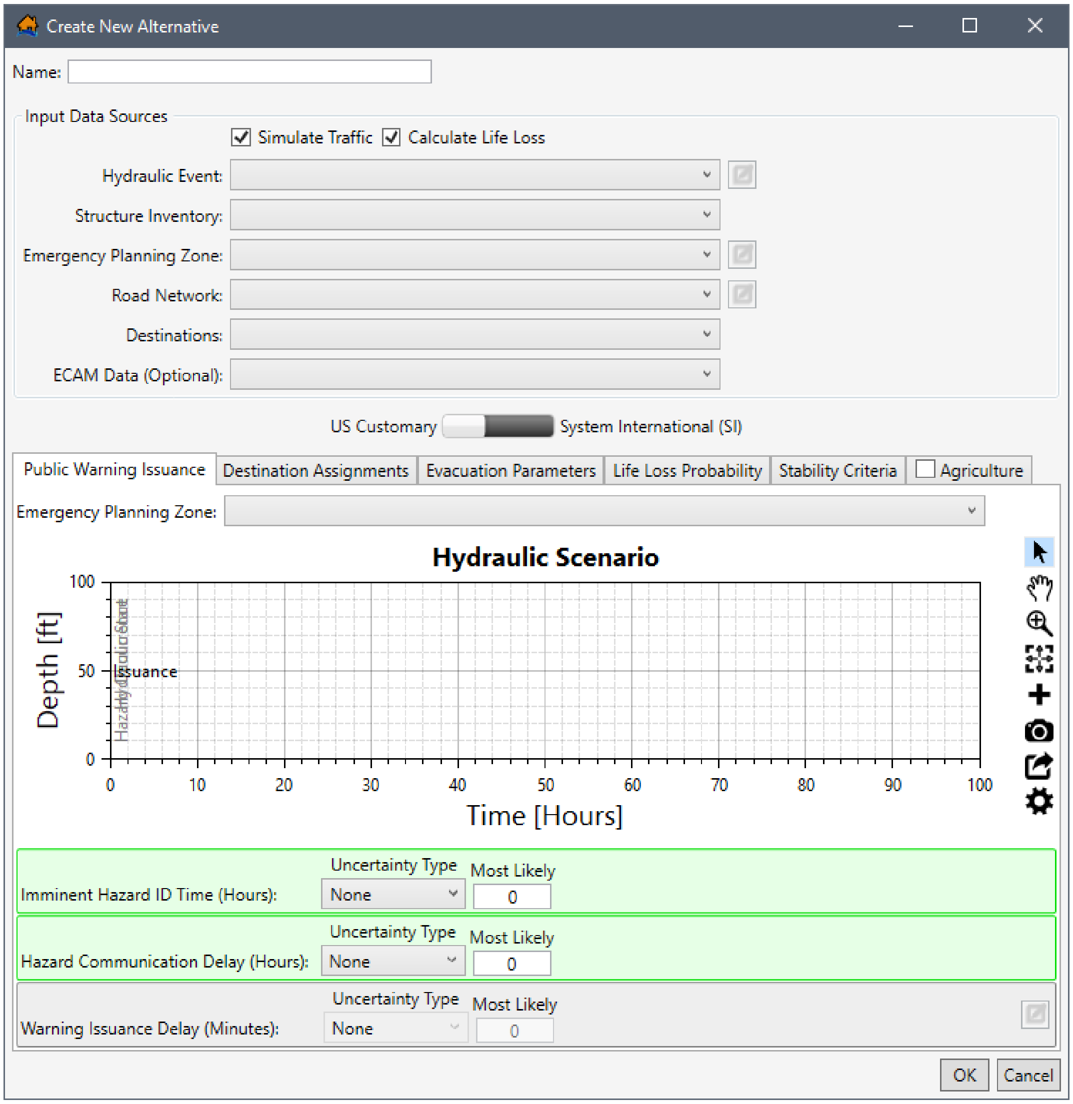


Figure 11-2. Create New Alternative Dialog Box

11.1.1 Define Input Data Sources

From the **Input Data Sources** panel (Figure 11-3), select the desired simulation compute options and define the desired input datasets for the new alternative.

1. By default, LifeSim will create a traffic simulation of the evacuation. To turn off the traffic simulation option, from the **Input Data Sources** panel, uncheck ☐ the **Simulate Traffic** checkbox (Figure 11-3).

2. By default, the **Calculate Life Loss** (Figure 11-3) option is turned on. To turn off the life loss calculation option, from the **Input Data Sources** panel, uncheck ☐ the **Calculate Life Loss** checkbox (Figure 11-3).

NOTE: If **Simulate Traffic** and **Calculate Life Loss** are turned off, then the road network and destinations datasets do not need to be defined.

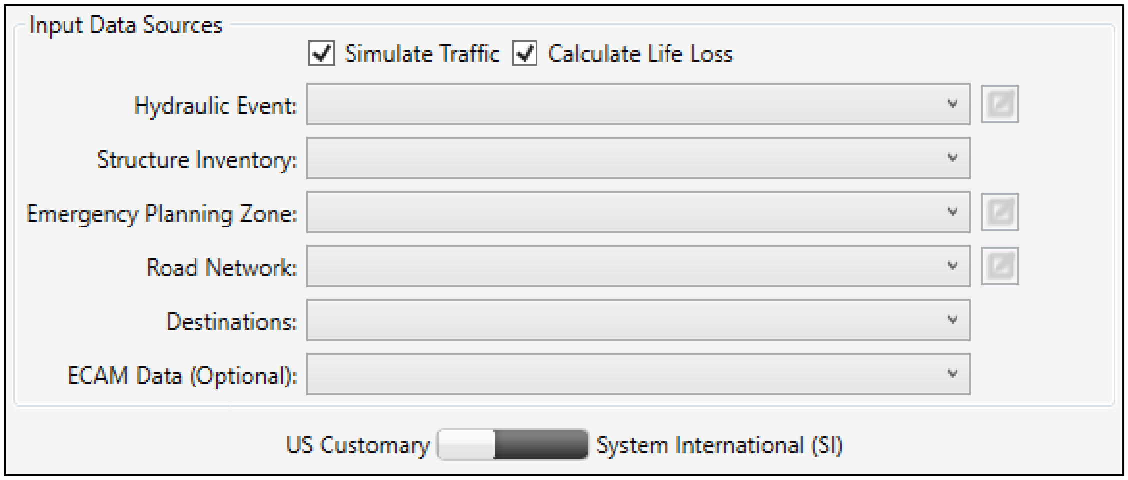


Figure 11-3. Create New Alternative Dialog Box – Input Data Sources

3. From the **Input Data Sources** panel (Figure 11-3), from the **Hydraulic Event** list, select a hydraulic dataset. If the selected hydraulic dataset requires editing, from the **Create New Alternative** dialog box (Figure 11-3), click , and the **Select Hydrograph and Timing Information** dialog box opens (see Chapter 4, Section 4.4 for further information).

4. From the **Structure Inventory** list (Figure 11-3), select a structure inventory.

5. From the **Emergency Planning Zone** list (Figure 11-3), select an EPZ dataset. If the selected EPZ dataset requires editing, from the **Create New Alternative** dialog box (Figure 11-3), click , and the **Emergency Planning Zone Editor** opens (see Chapter 6, Section 6.2 for further information).

6. When simulating traffic (**Simulate Traffic** checkbox is selected) select the appropriate road network dataset from the **Road Network** list, from the **Input Data Sources** box (Figure 11-3). If the selected road network dataset requires editing, from the **Create New Alternative** dialog box (Figure 11-3), click , and the **Edit Road Network** dialog box opens (see Chapter 7, Section 7.3 for further information).

7. When simulating traffic select the appropriate destinations dataset from the **Destinations** list, from the **Input Data Sources** box (Figure 11-3).

8. Below the **Input Data Sources** panel, select either **US Customary** or **System International (SI)** for the alternative. The **Public Warning Issuance**, **Destination Assignments**, **Evacuation Parameters**, **Life Loss Probability**, **Stability Criteria**, and **Agriculture** tabs, at the bottom of the **Create New Alternative** dialog box (Figure 11-2), provide the parameters for defining the simulation compute options. Section 11.1.2 describes the steps for defining the compute options for an alternative, for each tab in the **Create New Alternative** dialog box (Figure 11-2).

9. Click **OK**, the **Create New Alternative** dialog box (Figure 11-2) closes. The alternative is created and appears in the **Study Tree** under **Alternatives** in LifeSim main window (Figure 11-1).

11.1.2 Compute Options for an Alternative

LifeSim can analyze a variety of flood damages and impacts through a variety of computational methods. Flood damages and impacts as well as the methods used to estimate the impacts are selected as part of an alternative definition. For example, when simulating traffic and life loss, then users must define warning issuance scenarios, destination assignments, evacuation parameters, life loss probability functions, and stability threshold functions. The following sections describe the steps for defining the compute options for an alternative individually for each tab in the **Create New Alternative** dialog box (Figure 11-4).

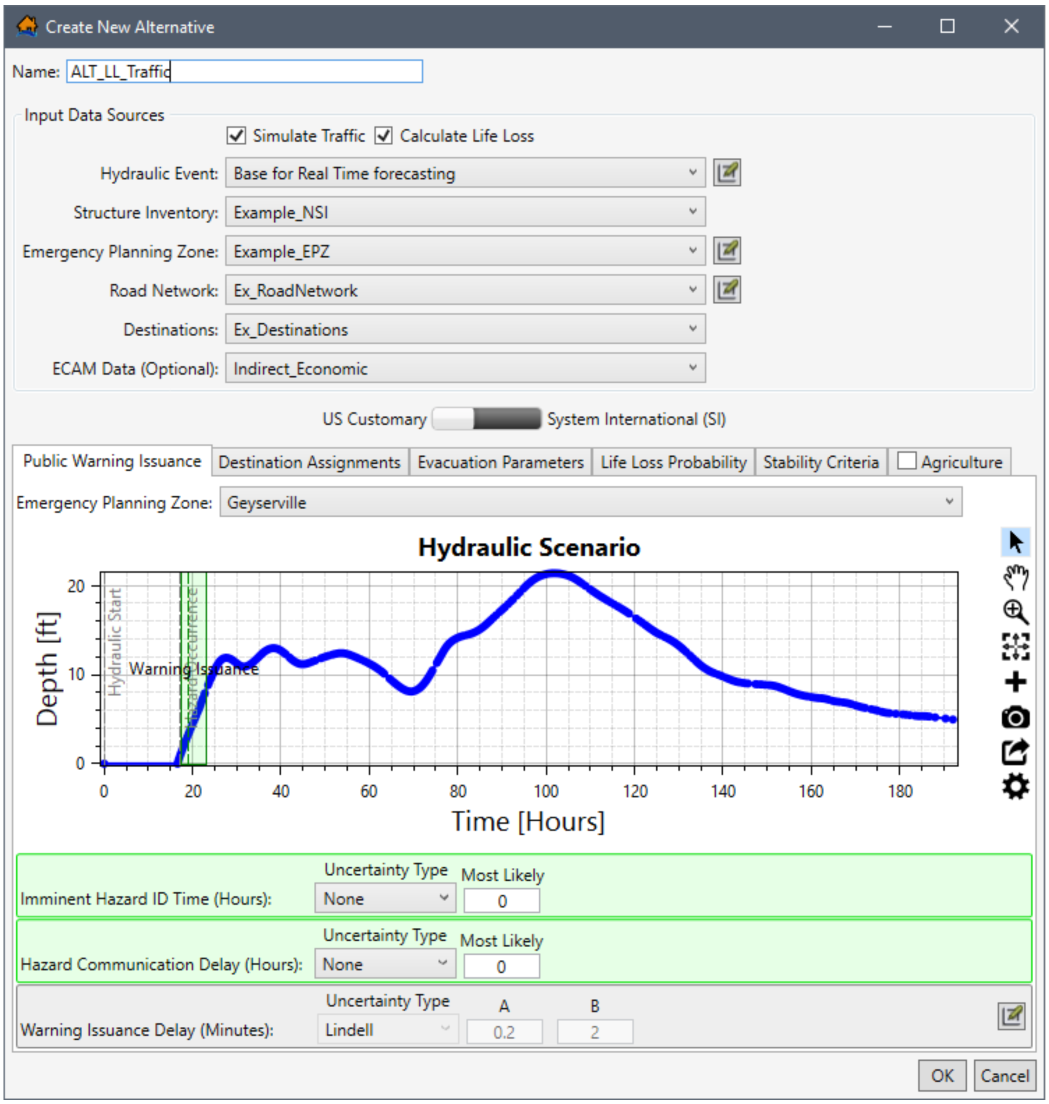


Figure 11-4. Create New Alternative – Public Warning Issuance Tab

Note, for the purposes of the discussion the units listed in this section will be in US Customary units. However, the **Create New Alternative** dialog box (Figure 11-4) allows users to select either **US Customary** or **System International (SI)**.

Public Warning Issuance

When simulating life loss, with or without simulating traffic or ECAM, the warning issuance times must be defined. To define the warning issuance scenarios:

1. From the **Create New Alternative** dialog box (Figure 11-4), click the **Public Warning Issuance** tab (Figure 11-4).

2. The **Emergency Planning Zone** list is a list of all the emergency planning zones in the selected EPZ dataset (Figure 11-3). Therefore, if a selected EPZ dataset has multiple zones then the list will contain the names of the multiple zones.

The public warning issuance scenario has three parameters. Each parameter can be defined with uncertainty and is used to define when warnings are issued to the population by zone in the emergency planning zone (EPZ) (Figure 11-5).

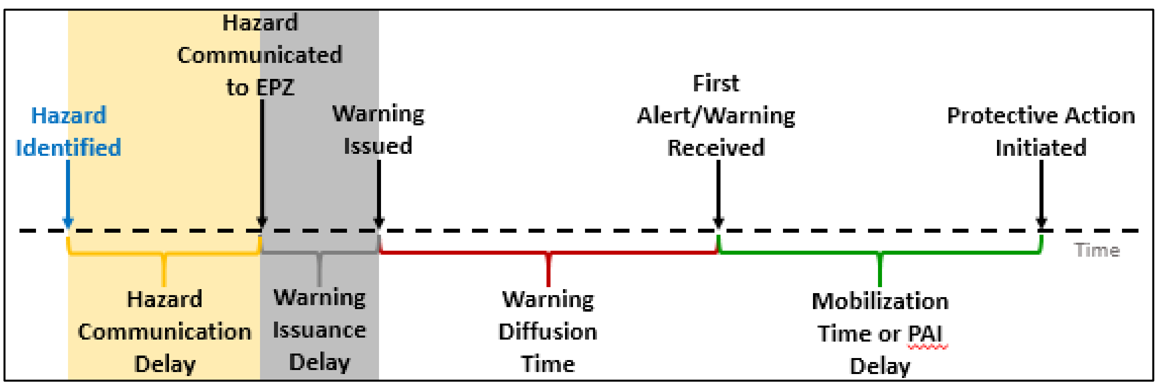


Figure 11-5. Conceptual diagram illustrating the flood warning and evacuation timeline. Highlighted sections relevant to warning issuance time are: Hazard Identified Relative Time (blue); Hazard Communication Delay (light yellow); and, Warning Issuance Delay (light grey).

3. The **Imminent Hazard ID Time (Hours)** parameter (Figure 11-4) is the period of time, in hours, between when the hazard is identified (ID) and the hazard occurs as defined by the selected hydraulic data. See section 4.3 for more information on the hazard occurrence time. From the **Uncertainty Type** dropdown menu (Figure 11-4), select and define the uncertainty in imminent hazard ID time. The default is **None**, however, the user can choose **Triangular** (minimum, maximum), **Normal** (standard deviation), and **Uniform** (minimum, maximum).

4. The **Hazard Communication Delay (Hours)** parameter (Figure 11-4) is the period of time between when the hazard is identified to when the EPZ representatives or Emergency Management Agency (EMA) is alerted. From the **Uncertainty Type** dropdown menu (Figure 11-4), select and define the uncertainty. The default is **None**,

however, the user can choose **Triangular** (minimum, maximum), **Normal** (standard deviation), and **Uniform** (minimum, maximum).

5. The **Warning Issuance Delay** parameter (Figure 11-4) is the time it takes from when the emergency managers receive the notification of the imminent hazard to when they issue an evacuation order to the public.

   The **Warning Issuance Delay** is part of the warning issuance process, but it is defined in the emergency planning zone (EPZ) dataset. However, if the warning issuance delay parameters require editing, click , to open the **Emergency Planning Zone Editor** and edit the warning issuance delay (see Chapter 6, Section 6.2 for further information). NOTE: By default, the **Emergency Planning Zone Editor** opens to the **Warning Issuance Delay** tab for the first EPZ zone in the **Emergency Planning Zone** list (in the **Emergency Planning Zone Editor**). In other words, the EPZ zone selected from the **Create New Alternative** (e.g., *Geyserville* in Figure 11-4) does <u>not</u> modify the EPZ selected in the opened **Emergency Planning Zone Editor**.

6. Either click **OK** to save the alternative and close the **Create New Alternative** dialog box (Figure 11-4) or select the next desired tab and continue to define the new alternative.

Destination Assignments

When simulating traffic (**Simulate Traffic** checkbox is selected, Figure 11-3) the user needs to define the destinations for the emergency planning zones; from the **Create New Alternative** dialog box (Figure 11-2), click the **Destination Zone Assignment** tab (Figure 11-6).

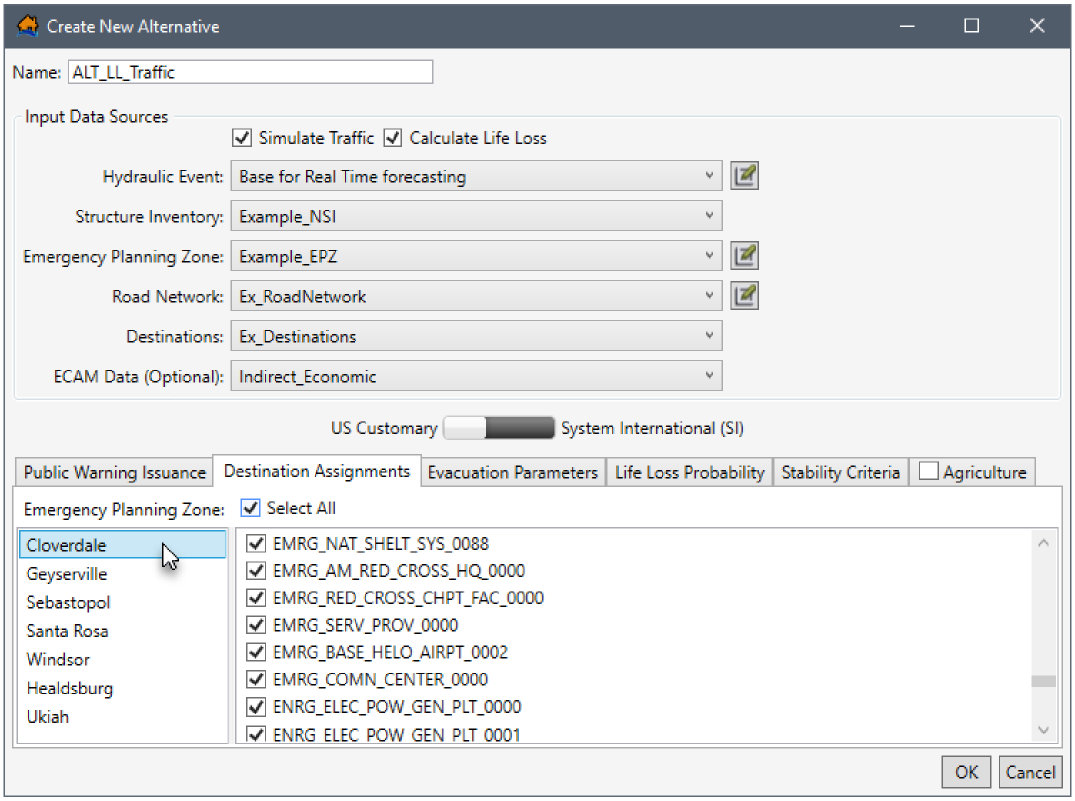


Figure 11-6. Create New Alternative Dialog Box – Destination Zone Assignments Tab

The **Destination Assignments** tab (Figure 11-6) is divided into two panels. The left panel, lists all of the emergency planning zones in the selected EPZ dataset (e.g., *Example_EPZ* in Figure 11-6). Therefore, if a selected EPZ dataset has multiple zones then the list will contain the names of the multiple zones. The right panel, provides a list of all of the destination names in the destination's dataset (e.g., *Ex_Destinations* in Figure 11-6).

The user can define unique (for multiple zones when present) destinations for selected EPZs. For example, if the warning message content for a particular EPZ dictates specific destination locations those locations can be defined and people within the EPZ will attempt to evacuate to one of the specified destination locations. To assign specific destinations to unique or multiple zones:

1. From the left panel, select the desired EPZ name (e.g., *Cloverdale* in Figure 11-6).

2. From the list of destinations in the right panel, select an individual destination or multiple destinations by checking ☑ the checkbox.

3. By default, all the destinations are selected, to unselect all destinations, click **Select All** (Figure 11-6). Alternatively, to reselect all destinations, click **Select All**.

4. Continue through the list of zones and list of destinations to assign the correct destinations to the correct zones.

5. Once all desired destinations have been assigned, either click **OK** to save the alternative and close the **Create New Alternative** dialog box (Figure 11-4) or select the next desired tab and continue to define the new alternative.

Evacuation Parameters

When simulating life loss the **Evacuation Parameters** tab (Figure 11-7) allows the user to adjust the default traffic evacuation simulation parameters in LifeSim. The **Evacuation Parameters** tab (Figure 11-7) contains two panels.

The **Evacuations Parameters** panel on the left (Figure 11-7), contains a list of traffic simulation parameters with defaults (Table 11-1). All available evacuation parameters are set with defaults, but the user can adjust the parameters by modifying the default values.

The **Depth Driven Willingness To Enter Flooded Road** panel on the right (Figure 11-7), is new to LifeSim Version 2.0. From the **Depth Driven Willingness To Enter Flooded Road** panel (Figure 11-7), users can define probability density functions for low clearance (cars) and high clearance (trucks) vehicles willingness to enter (ford) flooded roads. During the evacuation simulation vehicles encountering a flooded road will either attempt to ford the road or will attempt to turn around depending on the defined willingness to enter functions. The LifeSim Version 2.0 Technical Reference Manual, Chapter 6, provides extensive detail regarding evacuation parameters and depth driven willingness to enter flooded roads.

From the **Depth Driven Willingness To Enter Flooded Road** panel (Figure 11-7), users can define probability density functions for low clearance (cars) and high clearance (trucks) vehicles willingness to enter (ford) flooded roads. During the evacuation simulation vehicles encountering a flooded road will either attempt to ford the road or will attempt to turn around depending on the defined willingness to enter functions.

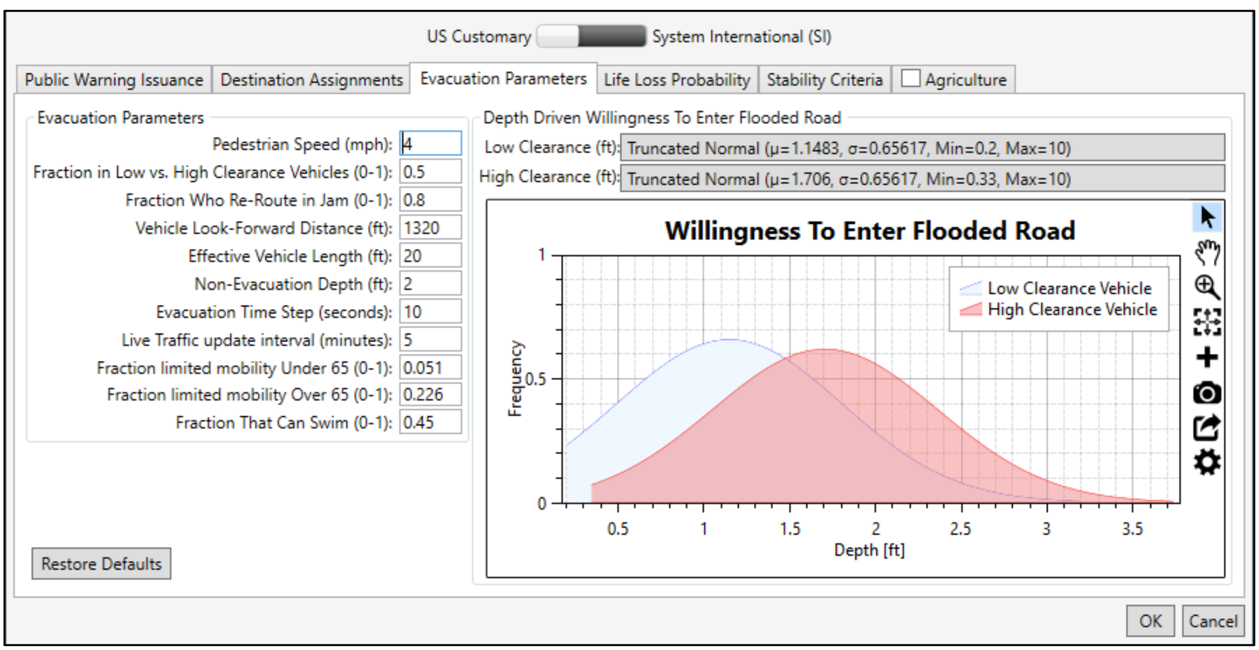


Figure 11-7. Create New Alternative Dialog Box – Evacuation Parameters Tab

Table 11-1. Evacuation Parameters Best Practice Range, Default Values and Definitions

Parameter	Units	Range	Default	Definition
Pedestrian Speed	mph	2 to 5	4	Speed of people evacuating on foot.
Low vs. High Clearance Vehicles	fraction	0 to 1	0.5	Fraction of people who will be evacuating in low (cars) vs. high clearance (trucks) vehicles.
Re-Route in Jam	fraction	0 to 1	0.8	Fraction of vehicles that will attempt to re-route when a traffic jam is reached.
Vehicle Look-Forward Distance	ft		1320	The distance in front of a vehicle that is used to define the traffic density.
Effective Vehicle Length	ft	20 to 25	20	The effective length of a vehicle which determines load limits.
Non-Evacuation Depth	ft	0 to 3	2	Depth at a structure where people will no longer attempt to evacuate away from the structure.
Evacuation Time Step	seconds	1 to 30	10	The timestep of the traffic simulation engine (smaller values = more accuracy but longer simulation compute times).
Live Traffic Update Interval	minutes	1 to 10	5	Define how often the current traffic conditions gets updated for vehicles that re-route when reaching a traffic jam (lower values increase simulation times).
Limited Mobility Under 65	fraction	0 to 1	0.051	Defines the faction of population under 65 years old that has limited mobility.
Limited Mobility Over 65	fraction	0 to 1	0.22	Defines the faction of population over 65 years old that has limited mobility.
Can Swim	fraction	0 to 1	0.45	Defines the fraction of population that are adept at swimming.

To edit the default traffic evacuation simulation parameters:

1. From the **Evacuation Parameters** panel (Figure 11-7), edit default evacuation parameters by selecting the desired cell and entering in the desired value. For example, double-click the **Non-Evacuation Depth (ft)** cell to highlight the default value and type in a depth in feet appropriate for the study.

2. From the **Depth Driven Willingness To Enter Flooded Road** panel (Figure 11-7), select desired clearance (e.g., **High Clearance** in Figure 11-8) button to open the distribution selection window.

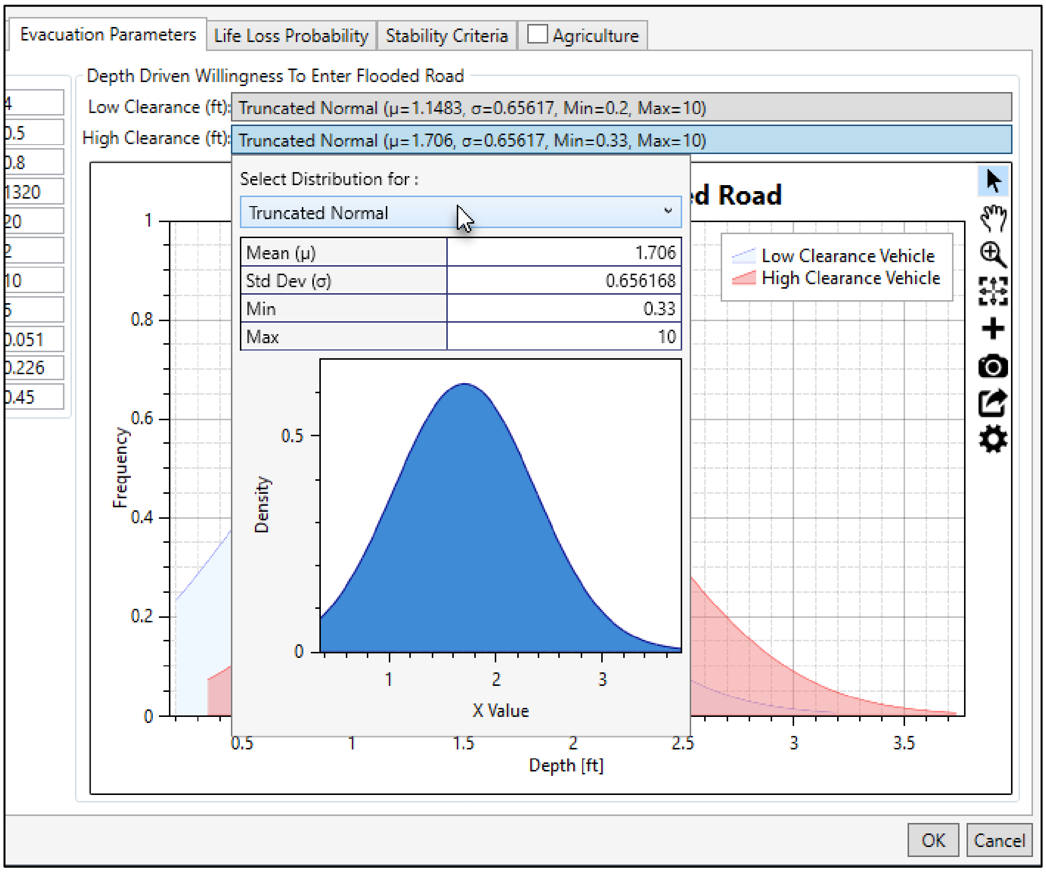


Figure 11-8. Evacuation Parameters Tab – Depth Driven Willingness To Enter Flooded Road Panel

3. From the distribution selection window (Figure 11-8), from the **Select Distribution for** dropdown list, select the desired function (e.g., ***Normal*** in Figure 11-9).

4. The selected distribution updates the distribution selection window (Figure 11-9). Enter the parameters for the selected distribution in the table provided. Close the distribution selection window by clicking anywhere in the **Create New Alternative** dialog box (Figure 11-7). The **Willingness To Enter Flooded Road** plot updates dynamically (on-the-fly) with the entered probability density function. Review Section 3.5 for instructions on using the general plotting tools (located on the right side of the plot window) and for customizing plots.

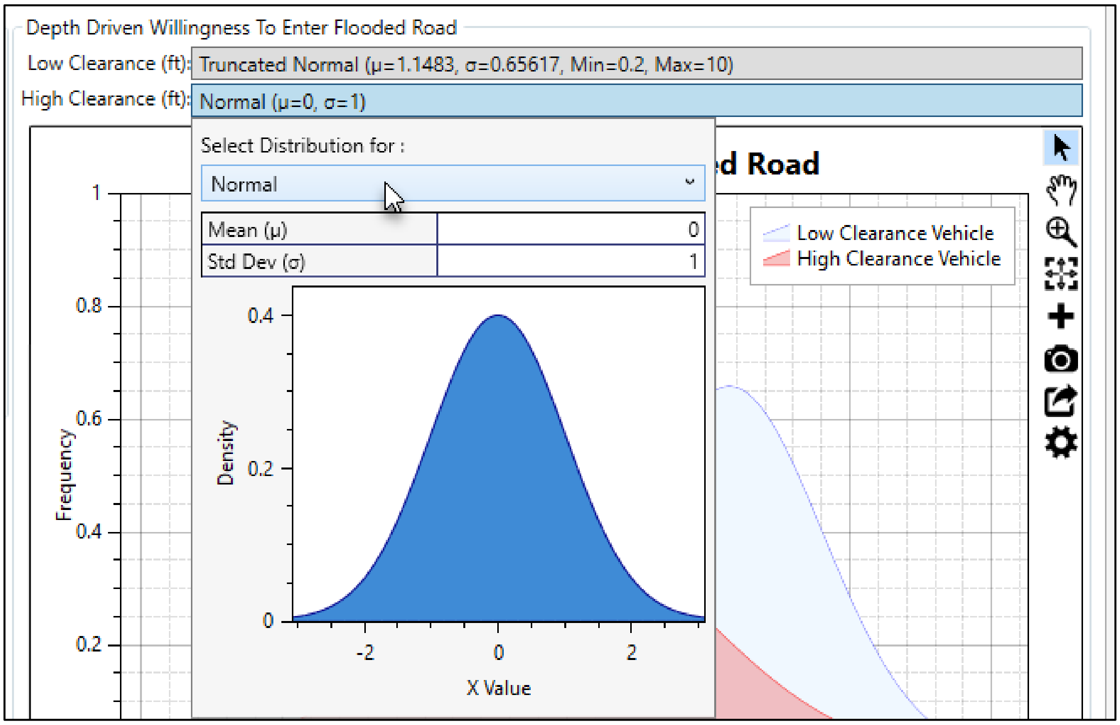


Figure 11-9. Depth Driven Willingness To Enter Flooded Road Panel – Example Distribution Selected

5. To restore edited parameter back to the default values, click **Restore Defaults** (Figure 11-7). **NOTE:** The **Restore Defaults** button returns both the evacuation parameters and the willingness to enter flooded roads functions back to defaults.

6. Either click **OK** to save the alternative and close the **Create New Alternative** dialog box (Figure 11-4) or select the next desired tab and continue to define the new alternative.

Life Loss Probability

The **Life Loss Probability** tab (Figure 11-10), allows the user to adjust the default fatality rate curves that are used to define probability of life loss for the high and low hazard zones. In LifeSim Version 1.0, default fatality rate curves were developed for three zones: chance, compromised and safe. However, in LifeSim Version 2.0, the original three zones were replaced with two new zones: low and high hazard. The LifeSim Version 2.0 Technical Reference Manual, Chapter 9, Section 9.2, provides a detailed explanation regarding the development of the two new zones and the fatality rate functions for the new zones.

LifeSim Version 2.0 characterizes each agent as either being in high hazard or low hazard conditions as described below:

High Hazard Zone: Refers to those conditions where the stability criteria or submergence criteria of the person, the vehicle (if caught while evacuating), or the structure (if not mobilized) has been exceeded. When the stability (or submergence) criteria has been exceeded survival depends largely on chance.

Low Hazard Zone: Refers to those conditions where the person or group of people are exposed to relatively calm floodwaters, where the stability of the individual, group, or shelter is not at risk. A hazard exists due to the potential for bad things to happen when people come in contact with water in locations not meant for such an interaction.

To modify the default fatality rate curves:

1. From the **Create New Alternative** dialog box (Figure 11-2), select the **Life Loss Probability** tab (Figure 11-10). The **Life Loss Probability** tab (Figure 11-10) provides a dropdown menu to select the desired zone (e.g., **High Hazard** in Figure 11-10) and view or modify the curve data.

   The **Relative Frequency of Exceedance** (X-Axis) represents the probability of a receiving a particular fatality rate. The **Proportional Life Loss** (Y-Axis) represents the fatality rate. Using the default high hazard data (Figure 11-10), if 100 people were in the high hazard zone 47 of them would have a 100 percent fatality rate.

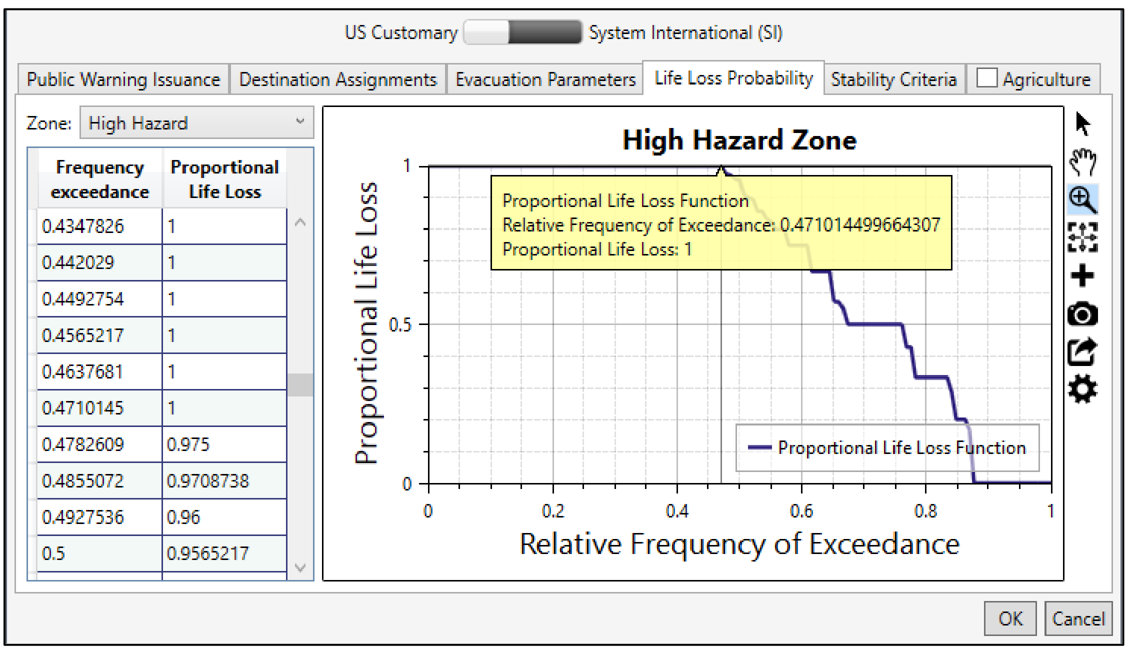


Figure 11-10. Create New Alternative Dialog Box – Life Loss Probability Tab

2. The default life loss probability functions can be edited manually or copied from another application (e.g., Microsoft Excel®) and pasted into the table (i.e., entire table, individual cells, columns, rows). From the table, right-click a cell (or group of selected cells) to open the shortcut menu (Figure 5 27), from the shortcut menu the user can add, insert, and delete row(s); select all; copy or copy with headers; and paste.

Errors in the life loss probability function are highlighted in red and a tooltip displays a message describing the error (Figure 5 28). Right-click on a header row to open the shortcut menu (Figure 5 28) to sort the table in ascending or descending order or clear the table sorting.

3. The **Hazard Zone** plot (Figure 11-10) provides a live update for the data entered in the table. Review Section 3.5 for instructions on using the general plotting tools (located on the right side of the plot window) and for customizing plots.

4. When done evaluating/modifying the life loss probability for the two zones, either click **OK** to save the alternative and close the **Create New Alternative** dialog box (Figure 11-4) or select the next desired tab and continue to define the new alternative.

Stability Criteria

The **Create New Alternative** dialog box (Figure 11-2), **Stability Criteria** tab (Figure 11-11) allows users to adjust the default stability criteria for three groups: human stability and low and high clearance vehicles. The user can adjust the default stability criteria thresholds (depth, velocity, and functional) for the three groups. The LifeSim Version 2.0 Technical Reference Manual provides more information on the human and vehicle stability criteria.

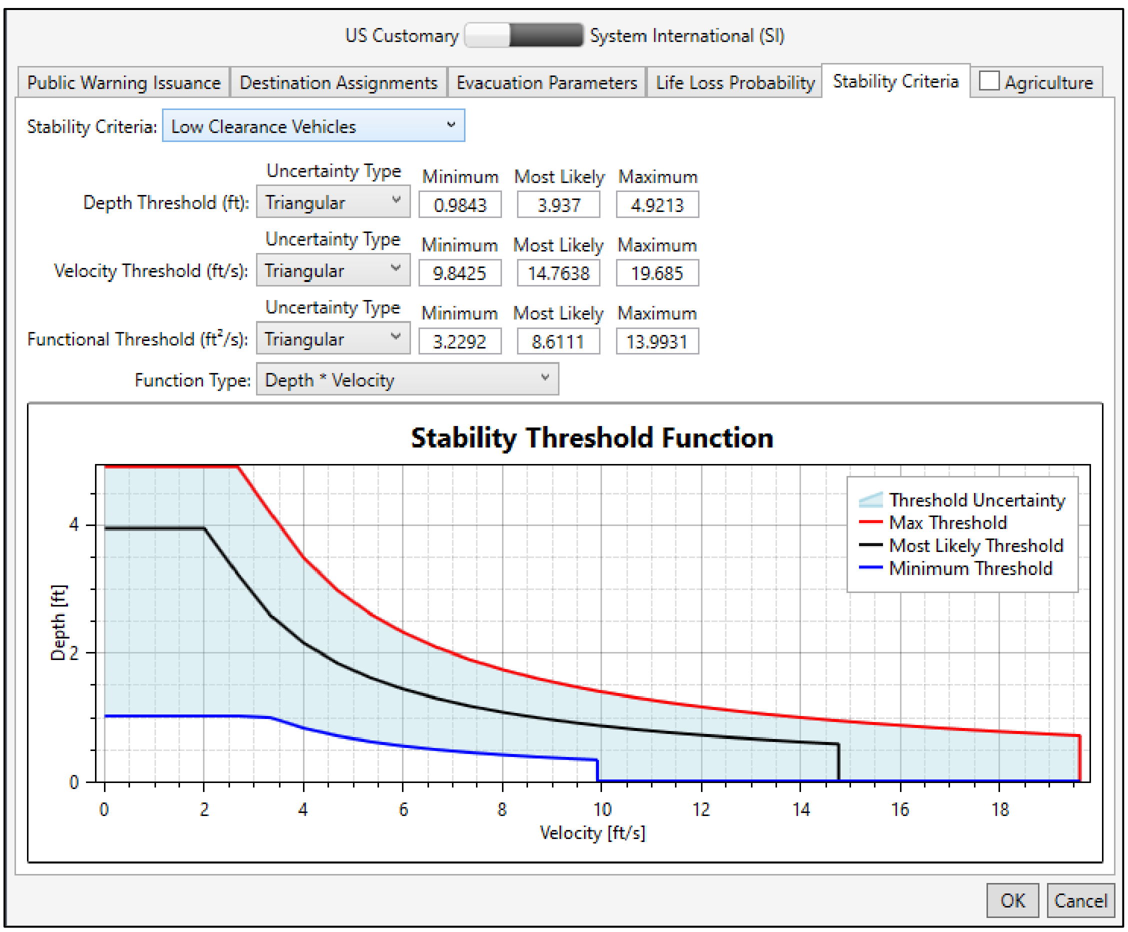


Figure 11-11. Create New Alternative Dialog Box – Stability Criteria Tab

To modify the default functions:

1. From the **Create New Alternative** dialog box (Figure 11-2), **Stability Criteria** tab (Figure 11-11). From the **Stability Criteria** dropdown list, select the desired group (e.g., **Low Clearance Vehicles** in Figure 11-11).

2. The user can modify the uncertainty about the three thresholds for the selected stability criteria group. From the **Uncertainty Type** dropdown menu, select the desired uncertainty type and define the uncertainty parameters for the threshold. Modify the listed stability criteria parameters (Figure 11-11) by entering values into the fields manually.

3. The **Functional Threshold** has an additional parameter for selecting the function type, from the **Function Type** list (Figure 11-11), as either **Depth*Velocity** or **Depth*(Velocity Squared)**.

4. When done evaluating/modifying the stability criteria parameters for all three groups (humans, low and high clearance vehicles), either click **OK** to save the alternative and close the **Create New Alternative** dialog box (Figure 11-4) or select the next desired tab and continue to define the new alternative.

Agriculture

The **Create New Alternative** dialog box (Figure 11-2), **Agriculture** tab (Figure 11-12) allows users set the agriculture inventory for the alternative and enter a specific hydraulic event starting date for calculating the agricultural flood damages.

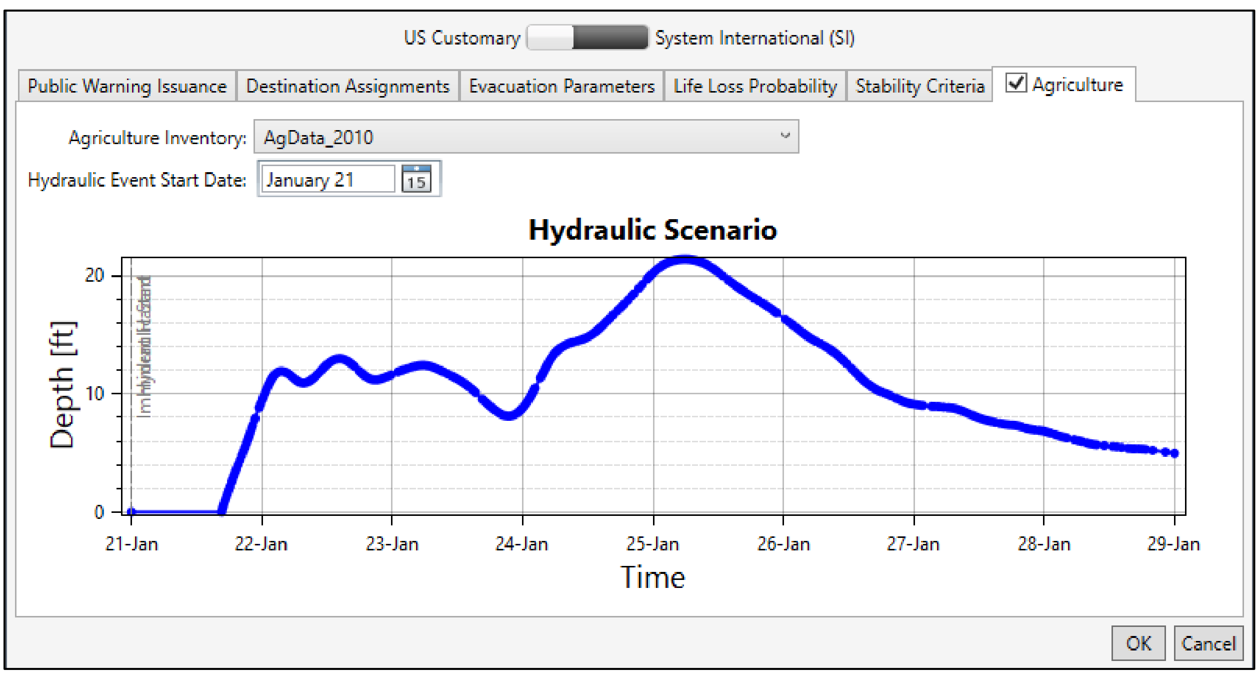


Figure 11-12. Create New Alternative Dialog Box – Agriculture Tab

To calculate agriculture damages for the created alternative:

1. From the **Create New Alternative** dialog box (Figure 11-2), select the **Agriculture** tab (Figure 11-12).

2. Turn on the agricultural damages calculation option by checking ☑ the checkbox in the header of **Agriculture** tab (Figure 11-12). By default, calculating agriculture damages is turned off (checkbox is unchecked ☐).

3. From the **Agriculture Inventory** list (Figure 11-12), select the desired dataset (e.g., ***AgData_2010*** in Figure 11-12).

4. From the **Hydraulic Event Start Date** box (Figure 11-12), click the button to open the date selector window and set the desired hydraulic event starting date (e.g., ***January 21*** in Figure 11-12).

   The **Hydraulic Scenario** plot (Figure 11-12) updates by starting the first timestep in the selected hydraulic dataset (e.g., ***Base for Real Time forecasting*** in Figure 11-4) based on the selected **Hydraulic Event Start Date**.

5. Click **OK** to save the alternative and close the **Create New Alternative** dialog box (Figure 11-4) or select the next desired tab and continue to define and/or edit the alternative.

11.2 Edit an Alternative

To edit an alternative, from the LifeSim main window (Figure 11-1), from the **Study** tab, from the **Study Tree**, from the **Alternatives** folder, right-click on an alternative name (e.g., ***ALT_LL_Traffic*** in Figure 11-13). From the shortcut menu (Figure 11-13), click **Edit**.

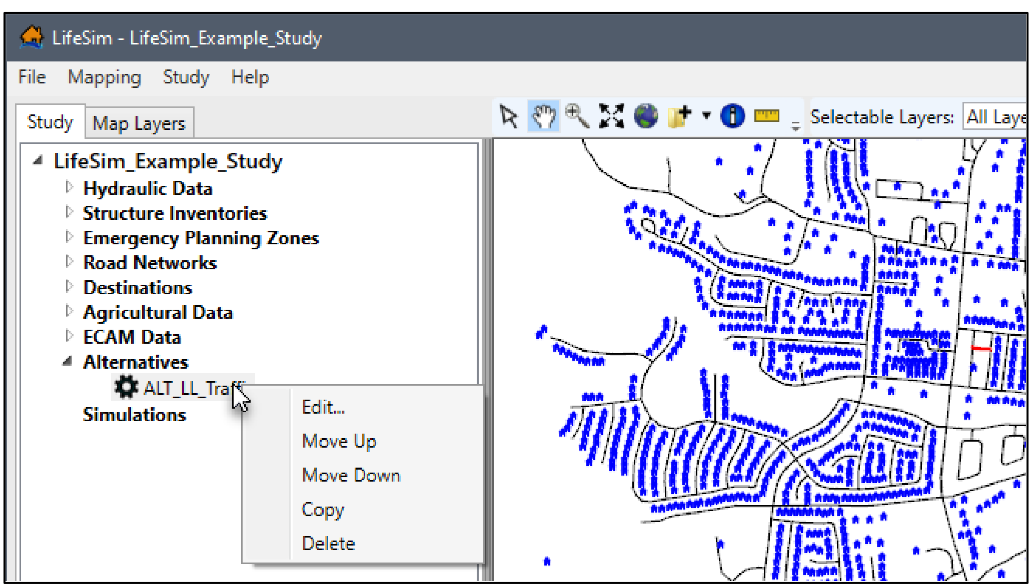


Figure 11-13. LifeSim Study Tree – Individual Alternative – Shortcut Menu

Alternatively, from the LifeSim main window, from the **Study** menu, point to **Alternatives** (Figure 11-1), point to **Edit**, click on an alternative name. Either way the **Alternative Editor** will open (similar to Figure 11-4). Section 11.1 for more information regarding the editable fields in the **Alternative Editor**.

11.3 Copy an Alternative

To copy an alternative, from the LifeSim main window (Figure 11-1), from the **Study** tab, from the **Study Tree**, from the **Alternatives** folder, right-click on an alternative name. From the shortcut menu (Figure 11-13), click **Copy**. Another way is from the LifeSim main window, from the **Study** menu, point to **Alternatives**, point to **Copy**, click on an alternative name.

Either way, the **Name of New Study Item** dialog box (Figure 5-19) opens. Enter a new name in the **Name** box. Click **OK**, the **Name of New Study Item** dialog box (Figure 5-19) closes and the copied alternative is added to the **Study Tree** under the **Alternatives** folder.

11.4 Delete an Alternative

To delete an alternative, from the LifeSim main window (Figure 11-1), from the **Study** tab, from the **Study Tree**, from the **Alternatives** folder, right-click on an alternative name. From the shortcut menu (Figure 11-13), click **Delete**. Another way is from the LifeSim main window, from the **Study** menu, point to **Alternatives**, point to **Delete**, click on an alternative name. Either way a **Confirm Delete** window (Figure 11-14) opens asking for confirmation to delete the selected alternative. Click **Yes** to confirm the deletion. The **Confirm Delete** window (Figure 11-14) closes, and the selected alternative is deleted, and no longer displays in the **Study Tree** under the **Alternatives** folder.

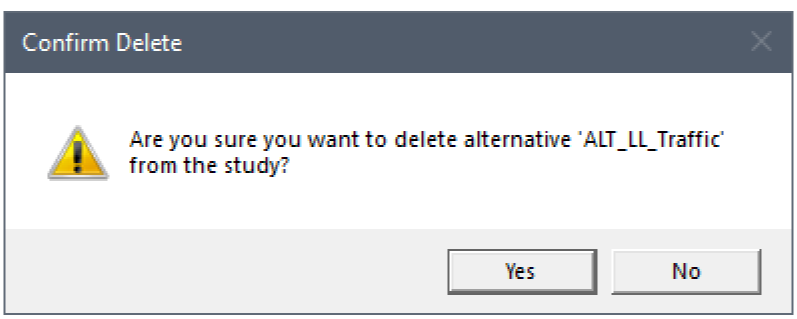


Figure 11-14. Confirm Delete Window

CHAPTER 12

Simulations

A LifeSim **simulation** is a mathematical representation of the behavior of a system, given inputs and initial conditions. It applies the various study inputs (e.g., hydraulic scenario) to simulate the evacuation process and calculates consequence results based on the inputs. LifeSim simulations are created for the purpose of running and viewing life loss and damage estimation results for multiple alternatives. A common application for LifeSim would be to inform a risk analysis where consequence estimates will be developed for multiple hypothetical events for current and future conditions. Other applications include planning alternatives to reduce consequences (e.g., non-structural measures).

12.1 Create a Simulation

Now that all of the building blocks have been added and configured (imported datasets, configured alternatives), the user can create an LifeSim simulation.

1. From the LifeSim main window (Figure 12-1), from the **Study** tab, from the **Study Tree**, right-click on **Simulations**, from the shortcut menu, click **Create New Simulation**. Another way is from the LifeSim main window, from the **Study** menu (Figure 12-1), point to **Simulations**, click **Create New**.

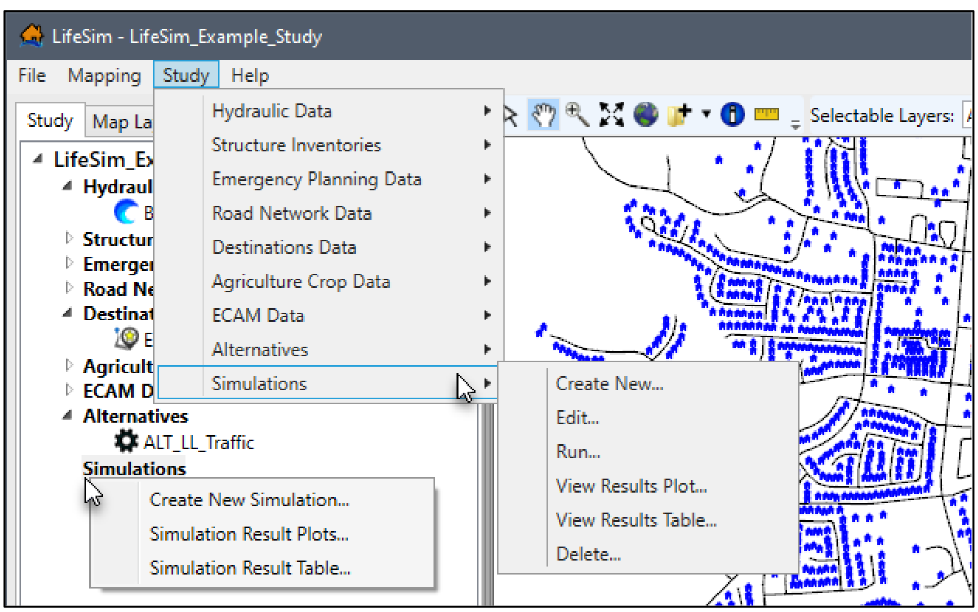


Figure 12-1. LifeSim Main Window – Study Menu Simulations Sub-menu – Simulations Study Tree Shortcut Menu

2. Either way, the **Create New Simulation** dialog box (Figure 12-2) opens.

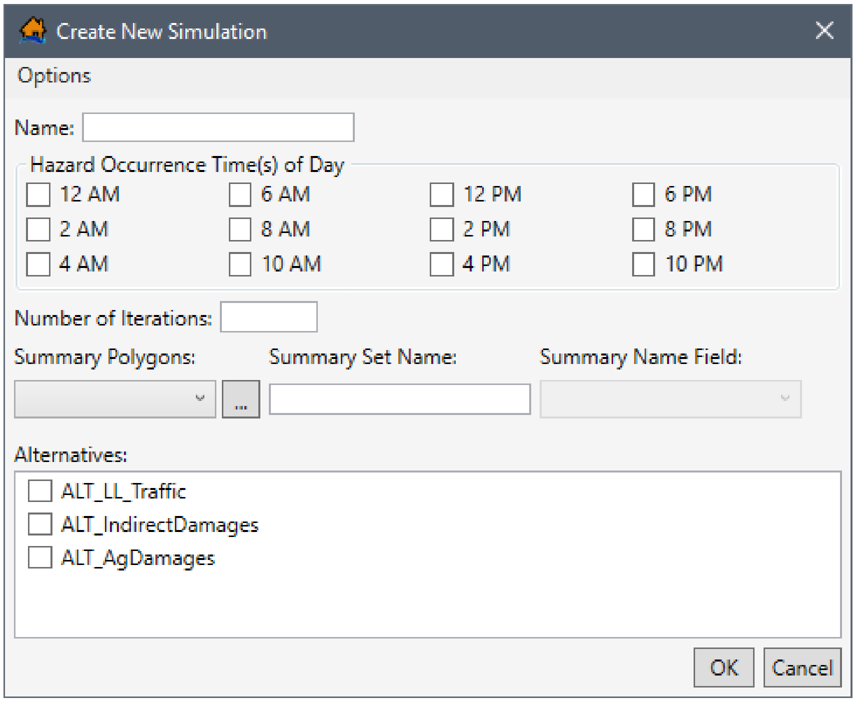


Figure 12-2. Create New Simulation Dialog Box

3. Enter a name for the simulation in the **Name** box (Figure 12-2).

4. In the **Hazard Occurrence Time(s) of Day** box (Figure 12-2) select the imminent hazard times for the day (check the ☑ checkbox for each time of day that is required in the simulation). The user must select at least one time of day. For each time of day selected, a new simulation will be calculated using the selected alternatives. For example, if *2 AM* is selected then the selected alternatives will be simulated for the entered number of iterations with the hazard occurring at *2 AM*.

5. In the **Number of Iterations** box (Figure 12-2) enter the number of iterations for the simulation. The number of iterations cannot be greater than 1,000,000 (one million).

6. The **Summary Polygons** list (Figure 12-2) allows the user to set the attribute field containing unique summary zones. The unique summary zones are used to group and summarize the simulation results. The zones are represented by polygons. Emergency planning zones are commonly used but any polygon shapefile (e.g., river mile from dam breach) can be used.

   Select the **Summary Polygons** from the list (Figure 12-3). Alternatively, click ... to open an **Open** browser window and navigate to the directory where the desired polygon shapefile is located. From the **Open** browser window, select the desired polygon shapefile and click **Open**. The **Open** browser closes and the **Summary Polygons** list (Figure 12-2) updates with the desired shapefile.

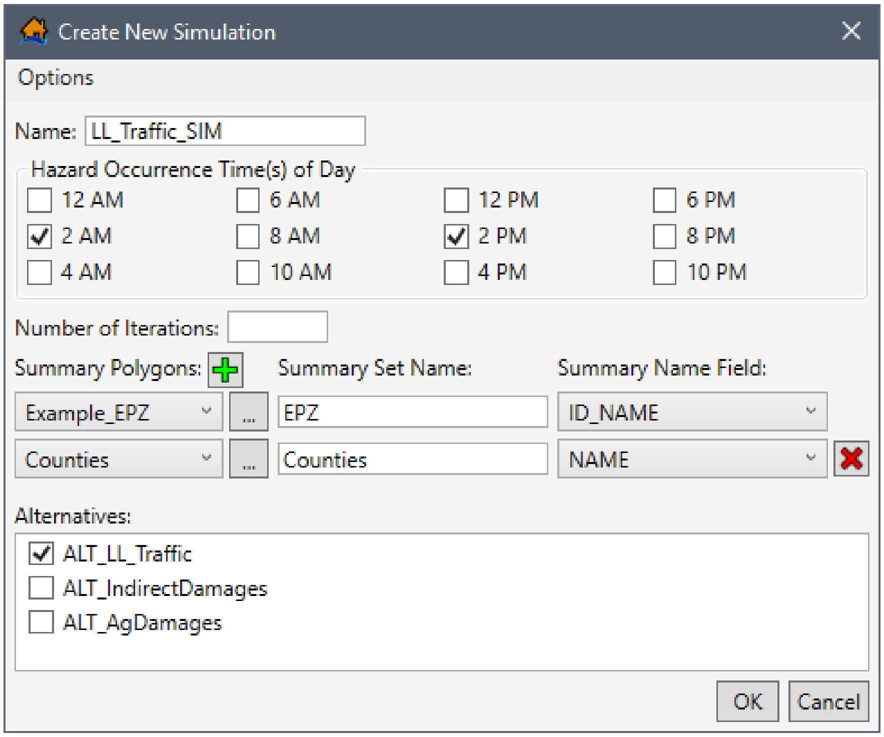


Figure 12-3. Example – Create New Simulation

7. Enter the **Summary Set Name** for the zones (e.g., *EPZ* in Figure 12-3). The name entered is used for display purposes.

8. From the **Summary Name Field** list (Figure 12-3) select the desired field containing the unique summary zone names. The **Summary Name Field** list is populated based on the selected **Summary Polygons**. If the user does not select a field containing unique names for each zone then a LifeSim message window opens (Figure 12-4) requesting the user to select a field containing unique names.

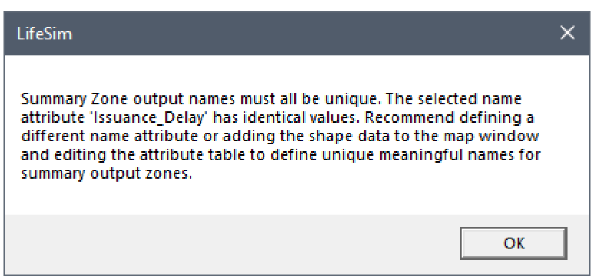


Figure 12-4. LifeSim Message Window

9. Additional summary zone sets can be added or removed from a simulation. Click the **Add** button (Figure 12-3) to add and define another summary set. Click the **Delete** button (Figure 12-3) to delete an added summary set.

10. From the **Alternatives** box (Figure 12-3), check the ☑ checkbox to select the alternatives to use in the simulation by clicking on the alternative name. The user must select at least one alternative. Multiple alternatives can be selected.

11. Once all of the information for the simulation has been entered, click **OK**, the **Create New Simulation** dialog box (Figure 12-2) will close. From the LifeSim main window (Figure 12-1), from the **Study Tree**, under the **Simulations** folder, the name of the simulation is displayed.

12.2 Simulation – Computation Engine Options

Computation Engine Options dialog box (Figure 12-5) allows the user to set the number of threads for the simulation. LifeSim is designed to take advantage of multiple processors, often called parallel processing. The number of threads for the simulation defaults to the maximum number of available processors on a user's computer. The user has the option of modifying the number of threads between one and the maximum number. Sometimes with very large datasets memory limits will be exceeded when a large number of threads are defined. It is recommended for very large datasets to reduce the number of threads if memory limits are reached.

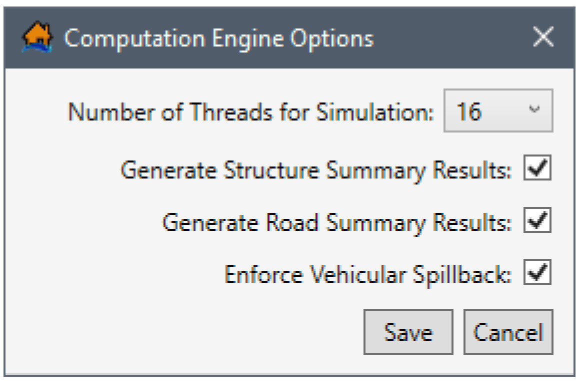


Figure 12-5. Computation Engine Options Dialog Box

In addition to setting the number of threads, the **Computation Engine Options** dialog box (Figure 12-5) allows users to turn on or off the following compute options: enforce vehicular spillback and/or generate summary results for structures and/or roads.

When the **Generate Structure Summary Results** and/or **Generate Road Summary Results** (Figure 12-5) options are turned on then users can create hydraulic summaries for structures. The structure and road summary results can be useful in identifying potential issues in the structure inventory or road network. For example, issues such as structures that are inundated too quickly due to inaccurate structure locations, or bridges that have water under them prior to the flood event.

When **Enforce Vehicular Spillback** (Figure 12-5) is turned on, the effective vehicle length is used to make sure the number of vehicles on a road segment is physically possible given the number of lanes, length of the road segment and space taken by the average vehicle. When vehicular spillback is enforced and too many vehicles are on a road segment at the end of a timestep, then the spillback function pushes the last vehicle to enter the road back onto the previous road segment. The pushback (or spillback) continues until each road segment is at or below capacity.

To access and modify the computation engine options:

1. From the **Create New Simulation** dialog box (Figure 12-2), or from the **Simulation Editor** (Figure 12-7), from the **Options** menu, click **Computation Engine Options**. The **Computation Engine Options** dialog box (Figure 12-5) will open.

2. From the **Number of Threads for Simulation** list, select the number of threads that will be available for the simulation (e.g., *16* in Figure 12-5).

3. Check the ☑ checkbox to turn on the **Generate Structure Summary Results**, **Generate Road Summary Results** and/or **Enforce Vehicular Spillback** (Figure 12-5) options.

4. Click **Save**, the **Computation Engine Options** dialog box closes (Figure 12-5).

12.3 Edit a Simulation

To edit a simulation, from the LifeSim main window (Figure 12-1), from the **Study** tab, from the **Study Tree** (Figure 12-6), under **Simulations**, right-click on a simulation name. From the shortcut menu, click **Edit** (Figure 12-6). Another way is from the LifeSim main window, from the **Study** menu, point to **Simulations** (Figure 12-1), point to **Edit**, click on a simulation name. Either way the **Simulation Editor** opens (Figure 12-7). Sections 12.1 and 12.2 cover all of the data items that can be edited by the user.

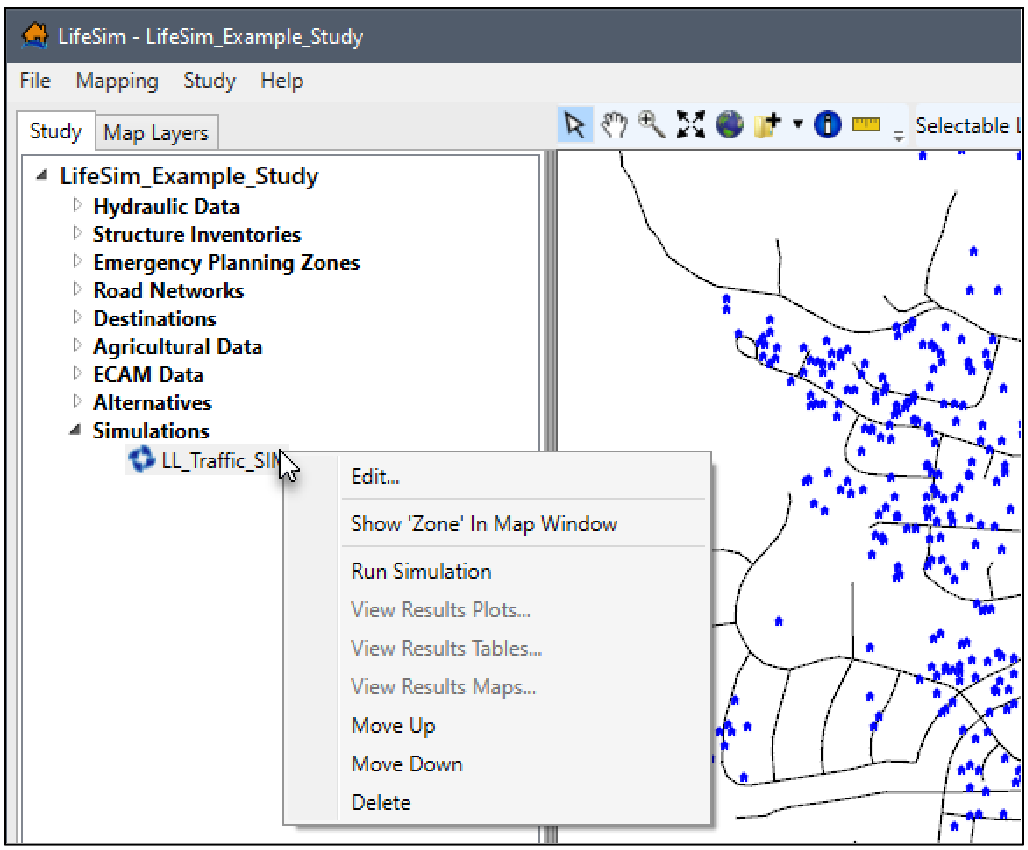


Figure 12-6. LifeSim Study Tree – Individual Simulation – Shortcut Menu

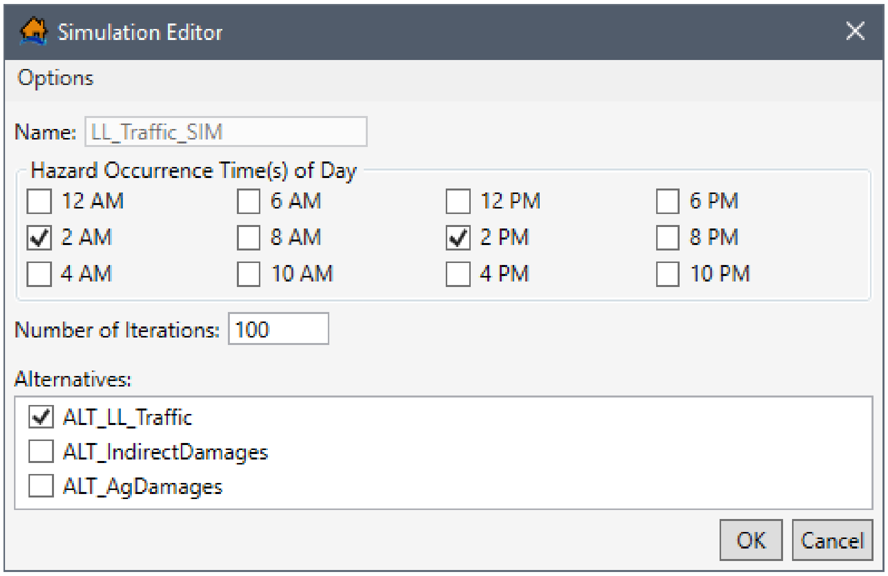


Figure 12-7. Example Simulation Editor

12.4 Delete a Simulation

To delete a simulation, from the LifeSim main window (Figure 12-1), from the **Study** tab, from the **Study Tree** (Figure 12-6), under **Simulations**, right-click on a simulation name. From the shortcut menu, click **Delete** (Figure 12-6). Another way is from the LifeSim main window, from the **Study** menu, point to **Simulations** (Figure 12-1), point to **Delete**, click on a simulation name.

Either way a **Confirm Delete Simulation** message window (Figure 12-8) opens. The window asks the user, "Are you sure you want to delete [*the selected simulation*] and all of its results" from the study. Click **Yes**, to delete the simulation and associated results files from the study and close the **Confirm Delete** message window (Figure 12-8). Alternatively, click **No**, to close the message window without deleting the selected simulation.

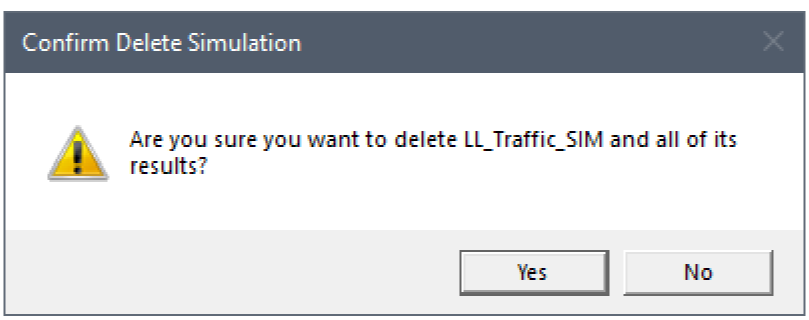


Figure 12-8. Confirm Delete Simulation Message Window

12.5 Compute a Simulation

To compute a simulation, from the LifeSim main window (Figure 12-1), from the **Study** tab, from the **Study Tree** (Figure 12-6), under **Simulations**, right-click on a simulation name. From the shortcut menu, click **Run Simulation** (Figure 12-6). Another way is from the LifeSim main window, from the **Study** menu, point to **Simulations** (Figure 12-1), point to **Run**, click on a simulation name.

The computation of the simulation will begin, from the LifeSim main window (Figure 12-1), a progress bar will appear (Figure 12-9) in the **Status Bar**, detailing the progress of the compute. Also, a **Simulation Progress** window (Figure 12-10) opens displaying status bars for the entire simulation (all alternatives) and for each alternative included in the simulation (Figure 12-10).

If the study data included in the selected alternatives contain errors, the compute of the simulation ends and error messages describing the issues with the alternative data display in the **Simulation Progress** window (Figure 12-11). Once the compute is complete, click **Close** to close the **Simulation Progress** (Figure 12-10) window.

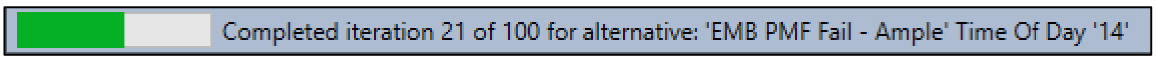


Figure 12-9. Progress Bar

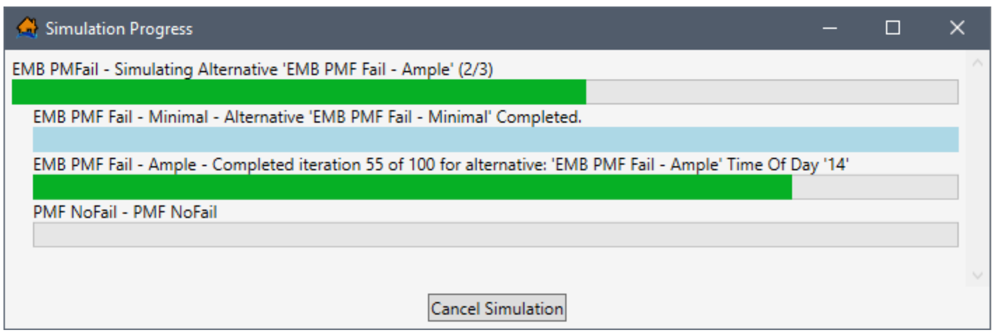


Figure 12-10. Simulation Progress Window

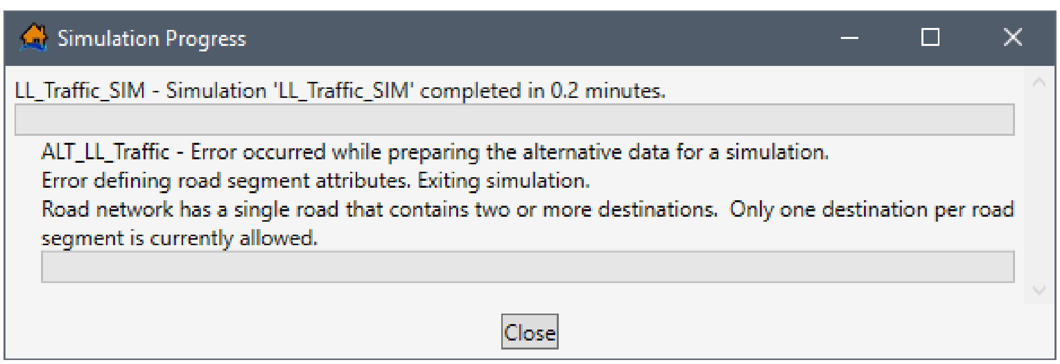


Figure 12-11. Example – Simulation Progress Window Error Message

CHAPTER 13

Viewing LifeSim Results

When a simulation has been created for an LifeSim study the simulation should be computed and the results reviewed. During a simulation compute, summary results for each iteration are written out to a database, the user can then generate detailed iteration results to view agent level simulation data such as animations and evacuation timing. Agent level refers to viewing results down to the vehicle or evacuating group level. The detailed iteration results are computed per iteration to save both storage space and computation time. Results include summary tabulated results by iteration, summary box and whisker plots, plot by iteration and plot by uncertainty. In addition, detailed output can be generated by iteration allowing the user to animate the simulation in the map window and view detailed iteration plot results.

13.1 Simulation Results Options

LifeSim Version 2.0 provides users with the ability to view summary results for all simulations or results for individual simulations. To view the simulation summary results for all simulations, from the LifeSim main window (Figure 13-1), from the **Study** tab, from the **Study Tree**, right-click **Simulations** (Figure 13-1), and click **Simulation Result Plots** or **Simulation Result Table** from the shortcut menu. Alternatively, from the **Study** menu, point to **Simulations** and click **View Results Plot** or **View Results Table** from the sub-menu (Figure 13-2). Section 13.2 provides an overview of the summary results for all simulations.

NOTE: All results plot windows contain the same general plotting tools (located on the right side of the plot window) used for customizing plots. Review Section 3.6 for instructions on using the general plotting tools.

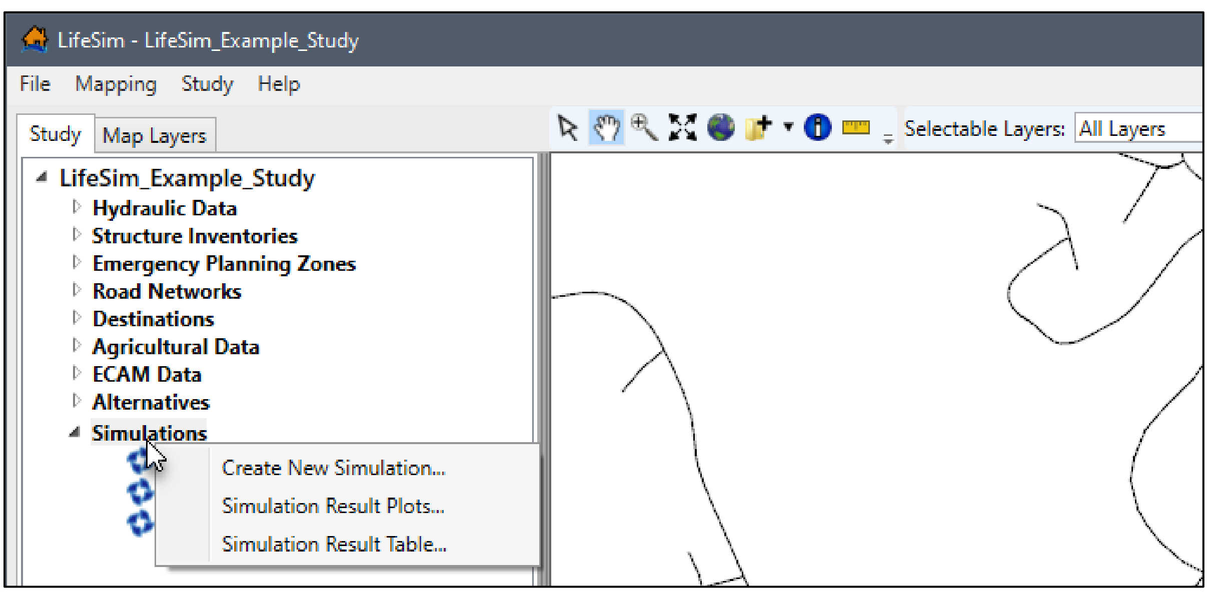


Figure 13-1. LifeSim Main Window – Simulations Study Tree Shortcut

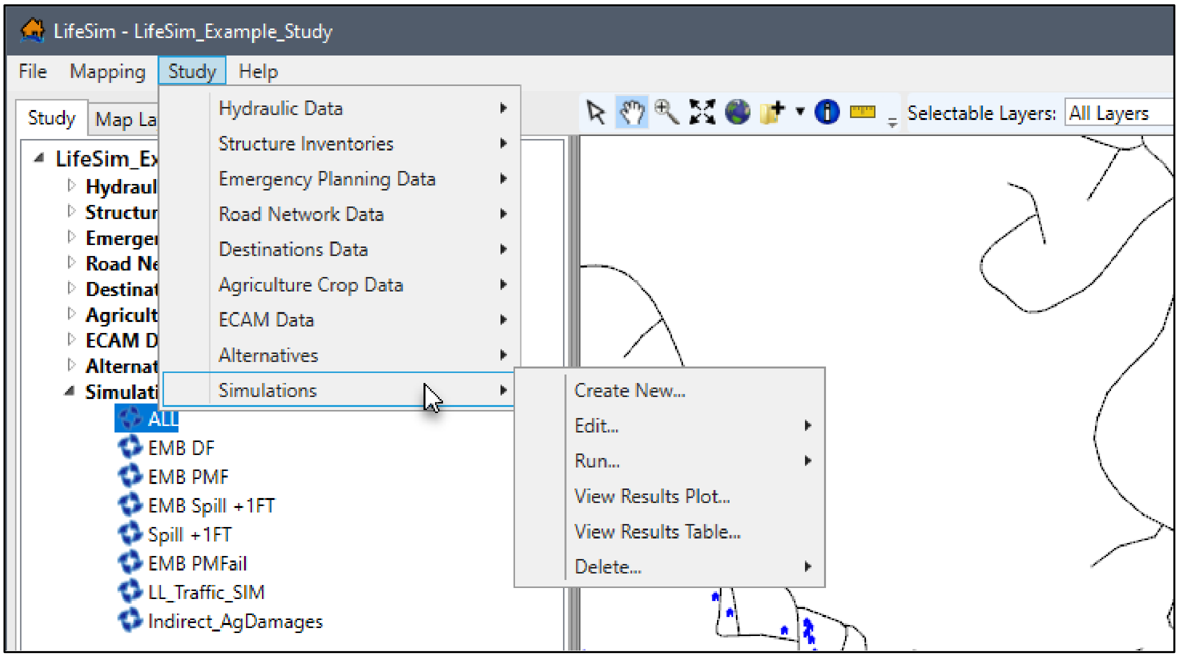


Figure 13-2. LifeSim Main Window – Study Menu Simulations Sub-menu

To view the simulation results for a specific simulation, from the LifeSim main window (Figure 13-3), from the **Study** tab, from the **Study Tree**, under **Simulations**, right-click on a simulation to view the shortcut menu commands (Figure 13-3). Sections 13.3, 13.4 and 13.5 provide details on viewing the simulation results in a plot, table, and in the map window, respectfully.

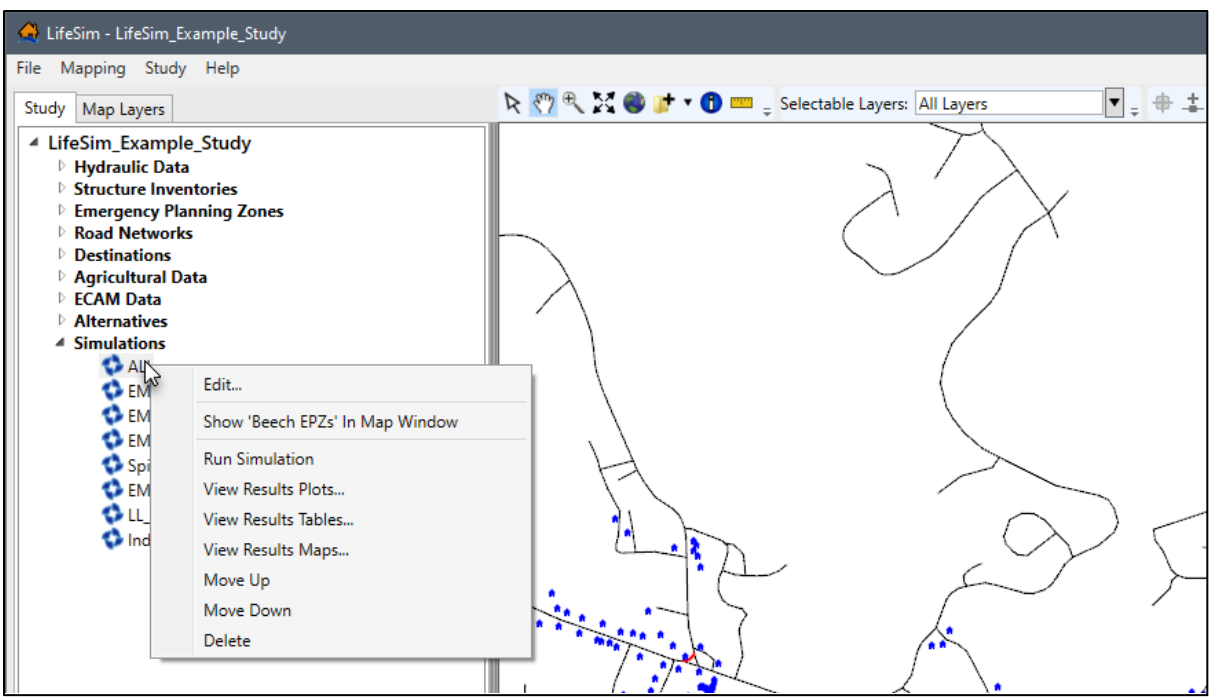


Figure 13-3. LifeSim Main Window – Individual Simulation – Study Tree Shortcut Menu

13.2 Open Summary Results

Users can view the simulation summary results in a table or in a plot window. To view the summary results, from the LifeSim main window (Figure 13-1), from the **Study** tab, from the **Study Tree**, right-click **Simulations** (Figure 13-1), and click **Simulation Result Plots** or **Simulation Result Table** from the shortcut menu. Alternatively, from the **Study** menu, point to **Simulations** and click **View Results Plot** or **View Results Table** from the sub-menu (Figure 13-2).

13.2.1 Simulation Result Plots

The **Simulation Result Plots** (Figure 13-1), **Study Tree | Simulations** shortcut menu command opens the **Results Summary Plots** (Figure 13-4). The **Results Summary Plots** (Figure 13-4) is divided into three main panels. The first panel located at the top left, is the **Simulation Results By Iteration** panel (Figure 13-4). The second panel located at the bottom left, is the **Simulation Results By Structure** panel (Figure 13-4). The third panel located on the right-side is the plot window containing two tabs: **Box and Whisker Plots** (Figure 13-4) and **Iteration Plots** (Figure 13-5).

However, there is a fourth optional panel, the **Filter** panel (Figure 13-6) is an expandable panel that is opened by clicking the **Filter** ⊞ expand button from the **Simulation Results By Iteration** panel. To close the optional **Filter** panel, click the **Filter** ⊟ collapse (Figure 13-6) button.

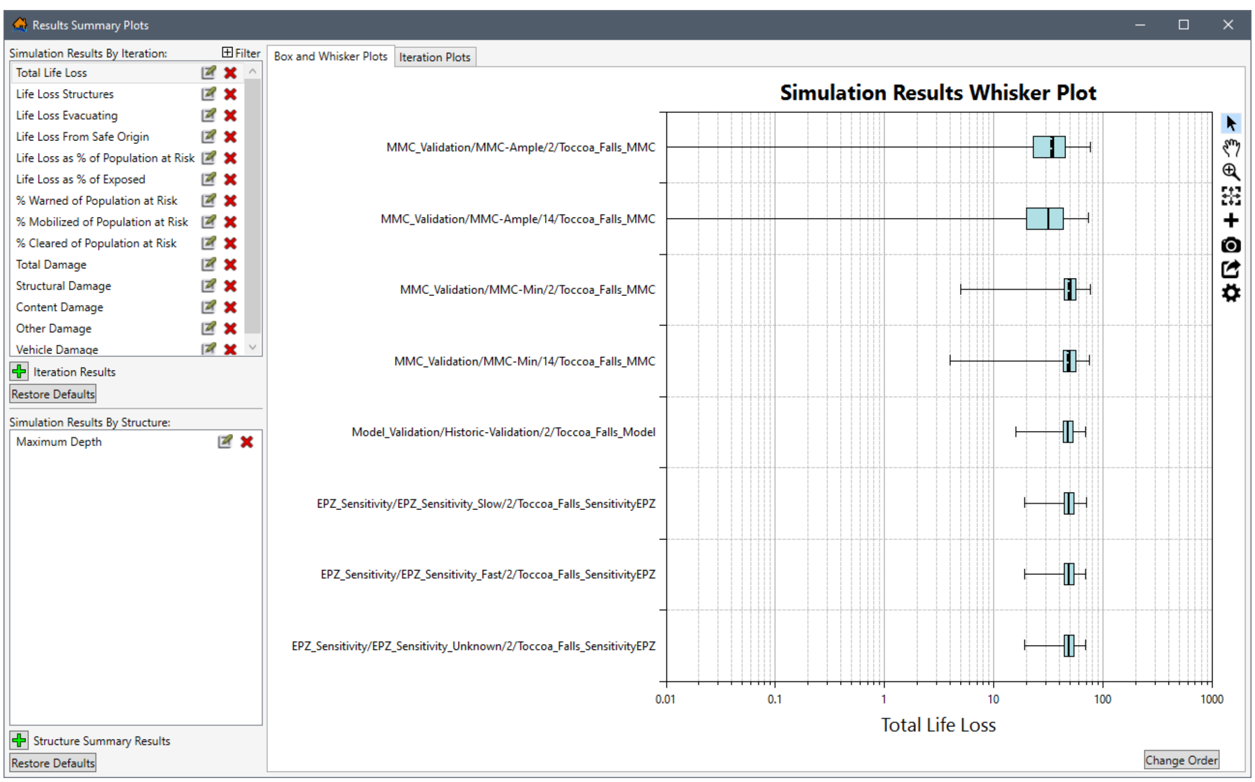


Figure 13-4. Example – Results Summary Plots – Box and Whisker Plots Tab

Figure 13-5. Example – Results Summary Plots – Iteration Plots Tab

Simulation Results Plot Options

The **Simulation Results By Iteration** panel (Figure 13-6), lists all of the default predefined iteration results. Select a specific iteration simulation results series (e.g., ***Total Life Loss*** in Figure 13-6) to set the results to display in the plot window.

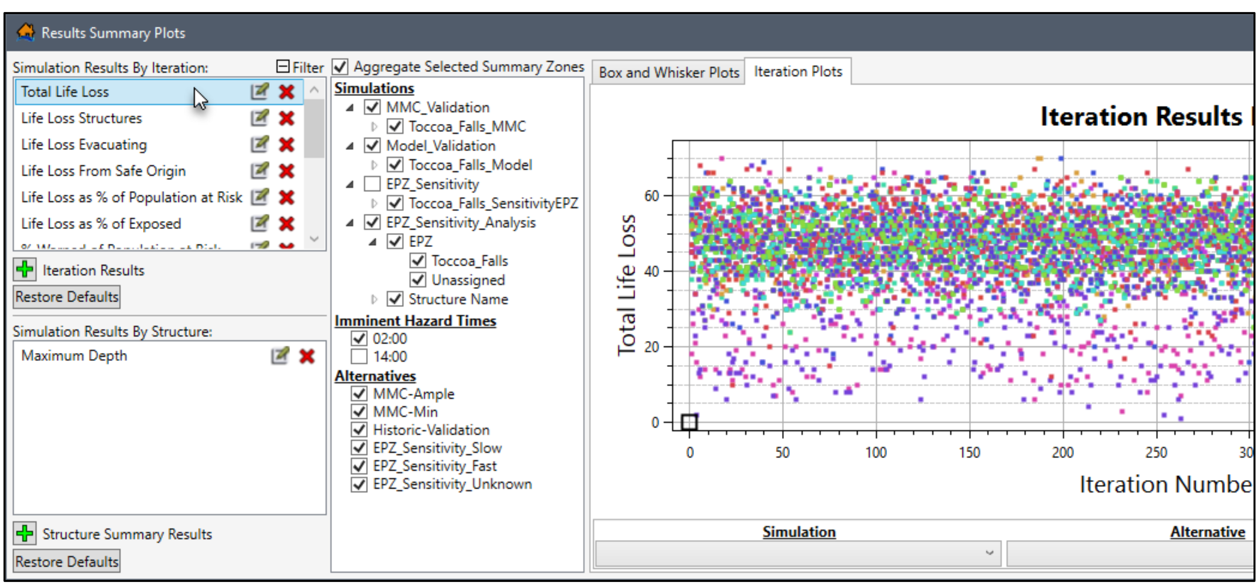


Figure 13-6. Example – Results Summary Plots – Expanded Filter Panel

Note, the **Simulation Results By Iteration** selection (e.g., ***Total Life Loss*** in Figure 13-6) updates the data displayed in the **Box and Whisker Plots** (Figure 13-4) and **Iteration Plots** (Figure 13-5) tabs. However, the **Simulation Results By Structure** selection (e.g., ***Maximum Depth*** in Figure 13-7) only updates the **Box and Whisker Plots**.

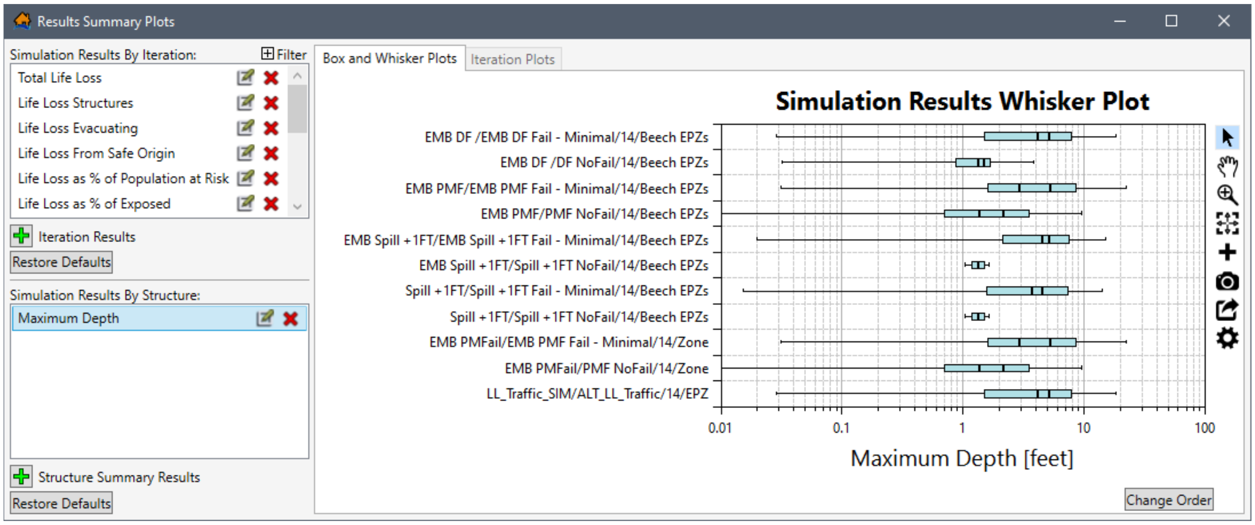


Figure 13-7. Example – Results Summary Plots – Simulation Results By Structure

The expandable **Filter** panel (Figure 13-6) contains three categories: **Simulations**, **Imminent Hazard Times**, and **Alternatives**; for limiting the simulation results displayed in the plot window by aggregated summary zones. Remove simulations from the plot window by unchecking the ☐ checkbox for the aggregated summary zone. For the example provided in in Figure 13-5, the results of the ***EPZ_Sensitivity*** simulation and ***14:00*** imminent hazard time has been removed from results displayed in the plot window.

In the **Simulations** category (Figure 13-6), the simulations contain aggregation summary zones. For the example provided in Figure 13-6, the ***EPZ*** and ***Structure Name*** are aggregation summary zones for the ***EPZ_Sensitivity_Analysis*** simulation. All fields in the summary zone, e.g., ***Toccoa_Falls*** and ***Unassigned*** in Figure 13-6) are listed to allow specific fields to be filtered from the plot window. The aggregation summary zones listed for a simulation depends on the summary polygons defined for a simulation from the **Create New Simulation** dialog box (review Section 12.1 for more information regarding the setup for a simulation).

Uncheck the ☐ checkbox to turn off the **Aggregate Selected Summary Zones** (Figure 13-6) to display the results for all fields in the summary zone individually by not grouping the simulation results by summary zone.

Create New Simulation Results

New iteration or structure summary results can be created and defined by clicking the add  **Iteration Results** or add **Structure Summary Results** (Figure 13-7) buttons, respectfully. The **Expression Dialog** (Figure 13-8) opens.

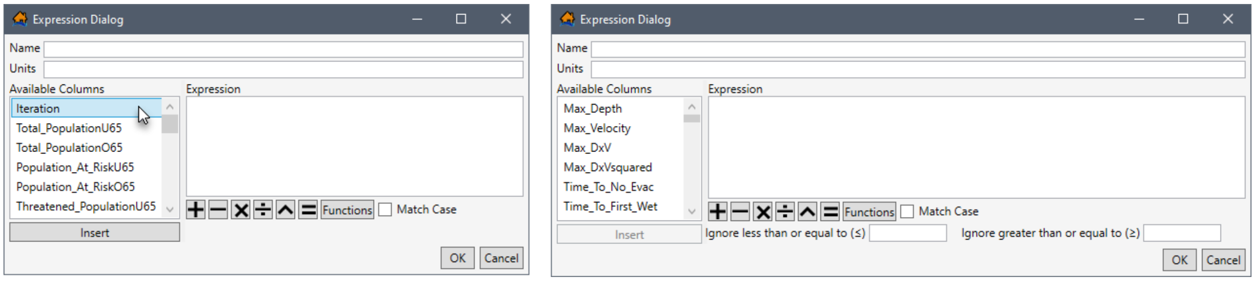


Figure 13-8. Expression Dialog – New Iteration Results (*left*) – New Structure Summary Results (*right*)

1. In the **Expression Dialog** (Figure 13-8), enter a name in the **Name** box (Figure 13-8), and if desired enter the units in the **Units** box (Figure 13-8).

2. To build the expression for the results in the **Expression** box (Figure 13-8), select a specific results column from the list of **Available Columns** (e.g., ***Iteration*** in Figure 13-8), and click **Insert**.

3. The **Functions** button (Figure 13-8) opens a new window providing a selection of all available functions in LifeSim. Select a function from the **Functions** list (e.g., ***IF*** in Figure 13-9), to view tips for using the function and click **Insert** to add the selected function to the **Expression** box (Figure 13-8) in the **Expression Dialog** (Figure 13-8).

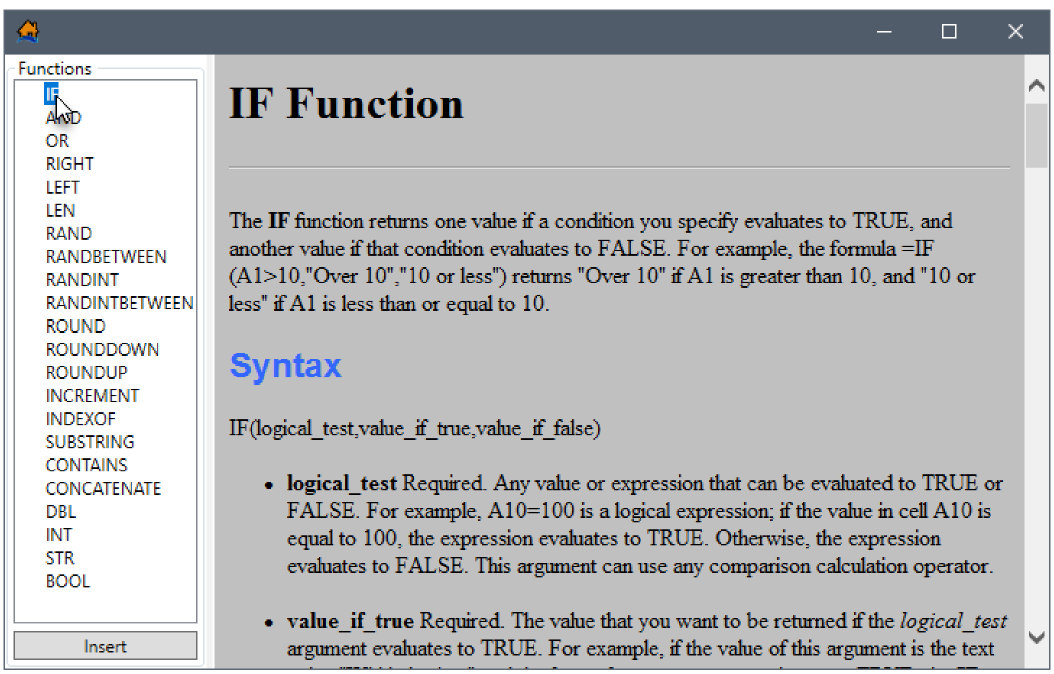


Figure 13-9. Functions

4. Turn on **Match Case** (Figure 13-8) by checking the checkbox ☑ to specify that the entered expression is case sensitive.

5. When creating a structure summary results expression, enter a value in the **Ignore less than or equal to (≤)** (Figure 13-8) box to set a minimum limit for the results.

Alternatively, enter a value in the **Ignore greater than or equal to (≥)** (Figure 13-8) box to set a maximum limit for the results.

6. Click **OK**, to close the **Expression Dialog** (Figure 13-8) and create the new iteration or structure summary results series.

Edit Existing Simulation Results

Existing iteration or structure summary results can be created and defined by clicking (Figure 13-7) for a specific results series (e.g., ***Total Life Loss*** in Figure 13-6). The **Expression Dialog** (Figure 13-10) opens for the selected results series. The **Expression** box (Figure 13-10) contains the default expression for the selected results series. Users can edit the default expression by following the same steps described for creating a new simulation results series.

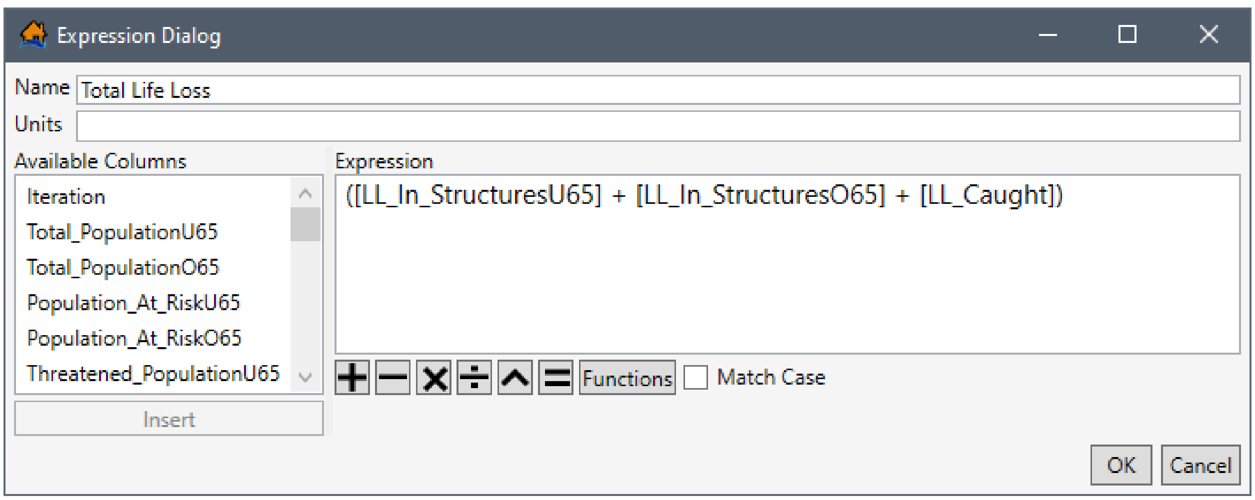


Figure 13-10. Example – Expression Dialog – Edit Existing

Restore Default Simulation Results

Edits made to existing default iteration results can be restored by clicking the **Restore Defaults** (Figure 13-7) button, located at the bottom of the **Simulation Results By Iteration** panel. The **Clear Existing** (Figure 13-11) message dialog box opens. To clear all edits made to existing default iteration results, delete all created iteration results series, and restore deleted default iteration results, click **Yes** (Figure 13-11). Alternatively, click **No** or **Cancel**, to close the **Clear Existing** (Figure 13-11) message dialog box without clearing edits made to existing default iteration results and without deleting newly created results.

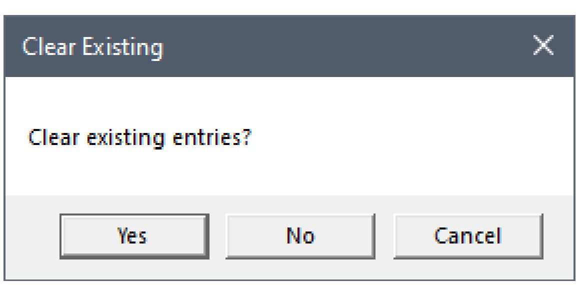


Figure 13-11. Clear Existing Message Dialog Box

If users edit the existing default structure summary results, from the bottom of the **Simulation Results By Structure** panel, click the **Restore Defaults** (Figure 13-7) button. The **Clear Existing** (Figure 13-11) message dialog box opens. To clear all edits made to existing default structure summary results, delete all created structure summary results series, and restore deleted default structure summary results click **Yes** (Figure 13-11). Alternatively, click **No**, or **Cancel**, to close the **Clear Existing** (Figure 13-11) message dialog box without clearing edits made to existing default structure summary results and without deleting newly created results.

Delete Existing Simulation Results

Existing iteration or structure summary results can be deleted by clicking the delete ✖ button (Figure 13-7) for a specific results series (e.g., ***Total Life Loss*** in Figure 13-6). Note, this cannot be undone for created summary results, but this action can be undone for deleted default summary results by clicking the **Restore Defaults** button.

Box and Whisker Plots Tab

In addition to the **Simulation Results By Iteration**, **Simulation Results By Structure**, and **Filter** panels (Figure 13-12), the **Box and Whisker Plots** tab contains a **Change Order** (Figure 13-12) button to further control how the results are displayed in the plot. Click the **Change Order** button (Figure 13-12) to open a window that allows the user to set the order of the results in the plot window.

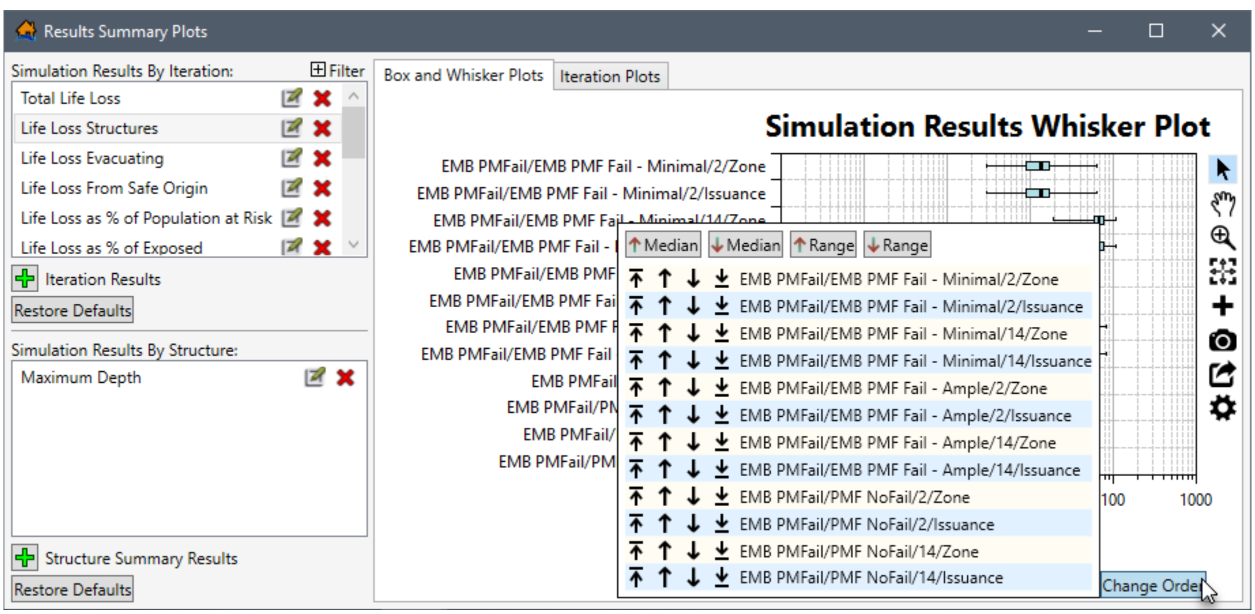


Figure 13-12. Results Summary Plots – Box and Whisker Plots Tab – Change Order Options

Click ⤒ to move the series to the top of the plot, click ↑ to move the series up one level, click ↓ to move the series down one level, or click ⤓ to move the series to the bottom of the plot. The ↑Median button orders all series based on the median value from low to high, the ↓Median button orders all series based on the median value from high to low. The ↑Range button order all series based on the range of values from low to high, and the ↓Range button orders all series based on the range of values from high to low.

Iteration Plots Tab

In addition to the **Simulation Results By Iteration** and **Filter** panels (Figure 13-13), the **Iteration Plots** tab contains a panel at the bottom of the plot window. The panel provides a **Simulation**, **Alternative**, **Imminent Hazard** dropdown lists; an **Iteration** box and a **Generate Detail** button (Figure 13-13). The purpose of the panel is to allow users to set the simulation, alternative, and imminent hazard to display in the plot. Then from the combined selection users can visualize each iteration result in a scatter plot to help the user identify which iterations calculated outlying values (e.g., ***Iteration Number: 90*** in Figure 13-13). These outlying values, or any iteration of interest, can then be selected to generate detailed output for that specific iteration, by clicking the **Generate Detail** button (Figure 13-13).

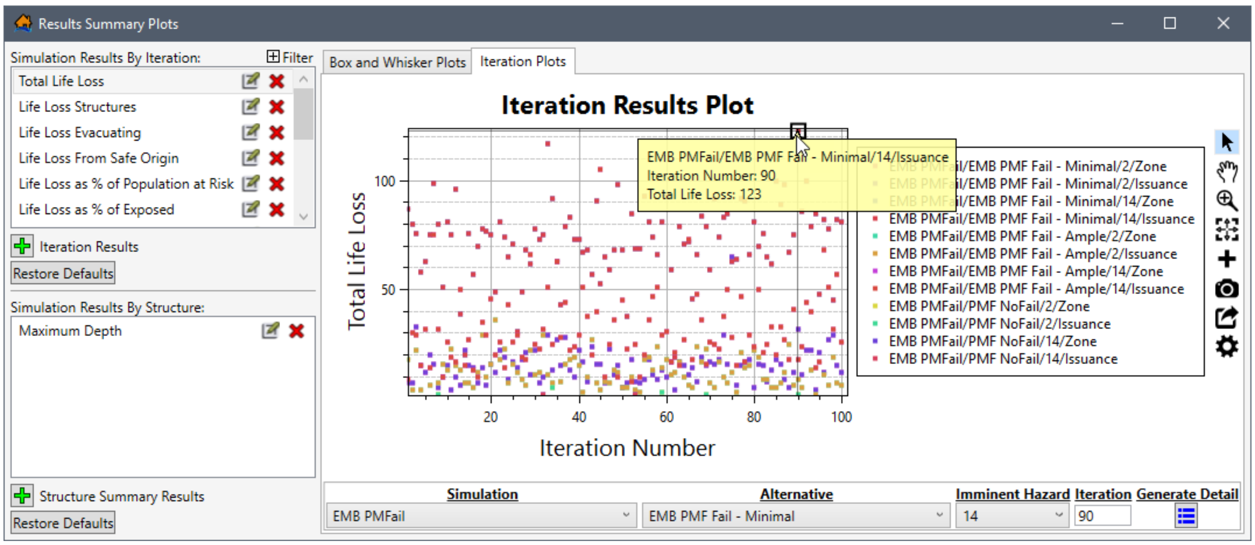


Figure 13-13. Results Summary Plots – Iteration Plots Tab – Example Iteration Selection

If detailed output is generated, LifeSim re-runs the specified iteration with the same random number sequence but this time much more information is saved. The detailed output generated contains information down to the evacuating group level regarding group size, location, and timing information during the simulation. The detailed output generated includes a database containing evacuation information for groups and their evacuation paths, structure inventory results shapefile, and a road network results shapefile. There is no limit to the number of iterations that LifeSim can generate a detailed output for, however physical memory could become an issue at some point. Detailed outputs cannot be generated concurrently.

The **Re-Generate Detailed Output** option re-generates the specified iteration's detailed output and is only available after a detailed output has been generated for a selected iteration. The **Re-Generate Detailed Output** option is best used when the simulation has been altered and a detailed output had already been created for specific iteration(s) and would need to be re-generated for the new simulation.

Refer to Sections 13.3.6, 13.4.6 and 13.5.2 for more information on viewing the results of the generated detailed output for iterations.

13.2.2 Simulation Result Table

The **Simulation Result Table** (Figure 13-1), **Study Tree | Simulations** shortcut menu command opens the **Results Summary Tables** (Figure 13-14). The **Results Summary Tables** (Figure 13-14) is divided into three main panels. The first panel located at the top left, is the **Simulation Results By Iteration** panel (Figure 13-14). The second panel located at the bottom left, is the **Simulation Results By Structure** panel (Figure 13-14). The third panel located on the right-side is the table containing the **Export and Copy Tools** and **Selection Tools** toolbars (Figure 13-14). Refer to Section 3.5 more information regarding the table toolbars.

However, there is a fourth optional panel, the **Filter** panel (Figure 13-15) is an expandable panel that is opened by clicking the **Filter** ⊞ expand button from the **Simulation Results By Iteration** panel. To close the optional **Filter** panel, click the **Filter** ⊟ collapse (Figure 13-15) button.

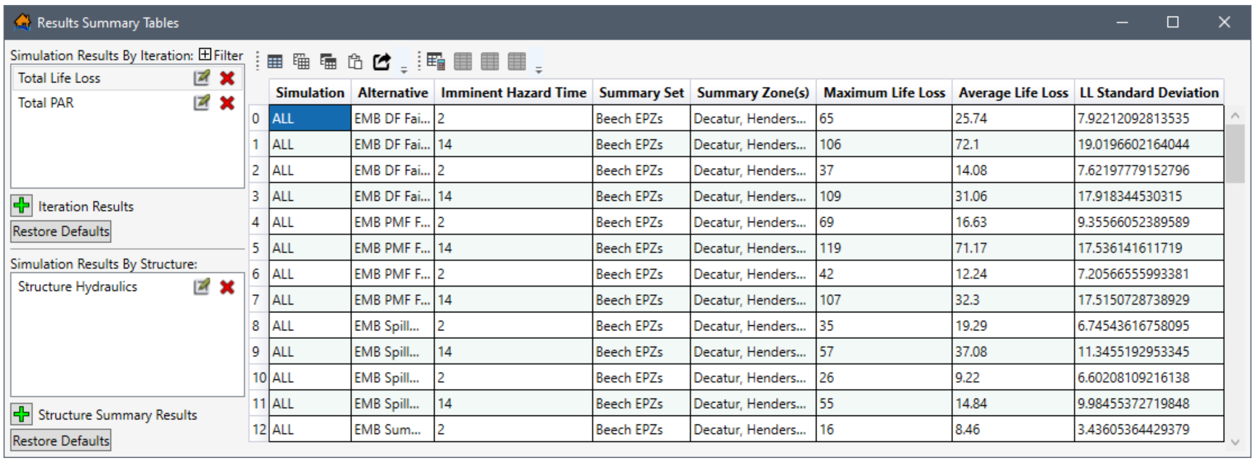


Figure 13-14. Example – Results Summary Tables

Results Summary Tables Options

The expandable **Filter** panel (Figure 13-15) contains three categories: **Simulations**, **Imminent Hazard Times**, and **Alternatives**; for limiting the simulation results displayed in the table by aggregated summary zones. Remove simulations simulation aggregation zones, imminent hazard times, and/or alternatives from the table by unchecking the ☐ checkbox.

In the **Simulations** category (Figure 13-15), the simulations contain aggregation summary zones. For the example provided in Figure 13-15, the ***EPZ*** is the aggregation summary zone for the ***Indirect_AgDamages*** simulation. All fields in the summary zone, e.g., ***Decatur***, ***Henderson*** and ***Unassigned*** in Figure 13-15) are listed to allow specific fields to be filtered from the table. The aggregation summary zones listed for a simulation depends on the summary polygons defined for a simulation from the **Create New Simulation** dialog box (review Section 12.1 for more information regarding the setup for a simulation).

Uncheck the ☐ checkbox to turn off the **Aggregate Selected Summary Zones** (Figure 13-6) to display the results for all fields in the summary zone individually by not grouping the simulation results by summary zone.

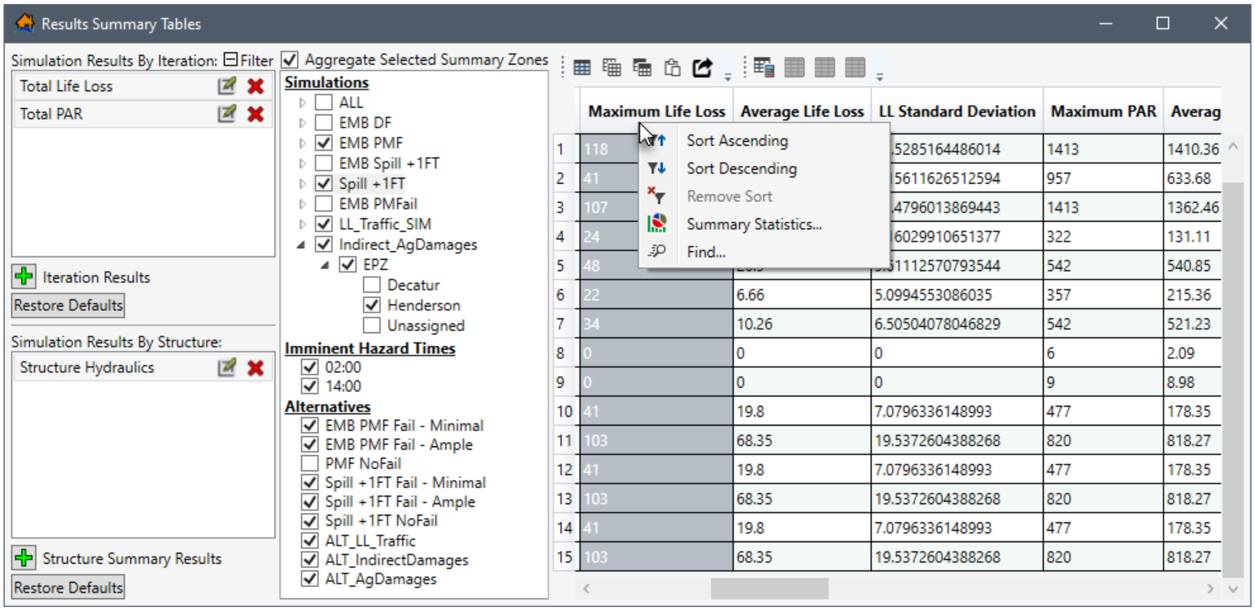


Figure 13-15. Example – Results Summary Table – Filter Panel – Header Shortcut Menu

From the table, right-click a header row in the table to access the shortcut menu (Figure 13-15) for the column. The shortcut menu allows users to sort the table, remove table sorting, open the **Summary Statistics** dialog box (Figure 13-16) for the column, or open the **Find In** dialog box (Figure 13-17) for the column.

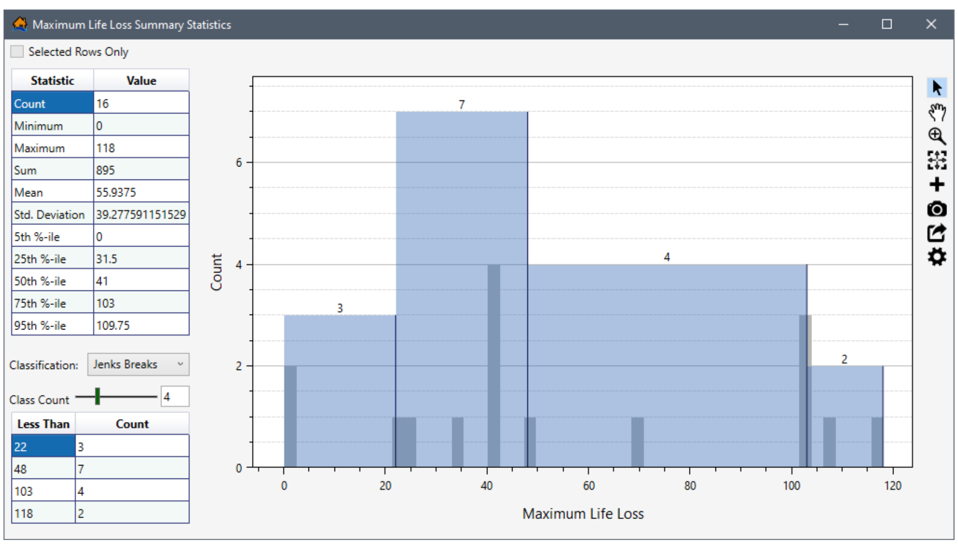


Figure 13-16. Example – Maximum Life Loss Column – Summary Statistics Dialog Box

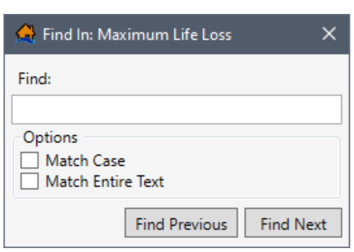


Figure 13-17. Find In Dialog Box

As described in Section 13.2.1, from the **Simulation Results By Iteration** or **Simulation Results By Structure** panels (Figure 13-14), users can click the button to create new simulation results, click the edit button to edit an existing simulation results, click the delete button to delete a specific simulations results, and click the **Restore Defaults** (Figure 13-14) button to restore all simulation results back to defaults.

Create or Edit Simulation Results

From the **Results Summary Tables** (Figure 13-14), open an existing iteration or structure summary results by clicking the edit button. Alternatively, create a new iteration or structure summary results by clicking the add **Iteration Results** or add **Structure Summary Results** (Figure 13-7) buttons, respectfully. Either way the **Expression Dialog** (Figure 13-18) opens.

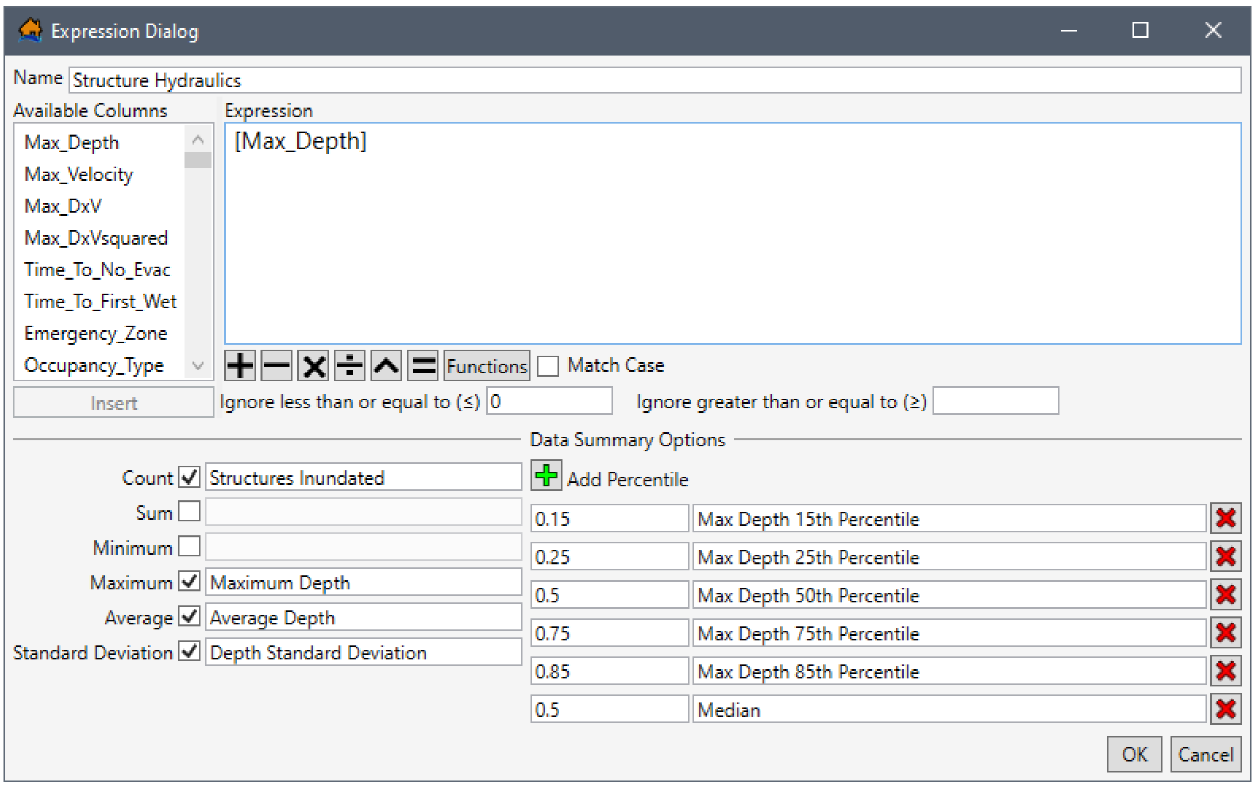


Figure 13-18. Example – Expression Dialog – Simulation Results Table – *Structure Hydraulics*

1. In the **Expression Dialog** (Figure 13-18), enter or edit the name of the simulation results in the **Name** box (Figure 13-18).

2. To edit or build the expression for the results in the **Expression** box (Figure 13-8), select a specific results column from the list of **Available Columns** (e.g., ***[Max_Depth]*** in Figure 13-18), and click **Insert**.

3. The **Functions** button (Figure 13-18) opens a new window providing a selection of all available functions in LifeSim. Select a function from the **Functions** list (e.g., ***IF*** in Figure 13-9), to view tips for using the function and click **Insert** to add the selected function to the **Expression** box (Figure 13-18) in the **Expression Dialog** (Figure 13-18).

4. Turn on **Match Case** (Figure 13-18) by checking the checkbox ☑ to specify that the entered expression is case sensitive.

5. Users can enter a value in the **Ignore less than or equal to (≤)** (Figure 13-18) box to set a minimum limit for the results. Alternatively, enter a value in the **Ignore greater than or equal to (≥)** (Figure 13-18) box to set a maximum limit for the results.

6. Default data summary options can be added to the summary results, from the **Data Summary Options** panel (Figure 13-18), check the checkbox ☑ to include a default summary option (e.g., ***Count*** in Figure 13-18) and enter the column header text for the new column (e.g., ***Structures Inundated*** in Figure 13-18).

7. Uncheck ☐ the default data summary option to remove the summary column from the table.

8. Also, from the **Data Summary Options** panel (Figure 13-18), users can add summary percentile columns. Click the **Add Percentile** button to add a value box (e.g., ***0.15*** in Figure 13-18) and column header text box (e.g., ***Max Depth 15th Percentile*** in Figure 13-18). Enter the desired percentile and column header in the provided boxes.

9. Click the delete button to delete a specific percentile from the table.

10. Click **OK**, to close the **Expression Dialog** (Figure 13-18) and create the new (or update an existing) iteration or structure summary results series.

13.3 View Simulation Results Plots

The **Simulation '*Simulation Name*' Results Plot** (Figure 13-19) provides a way to summarize the LifeSim iteration results for an alternative and compare them across different times of day. The results are summarized in the form of box and whisker plots to allow the user to quickly gain an understanding of the distribution of various results from the Monte Carlo simulation.

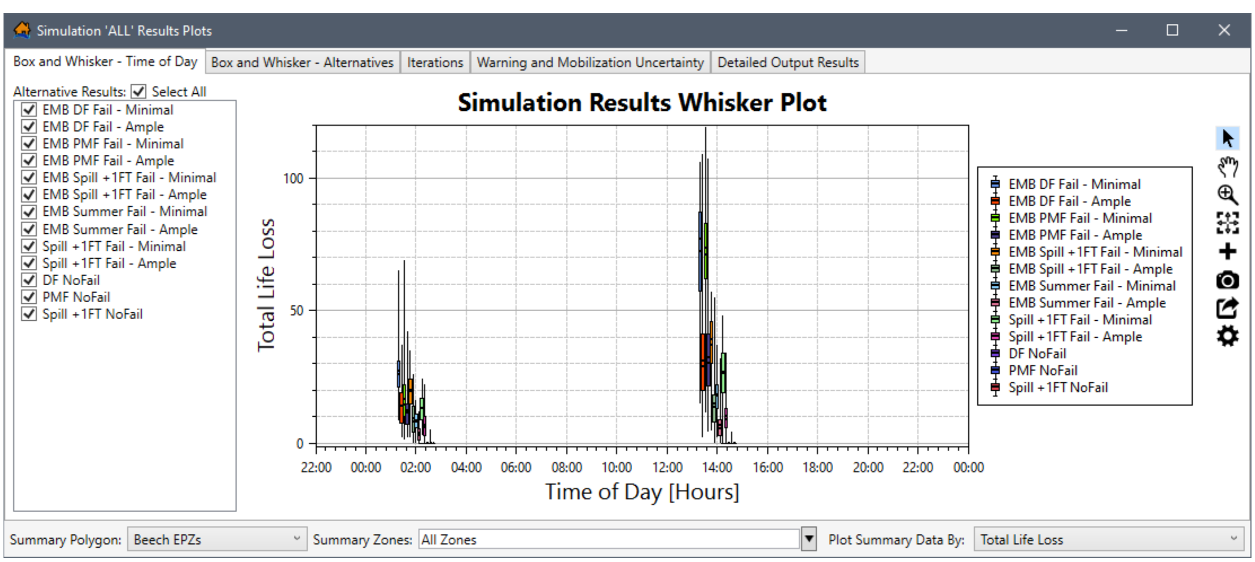


Figure 13-19. Example – Simulation '*ALL*' Results Plots – Box and Whisker – Time of Day Tab

Open the simulation result plots for a simulation, from the LifeSim main window (Figure 13-3), from the **Study** tab, from the **Study Tree**, under **Simulations**, right-click on a simulation (e.g., ***All*** in Figure 13-3), from the shortcut menu click **View Results Plots** (Figure 13-3). The **Simulation '*All*' Results Plot** (Figure 13-19) opens for the alternatives associated with the simulation.

13.3.1 Summarize Plot Results Options

The **Simulation Results Plots** (Figure 13-19) contains five tabs. Of the five tabs, the **Box and Whisker – Time of Day** (Section 13.3.2), **Box and Whisker – Alternatives** (Section 13.3.3), and the **Iterations** (Section 13.3.4) tabs includes the same selector panel at the bottom of the plot window. The purpose of the panel is to allow users to set the summary polygon, zones, and data to display in the plot window and the selection updates all three tabs. Review Section 3.6 for instructions on using the general plotting tools (located on the right side of the plot window) and for customizing plots.

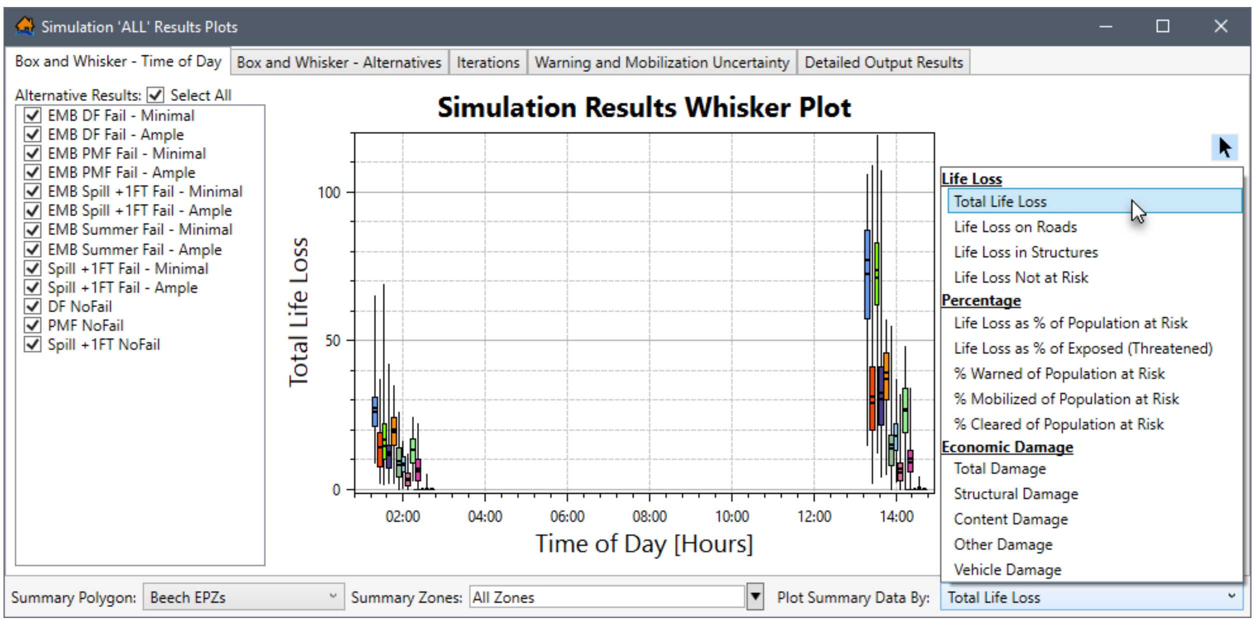


Figure 13-20. Example – Simulation Results Plots – Plot Summary Data By Options

The panel provides the following dropdown lists for summarizing the results in the plot window:

- **Summary Polygon** – The dropdown list for this summary option contains a list of all summary zones listed for the simulation. The **Summary Polygon** (Figure 13-20) list is based on the summary polygons defined for the simulation from the **Create New Simulation** dialog box (review Section 12.1 for more information regarding the setup for a simulation). Use the **Summary Polygon** (Figure 13-20) list to select a specific summary zone.

- **Summary Zones** – The dropdown list for this option contains the list of polygon zone names in the selected **Summary Polygon**. Select specific zone(s) by checking the checkbox ☑ for the zone(s), from the **Summary Zone** (Figure 13-20) list, to view the results for one or more areas in the study. Alternatively, select ***All Zones*** (default) to view results for all zones in the summary polygon.

- **Plot Summary Data By** – This dropdown menu contains a list of plot summary options grouped by **Life Loss**, **Percentage** and **Economic Damage** categories (Figure 13-20). The Simulation Results Whisker Plot updates the y-axis based on the **Plot Summary Data By** selection (e.g., ***Total Life Loss*** in Figure 13-20).

13.3.2 Time of Day Box and Whisker

By default, the **Simulation Results Plots** (Figure 13-19) opens to the **Box and Whisker – Time of Day** tab (Figure 13-19). The left side of the plot window provides an **Alternative Results** panel (Figure 13-19) containing a list of the alternative names included in the simulation. To view the results for specific alternative(s) in the simulation, check the checkbox(es) ☑ for the desired alternative(s). By default, the **Select All** (Figure 13-19) checkbox is checked ☑ to display all alternative results in the **Simulation Results Whisker Plot** (Figure 13-19).

In addition to the **Alternative Results** panel (Figure 13-19), the **Box and Whisker – Time of Day** tab contains a panel at the bottom of the plot window. The panel provides a **Summary Polygon**, **Summary Zones**, **Plot Summary Data By** lists (review Section 13.3.1). The purpose of the panel is to allow users to set the summary polygon, zones, and data to display in the **Simulation Results Whisker Plot** (Figure 13-19).

13.3.3 Alternatives Box and Whisker

From the **Simulation Results Plots** (Figure 13-21), select the **Box and Whisker – Alternatives** tab (Figure 13-21). The left side of the plot window provides the **Alternatives** and **Imminent Hazard Time** panels (Figure 13-21).

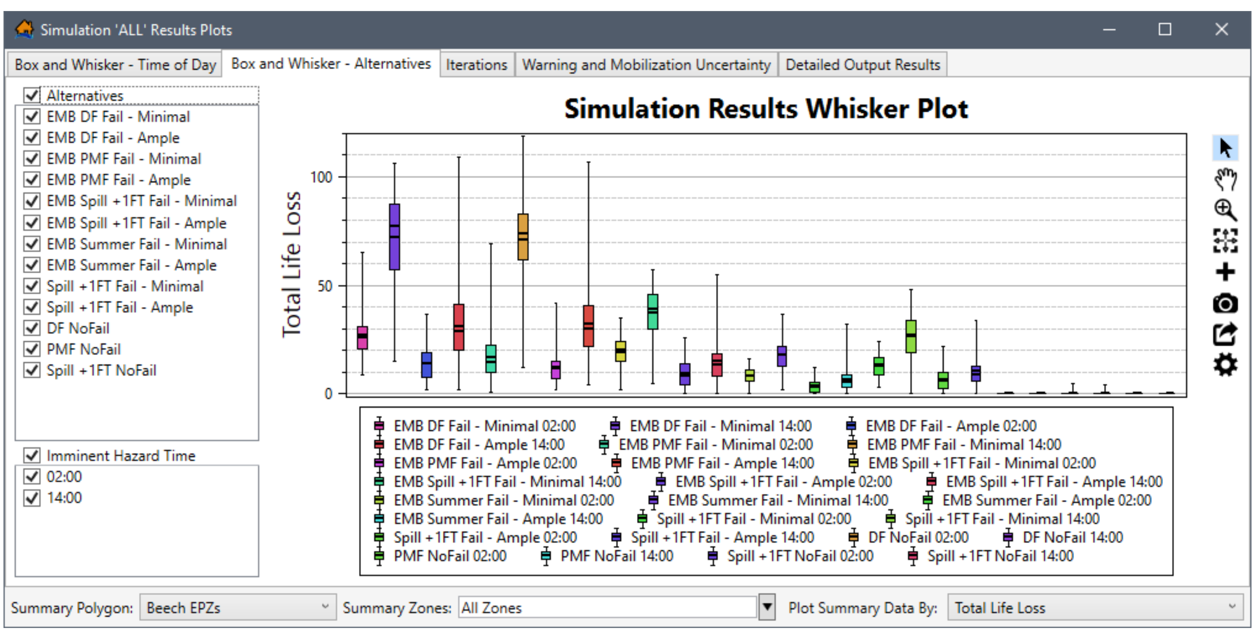


Figure 13-21. Example – Simulation '*ALL*' Results Plots – Box and Whisker – Alternatives Tab

The **Alternatives** panel (Figure 13-21) contains a list of the alternative names included in the simulation. To view the results for specific alternatives in the simulation, check the checkboxes ☑ for the desired alternatives. By default, the **Alternatives** (Figure 13-21) checkbox is checked ☑ to display all alternative results in the plot.

The **Imminent Hazard Time** panel (Figure 13-21) contains a list of all of the hazard occurrence time(s) of day selected from the **Create New Simulation** dialog box for the simulation (review Section 12.1 for more information regarding the setup for a simulation). By default, the **Imminent Hazard Time** (Figure 13-21) checkbox is checked ☑ to display all time(s) of day in the plot.

In addition to the **Alternatives** and **Imminent Hazard Time** panels (Figure 13-21), the **Box and Whisker – Alternatives** tab contains a panel at the bottom of the plot window. The panel provides a **Summary Polygon**, **Summary Zones**, **Plot Summary Data By** dropdown lists (review Section 13.3.1). The purpose of the panel is to allow users to set the summary polygon, zones, and data to display in the plot.

13.3.4 Results by Iteration

From the **Simulation Results Plots** (Figure 13-21), select the **Iterations** tab (Figure 13-22). Viewing results by iteration allows the user to visualize each iteration result in a scatter plot to help the user identify which iterations calculated outlying values. These outlying values, or any iteration of interest, can then be selected to generate detailed output for that specific iteration.

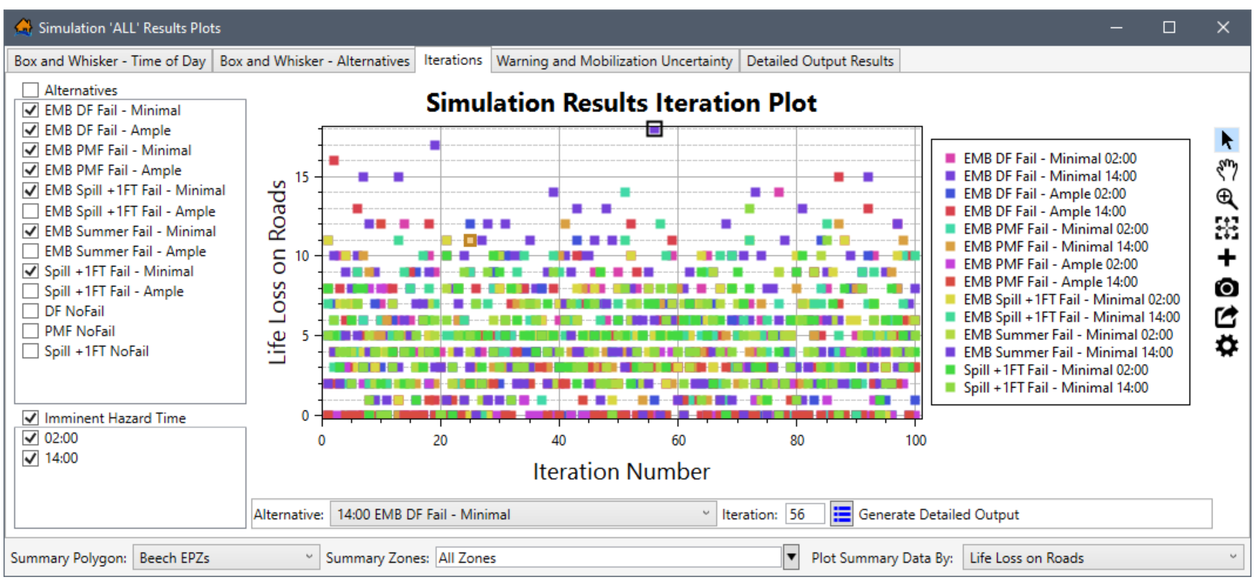


Figure 13-22. Example – Simulation '*ALL*' Results Plots – Iterations Tab

The left side of the plot window provides the **Alternatives** and **Imminent Hazard Time** panels (Figure 13-22). The **Alternatives** panel (Figure 13-22) contains a list of the alternative names included in the simulation. To view the results for specific alternatives in the simulation, check the checkboxes ☑ for the desired alternatives. By default, the **Alternatives** (Figure 13-21) checkbox is checked ☑ to display all alternative results in the plot.

The **Imminent Hazard Time** panel (Figure 13-22) contains a list of all of the hazard occurrence time(s) of day selected from the **Create New Simulation** dialog box for the simulation (review Section 12.1 for more information regarding the setup for a simulation). By default, the **Imminent Hazard Time** (Figure 13-21) checkbox is checked ☑ to display all time(s) of day in the plot.

In addition to the **Alternatives** and **Imminent Hazard Time** panels (Figure 13-22), the **Iterations** tab contains a panel at the bottom of the plot window. The panel provides a **Summary Polygon**, **Summary Zones**, **Plot Summary Data By** dropdown lists (review Section 13.3.1). The purpose of the panel is to allow users to set the summary polygon, zones, and data to display in the plot.

Detailed Output

Within the **Simulation Results Iteration Plot** (Figure 13-22), the **Iterations** tab contains an **Alternative** dropdown list, an **Iteration** box and a **Generate Detailed Output** button (Figure 13-22). The purpose of the detailed output panel is to allow users to identify iterations that calculated outlying values (e.g., ***Iteration Number: 56*** in Figure 13-22). These outlying values, or any iteration of interest, can be selected to generate detailed output for that specific iteration, by clicking the **Generate Detailed Output** button (Figure 13-22).

If detailed output is generated, LifeSim re-runs the specified iteration with the same random number sequence but this time much more information is saved. The detailed output generated contains information down to the evacuating group level regarding group size, location, and timing information during the simulation. The detailed output generated includes a database containing evacuation information for groups and their evacuation paths, structure inventory results shapefile, and a road network results shapefile. There is no limit to the number of iterations that LifeSim can generate a detailed output for, however physical memory could become an issue at some point. Detailed outputs cannot be generated concurrently.

The **Re-Generate Detailed Output** option re-generates the specified iteration's detailed output and is only available after a detailed output has been generated for a selected iteration. The **Re-Generate Detailed Output** option is best used when the simulation has been altered and a detailed output had already been created for specific iteration(s) and would need to be re-generated for the new simulation.

Refer to Sections 13.3.6, 13.4.6 and 13.5.2 for more information on viewing the results of the generated detailed output for iterations.

13.3.5 Warning and Mobilization Uncertainty

From the **Simulation Results Plots** (Figure 13-22), select the **Warning and Mobilization Uncertainty** tab (Figure 13-23). This tab allows the user to visualize the results of the sampled uncertainty to quickly identify parameters that are driving the consequences. For example, plotting by warning issuance could show a strong trend between when the warning is issued and the total life loss. These results can be extremely helpful in identifying alternatives for risk reduction measures.

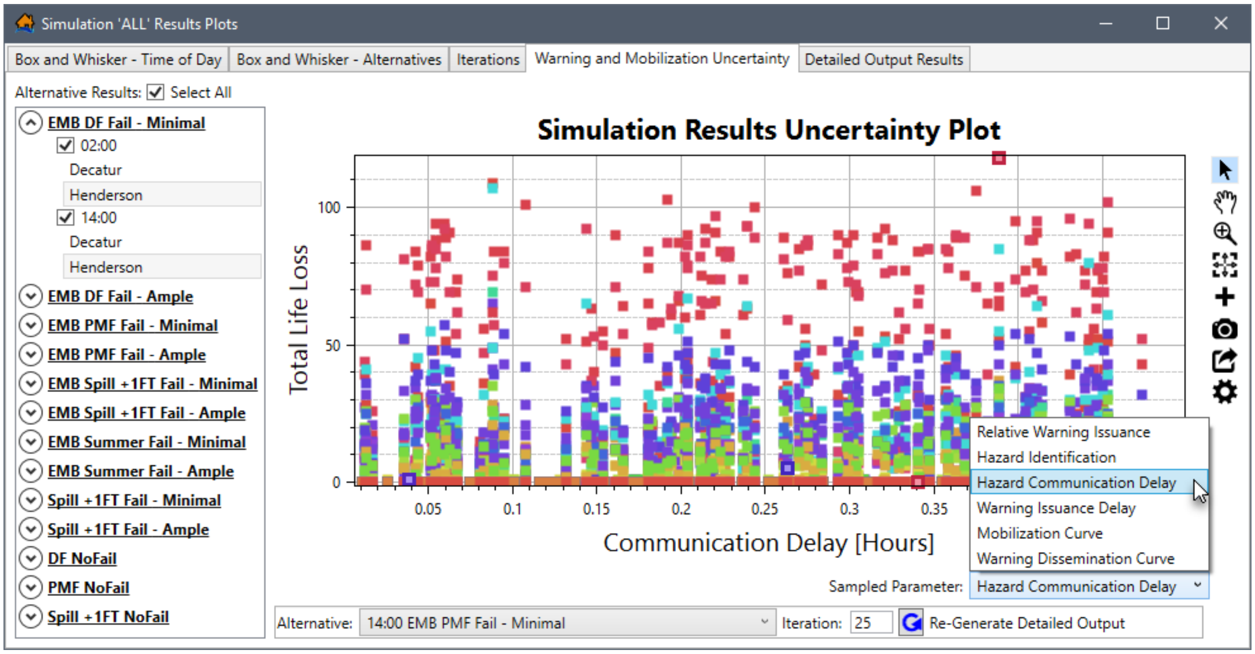


Figure 13-23. Example – Simulation '*ALL*' Results Plots – Warning and Mobilization Uncertainty Tab – Sampled Parameter Options

The left side of the **Warning and Mobilization Uncertainty** tab (Figure 13-23) provides an **Alternative Results** panel (Figure 13-23) containing a list of the alternative names included in the simulation. By default, the **Select All** (Figure 13-23) checkbox is checked ☑ to display all alternative results in the plot. Each alternative listed in the **Alternative Results** panel (Figure 13-23) contains the time(s) of day and zones. To expand the alternative, click ⌄; to collapse the alternative, click ⌃ (Figure 13-23).

Include a time of day by checking the checkbox ☑ (e.g., ***02:00*** and ***14:00*** are selected in Figure 13-23). Remove a time of day from the plot by unchecking ☐ the checkbox. Include specific zones in the plot by selecting the zone name(s). For the example provided in Figure 13-23, for both ***02:00*** and ***14:00***, ***Henderson*** is selected, but ***Decatur*** is not selected; therefore, only the results from ***Henderson*** is displayed in the plot.

Sampled Parameter Options

Within the **Simulation Results Uncertainty Plot** (Figure 13-23), the **Sampled Parameter** list (Figure 13-23), allows the user to summarize results by various parameters. From the **Sampled Parameter** list (Figure 13-23), users can select different parameters for viewing data in the plot. By default, the **Relative Warning Issuance** option is selected from the **Sampled Parameter** list (Figure 13-23) and displayed in the plot. For the example provided in Figure 13-23, the **Hazard Communication Delay** option (Figure 13-23) is selected and displayed in the plot.

For the **Mobilization Curve** and **Warning Dissemination Curve** options, the X-axis represents the sample that was used to define the curve. A value of zero would mean the curve was sampled at the lower bound of the uncertainty and a value of one would mean that the curve was sampled at the upper bound. For example, if the PAI curve maximum % initiated for an EPZ ranged between 85 and 95 percent and the X-axis value was zero, then the iteration had a maximum % initiated of 85 percent of the population for the EPZ.

Detailed Output

A panel located at the bottom of the **Simulation Results Uncertainty Plot** (Figure 13-23) contains an **Alternative** dropdown list, an **Iteration** box and a **Generate Detailed Output** button (Figure 13-23). The purpose of the detailed output panel is to allow users to identify iterations that calculated outlying values (e.g., ***Iteration Number: 25*** in Figure 13-23). These outlying values, or any iteration of interest, can be selected to generate detailed output for that specific iteration, by clicking the **Generate Detailed Output** button (Figure 13-22).

If detailed output is generated, LifeSim re-runs the specified iteration with the same random number sequence but this time much more information is saved. The detailed output generated contains information down to the evacuating group level regarding group size, location, and timing information during the simulation. The detailed output generated includes a database containing evacuation information for groups and their evacuation paths, structure inventory results shapefile, and a road network results shapefile. There is no limit to the number of iterations that LifeSim can generate a detailed output for, however physical memory could become an issue at some point. Detailed outputs cannot be generated concurrently.

The **Re-Generate Detailed Output** option (Figure 13-23) re-generates the specified iteration's detailed output and is only available after a detailed output has been generated for a selected iteration. The **Re-Generate Detailed Output** option is best used when the simulation has been altered and a detailed output had already been created for specific iteration(s) and would need to be re-generated for the new simulation.

Refer to Sections 13.3.6, 13.4.6 and 13.5.2 for more information on viewing the results of the generated detailed output for iterations.

13.3.6 View Detailed Output Results

When more information is desired for a specific iteration, detailed output can be generated. If detailed output is generated, LifeSim re-runs the specified iteration with the same random number sequence but this time much more information is saved. The detailed output generated contains information down to the evacuating group level regarding group size, location, and timing information during the simulation. The detailed output generated includes a database containing evacuation information for groups and their evacuation paths, structure inventory results shapefile, and a road network results shapefile. There is no limit to the number of iterations that LifeSim can generate a detailed output for, however physical memory could become an issue at some point. Detailed outputs cannot be generated concurrently.

From the **Simulation Results Plots** (Figure 13-22), select the **Detailed Output Results** tab (Figure 13-24). This tab allows the user to visualize the detailed output results generated for specific iterations of interest. The left side of the **Detailed Output Results** tab (Figure 13-24) provides a **Detailed Evacuation Results** panel (Figure 13-24) containing a list of all iterations with detailed output results. The iterations are grouped by alternative name and time of day.

To view the results for an iteration, from the **Detailed Evacuation Results** panel (Figure 13-24), select the desired iteration (e.g., ***Iteration : '97' – 02:00 Hazard*** in Figure 13-24). The **Time Evacuating Histogram** (Figure 13-24), **Cumulative Evacuation Outflow**, and **Cumulative Destination Arrivals** tabs display the results for the selected iteration.

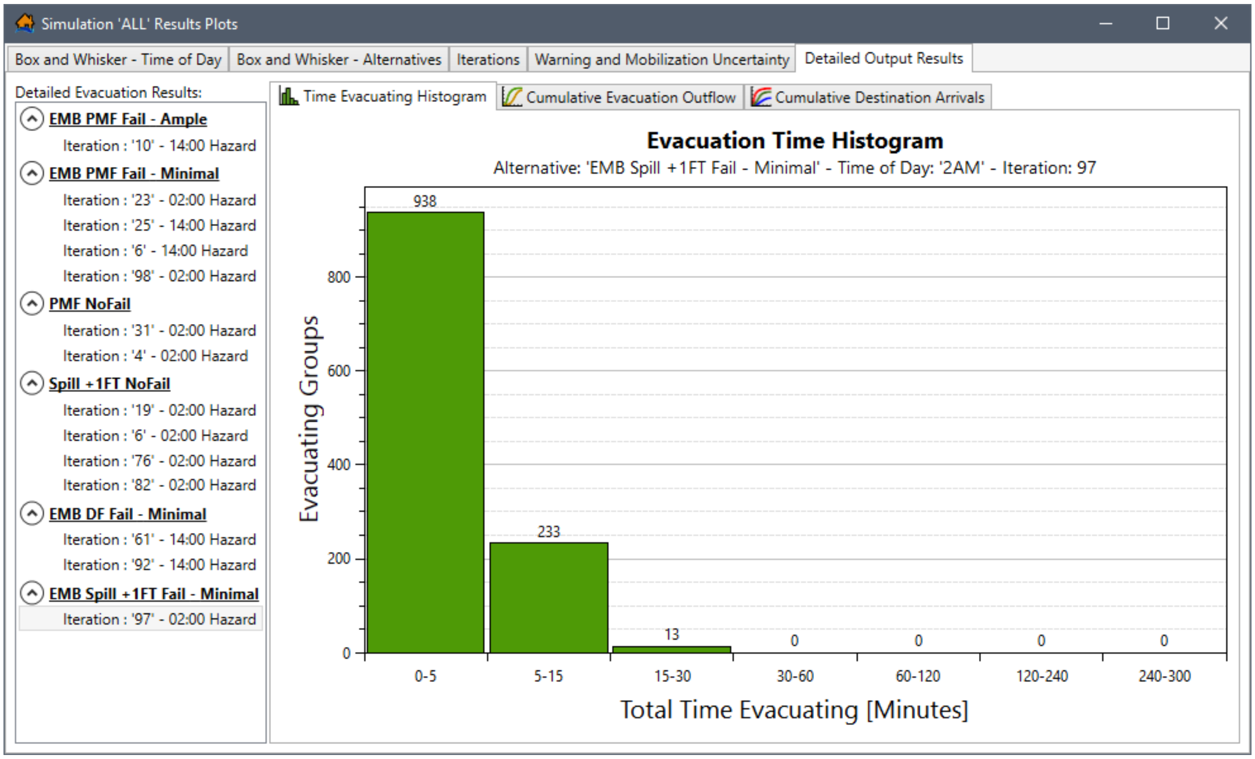


Figure 13-24. Example – Simulation '*ALL*' Results Plots – Detailed Output Results Tab – Time Evacuating Histogram Tab

Time Evacuating Histogram Tab

By default, the **Detailed Output Results** tab (Figure 13-24) opens to the **Time Evacuating Histogram** (Figure 13-24) tab. The time evacuating histogram, for the results of the iteration selected (e.g., ***Iteration : '97' – 02:00 Hazard*** in Figure 13-24), gives the user an indication for how long it takes groups to evacuate to a destination. This data is gathered from the detailed output table (review Section 13.4.6) as the difference between **TimeEnded** and **TimeMobilized**.

Cumulative Evacuation Outflow Tab

From the **Detailed Output Results** tab (Figure 13-24), select the **Cumulative Evacuation Outflow** tab (Figure 13-25). The evacuation outflow plot displays the cumulative flow of people that mobilize and reach safety over time. The plot can act as a good indicator of traffic congestion occurring during the evacuation. If the "Reached Safety" curve deviates from the "Mobilized" curve, this is an indication that traffic congestion is occurring.

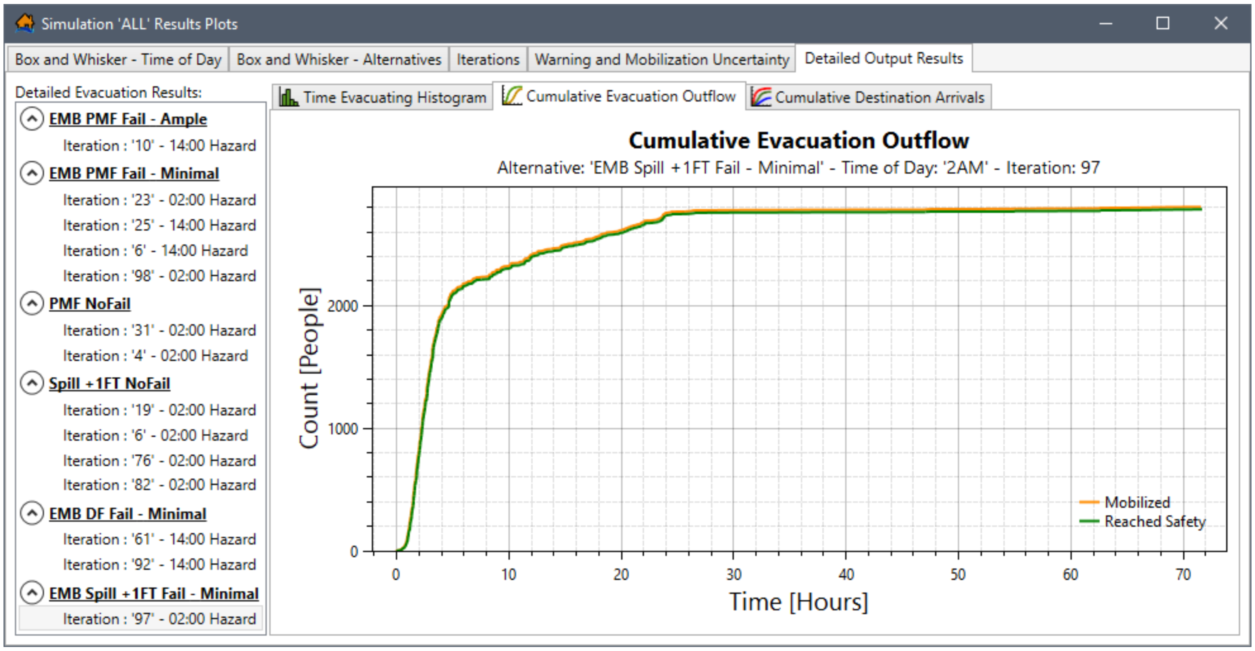


Figure 13-25. Example – Simulation '*ALL*' Results Plots – Detailed Output Results Tab – Cumulative Evacuation Outflow Tab

Cumulative Destination Arrivals Tab

From the **Detailed Output Results** tab (Figure 13-24), select the **Cumulative Destination Arrivals** tab (Figure 13-26). The cumulative destination outflow plot shows cumulative curves for the arrival of vehicles to each destination in the study.

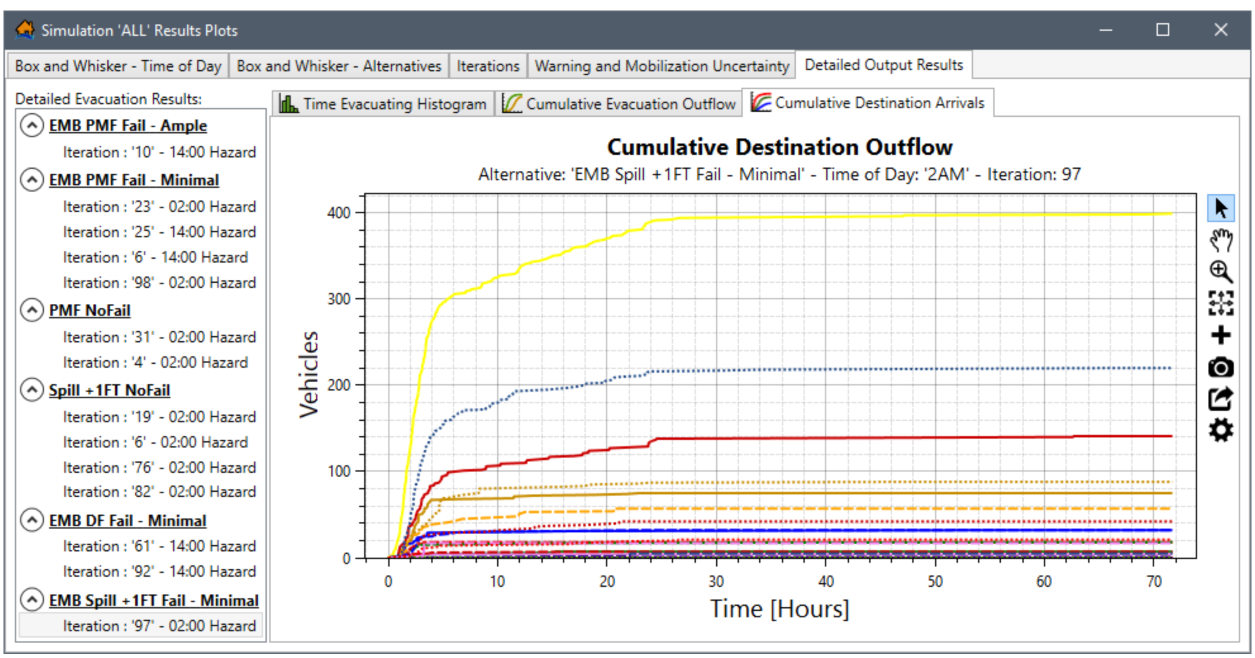


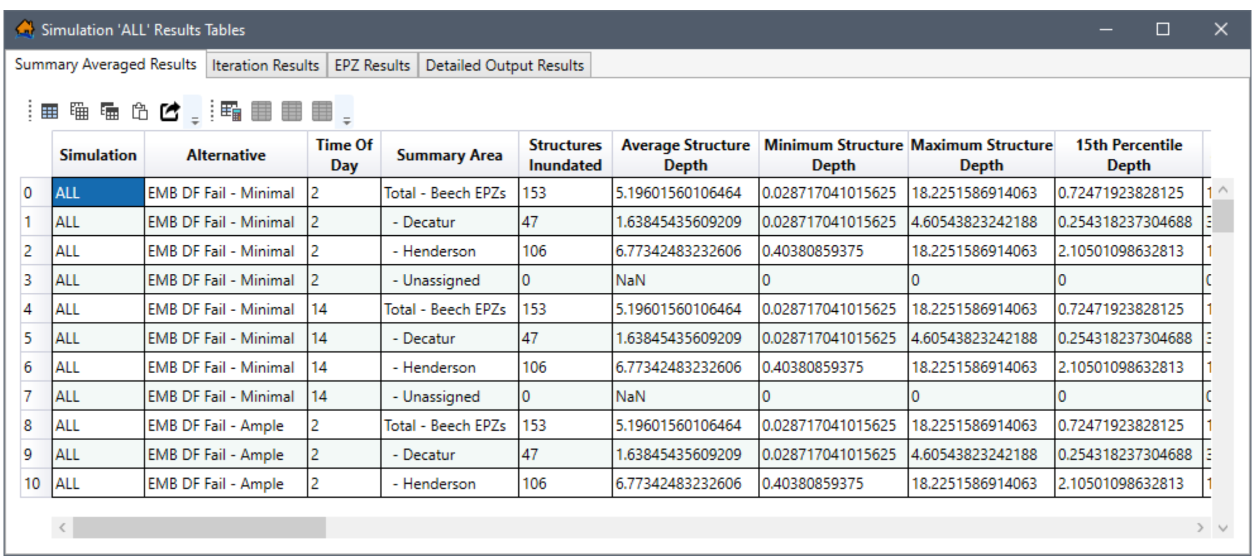
Figure 13-26. Example – Simulation '*ALL*' Results Plots – Detailed Output Results Tab – Cumulative Destination Arrivals Tab

The plot can be very useful in checking the validity of the traffic simulation results. If a destination is showing significantly more outflow than historical department of transportation data would indicate then the user should question the validity of the results and consider adjustments. If a destination that should be getting significant through traffic is getting none then the user should check the road network to make sure the roads are connected correctly (refer to Appendix B for tips on how to refine road network connectivity). Another way to check would be to animate the evacuation process to see the area of concern.

13.4 Simulation Results Table

Open the simulation results table for a simulation, from the LifeSim main window (Figure 13-3), from the **Study** tab, from the **Study Tree**, under **Simulations**, right-click on a simulation (e.g., ***All*** in Figure 13-3), from the shortcut menu click **View Results Tables** (Figure 13-3). The **Simulation '*All*' Results Tables** (Figure 13-27) opens for the alternatives associated with the simulation. By default, the **Simulation '*All*' Results Tables** (Figure 13-27) opens to the **Summary Averaged Results** tab; however, the simulation results tables also contains an **Iteration Results**, **EPZ Results**, and **Detailed Output Results** tabs (Figure 13-27). The **Export and Copy Tools** and **Selection Tools** toolbars (Figure 13-27) are located above the tabulated results in all four tabs. Refer to Section 3.5 more information regarding the table toolbars.

From the table, right-click a header row in the table to access the shortcut menu (Figure 13-15) for the column. The shortcut menu allows users to sort the table, remove table sorting, open the **Summary Statistics** dialog box (similar to Figure 13-16) for the column, or open the **Find In** dialog box (similar to Figure 13-17) for the column.



	Simulation	Alternative	Time Of Day	Summary Area	Structures Inundated	Average Structure Depth	Minimum Structure Depth	Maximum Structure Depth	15th Percentile Depth
0	ALL	EMB DF Fail - Minimal	2	Total - Beech EPZs	153	5.19601560106464	0.028717041015625	18.2251586914063	0.72471923828125
1	ALL	EMB DF Fail - Minimal	2	- Decatur	47	1.63845435609209	0.028717041015625	4.60543823242188	0.254318237304688
2	ALL	EMB DF Fail - Minimal	2	- Henderson	106	6.77342483232606	0.40380859375	18.2251586914063	2.10501098632813
3	ALL	EMB DF Fail - Minimal	2	- Unassigned	0	NaN	0	0	0
4	ALL	EMB DF Fail - Minimal	14	Total - Beech EPZs	153	5.19601560106464	0.028717041015625	18.2251586914063	0.72471923828125
5	ALL	EMB DF Fail - Minimal	14	- Decatur	47	1.63845435609209	0.028717041015625	4.60543823242188	0.254318237304688
6	ALL	EMB DF Fail - Minimal	14	- Henderson	106	6.77342483232606	0.40380859375	18.2251586914063	2.10501098632813
7	ALL	EMB DF Fail - Minimal	14	- Unassigned	0	NaN	0	0	0
8	ALL	EMB DF Fail - Ample	2	Total - Beech EPZs	153	5.19601560106464	0.028717041015625	18.2251586914063	0.72471923828125
9	ALL	EMB DF Fail - Ample	2	- Decatur	47	1.63845435609209	0.028717041015625	4.60543823242188	0.254318237304688
10	ALL	EMB DF Fail - Ample	2	- Henderson	106	6.77342483232606	0.40380859375	18.2251586914063	2.10501098632813

Figure 13-27. Example – Simulation '*ALL*' Results Tables – Summary Averaged Results Tab

13.4.1 Summary Averaged Results

By default, the **Simulation Results Tables** (Figure 13-27) opens to the **Summary Averaged Results** tab. The results in the **Summary Averaged Results** tab can be summarized by alternative, time of day, summary area by using the **Select Rows By Attribute** tool to define equations to select results.

Click , the **Select By Attribute** dialog box (Figure 13-28) opens. The **Available Columns** box (Figure 13-28) provides a list of all the result fields the user can select. The **Functions** button opens the LifeSim functions window (Figure 13-9) which provides the scripting commands and functions the user can use to filter results.

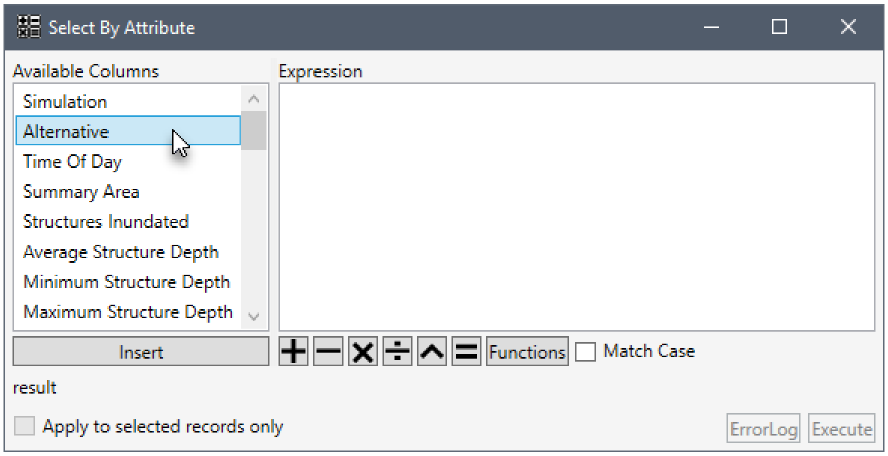


Figure 13-28. Select By Attribute

13.4.2 Iteration Results

From the **Simulation Results Tables** (Figure 13-27), select the **Iteration Results** tab (Figure 13-29). The left side of the tab provides an **Alternative Results** panel (Figure 13-29) containing a list of the alternative names included in the simulation. Each alternative listed in the **Alternative Results** panel (Figure 13-29) contains the time(s) of day included in the simulation. Click a specific time of day (e.g., ***02:00*** in Figure 13-29) for a specific alternative (e.g., ***EMB DF Fail – Minimal*** in Figure 13-29) to view the tabulated results. To expand the alternative, click ; to collapse the alternative, click (Figure 13-29).

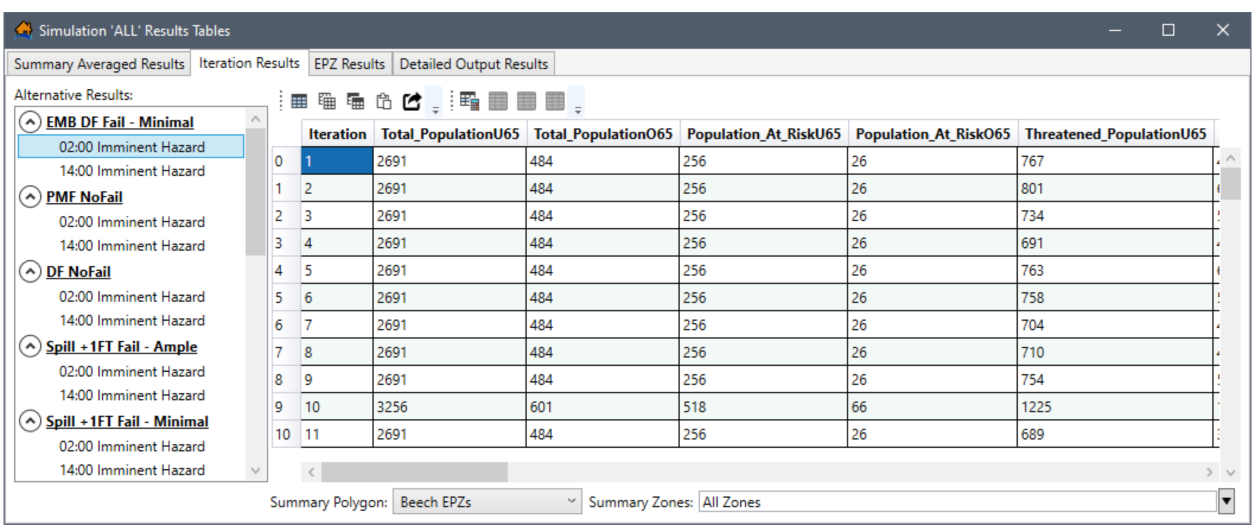


Figure 13-29. Example – Simulation '*ALL*' Results Tables – Iteration Results Tab

The panel at the bottom of the **Iteration Results** tab (Figure 13-29) provides the following dropdown lists for summarizing the results in the table:

- **Summary Polygon** – The dropdown list for this summary option contains a list of all summary zones listed for the simulation. The **Summary Polygon** (e.g., ***Beech EPZs*** in Figure 13-29) list is based on the summary polygons defined for the simulation from the **Create New Simulation** dialog box (review Section 12.1 for more information regarding the setup for a simulation). Use the **Summary Polygon** list to select a specific summary zone.

- **Summary Zones** – The dropdown list for this option contains the list of polygon zone names in the selected **Summary Polygon**. Select specific zone(s) by checking the checkbox ☑ for the zone(s), from the **Summary Zone** (Figure 13-29) list, to view the results for one or more areas in the study. Alternatively, select ***All Zones*** (default) to view results for all zones in the summary polygon.

13.4.3 EPZ Results

From the **Simulation Results Tables** (Figure 13-27), select the **EPZ Results** tab (Figure 13-30). The left side of the tab provides an **Alternative Results** panel (Figure 13-30) containing a list of the alternative names included in the simulation. Each alternative listed in the **Alternative Results** panel (Figure 13-30) contains the time(s) of day included in the simulation. Click a specific time of day (e.g., ***02:00*** in Figure 13-30) for a specific alternative (e.g., ***EMB Spill + 1FT Fail – Minimal*** in Figure 13-30) to view the tabulated results. To expand the alternative, click ⌄; to collapse the alternative, click ⌃ (Figure 13-30).

The panel at the bottom of the **EPZ Results** tab (Figure 13-30) contains an **Emergency Planning Zone** (Figure 13-30) dropdown list. The list allows the user to view the results one emergency planning zone (EPZ) at a time (e.g., ***Henderson*** in Figure 13-30).

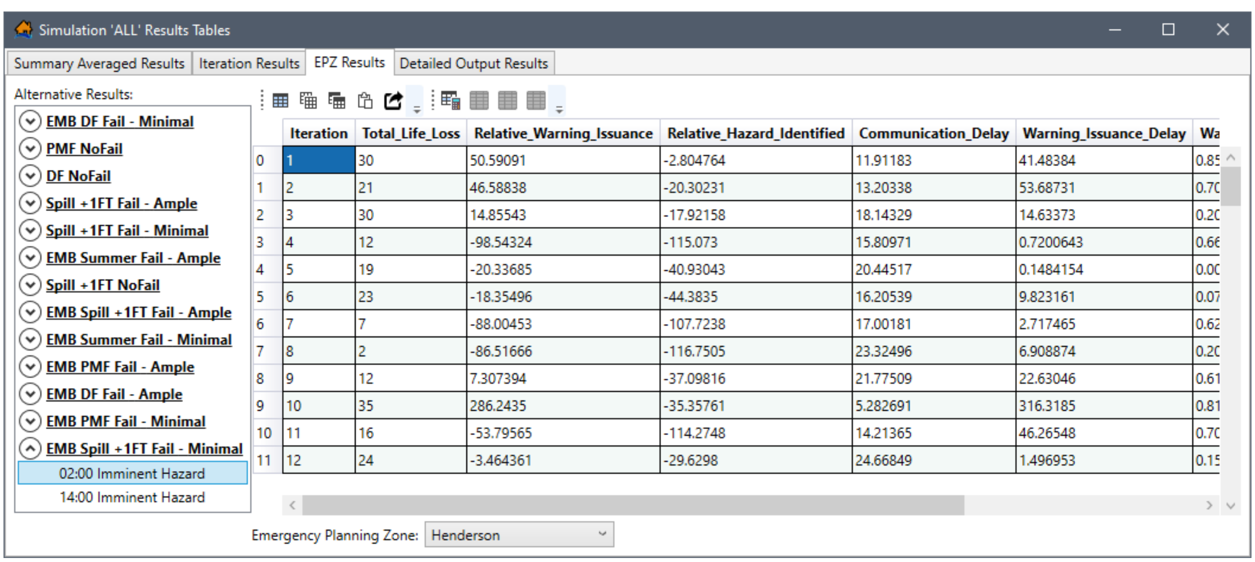
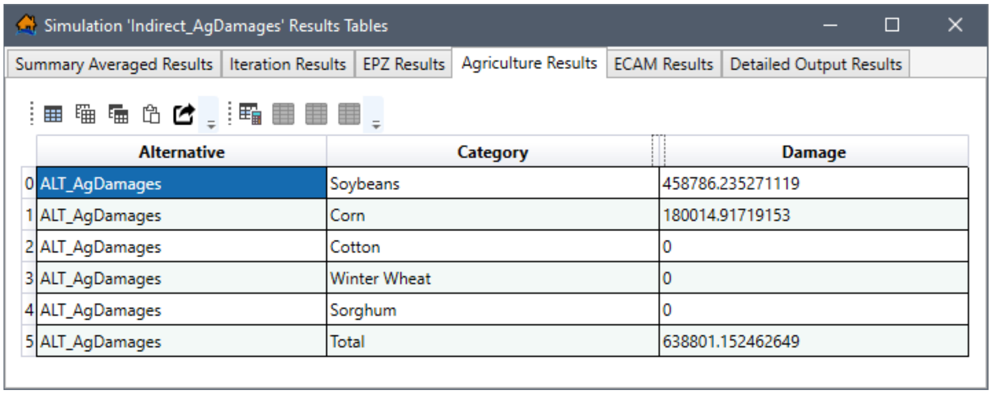


Figure 13-30. Example – Simulation '*ALL*' Results Tables – EPZ Results Tab

13.4.4 Agriculture Results

From the **Simulation Results Tables** (Figure 13-31), select the **Agriculture Results** tab (Figure 13-31). The agriculture damages are provided by alternative and crop category. Note, the **Agriculture Results** tab (Figure 13-31) is only available for simulations which include at least one alternative with an agriculture dataset and the agriculture compute option turned on at the alternative level (review Chapter 11, Section 11.1.2).


Simulation 'Indirect_AgDamages' Results Tables

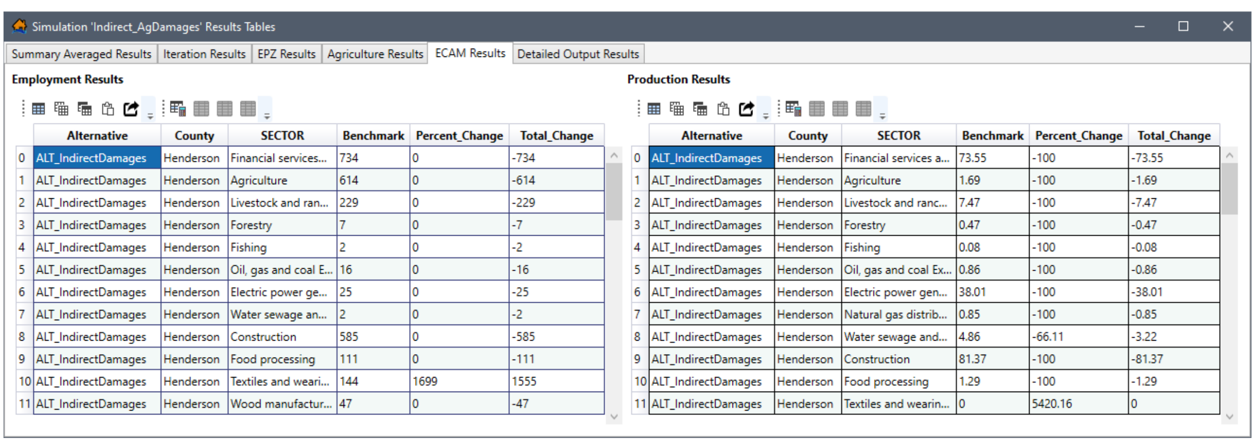
Summary Averaged Results | Iteration Results | EPZ Results | Agriculture Results | ECAM Results | Detailed Output Results

	Alternative	Category	Damage
0	ALT_AgDamages	Soybeans	458786.235271119
1	ALT_AgDamages	Corn	180014.91719153
2	ALT_AgDamages	Cotton	0
3	ALT_AgDamages	Winter Wheat	0
4	ALT_AgDamages	Sorghum	0
5	ALT_AgDamages	Total	638801.152462649

Figure 13-31. Example – Simulation Results Tables – Agriculture Results Tab

13.4.5 ECAM Results

From the **Simulation Results Tables** (Figure 13-31), select the **ECAM Results** tab (Figure 13-32). The ECAM results are provided in two tables, the **Employment Results**, and the **Production Results** (Figure 13-32) tables by alternative, county and ECAM sector. Note, the **ECAM Results** tab (Figure 13-32) is only available for simulations which include at least one alternative with an ECAM dataset (review Chapter 11, Section 11.1.2).


Simulation 'Indirect_AgDamages' Results Tables

Summary Averaged Results | Iteration Results | EPZ Results | Agriculture Results | ECAM Results | Detailed Output Results

Employment Results

	Alternative	County	SECTOR	Benchmark	Percent_Change	Total_Change
0	ALT_IndirectDamages	Henderson	Financial services...	734	0	-734
1	ALT_IndirectDamages	Henderson	Agriculture	614	0	-614
2	ALT_IndirectDamages	Henderson	Livestock and ran...	229	0	-229
3	ALT_IndirectDamages	Henderson	Forestry	7	0	-7
4	ALT_IndirectDamages	Henderson	Fishing	2	0	-2
5	ALT_IndirectDamages	Henderson	Oil, gas and coal E...	16	0	-16
6	ALT_IndirectDamages	Henderson	Electric power ge...	25	0	-25
7	ALT_IndirectDamages	Henderson	Water sewage an...	2	0	-2
8	ALT_IndirectDamages	Henderson	Construction	585	0	-585
9	ALT_IndirectDamages	Henderson	Food processing	111	0	-111
10	ALT_IndirectDamages	Henderson	Textiles and weari...	144	1699	1555
11	ALT_IndirectDamages	Henderson	Wood manufactur...	47	0	-47

Production Results

	Alternative	County	SECTOR	Benchmark	Percent_Change	Total_Change
0	ALT_IndirectDamages	Henderson	Financial services a...	73.55	-100	-73.55
1	ALT_IndirectDamages	Henderson	Agriculture	1.69	-100	-1.69
2	ALT_IndirectDamages	Henderson	Livestock and ranc...	7.47	-100	-7.47
3	ALT_IndirectDamages	Henderson	Forestry	0.47	-100	-0.47
4	ALT_IndirectDamages	Henderson	Fishing	0.08	-100	-0.08
5	ALT_IndirectDamages	Henderson	Oil, gas and coal Ex...	0.86	-100	-0.86
6	ALT_IndirectDamages	Henderson	Electric power gen...	38.01	-100	-38.01
7	ALT_IndirectDamages	Henderson	Natural gas distrib...	0.85	-100	-0.85
8	ALT_IndirectDamages	Henderson	Water sewage and...	4.86	-66.11	-3.22
9	ALT_IndirectDamages	Henderson	Construction	81.37	-100	-81.37
10	ALT_IndirectDamages	Henderson	Food processing	1.29	-100	-1.29
11	ALT_IndirectDamages	Henderson	Textiles and wearin...	0	5420.16	0

Figure 13-32. Example – Simulation Results Tables – ECAM Results Tab

13.4.6 Detailed Output Results

From the **Simulation Results Tables** (Figure 13-27), select the **Detailed Output Results** tab (Figure 13-33). This tab allows the user to visualize the detailed output results generated for specific iterations of interest. The left side of the **Detailed Output Results** tab (Figure 13-33) provides a **Detailed Evacuation Group Results** panel (Figure 13-33) containing a list of all iterations with detailed output results.

The iterations are grouped by alternative name and time of day. To expand the alternative, click ⌃; to collapse the alternative, click ⌃ (Figure 13-33). To view the results for an iteration, from the **Detailed Evacuation Group Results** panel (Figure 13-33), select the desired iteration (e.g., ***Iteration : '97' – 02:00 Hazard*** in Figure 13-33).

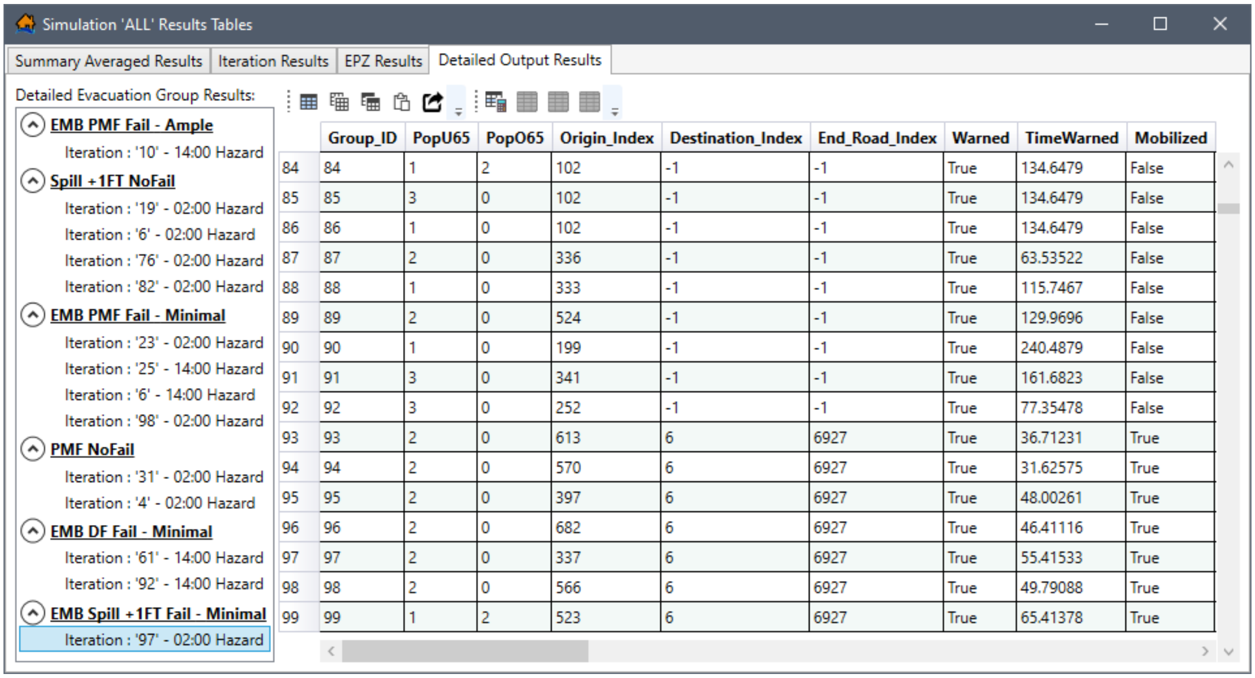


Figure 13-33. Example – Simulation '*ALL*' Results Tables – Detailed Output Results Tab

The functionality of the table is the same as the summary results table (Section 13.4). The detailed output table contains information about each evacuating group's behaviors during the iteration such as their origin, when they received the warning, when they mobilized, and where they evacuated to. This table can be very useful for finding specific information (e.g., the number of groups or people that did not get warned prior to getting trapped in their structure). The timing values for warning, mobilized, and ended are all relative to the imminent hazard time.

For every group that was defined for the iteration the detailed output table (Figure 13-33) contains the number of people, their origin (coincides with row number in structure attribute table), their destination (-1 means no destination was reached), warning information, mobilization information, and if they were caught evacuating.

13.5 Simulation Results in Map Window

Open the simulation results maps for a simulation, from the LifeSim main window (Figure 13-3), from the **Study** tab, from the **Study Tree**, under **Simulations**, right-click on a simulation (e.g., ***EMB Spill + 1FT***), from the shortcut menu click **View Results Maps** (Figure 13-3). The **Result Map Selector** dialog box (Figure 13-27) opens for the alternatives associated with the simulation.

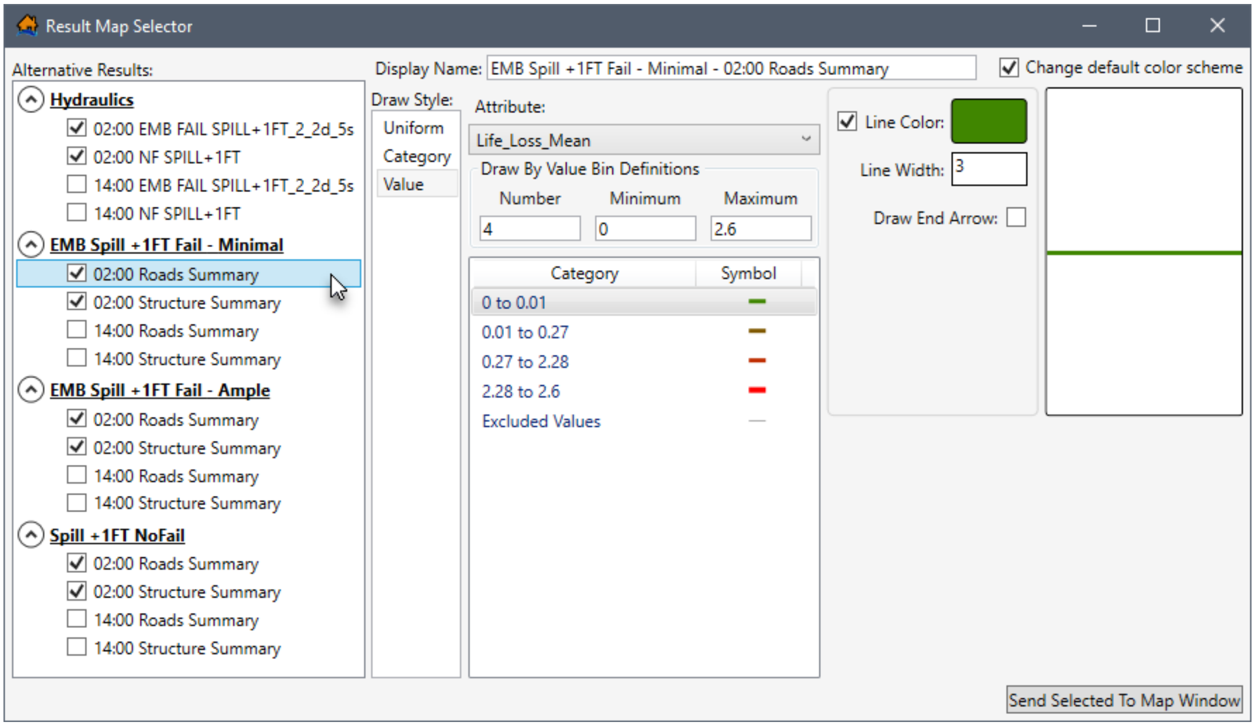


Figure 13-34. Example – Result Map Selector Dialog Box – Roads Summary

The left side of the **Result Map Selector** dialog box (Figure 13-33) provides an **Alternative Results** panel containing a list of the alternative names included in the simulation. Each alternative listed in the **Alternative Results** panel (Figure 13-33) contains the time(s) of day and results source (e.g., hydraulic data, roads summary, structure summary).

To expand the alternative, click ; to collapse the alternative, click (Figure 13-33). Include alternative results selection by checking the checkbox ☑ (e.g., ***Hydraulics 02:00 NF SPILL + 1FT*** in Figure 13-33). Uncheck ☐ the checkbox for an alternative result to exclude the results. Selected results are added to the map window when the **Send Selected To Map Window** button (Figure 13-33) is clicked.

The panel on the right side of the **Result Map Selector** dialog box (Figure 13-33) displays the default color scheme defined for the selected alternative results. Users can modify the default color scheme by clicking the checkbox ☑ to **Change default color scheme** (Figure 13-33). For more information regarding modifying the color scheme and drawing style, review Appendix A, Section A.8.1 in this manual.

NOTE: The roads and/or structure summary data is only included in the **Alternative Results** panel (Figure 13-33) if the simulation was set to **Generate Structure Summary Results** and/or **Generate Road Summary Results**, from the **Create New Simulation** dialog box for the simulation (review Section 12.2 for more information regarding the setup for a simulation). Furthermore, when detailed output results are generated for iterations, then the detailed output results (evacuation animation, road details and structure details) display in the **Alternative Results** panel (e.g., ***Iteration 97*** in Figure 13-35), and iterations are grouped by alternative name and time of day.

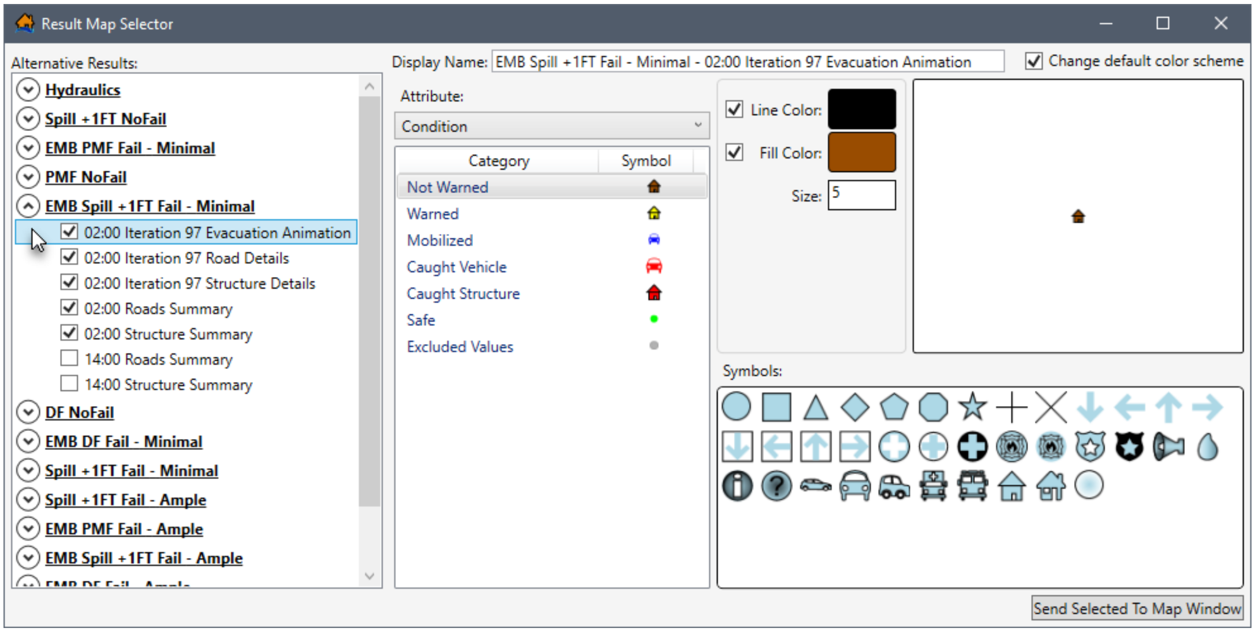


Figure 13-35. Example – Result Map Selector Dialog Box – Detailed Output Results

13.5.1 Color Scheme Options

The **Result Map Selector** dialog box (Figure 13-33) provides a default color scheme for the alternative results source (e.g., hydraulic data, roads summary, structure summary) listed in the **Alternative Results** panel (Figure 13-33). For the example provided in Figure 13-36, the hydraulic depth grid default color scheme is a continuous color ramp ranging from light blue to deep blue to identify hydraulic depths from shallow to deep.

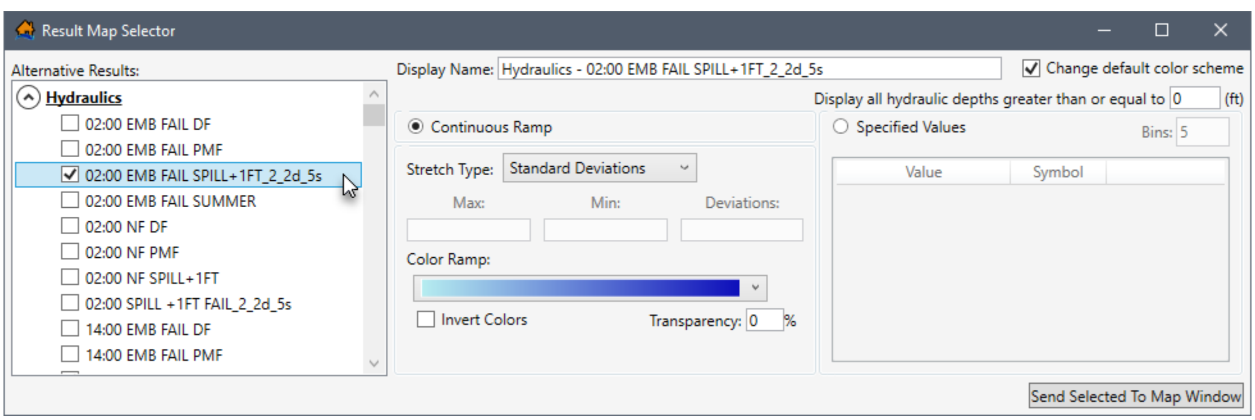


Figure 13-36. Example – Result Map Selector – Hydraulics Alternative Results

To modify the default color scheme for a specific alternative result, from the **Alternative Results** panel (Figure 13-36), select the desired results (e.g., ***Hydraulics 02:00 EMB FAIL SPILL + 1FT_2_2d_5s*** in Figure 13-36). Check the checkbox ☑ to **Change default color scheme** (Figure 13-36). The feature draw properties can now be modified for any results selected from the **Alternative Results** panel (Figure 13-36). For more information regarding modifying the color scheme and drawing style, review Appendix A, Section A.8.1 in this manual.

13.5.2 View Result Map in the Map Window

To view any of the alternative results in the map window, from the **Alternative Results** panel (Figure 13-36), check the checkbox ☑ for the results of interest (e.g., ***Hydraulics 02:00 EMB FAIL SPILL + 1FT_2_2d_5s*** in Figure 13-36), and click **Send Selected To Map Window** (Figure 13-36). The **Result Map Selector** dialog box (Figure 13-36) closes and the selected alternative results are added to the **Map Layers** tab (Figure 13-37) and display in the map window.

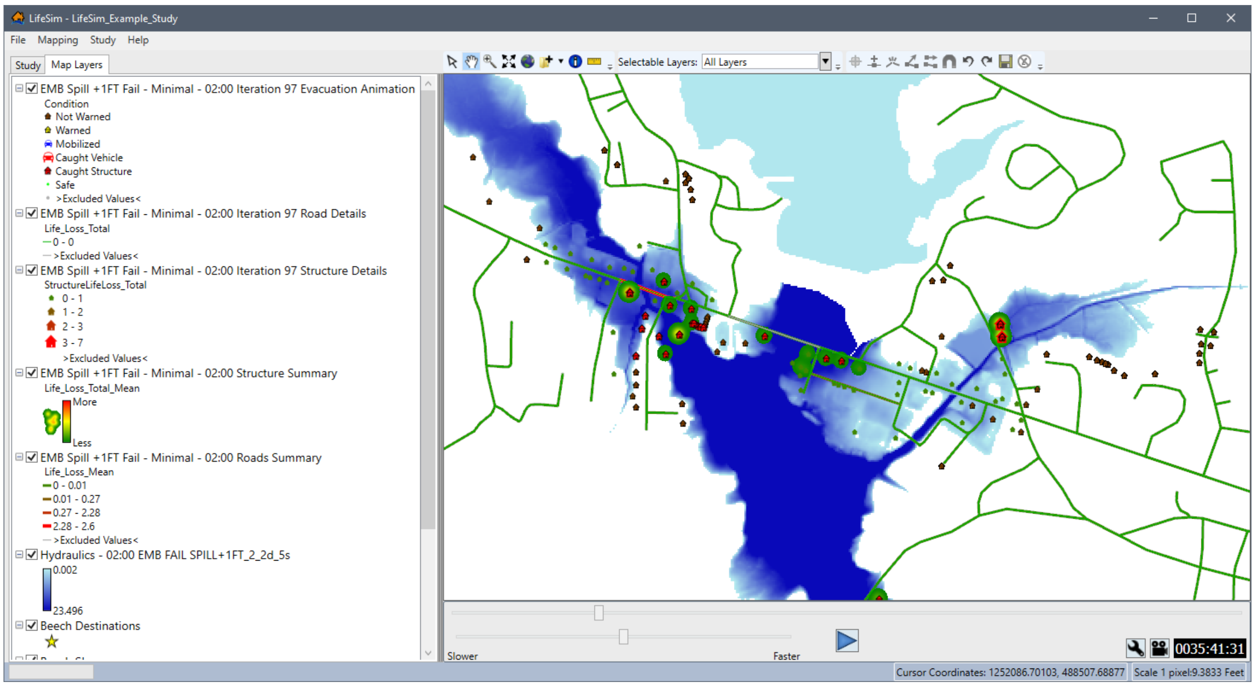


Figure 13-37. Example – Main Window – Map Layers Tab – Simulation Results Maps

Road and Structure Details

The road and structure summary or detailed output can be viewed for an iteration (e.g., ***Iteration 97 Road Details*** in Figure 13-37) or an alternative (e.g., ***EMB Spill + 1FT Fail – Minimal Structure Summary*** in Figure 13-37) from the map window. The road and structure information added to the map window for the detailed output displays life loss total (i.e., *Life_Loss_Total*) that occurred inside each structure or on each road segment and is color-coded and changes by size. For example, an increasing loss of life in a structure there is an increasing icon size and gray to red coloration. This output can be very useful for viewing the entire study area and highlight areas with the greatest risk for life loss in structures or on roads which require supplementary investigation.

Animate Alternative Results

Hydraulic results and detailed output results for iterations can be reviewed as an animation in the map window. The detailed output animation provides a dynamic view for the results of the iteration number selected (e.g., ***Iteration 97*** in Figure 13-37). The animation results can be used to review evacuation routes, warning issuance times, hydraulic data, and other simulation items. The animations can also be exported as video files to share. The map window displays an animation playback control in the animation toolbar (Figure 13-38), which functions similarly to video controls for viewing the animation results in the map window.

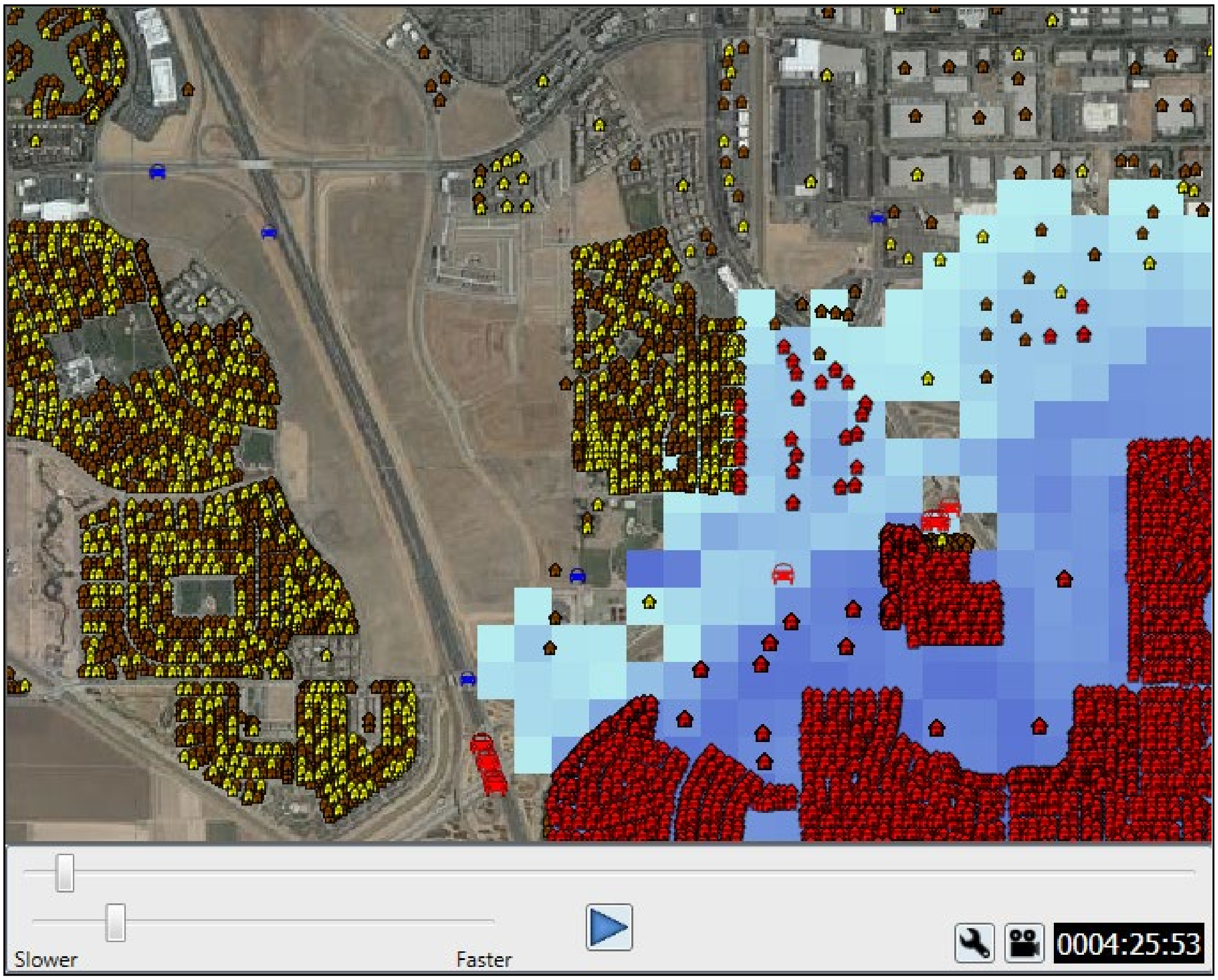


Figure 13-38. LifeSim Map Window – Animation Playback Toolbar – Evacuation and Hydraulic Animation – Example

The animation toolbar controls the evacuation animation and the hydraulic animation. To edit the animation speed, or animation window start and stop time, from the animation toolbar, click to open the **Animation Settings** dialog box (Figure 13-38). From the **Animation Settings** dialog box (Figure 13-38), set the animation window start time and end time and the slider settings. Click **OK** (Figure 13-38) to apply the animation settings to the animation playback and close the **Animation Settings** dialog box (Figure 13-38).

To export the animations as a video file to share, from the animation toolbar, from the animation toolbar, click to open the **Export Animation Wizard** dialog box (Figure 13-38). From the **Export Animation Wizard** dialog box (Figure 13-38), identify the **Output File** location and file name, the start and end animation window, the timer position and frame rate for the exported file. Click **OK** to create the video.

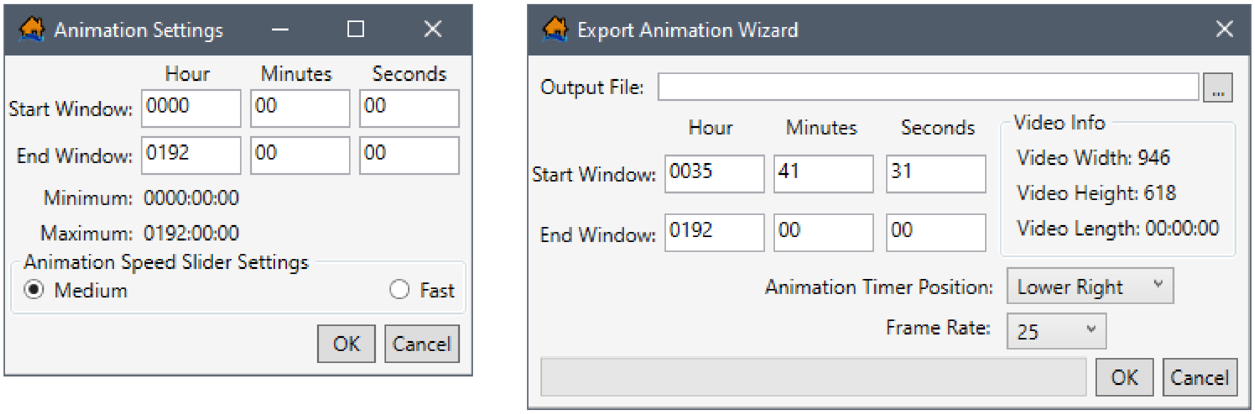


Figure 13-39. Animation Settings Dialog Box (*left*) – Export Animation Wizard Dialog Box (*right*)

Appendix A

Map Preferences

To help the user visualize the study and results, LifeSim has the capability to display map layers that represent the features of a study. The map layers displayed in the map window are also listed in the **Map Layers** tab (Figure A - 1). Map layers provide a geographical reference for the study area. The LifeSim GUI (Figure A - 1) describes the formats of the map layers, and provides commands that allow the user to configure, manage, and edit map layers.

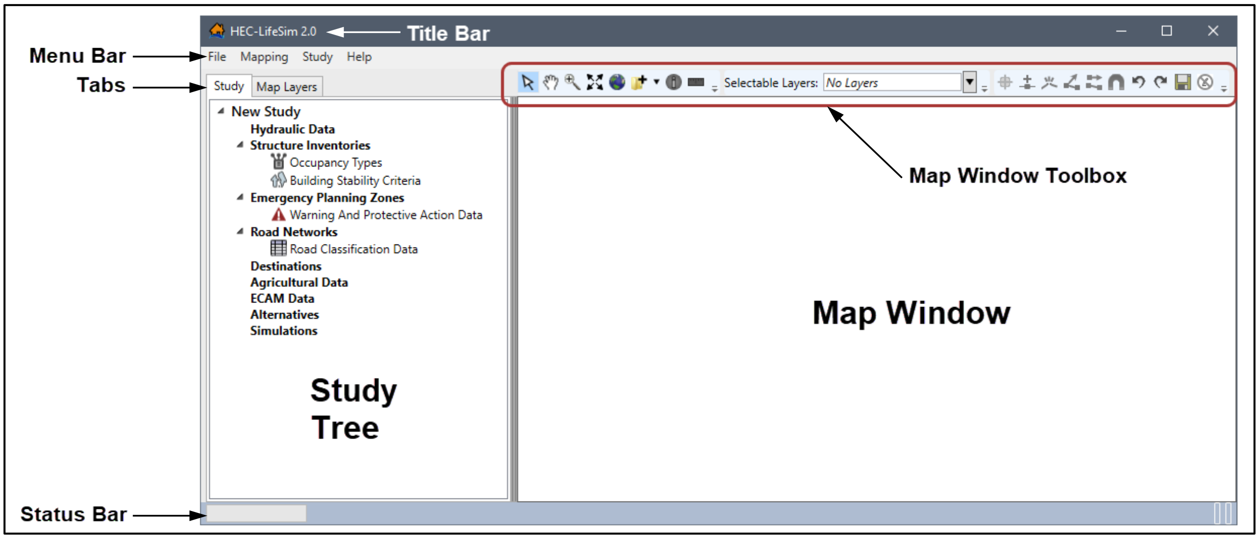


Figure A - 1. LifeSim Main Window

Map layers are optional and not necessary for computing and reviewing results (Chapter 13). However, to view the results of simulation animations (Chapter 13, Section 13.5.2) the map window is useful for helping to facilitate the review of simulation results and discussion with emergency managers. Furthermore, viewing map layers in the map window is a useful way to examine the validity of imported datasets. For example, a road network with missing connectivity or incorrect directions on one-way roads can lead to erroneous results. Therefore, editing the road network dataset can provide a reliable foundation for traffic simulation (Appendix B provides an example of editing an imported road network dataset in the map window).

A.1 Map Window

The map window (Figure A - 1) provides a way to graphically display LifeSim components. The rendering engine takes advantage of the computers graphics processing using Open Graphics Library - OpenGL (https://www.opengl.org/) to draw the map layers. All geographic information system (GIS) data is re-projected on the fly using GDAL (Geospatial Data Abstraction Library) (http://www.gdal.org/).

A.1.1 Map Window Toolbars

There are three toolbars located in the LifeSim main window **Map Window Toolbox** (Figure A - 1). The first is the **General Toolbar**, second is the **Selectable Layers Toolbar**, and third is the **Editor Toolbar**. Details about each toolbar are detailed in the following sections.

General Toolbar

The **General Toolbar** includes the **Select Tool (B)**, **Pan Tool (P)**, **Zoom In Tool (Z)**, **Zoom to full extents**, **Add Data**, **Query Features Tool (Q)**, and **Measure Distance Tool (M)**. These tools change the appearance of the cursor, as well as the functionality of the mouse cursor in the map window.

Select Tool (B)

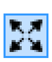
The **Select** tool allows the user to select elements displayed in the map window that are currently selected in the **Selectable Layers**. More details on **Selectable Layers** can be found in the next section. To select features, click on a feature or click, hold, and drag a rectangle around features to select them in the map window. Additionally, the **Select Tool** allows users to access shortcut menus which can be used to customize features when in editing mode.

Pan Tool (P)

The **Pan** tool allows the user to pan the map window; click the **Pan** tool and click, hold and move the mouse around the map window. Also, if the mouse has a wheel, then the user can pan the map window by using the wheel (click and hold the wheel to pan).

Zoom In Tool (Z)

The **Zoom In** tool allows the user to zoom in to areas of the map window. To zoom in, hold the mouse button down and outline the area of interest to enlarge. Also, if the mouse has a wheel then the user can zoom in or out using the wheel (wheel up zooms in and wheel down zooms out).



Fixed Zoom Out Tool (O)

The **Fixed Zoom Out** tool zooms out a fixed distance. Click the **Fixed Zoom Out** tool once and the LifeSim map window view will zoom out a fixed distance. Double-click on the **Fixed Zoom Out** tool and the map window view will zoom out twice the fixed distance.

Additionally, the user can utilize the shortcut menu option, right-click anywhere in the map window, to zoom in or out a fixed distance.

Zoom to full extents Tool

The **Zoom to full extents** tool is a quick way to re-center the map by zooming out to the full geo-graphical extent of the map layers in the map window. Web layers do not affect the map extents.

Add Data Tool

Click the **Add Data** tool**,** the **Browse for files** browser will open. This browser allows users to navigate to layers (including **.shp*, **.flt*, **.tif*, and **.vrt*) of interest and add the layer(s) to the map window.

If the user clicks the down arrow next to the **Add Data** tool a shortcut menu will appear which allows the user to add data, add web imagery, or create new data that is added to the map window.

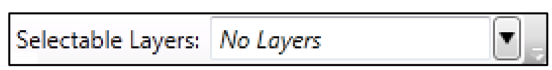
Query Features Tool (Q)

The **Query Features** tool identifies the geographic feature or place currently selected in the map window. Alternatively, multiple features can be identified by dragging a box over the desired area. Once a selection is made the **Feature Query** dialog box will open with the information presented in table format separated by **Feature** type. Only visible layers in the map window will be queried and the **Selectable Layers** option does not modify the **Query Features** tool.

Measure Distance Tool (M)

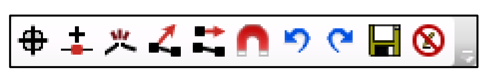
The **Measure Distance** tool allows the user to measure the distance along a specified line. With the tool selected, click to initiate the measure line. Each subsequent click will add a new point to the measure line. When finished, double-click on the location of the final point. The **Measure Results** dialog box will open and display the distance in the units the user has chosen (feet, miles, meters, or kilometers).

Selectable Layers Toolbar



Selectable Layers Toolbar allows the user to turn on and off map layers of a study using the **Select** tool from the **Selectable Layers Toolbar**. The list of options depends on the maps layers located in the **Map Layers** tab, except Internet (web) layers. For further explanation on the use of **Selectable Layers**, refer to Section A.13.1.

Map Window Editor Toolbar



The set of tools located in the **Editor Toolbar** become active after users initiate an editing session for a map layer. When a user right-clicks on a map layer in the **Map Layers** tab, from the shortcut menu (Figure A - 2), click **Edit**, and the tools in the **Editor Toolbar** become active. Only one map layer can be edited at one time. For lines and polygons, there are two editing modes: feature edit, and vertex edit. Clicking on a selected feature puts the user into the feature edit mode (move, delete, reverse), double-clicking on a selected feature puts the user in the vertex edit mode (add, delete, edit vertex points).

There are some tools available during an edit session which are not available from the shortcut menu (Figure A - 2) or the **Editor Toolbar**. Clicking **Delete** will delete all currently selected features or currently selected vertices depending on which mode the user is in. Clicking the **R** key will reverse the direction of a currently selected line feature. Also, right-clicking during an edit session allows the user to do multiple tasks depending on what is clicked.

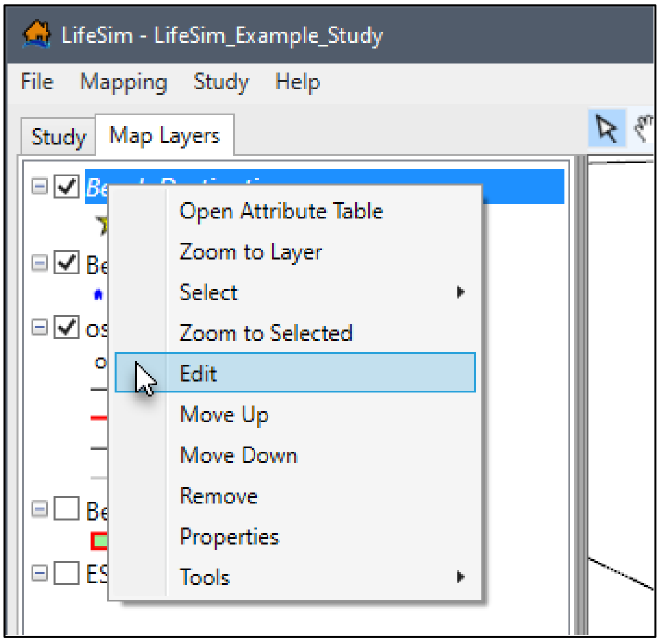
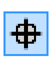
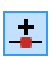


Figure A - 2. Individual Map Layer Shortcut Menu



Add new features Tool (A)

The **Add new features** tool allows the user to add a new feature to the editable map layer by left-clicking to add feature data. Selected features can be deleted by either clicking the **Delete** key, or right-click on the selected feature(s) and from the shortcut menu click **Delete Selected Features**.



Insert vertex Tool (I)

The **Insert vertex** tool allows the user to add a vertex to an existing line or polygon map layer. The **Insert vertex Tool** becomes available when the selected feature is in the vertex edit mode. As the cursor gets close to the line or polygon edge the map window will display what the updated feature will look like if a vertex is added at the cursor. Left-click to insert the new vertex.

Break line feature into two line features at a defined location Tool (K)

The **Break line feature** tool will split a line feature into two features at the point the user clicked on the line. The user needs to be in the feature edit mode for this tool to be enabled.

Continue from end (lines only) Tool

The **Continue from end (lines only)** tool continues a line feature from the end point of the line feature to a second point clicked in the map window and each new click adds a vertex point to the line feature. Double-click to finish the line or right-click and choose **Finish Editing Vertices**. The user needs to be in the vertex edit mode to enable this tool.

Add new line or polygon to existing feature Tool

The **Add new line or polygon to existing feature** tool will allow the user to add a polygon or line to an existing selected polygon or line feature. For example, to add a hole to a polygon, click to add points defining the hole and double-click to finish. The user needs to be in the vertex edit mode for this tool to be enabled.

Feature Snapping Settings Tool

The **Feature Snapping Settings** tool allows the user to define snapping rules when editing. Snapping will automatically move the desired action to the nearest snap point when the cursor is close. This can be very useful when trying to line up endpoints of line features or place points directly on top of other features. Enable snapping by clicking **Enable Snapping**. From the **Snap to Feature** list, select the desired feature to snap to, then select how snapping will be applied (all points, end points, or edges).

Undo Tool (Ctrl + Z)

The **Undo** tool allows the user to undo changes made during an edit session. Only enabled if there are edits that can be undone.

Redo Tool (Ctrl + Y)

The **Redo** tool allows the user to redo changes made. Only enabled if there are edits to redo.

Save Edits Tool (Ctrl + S)

The **Save Edits** tool will save all edits made during the editing session. Only enabled if edits have been made.

Stop Editing Tool

The **Stop Editing** tool will stop the edit session. At this point a new map layer can be selected for editing. If edits have been made, then the user will be prompted to save them before stopping the edit session.

Hot Keys

If a tool has a parentheses after the tooltip the letter in the parentheses represents a hot key (Table A - 1) to call on the tool without clicking on it. For example, for the **Query** tool if the user clicks **Shift-Q** over the map window the **Query** tool will be active. Note, for the hot keys to work for tools located in the **Editor Toolbar** users must first initiate an editing session for a map layer.

Table A - 1. Map Window Toolbars – Hot Keys

Icon	Tool Name	Hot Key(s)
	Select Tool	(Shift + B)
	Pan Tool	(Shift + P)
	Zoom In Tool	(Shift + Z)
	Fixed Zoom Out Tool	(Shift + O)
	Query Features Tool	(Shift + Q)
	Measure Distance Tool	(Shift + M)
	Add new features	(Shift + A)
	Insert vertex	(Shift + I)
	Break Line Feature	(Shift + K)
	Undo	(Ctrl + Z)
	Redo	(Ctrl + Y)
	Save Edits	(Ctrl + S)

A.1.2 Map Window Properties

To set the projection (coordinate system) associated with the LifeSim study, from the LifeSim main window (Figure A - 1), from the **Mapping** menu (Figure A - 3), click **Map Properties**. The **Map Properties** dialog box will open (Figure A - 4), from this dialog box the user can setup the projection for the LifeSim study.

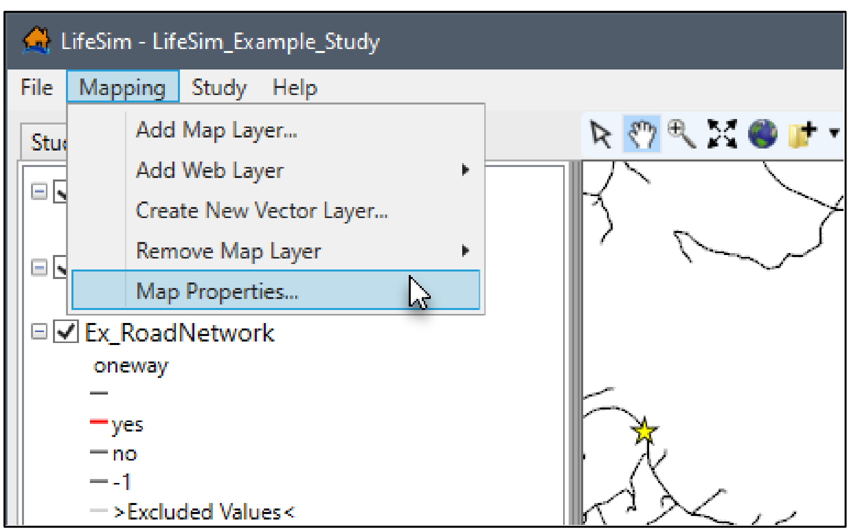


Figure A - 3. Mapping Menu

1. From the **Map Properties** dialog box (Figure A - 4), click **Import New Projection**, an **Open** browser will open (Figure A - 5).

2. From the **Open** browser (Figure A - 5), navigate to the projection file of interest (usually any shapefile associated with the study area) and select a projection File (**.prj*) and click **Open**.

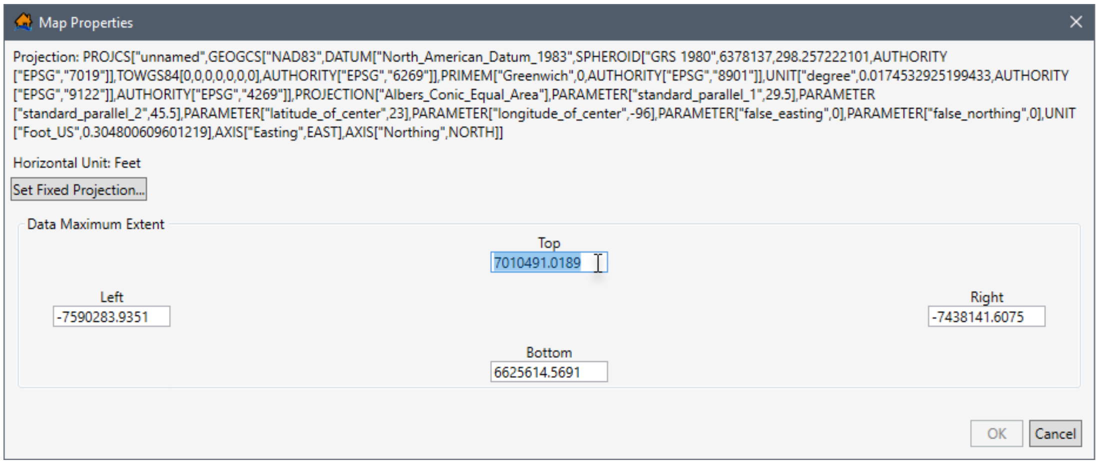


Figure A - 4. Map Properties Dialog Box

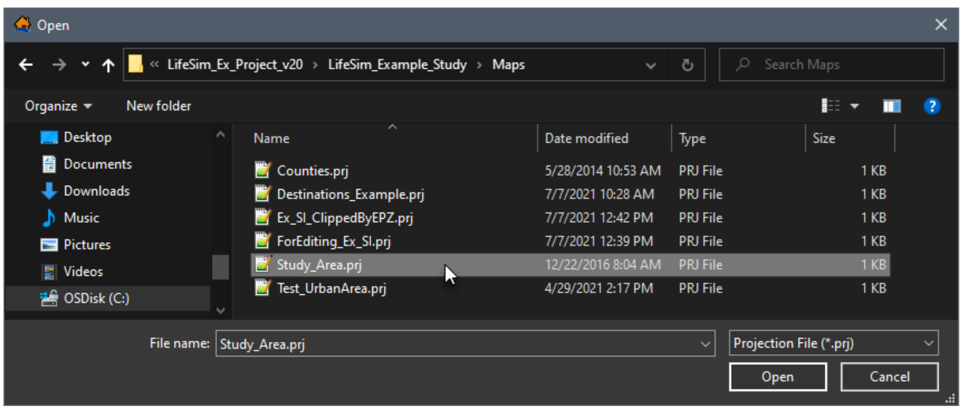


Figure A - 5. Open Browser

3. The **Open** browser will close (Figure A - 5) and the projection for the study will now be displayed in the **Map Properties** dialog box (Figure A - 6). Click **OK**, and the **Map Properties** dialog box will close (Figure A - 4).

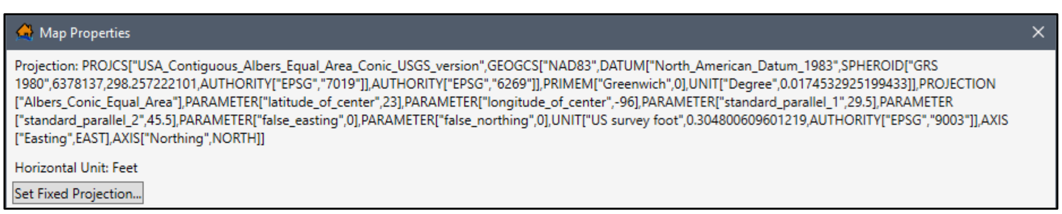


Figure A - 6. Map Properties Dialog Box - Projection

A.2 Map Layers Tab

The **Map Layers** menu (Figure A - 7) provides a tree view for the map layers that have been added to an LifeSim study. However, only selected () layers in the **Map Layers** tab will display in the map window (Figure A - 1). The user can turn map layers on or off, adjust the properties of the layers, and, modify the order of the layers for viewing in the map window.

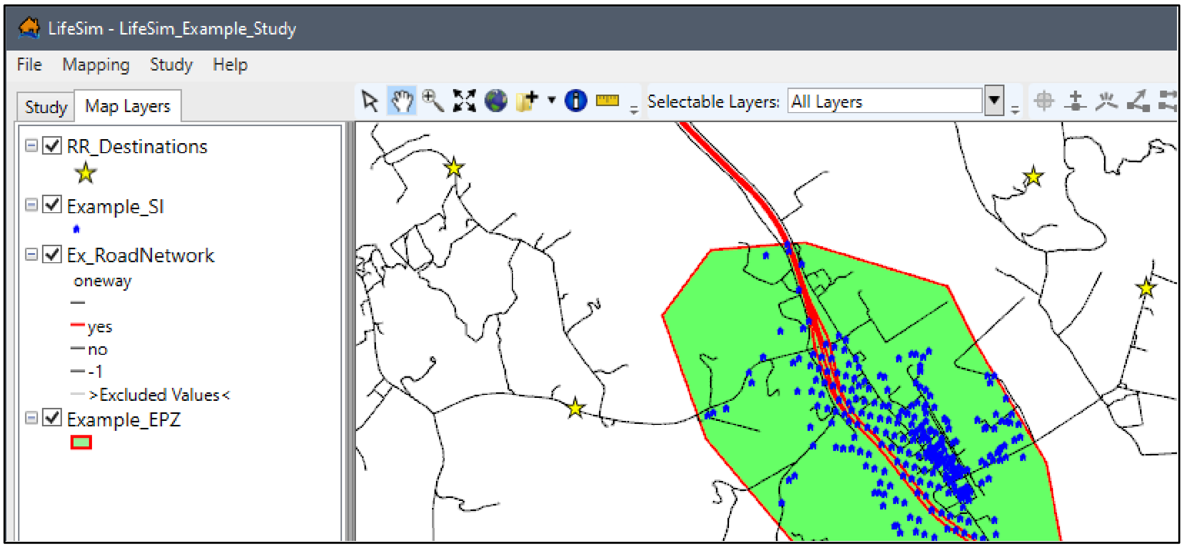


Figure A - 7. Map Layers Tab – Map Layers

A.3 Map Layers

Geographic Information System (GIS) files are the map layer formats supported by LifeSim and include: **.shp*, **.flt*, **.tif*, and **.vrt*. The user can add map layers, add background map layers, remove map layers, adjust a map layer's position in the view, edit a map layer's properties, zoom into a layer, and many other functions. The ability to manipulate the different map layers is specific for that map layer format.

A.3.1 Adding Map Layers

To add a map layer:

1. From the LifeSim main window (Figure A - 1), from the **Mapping** menu (Figure A - 3), click **Add Map Layer**. Also, from the **General Toolbar** (Section A.1.1), click . Either way, the **Browse for files** browser (Figure A - 8) will open.

2. From the **Browse for files** browser (Figure A - 8), navigate to the GIS file of interest that will be added as a map layer. Select the appropriate GIS file (**.shp*, **.flt*, **.tif*, or **.vrt*) and the **File name** box (Figure A - 8) will now display the selected filename.

3. Click **Open**, the **Browse for files** browser (Figure A - 8) will close, the selected dataset will now be displayed in the map window, and selected map layer name will appear in the **Map Layer Tree**.

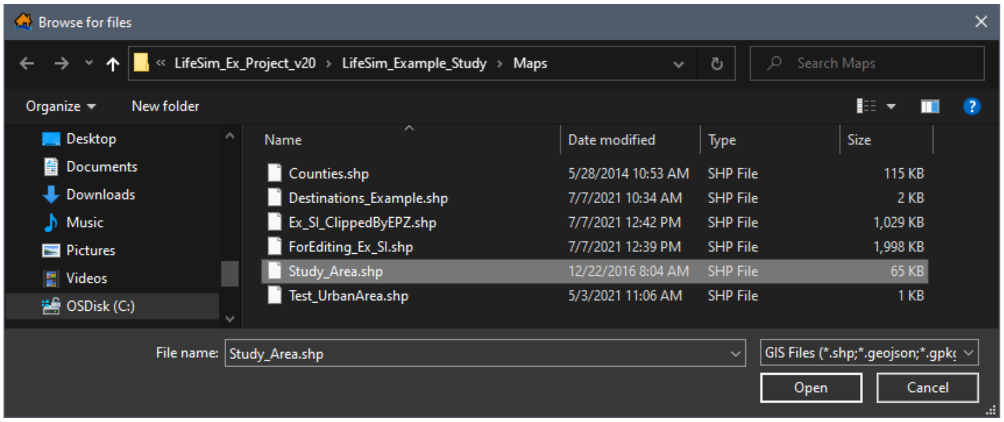


Figure A - 8. Browse for files Browser

A.3.2 Adding Background Internet Maps

Knowing where the study is located in reference to known geographic positions can be quite helpful with communication. There are two types of internet background maps, **OpenStreetMap** and **ESRI World Imagery** map and there are two ways to add these background internet maps.

To add a background internet map to the LifeSim study, from the **Mapping** menu (Figure A - 9) on the LifeSim main window (Figure A - 3), point to **Add Web Layer**, and select the internet map from the available list which would be a good background map for the study area. Also, from the **General Toolbar** (Section A.1.1), click the **Add Data** dropdown menu (Figure A - 9), point to **Add Web Imagery** (Figure A - 9), and select the internet map from the available list which would be a good background map for the study area. Either way, the selected background map displays in the map window (Figure A - 10).

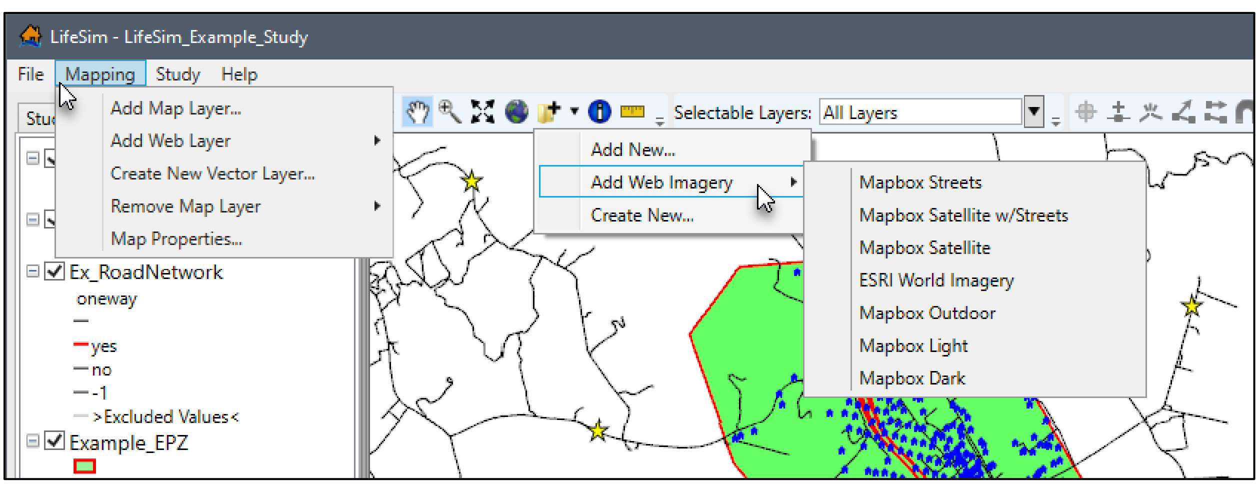


Figure A - 9. Mapping Menu – General Toolbar – Add Web Imagery Options

A.3.3 Map Layer Position in the Map Layer Tree

Once a map layer is added to an LifeSim study, the user can adjust the position of a map layer. Within the LifeSim framework, some map layers represent a particular LifeSim dataset (i.e., hydraulic data, structure inventories, emergency planning zones), while other map layers provide information.

The **Map Layer Tree** (Figure A - 10) displays the map layers from top to bottom, where the top-most layer overlays the next one down in the list and continues to the bottom layer. In other words, if a background internet map is added, from the **Add Data** dropdown menu (Figure A - 9), that map layer will automatically be added to the bottom of the **Map Layers Tree** (Figure A - 10), and will be underneath all layers listed above the internet map. At times it may be desirable to have various map layers drawn in a different order in the map window. To move map layers:

1. From the **Map Layers Tree** (Figure A - 7), right-click on the map layer of interest, from the shortcut menu (Figure A - 2), either select **Move Up** or **Move Down**.

2. To move the map layer of interest below other map layers listed in the **Map Layers Tree**, click **Move Down**. The selected map layer will move one step below its current position. Repeat until the map layer of interest is at the desired location within the **Map Layers Tree**.

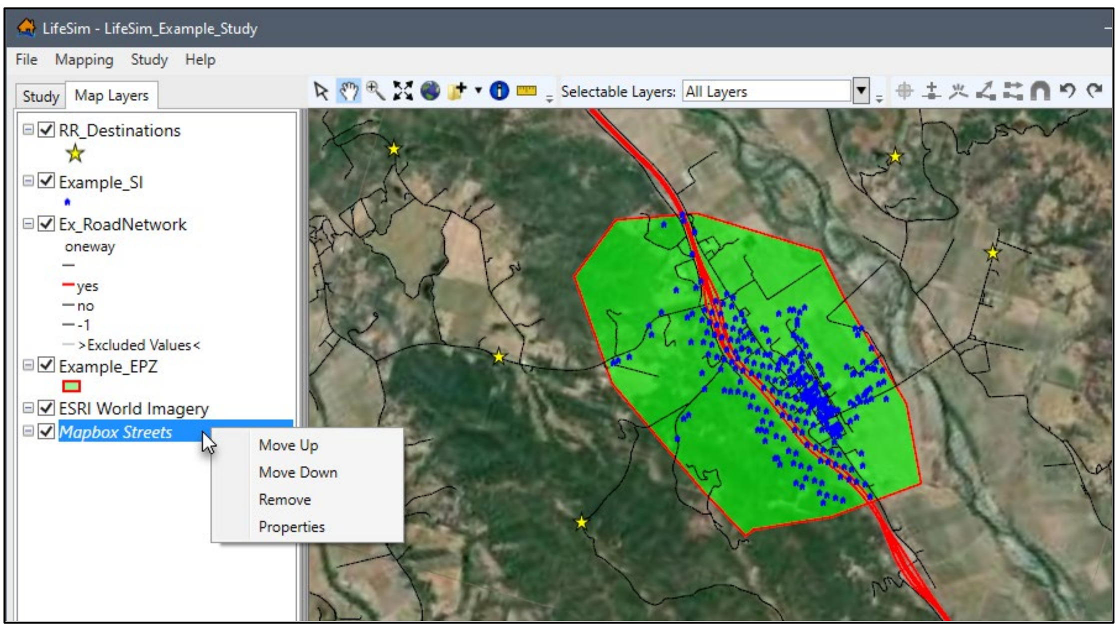


Figure A - 10. LifeSim Main Window – Map Layers Tree – Internet Map Shortcut Menu

3. To move the map layer of interest above other map layers listed in the **Map Layers Tree**, click **Move Up**. The selected map layer will move one step above its current position. Repeat until the map layer of interest is at the desired location within the **Map Layers Tree**.

Another way to move map layers in the **Map Layers Tree** is to drag and drop a map layer to the desired location in the **Map Layers Tree**. From the **Map Layers Tree**, click and hold the map layer of interest, move the cursor up or down the **Map Layer Tree** (the selected map layer will move in conjunction with the mouse). A bar will show the new location where the selected map layer will be moved (Figure A - 11), release the mouse, and the selected map layer will be in a new position on the **Map Layer Tree**.

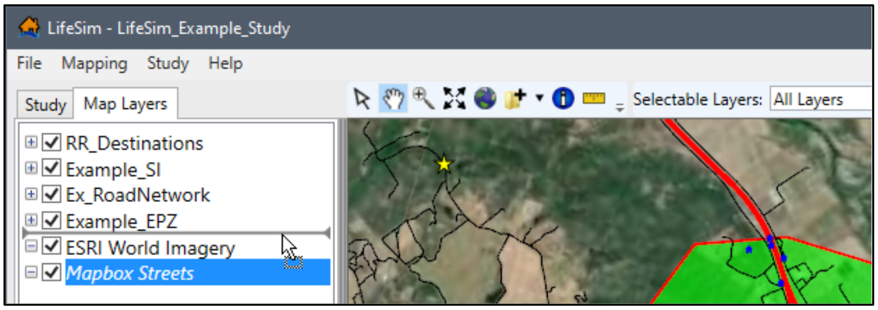


Figure A - 11. Moving a Selected Map Layer in the Map Layers Tree

A.3.4 Zoom to Layer

This option, will set the zoom of the map window (Figure A - 10) based on the extents of the selected map layer. From the **Map Layers Tree** (Figure A - 11), right-click on the map layer of interest, from the shortcut menu (Figure A - 2), click **Zoom to Layer**, the map window view will change to include the entire extent for the selected map layer.

A.3.5 Map Layer Select Options

There are three options (**Select All**, **Select by Polygon**, or **Reverse Selection**) for selecting map layer features from the individual map layer shortcut menu (Figure A - 12).

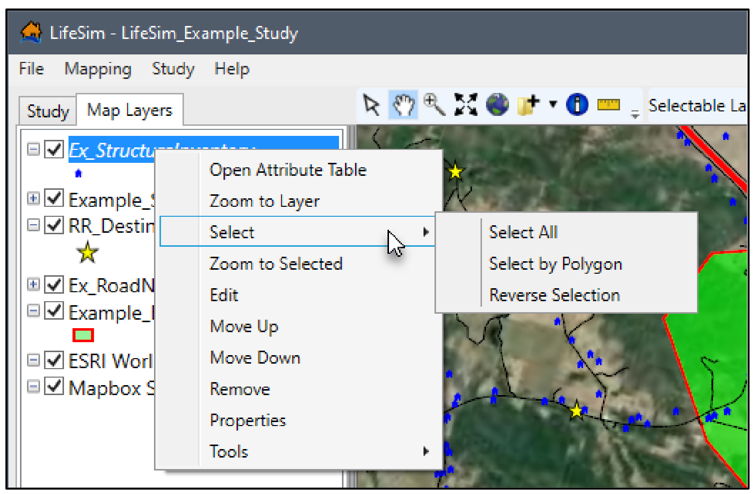


Figure A - 12. Individual Map Layer Shortcut Menu – Select

Note: If the map window tool, **Selectable Layers** (Section A.1.1), has been modified from **All Layers**, the **Select** commands might not be accessible. If the **Select** shortcut commands map layer, make certain the map layer is selected (☑) in the **Selectable Layers** dropdown menu and try the **Select** shortcut commands again. The following describes the **Select** sub-menu commands:

Select All	The **Select All** command will select all of the features within the map layer.
Select by Polygon	The **Select by Polygon** command will open the **Select by Polygon** dialog box (Figure A - 13).
Reverse Selection	The **Reverse Selection** command will select the opposite. For example, if all of the features are selected in the map layer then choosing **Reverse Selection** will de-select all of the features.

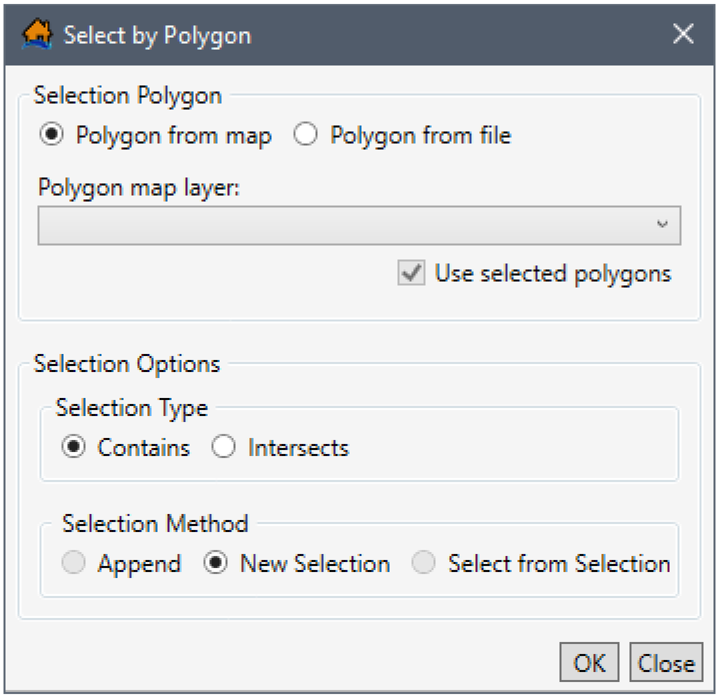


Figure A - 13. Select by Polygon Dialog Box

Select By Polygon

1. From the **Map Layers** tab, right-click on a map layer of interest (Figure A - 12), from the shortcut menu point to **Select**, click **Select by Polygon**. The **Select by Polygon** dialog box will open (Figure A - 13).

2. From the **Selection Polygon** box (Figure A - 13), the user must select either **Polygon from map** or **Polygon from file**.

   - **Polygon from map** (default): The user will need to select a polygon map layer that is part of the LifeSim study from the **Polygon map layer** list (Figure A - 13), which contains a list of the polygon map layers (e.g., the *Emergency Planning Zone(s)* map layer) currently in the **Map Layers** tab. Notice, there is an option in the **Polygon from map** section to **Use selected polygons** (Figure A - 13). The **Use selected polygons** option will only be available when the identified **Polygon map layer** has features selected or if a subset of a layer has been selected (using the **Select By Attribute** option is explained in Section A.4; or the map layer **Select** options are explained in Section A.3.5).

   - **Polygon from file**: This allows the user to select a polygon shapefile. When selected (Figure A - 14), from the **Polygon Shapefile** box (Figure A - 14), click , the **Select a Polygon Shapefile** browser will open. The user will need to navigate to the file of choice and select the desired polygon shapefile (**.shp*). Click, **Open**, the **Select a Polygon Shapefile** browser will close, and the name of the selected polygon shapefile will display in the **Polygon Shapefile** box (Figure A - 14).

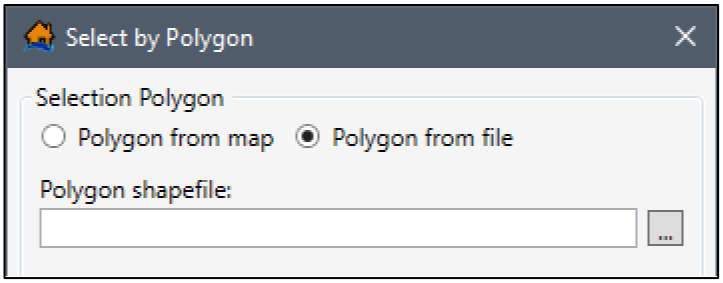


Figure A - 14. Select by Polygon Dialog Box – Polygon from file option

3. Next, from the **Selection Options** box (Figure A - 13), the user needs to set the selection type. The choices are **Contains** and **Intersects**:

    - **Contains** (default) – this option (Figure A - 13) will select features that are completely within a polygon that has been selected. In other words, this option will not select any features which intersected with the border of the selecting polygon.

    - **Intersects** – this option (Figure A - 13) will only select features which intersects the polygon that has been selected. This includes features that are completely contained in the selection polygon in addition to intersecting with the border of the selecting polygon. Note, this option will is not available for point shapefiles.

4. The last step, is the selections method, available options are **Append**, **New Selection**, and **Select from Selection**:

    - **Append** – this option (Figure A - 13) adds the selection to the already selected features. This option will only be available if there are selected features within the input map layer.

    - **New Selection** (default) – this option (Figure A - 13) will clear any previous selection made to the input map layer.

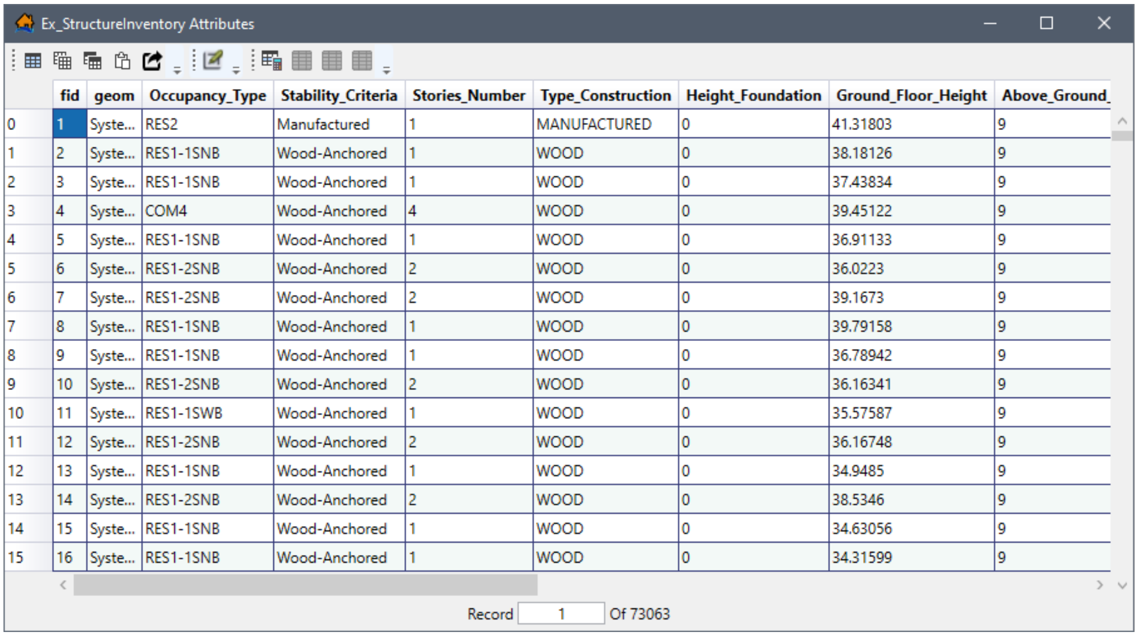
    - **Select from Selection** – this option (Figure A - 13) will only select features that are currently selected in the input map layer.

5. Click, **OK**, the **Select by Polygon** dialog box will close (Figure A - 13). Now, the selected map layer will have features selected in the map window corresponding to the preferences set in the **Select by Polygon** dialog box (Figure A - 13).

A.4 Map Layer Attributes Dialog Box

Map layer attributes can be accessed for each map layer and provides a table for the dataset referenced by the map layer. To access a specific map layer **Attribute** table:

- From the **Map Layers** tab, right-click on a map layer of interest, from the shortcut menu (Figure A - 2), click **Open Attribute Table**. The *Map Layer* **Attributes** dialog box will open (Figure A - 15). Editing of the map layer attributes is detailed in Section A.10 of this appendix.


Ex_StructureInventory Attributes

	fid	geom	Occupancy_Type	Stability_Criteria	Stories_Number	Type_Construction	Height_Foundation	Ground_Floor_Height	Above_Ground_
0	1	Syste...	RES2	Manufactured	1	MANUFACTURED	0	41.31803	9
1	2	Syste...	RES1-1SNB	Wood-Anchored	1	WOOD	0	38.18126	9
2	3	Syste...	RES1-1SNB	Wood-Anchored	1	WOOD	0	37.43834	9
3	4	Syste...	COM4	Wood-Anchored	4	WOOD	0	39.45122	9
4	5	Syste...	RES1-1SNB	Wood-Anchored	1	WOOD	0	36.91133	9
5	6	Syste...	RES1-2SNB	Wood-Anchored	2	WOOD	0	36.0223	9
6	7	Syste...	RES1-2SNB	Wood-Anchored	2	WOOD	0	39.1673	9
7	8	Syste...	RES1-1SNB	Wood-Anchored	1	WOOD	0	39.79158	9
8	9	Syste...	RES1-1SNB	Wood-Anchored	1	WOOD	0	36.78942	9
9	10	Syste...	RES1-2SNB	Wood-Anchored	2	WOOD	0	36.16341	9
10	11	Syste...	RES1-1SWB	Wood-Anchored	1	WOOD	0	35.57587	9
11	12	Syste...	RES1-2SNB	Wood-Anchored	2	WOOD	0	36.16748	9
12	13	Syste...	RES1-1SNB	Wood-Anchored	1	WOOD	0	34.9485	9
13	14	Syste...	RES1-2SNB	Wood-Anchored	2	WOOD	0	38.5346	9
14	15	Syste...	RES1-1SNB	Wood-Anchored	1	WOOD	0	34.63056	9
15	16	Syste...	RES1-1SNB	Wood-Anchored	1	WOOD	0	34.31599	9

Record 1 Of 73063

Figure A - 15. *Map Layer* Attributes Dialog Box

- **Note**: From the table, when a single cell is selected (e.g., row 1, column 1) the keyboard arrow keys (and the tab and enter keys) can be utilized to step through the table one cell at a time.

A.4.1 Attribute Export and Copy Tools

Select all cells Tool
This tool selects all cells in the table.

Copy Selected Cells Tool
This tool copies cells selected in the table. The information is copied to the clipboard and needs to be pasted to another application (e.g., Microsoft Excel®) for saving.

Copy Selected Cells with Table Headers Tool
This tool copies cells selected in the table and the table headers. The information is copied to the clipboard and needs to be pasted to another application (e.g., Microsoft Excel®) for saving.

Paste from Clipboard into Table
This tool pastes data copied from another table into the current table.

Export Table

The **Export Table** tool opens a **Save As** browser window (Figure A - 16), which allows the user to export the data in the table to a Microsoft Excel® (*.*xlsx*) file.

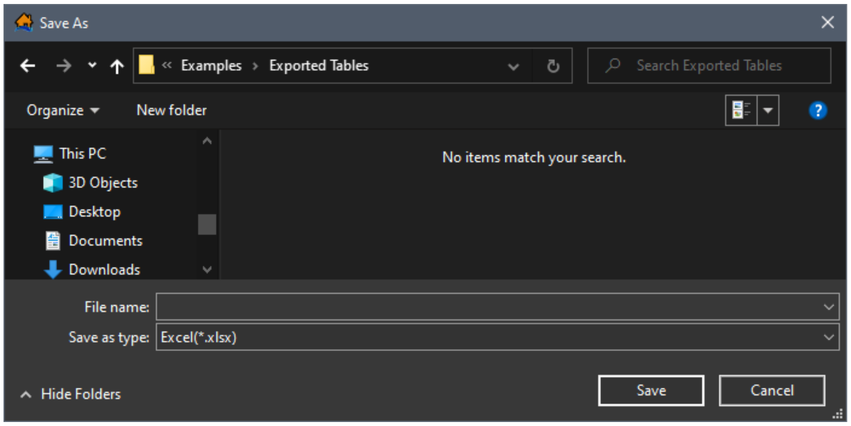


Figure A - 16. Example – Export Table Tool – Save As Browser

A.4.2 Attribute Edit Tool

Start Feature Edit Session

This tool enables editing for the map layer open in the attributes table. More information regarding editing map layers and map layer attributes, review Section A.10 and A.10.1 respectfully.

A.4.3 Attribute Selection Tools

Select Rows By Attribute Tool

Opens the **Select By Attribute** dialog box (Figure A - 17), which allows the user to perform queries, make selections, and locate features in the map window. When a feature is selected in the attribute table, that same feature will also be selected in the map window (and vice versa).

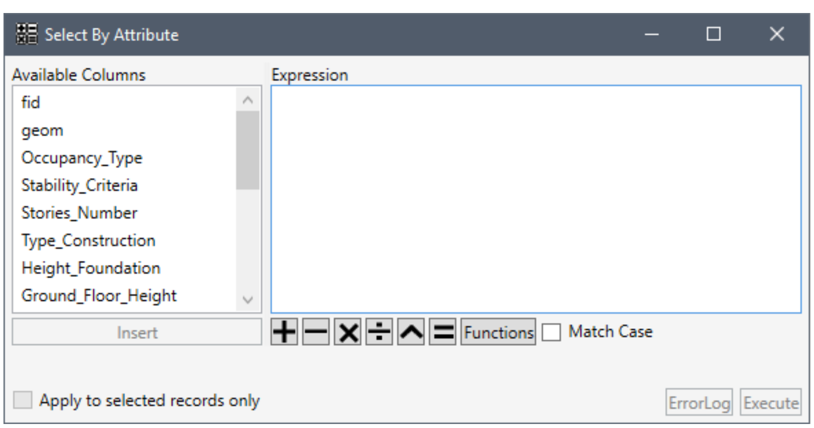


Figure A - 17. Select By Attribute Dialog Box

Show All Row Tool
Allows the user to show all rows (selected or unselected) in the attribute table for the map layer of interest.

Show Selected Rows Only Tool
Allows the user to only show features in the attribute table which were selected in the **Select By Attribute** dialog box (Figure A - 17).

Deselect All Rows Tool
Unselects all rows which were selected in the **Select By Attribute** dialog box (Figure A - 17). When the tool is selected all rows will be viewable and none will be highlighted in the attribute table (and in the map window).

A.4.4 Selecting Features/Rows by Attribute

To access and select specific attributes for a map layer:

1. From the **Map Layers** tab of the LifeSim main window (Figure A - 1), right-click on the map layer of interest. From the shortcut menu (Figure A - 2), click **Open Attribute Table**, the *Map Layer* **Attributes** dialog box opens (Figure A - 15).

2. Click , the **Select By Attribute** dialog box opens (Figure A - 17). The user by selecting certain attributes and using expressions can manipulate the selected map layer. The **Available Column** box (Figure A - 17) provides a list of all of the attributes fields of the selected map layer. Double-click on an attribute in the **Available Columns** box, this will add the attribute to the **Expression** box (Figure A - 19).

3. The **Functions** button (Figure A - 17) opens a new window providing a selection of all available functions in LifeSim. Select a function from the **Functions** list (e.g., ***IF*** in Figure A - 18), to view tips for using the function and click **Insert** to add the selected function to the **Expression** box (Figure A - 17).

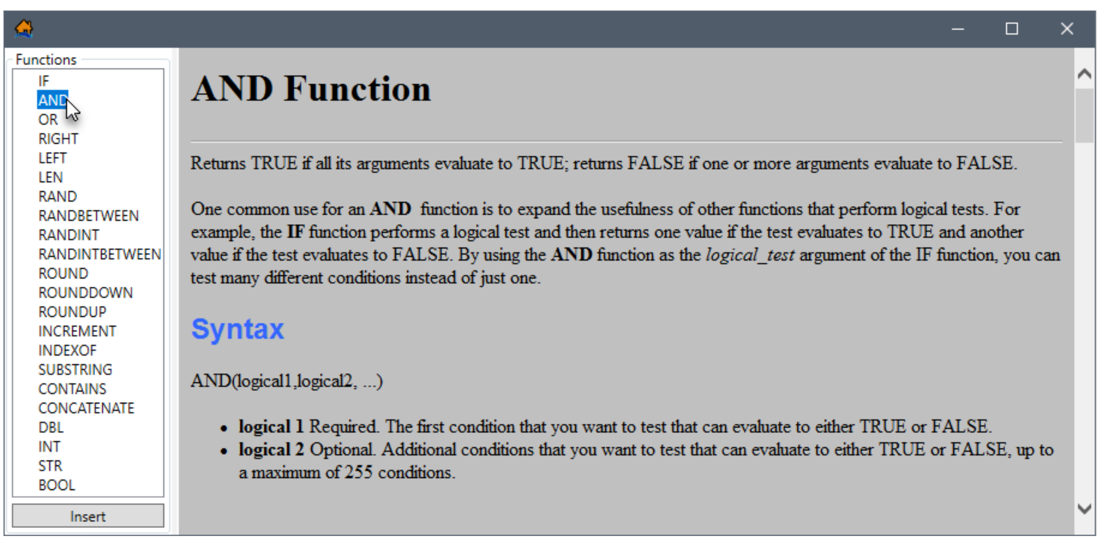


Figure A - 18. Functions

4. Figure A - 19 provides an example expression for selecting attributes in the table that satisfy the entered expression. The expression was entered by selecting ***AND***, from the **Functions** list (Figure A - 18); *Stories_Number* and *DayU65Population*, from the **Available Columns** (Figure A - 17) list; and has set *Stories_Number* equal to *1*, and *DayU65Population* greater than *100*.

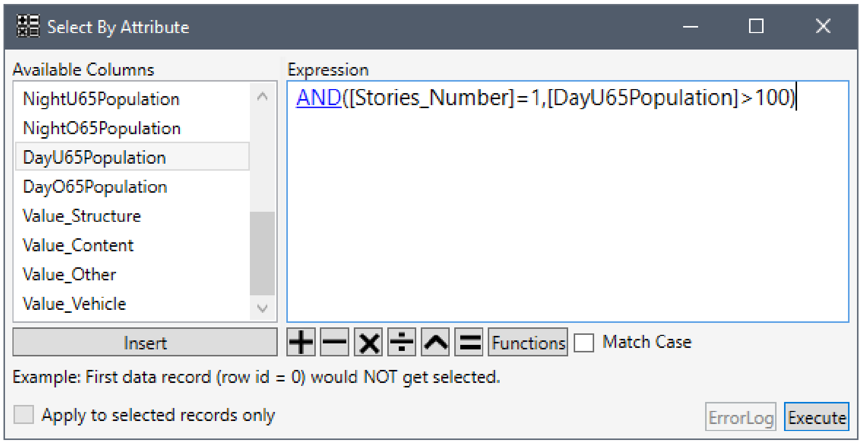


Figure A - 19. Select By Attribute Dialog Box – Example Expression

A message appears below the **Expression** box (Figure A - 19) *Example: First data record (row id = 0) would NOT get selected*. In other words, the first row does not satisfy the expression; therefore, the first row in the attributes table will not become selected when the **Execute** (Figure A - 19) button is clicked. On the other hand, if the first row did satisfy the expression then the message would say, *Example: First data record (row id = 0) WOULD get selected*, and the first row in the attribute table would become selected when the **Execute** (Figure A - 19) button is clicked.

5. If the entered expression contains errors the message section below the **Select from** *Map Layer Name* **where** box (Figure A - 19) will display the issues with the expression. The message will say **Tree contains errors** until the expression is completed correctly.

6. Another resource providing help to complete an expression is the **ErrorLog** which is available when an expression is incorrect. Click **ErrorLog**, the **Errors Encountered in Expression** dialog box will open (Figure A - 20). Information is provided on what the possible issues are with the defined expression.

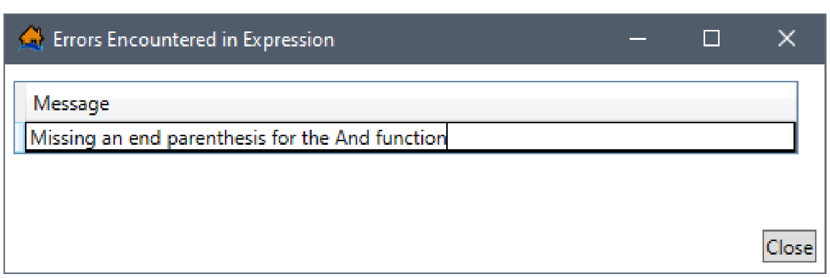
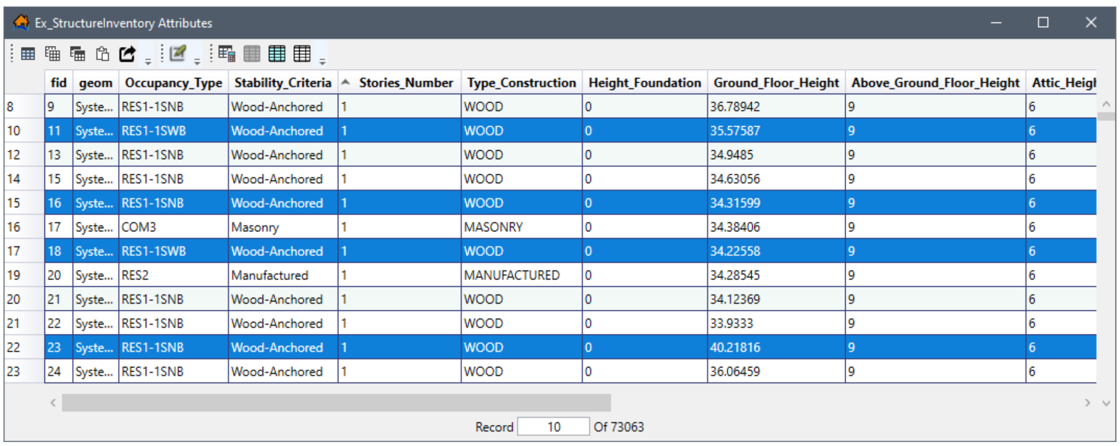


Figure A - 20. Errors Encountered in Expression Dialog Box

7. Once the expression is correct, click **Execute** to run the expression (Figure A - 19). The **Select By Attribute** dialog box will close (Figure A - 19) and the rows in the table on the *Map Layer* **Attributes** dialog box (Figure A - 21) where the expression is *True* will be highlighted in dark teal. Also, the **Show Selected Rows Only** () and **Deselect All Rows** () icons are now active.

8. After a selection is made in the table (Figure A - 21), the feature is not only highlighted in the table, but is also highlighted in the map window (Figure A - 1) for the selected map layer. For example, in Figure A - 22, features are highlighted in the map window following the successful execution of the defined expression. The highlighted features will be differentiated in bright teal (as opposed to the original blue color). Double-clicking in the map window will deselect the features in the map window and in the *Map Layer* **Attributes** dialog box (Figure A - 21).



	fid	geom	Occupancy_Type	Stability_Criteria	Stories_Number	Type_Construction	Height_Foundation	Ground_Floor_Height	Above_Ground_Floor_Height	Attic_Heigh
8	9	Syste...	RES1-1SNB	Wood-Anchored	1	WOOD	0	36.78942	9	6
10	11	Syste...	RES1-1SWB	Wood-Anchored	1	WOOD	0	35.57587	9	6
12	13	Syste...	RES1-1SNB	Wood-Anchored	1	WOOD	0	34.9485	9	6
14	15	Syste...	RES1-1SNB	Wood-Anchored	1	WOOD	0	34.63056	9	6
15	16	Syste...	RES1-1SNB	Wood-Anchored	1	WOOD	0	34.31599	9	6
16	17	Syste...	COM3	Masonry	1	MASONRY	0	34.38406	9	6
17	18	Syste...	RES1-1SWB	Wood-Anchored	1	WOOD	0	34.22558	9	6
19	20	Syste...	RES2	Manufactured	1	MANUFACTURED	0	34.28545	9	6
20	21	Syste...	RES1-1SNB	Wood-Anchored	1	WOOD	0	34.12369	9	6
21	22	Syste...	RES1-1SNB	Wood-Anchored	1	WOOD	0	33.9333	9	6
22	23	Syste...	RES1-1SNB	Wood-Anchored	1	WOOD	0	40.21816	9	6
23	24	Syste...	RES1-1SNB	Wood-Anchored	1	WOOD	0	36.06459	9	6

Record 10 Of 73063

Figure A - 21. ***Map Layer*** **Attributes Dialog Box – Results of Executed Expression**

9. **Show Selected Rows Only** ( ) will only display the highlighted features in the table on the *Map Layer* **Attributes** dialog box (Figure A - 21). Clicking (**Show All Rows**), will bring back all of the records for the selected map layer.

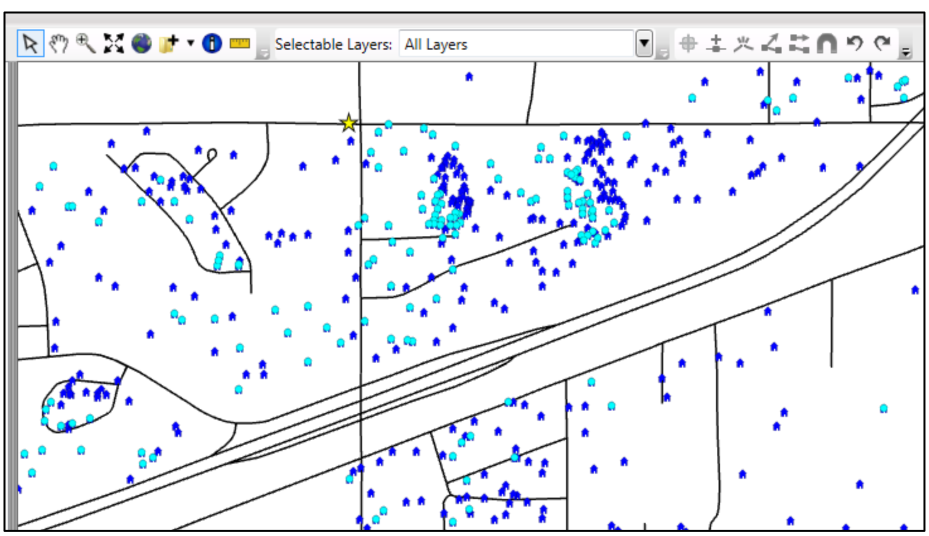


Figure A - 22. Map Window – Displaying Selected Features

A.4.5 Attribute Table Shortcut Menu

Other functions and commands are available from the *Map Layer* **Attributes** dialog box (Figure A - 21). If the user right-clicks on a selected row in the table, a shortcut menu will display, with one option - **Zoom to Selected** (Figure A - 23). **Zoom to Selected** will zoom the map window view to the features selected in the table (Figure A - 22).

When the user right-clicks on selected column(s) in the table, a shortcut menu will display (Figure A - 23) with several available options - **Sort Ascending**, **Sort Descending**, **Remove Sort**, **Summary Statistics**, **Find**, **Field Calculator**, and **Delete Column**. Other options are available when the user is in edit mode (described in Section A.10.1).

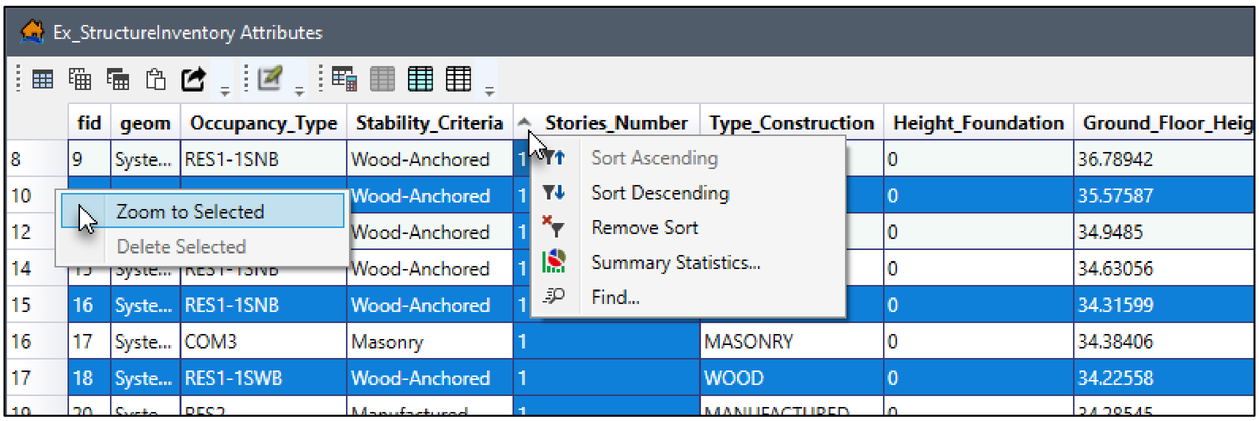


Figure A - 23. Attributes Table – Selected Columns Shortcut Menu

Sort Ascending Sorts the selected column in ascending (i.e., low to high; A – F) order. After clicking **Sort Ascending**, an up arrow displays in the column header (e.g., ***Stories_Number*** in Figure A - 23) to show the sorting of the selected column. Double-clicking on a selected column will reverse the current sort.

Sort Descending Sorts the selected column in descending (i.e., high to low; F – A) order. After clicking **Sort Descending**, a down arrow displays in the column header to show the sorting of the selected column. Double-clicking on a selected column will reverse the current sort.

Remove Sort **Remove Sort** does not become active until either **Sort Ascending** or **Sort Descending** has been clicked. **Remove Sort** will revert the selected column back to its original state.

Summary Statistics When clicked the *Attribute* **Summary Statistics** dialog box will open (Figure A - 24), providing the summary statistics for the selected column in the table (Figure A - 21). If the selected column have numeric values then the summary statistics will include indicative statistics of the data and a histogram (similar to Figure 13-16). If the selected columns have text data then the summary statistics will have a list of all the unique text values and the frequency of those text values along with a pie chart of the most frequent text values. When rows in the table on the *Map Layer*

Attributes dialog box (Figure A - 21) are selected, the **Selected Rows Only** option (Figure A - 24) becomes available in the *Attribute* **Summary Statistics** dialog box (Figure A - 24). If the user clicks this option, the summary statistics displayed are only for the selected rows in the *Map Layer* **Attributes Table** (Figure A - 21). The pie chart (Figure A - 24) will lump smaller counts into a new unique value and count entitled *Other* if there are more than four unique items associated with the selected column.

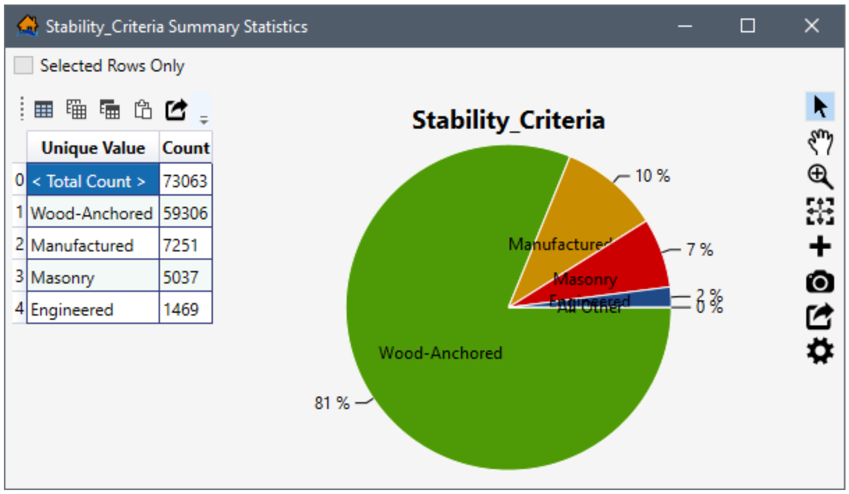


Figure A - 24. ***Attribute*** **Summary Statistics Dialog Box**

Find The **Find** option is only available when the entire table on the *Map Layer* **Attributes** dialog box (Figure A - 21) is available. If the user has selected **Show Selected Rows Only** (Figure A - 21), then the **Find** option will not be available. When **Find** is clicked, the **Find In:** *Attribute* dialog box (Figure A - 25) will open. The user can search for specific attributes by entering the attribute name in the **Find** box (Figure A - 25).

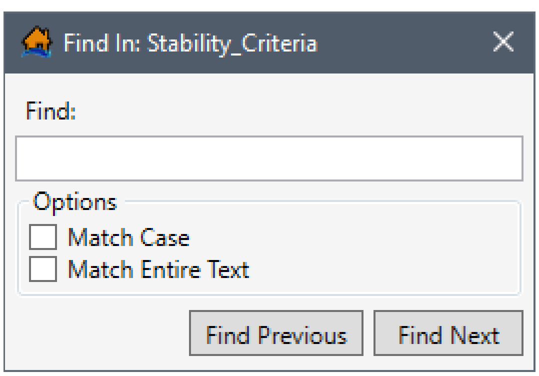


Figure A - 25. **Find In:** ***Attribute*** **Dialog Box**

A.5 Create New Map Layers

New map layers can be created in LifeSim using the **Create New Vector Features File** dialog box (Figure A - 26). The new map layers created are shapefiles (**.shp*), which store non-topological geometry and attribute information for the spatial features of a dataset.

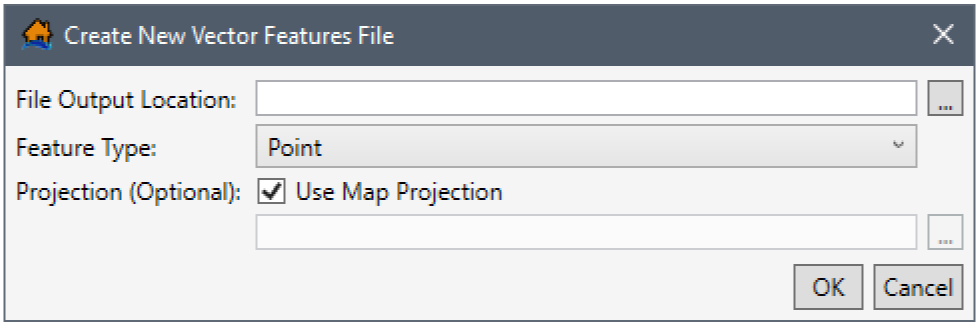


Figure A - 26. Create New Vector Features File Dialog Box

To create a new map layer:

1. From the LifeSim main window (Figure A - 1), from the **Mapping** menu (Figure A - 27), click **Create New Vector Layer**. Another way is from the **General Toolbar** (Figure A - 1), from **Add Data** ( ); from the shortcut menu (Figure A - 28), click **Create New**. Either way, the **Create New Vector Features File** dialog box (Figure A - 26) will open.

2. From the **File Output Location** box (Figure A - 26) click , a **Save As** browser will open (Figure A - 29), navigate to the desired directory for storing the shapefile (**.shp*). Enter a name (required) in the **File name** box (Figure A - 29) for the new map layer. Click **Save**, the **Save As** browser will close (Figure A - 29) and pathname and filename of the new map layer will display in the **File Output Location** box (Figure A - 26).

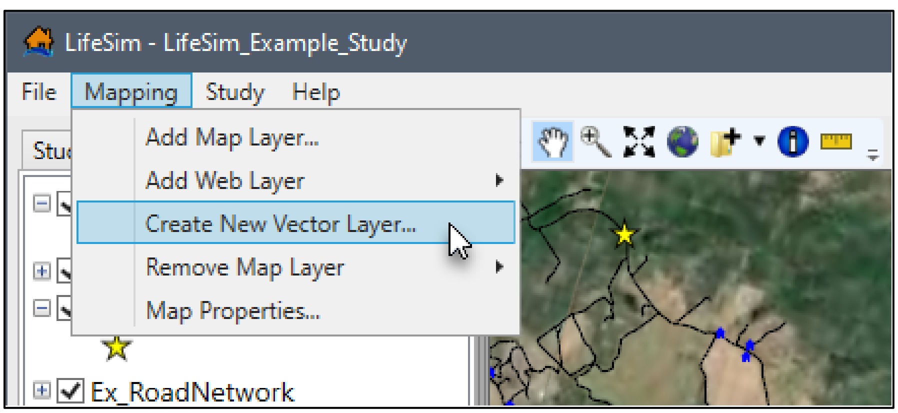


Figure A - 27. Mapping Menu

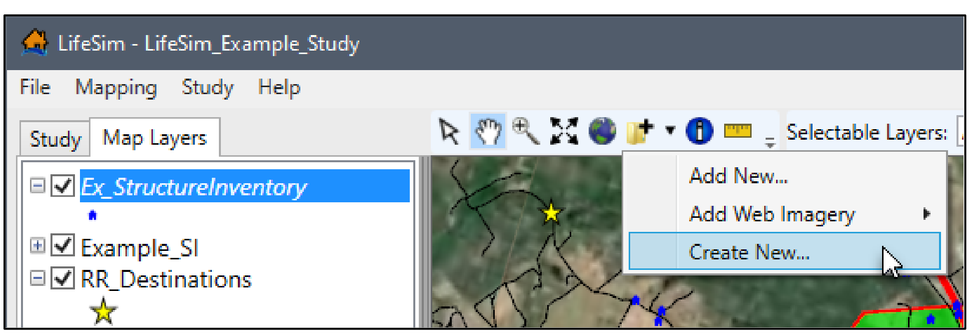


Figure A - 28. General Toolbar – Add Data Shortcut Menu

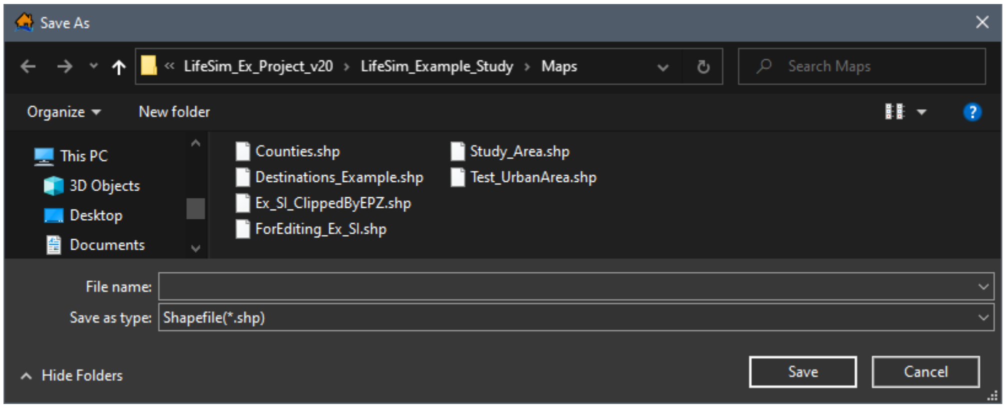
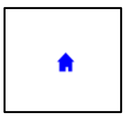
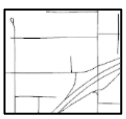


Figure A - 29. Save As Browser

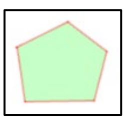
3. From the **Feature Type** list (Figure A - 26) select the geographic features of the shapefile which can be represented as points, polylines (lines), or polygons (areas).



Point
Features that are too small to be represented by lines or polygons. Examples include structure inventories and destinations.



Polyline
Features that represent the shape and location of geographic objects, such as the road networks.



Polygon
A set of many-sided area features which represent the shape and location of a feature or set of features, such as emergency planning zone(s).

4. Users have the option to select a projection for the new map layer from another map layer. To add a reference map coordinate system (create a projection), from the **Projection** box (Figure A - 26), click , an **Open** browser will open (Figure A - 5). Navigate to the location of the projection file that contains the projection that will be used for the new map layer (**.prj*), select the file and click **Open**. The **Open** browser will close (Figure A - 5), the **Create New Vector Features File** dialog box (Figure A - 26), **Projection** box will now contain the file location of the selected projection file.

5. Click **OK**, the **Create New Vector Features File** dialog box will close (Figure A - 26). The LifeSim **Map Layers** tab (Figure A - 1) will now contain the newly created map layer (at the bottom of the **Map Layers** tab).

6. The next step after creating a new map layer is to add new features. Adding new features to the map layer will be discussed in detail in Section A.10.

A.6 Remove Map Layers

Removing map layers for an LifeSim study:

1. From the LifeSim main window (Figure A - 1), from the **Mapping** menu, point to **Remove Map Layer** (Figure A - 30), from the list of map layers click on the map layer that is to be removed from the LifeSim study. The map layer is removed.

2. Another option is, from the LifeSim main window, from the **Map Layers** tab (Figure A - 1), right-click on the map layer of interest, from the shortcut menu (Figure A - 2), click **Remove**. The selected map layer will be removed from the LifeSim study. **Remove** will allow the user to select multiple map layers for removal.

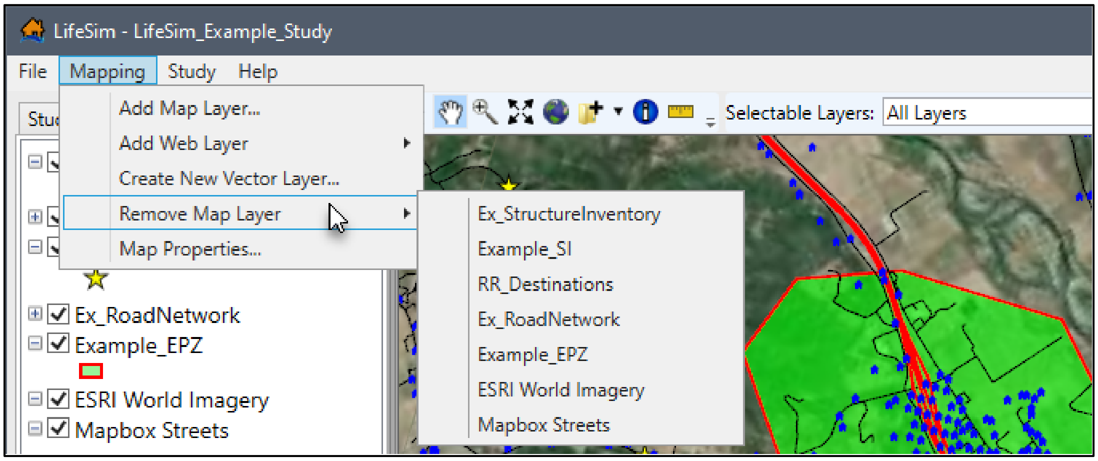


Figure A - 30. Mapping Menu – Remove Map Layer

A.7 Map Layer Tools

In LifeSim there are tools available to perform operations on map layers that can be useful in preparing data and analyzing results (e.g., exporting shapefiles, clipping features, or creating buffer polygons). Currently tools are only available for points, lines, and polygons shapefiles but will be expanded in the future to include gridded data. From the **Map Layers** tab on the LifeSim main window (Figure A - 1), right-click on a map layer of interest. From the shortcut

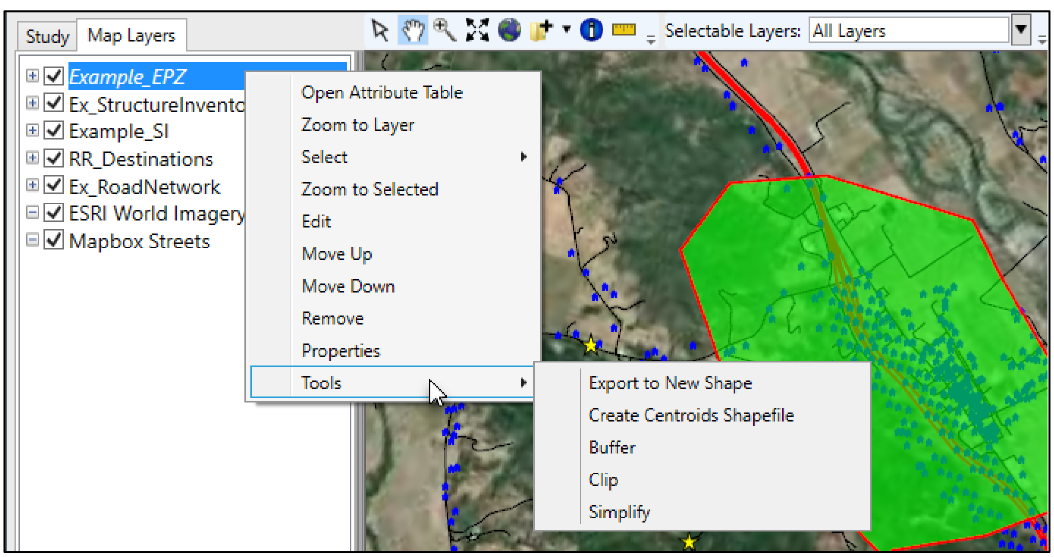


Figure A - 31. Individual Map Layer Shortcut Menu – Tools Sub-menu Commands

menu, point on **Tools**, from the submenu (Figure A - 31) are several commands that are available for editing map layers. These commands change according to the feature type (points, polylines, or polygons).

The following is an overview of each command:

Export to New Shape Exports the selected map layer as a new shapefile (i.e., output shapefile), saved with a name provided by the user. There are options to save the new shapefile with the map window projected coordinate system or the original coordinate system set for the selected map layer. Another alternative is to export only selected features for the map layer, when a subset of the map layer's features has been selected. This command is available for points, polylines, and polygon feature types. See Section A.7.1 for further details on the use of this command.

Buffer Creates polygons around features within the selected map layer, for a set distance around the object called a buffer (Figure A - 32). There is an option to merge the buffered features in the created polygon shapefile. If the merge option is selected all overlapping buffer polygons will be merged into one features and all attribute information will be removed from all of the new buffered features of the created polygon shapefile. This command is available for points, polylines, and polygon feature types. See Section A.7.3 for further details on the use of this command.

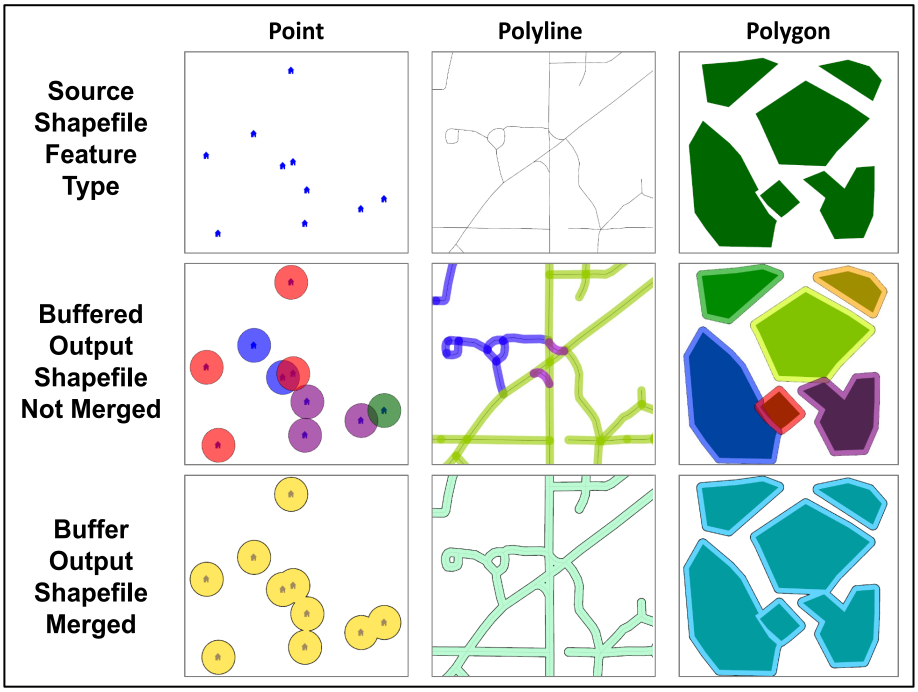


Figure A - 32. Buffer Tool Options – Illustration by Feature Type

Create Centroids Shapefile

Calculates the center point for the polygon(s) in the selected map layer and creates a centroids shapefile. An additional option allows the user to only include selected features in the centroid's shapefile. This command is only available for polygon feature types. See Section A.7.2 for further details on the use of this command.

Clip

Clip can be utilized to extract features from the selected map layer, using a polygon map layer or polygon shapefile to clip the input map layer features to within the polygon boundaries (Figure A - 33). This command is available for polylines and polygon feature types. See Section A.7.4 for further details on the use of this command.

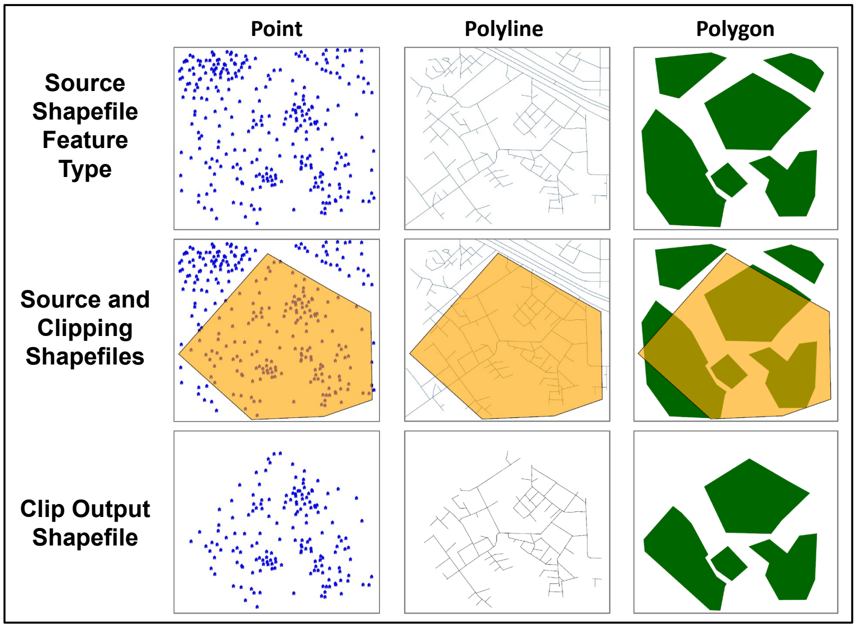


Figure A - 33. Clip Tool – Illustration by Feature Type

Simplify

This tool is utilized for simplifying the source polyline or polygon map feature by eliminating points based on the **Simplification Method** chosen. There are three methods to choose from (**Visalingam Whyatt**, **Lang**, and **Douglas Peucker**). Based on the **Simplification Method** chosen the dialog box will change to accommodate the parameters for the method of choice. This command is available for polylines and polygon feature types. See Section A.7.5 for further details on the use of this command.

A.7.1 Export to New Shapefile

To export a selected map layer to a new shapefile:

1. From the **Map Layers** tab on the LifeSim main window (Figure A - 1), right-click on the map layer of interest, from the shortcut menu (Figure A - 31), click **Export to New Shape**. The **Export To Shapefile** dialog box will open (Figure A - 34).

2. From the **Output Shapefile** box (Figure A - 34), click , a **Save As** browser will open (Figure A - 29). Navigate to the desired location for storing the new shapefile (**.shp*), enter a name (required) in the **File name** box (Figure A - 29) for the new shapefile.

3. Click **Save**, the **Save As** browser will close (Figure A - 29), the **Export To Shapefile** dialog box (Figure A - 34) will now display the path and name for the new shapefile to be saved in the **Output Shapefile** box.

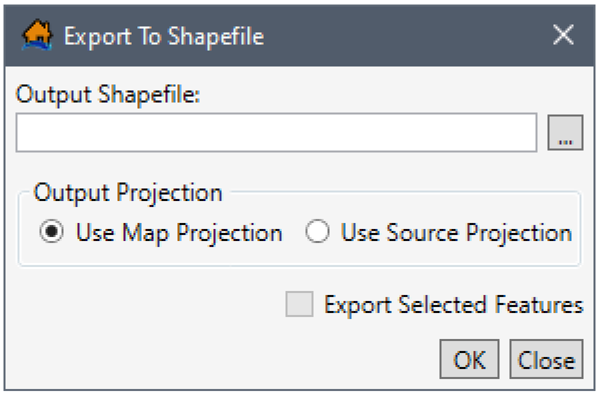


Figure A - 34. Export to Shapefile Dialog Box

4. Next, the user needs to select the projection for the new shapefile. From the **Output Projection** box (Figure A - 34) the available options: **Use Map Projection** or **Use Source Projection**.

   - **Use Map Projection** – this will set the new shapefile's projected coordinate system to that of the LifeSim study.

   - **Use Source Projection** – this will set the new shapefile's projected coordinate system to that of the selected map layer.

5. An optional step, **Export Selected Features** (Figure A - 34) allows the user to save the new shapefile with only features that have been selected for the selected map layer (i.e., using the **Select By Attribute** option explained in Section A.4; or the **Select** options explained in Section A.3.5).

6. Click **OK**, the **Verify Add Results** message window will open (Figure A - 35) asking the user if the newly created shapefile should be added to the map window (Figure A - 1). Click **Yes**, to add the new shapefile to the map window, and the **Verify Add Results** message window will close (Figure A - 35).

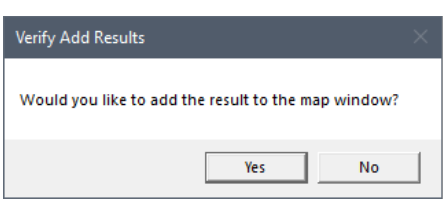


Figure A - 35. Verify Add Results Message Window

A.7.2 Create Centroids Shapefile

To create a centroids shapefile from a selected map layer (polygon shapefile):

1. From the **Map Layers** tab, on the LifeSim main window (Figure A - 1), right-click on the map layer of interest, from the shortcut menu (Figure A - 31), click **Create Centroids Shapefile**. The **Create Centroids** dialog box will open (Figure A - 36).

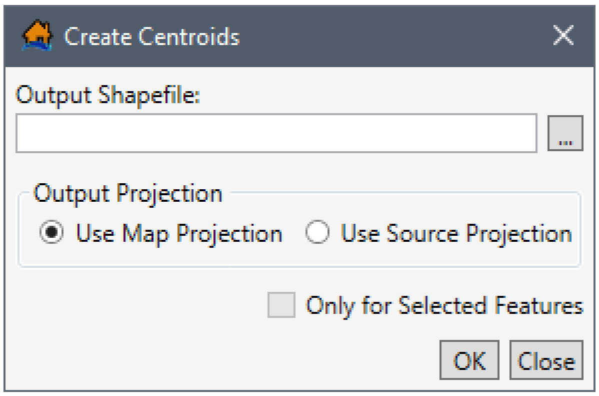


Figure A - 36. Create Centroids Dialog Box

2. From the **Output Shapefile** box (Figure A - 36), click , a **Save As** browser will open (Figure A - 29). Navigate to the desired location for storing the new centroids shapefile (**.shp*), enter a name (required) in the **File name** box (Figure A - 29) for the new centroids shapefile.

3. Click **Save**, the **Save As** browser will close (Figure A - 29), the **Create Centroids** dialog box (Figure A - 36) will now display the path and shapefile name for the new centroids shapefile to be saved in the **Output Shapefile** box.

4. Next, the user needs to select the projection for the new centroids shapefile. From the **Output Projection** box (Figure A - 36) the available options: **Use Map Projection** or **Use Source Projection**.

   - **Use Map Projection** – this will set the new centroids shapefile's projected coordinate system to that of the LifeSim study.

   - **Use Source Projection** – this will set the new centroids shapefile's projected coordinate system to that of the selected map layer.

5. An optional step, **Only for Selected Features** (Figure A - 36) allows the user to save the new shapefile with only features that have been selected for the selected map layer (i.e., using the **Select By Attribute** option explained in Section A.4; or the **Select** options explained in Section A.3.5).

6. Click **OK**, the **Verify Add Results** message window will open (Figure A - 35) asking the user if the newly created shapefile should be added to the map window (Figure A - 37).

Click **Yes**, to add the new shapefile to the map window, and the **Verify Add Results** message window will close (Figure A - 35).

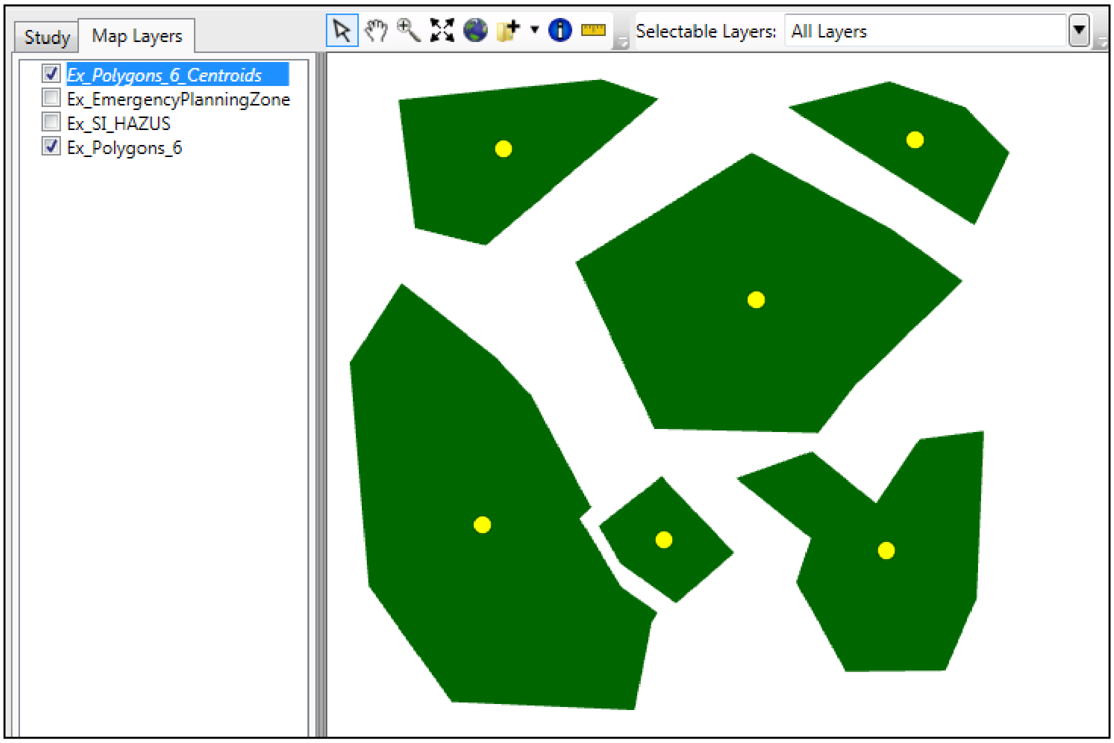


Figure A - 37. Map Window Displaying Centroid Shapefile

A.7.3 Buffer

To create a new shapefile that is based one set distance of a selected map layer:

1. From the **Map Layers** tab on the LifeSim main window (Figure A - 1), right-click on the map layer of interest, from the shortcut menu (Figure A - 31), click **Buffer**. The **Create Buffer Polygon(s)** dialog box will open (Figure A - 38).

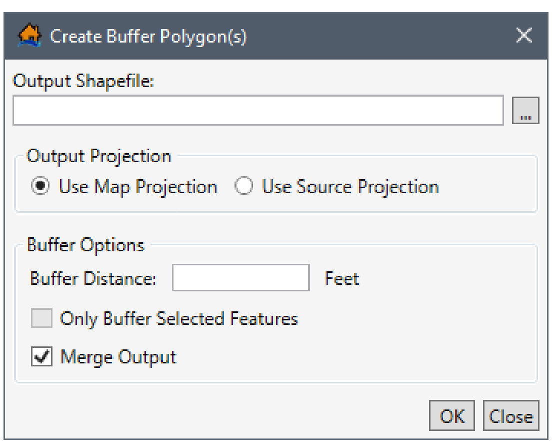


Figure A - 38. Create Buffer Polygon(s) Dialog Box

2. From the **Output Shapefile** box (Figure A - 38), click , a **Save As** browser will open (Figure A - 29). Navigate to the desired location for storing the new shapefile (**.shp*), enter a name (required) in the **File name** box (Figure A - 29) for the new shapefile.

3. Click **Save**, the **Save As** browser will close (Figure A - 29), the **Create Buffer Polygon(s)** dialog box (Figure A - 38) will now display the path and shapefile name for the new shapefile to be saved in the **Output Shapefile** box.

4. Next, the user needs to select the projection for the new centroids shapefile. From the **Output Projection** box (Figure A - 38) the available options: **Use Map Projection** or **Use Source Projection**.

   - **Use Map Projection** – this will set the new shapefile's projected coordinate system to that of the LifeSim study.

   - **Use Source Projection** – this will set the new shapefile's projected coordinate system to that of the selected map layer.

5. The user will need to enter the buffer distance for the new shapefile. In the **Buffer Options** box (Figure A - 38), in the **Buffer Distance** box enter the buffer distance in feet (e.g., *50*).

6. An optional step, **Only Buffer Selected Features** (Figure A - 38) allows the user to save the new shapefile with only features that have been selected for the selected map layer (i.e., using the **Select By Attribute** option explained in Section A.4; or the **Select** options explained in Section A.3.5).

7. Click **OK**, a **Verify Add Results** message window will open (Figure A - 35) asking the user if the newly created shapefile should be added to the map window (Figure A - 1). Click, **Yes**, to add the new shapefile to the map window, and the **Verify Add Results** message window will close (Figure A - 35).

A.7.4 Clip Tool

To create a shapefile of a selected map layer that has been clipped (points and polygons):

1. From the **Map Layers** tab on the LifeSim main window (Figure A - 1), right-click on the map layer of interest, from the shortcut menu (Figure A - 31), click **Clip**. The **Clip by Polygon** dialog box will open (Figure A - 39).

2. From the **Output Shapefile** box (Figure A - 39), click , a **Save As** browser will open (Figure A - 29). Navigate to the desired location for storing the new shapefile (**.shp*), enter a name (required) in the **File name** box (Figure A - 29) for the new shapefile.

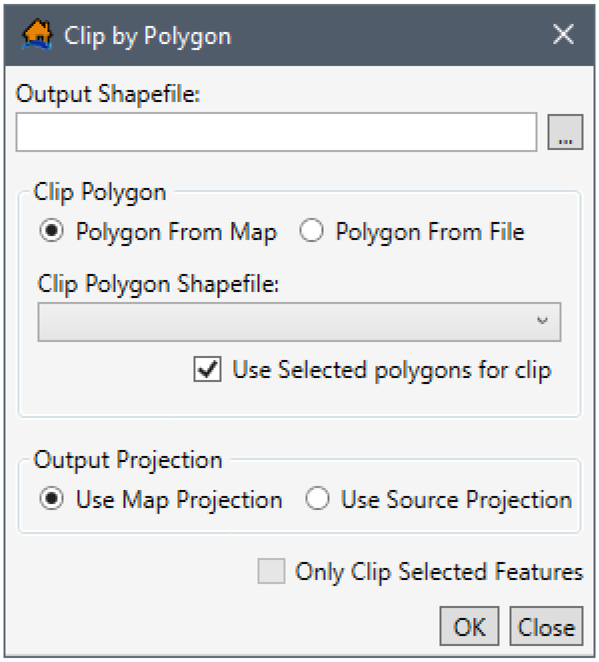


Figure A - 39. Clip by Polygon Dialog Box

3. Click **Save**, the **Save As** browser will close (Figure A - 29), the **Clip by Polygon** dialog box (Figure A - 39) will now display the path and shapefile name for the new shapefile to be saved in the **Output Shapefile** box.

4. Next, the user needs to clip the selected map layer based on polygon shapefiles that are part of the LifeSim study or from other polygon shapefiles. The available options are: **Polygon From Map** or **Polygon From File**.

   - **Polygon From Map**: From the **Clip Polygon** box (Figure A - 39), from the **Clip Polygon Shapefile** list (Figure A - 39), all polygon map layers (e.g., the *Emergency Planning Zone(s)*) imported in the current LifeSim study will be available. Select the desired polygon map layer to clip the selected map layer. An optional feature is **Use Selected polygons for clip** (Figure A - 39) which is only available when the selected map layer has features selected (i.e., using the **Select By Attribute** option explained in Section A.4, or the **Select** options explained in Section A.3.5).

   - **Polygon From File**: From the **Clip Polygon** box (Figure A - 40), from the **Clip Polygon Shapefile** box, click , the **Please select a clip polygon shapefile** browser will open. Navigate to the file of choice and select the desired polygon shapefile (**.shp*). Click **Open**, if the selected file is a polygon shapefile, the **Please select a clip polygon shapefile** browser will close. On the **Clip by Polygon** dialog box (Figure A - 40), the selected file's path and name will display in the **Clip Polygon Shapefile** box (Figure A - 40). However, if the file selected is not a polygon shapefile, then, a warning message window will open (Figure A - 41).

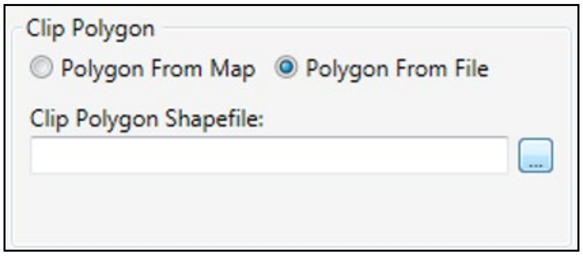


Figure A - 40. Clip by Polygon Dialog Box – Polygon From File Option

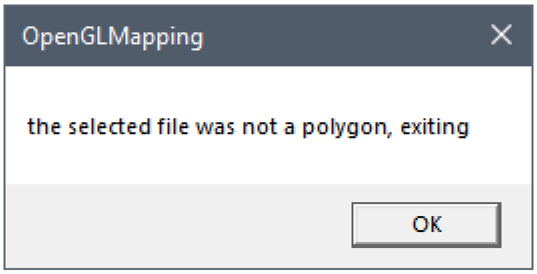


Figure A - 41. OpenGLMapping Warning Message

5. Next, the user needs to select the projection for the new centroids shapefile. From the **Output Projection** box (Figure A - 39) the available options: **Use Map Projection** or **Use Source Projection**.

   - **Use Map Projection** – this will set the new shapefile's projected coordinate system to that of the LifeSim study.

   - **Use Source Projection** – this will set the new shapefile's projected coordinate system to that of the selected map layer.

6. An optional step, **Only Clip Selected Features** (Figure A - 39) allows the user to save the new shapefile with only features that have been selected for the selected map layer (i.e., using the **Select By Attribute** option explained in Section A.4; or the **Select** options explained in Section A.3.5).

7. Click **OK**, a **Verify Add Results** message window will open (Figure A - 35) asking the user if the newly created shapefile should be added to the map window (Figure A - 1). Click **Yes**, to add the new shapefile to the map window, and the **Verify Add Results** message window will close (Figure A - 35).

A.7.5 Simplify – Polygon Map Layer

To create a shapefile of a selected polygon map layer that has been simplified (polylines and polygons):

1. From the **Map Layers** tab on the LifeSim main window (Figure A - 1), right-click on the map layer of interest, from the shortcut menu (Figure A - 31), click **Simplify**. Based on the feature type of the selected map layer, either the **Simplify Polygon(s)** dialog box (Figure A - 42) or the **Simplify Line(s)** dialog box (similar to Figure A - 42) will open.

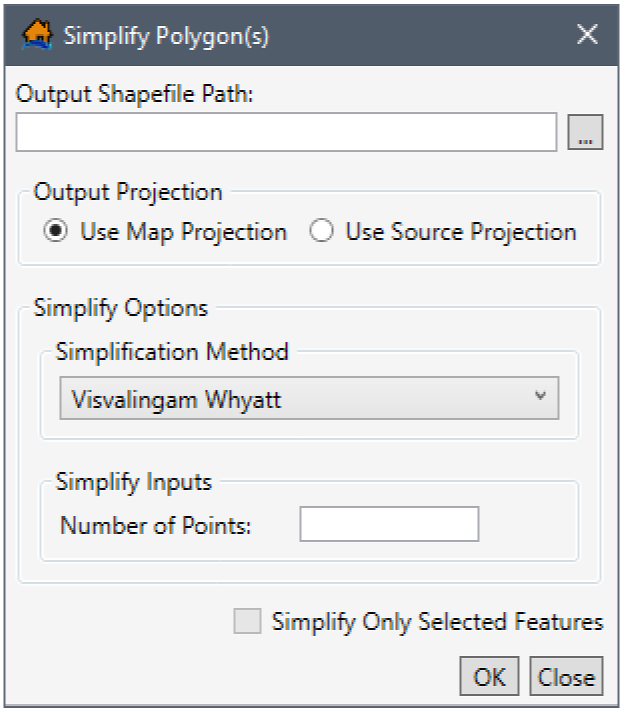


Figure A - 42. Simplify Polygon(s) Dialog Box

2. From the **Output Shapefile** box (Figure A - 42), click , a **Save As** browser will open (Figure A - 29). Navigate to the desired location for storing the new shapefile (**.shp*), enter a name (required) in the **File name** box (Figure A - 29) for the new shapefile.

3. Click **Save**, the **Save As** browser will close (Figure A - 29), the **Simplify Polygon(s)** dialog box (Figure A - 42) will now display the path and shapefile name for the new shapefile to be saved in the **Output Shapefile** box.

4. Next, the user needs to select the projection for the new centroids shapefile. From the **Output Projection** box (Figure A - 42) the available options: **Use Map Projection** or **Use Source Projection**.

   - **Use Map Projection** – this will set the new shapefile's projected coordinate system to that of the LifeSim study.

   - **Use Source Projection** – this will set the new shapefile's projected coordinate system to that of the selected map layer.

5. The user must select method for simplifying the selected map layer. From the **Simplify Options** box (Figure A - 42), from the **Simplification Method** list (Figure A - 42), select the appropriate simplification method. Each method has its own set of inputs. The simplification methods are briefly described:

   - **Visalingam Whyatt** – An area based algorithm to eliminate points based on each individual points effective area (area within the polygon the point effects). The point(s) which affect the polygon the least will be removed first. The user must enter the maximum number of points (greater than four and less than two points than the total number of points in the polygon) to be eliminated (**Number of Points** box, Figure A - 42). If the number of points entered is greater than the total number of points in the polygon then an identical copy of the polygon will be made.

- **Lang** – A perpendicular distance based algorithm to eliminate points based on the perpendicular distance of the points. The perpendicular distance is compared to a user entered tolerance limit for intermediate points between the end point and the entered number of look ahead points. The algorithm works based on the simplify inputs (**Number of Points** box, Figure A - 42), tolerance and look ahead points that the user enters. The **Tolerance** value must be greater than zero and the **Look Ahead Points** number must be less than or equal to the total number of points in the polyline or polygon shapefile. The point(s) which have a perpendicular distance less than the **Tolerance** limit for a set number of **Look Ahead Points** will be eliminated until there are no more intermediate points. If the number of **Look Ahead Points** entered is greater than the total number of points in the polygon or polyline shapefile then an identical copy of the selected shapefile will be made.

- **Douglas Peucker** – Similar to the **Lang** method, in that the points are eliminated based on the points distance from a simplification line. The simplification line is a straight line between key points. Key points are set for points greater than the **Tolerance** level defined by the user. The **Tolerance** level is the **Simplify Inputs** for the **Douglas Peucker** method, and the value must be greater than zero. In the end only the key points will remain after this simplification method is used.

6. An optional step, **Simplify Only Selected Features** (Figure A - 42) allows the user to save the new shapefile with only features that have been selected for the selected map layer (i.e., using the **Select By Attribute** option explained in Section A.4; or the **Select** options explained in Section A.3.5).

7. Click **OK**, a **Verify Add Results** message window will open (Figure A - 35) asking the user if the newly created shapefile should be added to the map window (Figure A - 1). Click **Yes**, to add the new shapefile to the map window, and the **Verify Add Results** message window will close (Figure A - 35).

A.8 Map Layers Properties

The properties of map layers can be viewed or edited in the *Map Layer* **Properties** dialog box (Figure A - 43) of the selected map layer. Depending on the map layer feature type (point, polyline, polygon, or gridded data) the options available in the *Map Layer* **Properties** dialog box (Figure A - 43) will vary accordingly. The *Map Layer* **Properties** dialog box (Figure A - 43) is

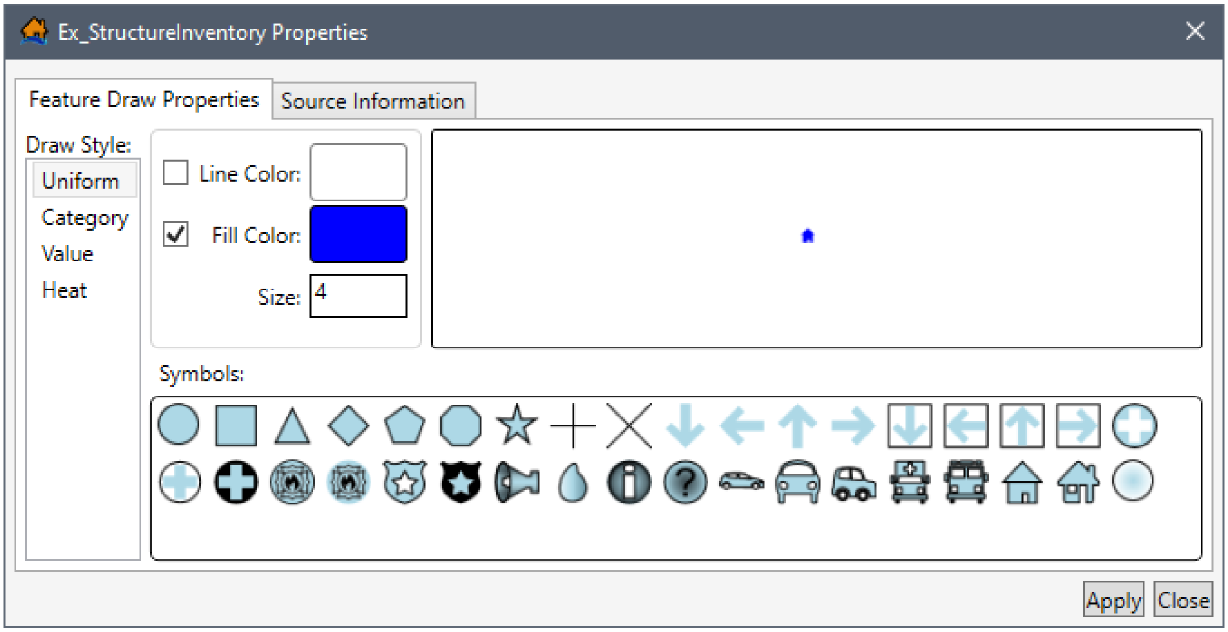


Figure A - 43. ***Map Layer*** **Properties Dialog Box**

organized into two tabs which allow the user to view and modify map window (Figure A - 1) display features (**Feature Draw Properties** tab) and to view the map layer source information (**Source Information** tab).

A.8.1 Feature Draw Properties Tab

The **Feature Draw Properties** tab (Figure A - 43) provides the user with the ability to edit map layer symbology (type, color, size, and other options).

Draw Style: **Uniform** Only the symbol can be modified directly. In other words, all features will be drawn with the same symbology.

Category The symbol(s) can be edited by unique attribute names or values. For example, one-way roads can be drawn differently than two-way roads.

Value The symbol(s) can be modified for a range of attribute values. For example, structures can be drawn differently for larger populations than smaller populations.

Heat The symbol(s) can be displayed as a heat map for a range of attribute values. For example, structures can be drawn differently for lower life loss than higher life loss.

Line/Fill Color: The user can modify the color for **Line** or **Fill** color setting. For example, to change line color, click on **Line Color**, the **Select a Color** dialog box will open (Figure A - 44). Properties are:

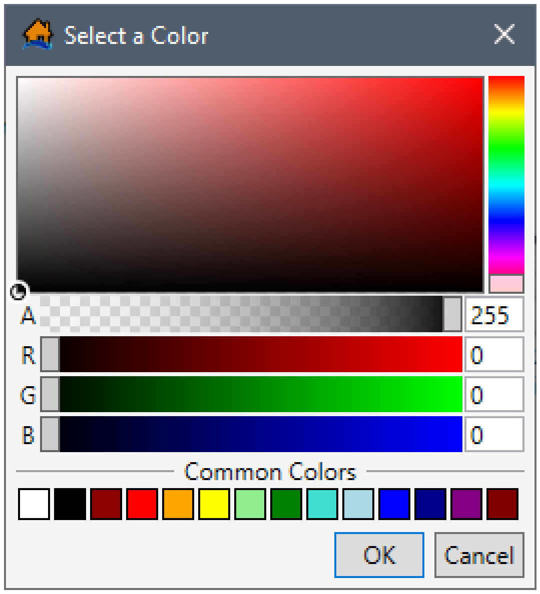


Figure A - 44. Select a Color Dialog Box

Preview This display box provides a preview of the selected color.

Color Pallet This display area (Figure A - 44) provides saturation (amount of black) options for the chosen color in the **Color Bar**. Notice the white-black circle in the **Color Bar** display area, this circle shows the location of the selected color. Click anywhere inside the **Color Pallet** area and the white circle will be moved to the new location, and the **Color Properties** (Figure A - 44) values will change automatically.

Color Bar This vertical slider bar (Figure A - 44) allows the user to choose the color from the options available (ROYGBIVR).

Transparency The **Transparency** horizontal slider bar (Figure A - 44) changes the transparency of the color (also known as ***A***, for Alpha). For maximum color (zero transparency of the color) slide the bar completely to the left (*Alpha* will change to 255). For zero color (or maximum transparency of the color) slide the bar completely to the right (*Alpha* will change to 0).

Color Properties Provides value boxes (Figure A - 44) for ***A*** (Alpha, measure of opacity), ***R*** (Red), ***G*** (Green), ***B*** (Blue). Each box is a measure (from 0 to 255) of opacity or color. The maximum value is 255 and minimum value is zero. Modifying these numbers also affects the saturation (amount of black, lower the value more black).

Example **Color Properties** values (for all examples the Alpha value = 255):

- Bright Red – Red 255, Green 0, Blue 0
- White – Red 255, Green 255, Blue 255
- Bright Yellow – Red 255, Green 255, Blue 0
- Bright Teal – Red 0, Green 255, Blue 255

Selected Colors Provides twelve commonly used colors; click on a color and the **Select a Color** dialog box (Figure A - 44) will update with the color chosen.

Symbol Size Line Width: The **Symbol Size** (for point map layers) or **Line Width** (for polyline of polygon map layers) is a number representing the size/width (Figure A - 43) of the symbol/line being displayed. The numerical value of size/width ranges from 1 to 50 and the smaller the number the smaller/thinner the symbol/line will be.

Default **Symbol Size** or **Line Width** examples:

- **Structure Inventories**, Symbol = 4
- **Destinations**, Symbol = 11
- **Road Networks**, Line Width = 1
- **Emergency Planning Zones**, Line Width = 2

Uniform Draw Style

To adjust symbol type, symbol size, and the symbol line and fill color for a point map layer:

1. From the **Map Layers** tab on the LifeSim main window (Figure A - 1), right-click on the point map layer of interest, from the shortcut menu (Figure A - 2), click **Properties**. The *Map Layer* **Properties** dialog box for the selected map layer will open (Figure A - 43).

2. The first option available to the user to modify the point map layer is to change the **Draw Style** definitions. For example, the default setting for point (as well as polygon, and polyline) map layer properties is **Draw Style**, ***Uniform*** (Figure A - 43). At this point keep the default **Draw Style**, ***Uniform*** setting.

3. To change the symbol type from the default blue house () select a new symbol from the available **Symbols** box (Figure A - 45). Selecting a new symbol will automatically adjust the symbol preview (symbol in the top-right box of the **Properties** dialog box).

4. Click **Apply**, all of the features in the map layer will be updated to the newly selected symbol (once **Apply** is clicked this change cannot be undone).

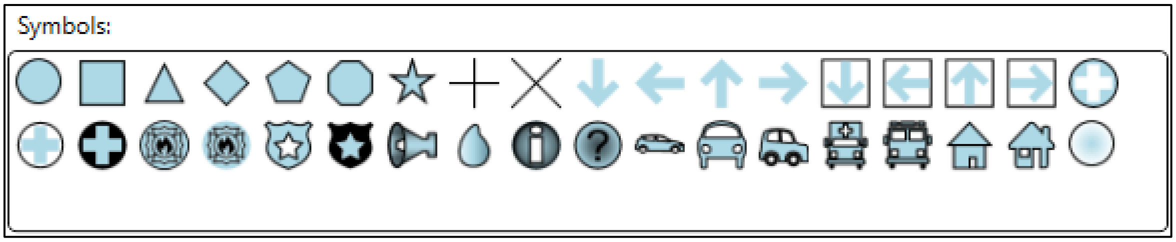


Figure A - 45. ***Map Layer*** **Properties Dialog Box – Symbols**

5. To adjust the map symbol line color, select **Line Color** (Figure A - 43), click the **Line Color** preview box. The **Select a Color** dialog box will open (Figure A - 46).

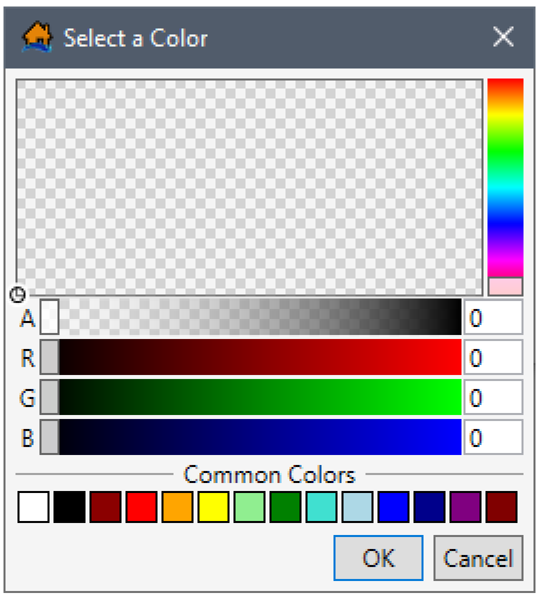


Figure A - 46. Select a Color Dialog Box – Transparency (maximum)

6. For example, in Figure A - 46, the **Select a Color** dialog box has a large checkered box, and other items that are similar to the *Map Layer* **Properties** dialog box (Figure A - 43).

7. Modify the **Selected Color** in the **Select a Color** dialog box (Figure A - 46) until the desired color is achieved. Click **OK**, the **Select a Color** dialog box will close (Figure A - 46). The user will be returned to the *Map Layer* **Properties** dialog box (Figure A - 43) where the new color will be displayed in the **Line Color** box and displayed in the symbol preview box.

8. Repeat Steps 5 through 7, for the **Fill Color** (Figure A - 43) if the **Fill Color** needs to be modified.

9. To modify the size of the symbol to be displayed in the map window, from the **Size** box (Figure A - 43) enter the new size.

    - For polylines and polygon shapefile the **Size** box is call **Line Width** and modifies the line (or polygon outline) width.

- An extra option for a polyline *Map Layer* **Properties** dialog box (Figure A - 43) is the option for adding an arrow at the end of lines. To add an arrow at the end of the lines select the **Draw End Arrow** option located under the **Line Width** box.

10. Once editing of the symbol type, symbol size, and the symbol line and fill color for a point map layer is complete, click **Apply** the *Map Layer* **Properties** dialog box will close (Figure A - 43).

Category Draw Style

To adjust **Draw Style**, *Category* map layer properties options for a point map layer:

1. From the **Map Layers** tab, from the LifeSim main window (Figure A - 1), right-click on the point map layer of interest, from the shortcut menu (Figure A - 2), click **Properties**. The *Map Layer* **Properties** dialog box for the selected map layer will open (Figure A - 43).

2. The first option available to the user to modify the point map layer is to change **Draw Style**. From the **Draw Style** list (Figure A - 43), the default value is ***Uniform*** (Figure A - 43). From the list the user can select other options.

3. From the **Draw Style** list (Figure A - 43), select ***Category***, now the **Attribute** list (Figure A - 47) is available. The **Attribute** list (Figure A - 47) contains all of the attributes contained in the selected map layer.

4. From the **Attribute** list (Figure A - 47) select the attribute of interest to draw the map layer by.

5. For the attribute that is selected, from the **Category** and **Symbol** table (Figure A - 48), the table will update with the range of values (text or numerical) for the selected attribute. Figure A - 48 provides an example for the ***Type_Construction*** attribute selection with the range of values equal to *Manufactured*, *Wood*, *Masonry*, *Concrete*, and *Steel*. By default, the *Excluded Values* category is included at the bottom of the table with a default color and size.

   The example shown in Figure A - 48 also displays the default symbol assignment for the range of values present. These defaults can be modified by the user and must be edited individually.

6. To change a symbol, select a **Category**/**Symbol** row in the table (Figure A - 48), and the selected symbol will be displayed in the **Preview** box (Figure A - 49). From the **Symbols** box (Figure A - 49) click on the desired new symbol. The **Preview** box and the symbol in the **Category**/**Symbol** table (Figure A - 48) will automatically update with the newly selected symbol.

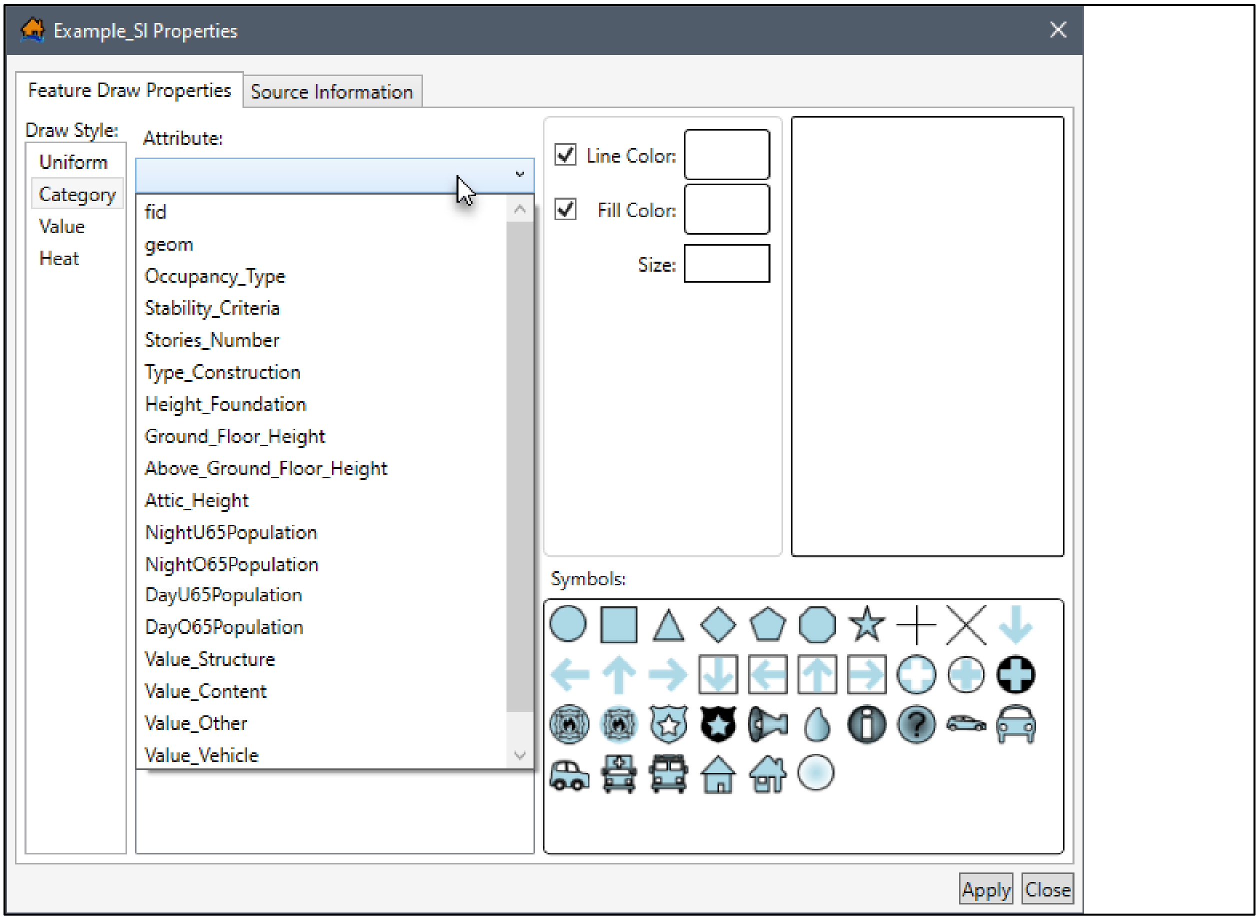


Figure A - 47. ***Map Layer*** **Properties Dialog Box – Attribute List - Category**

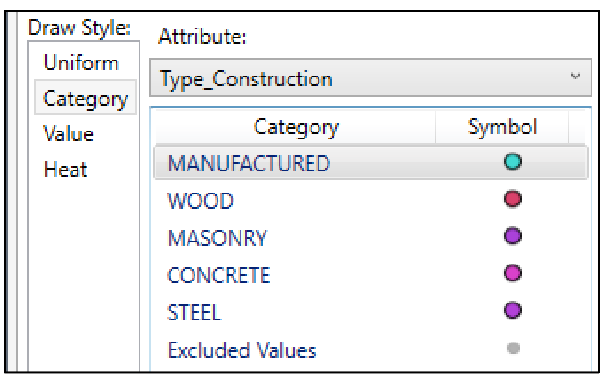


Figure A - 48. ***Map Layer*** **Properties Dialog Box – Value and Symbol Table**

7. To adjust the map symbol line color, select **Line Color** (Figure A - 43), click **Line Color** preview box. The **Select a Color** dialog box will open (Figure A - 46).

8. For details on the **Select a Color** dialog box see Section A.8.1. For example, in Figure A - 46, the **Select a Color** dialog box has a large checkered box, and other items that are similar to the *Map Layer* **Properties** dialog box (Figure A - 43).

9. Select the desired color in the **Select a Color** dialog box (Figure A - 46). Click **OK**, the **Select a Color** dialog box will close. The user will be returned to the *Map Layer*

Properties dialog box (Figure A - 43) where the new color will be displayed in the **Line Color** box and displayed in the symbol preview box.

10. Repeat Steps 6 through 9 for all of the values listed in the **Category**/**Symbol** table (Figure A - 48), which need to be changed.

11. Once editing of the **Draw Style** ***Category*** for a point map layer is complete, click **Apply** to view the changes in the map window. Note, make sure the map layer is selected to be viewed in the map window and is above other map layers).

    Once the desired drawing style is achieved, click **Apply** and then click **Close**, the *Map Layer* **Properties** dialog box will close (Figure A - 43).

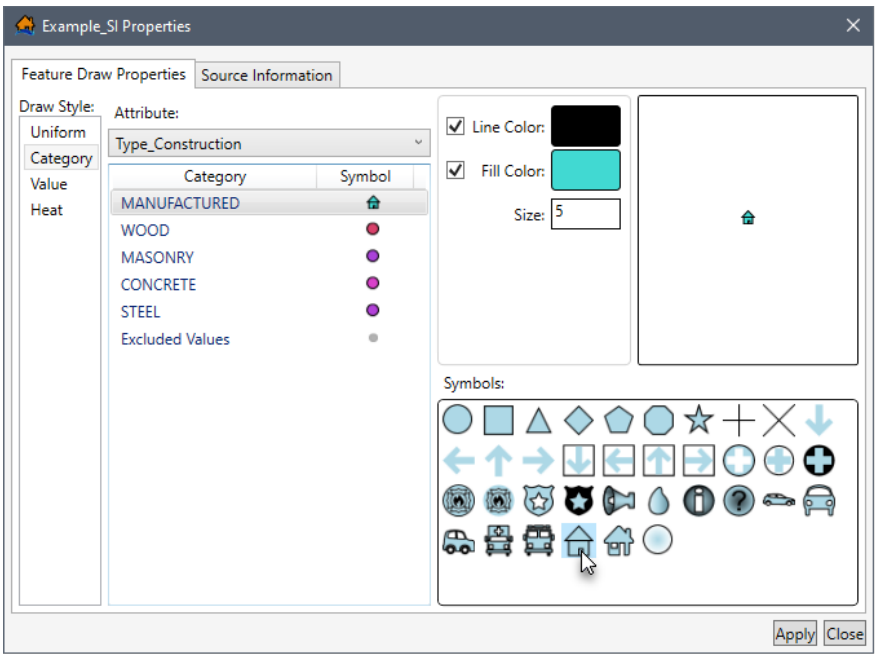


Figure A - 49. ***Map Layer*** **Properties Dialog Box – Symbol Preview Box**

12. Editing the **Draw Style** *Category* for polyline and polygon shapefiles is mainly the same as point shapefiles. The only difference in the previously mentioned options is that users can change the **Line Width** (polylines and polygons) instead of **Size** (points), and there is an added option to **Draw End Arrow** (polylines only).

Value Draw Style

Another option for modifying a map layer's properties is to adjust the **Draw Style** option to ***Value*** for a specific attribute.

1. From the **Map Layers** tab on the LifeSim main window (Figure A - 1), right-click on the point map layer of interest, from the shortcut menu (Figure A - 2), click **Properties**. The *Map Layer* **Properties** dialog box (Figure A - 43) for the selected map layer will open.

2. The first option available to the user to modify the point map layer is to change **Draw Style** option. From the **Draw Style** list, the default value is ***Uniform*** (Figure A - 43). From the list the user can select other options. From the **Draw Style** list, select ***Value***, now the **Attribute** list (Figure A - 50) is available. The **Attribute** list contains all the numerical attributes contained in the selected map layer.

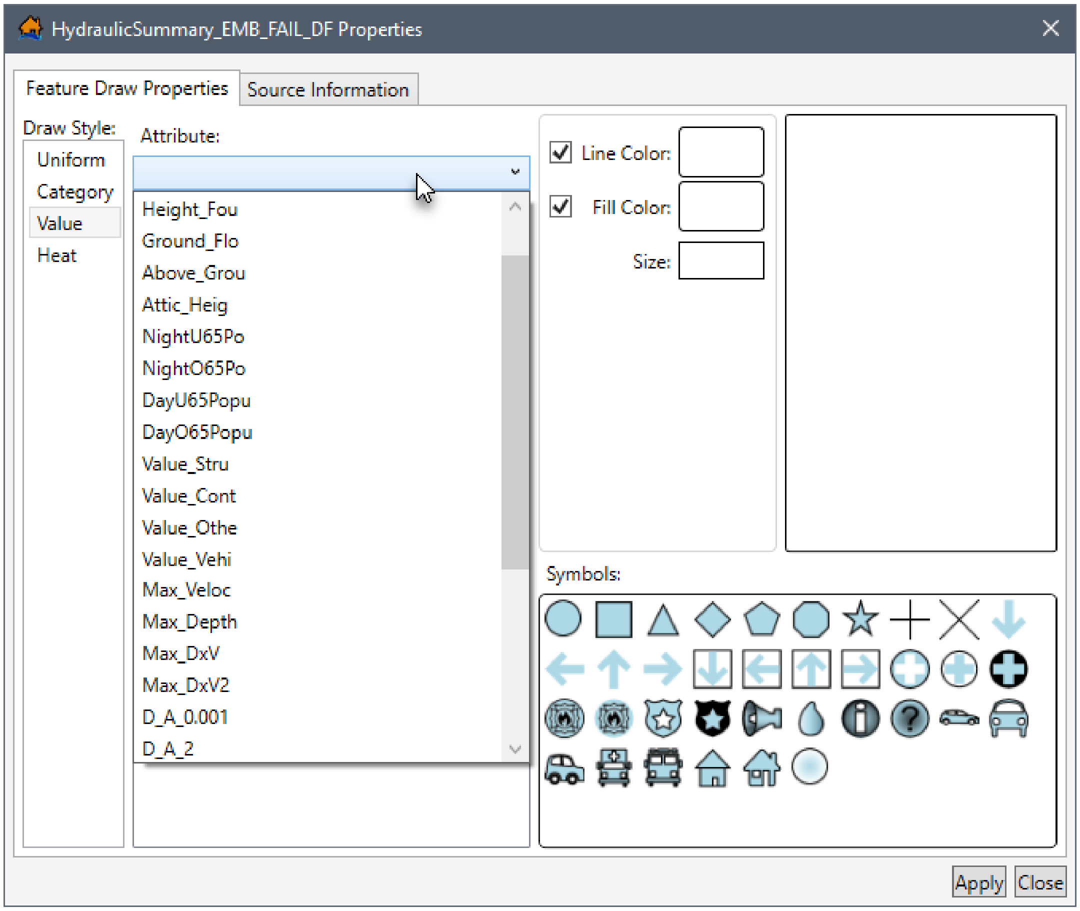


Figure A - 50. ***Map Layer* Properties Dialog Box – Attribute List – Value**

3. From the **Attribute** list (Figure A - 50) select the attribute of interest to draw the map layer by. Notice that the list of **Attributes** for **Draw Style** ***Value*** (Figure A - 50) is limited to only the ones with numerical values.

4. For the attribute that is selected, the parameters in the **Draw By Value Bin Definitions** box (Figure A - 51) become available and will update with the range of quantitative values for the selected attribute.

   The values entered for **Number**, **Minimum** and **Maximum** are used to define bins in equal intervals between the **Minimum** and **Maximum** values for the selected attribute. The total number of breaks (set at equal intervals by default) are used to determine the range of values (in the **Category/Symbol** table) by the value entered in the **Number** box (Figure A - 51).

The ***Excluded Values*** row (Figure A - 51) in the **Category/Symbol** table represents all values that are less than the **Minimum** or greater than the **Maximum** values. By default, the **Minimum** and **Maximum** values are set to the minimum and maximum values of the data, so no features are excluded.

Figure A - 51 provides an example with the default, and three examples that modify the **Number** and the **Maximum** value fields and how those changes modify the bins displayed in the **Category**/**Symbol** table.

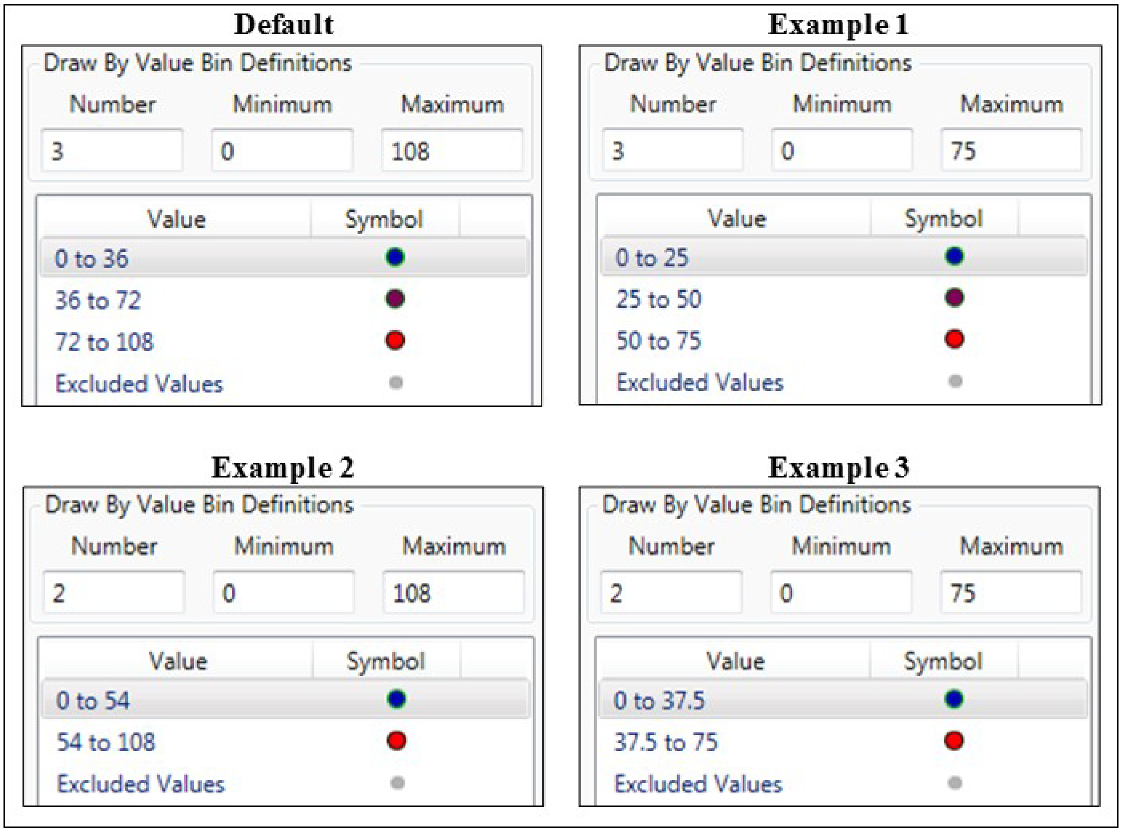


Figure A - 51. *Map Layer* Properties Dialog Box – Draw by Value Bin Definitions Box

5. Individual bin ranges can be adjusted in the **Category**/**Symbol** table (Figure A - 51) by entering the maximum value for each range, manually. To enter the maximum value manually, double-click a **Category/Symbol** row, the row is now editable. Then over-write the current value and the range will update (maximum value for the over-written value and the minimum value for the next row in the **Category**/**Symbol** table).

   - A hint for users is that sometimes the best way to determine the **Draw By Value Bin Definitions** is to review the summary statistics from the *Map Layer* **Attributes** dialog box (Figure A - 15), from the shortcut menu (Figure A - 23), click **Cal. Statistics**. For the selected attribute (i.e., *NightO65Population*), the *Attribute* **Summary Statistics** dialog box will open (Figure A - 52). Section A.4.5 provides a detailed description.

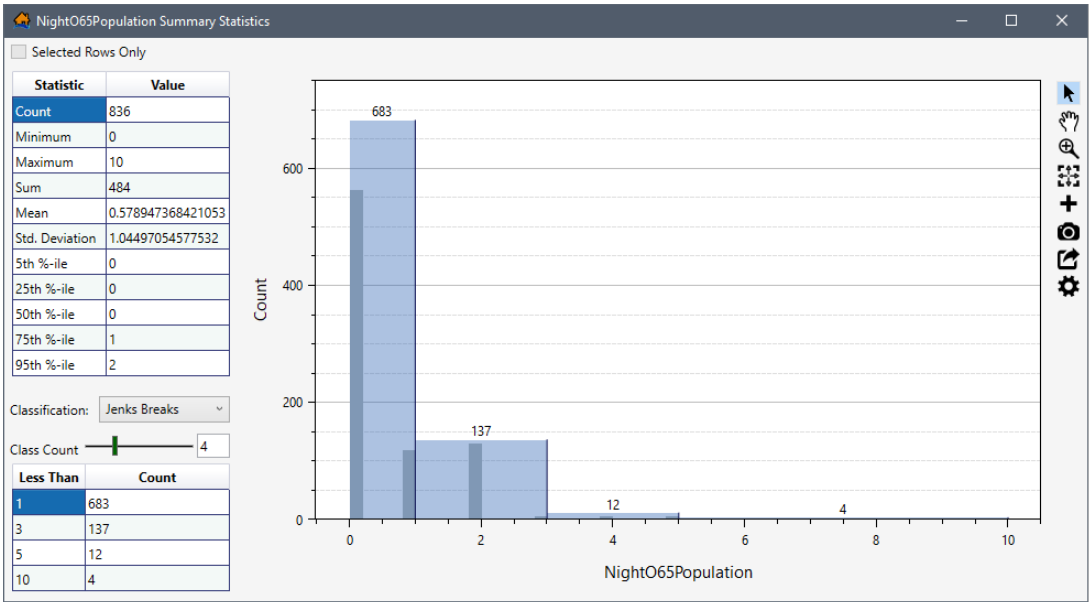


Figure A - 52. ***Attribute*** **Summary Statistics Dialog Box**

6. Also, the examples shown in Figure A - 51 present the default symbol assignments for the range of values. These defaults can be modified by the user, for the selected **Category**/**Symbol** in the table. Each **Category**/**Symbol** the user wishes to modify from the default must be edited individually.

7. To change a symbol, select a **Category**/**Symbol** row in the table (Figure A - 51), and the selected symbol will be displayed in the **Preview** box (Figure A - 49). From the **Symbols** box (Figure A - 45) click on the desired new symbol. The **Preview** box (Figure A - 49) and the symbol in the **Category**/**Symbol** table (Figure A - 51) will automatically update with the newly selected symbol.

8. To adjust the map symbol line color, select **Line Color** (Figure A - 45), click **Line Color** preview box. The **Select a Color** dialog box will open (Figure A - 46).

9. For details on the **Select a Color** dialog box see Section A.8.1. For example, in Figure A - 46, the **Select a Color** dialog box has a large checkered box, and other items that are similar to the *Map Layer* **Properties** dialog box (Figure A - 43).

10. Modify **Selected Color** in the **Select a Color** dialog box (Figure A - 46) until the desired color is achieved. Click **OK**, the **Select a Color** dialog box will close. The user will be returned to the *Map Layer* **Properties** dialog box (Figure A - 43) where the new color will be displayed in the **Line Color** box and displayed in the symbol preview box.

11. Repeat Steps 8 through 10 for all of the values listed in the **Category**/**Symbol** table (Figure A - 51), which need to be changed.

12. Once editing of the **Draw Style** ***Value*** for a point map layer is complete, click **Apply** and the *Map Layer* **Properties** dialog box will close (Figure A - 43).

Heat Draw Style

Another option for modifying a map layer's properties is to adjust the **Draw Style** option to create a *Heat* map for a specific attribute. Note, the **Heat** draw style is only available for point map layers.

1. From the **Map Layers** tab on the LifeSim main window (Figure A - 1), right-click on the point map layer of interest, from the shortcut menu (Figure A - 2), click **Properties**. The *Map Layer* **Properties** dialog box (Figure A - 43) for the selected map layer will open.

2. The first option available to the user to modify the point map layer is to change **Draw Style** option. From the **Draw Style** list, the default value is ***Uniform*** (Figure A - 43). From the list the user can select other options. From the **Draw Style** list, select ***Heat***, the **Properties** dialog box updates (Figure A - 53).

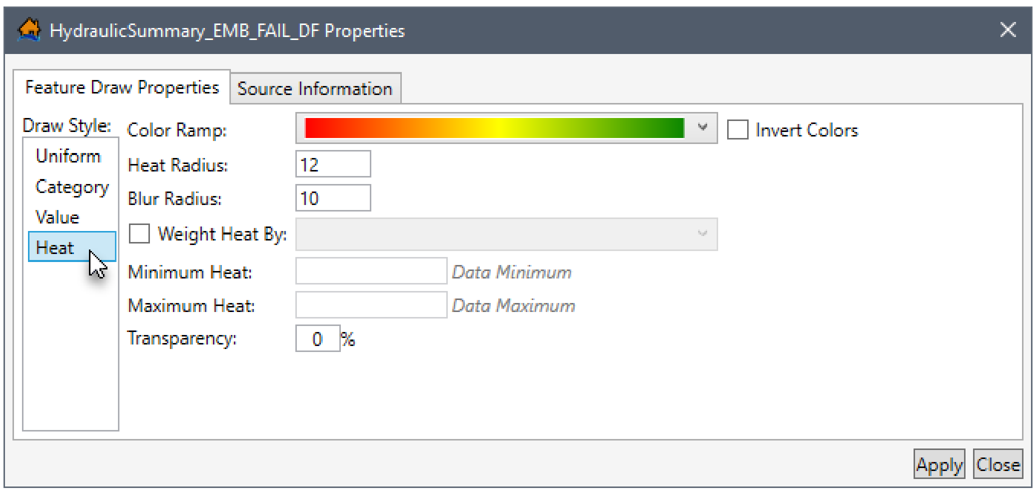


Figure A - 53. ***Map Layer* Properties Dialog Box – Draw Style – Heat**

3. From the **Color Ramp** list (Figure A - 54) users can select a color ramp for the heat map. The default color ramp is red to green (Figure A - 53), the user can change this from the **Color Ramp** list (Figure A - 58).

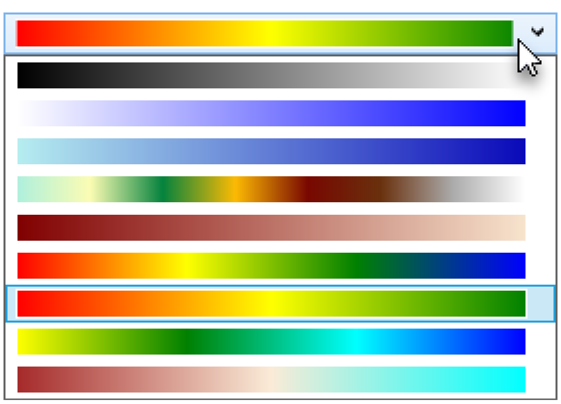


Figure A - 54. Draw Style – Heat – Color Ramp

4. If the **Color Ramp** selection is not correct then select **Invert Colors** (Figure A - 53) to invert the color ramp colors representation. For the example provided in Figure A - 55, the default color ramp (red = ***More***, green = ***Less***) was inverted (red = ***Less***, green = ***More***).

Figure A - 55. Example – Invert Colors

5. By default, the **Heat Radius** is set to ***12*** (Figure A - 53). Set a different radius by entering a different value in the **Heat Radius** box.

6. By default, the **Blur Radius** is set to ***10*** (Figure A - 53). Set a different radius by entering a different value in the **Blur Radius** box.

7. Users can weight the heat map by an attribute in the all the numerical attributes contained in the selected map layer. To add a weight to the heat map, check ☑ the **Weight Heat By** checkbox (Figure A - 56) to enable the list of attributes. From the list (e.g., similar to Figure A - 50), select the desired attribute (e.g., ***Max_Depth*** in Figure A - 56). The **Minimum** and **Maximum Heat** values are set by default using the data minimum and maximum, respectfully. Set a different minimum and/or maximum heat by entering a different value in the **Minimum** and/or **Maximum Heat** boxes (Figure A - 56).

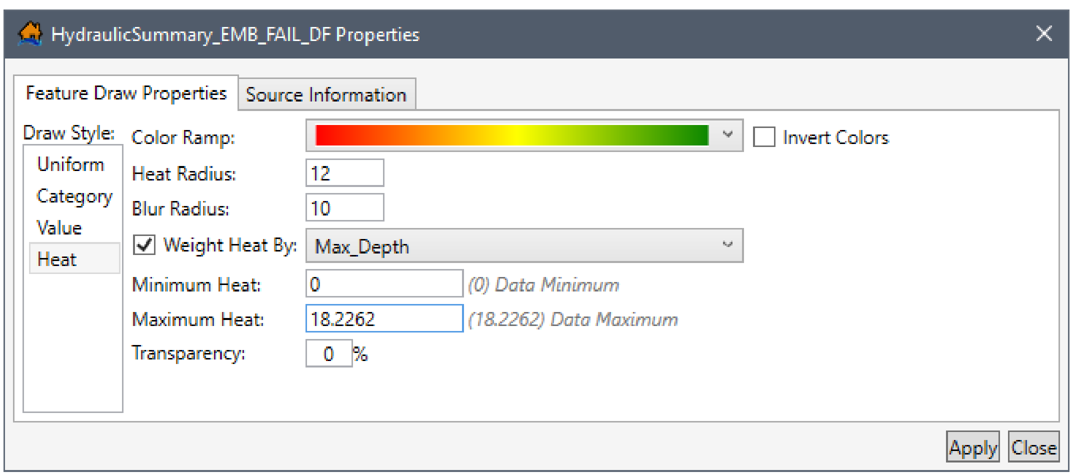


Figure A - 56. Example – Draw Style – Heat – Weight Heat By

8. The transparency of the heat map can be changed (from the default value of fully opaque). In the **Transparency** box (Figure A - 53), the user can enter a value from zero to 100 (percent) to increase the transparency of the map layer (the maximum value of 100 percent is maximum translucency).

9. Once editing of the symbology (type, color ramp, and other options) for a point map layer is complete, click **Apply**, and the selected gridded map layer is updated.

A.8.2 Source Information Tab

The **Source Information** tab (Figure A - 57) provides the:

- **Projection** coordinate system associated with the map layer,
- **Horizontal Units** of the map layer, and
- **Extent** of the map layer.

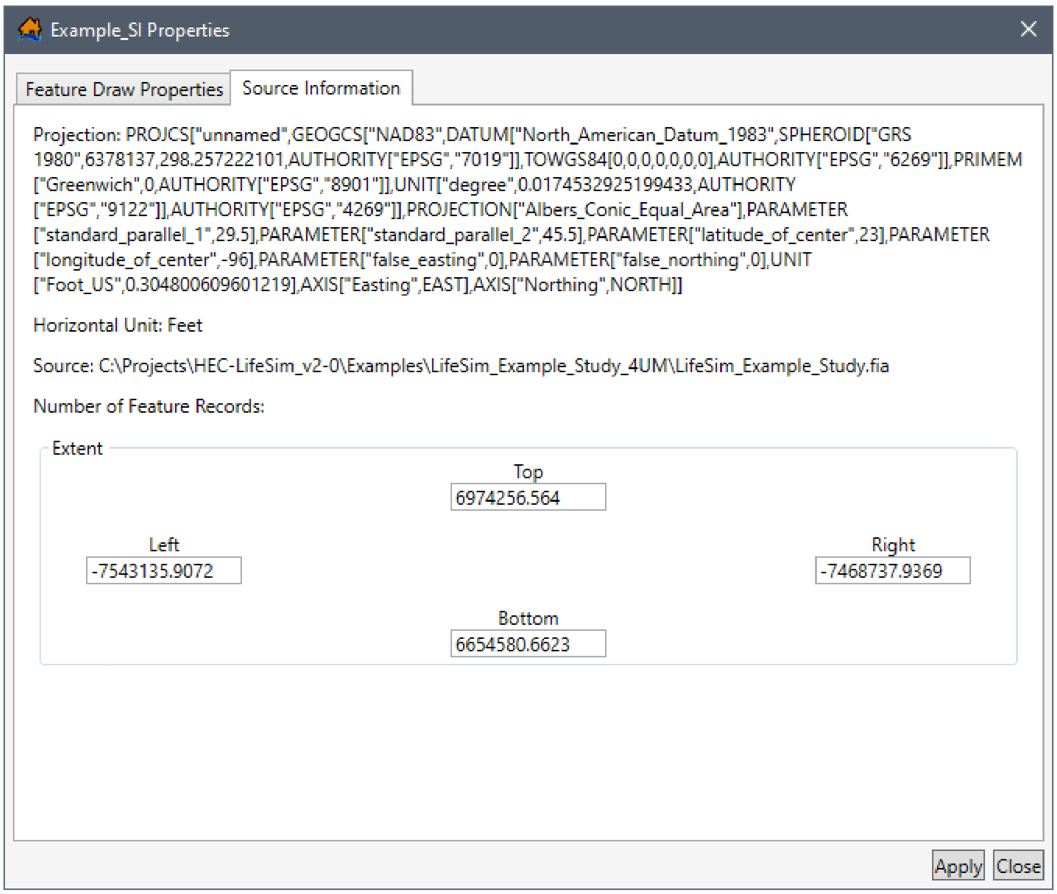


Figure A - 57. ***Map Layer*** **Properties Dialog Box – Source Information Tab**

A.9 Gridded Map Layer Properties

The properties of gridded map layers (**.tif*) can be viewed or edited in the *Map Layer* **Properties** dialog box (Figure A - 58) of a selected gridded map layer. The *Map Layer* **Properties** dialog box (Figure A - 58) is organized into two tabs which allow the user to view and modify map window (Figure A - 1) display features (**Raster Draw Properties** tab) and to view the map layer source information (**Source Information** tab).

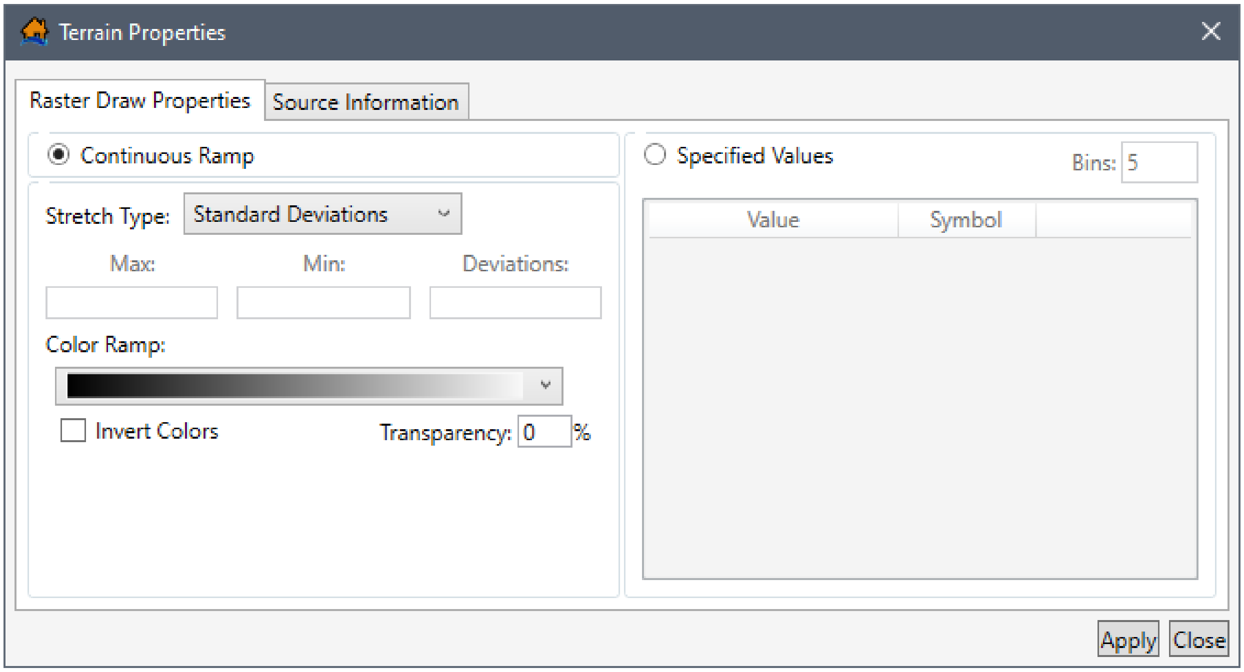


Figure A - 58. ***Map Layer*** **Properties Dialog Box – Gridded Map Layers**

A.9.1 Raster Draw Properties Tab

The **Raster Draw Properties** tab (Figure A - 58) provides the user with the ability to edit gridded map layer symbology (type, color or color ramp, and other options). There are three sections in the **Raster Draw Properties** tab which are:

- **Continuous Ramp**
- **Stretch Type** and **Color Ramp**
- **Specified Values**

Continuous Ramp

The **Stretch Type** list (Figure A - 58) contains four options:

- **None**
- **Minimum Maximum**
- **Standard Deviations**
- **Histogram-Equalization**

Depending on the selection the **Min** and **Max** or **Deviations** boxes (Figure A - 58) will become active.

Color Ramp

The **Color Ramp** list (Figure A - 59) contains nine options for the user to select for the color ramp of the gridded map layer.

There are two additional options for modifying the **Color Ramp**. The first is **Invert Colors**, which when selected will invert the selected color ramp scheme. The second is **Transparency**, the user will enter a transparency value (percentage) to modify the translucency of the selected color scheme.

Lower numbers are higher opacity and higher numbers are more transparent (one hundred effectively removes the map layer from view).

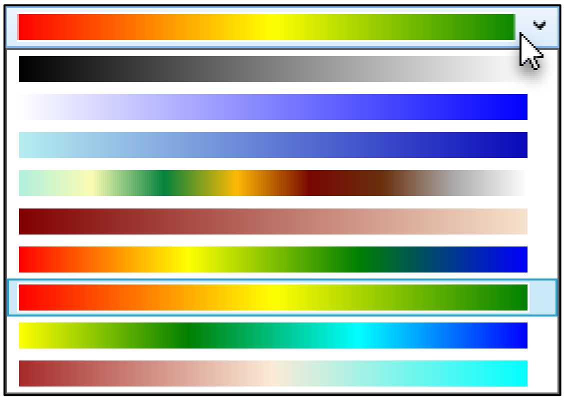


Figure A - 59. ***Map Layer*** **Properties Dialog Box – Gridded Map Layers Color Ramp Options**

Specified Values

Specified Values allow the user to set the number of **Bins** which updates the **Value** and **Symbol** table (Figure A - 58).

The preset **Symbol** color scheme ranges from black to white (low to high values). To change the preset symbol color click on the color box and the **Select a Color** dialog will open (Figure A - 44). The available options for the **Select a Color** dialog box (Figure A - 44) are very similar to what is described in Section A.8.1.

Continuous Ramp Rendering

The **Continuous Ramp** rendering options for gridded map layers and allows the user to adjust the symbology (type, color or color ramp, and other options) for a selected gridded map layer:

1. From the *Map Layer* **Properties** dialog box for the selected gridded map layer (Figure A - 58), from the **Stretch Type** and **Color Ramp** panel, users must decide which stretch type to select from the **Stretch Type** list (Figure A - 58):

   - **None** – The colors in the selected color ramp will be linearly distributed based on the cell values relation to the minimum and maximum data values.

   - **Minimum Maximum** – This option activates the **Min** and **Max** boxes (Figure A - 58). The selected color ramp will be stretched between the defined minimum and maximum values. This can be useful when the user wants data to be stretched over a very small data range. The user must use the default values or input different values. If the default minimum and maximum data values are chosen the gridded data will render the same as the **None** option.

   - **Standard Deviations** (default) – This option activates the **Deviations** box (Figure A - 58). The default **Deviations** value is *2*, users can delete and enter different deviations in the box. The ramp is stretched by the number of deviations from the mean data value.

- **Histogram-Equalization** – If the user selects this option and no histogram information exists for the gridded data, a **Histogram Does Not Exist** message window will open (Figure A - 60) asking the user for confirmation to create a histogram for the selected raster/gridded map layer.

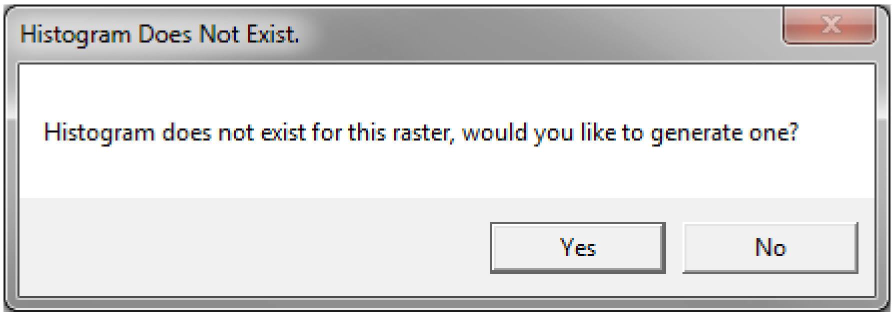


Figure A - 60. Histogram Does Not Exist Message Window

2. The default color ramp is black fading to white, the user can change this from the **Color Ramp** list (Figure A - 58). To determine if the color ramp scheme is correct, click **Apply** (Figure A - 58) and view the updated gridded map layer in the map window (Figure A - 1). Once **Apply** is clicked this change cannot be undone.

3. If the **Color Ramp** selection is not correct (e.g., higher water velocity with green colors and lower flow velocity with red colors) then select **Invert Colors** (Figure A - 58). Click **Apply**, and view the updated gridded map layer in the map window (Figure A - 1) to confirm the color scheme selection.

4. The transparency of the entire gridded/raster map layer can be changed (from the default value of fully opaque). In the **Transparency** box (Figure A - 58), the user can enter a value from zero to 100 (percent) to increase the transparency of the map layer (the maximum value of 100 percent is maximum translucency).

5. Once editing of the symbology (type, color or color ramp, and other options) for a gridded map layer is complete, click **Apply**, and the selected gridded map layer is updated.

Rendering with Specified Values

To use **Specified Values** for rendering gridded map layers and adjust the symbology (type, color or color ramp, and other options) for the selected gridded map layer:

1. From the *Map Layer* **Properties** dialog box (Figure A - 58) for the selected gridded map layer select **Specified Values** (Figure A - 61), the **Continuous Ramp** selection (Figure A - 58) is no longer available, while the **Specified Values** parameters becomes available.

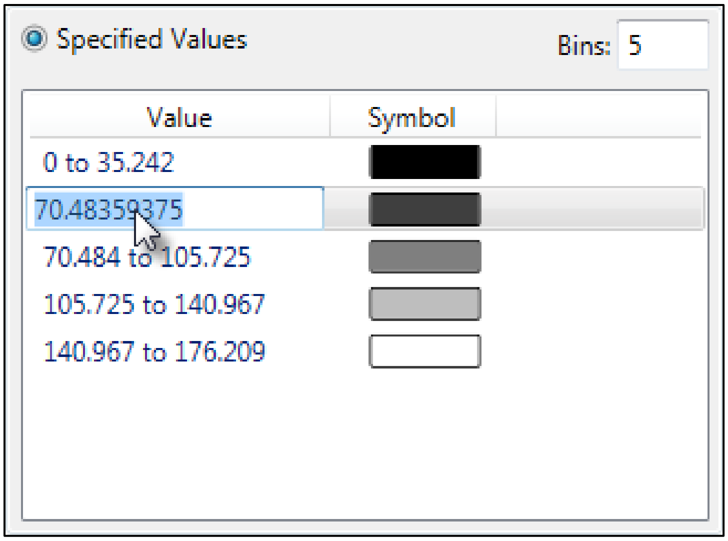


Figure A - 61. ***Map Layer*** **Properties Dialog Box – Gridded Map Layer – Specified Values**

2. For **Specified Values**, the user needs to enter the number of bins. The default number of bins is five. Values entered in the **Bins** box (Figure A - 61) cannot be less than one or greater than fifty. When the number of bins is entered, the **Value** and **Symbol** table (Figure A - 61) is modified with a range of quantitative values (set at equal numbers) based on the number of bins specified and the total range of values in the map layer.

3. Individual bin ranges can be adjusted in the **Value**/**Symbol** table (Figure A - 61) by entering the maximum value for each range, manually. To enter the maximum value manually, double-click a **Value/Symbol** row, the row is now editable. Then over-write the current value and the range will update (maximum value for the over-written value and the minimum value for the next row in the **Value**/**Symbol** table).

4. The default color ramp is black fading to white, the user can change this from the **Color Ramp** list (Figure A - 59). To determine if the color ramp scheme correct, click **Apply** (Figure A - 58) and view the updated gridded map layer in the map window (Figure A - 1). Once **Apply** is clicked this change cannot be undone.

5. If the **Color Ramp** selection turns not to be correct (e.g., higher water velocity with green colors and lower flow velocity with red colors) then select **Invert Colors** (Figure A - 58). Click **Apply**, and view the updated gridded map layer in the map window (Figure A - 1) to confirm the color scheme selection.

6. To adjust the color for a range of values, in the **Value** and **Symbol** table (Figure A - 61) click on a specific color in the table. The **Select a Color** dialog box will open (Figure A - 46).

   For details on the **Select a Color** dialog box see Section A.8.1. For example, in Figure A - 46, the **Select a Color** dialog box has a large checkered box, and other items that are similar to the *Map Layer* **Properties** dialog box (Figure A - 43). For the default **Line Color** setting the **Opacity** is set to maximum and that is why the **Selected Color** is empty (transparent).

7. Modify **Selected Color** in the **Select a Color** dialog box (Figure A - 46) until the desired color is achieved. Click **OK**, the **Select a Color** dialog box will close. The user will be returned to the *Map Layer* **Properties** dialog box (Figure A - 58) where the new color will be displayed in the **Value** and **Symbol** table (Figure A - 61).

8. Repeat Steps 6 through 7 for all of the values listed in the **Values**/**Symbol** table (Figure A - 61), which need to be changed.

9. The transparency of the entire gridded/raster map layer can be changed (from the default value of fully opaque). In the **Transparency** box (Figure A - 58), the user can enter a value from zero to 100 (percent) to increase the transparency of the map layer (the maximum value of 100% is maximum translucency).

10. Once editing of the symbology (type, color or color ramp, and other options) for a gridded map layer is complete, click **Apply**, selected gridded map layer is updated.

A.9.2 Source Information Tab

The **Source Information** tab for gridded map layers (similar to Figure A - 57) provides the:

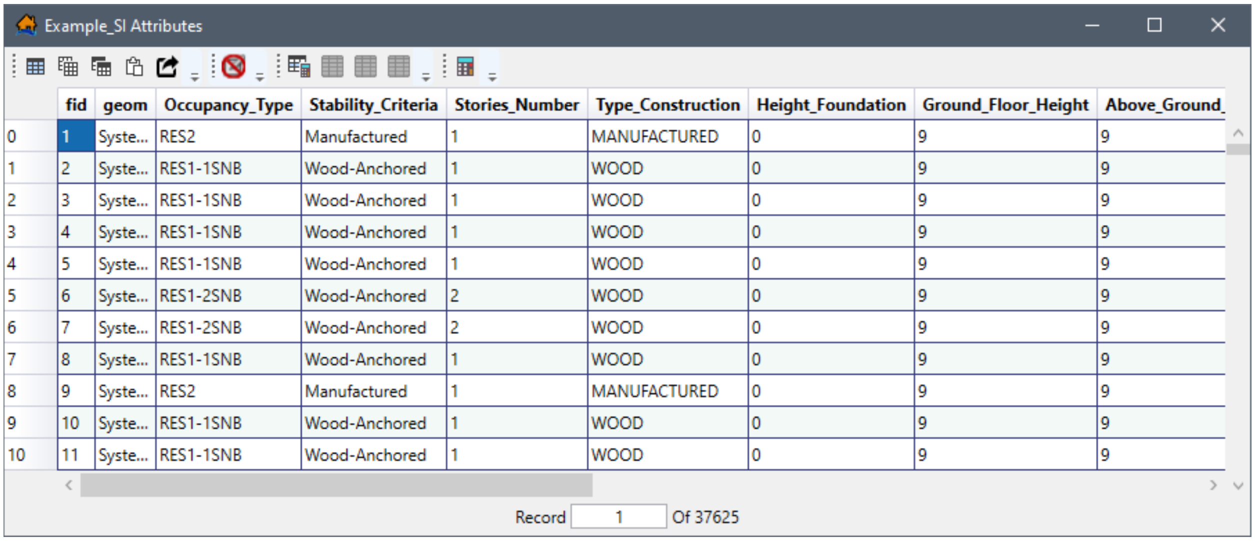
- **Projection** coordinate system associated with the map layer,
- **Horizontal Units** of the map layer, and
- **Extent** of the map layer.

A.10 Editing Map Layer Features and Attributes

As explained in Section A.1.1, the map window provides several editing tools in the **Editor Toolbar**. Additional map layer editing can be accomplished from the *Map Layer* **Attribute** dialog box (Figure A - 62). The editing capabilities of the map window are described in Section A.10.2 for newly created shapefiles, Section A.11 for point map layers, and Section A.12 for polyline and polygon map layers. This section describes editing map layers from the attributes table.

The map window (Figure A - 1) and *Map Layer* **Attribute** dialog box (Figure A - 62) editing tools are designed to facilitate in adjusting the geometric and attribute data directly within the software to increase overall productivity. For example, moving structures or editing roads for use in the simulation can be done quickly and easily from the LifeSim software interface.

It is important to note, that although more than one attribute table can be open at one time, only one map layer can be edited at a time. Also, it is beneficial to save often while editing a map layer.



	fid	geom	Occupancy_Type	Stability_Criteria	Stories_Number	Type_Construction	Height_Foundation	Ground_Floor_Height	Above_Ground_
0	1	Syste...	RES2	Manufactured	1	MANUFACTURED	0	9	9
1	2	Syste...	RES1-1SNB	Wood-Anchored	1	WOOD	0	9	9
2	3	Syste...	RES1-1SNB	Wood-Anchored	1	WOOD	0	9	9
3	4	Syste...	RES1-1SNB	Wood-Anchored	1	WOOD	0	9	9
4	5	Syste...	RES1-1SNB	Wood-Anchored	1	WOOD	0	9	9
5	6	Syste...	RES1-2SNB	Wood-Anchored	2	WOOD	0	9	9
6	7	Syste...	RES1-2SNB	Wood-Anchored	2	WOOD	0	9	9
7	8	Syste...	RES1-1SNB	Wood-Anchored	1	WOOD	0	9	9
8	9	Syste...	RES2	Manufactured	1	MANUFACTURED	0	9	9
9	10	Syste...	RES1-1SNB	Wood-Anchored	1	WOOD	0	9	9
10	11	Syste...	RES1-1SNB	Wood-Anchored	1	WOOD	0	9	9

Figure A - 62. ***Map Layer*** **Attribute Dialog Box – Active Editing Session**

A.10.1 Editing in the Attribute Table

Each map layer (shapefile) has an attribute table, which displays the dataset for the selected map layer. From the *Map Layer* **Attribute** dialog box (Figure A - 62), there are editing tools that allow the user to edit the attributes.

- From the **Map Layers** tab on the LifeSim main window (Figure A - 1), right-click on the map layer of interest, from the shortcut menu (Figure A - 2), click **Open Attribute Table**. The *Map Layer* **Attributes** dialog box (Figure A - 62) will open for the selected map layer. For non-editing attribute table functionality refer to Section A.4.

- Note, when a single cell in the attribute table (Figure A - 62) is selected (e.g., row 1, column 1), the keyboard arrow keys (and the tab and enter keys) can be utilized to step through the table one cell at a time (Figure A - 62).

Attribute Editing Tools

Open Field Calculator Tool
This tool allows the user to create a new field or update an existing field with a calculated result based on a specified expression. The **Open Field Calculator** tool is only enabled if feature editing has been enabled.

Start Feature Edit Session Tool

Start Feature Edit Session will start a feature editing session which allows the user to edit cells and add/delete columns in the attribute table, as well as edit the feature data. Once an editing session has been started the **Start Feature Edit Session** will be replaced with **Stop Feature Edit Session**.

Stop Feature Edit Session Tool

Stop Feature Edit Session will end an active editing session of the attribute table. Once an editing session has been stopped the **Stop Feature Edit Session** will be replaced with **Start Feature Edit Session**.

Editing Attribute Table Shortcut Menu

While editing mode is active, selected column(s) in the attribute table have several editing options. Right-click on a column, a shortcut menu will appear (Figure A - 63), from the shortcut menu the user can sort the column, calculate statistics, do a find for defined items, a field calculator is provided, and the selected column can be deleted.

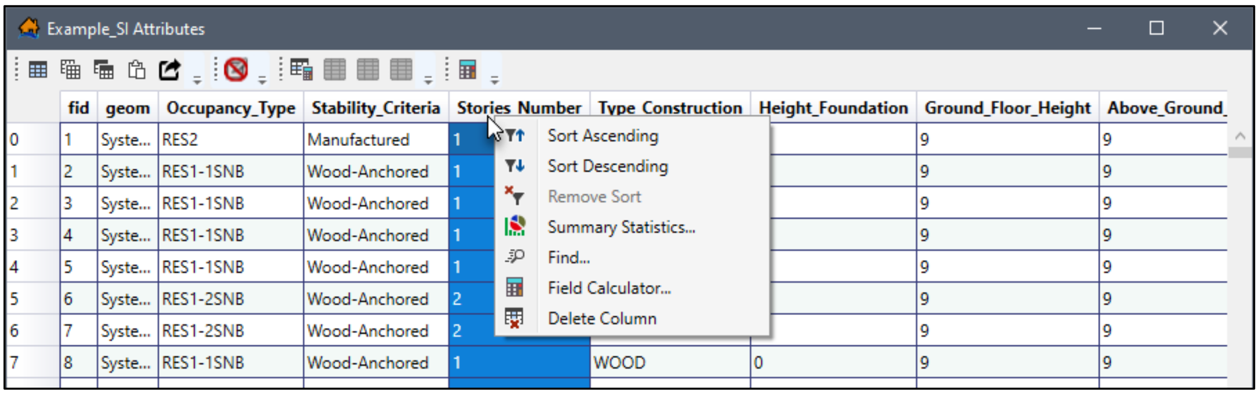


Figure A - 63. Attribute Table – Column Shortcut Menu – Active Editing Session

Other options which become available in edit mode are the standard cut/copy/paste functionality. Information in a selected cell can be edited or replaced by typing directly into the selected cell. However, depending on the column type: **String** (e.g., "Masonry"), **Integer** (e.g., 10), or **Double** (e.g., 0.5); only like types can be entered. In other words, only integers can be entered into integer cell(s) or column(s), with the exception of string columns which can contain integers or doubles. To identify the type of data a column can contain, hover over the column header (Figure A - 64) for the tooltip that contains the type.

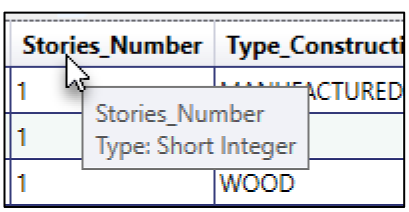


Figure A - 64. Attribute Table – Column Tooltips

Furthermore, depending on the character limit an entered value could be truncated by limits. For example, if *ManufacturedLogCabin* is entered into cell that has a character limit of 12 characters, then after **Stop Feature Edit Session** is clicked, then the extra characters beyond the character limit will automatically be removed and only *Manufactured* will remain.

Field Calculator This menu command functions similarly to **Open Field Calculator**, the difference is any action will be done on the selected column and new fields cannot be created.

Delete Column Deletes the selected column.

Attribute Field Calculator

The **Field Calculator** (Figure A - 65) can be utilized to add new calculated fields or update existing fields in the selected column during an active editing session.

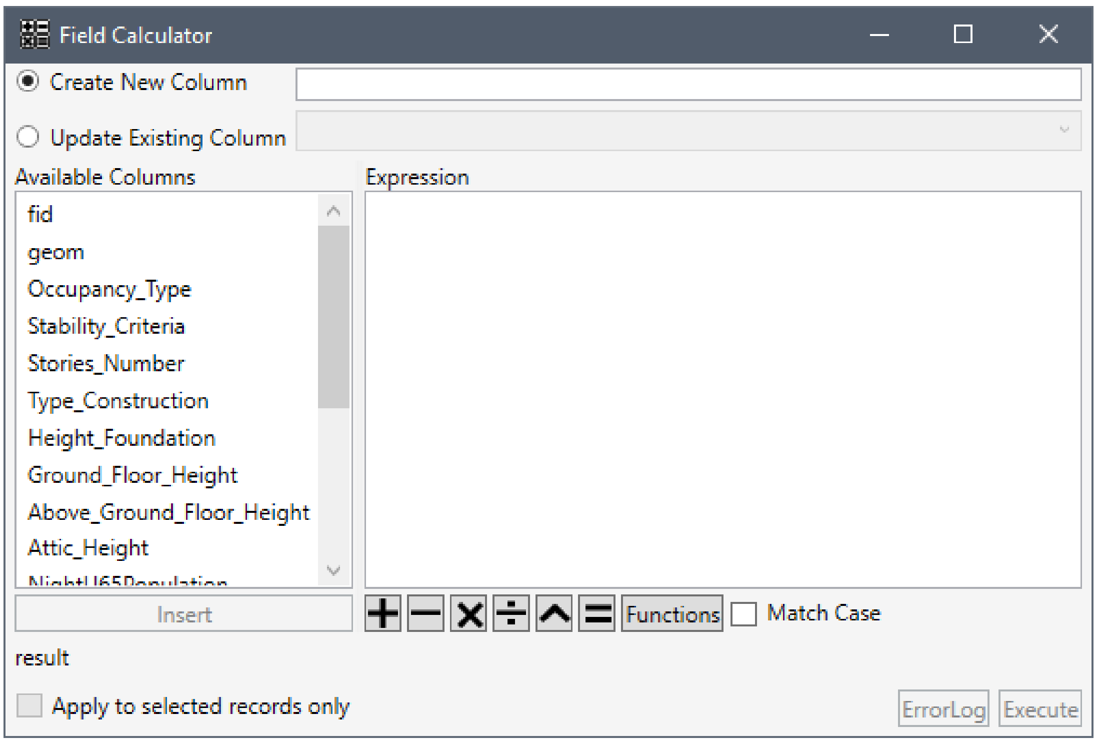


Figure A - 65. Field Calculator Dialog Box

To create a new field:

1. From the **Map Layers** tab on the LifeSim main window (Figure A - 1), right-click on the map layer of interest, from the shortcut menu (Figure A - 2), click **Open Attribute Table**. The *Map Layer* **Attributes** dialog box will open (Figure A - 62).

2. Start an editing session by clicking **Start Feature Edit Session**. Alternatively, users can activate an editing session by right-clicking on the map layer of interest, from the shortcut menu (Figure A - 2), click **Edit**. The user can now edit the map layer of interest, and only one map layer/map attribute table can be edited at one time.

3. From the *Map Layer* **Attributes** dialog box (Figure A - 62) click **Open Field Calculator**, the **Field Calculator** dialog box will open (Figure A - 65).

4. To add a new calculated field for the selected map layer, select **Create New Column** (Figure A - 66) and enter a name for the new column (has a limit of ten characters).

5. For the new field create an expression in the **Expression** box (Figure A - 66). For example, in Figure A - 66, a new field *FoundHtNew* has been created. To create the expression for the new field based on an existing field, double-click on the existing field of interest from the **Available Columns** box (e.g., *Height_Foundation* in Figure A - 66) to add the field name to the **Expression** box. The **Functions** button (Figure A - 66) opens the **Functions** window (Figure A - 18) which provides commands and functions to use in creating the executable expression.

   An example **Expression** for creating the new column, *FoundHtNew*, is displayed in Figure A - 66, *[Height_Foundation]+0.2*. The result of the expression is displayed at the bottom of the **Field Calculator** dialog box (Figure A - 66) *Example: First data record (row id = 0) would be equal to '0.2'*.

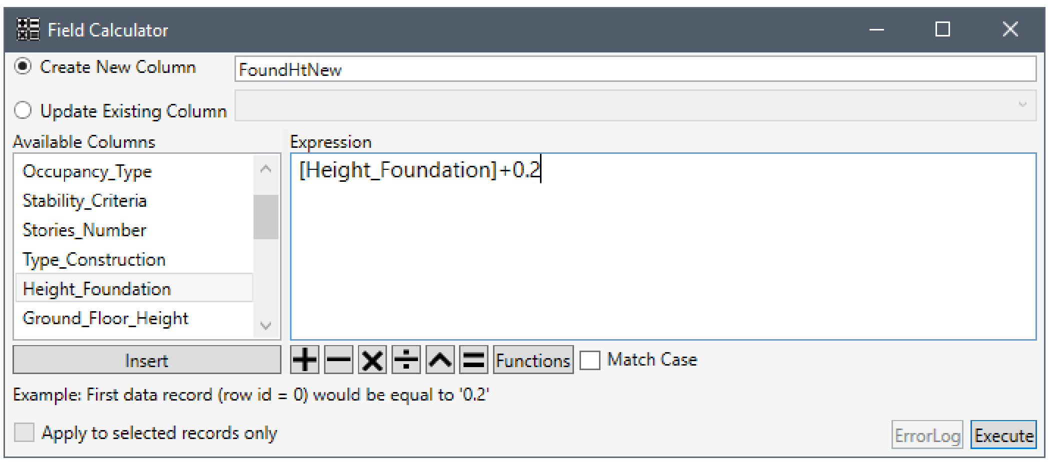


Figure A - 66. Field Calculator Dialog Box – Create New Column

   Note: when creating a new attribute field that is a string field (e.g., text, symbols, numbers, or any sequence of characters) then LifeSim requires that the text in the executable expression box be contained in double quotation marks.

6. **Match Case** (Figure A - 66) when selected, means that the entered expression will be case sensitive.

7. If the user selects **Apply to selected records only** (Figure A - 66), then the expression will only be applied to features selected in the *Map Layer* **Attributes** dialog box (Figure A - 62), and will not alter features that were not selected. The **Apply to selected records only** option will only be available for existing columns and will not be enabled at the time a new field is created. Also, if rows are not selected in the attribute table, then The **Apply to selected records only** option will be disabled.

8. Click **Execute**, the **Field Calculator** dialog box (Figure A - 65) close. The user will be returned to the *Map Layer* **Attributes** dialog box (Figure A - 62) where a newly calculated field will be present (e.g., *FoundHtNew*). New fields will be added to the end of the attribute table (Figure A - 62).

9. Once editing is complete, from the *Map Layer* **Attributes** dialog box (Figure A - 62), click **Stop Feature Edit Session**. Alternatively, from the **Map Layers** tab on the LifeSim main window (Figure A - 1), click **Save** from the **Editor Toolbar**.

Updating an existing field will overwrite the original data in the field, so caution must be taken when using this option. However, changes made to existing fields can be undone by clicking the undo button from the Map Window's Editing Toolbar (Section A.1.1) . To edit an existing field:

1. From the **Map Layers** tab on the LifeSim main window (Figure A - 1), right-click on the map layer of interest, from the shortcut menu (Figure A - 2), click **Open Attribute Table**. The *Map Layer* **Attributes** dialog box will open (Figure A - 62).

2. Start an editing session by clicking **Start Feature Edit Session**. Alternatively, users can activate an editing session by right-clicking on the map layer of interest from the **Map Layers** tab on the LifeSim main window, from the shortcut menu (Figure A - 2), click **Edit**. The user can now edit the map layer of interest, and only one map layer/map attribute table can be edited at one time.

3. From the *Map Layer* **Attributes** dialog box (Figure A - 62) click **Open Field Calculator**, the **Field Calculator** dialog box will open (Figure A - 65).

4. To update an existing field for the open map layer, select **Update Existing Column** (Figure A - 67). From the **Update Existing Column** list (Figure A - 67), select the field that needs to be updated (e.g., *FoundHtNew* in Figure A - 67).

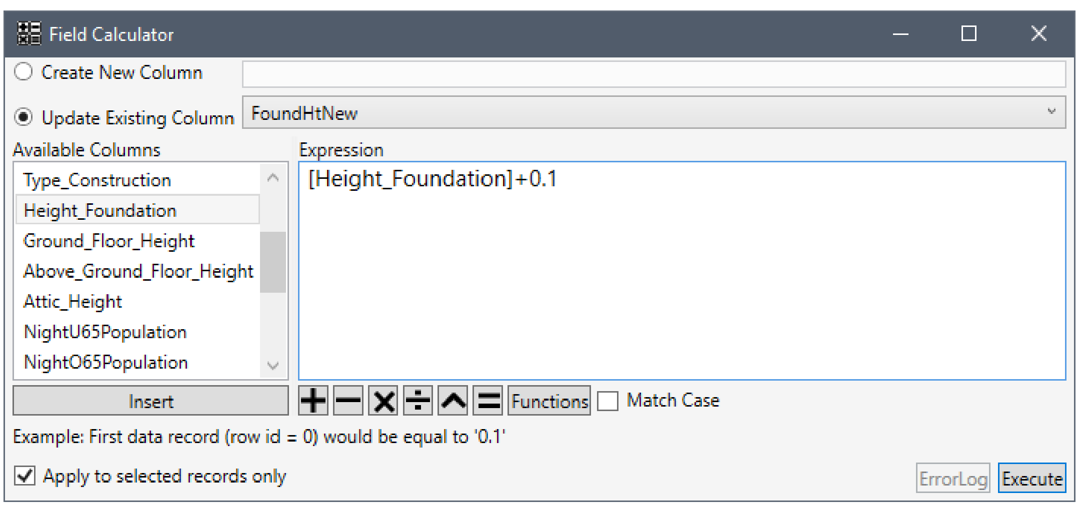


Figure A - 67. Field Calculator Dialog Box – Update Existing Column

5. Create an expression in the **Expression** box (Figure A - 67). To create the expression to update the existing field based on another existing field, double-click on the field of interest from the **Available Column** box (e.g., *Height_Foundation* in Figure A - 67) to add the field name to box below the **Available Column** box. The **Functions** button

(Figure A - 67) opens the **Functions** window (Figure A - 18) which provides commands and functions to use in creating the executable expression.

An example expression for the field *FoundHtNew*, is displayed in Figure A - 67, *[Height_Foundation]+0.1*. The result of the entered expression is shown at the bottom of the **Field Calculator** dialog box (Figure A - 67) *Example: First data record (row id = 0) would be equal to '0.1'.*

Note: when editing an existing column that is a string field (e.g., text, symbols, numbers, or any sequence of characters) then LifeSim requires that the text in the executable expression box be contained in double quotation marks.

6. **Match Case** (Figure A - 67) when selected, means that the expression will be case sensitive.

7. If the user selects **Apply to selected records only** (Figure A - 67), then the expression will only be applied to features selected in the *Map Layer* **Attributes** dialog box (Figure A - 62), and will not alter features that were not selected.

8. Click **Execute**, the **Field Calculator** dialog box (Figure A - 65) close. The user will be returned to the *Map Layer* **Attributes** dialog box (Figure A - 62) where the updated field will be present (e.g., *FoundHtNew*).

9. For the updated field, the user can click undo from the *Map Window Editor Toolbar* (Section A.1.1) to revert back to the original stated of the selected field. The **Redo** button will then become active and the user can redo the change made to the field.

10. Once adding a new field has been complete, from the *Map Layer* **Attributes** dialog box (Figure A - 62), click **Stop Feature Edit Session**. Alternatively, from the **Map Layers** tab on the LifeSim main window (Figure A - 1), click **Save** from the **Editor Toolbar**.

A.10.2 Adding Features to a New Shapefile

After creating a new shapefile (described in Section A.5) the user can add features by:

1. From the **Map Layers** tab on the LifeSim main window (Figure A - 1), right-click on the newly created map layer. From the shortcut menu (Figure A - 2), click **Edit**. The selected map layer is now in an active editing session and the **Editor Toolbar** is active (Figure A - 9). Review Section A.1.1 for an overview of the map window **Editor Toolbar** and Table A - 1 which provides the list of hot keys.

2. To add new features to the selected map layer, from the **Editor Toolbar**, click **Add new features** . Notice, in the map window the cursor will change to (note the dot in the center of the cursor). Click inside the map window and new points will be placed at the center of the cross-hairs (where the dot is located). Review Section A.11 for more information added or editing point features. Click to undo newly added features.

Polylines and polygons start at the first click and each subsequent mouse click will create a new vertex point in the polyline or polygon. To end the polyline or polygon double-click the mouse at the desired location. Review Section A.12 for more information adding or editing polyline or polygon features.

Note: When a new feature is added a new row is added to the bottom of the map layer's **Attribute** table with empty cells for text columns and zero's for numeric columns.

3. Once adding new features is complete, from the **Editor Toolbar** click **Save Edits**, and then click **Stop Editing**. If the editing session is ended before **Save Edits** has been clicked, a **Save Map Edits** message window will open (Figure A - 68) asking the user if the edits need to be saved. Click **Yes** to save the edits made to the map layer, and the **Save Map Edits** message window will close (Figure A - 68). If the user did not intend to exit the editing session, click **Cancel** to return to the editing session.

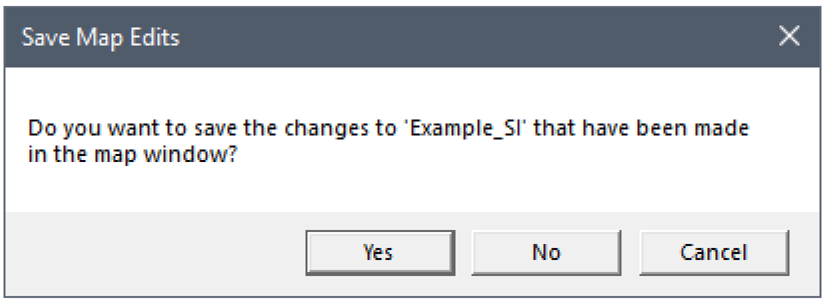


Figure A - 68. Save Map Edits Message Window

A.11 Editing Point Map Layers

The user can remove, move, or add points in newly created map layers (described in Section A.5) or existing point map layers.

A.11.1 Add New Point Features

To add new point features:

1. From the **Map Layers** tab on the LifeSim main window (Figure A - 1), right-click on the point map layer (newly created or existing). From the shortcut menu (Figure A - 2), click **Edit**. The selected map layer is now in an active editing session and the **Editor Toolbar** is active.

2. To add new point features to the selected map layer, from the **Editor Toolbar**, click **Add new features Map Window**. Notice, in the map window the cursor will change to ⊞ (note the dot in the center of the cursor). Click inside the map window and new points will be placed at the center of the cross-hairs (or wherever the dot is located; Figure A - 69a).

   With snapping enabled (refer to Section A.13.1 for more information), when the cursor gets close to the snap to feature map layer the dot in the middle of the cursor will "snap" to the snap to feature and new points will be placed at the snap point. Click at the desired location and a point will be added with default properties (Figure A - 69b).

3. Continue clicking in the map window to add new points at desired locations.

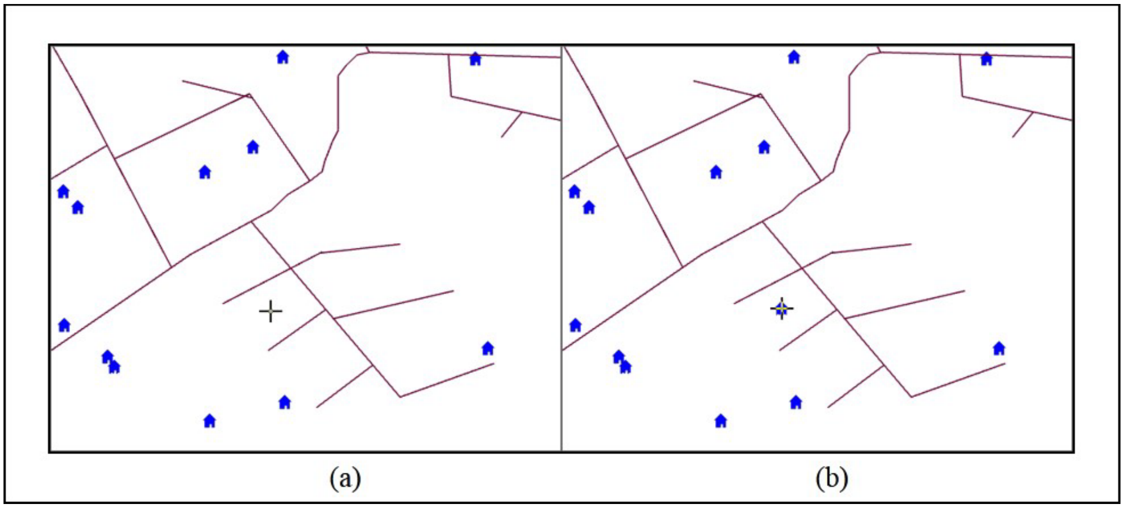


Figure A - 69. Adding New Point Features

4. Once the user is done adding new point features, from the **Editor Toolbar** click **Save Edits**, and then click **Stop Editing**. If the editing session is ended before **Save Edits** has been clicked, a **Save Map Edits** message window will open (Figure A - 68) asking the user if the edits need to be saved. Click **Yes** to save the edits made to the map layer, and the **Save Map Edits** message window will close. If the user did not intend to exit the editing session, click **Cancel** to return to the editing session.

A.11.2 Move Point Features

To move an already existing point (or points) in the map window:

1. From the **Map Layers** tab on the LifeSim main window (Figure A - 1), right-click on a map layer. From the shortcut menu (Figure A - 2), click **Edit**. The selected map layer is now in an active editing session and the **Editor Toolbar** is active.

2. From the **General Toolbar**, click the **Select** tool; to select features click on the feature or click, hold, and drag a rectangle around the features to select them in the map window (Figure A - 70A and B).

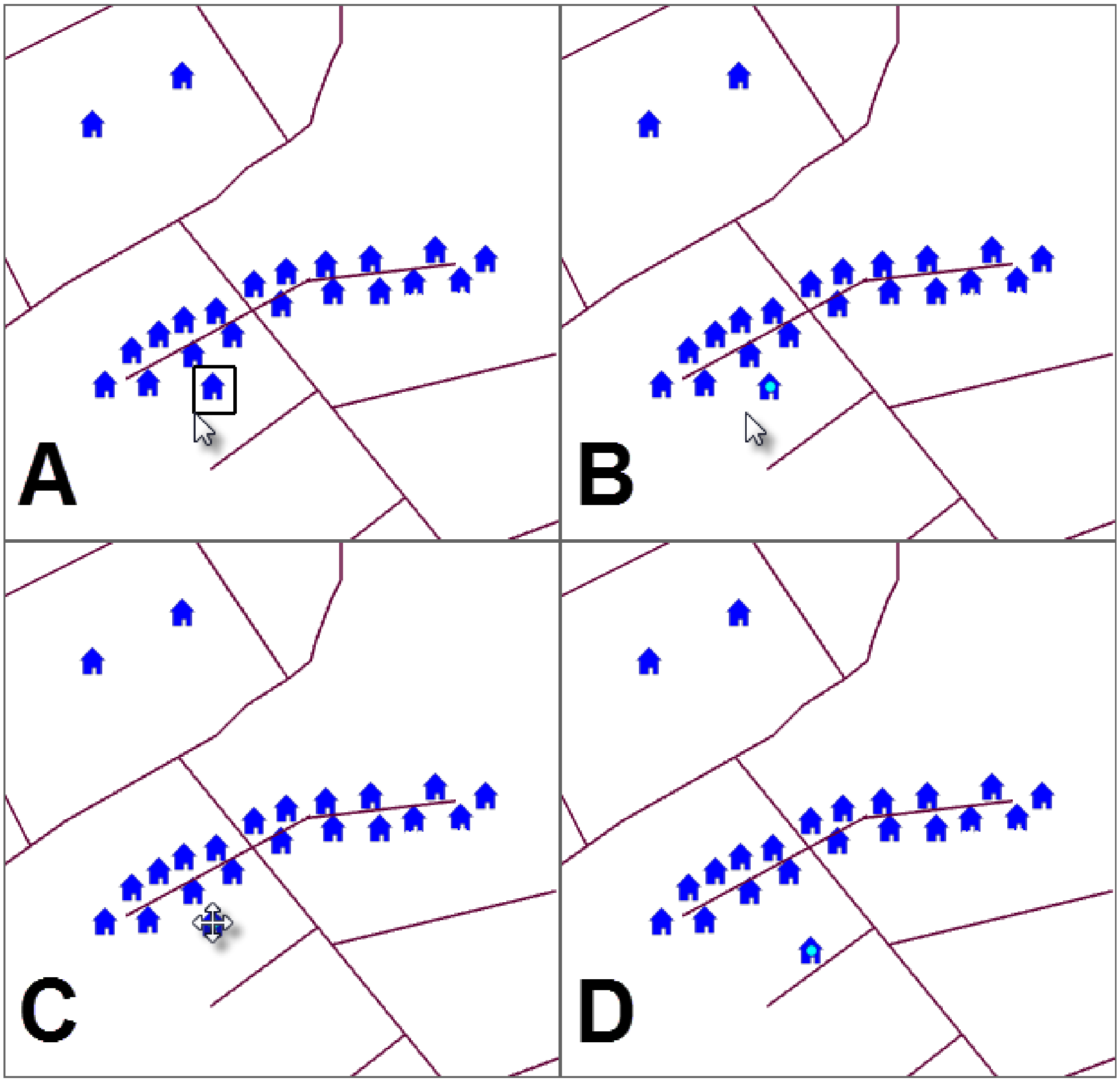


Figure A - 70. Select and Move Points

Note, if more than one map layer is displayed in the map window and the user only wants one map layer to be selected (e.g., map layer being edited) use the **Selectable Layers** list to set the selectable layers as explained in Section A.1.1.

3. With the point selected, click and drag the point to the desired location (Figure A - 70, C and D). The user can move also multiple points.

4. Once the user is done moving point features, from the **Editor Toolbar** click **Save Edits**, and then click **Stop Editing**. If the editing session is ended before **Save Edits** has been clicked, a **Save Map Edits** message window will open (Figure A - 68) asking the user if the edits need to be saved. Click **Yes** to save the edits made to the map layer, and the **Save Map Edits** message window will close. If the user did not intend to exit the editing session, click **Cancel** to return to the editing session.

A.11.3 Delete Point Features

To remove an already existing point (or points) in the map window:

1. From the **Map Layers** tab on the LifeSim main window (Figure A - 1), right-click on the map layer that represents the new shapefile that has been created. From the shortcut menu (Figure A - 2), click **Edit**. The selected map layer is now in an active editing session and the **Editor Toolbar** is active.

2. From the **General Toolbar**, click the **Select** tool; select the point(s) to remove (Figure A - 70A). The point(s) are highlighted (Figure A - 70B) in the map window, use the **Delete** key to delete the selected point feature(s). Another option for deleting the selected points, is right-click in the map window, from the shortcut menu (Figure A - 71) click **Delete Selected Features**. Either way the selected features are deleted from the selected map layer.

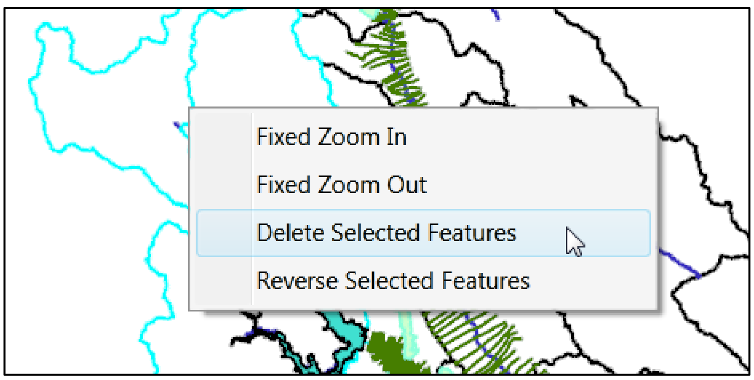


Figure A - 71. Map Window – Selected Feature - Shortcut Menu

3. Once the user is done deleting point features, from the **Editor Toolbar** click **Save Edits**, and then click **Stop Editing**. If the editing session is ended before **Save Edits** has been clicked, a **Save Map Edits** message window will open (Figure A - 68) asking the user if the edits need to be saved. Click **Yes** to save the edits made to the map layer, and the **Save Map Edits** message window will close. If the user did not intend to exit the editing session, click **Cancel** to return to the editing session.

A.12 Editing Polyline or Polygon Map Layers

The user can remove, move, or add vertices in polyline or polygon map layers (shapefiles).

A.12.1 Add Vertex Points

To add new vertex points:

1. From the **Map Layers** tab on the LifeSim main window (Figure A - 1), right-click on a map layer. From the shortcut menu (Figure A - 2), click **Edit**. The selected map layer is now in an active editing session and the **Editor Toolbar** is active.

2. From the **General Toolbar**, click the **Select** tool; double-click on a specific polyline or polygon feature in the selected map layer. Note, if more than one map layer is displayed in the map window and the user only wants one map layer to be selected use the **Selectable Layers** list (Section A.1.1) to set the map layer to be edited.

3. When the user double-clicks on a polyline or polygon feature, the vertex points (black squares) will display and the user will be in the vertex edit mode. At this point depending on the feature (polyline or polygon) several map window editing tools will become available in the **Editor Toolbar** (Figure A - 72).

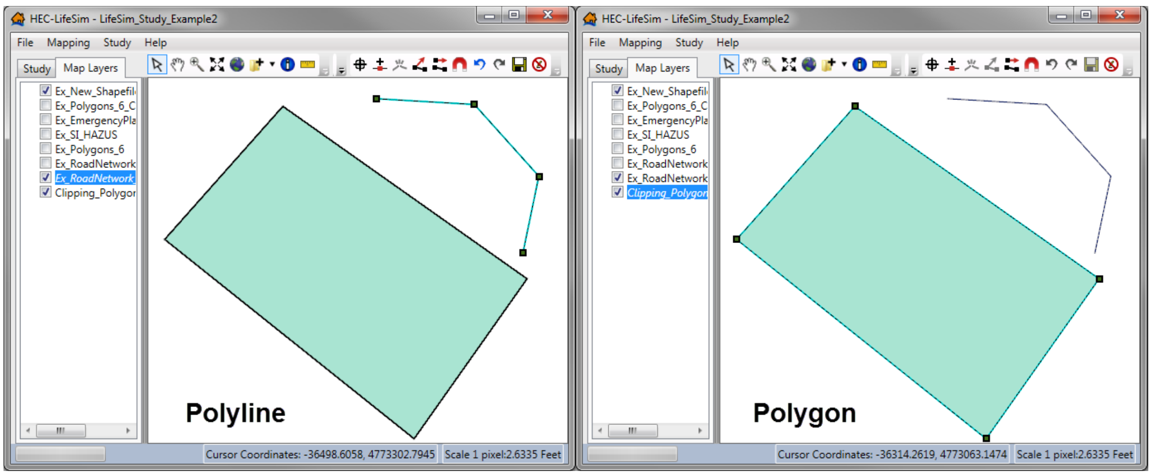


Figure A - 72. Vertex Editing Mode – Polyline or Polygon Features

4. From the **Editor Toolbar**, click **Insert Vertex** (±), locate on the map window where the user want to add a new vertex point, click the location, and a new vertex point will be added. Alternatively, the user can add a new vertex in the vertex edit mode, by right-clicking on the location where a vertex point is to be added, from the shortcut menu (Figure A - 73), click **Insert Vertex**, and a new vertex point will be added.

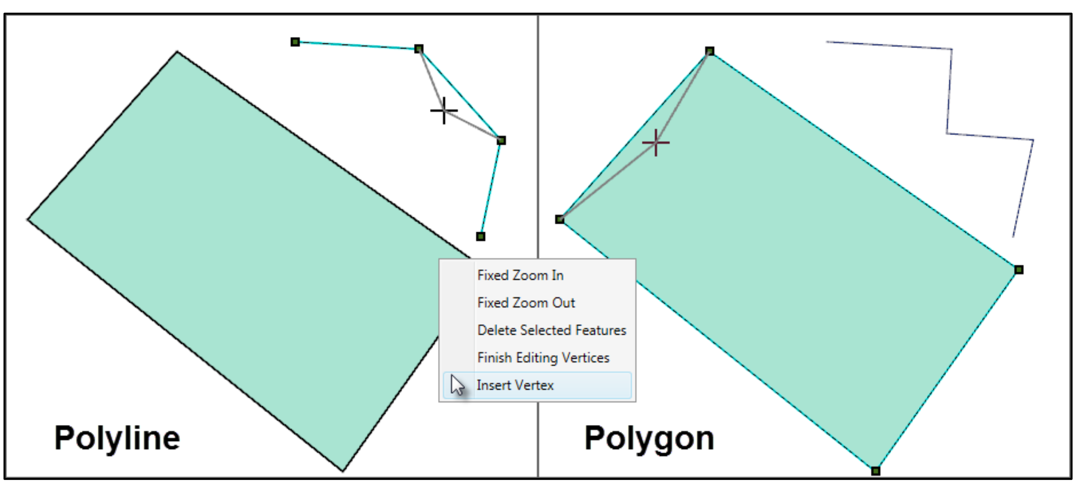


Figure A - 73. Vertex Edit Mode Shortcut Menu

5. Once the user is done adding vertex points, from the **Editor Toolbar** click **Save Edits**, and then click **Stop Editing**. If the editing session is ended before **Save Edits** has been

clicked, a **Save Map Edits** message window will open (Figure A - 68) asking the user if the edits need to be saved. Click **Yes** to save the edits made to the map layer, and the **Save Map Edits** message window will close. If the user did not intend to exit the editing session, click **Cancel** to return to the editing session.

A.12.2 Delete Vertex Points

To delete vertex points:

1. From the **Map Layers** tab on the LifeSim main window (Figure A - 1), right-click on a map layer. From the shortcut menu (Figure A - 2), click **Edit**. The selected map layer is now in an active editing session and the **Editor Toolbar** is active.

2. From the **General Toolbar**, click the **Select** tool; double-click on a specific polyline or polygon feature in the selected map layer. **Note**, if more than one map layer is displayed in the map window and the user only wants one map layer to be selected use the **Selectable Layers** list (Section A.1.1) to set the map layer to be edited.

3. To remove a vertex or vertices points, select the vertex or vertices points to be deleted, click the **Delete** key. Another way, is right-click on the vertex point that is to be deleted, from the shortcut menu (Figure A - 73), click **Delete Selected Features**. Either way, the vertex point is deleted.

4. Once the user is done deleting vertex points, from the **Editor Toolbar** click **Save Edits**, and then click **Stop Editing**. If the editing session is ended before **Save Edits** has been clicked, a **Save Map Edits** message window will open (Figure A - 68) asking the user if the edits need to be saved. Click **Yes** to save the edits made to the map layer, and the **Save Map Edits** message window will close. If the user did not intend to exit the editing session, click **Cancel** to return to the editing session.

A.12.3 Edit a Line Feature – Shorten

The user can shorten the length of line features in newly created (described in Section A.5) or existing polyline map layer by:

1. From the **Map Layers** tab on the LifeSim main window (Figure A - 1), right-click on a map layer. From the shortcut menu (Figure A - 2), click **Edit**. The selected map layer is now in an active editing session and the **Editor Toolbar** is active.

2. From the **General Toolbar**, click the **Select** tool; click on a specific line feature in the selected map layer. Note, if more than one map layer is displayed in the map window and the user only wants one map layer to be selected use the **Selectable Layers** list (Section A.1.1) to set the map layer to be edited.

3. Once the line feature to be broken is selected, from the **Editor Toolbar**, select **Break line feature** , the curser will change to ⊞ (note, the dot which follows the cursor along the selected line feature). **Note**, if more than one map layer is displayed in the map

window and the user only wants one map layer to be selected use the **Selectable Layers** list (Section A.1.1) to set the map layer to be edited.

4. Hover the cursor over the exact location to break the selected line feature, and a dot will appear on the line feature. Move the dot to the location where the line will be broken (i.e., at the edge of an Emergency Planning Zone polygon map layer). Figure A - 74A, provides an example of what the user might visualize.

5. Wherever the dot is located on the line feature, when the mouse is clicked, that is where the break in the line will occur (Figure A - 74B). As an example, the line section outside of a Emergency Planning Zone (EPZ) can be deleted if the user breaks the line at the edge of the EPZ, and then selects the line outside of the EPZ. Click the **Delete** key or right-click, from the shortcut menu (Figure A - 74C), click **Delete Selected Features**, which will delete the select line section (Figure A - 74D).

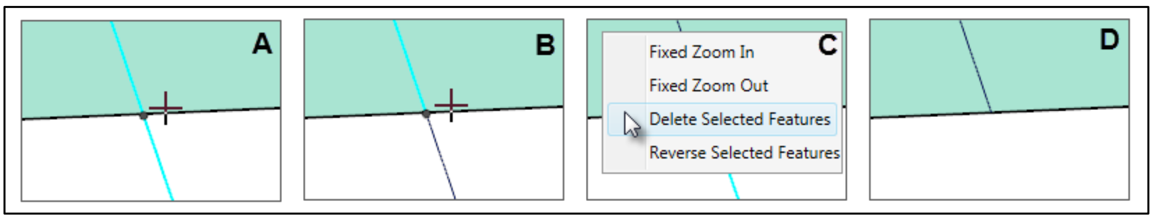


Figure A - 74. Break Line Feature Tool

6. Once the user is done modifying map layer features, from the **Editor Toolbar** click **Save Edits**, and then click **Stop Editing**. If the editing session is ended before **Save Edits** has been clicked, a **Save Map Edits** message window will open (Figure A - 68) asking the user if the edits need to be saved. Click **Yes** to save the edits made to the map layer, and the **Save Map Edits** message window will close. If the user did not intend to exit the editing session, click **Cancel** to return to the editing session.

A.12.4 Edit a Line Feature – Lengthen

The user can lengthen a line feature in newly created (described in Section A.5) or existing polyline map layer by:

1. From the **Map Layers** tab on the LifeSim main window (Figure A - 1), right-click on a map layer. From the shortcut menu (Figure A - 2), click **Edit**. The selected map layer is now in an active editing session and the **Editor Toolbar** is active.

2. From the **General Toolbar**, click the **Select** tool; click on a specific line feature in the selected map layer. Note, if more than one map layer is displayed in the map window and the user only wants one map layer to be selected use the **Selectable Layers** list (Section A.1.1) to set the map layer to be edited.

3. From the **Editor Toolbar**, click **Continue from end (lines only)**, the curser will change to , with a line connected to the end of the selected polyline feature (Figure A - 75). Click the cursor at the desired location to continue the line and double-click to end the

lengthened line. Note, the user can reverse the line from a shortcut menu, click **Reverse Selected Features** (Figure A - 75).

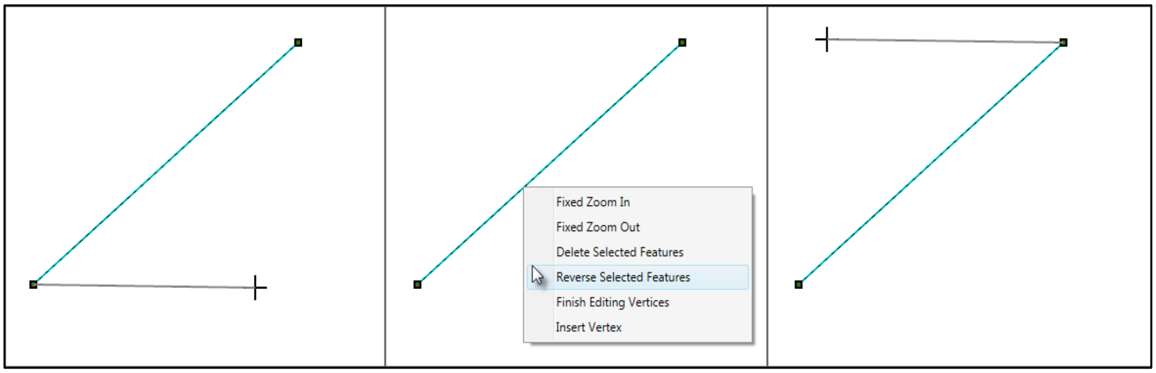


Figure A - 75. Lengthen a Line Feature

4. Once the user is done modifying map layer features, from the **Editor Toolbar** click **Save Edits**, and then click **Stop Editing**. If the editing session is ended before **Save Edits** has been clicked, a **Save Map Edits** message window will open (Figure A - 68) asking the user if the edits need to be saved. Click **Yes** to save the edits made to the map layer, and the **Save Map Edits** message window will close. If the user did not intend to exit the editing session, click **Cancel** to return to the editing session.

A.12.5 Adding New Features

Adding new features:

1. From the **Map Layers** tab on the LifeSim main window (Figure A - 1), right-click on a map layer. From the shortcut menu (Figure A - 2), click **Edit**. The selected map layer is now in an active editing session and the **Editor Toolbar** is active.

2. From the **General Toolbar**, click the **Select** tool; click on a specific line feature in the selected map layer. Note, use the **Selectable Layers** list (Section A.1.1) to set the map layer to be edited as the only selectable map layer.

3. From the **Editor Toolbar**, click **Add new line or polygon to existing feature** ( ), the curser will change to (note the dot in the center of the cursor). Click the desired location, where the user will begin drawing the new line or polygon to the existing selected feature. Each subsequent click of the cursor will create a new vertex point to the polyline or polygon (Figure A - 76). Double-click to complete the new addition to the existing feature.

4. Once the user is done modifying map layer features, from the **Editor Toolbar** click **Save Edits**, and then click **Stop Editing**. If the editing session is ended before **Save Edits** has been clicked, a **Save Map Edits** message window will open (Figure A - 68) asking the user if the edits need to be saved. Click **Yes** to save the edits made to the map layer, and the **Save Map Edits** message window will close. If the user did not intend to exit the editing session, click **Cancel** to return to the editing session.

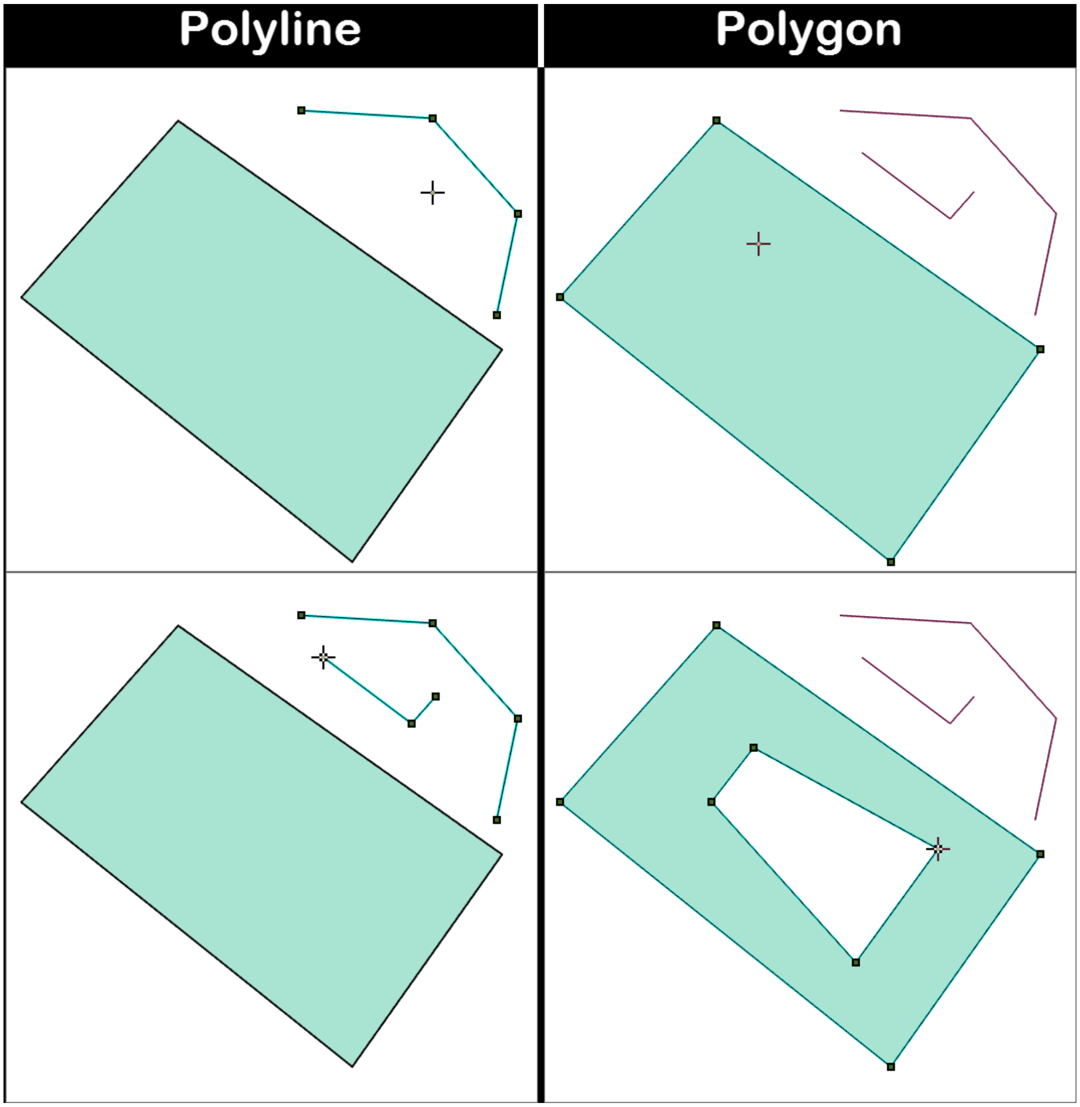


Figure A - 76. Add New Line or Polygon - Existing Feature Tool

A.13 Create Destination Locations Example

The following sections gives an example of creating a destination location shapefile for use in LifeSim. Destinations data is required in a LifeSim study to represent possible evacuation locations during flooding events. The destination data is represented as points in space, LifeSim will find the nearest road from a destination point and consider that the evacuation location. In other words, destination points do not need to connect with a road for LifeSim simulations to be successful.

A.13.1 Add Features to the Destination Locations Map Layer

After creating the destination shapefile (described in Section A.5) the user can add features with the **Feature Snapping Settings** tool from the **Editor Toolbar** by:

1. From the **Map Layers** tab on the LifeSim main window (Figure A - 1), right-click on a map layer (e.g., *Example_Destinations*). From the shortcut menu (Figure A - 2), click **Edit**. The selected map layer is now in an active editing session and the **Editor Toolbar** is active.

2. This step is optional as LifeSim will find the nearest road from a destination point and consider that the evacuation route for the destination point. From the **Editor Toolbar**, click **Feature Snapping Settings**, a **Snapping Settings** window will display (Figure A - 77). From the **Snap To Layer** list, the user will select a map layer to base the newly added feature points to snap to (e.g., *Example_Roads*).

    - If the **Snap To Feature** is a polyline map layer then all snapping options will be available. Also, the map layer selected (e.g., *Example_Roads*) needs to be active in the LifeSim study.

3. To add new features, from the **Editor Toolbar**, click (**Add new features Map Window**). Notice, in the map window the cursor will change to (there will be a dot in the center of the cursor). Click inside the map window and new points will be placed at the center of the cross-hairs (Figure A - 78 a).

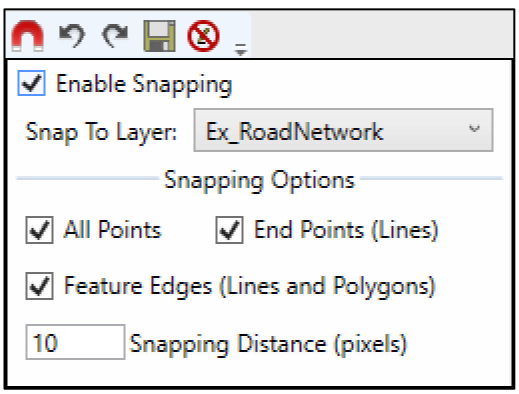


Figure A - 77. Snapping Settings Window

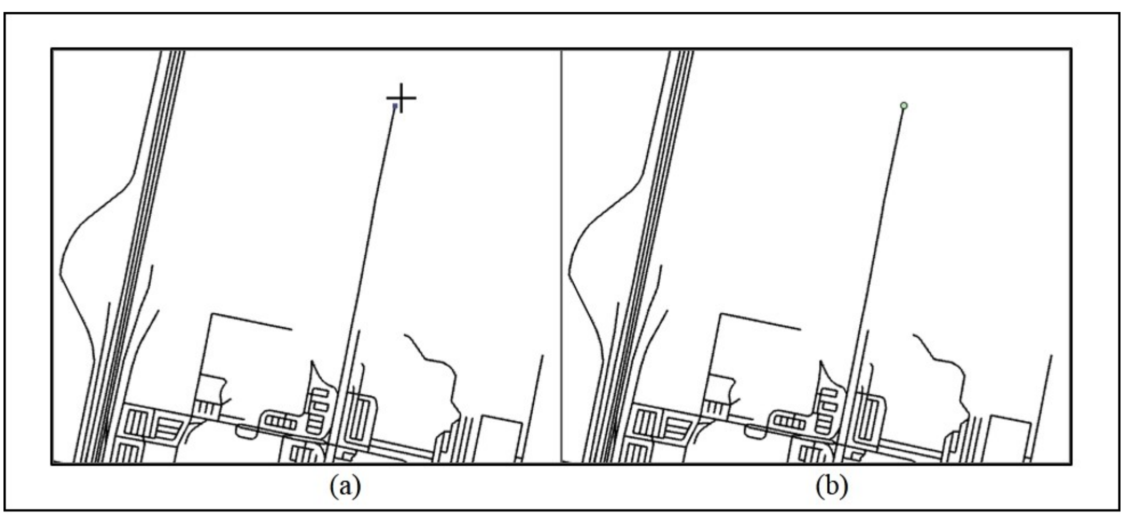


Figure A - 78. Add New Features Tool

- If snapping is enabled in Step 2 (if desired), then when the cursor gets close to the snap to feature map layer the dot in the middle of the cursor will "snap" to the snap to feature and new points will be placed at the snap point. Click at the desired location and a point will be added with default properties (Figure A - 78b).

4. Destinations typically are viewed in the map window with a star symbol. Next the properties for the newly created destinations map layer (e.g., *Example_Destinations*) should be modified from the default general point layer symbol. Alternatively, users can modify the map layer properties before adding any new points to the newly created map layer.

5. From the **Map Layers** tab on the LifeSim main window (Figure A - 1), right-click on the point map layer of interest (e.g., *Example_Destinations*), from the shortcut menu (Figure A - 2) click **Properties**, the *Map Layer* ***P*****roperties** dialog box will open (Figure A - 43).

6. To change the symbol type, from the *Map Layer* **Properties** dialog box (Figure A - 43), from the **Symbols** box (Figure A - 45), select a new symbol.

7. Change the **Size** of the symbol (e.g., size *11*) by entering the number in the **Size** box (Figure A - 43).

8. Change the fill color, click the box next to **Fill Color** (Figure A - 43), the **Select a Color** dialog box will open (Figure A - 44). From the **Select a Color** dialog box choose a color from the **Common Colors** box (Figure A - 46). Click **OK**, the **Select a Color** dialog box will close, and the user will be returned to the *Map Layer* **Properties** dialog box (Figure A - 43).

9. From the *Map Layer* **Properties** dialog box (Figure A - 43), click **Apply** button and check to see if the desired changes were made to the map layer in the map window (Figure A - 1). Once the user has entered all of the desired features, click **Close**, changes will be saved to the map layer (e.g., *Example_Destinations*) and the *Map Layer* **Properties** dialog box will close (Figure A - 43).

A.13.2 Add Attributes to the Destinations Locations Map Layer

After all of the necessary destination points have been added to the map layer (e.g., *Example_Destinations*), as described in Section A.13.1, the attributes for the new map layer must be entered by:

1. From the **Map Layers** tab on the LifeSim main window (Figure A - 1), right-click on a map layer (e.g., *Example_Destinations*). From the shortcut menu (Figure A - 2), click **Open Attribute Table**. The *Map Layer* **Attribute Table** dialog box will open (Figure A - 62) in an active editing session.

2. Click **(Open Field Calculator)**, the **Field Calculator** dialog box will open (Figure A - 65), which has two options: **Create New Column** or **Update Existing Column**. The **Create New Column** is selected by default.

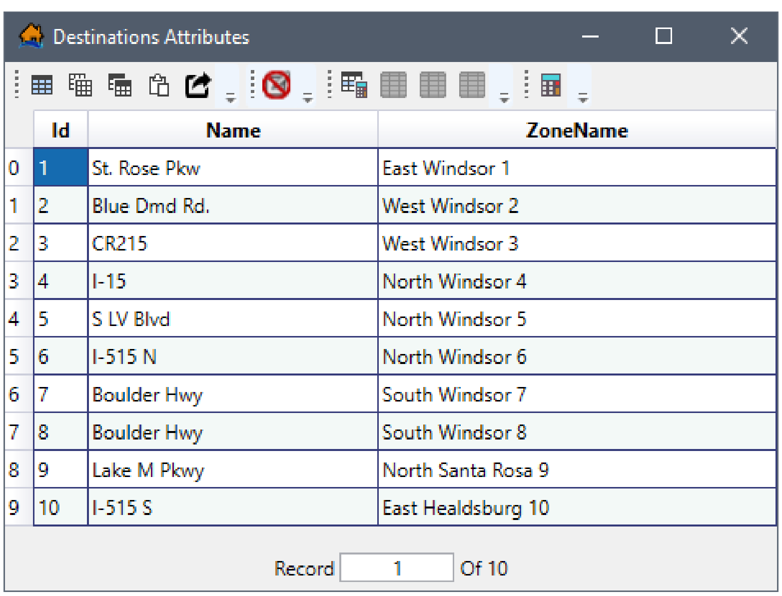
3. Enter a name (e.g., *Name*) in the **Create New Column** box (Figure A - 65), the attribute name is limited to ten characters.

4. When the expression for the new field is entered in the **Expression** box (Figure A - 66), and that entered expression is executed (**Execute** button is clicked), the new column is created and the **Field Type** for the column is set. Depending on the expression, the **Field Type** will be either: ***String*** (e.g., "Masonry"), ***Integer*** (e.g., 10), or ***Double*** (e.g., 0.5). Furthermore, the total number of characters entered in the **Expression** box (Figure A - 65) will set the character limit for the new column (e.g., "Masonry" will set the character limit for the new column to 7 characters).

   Note: when creating a new attribute field that is a string field (e.g., text, symbols, numbers, or any sequence of characters) then LifeSim requires that the text in the executable expression box be contained in double quotation marks.

5. Click **Execute** (Figure A - 65) to create the new field (e.g., *Name*). The **Field Calculator** dialog box (Figure A - 65) will close and the new field will be added to the **Attributes** table (Figure A - 62).

6. Now the names of the destinations need to be entered. One way to add names to destinations is by using the cardinal directions (Figure A - 79). Utilizing the cardinal directions for the study area to name destinations can be very useful for projects that contain multiple emergency planning zones. Also, using cardinal directions can be useful when completing the **Destination Assignments** tab of the **Alternative Editor** (Chapter 11) when creating alternatives.



	Id	Name	ZoneName
0	1	St. Rose Pkw	East Windsor 1
1	2	Blue Dmd Rd.	West Windsor 2
2	3	CR215	West Windsor 3
3	4	I-15	North Windsor 4
4	5	S LV Blvd	North Windsor 5
5	6	I-515 N	North Windsor 6
6	7	Boulder Hwy	South Windsor 7
7	8	Boulder Hwy	South Windsor 8
8	9	Lake M Pkwy	North Santa Rosa 9
9	10	I-515 S	East Healdsburg 10

Figure A - 79. Example Attributes Table – Destination Locations

7. Alternatively, the names of the destinations could be entered based on map layers in the map window (Figure A - 1). To use this option, select the first row of the table (Figure A - 79), right-click on the row identification number, from the shortcut menu, click **Zoom to Selected**. The map window (Figure A - 1) will zoom to the designated destination location.

   If an **OpenStreetMap** web layer has been added to the map window (Section A.3.2) then the **OpenStreetMap** web layer can be utilized to name the selected destination. For example, in Figure A - 79, the *Name_2* column, and from Figure A - 80, the *Name* column provide two examples for a completed table utilizing the **OpenStreetMap** web layer to name the created destinations.

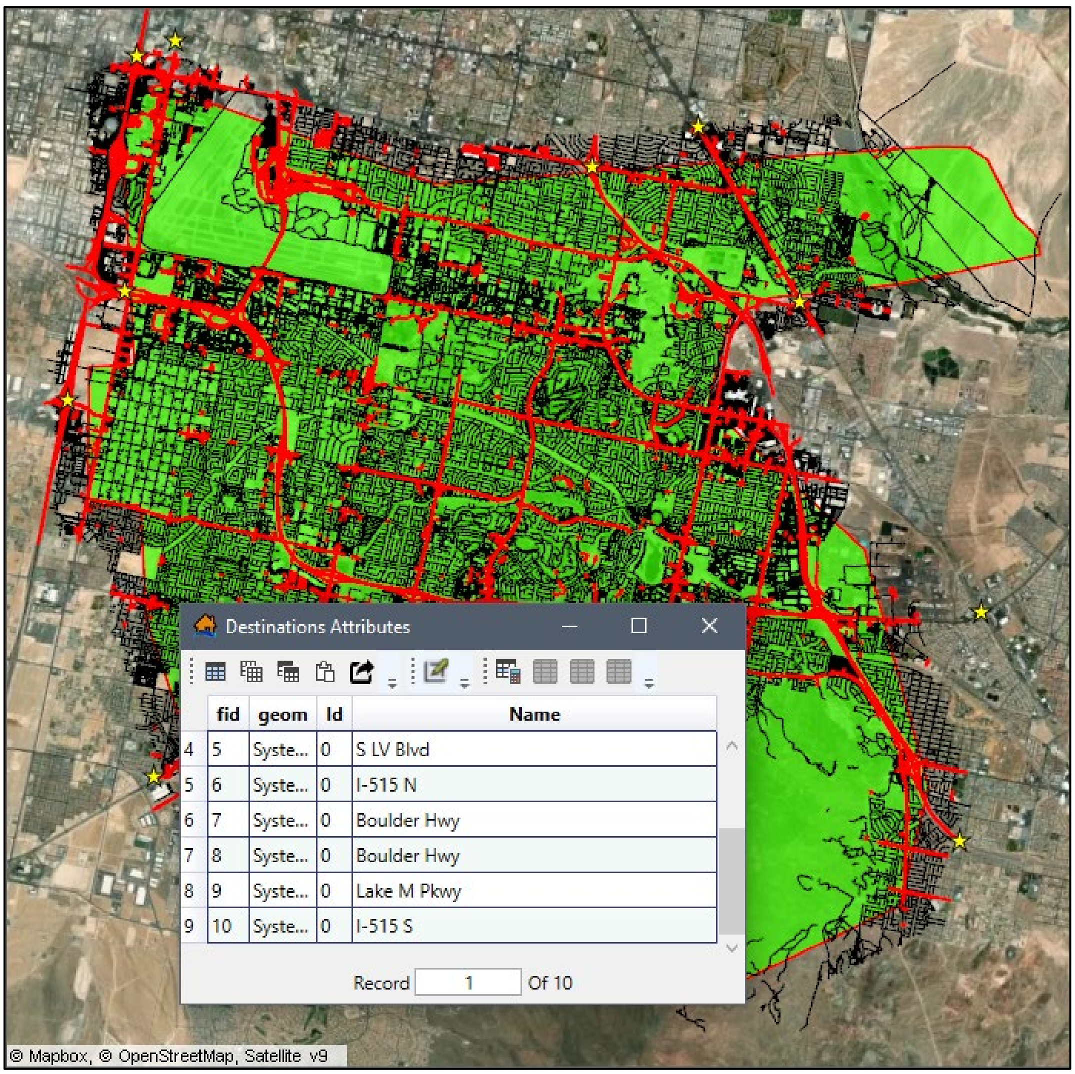


Figure A - 80. Completed Attribute Table – Map Window View

8. Manually type (or copy and paste) an appropriate name (e.g., *I-515 N*; *North Windsor 6)* in the newly created column for each destination that has been added (Figure A - 80).

9. Once the user is done adding new features, from the **Editor Toolbar** click **Save Edits**, and then click **Stop Editing**. If the editing session is ended before **Save Edits** has been clicked, a **Save Edits** message window will open (Figure A - 68) asking the user if the edits need to be saved. Click **Yes** to save the edits made to the map layer, and the **Save Edits** message window will close. If the user did not intend to exit the editing session, click **Cancel** to return to the editing session.

Appendix B

Editing Road Network Data for use in LifeSim

The road network is an important part of the LifeSim computations when simulating the evacuation process. A road network with missing connectivity or incorrect directions on one-way roads can lead to erroneous results. This appendix provides some methods for editing the road network dataset to provide a reliable foundation for traffic simulation. This appendix makes use of the editing functionality in the map window of the LifeSim main window (Figure B - 1). The map window functionality and a full list of the tools available can be found in Appendix A. The road network used in the following examples was obtained from OpenStreetMap (https://www.openstreetmap.org/). This appendix focuses on two aspects of road network improvement:

- Verifying network connectivity.
- Verifying one-way street direction.

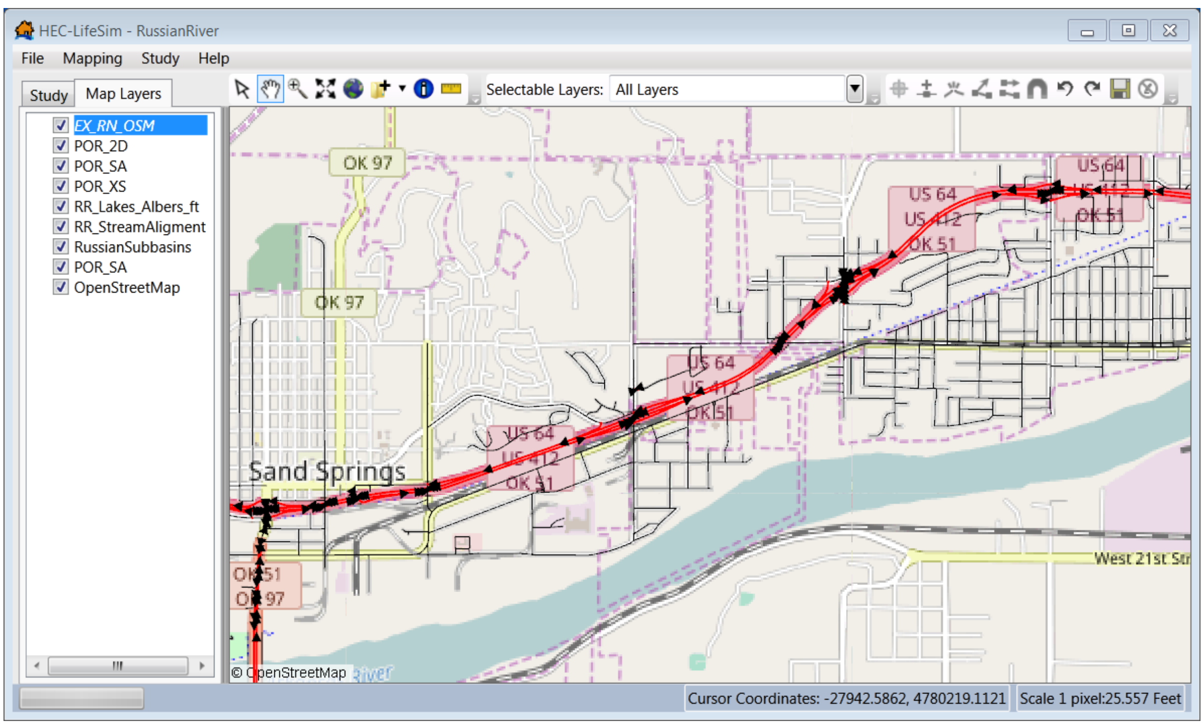


Figure B - 1. LifeSim Main Window – Map Window – Road Network Layer

B.1 Overview

The road network connectivity defines how vehicles can traverse from one road segment to the next road segment in LifeSim. Roads are considered connected if the individual road segments share an end point.

If two roads do not share an end point then a vehicle cannot move between the two roads. Conversely, if two roads share an end point a vehicle can move between the roads even if one is a bridge and the other is the road going underneath. The first step is to check for disconnect between road segments (see Section B.1.1 and Section B.1.2). Another step to verify the direction on one-way streets (refer to Section B.2).

B.1.1 Disconnect Invalid Road Segment Connections

Invalid road segment connections can occur several ways. For example, it is common for overpasses or underpasses to have roads connected at the overpass intersection. In LifeSim, this connection would imply that vehicles can jump from an overpass down onto a highway as if it were a standard intersection.

For example, in Figure B - 2, in an off-ramp that goes underneath a highway is displayed. The road segment that goes underneath the highway shares an endpoint with the highway. The invalid road segment connection would allow vehicles to immediately jump from the highway to the underpass rather than take the off ramp, which would affect simulation results (i.e., destination arrival times, evacuation times, evacuation outflow).

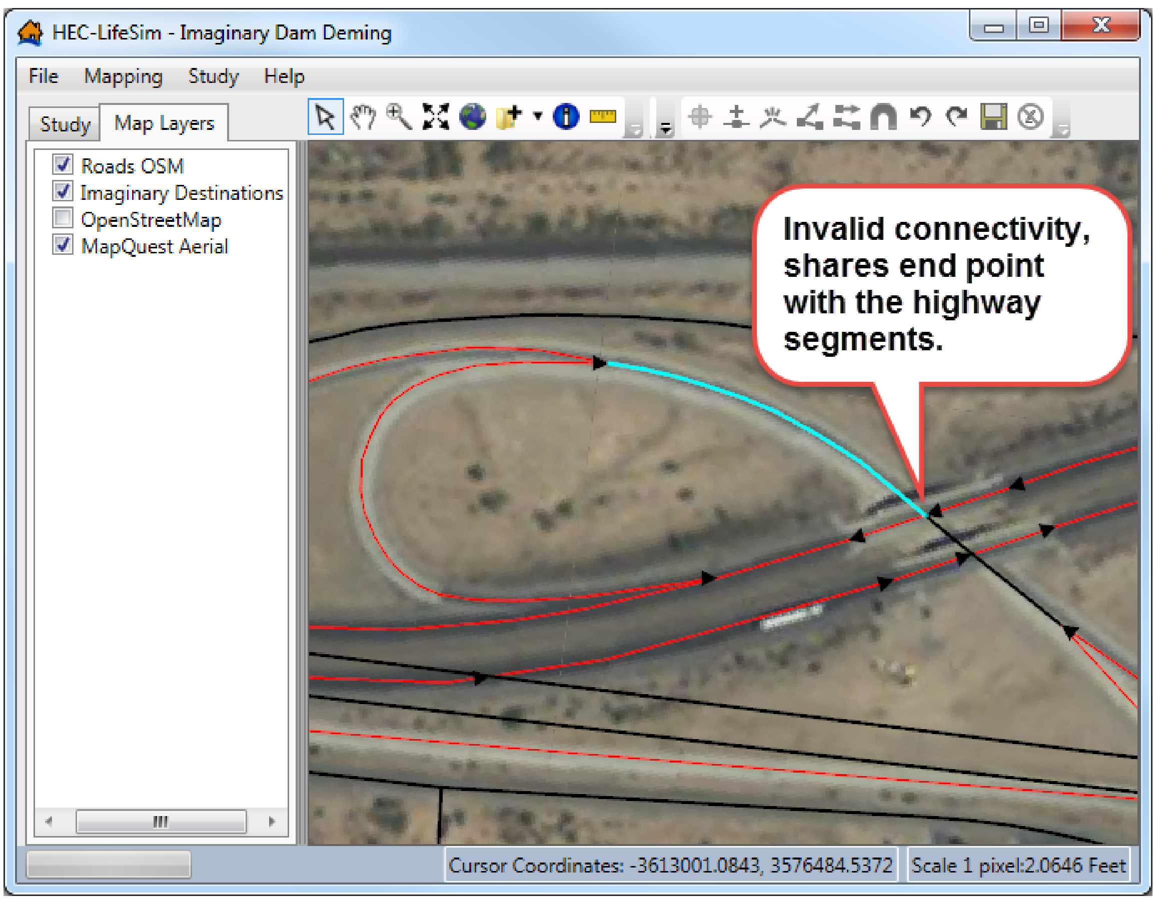


Figure B - 2. Road Network with Bad Connectivity

The following steps can be used to fix road network connectivity issues:

1. To add a road network layer to the map window, from the **Study** tab, from the **Road Networks** folder, right-click on a road network dataset. From the shortcut menu, click **Show In Map Window**, the select road network dataset will now display in the map window of the LifeSim main window (Figure B - 1).

2. To start an edit session, from the **Map Layers** tab (Figure B - 2), right-click a road network layer, from the shortcut menu click **Edit**. Another way to start an edit session is from the **Map Layers** tab (Figure B - 2), right-click on a road network layer, from the shortcut menu click **Open Attribute Table**. The *Map Layer* **Attributes** dialog box will open (Figure B - 3), click (**Start Feature Edit Session**), close the *Map Layer* **Attributes** dialog box. Either way, the **Edit Toolbar** (Figure B - 4) on the LifeSim main window (Figure B - 4) now becomes active.

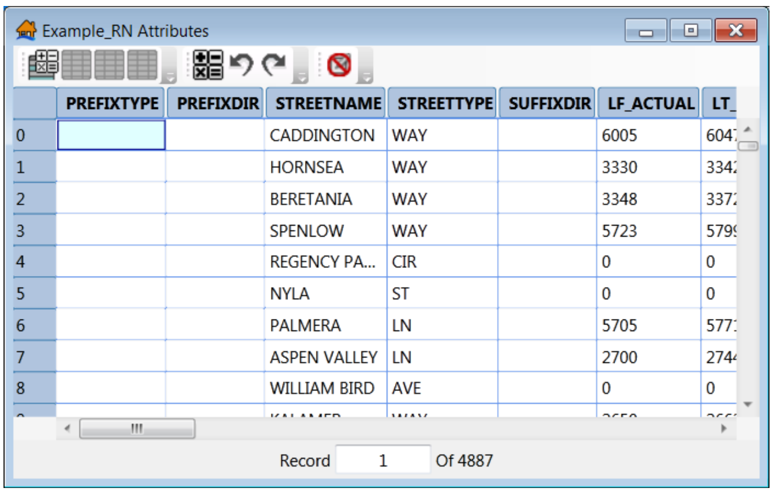


Figure B - 3. *Map Layer* Attributes Dialog Box

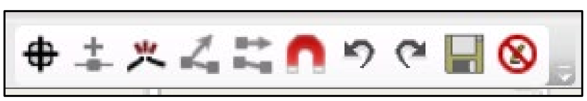


Figure B - 4. Edit Toolbar

3. Use the **Break Line Feature** tool (hot key **K**) to break the connected line segment away from the highway segment (Figure B - 5).

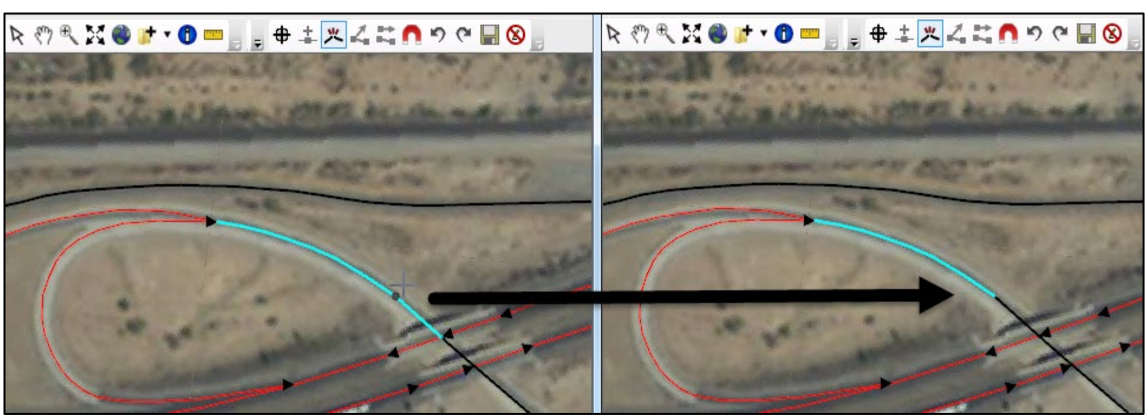


Figure B - 5. Break Line Feature Tool – Break a Road Segment into Two Segments

4. Now select the segment that shares an end point with the highway segment using the **Select** tool and delete the selected segment. Either right-click on the segment and from the shortcut menu click **Delete Selected Features** (Figure B - 6), or press **Delete** key.

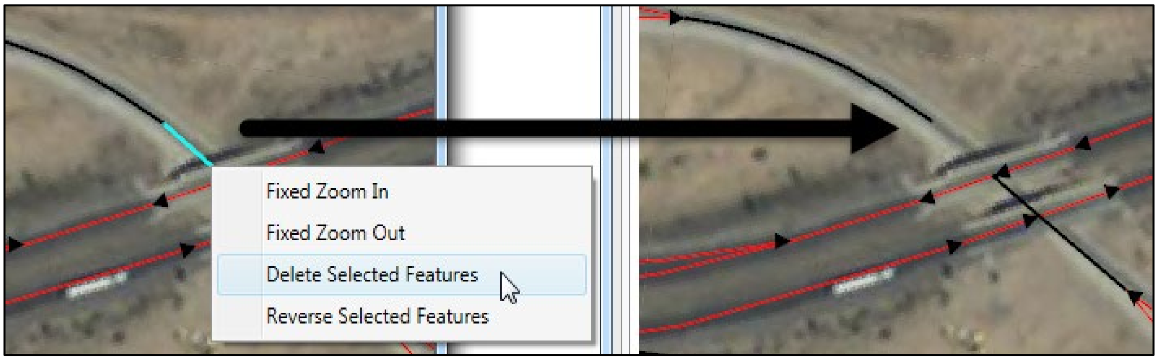


Figure B - 6. Road Network – Delete Undesired Road Segment

5. Now select the **Snapping** tool (Figure B - 4) to make sure that when re-connecting the line segment it will be properly connected by having the network points directly on top of each other. Click , a **Snapping** window (Figure B - 7) will open.

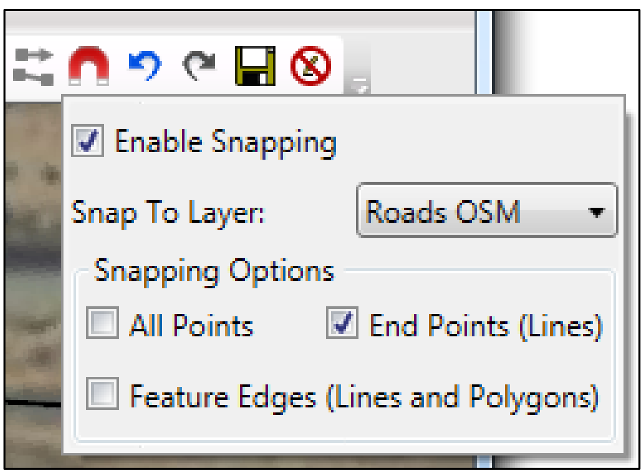
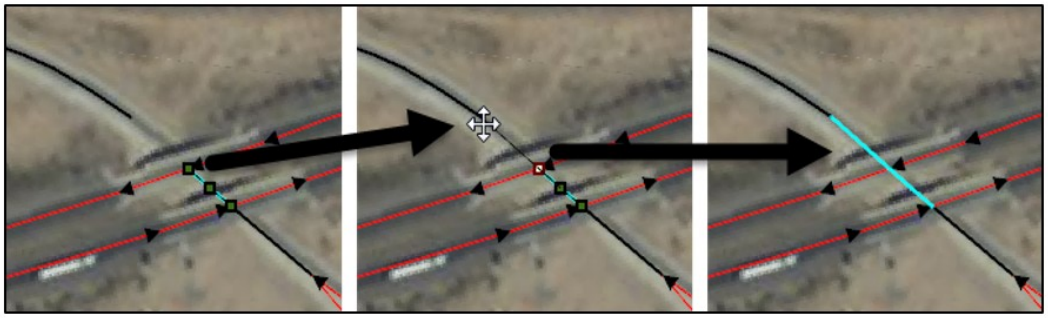


Figure B - 7. Snapping Options – Road Network – Reconnect Road Segments

6. Select **Enable Snapping** (Figure B - 7), from the **Snap To Layer** list, select the road network layer that will be edited (e.g., *Roads OSM*), and then select **End Points (Lines)** (Figure B - 7).

7. After snapping has been enabled, click , and double-click the other road segment that is connected to the highway end point to enter vertex editing mode (Figure B - 8a). Next, move the mouse over the vertex that will be connected, left-click, hold and drag to move the vertex towards the disconnected road segment. Once close enough, the indicator line will snap to the endpoint of the open road segment (Figure B - 8b). Release the left-click to finish moving the vertex. Either left-click away from the line segment or right-click and click **Finish Editing Vertices** from the shortcut menu to exit the editing session (Figure B - 8c).



(a) Enter Vertex Edit (b) Reconnect Roads (c) Exit Vertex Edit

Figure B - 8. Process for Reconnecting Road Segments – Better Network Connectivity

8. Now, the road segment does not share an end point with the highway road segments and in LifeSim vehicles will have to, appropriately, take the highway off-ramp to get to the underpass (Figure B - 9).

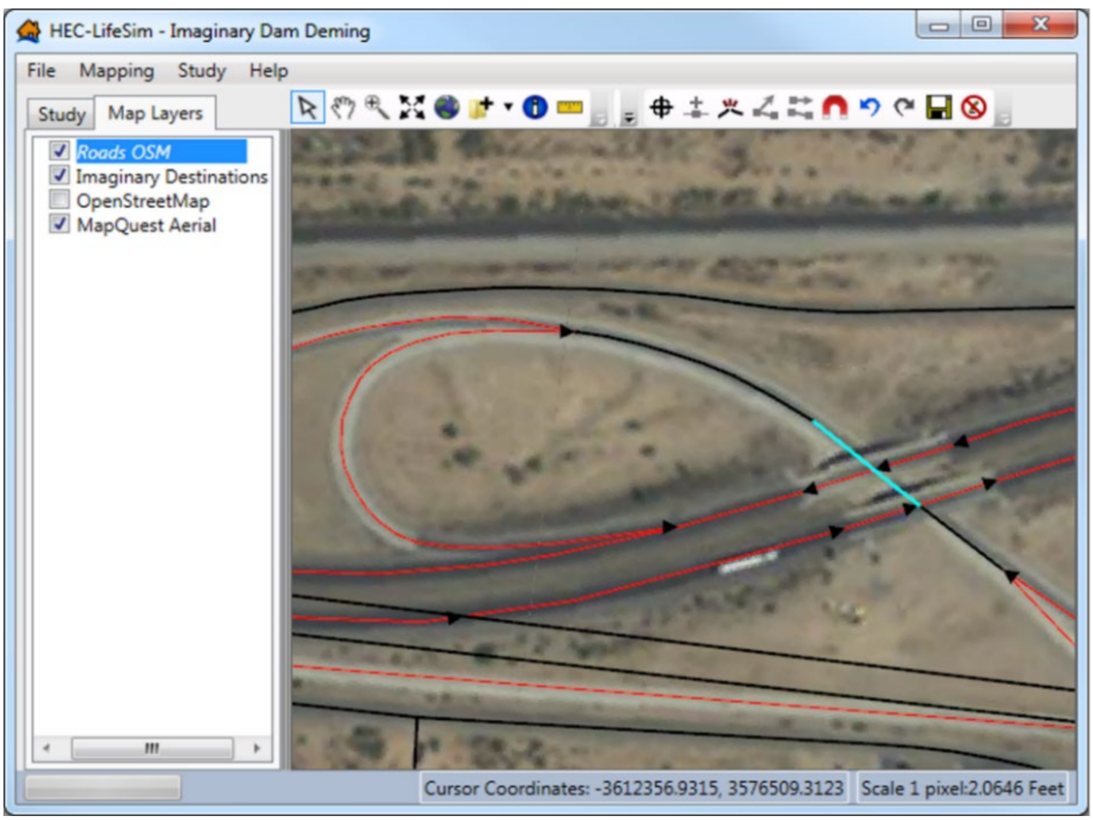


Figure B - 9. Better Network Connectivity – After Road Network Edits

9. This process should be repeated for any other roads in the road network that should not be connected.

10. Remember to save often (from the **Edit Toolbar** (Figure B - 4) click).To avoid spending an hour editing the road network and then get surprised with a computer restart or worse an unexpected error in the software.

B.1.2 Roads that should be Connected

Road segments that represent a continuous road can sometimes lack a common endpoint (Figure B - 10). When this happens, vehicles in LifeSim cannot move from one segment to the next when in reality the vehicles clearly should be able move freely.

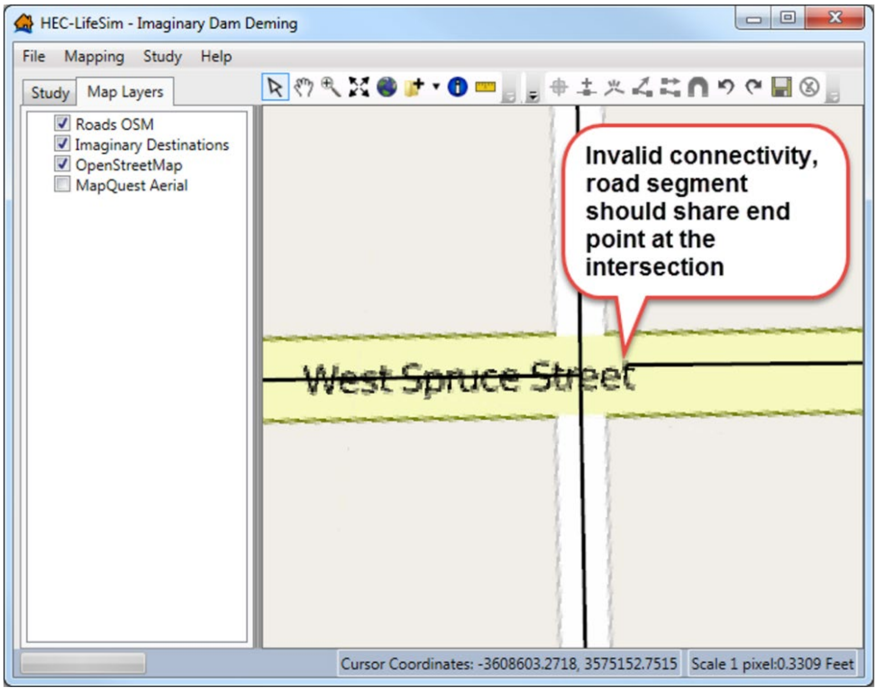


Figure B - 10. Invalid Connectivity – Road Segment End Point Interferes with Other Roads Connectivity

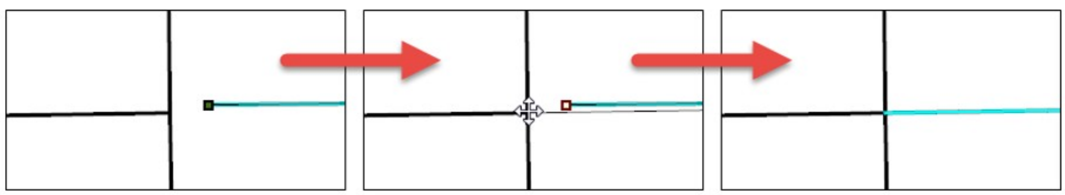
The following steps can be used to fix the connectivity issue:

1. A tool that can aid in finding connectivity issues is to display a shapefile that shows locations where a line segment's endpoint is not shared with another line segment. From the LifeSim main window (Figure B - 1), from the **Study** tab, from the **Road Networks** folder, right-click on a road network dataset. From the shortcut menu, click **Show Dangles in Map Window**, a shapefile that contains dangle locations will display in the map window of the LifeSim main window (Figure B - 1).

2. To connect the road segment at the West Spruce Street intersection, from the **Map Layers** tab (Figure B - 2), right-click a road network layer, from the shortcut menu click **Edit**. Another way to start an edit session is from the **Map Layers** tab (Figure B - 2), right-click on a road network layer, from the shortcut menu click **Open Attribute Table**. The *Map Layer* **Attributes** dialog box will open (Figure B - 3), click (**Start Feature Edit Session**), close the *Map Layer* **Attributes** dialog box. Either way, the **Edit Toolbar** (Figure B - 4) on the LifeSim main window (Figure B - 1) now becomes active.

3. Now turn on snapping, click, from the **Edit Toolbar** (Figure B - 4), a **Snapping** window will open (Figure B - 7). Make sure that the re-connected line segment is properly connected by having the network points directly on top of each other. From the Snapping window (Figure B - 7), click **Enable Snapping**, from the **Snap To Layer** list select the road network that is currently being edited (e.g., *Roads OSM*), and then select **End Points (Lines)**.

4. Click, double-click the road segment that is not connected to the intersection to enter vertex editing mode (Figure B - 11a). Next, move the mouse over the vertex to connect, left-click hold and drag to move the vertex towards the intersection. Once close enough, the indicator line will snap to the end point of the intersection (Figure B - 11b). Release the left-click to finish moving the vertex. Either left-click away from the line segment or right-click and click **Finish Editing Vertices** from the shortcut menu to exit the editing session (Figure B - 11c).



(a) Enter Vertex Editing (b) Connect Roads (c) Exit Vertex Editing

Figure B - 11. Process -Connect Road Segments – Better Network Connectivity

5. Now, the road segment that was previously disconnected will now allow traffic to continue on *West Spruce Street* (Figure B - 12).

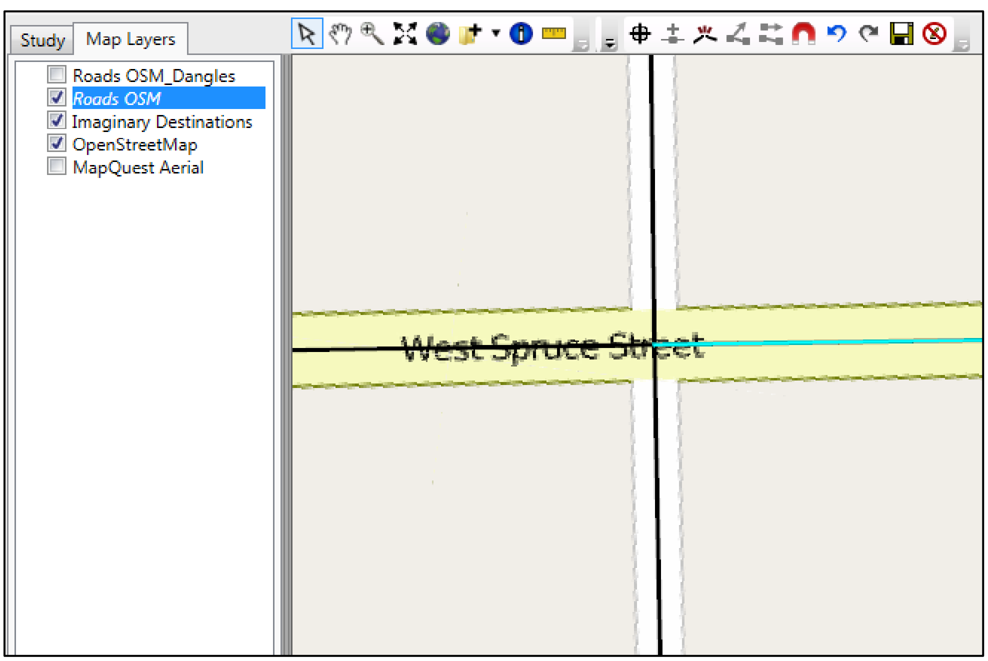


Figure B - 12. Example – Better Network Connectivity – Following Road Network Edits

6. This process should be repeated for any other roads in the road network that should be connected.

7. Remember to save often (from the **Edit Toolbar** (Figure B - 4) click).To avoid spending an hour editing the road network and then get surprised with a computer restart or worse an unexpected error in the software.

B.2 Verify One-Way Street Direction

LifeSim's traffic simulation engine allows road segments to either be one-way or two-way road segments. If a road is considered one-way, then vehicles can only travel in one direction along the road segment. The direction is defined from the first point in the road segment to the last point in the road segment and can be defined by enabling arrows to be drawn at the end of one-way roads through the road network map layer properties. For details on how to use the map layer properties refer to Section A.8 of Appendix A. Sometimes a road segment that is defined as one-way will be pointing in the wrong direction. This would mean that vehicles would have to find a way around the incorrectly oriented one-way segment. The following example (Figure B - 13) shows a case where a one-way highway segment is pointing in the wrong direction and needs to be fixed.

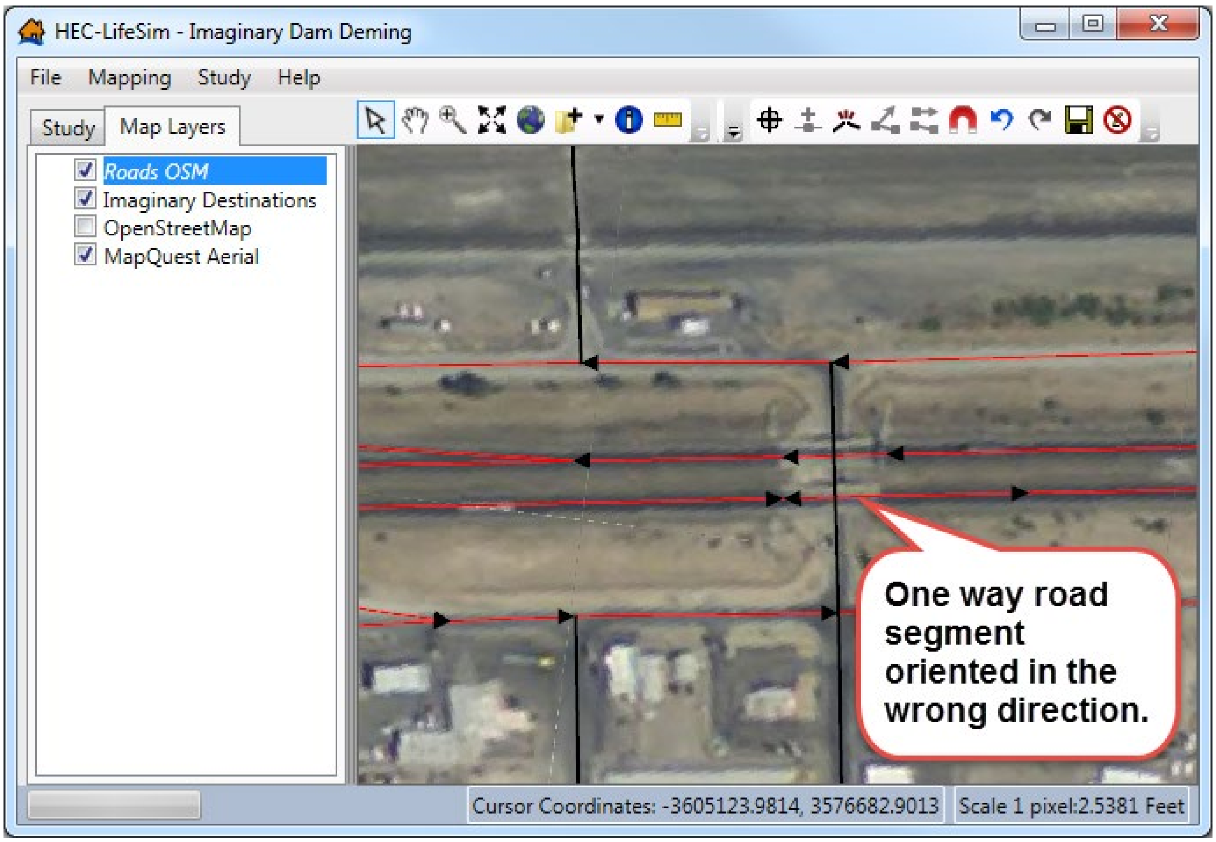


Figure B - 13. Example – One-Way Road Segment in LifeSim – Set in the Wrong Direction

To fix the orientation of a one-way road segment which is pointing in the wrong direction follow the steps below:

1. A tool that can aid in finding connectivity issues is to display a shapefile that shows locations where a line segment's endpoint is not shared with another line segment. From the LifeSim main window (Figure B - 1), from the **Study** tab, from the **Road Networks** folder, right-click on a road network dataset. From the shortcut menu, click **Show Dangles in Map Window**, a shapefile that contains dangle locations will display in the map window of the LifeSim main window (Figure B - 1).

2. From the **Map Layers** tab (Figure B - 2), right-click a road network layer, from the shortcut menu click **Edit**. Another way to start an edit session is from the **Map Layers** tab (Figure B - 2), right-click on a road network layer, from the shortcut menu click **Open Attribute Table**. The *Map Layer* **Attributes** dialog box will open (Figure B - 3), click (**Start Feature Edit Session**), close the *Map Layer* **Attributes** dialog box. Either way, the **Edit Toolbar** (Figure B - 4) on the LifeSim main window (Figure B - 1) now becomes active.

3. Once edit mode has been initiated, click , and select the road segment that is oriented in the wrong direction. With the correct segment selected, either press the **R** key, or right-click, and from the shortcut menu click **Reverse Selected Features** (Figure B - 14) to reverse the direction of the line segment.

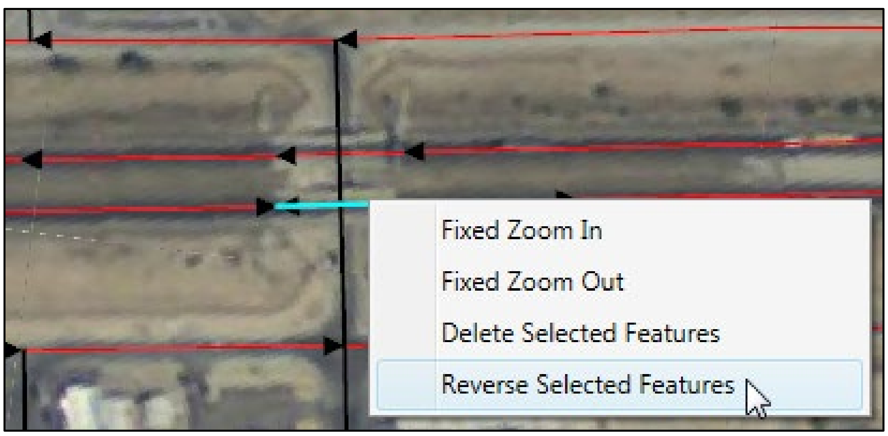
   

   Figure B - 14. Example – Reverse Selected Features – Shortcut Menu

   Note, reversing the direction of selected features applies to all features that are currently selected. Make sure that selected line features only contain lines whose direction is desired to be reversed.

4. Now, the one-way road segment that was previously oriented in the wrong direction (against the flow of traffic) will allow traffic to continue in the appropriate direction (Figure B - 15).

5. This process should be repeated for any other roads in the road network that are defined as one-way roads and are oriented in the wrong direction.

6. Remember to save often (from the **Edit Toolbar** (Figure B - 4) click).To avoid spending an hour editing the road network and then get surprised with a computer restart or worse an unexpected error in the software.

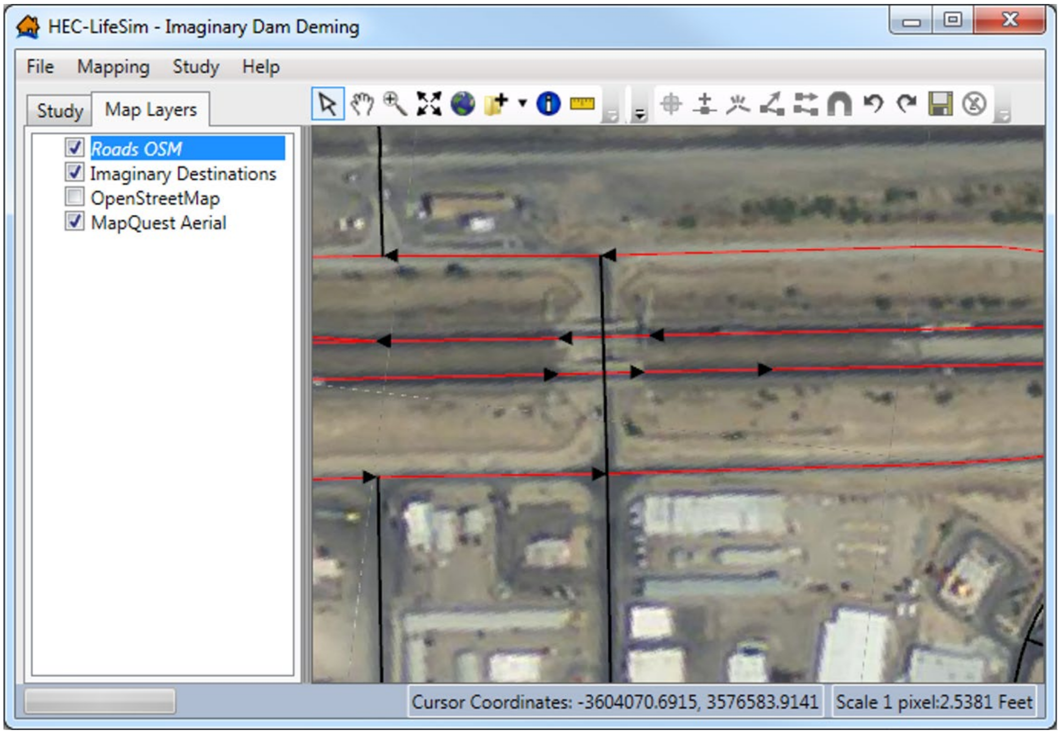


Figure B - 15. Example - Better Network Connectivity – Following Road Network Edits

Appendix C

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Appendix D

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